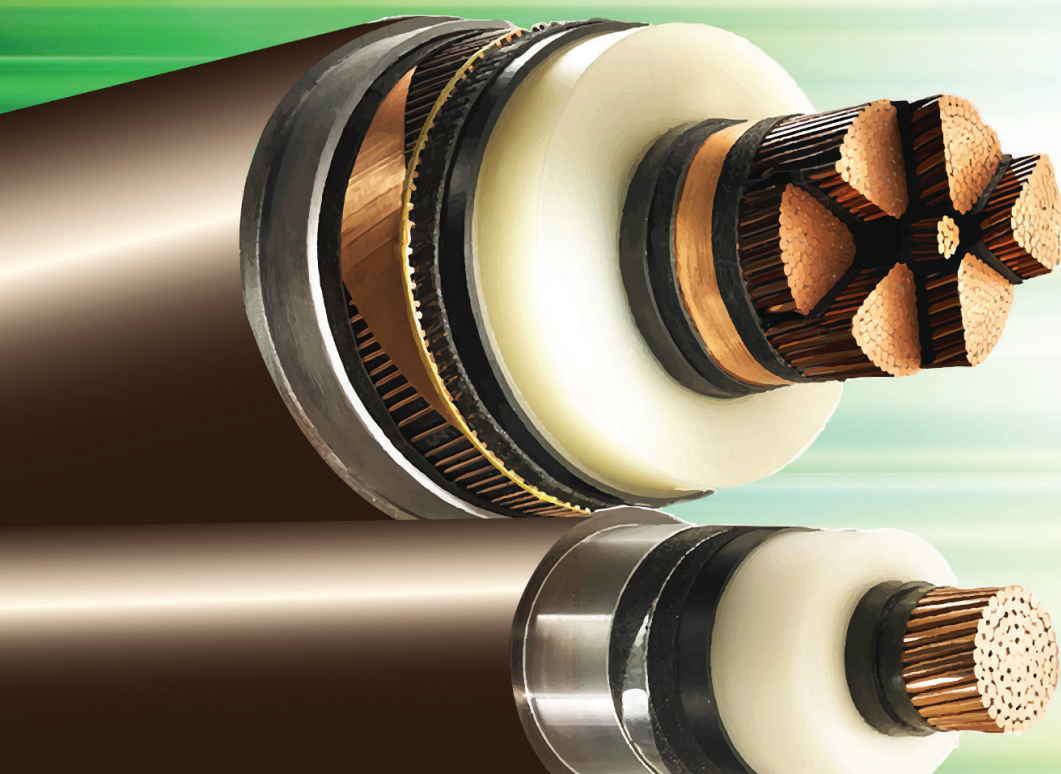


B1

TECHNICAL BROCHURE

Insulated cables



Power cable rating examples for calculation tool verification

Reference: 880



September 2022

Power cable rating examples for calculation tool verification

WG B1.56

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"The calculation of the temperature rise and load capability of cable systems"

J.H. Neher and M.H. McGrath

Power Apparatus and Systems, Part III, America Institute of Electrical Eengineers, Volume 76, no. 3, pp 752-764, October 1957



J.H. Neher



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Figure 1: Writers of the paper that formed the basis of the IEC 60287 standard

Acknowledgement

The topic of current ratings of power cables has been studied for a very long time. It was, is and will remain to be important that the current rating of power cables is calculated such that it is trustworthy. Many have worked to reach that goal increasingly well, building on the shoulders of experts before them. Let us remember the two experts who laid the foundations of the current rating calculations we have today, Mr. J.H. Neher and Mr. M.H. McGrath. Their contributions, refer to Figure 1 and [1], formed the basis of the standard IEC 60287 which is still the key reference document in nearly all of todays current rating calculations.

The working group would first of all like to express their sincere thanks to the external reviewers of this report, Mr. Coates and Mr. Dorison. Furthermore, the working group would like to thank the IEC TC20 WG19 for the good discussions we have had regarding our findings. Lastly, a special thanks to the many SC B1 members whose comments to our draft report helped us in the finalization of this document.
Thank you!

Executive Summary

The current ratings of transmission and distribution power cables for land and submarine environments are considered in this Technical Brochure (TB). Immediate previous work on this subject was reported in TB 640, the CIGRE “*Guide for current rating calculations of insulated cables*” [2]. As was recommended in that report and as experienced in the industry, there was a clear need to develop a method with which tools such as calculation spreadsheets, calculation techniques or calculation software could be verified to work consistently and in accordance with the IEC standards [3] , [4]. This Technical Brochure addresses that need.

The need to verify calculation tools, techniques and software comes from the difficulty in setting-up a calculation according to the IEC 60287 standards, and the requirement to deliver consistent and IEC-compliant calculations to the market place.

When a cable specialist sets up rating calculations using the IEC standards, the specialist needs to make several decisions regarding selection of correct equations and the choice of parameters. Often there is no clear guidance provided by the standards on how to proceed and this may lead to different calculation results for similar cable installations depending on the engineer and the tool used to make the calculation.

Utilities, project developers and others who are interested in buying a power cable from the market place to fulfill the requirements of a specific project, want to receive a product which has the ability to transmit a certain amount of electric power at a certain voltage level and therefore has a reliable current rating. To ensure the current rating is consistently deduced, calculations are usually based on relevant IEC standards. It is not possible to test the current rating as it strongly depends on the properties of the cable environment. Therefore the only information ensuring that a certain current rating is achieved in a project, is obtained from a current rating calculation.

Current rating calculations are therefore an important aspect to consider when a new power cable is required, and technical specifications often detail what calculations should be performed. Similar considerations apply when reviewing the current ratings of existing cable circuits. Thus, there is a need for guidance on how to interpret the IEC standards to ensure that current ratings are consistent regardless of the specialist carrying out the calculation and regardless of the tool used to perform the calculation.

Solution

In this TB, a verification procedure is reported to allow the validation of existing calculation tools and to ensure that current ratings can be computed consistently and in accordance with IEC standards. Eleven benchmark examples have been prepared by the experts in CIGRE WG B1.56, which together with some guidelines, form the central part of this brochure. These examples allow the user of the brochure to check and verify their final current rating as well as to compare the starting points, cable and installation parameters and calculated intermediate values. It should be pointed out that the interpretation of the IEC standards and the guidance given in this TB reflect the views of the experts of this WG and, in their opinion, reflect the current best practice in the calculation of current ratings.

Developing the guidance and case studies proved to be time consuming and difficult. The WG started with a simple case for which initially nearly all experts obtained different answers. Through discussion a consensus was reached on a number of points including interpretation of the standard, calculation methodology and input parameters until it was possible for all of the WG members to consistently produce the same results.

During the case study analysis a number of problems were identified with the IEC standard itself and these were referred to the IEC working group for possible amendment of the standard. These problems in part existed because the standard was developed before widespread availability of computers and its aim was to provide the equations that could be used in hand calculations. This led to many simplifications, which are no longer required. Additionally, several equations are only applicable to older types of cables, which may be in service today but are no longer manufactured. Many new cable constructions require special interpretation of the standard equations as gaps can be found in the IEC procedures.

The guidance was discussed at length in the WG meetings and with the IEC working group developing and maintaining the standards, who will further consider the findings and recommendations. The guidance aims to enhance and improve the IEC standards without changing them significantly.

This TB therefore is an 'add-on' to the IEC standards, providing guidance, updates and improvements with the goal of helping cable specialists to obtain the same results for identical cases. It allows a fair comparison between different cable suppliers, and a sound basis for current rating calculations and future innovations in calculation techniques and methodologies.

Future Work

The existing WG has agreed to continue its effort with a number of more complex and realistic case studies for which further calculation and interpretation difficulties must be overcome. The results are expected to be delivered in 2022, and will further ensure that engineers calculating current ratings provide the best possible answers.

Contents

Executive Summary	4
1. Introduction.....	11
1.1 Background.....	11
1.2 Key benefits	12
1.3 This brochure and the standards	12
1.4 This brochure and other work on CABLE current ratings	13
1.5 Referencing this brochure in technical specifications.....	13
1.6 How to use this brochure.....	14
2. Guidance	16
2.1 Overview.....	16
2.2 Rounding and accuracy	16
2.3 The standard to be used in the current rating calculations	17
2.4 Common pitfalls.....	18
2.5 Iteration in calculations.....	20
2.6 Simplifications in the IEC standard.....	20
2.7 Current rating calculations of HVDC cables	22
2.8 Deduction of dimensions (Interpretation of datasheet values)	23
2.9 Properties of materials	26
2.10 Electrical Capacitance.....	29
2.11 AC and DC Conductor resistance calculations.....	31
2.12 calculation of losses in metallic screens, sheaths and armours	39
2.13 Calculations for sheaths and screens	46
2.14 Calculations for armour	50
2.15 Thermal resistance calculations	56
2.16 Thermal resistance of Fillers in three core submarine cables.....	57
3. Interpretation of differences	60
4. Introductory Case study 0.....	61
4.1 Introduction.....	61
4.2 CASE #0: Input data	62
4.3 CASE #0: Guidelines for the iterative calculation.....	66
4.4 CASE #0-1: Results independent of the temperature.....	69
4.5 case #0-1: Results dependent on the temperature	72
4.6 Case#0-1: Possible variations	76
4.7 Case #0-2: Sub-case study with touching HDPE ducts.....	83
4.8 case #0-3: Sub-case study with PVC ducts in flat formation embedded in concrete	90
4.9 case #0-4: Sub-case study with cables laid in free air directly exposed to solar radiation	102
4.10 case #0-5: Sub-case study with cables in an unfilled trough	115
4.11 case #0: Summary table	125
5. Case study 1: Direct buried 132kV cables	131
5.1 132 kV cables in direct buried trefoil formation	131
5.2 132 kV cables in directly buried flat formation.....	146
6. Case study 2: a 30kV submarine array cable.....	164
6.1 Introduction.....	164
6.2 Calculation of lay-up factor of the cores	165
6.3 Calculation of the conductor AC resistance at operation temperature	165
6.4 Dielectric losses	167
6.5 Loss factor for sheath	168
6.6 Loss factor for armour	172
6.7 Thermal resistance T_1 between conductor and screen.....	174
6.8 Thermal resistance T_2 of the sheath around each core, fillers and bedding.....	176
6.9 Thermal resistance T_3 of outer covering	177
6.10 External thermal resistance T_4	178
6.11 Permissible current rating	178

6.12	Calculation of losses.....	179
7.	Case study 3: A 230kV HPFF cable	180
7.1	Introduction.....	180
7.2	Physical dimensions of cable system	181
7.3	Calculation of the conductor AC resistance at operation temperature	182
7.4	Dielectric losses	184
7.5	Loss factor for taped screen	184
7.6	Loss factor for skid wire	186
7.7	Loss factor for pipe loss increment	188
7.8	Thermal resistance T_1 between conductor and screen.....	188
7.9	Thermal resistance T_2 between the cable surface and inside of pipe.....	189
7.10	Thermal resistance T_3 of outer covering (pipe coating).....	189
7.11	External thermal resistance T_4	190
7.12	Permissible current rating	192
7.13	Calculation of losses.....	193
7.14	Calculation of cable temperatures	193
8.	Case study 4: A 33kV land cable	195
8.1	Introduction.....	195
8.2	Installation conditions.....	196
8.3	Calculation of the conductor AC resistance at operation temperature	196
8.4	Dielectric losses	198
8.5	Loss factor for Screen.....	199
8.6	Thermal resistance T_1 between conductor and screen.....	201
8.7	Thermal resistance T_2 between screen and armour	203
8.8	Thermal resistance T_3 of outer covering	203
8.9	External thermal resistance T_4	204
8.10	Permissible current rating	204
9.	Case study 5: A 400kV LPOF Cable.....	206
9.1	Introduction.....	206
9.2	400 kV LPOF cable in trefoil	206
9.3	400kV LPOF cable in Flat formation.....	219
10.	Case study 6: A 400kV single core AC submarine cable circuit	238
10.1	Introduction.....	238
10.2	Calculation of the conductor AC resistance at operation temperature	239
10.3	Dielectric losses	242
10.4	Loss factor for screen, lead alloy sheath and armour.....	243
10.5	Thermal resistance T_1 between conductor and screen.....	255
10.6	Thermal resistance T_2 between lead alloy sheath and armour.....	257
10.7	Thermal resistance T_3 of outer covering	258
10.8	External thermal resistance T_4	258
10.9	Permissible current rating	259
10.10	Calculation of losses.....	260
11.	Case study 7: A 320kV HVDC submarine bipole	262
11.1	Introduction.....	262
11.2	Introduction to rating calculation of HVDC cables	263
11.3	Calculation of the conductor DC resistance at operation temperature	265
11.4	Thermal resistance T_1 between conductor and the screen.....	265
11.5	Thermal resistance T_2 between lead sheath and the armour.....	267
11.6	Thermal resistance T_3 outer serving.....	268
11.7	External thermal resistance T_4	269
11.8	Permissible current rating-thermally limited.....	269
11.9	Calculation of temperature drop across the insulation.....	270
11.10	Calculation of field limited conductor temperature:.....	270
11.11	Permissible current rating-field limited	270
12.	Case study 8: A 220kV 3-core submarine export cable	271
12.1	Introduction.....	271
12.2	Core layup effect.....	272

12.3	Calculation of the conductor AC resistance at operation temperature	272
12.4	Dielectric losses	274
12.5	Loss factor for sheath.....	275
12.6	Loss factor for armour	278
12.7	Thermal resistance T_1 between conductor and screen.....	278
12.8	Thermal resistance T_2 between screen and armour	280
12.9	Thermal resistance T_3 of outer covering	282
12.10	External thermal resistance T_4	282
12.11	Current rating.....	283
12.12	Iteration	284
12.13	Final result	287
12.14	Calculation of losses.....	287
13.	Case study 9: A 110kV retrofitted cable	288
13.1	Introduction.....	288
13.2	Calculation of the conductor AC resistance at operation temperature	290
13.3	Calculation of Core layup factor.....	291
13.4	Dielectric losses	292
13.5	Loss factor for sheath.....	293
13.6	Loss factor for armour	296
13.7	Thermal resistance T_1 between conductor and screen.....	298
13.8	Thermal resistance T_2 between sheath and armour	300
13.9	Thermal resistance T_3 of outer covering	301
13.10	External thermal resistance T_4	302
13.11	Current rating.....	304
13.12	Iterative calculation of Current rating	305
13.13	Final result	309
13.14	Calculation of losses.....	309
14.	Case study 10: A 10kV three core PILC cable.....	311
14.1	Introduction.....	311
14.2	Calculation of the conductor AC resistance at operation temperature	313
14.3	Dielectric losses	315
14.4	Loss factor for lead sheath.....	317
14.5	Loss factor for the armouring.....	319
14.6	Thermal resistance T_1 between conductor and screen.....	321
14.7	Thermal resistance T_2 between sheath and armour	322
14.8	Thermal resistance T_3 of outer covering	322
14.9	External thermal resistance T_4	323
14.10	Permissible current rating	323
14.11	Calculation of losses.....	324
APPENDIX A. Mechanical neutral line deduction.....		328
15.	References	330

Figures and Illustrations

Figure 1: Writers of the paper that formed the basis of the IEC 60287 standard.....	3
Figure 2: Difference between axial distance and spacing between circuits.....	19
Figure 3: A typical datasheet.....	25
Figure 4: Representation of an unscreened belted two core cable	30
Figure 5: Representation of an unscreened belted three core cable	30
Figure 6: Definition of the length of lay.....	32
Figure 7: Comparison of IEC 60228 and IEC TR 62602 conductor resistance as a function of size, for class 2 stranded conductors	33
Figure 8: Difference between IEC 60228 and IEC TR 62602 conductor resistances shown in Figure 7.....	34
Figure 9: Example of the assembly of the segments of a Milliken conductor	38
Figure 10: Calculation of losses for one and three core cables.....	40
Figure 11: The loss calculations for single core cables without armour, solid bonded	40
Figure 12: The loss calculations for single core cables with non-magnetic armour, solid bonded.....	41
Figure 13: The loss calculations for single core cables with magnetic armour, solid bonded	41
Figure 14: The loss calculations for single core cables without armour, single point bonded or balanced cross bonded	42
Figure 15: The loss calculations for single core cables without armour, unbalanced cross bonded.....	42
Figure 16: The loss calculations for three core cables (SL type) without armour, solid bonded	43
Figure 17: The loss calculations for three core cables with non-magnetic armour, solid bonded	43
Figure 18: The loss calculations for three core cables with magnetic armour, solid bonded	44
Figure 19: The loss calculations for three core cables with a common sheath, solid bonded	44
Figure 20: The loss calculations for pipe type cables, solid bonded	45
Figure 21: Annular and spiralling sheath corrugation	47
Figure 22: Various forms / shapes of corrugation	47
Figure 23: Definitions in case of corrugated sheaths.....	47
Figure 24: Close up of corrugation type 3	48
Figure 25: Cable datasheet as used for case #0	63
Figure 26: Cross sectional drawing of cable for case #0.....	63
Figure 27: Illustration of the method of images	94
Figure 28: Cable datasheet 1x1200 mm ² CU/XLPE/CWS/PE 76/132 (145) kV	131
Figure 29: Cable datasheet 3x630 mm ² CU/XLPE/PB/SC PE/PPY/SWA/PPY 18/30 (36) kV.....	164
Figure 30: Cable data sheet: 1266,8mm ² pipe-type cable with laminated paper polypropylene insulation and stainless steel skid wires	180
Figure 31: Pipe-type cable trench configuration and dimensions.....	181
Figure 32: 33kV land cable, extruded, single core, copper wires screen	195
Figure 33: 400kV LPOF cable drawing (trefoil).....	206
Figure 34: 400 kV LPOF cable installation detail	207
Figure 35: 400 kV LPOF cable installation detail (flat formation)	221
Figure 36: Indicative cable drawing – 1x800 mm ² CU/XLPE/CWS/sc PE/FSWA/PPY 230/400 (420) kV	238
Figure 37: 320kV HVDC indicative cable drawing	262
Figure 38: Installation of two poles under seafloor	263
Figure 39: Distribution of electric field across the insulation at no-load and full-load conditions, specific for this cable.....	264
Figure 40: Cable datasheet 220kV 3x1000mm ² Submarine	271
Figure 41: Generic drawing of a 110kV cable with steel wire armour used for retrofitting.....	288
Figure 42: Cable data for a 110kV cable with steel wire armour used for retrofitting.....	289
Figure 43: Cable datasheet 3x95 mm ² AL/PILC/PVC 8/10 (12) kV	312

1. Introduction

1.1 Background

This Technical Brochure (TB) is the final report of the CIGRE Working Group (WG) B1.56 and discusses the verification of current rating calculation tools. It has been produced based on the recommendation of WG B1.35 as published in CIGRE TB 640, the '*Guide for rating calculations of insulated cables*' in 2015 and the reader should note that the detailed explanations, references and symbol list provided in TB 640 are also relevant here [2].

The report has been completed in the period 2016 to 2022 and provides detailed guidance and case studies with the goal of verifying techniques and tools used for the calculation of the current rating of a power cable. The report is intended to be used by cable specialists who perform calculations themselves or request current rating calculations from others. In this TB the term current rating will be used to refer to the current-carrying capacity of a power cable. Note that in American English the term used is usually 'ampacity', the combination of Ampère-capacity.

The current rating depends not only upon the electrical and thermal parameters of the cable itself but also upon the thermal parameters of the environment in which it is laid. It is well known that the different thermal resistivities of soil (chiefly depending on the density and moisture content) may strongly vary along the route and unless extensive measurements are taken, these parameters cannot be known with great detail.

It is also worth noting the ambiguity in the forecast of soil temperatures, both for the variations in environmental parameters and for the possible presence of external heat sources nearby to the route [2]. From an engineering standpoint, this justifies the fact that the current rating of a cable is not a unique value but it is more accurately a range. This range must arise from an engineering awareness of the current rating concept and not from arbitrary application of IEC 60287.

With the help of this TB, the user can verify a calculation technique, calculation tool or software tool before using it, providing a good basis for a reliable power cable current rating, as was recommended in CIGRE TB 640 [2].

Clarity on current rating calculations is much needed in the energy sector. Many in the cable community know examples where in tender evaluations for new cable projects, very similar (sized and constructed) power cables appear to have very different current ratings, sometimes with significant variations. Additionally, different commercially available tools and company spreadsheets provide different current ratings for similar cables in similar environments, sometimes varying depending on the engineer using the tool or technique.

This is illustrated once more by the experience of the Working Group who created this brochure. Having been selected as subject matter experts of their respective countries, calculation results for the first case studies still showed significant variations often with 15 different answers from 15 different countries. Through discussion a consensus was reached on the starting points, calculation method and final results and this formed the basis for the guidance in this Technical Brochure.

It thus became clear that performing current rating calculations is not straightforward, even with the IEC standards [1], [4] and the CIGRE TB 640 [2] to hand. The main reason for this lack of clarity is the difficulty in the interpretation of the applicability of the IEC formulae, especially because there are various details in the standards that are difficult to adapt correctly to modern cable designs and where additional guidance is required on how to use the calculation procedures outlined in the standards.

The existing IEC standards date from a time when hand calculations were customary. Today engineers use computers to calculate current ratings for power cables, whether this is using automated tools, spreadsheets, own or commercial software. The additional functionality leads to additional choices to be made on how to proceed, which may vary from tool to tool and from engineer to engineer.

The Working Group has agreed on the current rating and calculation method for several different case studies. This achievement was only possible by agreeing on 49 points of guidance when making the calculations. Some of these points are simple and straightforward, and some are more complex and elaborate, however this guidance is much needed to bring the clarity on how to calculate consistent current ratings for different power cables.

The agreed points of guidance, which provides the necessary clarity of performing consistent calculations, are provided in this TB as 'Guidance Points'. These 'Guidance Points' are accordingly referenced and marked as clarification to IEC, an addition to IEC or following a different approach than IEC. Though the guidance thus adds to the existing IEC standards, by following the guidance points the IEC basis is adhered to while removing the ambiguities where inconsistencies between calculations arise. According to the WG, all defined guidance points are required to resolve the calculation issues which would otherwise arise and would otherwise have to be resolved by individual cable specialists and software tools / calculation techniques.

1.2 Key benefits

The agreement between experts on how to calculate the current rating of selected power cables has a great benefit. With the guidance and case studies in this brochure, current rating calculation techniques and tools can be checked and verified. When discrepancies are found, these can be examined so that everyone in the industry can ensure they are using a consistent tool to calculate the current rating of the selected types of power cables. Given correctly working tools:

- **Fair comparison of calculations can be made in the contract-award phase of projects**
With the same assumptions and input parameters, and with the verified calculation technique presented in this technical brochure, current ratings for similar power cables (sizes and constructions) should be the same. Therefore when evaluating the bids of suppliers this should remove ambiguities in the calculation methods between bidders enabling a fair comparison.
- **Trust is provided in current rating calculations performed by different companies**
The use of verified calculation tools and software, both in design and engineering phases and during re-engineering or re-consideration during the operational life of power cables, prevents interpretation differences leading to discrepancies between calculations by different engineers or companies.
- **Confidence can be built up in a specific calculation technique, calculation tool or software**
Own, in-house or commercial techniques, tools or software can all be verified against this TB, clarifying their performance. Tools may be updated based on the findings, building up confidence. With this confidence in place, more challenging problems can be faced from a sound foundation.

1.3 This brochure and the standards

The key standards used in this technical brochure are the IEC 60287 series [3], [4]. Despite various issues encountered in applying the IEC methodology, these standards provide a widely accepted method to calculate the current rating of power cables, to which many engineers and scientists have contributed in the past, and will contribute to in the future.

The WG considered many details in the IEC 60287 standards and as a result identified a number of issues or 'gaps'. All findings were shared in a full-day meeting with the relevant IEC Working Group (WG19), which maintains the IEC 60287 series of standards. It was agreed that the issues identified by WG B1.56 will be taken into account in the future updates of the standards.

Changing and updating standards takes time and the user should note that it may take several years before an update is released that includes the recommendations of this WG. It is likely that not all of the considerations will be able to be included in the next planned update of the IEC standards, due to the timescales involved.

This TB is based on the IEC standards and closes gaps, removes interpretation issues as required to come to unambiguous results for the selected cable types. As such, this TB aims to enhance, complete and update the IEC standards where needed, sometimes with simple additions and sometimes with more involved changes. This TB is to be used as a complementary add-on to the IEC standards.

1.4 This brochure and other work on CABLE current ratings

This TB provides guidance and information on how to calculate the current rating of power cables and reflects the views of the experts of CIGRE WG B1.56 on how to interpret the IEC 60287 series of standards [3], [4]. The working group believes this is a correct and trustworthy basis for current rating calculations.

A guide for rating calculations of insulated cables

A reference is made to CIGRE TB 640 [2], which is a guideline for current rating calculations, discussing aspects such as how to obtain the starting points, how to perform calculations in difficult or special situations and to learn how others calculate current ratings. Together with the IEC standards, this document is intended to provide engineers with a sound understanding of how to perform current rating calculations in specific situations.

Armour losses

Another reference is made to currently ongoing work in CIGRE WG B1.64 on the calculation of the armour losses. It is anticipated that this WG will propose an update on how to improve the evaluation of these losses. Similar improvements to the calculation methods may be happening in other areas as well.

Innovations in calculating techniques

Innovation may lead to the improvement of the calculation methodologies, however, this goal is different from the goal of standardisation of computations which is focused on in this brochure. Standardised calculations have the aim of providing a fair comparison for companies to select power cables between suppliers, and to provide clarity on the current rating of a cable system. It should be noted that in some places this TB applies a conservative approach, which in future may become outdated.

It is recommended that calculations are made according to the most recent IEC standard supported by the methodology outlined within this TB. Additionally, a calculation could be made with a new, innovative methodology, which must then also be described in full when the calculation results are provided. This allows innovations in calculation methods to be taken into account, while maintaining the opportunity for a fair comparison to be made between different suppliers.

Future work

A last reference is made to future work of this WG. The existing WG has agreed to continue its efforts towards verification of the case studies for more complex situations. These more complex case studies will probably result in the analysis of cases which are not covered by the IEC standards, but which represent very realistic situations. Based on the current agreement between the experts on how to perform the calculations, the group will continue their work until a new report is issued, most probably in 2022-2023.

1.5 Referencing this brochure in technical specifications

To ensure the correct relation between IEC standards and this CIGRE Technical Brochure, it is proposed to use the following wording in a technical specification for a power cable project:

"All current rating calculations shall be performed with a verified current rating calculation tool and on the basis of IEC 60287. Where CIGRE TB 880 provides additional guidance compared to IEC 60287, this shall be followed. If the TB 880 provides a different interpretation of the IEC calculations, it is suggested to follow the TB 880 guidance"

Such wording ensures that a fair comparison can be made between bidders on a power cable project. There is a risk that if only the IEC standards are referenced, that suppliers may interpret them so as to give their bid preference.

Evidence could be requested to show which calculation tool and version has been used, and to show how the tool aligns with the guidance provided in this TB, for example in the form of a verification report. This will provide clarity on the performance of the tool and subsequently the quality of the calculated current rating.

1.6 How to use this brochure

In order to verify a calculation technique, tool or software, the following procedure is recommended:

1. **Read the guidance in Section 2.** Verify that no obvious errors have been made, and that the interpretation of the tool is aligned with the provided guidance.
2. **Calculate the current rating of the first case study** reported in Section 4 with the selected tool. This is a simple case study, which is calculated with many different variations and the method of performing the calculations is clarified in detail. The case study also shows how different cable environments lead to different calculation results.
3. **Select a number of case studies and perform the case study calculations.** It is noted that the case studies comprise many different cable types and installation conditions, refer to Table 1-1 for details. For some tools, a subset (e.g. power cables for distribution voltages) may be adequate, however for many tools it should be possible to calculate all or most of the case studies. Case studies have been selected on the basis that each follows a different set of calculations and therefore the more case studies that are used, the better the verification will be. Where possible it is recommended that the full set of verification case studies are used.
4. **Evaluate the findings using Section 3,** and update / improve the tool as required. Strive to find the same answers as given in this TB.
5. **Report** the similarity of the tool's results against the results reported in this TB in full detail for later reference / evidence. Provide the verification report with the calculated current rating.

It is noted that in this TB, all calculations are performed using a large number of digits, usually a minimum of 10 significant digits as described in Section 2.2. The only reason for this, is to allow the user to easily find errors and deviations. Deviations should be considered (see Section 3) and may lead to improvements. In one situation, an error can result in only a minute difference, while in another case the difference can become significant. The user is advised not to consider the 10 significant digit calculation results as a realistic accuracy. Guidance on the accuracy of the calculated current ratings is given in Guidance Point 2.

Table 1-1: Overview of case studies included in this brochure

Case Study	Description	Brochure Section
0 Intro	• 132 kV XLPE insulated power cable with smooth aluminium sheath • Trefoil and flat formation • Cables directly buried, in ducts, in concrete troughs, in backfill, in free air with and without solar radiation • Single end bonded	4
1	• 132 kV XLPE insulated power cable with copper wire screen and aluminium foil • Direct buried • Trefoil, both ends bonded • Flat formation, cross bonded	5
2	• 30 kV three core submarine power cable with XLPE insulation (array cable) • Buried in the sea floor • Both ends bonded	6
3	• 230 kV pipe-type cable comprising 3 cables in a steel pipe • Typical North-American practice • Buried in a trench surrounded by a special backfill • Using a load factor	7
4	• 33 kV XLPE insulated land cable with less than 50% covering wire screen • Direct burial in trefoil	8
5	• 400 kV single core low pressure oil filled land cable with corrugated aluminium sheath • Direct burial with flat and trefoil formations • Single end bonded	9
6	• 400 kV single core XLPE power cable for submarine applications • Direct buried in the sea floor in flat formation • Both ends bonded	10
7	• 320 kV HVDC cable for submarine applications • Limitation on the temperature difference over the insulation material • Direct buried in the sea floor in flat formation	11
8	• 220 kV three core submarine cable (typical export cable) • Directly buried in the sea floor	12
9	• 110 kV XLPE insulated three core cable in a PE-covered steel pipe • Armoured cable for land applications • Directly buried • Both ends bonded	13
10	• 10 kV, three core paper-insulated lead-sheathed power cable (PILC) with sector shaped conductors for land applications • Directly buried • Both ends bonded	14

2. Guidance

2.1 Overview

In this section, guidance has been provided in order to assist the user in performing calculations for the case studies in accordance with the methods recommended by the experts of this WG.

As a starting point, the IEC current rating standards [3], [4] are followed to perform the calculations. However, there are situations in which the standards are not applicable, give an ambiguous answer on how to perform the calculations, or where the calculations recommended by IEC are not fully correct. Guidance is provided to the user on how to proceed in these situations. This also means that some guidance points will clarify the existing standards, some guidance points will add calculations or assessments to the existing standards, and some guidance points will adapt an aspect of the existing standards.

With this guidance in place, the current rating calculations for the case studies can be performed and should lead to the same result, regardless of the person making the calculations or of the tool used.

The guidance in this Section has led CIGRE WG B1.56 to also make a number of recommendations to the IEC for improvements to the standards. These recommendations have been provided separately to the IEC, including through direct discussion with IEC TC20 WG 19 and the IEC are considering these recommendations for inclusion in future updates to the standards.

When conducting current rating calculations and when verifying calculation tools, various general aspects need to be clarified. In the following sections various topics will be discussed starting with more generic topics as rounding and accuracy, the standards to use, common pitfalls and the way to perform iterative calculations. Subsequently, the simplifications in the standards and the application of the standard to HVDC cables are discussed. This is followed by sections on the deduction of input parameters, namely cable dimensions, material properties, electrical capacitance and electrical resistance. Calculations of losses in screens, sheaths and armours, and details on these calculations, plus details on the calculations for thermal resistance conclude this chapter.

2.2 Rounding and accuracy

A first guidance point relates to the rounding of numbers, especially for the intermediate results.

Guidance Point 1: Rounding errors in the intermediate numbers

Issue	Rounding errors in the intermediate numbers may lead to an increasing calculation error, especially in the case of iterative calculations.
Guidance	Do not round intermediate calculation results, nor the value of constants like ' π ' (Pi) inside a calculation.
Notes	This report shows the effect of rounding numbers in various places. All case studies contain in all calculations at least 10 significant digits to be able to check results.
Relation to IEC	This is not related to IEC.

When conducting a current rating calculation or setting up a calculation tool, the precision of floating point numbers (real numbers) requires attention. The number of significant digits differs between 32 and 64 bit compilers and possibly depends on other hardware and software aspects. As an example, the number of significant digits will not be more than 7 with single precision floating point numbers (*float* in C or Java, *real*4* in Fortran), while 16 digits are expected with double-precision floating point numbers (*double* in C or Java, *real*8* in Fortran). It is noted that commonly used spreadsheets can also produce unexpected results when a large number of significant figures is used.

In this brochure, all results are given to a high number of digits (often 10 digits) to enable intermediate results to be verified. Differences can appear in e.g. the 5th decimal place, and verification up to these details can aid the user in pinpointing the origin of the differences. The workgroup has been able to

spot and remove errors in calculations in this way which in some cases may only lead to small differences, but which in other cases can become quite influential.

How to interpret and how to deal with differences found, also with minute differences, is discussed in section 3.

Although this technical brochure provides results often in 10 significant digits, it does not mean that the current rating is actually known with this level of accuracy. Guidance Point 2 provides guidance on the accuracy of rating calculations.

Guidance Point 2: Accuracy of rating calculations

Issue	What is the accuracy of the calculated current rating?
Guidance	The calculated current rating will be a worst case value providing all input parameters are set correctly and that the guidance in this TB is followed. Input parameters will have a limited accuracy and this will lead to a limited accuracy of the end result. Care should be taken to accommodate inaccuracies in input parameters in the cable design, a discussion is given in CIGRE TB 640 [2]. When considering the accuracy of the calculations only, it is advised to round the final current rating as follows: • Round down to the nearest 1 A for ratings up to 200 A (example: 156.7 A becomes 156 A) • Round down to the nearest 5 A for ratings from 200 to 500 A (example: 459.3 A becomes 455 A) • Round down to the nearest 10 A for ratings over 500 A (example: 1234.9 A becomes 1230 A)
Notes	Although a calculation can be made to great precision (e.g. 10 digits), the resulting current rating is not that accurate. Approximations are inherently made when applying the IEC standard approach since, for example, various calculation routines are based on experiments made many decades ago which have limits to their accuracy. The largest inaccuracy is usually related to the values used for the soil thermal parameters. It is often worth spending more time and effort to understand these parameters better, rather than to spend time on improving calculation routines. Refer to TB 640, section 2.3 [2] for a thorough discussion on this, and Section 4.8.7 in this TB for a sensitivity analysis on the input parameters of a current rating calculation, but note that the results of such a sensitivity analysis are case dependent.
Relation to IEC	This is an addition to IEC.

2.3 The standard to be used in the current rating calculations

The standard to be used in the current rating calculations is the latest version of the IEC 60287 [3], [4] and this should be accompanied with the guidance given in this TB. Calculation tools should adhere to the latest version of the IEC standards, and hence must be updated regularly, refer to Guidance Point 3.

Note that if the standard does not cover a specific situation which is to be calculated, CIGRE TB 640 [2] may give information on how to proceed.

Guidance Point 3: The standard to be used

Issue	Standards are updated regularly leading to tools based on different versions of standards.
Guidance	The standards for current rating calculations (including IEC 60287 [3], [4]) are frequently updated. It should be ensured that the latest version of the standard is used by the calculation tool. 'Redline' versions are very useful for checking the changes in the standards. When developing a calculation tool, it is important to give version numbers to the calculation tool and to record and report the version of the IEC standards and CIGRE guidelines used in the calculations.
Notes	This TB uses the latest standards as a basis at the time of writing this document, refer to [3], [4]. As standards and guidelines are frequently updated, it is important that the calculation tool is updated too. The tool should give a version number and record the date of the latest update which should be changed when a CIGRE guideline or IEC standard is updated or renewed.
Relation to IEC	This is an addition to IEC, following the same approach.

2.4 Common pitfalls

This section reviews potential sources of errors, not only coded by the developers of the calculation tools, but also made by users and identifies a number of common pitfalls that must be avoided.

Guidance Point 4: Avoidance of common errors and pitfalls

Issue	Avoiding common pitfalls made when performing current rating calculations.
Guidance	Make sure common mistakes are avoided by verifying the calculation tool against those listed in this section. Ensure that the user of the tool interprets the variables in the correct way.
Notes	This section provides a number of common pitfalls.
Relation to IEC	This is an addition to IEC.

General mistakes:

- IEC Standards mix length units: diameters and thickness are expressed in millimetres (mm), while the engineer usually relies on depths and axial distances expressed in metres (m) given in a dimensional drawing. The outer diameter of the cable D_e^* used in the calculation of the external thermal resistance of cable in still air (Section 4.2.1.1 of [4]) has an asterisk to remind that the variable is expressed in meters. The confusion is also possible in the calculation of $u = 2L/D_e$ for T_4 of buried cables (Section 4.2.2 of [4]).
- Confusion is possible for developers or users concerning the angles provided in degrees and provided in radians for trigonometric calculations (e.g. the default unit used by Microsoft Excel is radian).
- Confusion is possible between natural logarithm (base e) and the base 10 logarithm.

Mistakes when using the IEC standard:

- In some cases, the IEC standard provides relevant information in another section than where that information is needed. For example, in the case of three single-core cables in touching trefoil with metal sheath (Section 4.2.4.3.2 of [4]), the section describes the calculation of the external thermal resistance T_4 , with the following remark: "In this case, the thermal resistance of the serving over

the sheath or armour, T_3 , as calculated by the method given in 4.1.4 shall be multiplied by a factor of 1,6". However, there is no mention about this in the related section of T_3 .

- Attention must be paid to some variables such as the conductor diameter d_c . Depending on the context, the inner semiconducting layer may be included as part of the conductor diameter (calculation of the capacitance C according to [3], Section 2.2) or not included as part of the conductor diameter (calculation of the thermal resistance T_1 according to [4], Section 4.1.2.1). There is no possibility of confusion for the engineer, but the use of the d_c variable with different values may provide a source for error for a developer of a calculation tool.
- Confusion is possible with the values of electrical resistivity of aluminium, which are different for the conductors ($2,8264 \cdot 10^{-8} \Omega \cdot m$) and the sheaths ($2,84 \cdot 10^{-8} \Omega \cdot m$).
- The calculation of the thermal resistance between conductor and sheath (T_1) and between sheath and armour (T_2) includes geometric or screening factors ($G, \bar{G}, K$) for belted or screened cables. The factors can be obtained graphically or from the formula given in the standard. Many discrepancies between users may occur at this stage. It is recommended to use the formula given in the standard to calculate these factors.

Common mistakes in cable configuration parameters:

- IEC standards consider the L parameter as the depth of the centre of a circuit. The user must ensure the correct depth is used, rather than other possible and sometimes more intuitive values such as: bottom of the trench, top of the upper cable, etc.
- In the case of cables in ducts, the centre of the duct is higher than the centre of the pulled cable. It is the depth of the centre of the duct, or duct formation if in trefoil, that is used as the circuit depth. The heat source is the duct (with respective diameter and depth).
- Some calculation tools truncate input values and this influences the verification or comparison of the case studies. For example, the depth of the circuit or the axial distance between circuits can be truncated with an accuracy of a centimetre. Practically, the millimetre accuracy cannot be achieved, but in order to allow a comparison of the intermediate parameters between a calculation tool and the case studies in this TB, truncation is not used. Refer also to the guidance given in Section 2.2.
- Similarly, the horizontal distances between cables and circuits shall ensure a certain homogeneity. Mixing axial distances (distance between cable centres) and separation distances (free space between cable, ducts, backfill or concrete bank) increases the risk of confusion, refer to Figure 2.
- When performing calculations for the mutual heating of the cables, the distances between the pairs of cables should be used rather than the distances between the circuits.

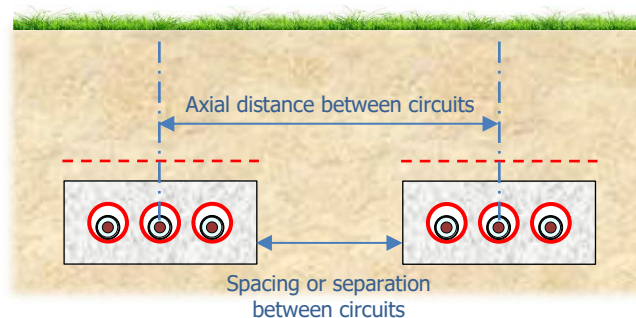


Figure 2: Difference between axial distance and spacing between circuits

2.5 Iteration in calculations

In many current rating calculations, iterations are required to come to the final result. An example is the starting temperature used to calculate the resistance of a metallic sheath to obtain the losses. The losses are needed to calculate the current rating, which subsequently can be used to calculate the temperature distribution within the power cable. The resulting sheath temperature may be different from the starting value, and hence iterative calculations must be performed.

The iterations are made on conductor temperature and on current. Additionally for cables in air, iterations are made on the air gap temperature.

The precise way to iterate is illustrated in various case studies in this TB. In various cases, a relaxation of parameters in the iteration process is required to reach convergence more quickly. An extensive illustration on how to make iterative calculations is provided in Section 4. Iterations must be made until a certain convergence is reached, refer to Guidance Point 5.

Guidance Point 5: Iterations - criteria for convergence

Issue	When do you stop with iterations in calculations?
Guidance	Iterations should be made at least up to reaching the following convergence between iteration steps: • Difference between the starting and final temperatures: ≤ 0.001 K • Difference between the starting and final current ratings: ≤ 0.001 A
Notes	When an automated calculation method is used, there is no need to relax these values. Note that stricter convergence criteria up to 10^{-10} were used in this TB to allow easier location of errors.
Relation to IEC	This is an addition to IEC.

2.6 Simplifications in the IEC standard

The IEC standard [3], [4] contains three key simplifications, as addressed in Guidance Point 6, Guidance Point 7 and Guidance Point 8. These simplifications have been created in the past most importantly to allow hand calculation of the current ratings. Nowadays, the requirement to perform calculations by hand has disappeared and with it the reason to use such simplifications.

In addition, the power cable designs have changed, which leads to situations where the simplifications do not hold true anymore. This is shown in various case studies in this TB.

To avoid problems of validity with the existing simplifications, it is recommended not to simplify at all. This leads to the below points of guidance.

Guidance Point 6: Eddy current losses

Issue	When to take eddy current losses into account?
Guidance	How to deal with the simplifications in the calculations of these eddy current losses? Eddy current losses should not be neglected, and should always be taken into account. Contrary to the IEC 60287 [3], this means that these eddy current losses must be calculated in each situation when there is a continuous metallic sheath. For this, refer to section 2.6. In the case of solid bonding, the eddy current losses must be multiplied by a factor F, refer to Guidance Point 31. Some simplifications are included in the calculation of eddy current losses according to section 2.3.6 of [3], namely: • Note 1 in Section 2.3.6.1

Notes	Below note 3, where $\Delta 1$ and $\Delta 2$ are neglected if $m \leq 0.1$ Although these simplifications result in small errors, it is recommended not to make them if it is not necessary to do so. If eddy current losses are ignored the current rating will be optimistic. The effect is not always negligible. In Case Study 0, an example of this can be found. In the case of circulating currents and eddy currents, circulating currents will reduce eddy current losses leading to a correction factor. Guidance on this can be found in Guidance Point 31.
Relation to IEC	This is an approach differing from the approach of IEC.

Guidance Point 7: Dielectric Losses

Issue	When to take dielectric losses into account?
Guidance	Dielectric losses should not be neglected, and should always be taken into account. Contrary to the IEC 60287 (refer to the note in table 3 of [3]), the dielectric losses should be calculated regardless of the voltage level.
Notes	If the calculation result shows that the dielectric losses are negligible this guidance point has no effect, if the calculation shows that the dielectric losses are not negligible, this guidance point ensures no wrong result is achieved.
Relation to IEC	This is an approach differing from the approach of IEC.

Guidance Point 8: Simplification formulas for T_4

Issue	What formula for the calculation of T_4 to use (IEC 60287 [4], Section 4.2.3.3.2 and further)?
Guidance	Although cables may be <u>equally loaded</u>, they typically do not have <u>equal losses</u>, which is of key importance in the current rating calculations. The reason for this is that there are temperature differences between the various cable layers in a 3-phase system, and therefore there will be differences in the losses in the conductors, screens, sheaths and armours. Differences in the sheath losses also arise because of the relative position of the cable phases. <u>Non touching cables</u> The user is recommended to use Section 4.2.3.2 of [4], the general method for unequally loaded cables, to calculate T_4 in the case of both equally and unequally loaded non-touching cables. Do not use the simplified formulae in 4.2.3.3. <u>Touching cables</u> The principle of superposition does not apply for touching situations. In this case the user is recommended to use Section 4.2.4 of [4] (including correction factors for T_1 and T_3) to ensure that the disturbed heat fields are taken into account. Where there is a combination of touching and non-touching cables, T_4 should be calculated using Section 4.2.4 for circuits with touching cables and the method of Section 4.2.3.2 should be applied to deal with mutual heating between circuits. A simplification is comprised in the calculation of T_4 which is the use of $2 \cdot u$ instead of $u + \sqrt{u^2 - 1}$, where u is larger than 10. Although this simplification leads only to small errors, it is recommended to use the full equation.

Notes	It is good practice to ask whether the losses are equal, rather than whether the loading is equal, as the current rating calculation considers heat which is created by the losses. The findings of the working group have been delivered to the IEC for further consideration. Note that simplified formulae in [4] sections 4.2.3.3.2, 4.2.3.3.3 and 4.2.3.3.4 of T₄ were directly derived from the general formula 4.2.3.2 therefore the original formulae can be used. It is noted that caution is needed when cables are closely spaced but not (yet) touching. In such cases, the spacing may not be large enough to use the superposition principle. The improvement this guidance point offers may be limited.
Relation to IEC	This is an approach differing from the approach of IEC.

2.1 Current rating calculations of HVDC cables

In the case of HVDC cables, there are two limitations that must be observed. One is the maximum conductor temperature, similarly as in the AC case and the other is the limitation of the electric field strength in the insulation. The latter can be shown to be equivalent to limiting the temperature drop across the insulation.

In an HVDC cable, the electrical field is determined by the resistive properties (governed by the specific electric resistivity ρ) of the insulation material, contrary to a HVAC cable where this is determined by capacitive properties (governed by the dielectric permittivity ϵ). The specific thermal resistivity is significantly dependent on temperature, and because this temperature is not constant throughout the insulation material, the electrical field is not constant and changes with changing cable loading situations. In order to ensure that the electrical fields in the insulation material remain within limits, the temperature difference over the insulation material is limited. This limit may be more stringent than the limit on the maximum conductor operating temperature with typical values between 10 and 15 K.

It is noted that the maximum temperature difference over the insulation material of a specific cable is an important parameter during type testing, and should not be surpassed during cable operation.

To take this additional limitation into account, the following guidance is given.

Guidance Point 9: HVDC cable current rating calculation

Issue	How to calculate the current rating of an HVDC cable?
Guidance	A second temperature limit should be taken into account on the temperature difference over the insulation material excluding semiconducting screens. The precise limit is defined in the type test of the specific HVDC cable under consideration and should be supplied by the manufacturer. Both the limits on the operating temperature and the temperature difference over the insulation material must be obeyed at all times.
Notes	The limit on the temperature difference over the insulation material results from electric field control in an HVDC cable. In a type test, the cable is tested for a certain maximum temperature difference over the insulation material, which should not be surpassed during cable operation.
Relation to IEC	This is an addition to IEC.

Guidance Point 10: HVDC cable effect of harmonics

Issue	How to treat harmonics from the converters
Guidance	When AC is converted to DC there will be some harmonics in the DC current. These harmonics will induce losses in screen (λ_1) and give a some skin and proximity effects (y_s and y_p) to the conductor resistance. In modern converters these harmonics have low amplitude compared to transmitted current while in older version of converters the harmonics might have a significant impact to the rating. Today IEC 60287 do not have any formula of how to include losses from converters and in most cases the harmonics are not known for the cable designer so the suggestion is to put the factors to 0. I.e. $\lambda_1 = 0$, $y_s = 0$, $y_p = 0$.
Notes	If harmonics are known it is always a good approach to include the effect of them.
Relation to IEC	This is a clarification to IEC.

The workgroup further notes that DC conduction losses through the cable insulation are believed to be negligible, unless otherwise stated by the manufacturer.

2.2 Deduction of dimensions (Interpretation of datasheet values)

When setting up a current rating calculation, one of the first considerations concerns what dimensions should be used. These dimensions can result from:

- **Factory acceptance test results**
This data provides actual measured dimensions of how the cable was made. It should be noted that these measurements may have been done on a limited amount of cross sections and thus, variation is possible. This is seen as the best set of dimensional data that could be available.
- **Cable factory data sheets**
These sheets provide a good description of the cable design, but do not necessarily reflect the manufacturing tolerances or the actual deviation from the design of the cable system to that which is actually put into production.
- **Specifications and industry standards**
Generally, cables of a given voltage class and conductor size will have characteristics falling within the parameters of industry standards and specifications. If better sources of information are not available due to lost or unavailable records, this may be the only alternative.

A further discussion on these sources of data is provided in Section 3.1 of [2].

The case studies in this TB have been made based upon available, and thus realistic, cable factory data sheets. These data sheets typically have a number of inherent problems for which guidance is needed. The following general applicable guidance is used to deduce the dimensions and parameters for calculating the current rating from a cable data sheet.

It is noted that while the international agreed unit for length is meters, the common working dimensions and the relevant standard (IEC 60287) uses millimeters for cable components, and therefore this unit is used in the Technical Brochure. This avoids large discrepancies.

Guidance Point 11: Cable data sheet - Dimensions not matching up

Issue	Commonly, the dimensions given in a data sheet, like thicknesses and diameters, do not add up and there seem to be empty spaces in the cable design.
Guidance	The diameter is considered the more important number compared to the thickness. The cable geometry used in calculations should be based on the diameters if given, and the thicknesses should be deduced from these. If the diameter is not given but the thickness is, the thickness should be used.
Notes	Certain layers are applied by extrusion, so the control parameter is the diameter of the cable being produced. Other material are applied as tapes, so the thickness is easier to be controlled in this case. In all cases, the nominal value refers to a typical value upon which tolerances and deviations are possible up to a certain degree. The user may not know which layer is controlled by thickness and which by the diameter. Therefore, as a clear recommendation, the WG recommends to use the diameter as a controlling factor.
Relation to IEC	This is an addition to IEC.

Guidance Point 12: Cable data sheet - Wire screen cross-section

Issue	The given total cross section of the wire screen is different from the area that can be deduced from the diameter and the number of wires in the wire screen.
Guidance	The cross sectional area that can be deduced from the diameter and number of wires should be used.
Notes	The given cross-section on a manufacturer's data sheet is usually a nominal value based upon the minimum area for fault current capability and may differ from the actual total cross-sectional area of the wire screen.
Relation to IEC	This is an addition to IEC.

Guidance Point 13: Cable data sheet - Swelling tape location

Issue	It is not defined where the swelling tape is located around a wire screen.
Guidance	It should be assumed that 50% of the swelling tape is above, and 50% of the swelling tape is below the wire screen.
Notes	This is an assumption to use in case no further information is available.
Relation to IEC	This is an addition to IEC.

Guidance Point 14: Position of glue in a metal foil water barrier

Issue	It is not defined where the glue is located in a metal foil water barrier.
Guidance	Assume that the glue is on the outside of the metal foil water barrier, not on the inside.
Notes	This is an assumption to use in case no further information is available.
Relation to IEC	This is an addition to IEC.

2.2.1 Example

To illustrate how the above guidance works in practice where non-ideal information is provided, consider the data sheet in Figure 3. Note that this example is different from any of the case studies provided in this TB.

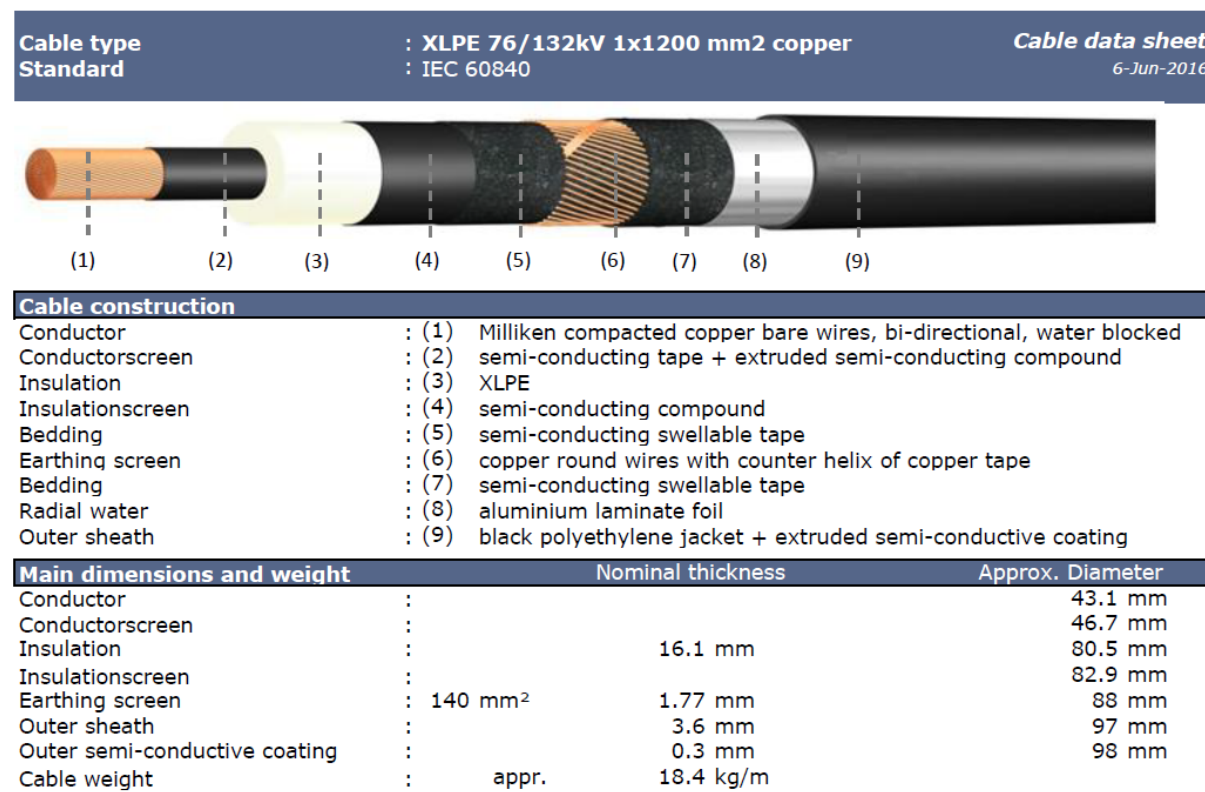


Figure 3: A typical datasheet

From the data sheet in Figure 3, and the guidance points 11 to 14, the dimensions for the current calculation are deduced as shown in Table 2-1. In this table, the order of deduction shows the order in which the various dimensions are derived.

The example shows that some assumptions on the cable construction details are required to deduce the exact cable geometries for calculation purposes. When guidance points 11 to 14 are followed, these assumptions will be largely similar in realistic cases. The assumptions should be reported together with the resulting current rating.

In practice, the assumptions might be avoidable by contacting the manufacturer to learn details or to request an updated datasheet. This might however not always be possible, and in that case, the above guidance can be followed.

Table 2-1: Example dimensional calculation

Cable part	Given outer diameter [mm]	Diameter for calculation [mm]	Order of deduction	Argumentation
Conductor (1)	43.1	43.1	A	The diameter is given
Conductor screen (2)	46.7	46.7	B	The diameter is given. The conductor screen thickness is calculated to be 1.8 mm
Insulation (3)	80.5	80.5	C	The diameter is given and is leading over the thickness. Guidance point 10. The insulation thickness is calculated to be 16.9 mm
Insulation screen (4)	82.9	82.9	D	The diameter is given. The insulation screen thickness is calculated to be 1.2 mm
Bedding / swelling tape part 1 (5)	-	84.46	F	Deduced from the wire screen diameter (88 mm) minus twice the wire thickness (2x1.77 mm). The thickness of the swelling tape layer is 0.78 mm
Wire screen (6)	88	88	E	The diameter is given and is leading over the thickness. Guidance point 10
Swelling tape part 2 (7)	-	89.56	G	The swelling tape is assumed to be 50% under, 50% over the wire screen, guidance point 12. The thickness of the swelling tape is thus again 0.78 mm
Aluminium foil (8)	-	89.96	H	The aluminium foil thickness is not provided. An assumption has to be made. This assumption is that the foil has a thickness of 0.2 mm, and has a 0.05 mm glue layer on the outside. From this the outer diameter is deduced
Glue layer (8)	-	90.06	I	Refer to the previous point. The glue's properties are taken as equal to the properties of the jacket
PE sheath (9)	97	97	J	The diameter is given (guidance point 10). The thickness can be calculated to be 3.47 mm
Semiconductive coating (9)	98	98	K	The outer diameter is given (guidance point 10). The thickness of the semiconductive coating is 0.5 mm

2.3 Properties of materials

In this TB, the IEC standards are followed closely with respect to the material properties. As mentioned in [2], the values provided in the standard are either internationally agreed or are generally accepted in practice. Further, it is noted that these material properties represent not only the quality of new cables, but are also applicable to cables after a long period of use. As such, the values include a margin

to take into account material ageing effects. As the properties are well considered, it is advised to use the material properties as reported in the IEC standards¹.

There are various materials and combinations of materials used in the construction of modern power cables for which information is missing in the standards [3], [4]. In some cases, simplifications in the standard exist in which different materials are combined, all having the same properties. However, with this approach sometimes important differences from reality are introduced. Therefore, the guidance of the working group is to not make these simplifications, and to utilise specific material properties, refer to Guidance Point 15.

Guidance Point 15: Cable data – Layers of materials with different properties

Issue	Combination of layers of different materials into a single layer.
Guidance	Do not combine layers of different materials in a cable, but take the effect of the different layers with different properties into account in the calculations.
Notes	Small layers can have a significant effect especially when the properties are unfavourable. To ensure a trustworthy result, the full detail of layers must be comprised in the calculations.
Relation to IEC	This is an addition to IEC.

Throughout all case studies, the properties as listed in [3] and [4] plus the additional properties as given in Table 2-2 and Table 2-3 are used.

Table 2-2: Additional electrical resistivity values

Material	Electric resistivity [$\Omega \cdot m$] at 20 °C	Rationale
<i>Copper for usage in sheaths and screens</i>	Value: Same as copper for usage in conductors according to [3]	In [3], the electrical resistivity of copper used other than in the conductor is not provided. It is recommended to use the same properties for copper used elsewhere in the cable system.

¹ It is noted that the IEC provides an opportunity to use values for material properties other than those quoted in the standard, where it is proven that these other values are representative. In such a case, the outcome of a current rating study should be twofold: one using the IEC material properties and corresponding current rating, and one using the manufacturer's material properties and corresponding current rating.

Table 2-3: Additional thermal resistivity values

Material	Thermal resistivity [Km/W]	Rationale
<i>Extruded semiconductive screens around XLPE insulation</i>	Value: 2.5	This value is not given in [4], but is given in [5]. In order to align the various design calculations it is proposed to use this value in current rating calculations too.
<i>Water blocking tapes</i>	Value: 12	Water blocking tapes have high thermal resistivity and therefore must be taken into account. The value of 12 results from [6].
<i>Copper woven fabric tapes</i>	Value: 6.0	This tape is a fibrous tape for which the entry of fibrous materials in [4] is used. The copper inside the tape has no thermal effect. The electrical resistance of thin Cu wires is neglected because it is much higher than the resistance of the adjacent metallic screen or sheath.
<i>Woven nylon tape</i>	Value: 6.0	This tape is a fibrous tape for which the entry of fibrous materials in [4] is used. If the tape is drenched in e.g. bitumen, the thermal resistivity of the bitumen may influence the material resistivity. In both cases, the thermal resistivity is 6 Km/W.
<i>Metallized paper tapes</i>	Value: 5.0 to 6.0	Assume a thermal resistivity equal to the base material (the paper), which is dependent on the cable type as per IEC [4].
<i>Carbon black paper tapes</i>	Value: 5.0 to 6.0	Assume a thermal resistivity equal to the base material (the paper), which is dependent on the cable type as per IEC [4].
<i>Polypropylene yarn</i>	Value: 6.0	Polypropylene yarn is a fibrous material and hence the value for fibrous materials as per IEC [4] is to be used.

The following further guidance points can be given regarding material properties:

Guidance Point 16: Cable data - Material properties listed in IEC 60287

Issue	Clarity over whether the IEC value for the material property or the manufacturer's value should be used.
Guidance	The IEC value should be used for the material property, as given in [3], [4].
Notes	Only with evidence in place, should other values be used. Where this is done, the IEC 'base case' should be calculated too to allow consistent comparison between the rating results.
Relation to IEC	This is a clarification of IEC.

Guidance Point 17: Cable data - Material properties not listed in IEC 60287

Issue	Clarity over which material property should be used.
Guidance	The values provided in Table 2-2 and Table 2-3 should be used.
Notes	These values have a background, as listed in the table, and as such it can be verified if the value holds true in the case under consideration. All case studies use the values as provided in Table 2-2 and Table 2-3.
Relation to IEC	This is an addition to IEC.

Guidance Point 18: Cable data - Type of XLPE is unknown

Issue	It is often not known whether the XLPE is 'filled' or 'unfilled'.
Guidance	Always assume 'unfilled' XLPE unless it is specified the XLPE is 'filled'.
Notes	None.
Relation to IEC	This is an addition to IEC.

2.4 Electrical Capacitance

2.4.1 Electrical capacitance of unscreened belted cables

The electrical capacitance for unscreened belted cables is not provided in the standard [3] and must be calculated in a different way given the fact that dielectric losses should be calculated according to Guidance Point 7. Calculations provided in [7] are proposed in this case, leading to the formula in Guidance Point 19.

Guidance Point 19: Cable data - Electrical capacitance of two- and three core unscreened belted power cables

Issue

What is the electrical capacitance of an unscreened belted power cable?

Guidance

Use the additional formulae below:

Two-core belted cable

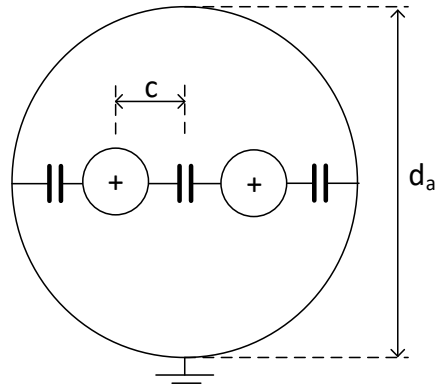


Figure 4: Representation of an unscreened belted two core cable

Capacitance per phase:

$$C = \frac{\epsilon}{18 \cdot \ln \left(\frac{2c \left(\left(\frac{d_a}{2} \right)^2 - c^2 \right)}{\left(\frac{d_c}{2} \right) \left(\left(\frac{d_a}{2} \right)^2 + c^2 \right)} \right)} \cdot 10^{-9}$$

Three-core belted cable

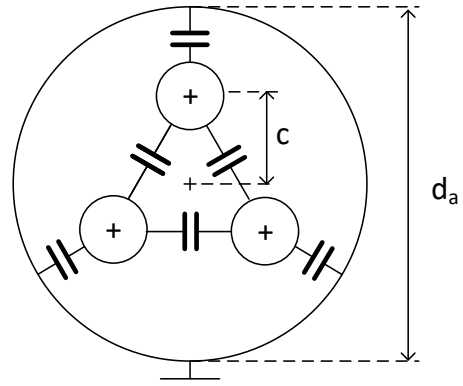


Figure 5: Representation of an unscreened belted three core cable

Capacitance per phase

$$C = \frac{\epsilon}{18 \cdot \ln \left(\frac{3c^2 \left(\left(\frac{d_a}{2} \right)^2 - c^2 \right)^3}{\left(\frac{d_c}{2} \right)^2 \left(\left(\frac{d_a}{2} \right)^6 - c^6 \right)} \right)} \cdot 10^{-9}$$

With:

C	capacitance per core	[F/m]
d _c	external diameter of conductor or conductor screen if present	[mm]
d _a	external diameter of belt insulation	[mm]
c	distance between the axes of conductors and the axis of the cable for two and three-core cables	[mm]
ε _r	relative permittivity of insulation	[-]

note: it is assumed that permittivity of the fillers and the belt insulation are equal to the permittivity of the insulation.

For sector shaped conductors:

$$d_c = d_x$$

$$c = 0,55 \cdot r_1 + 0,29 \cdot t$$

With:

d _x	diameter of a circular conductor having the same cross-sectional area and degree of compaction as the shaped one	[mm]
r ₁	circumscribing radius of two or three-sector shaped conductors	[mm]
t	thickness of insulation between conductors	[mm]

Notes

This formula is taken from [7].

Relation to IEC

This is an addition to IEC.

2.4.2 Effect of the lay length on the electrical capacity of three core cables

In a three core cable where individual cores are twisted together, the electrical capacitance of the cable is larger than the electrical capacitance of the individual cores. This is because the cores are longer due to the twisting.

The IEC 60287 [3] specifies that the capacitance per core results from the calculations, though in datasheets often the capacitance per cable is given.

If the electrical capacitance is not given by the manufacturer, the value must be calculated manually, and the lay length must be taken into account by the user, using the same formula as given in section 2.5 for the electrical resistance of conductors. The electrical capacitance of the cable can then be found as the electrical capacitance of an individual core multiplied by the lay length factor, refer to Guidance Point 20.

Guidance Point 20: Lay length and electrical capacitance in three-core cables

Issue	When and how to compensate the electrical capacitance for laid up cores?
Guidance	If the electrical capacitance per core or per cable is provided on the data sheet this information can be used best. If not provided, the electrical capacitance per core can be calculated according to the formulae in the standards [3] or the formulae in Guidance Point 19. The electrical capacitance per cable is required for current rating calculations, and is given by the 'lay length factor' multiplied by the electrical capacitance of the core. The lay length factor is: $F_{lay\ length} = \sqrt{1 + \left[\frac{\pi \cdot Lay_diameter}{Length_of_lay} \right]^2}$ In which: <i>Length_of_lay</i> axial cable length over which the cores make one full helical turn [m] <i>Lay_diameter</i> diameter of the mechanical neutral assembly. This is equal to 1,29 times the core diameter. Refer to appendix A. [m] A typical lay length factor is 1.01.
Notes	Refer to section 2.5 for a description of the lay length factor.
Relation to IEC	This is an addition to IEC.

2.5 AC and DC Conductor resistance calculations

For the majority of cable rating calculations it will be possible to obtain the dc resistance of the conductor from the IEC 60228 [8] or IEC TR 62602 [9] standards. For some cases however, further guidance is required to calculate the AC or DC resistance of conductors in a consistent way. This guidance relates to the selection of the method to determine the DC resistance of the conductor and the skin and proximity effect factors. Some special constructions and cable types deserve some specific attention when calculating the AC resistance, and the lay length is again of importance here, as previously introduced in Section 2.4.2.

2.5.1 Selection of the Conductor Resistance

The conductor resistance is an important parameter for calculating the rating of a power cable as all losses within the cable are related to the conductor I^2R losses. The resistance is a function of the cross-sectional area, which includes the total cross-sectional area of the wire strands in the conductor, and

the electrical resistivity of the conductor metal alloy with additional considerations for the lay length of wires comprising the conductor (except in the case of a solid conductor).

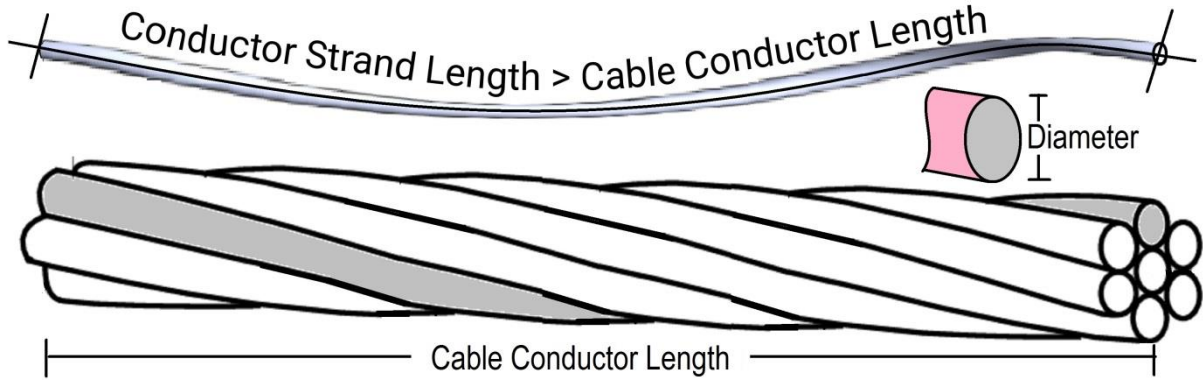


Figure 6: Definition of the length of lay

When considering the individual greyed strand in Figure 6, the resistance of the strand can be calculated with:

$$R_{DC,strand} = \frac{\rho_{strand} \cdot F_{lay\ length}}{A_{strand}} \quad (\Omega/Unit\ length)$$

Where ρ_{strand} is the electrical resistivity [Ωm] of the material the strand is made of at a specific temperature, $F_{lay\ length}$ is the lay length factor of the strand and A_{strand} is the cross sectional area of the strand perpendicular to the axis of the strand. The lay length factor is the same as given in Guidance Point 20 where it was used for use in a three core cable.

Typically, stranded conductors comprise multiple layers of strands, which all have a different lay diameter and also typically have different lengths of lay. Note that also the direction of lay may change from layer to layer. A direct calculation of the electrical resistance of a stranded conductor thus becomes involved and requires information that is normally not available for the user calculating the current rating.

An average lay length factor for all strands in a conductor can be defined. Then, when a conductor is build up from strands of uniform dimensions, the DC resistance of a conductor can be calculated to be:

$$Resistance = \frac{\rho_{conductor} \cdot \overline{F_{lay\ length}}}{A_{conductor}} = \frac{4 \cdot \rho_{conductor} \cdot \overline{F_{lay\ length}}}{N \cdot \pi \cdot D_{strand}^2 \cdot 10^{-6}}$$

where $\rho_{conductor}$ is the electrical resistivity of the conductor material [Ωm] at a specific temperature, $\overline{F_{lay\ length}}$ is the lay length factor averaged over all strands in the conductor, $A_{conductor}$ is the cross sectional area of the conductor, D_{strand} [mm] is the diameter of an individual strand in the conductor and N is the number of strands in the conductor. It is noted that in this formula it is assumed that the current does not pass from one wire to another.

If the lay length factor averaged over all strands in the conductor is unknown, a typical value of 1.05 can be used both for copper and aluminium.

The electrical resistivity of the conductor material which is used can vary between manufacturers and between projects as a function of the precise alloy or material composition chosen, as a function of the extent of material handling which could impact annealing, and as a function of the lay length of the individual strands in the conductor.

To correctly consider all manufacturing variances, industry standards such as IEC 60228 [8] or the IEC TR 62602 [9] provide tables of maximum DC conductor resistance as a function of conductor size for plain, stranded and segmental designs. These are used for design purposes in calculations and as a performance criteria in manufacturing specifications. By providing the maximum electrical resistance a conductor can have, these documents limit the variation in electrical resistance between conductors of

the same size. The IEC 60228 deals with conductors having metric sizes while the IEC TR 62602 deals with conductors having American Wire Gauge (AWG) or kcmil sizes.

Manufacturers can also measure the DC resistance of a given conductor and provide this value to the user for the purpose of calculating ratings and losses. However, these measured values are not generally known until the cable is manufactured and usually after an owner has purchased the cable. Manufacturers generally do not publish measured resistance values as these may be proprietary. Furthermore, measuring the DC resistance is often done on samples of limited length, and the variation of material resistivity with length, lay length variations and strand dimension variations are all not comprised in such a single measurement value. Variation in the resistance over the cable length thus will exist and will need to be considered in a current rating calculation.

Summarizing, there can be three alternatives to selecting a conductor resistance for calculating current ratings. For the purposes of rating calculations as discussed in this TB, the order of preference for selecting the conductor resistance is as follows:

1. Using a standardized conductor resistance from IEC 60228 or IEC TR 62602

This is the preferred and standard approach and will also align with the electrical resistance mentioned on the cable data sheet.

For cable conductors having metric sizes, the appropriate standard is IEC 60228 [8]. For cables having AWG or kcmil conductor sizes different publications give slightly different resistance values for the same conductor size. Because of this, it is recommended that the manufacturer's data sheet is used when available. If not available, the values given in IEC TR 62602 [9] should be used.

The values of IEC 60228 and IEC TR 62602 are not exactly the same meaning that it is relevant which standard is chosen to define the conductor. This is illustrated in Figure 7 and Figure 8²

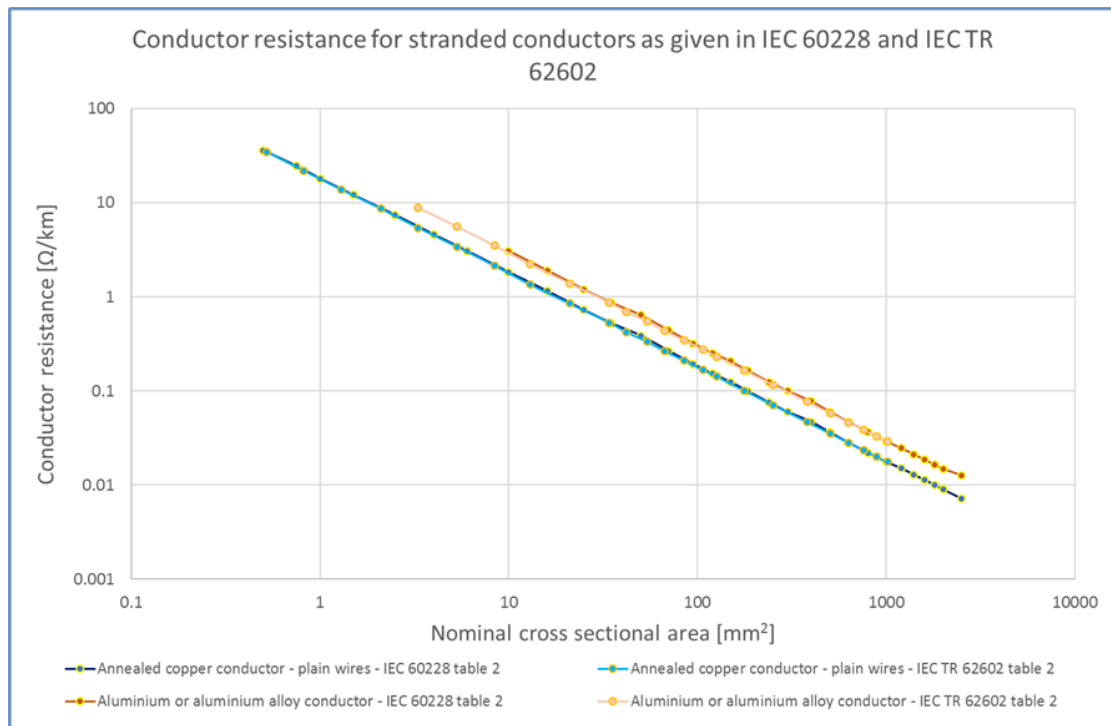


Figure 7: Comparison of IEC 60228 and IEC TR 62602 conductor resistance as a function of size, for class 2 stranded conductors

² To deduce the difference between IEC 60228 and IEC TR 62602, the datapoints from IEC TR 62602 were linearly interpolated to yield the resistance value at the cross sections reported in IEC 60228. Only interpolation was used, no extrapolation.

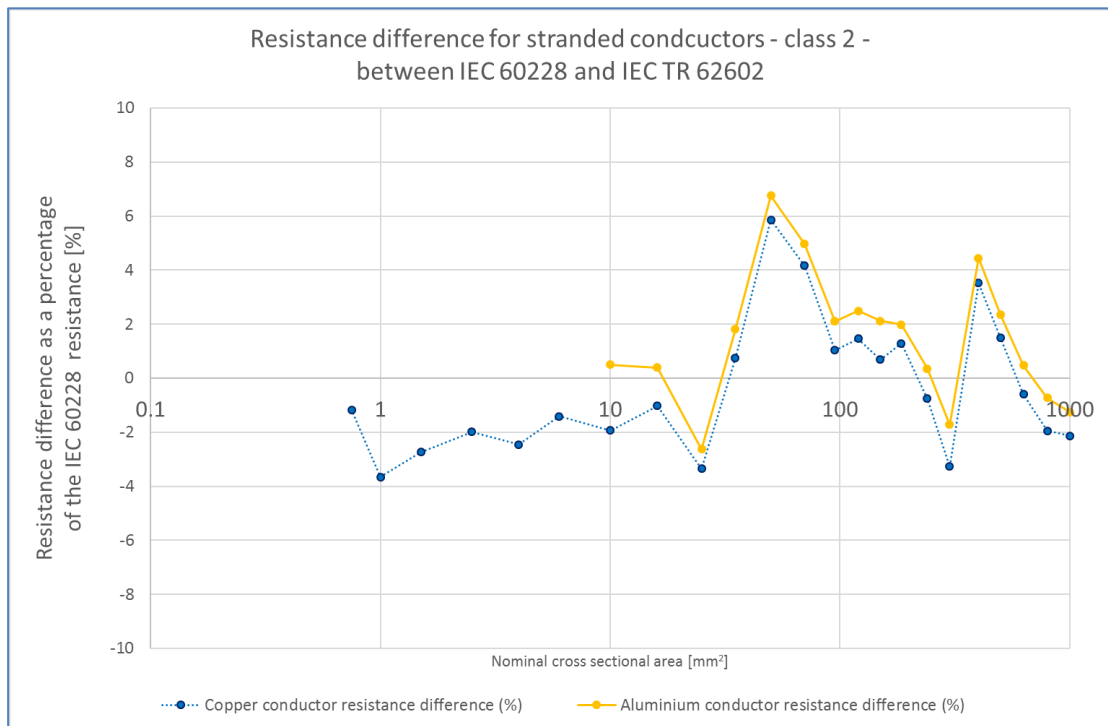


Figure 8: Difference between IEC 60228 and IEC TR 62602 conductor resistances shown in Figure 7

A disadvantage to standardized resistance tables is that the manufacturer is allowed tolerances to achieve the conductor resistance performance requirements and therefore the details of the conductor design are not precisely specified by using the standardized values. This also means that the 'nominal' cross sectional area as mentioned in the standards is merely a name and not a true representation of the actual cross section. Nevertheless, the tabulated electrical resistance is a commitment of the manufacturer which will not be exceeded by the actual electrical resistance of the conductor. A potential tolerance would be on the safe side for the current rating purposes.

Note that non-standard conductor sizes (ones not included in the standardized tables) require interpolation to find the corresponding resistance for calculations. Linear interpolation is recommended for this.

The tables in the standards sometimes treat the electrical resistance differently for multiple-core cables such as triplex³ or three core cable constructions. In these constructions, the cores are twisted effectively resulting in a longer core conductor length relative to the main cable axis. Effectively, this increases the resistance of each core. Correction factors for the lay length of three core cables are not required where conductors comply to IEC 60228 but they are required for AWG/kcmil conductors which comply to IEC TR 62602 and values are given in that publication. Also refer to Guidance Point 21.

2. Using a measured electrical resistance for a specific cable

This is not a preferred approach and may not be representative for the full cable.

Measured resistance values could be used to calculate the current rating. However, resistance measurements quoted by a manufacturer may have been made on a short sample and hence, because of manufacturing tolerances, may not represent the electrical resistance representing the total production length. Using measured values could also lead to it being necessary to recalculate the current rating of a circuit if a fault were to occur and a new section of replacement cable has to be installed.

³ A triplex cable construction is a cable circuit in trefoil where the cables are twisted together; a twisted trefoil construction.

It is recommended that measured values are only used for non-standard conductor sizes. In such cases the maximum value declared by the manufacturer should be used.

3. Using a calculation based on cross-sectional area, resistivity and lay length factor

This is not a preferred approach and may yield an approximate value only.

This approach is universally available for the calculation of any cable size. However, the calculation is based on a specific value of the conductor material electrical resistivity and is based on the lay length factors of either each individual stranding layer in a conductor or on an averaged lay length factor. These details may not be known making this approach problematic.

In special cases however this may be the only approach possible.

Except where specifically discussed, this TB will use resistance values either directly or using a linear interpolated value from the IEC 60228 [8] table, leading to Guidance Point 21. An example is provided for calculating the resistance given a cross sectional area, known electrical resistivity and an assumed average conductor stranding factor.

Guidance Point 21: Conductor electrical resistance calculation

Issue	Clarity on what method to use to define the electrical resistance of the conductor.
Guidance	The following order of preference is recommended when defining the DC resistance of the conductor: 1. Base the calculation on the standardised DC resistance tables, given in IEC 60228 [8] or TR 62602 [9] 2. Use the measured value for the cable under consideration 3. Base the calculation on a cross sectional area, resistivity and an average lay length factor. If the lay length factor averaged over all strands in the conductor is unknown, a typical value of 1.05 can be used.
Notes	The different merits, drawbacks and points of attention are clarified in the text. Approaches 2 and 3 are not preferred. All calculations in this TB follow point no. 1 above. For triplex or three core cables, the lay length factor might have to be taken into account in three core cable systems, refer to Guidance Point 23.
Relation to IEC	This is an addition to IEC.

In the case of sector shaped conductors, which can be found in power cables for distribution, there is a specific issue requiring the guidance given in Guidance Point 22.

Guidance Point 22: Selection of equivalent conductor diameter of sector shaped conductor

Issue	Clarity over what should be done if the calculation of a sector shaped conductor diameter according to section 2.1.4.2 of [3] is ambiguous.
Guidance	To determine d_x according to 2.1.4.2 of [3], the IEC 60228 [8] can be used. The minimum and maximum diameters should be found, as reported in appendix C of IEC 60228, which are related to the same cross section as the cable under consideration, and the diameter giving the worst case rating should be used.
Notes	Parameter d_x influences the current rating calculation in multiple places as the AC conductor resistance, the dielectric losses, the armour / tape losses and the thermal resistance. It should be ensured that the diameter is such that a worst case rating is found. This is not an obvious choice and hence an assessment is required.
Relation to IEC	This is an addition to IEC.

2.5.2 Accounting for the lay length in three core cables

The fact that triplex or three core cables are twisted together leads to the possible need of a correction factor to take into account the longer length of the cores. This correction factor is the lay length factor as was already given in Guidance Point 20.

If the manufacturer provides an electrical resistance value according to the IEC 60228 [8], the electrical resistance value must be valid for the cable, meaning that the lay length factors should already be included. This holds for any type of power cable, including three-core constructions and cables that are delivered as triplex constructions. Correction factors for lay length are not required where conductors comply with IEC 60228 but they are required for AWG/kcmil conductors to IEC TR 62602 and are given in that publication.

If the electrical resistance of a power cable is not provided by the manufacturer, the lay lengths must be taken into account manually. Then, the electrical resistance calculated on the basis of the conductor cross section must be multiplied by the lay length factor to take the effect of the twisting into account, refer to Guidance Point 23.

Guidance Point 23: The lay length and the electrical resistance in three core- or triplex cables

Issue	Consideration of the length of lay when calculating the electrical resistance of a laid up cable.
Guidance	When the IEC 60228 [8] is used, use the electrical resistance as quoted by the manufacturer. These values must take into account the lay length factor and thus the electrical resistance must be representative for the finished cable. However, do calculate the lay length factor (refer to the notes of this guidance point). When the IEC 60228 is not used, the electrical resistance for the cable core may need to be corrected with the lay length factor to yield the electrical resistance for the cable. When using the IEC TR 62602 [9] the user should verify if a lay length factor is used. If not, the below factor is recommended. The cable core resistance can also be derived from other tables, measurements or calculations, refer to Guidance Point 21. Subsequently, this value must be corrected with a 'lay length factor' to find the electrical resistance for the laid up assembly, which will be larger. The lay length factor is: $F_{lay\ length} = \sqrt{1 + \left[\frac{\pi \cdot lay_diameter}{Length_of_lay} \right]^2}$ In which:

Notes	<i>Length_of_lay</i>	axial cable length over which the cores make one full helical turn	[m]
	<i>Lay_diameter</i>	diameter of the mechanical neutral assembly. This is equal to 1,29 times the core diameter. Refer to appendix A.	[m]
	A typical lay length factor is 1.01.		
	If the manufacturer is not obliged to follow IEC 60228 [8], the definition of electrical resistance may be different. In that case, it should be checked carefully if compensation for the length of lay is required as per the guidance above.		
Relation to IEC	The lay length factor must always be calculated or assumed, as it is required also for the calculation of the thermal resistances in the power cable.		
	This is an addition to IEC.		

2.5.3 Impact of the Separator Tape between Segments of the Conductors on Skin and Proximity Effects

Conductor construction has a significant impact on the AC resistance of the conductor, which includes the DC resistance at operating temperature plus the impact of the self-inductance of a conductor and, specifically, the relative reactance of each wire strand within the conductor (skin effect), and the mutual inductance between adjacent cable phases (proximity effect). Both phenomena influence the distribution of AC current, which consequentially impacts the losses of the conductor. The calculation of specific interactions among strands within a conductor is extremely complex and requires knowledge of the conductor construction that is generally unknown when performing cable ratings.

To simplify these calculations, skin and proximity effect factors are associated with each conductor construction that are then incorporated into the rating methodology. A goal of conductor design is to optimize the current distribution across the cable conductor cross-section to minimize heating losses and therefore maximize the cable rating.

One type of construction that is often used for power cables having large conductors is a Milliken conductor⁴ where three or more segments, lightly insulated from each other, comprise the completed conductor; four- or five-segment conductors are common. Milliken conductors are manufactured by stranding a segment together and then passing it through a die that gives the segment a sector shape (120° for three-segment, 90° for four-segment, 72° for five-segment and 60° for a six-segment conductors, respectively. That is $360^\circ/n$, where n is the number of segments). The individual segments are then brought together to form a generally round conductor that is the basis for applying additional layers of the power cable (conductor screen, insulation, insulation screen, etc.). Figure 9 below illustrates the assembly of the segments.

⁴ IEC 60050-461 defines a Milliken conductor as a "stranded conductor comprising an assembly of shaped stranded conductors, lightly insulated from each other"



Figure 9: Example of the assembly of the segments of a Milliken conductor

The k_s and k_p factors associated with Milliken conductors are lower than for other conductor constructions. Manufacturers select the number of segments and some manufacturers incorporate a tape separator between adjacent segments in order to make them electrically insulated. As the difference of potential between segments is very low, one or two layers of paper or plastic tape are sufficient. The number of segments affects the overall current distribution in the conductor, and the presence of the separator tape provides better electrical isolation between adjacent segments, potentially improving the current distribution over a segmental conductor that is not configured with the separating tapes. The standards do not make a distinction between the number of conductor segments when arriving at the DC resistance. The distinction is made later when the AC resistance is calculated, depending to the design of the conductor (specific k_s and k_p values are applied). Choosing specific k_s and k_p values means choosing for a Milliken conductor including the presence of tapes between the segments of a conductor.

2.5.4 Considerations for Selecting Skin and Proximity Effect Factors

The selection of the skin (k_s) and proximity (k_p) effect factors is most often based on recommendations in IEC 60287 [3]. However, there are differences in the values in IEC 60287 [3] and CIGRE Technical Brochure 272 [10]. Beginning with the 2014 edition, the IEC 60287 values for Milliken conductors are conservative, while TB 272 provides more options for selecting these coefficients.

Since the new standard values for the skin and proximity effect factors for Milliken conductors are different for all plastic insulated cables, the calculated ratings will be different even when the other factors are held constant, without any physical change in cable construction or assumed installation conditions.

While the IEC 60287 standard should be followed as a recommended practice, appropriate selection of skin and proximity effect factors from TB 272 can be used under the right conditions. This leads to Guidance Point 24.

Guidance Point 24: Factors for skin and proximity effect

Issue	Clarity over which information source should be used (IEC values or CIGRE TB 272 values) for skin and proximity effect factors k_s and k_p .
Guidance	Where both IEC [3] and CIGRE [10] provide values for the same situation, the IEC values should be used. Where IEC does not provide values but CIGRE does for a certain (typically more specific) situation, the value from CIGRE TB 272 should be used Note that the values might change when a document is updated, refer to Guidance Point 3.
Notes	
Relation to IEC	This is an addition to IEC.

A specific situations exists in which the AC resistance of conductors must be changed compared to the standard, which is the case for armoured cables and for cables with sector shaped conductors. Refer to Guidance Point 25.

Guidance Point 25: AC conductor resistance in magnetic armoured cables

Issue	Clarity over how to calculate skin and proximity effect in single phase or three phase magnetic armoured or shielded cables.
Guidance	For these cables, the AC resistance of the conductor should be calculated with the following formula: $R = R' [1 + 1,5(\gamma_s + \gamma_p)]$
Notes	The skin and proximity effects are thus increased with a factor of 1,5. The armour wires will electrically influence the skin and proximity effects in the cable conductor. When the IEC standard [3] is followed, this effect is not taken into account, which is incorrect. The guidance is aligned with section 2.1.5 of [3] bringing the same effect in account in pipe type cable systems. This is applicable to cables in magnetic pipes (Section 13) or to cables with magnetic steel tape reinforcement (Section 14). The inclusion of the factor 1.5 is a conservative approach and is subject of ongoing further studies. As the effect is present, but there currently is no better number, the factor of 1.5 is adhered to. A last note is that CIGRE WG B1.64 is currently considering the calculation of the armour losses. It is anticipated that this WG may propose an update on how to improve the evaluation of these losses.
Relation to IEC	This is an approach differing from the approach of IEC.

2.6 calculation of losses in metallic screens, sheaths and armours

When calculating losses in power cables, discussions arise on how to calculate losses in the metallic screens, sheaths and armours. Many of these discussions originate from the fact that in today's cable designs, several of these layers may be present, and may be separated electrically and thermally. This complexity is not integrated in the IEC documents [3], [4], leading to problems with the interpretation of the standards.

The working group has prepared charts showing which loss calculation must be performed for a specific cable design.

It is noted that the calculations utilize the standardized loss formulae. A separate CIGRE working group is currently considering possible improvements in evaluation of the losses in three-core submarine cables. As these improvements will be dependent on the specific cable design, the possible improvements will be applicable to the selected designs only (potentially leading to a reduction of calculated losses, and hence an increase of current rating)

The loss calculations for one and three core cables are identified in Figure 10. Subsequently, the various design dependent calculations are detailed in Figure 11 to Figure 20. It is noted that the cases that are marked to be not included refer to situations which either do not exist or are very rare.

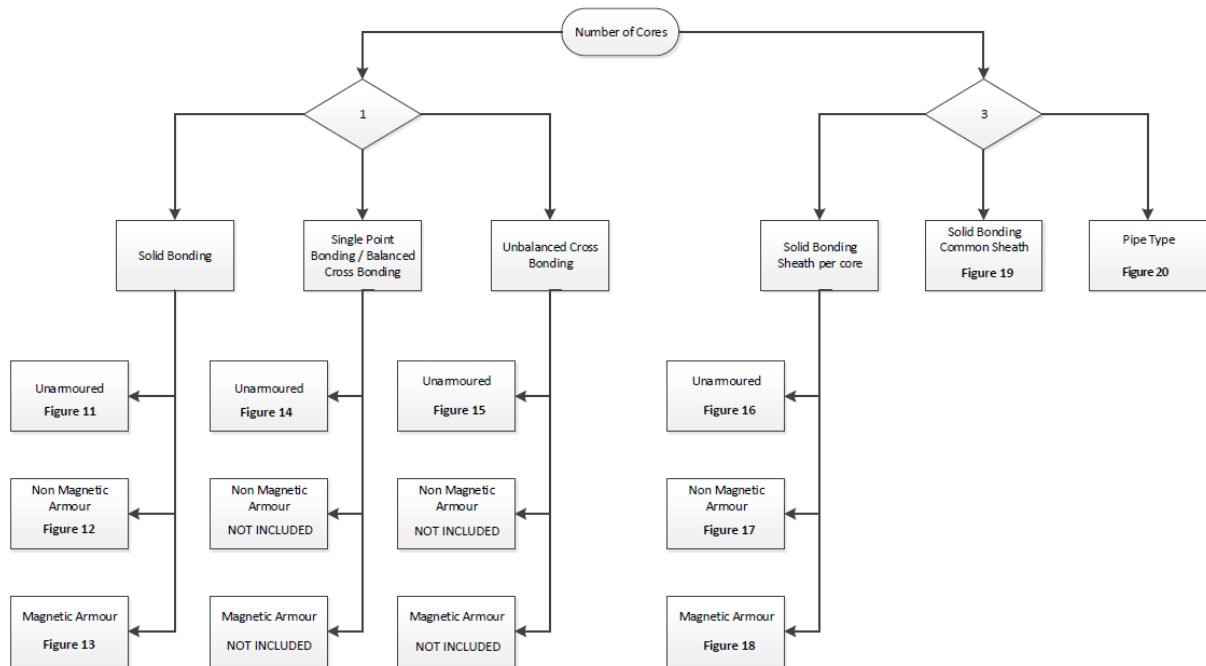


Figure 10: Calculation of losses for one and three core cables

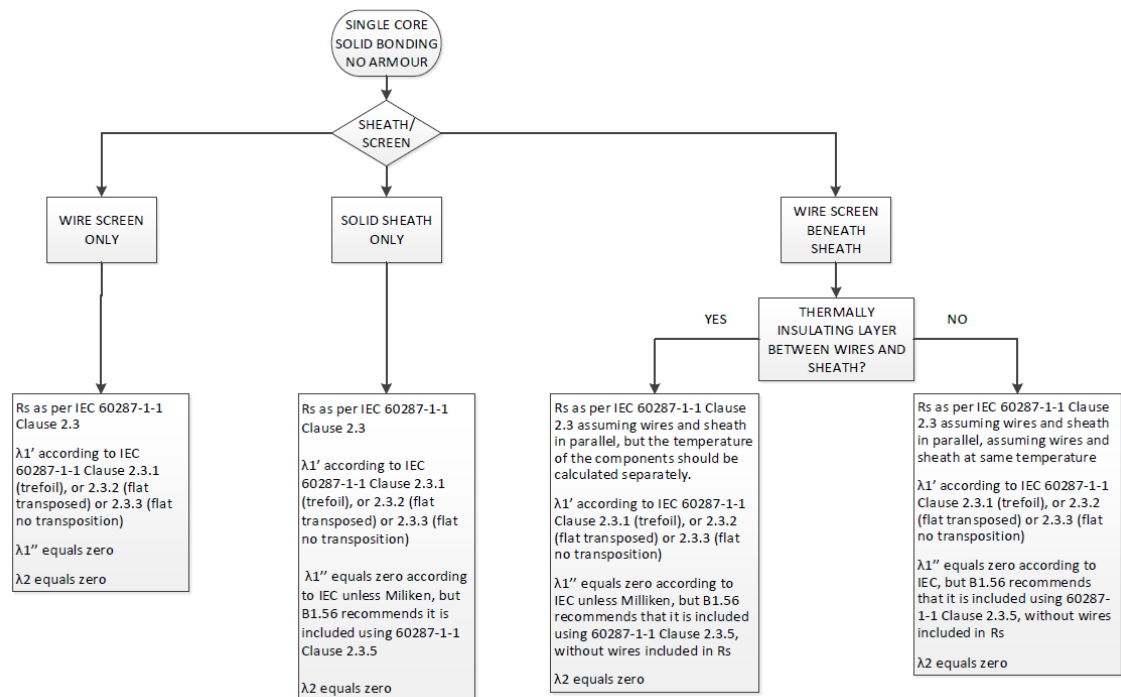


Figure 11: The loss calculations for single core cables without armour, solid bonded

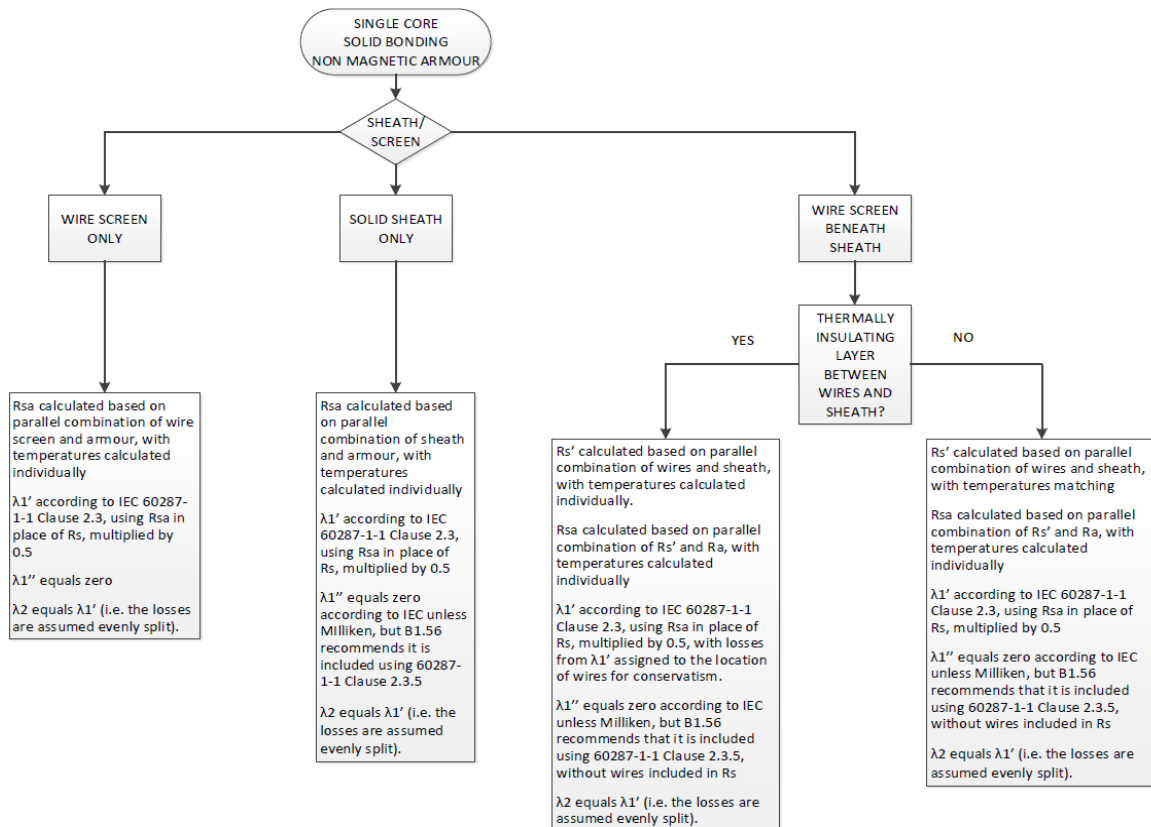


Figure 12: The loss calculations for single core cables with non-magnetic armour, solid bonded
 Note: the workgroup follows IEC in the sense that the losses are assumed to be evenly split (between λ_1' and λ_2).

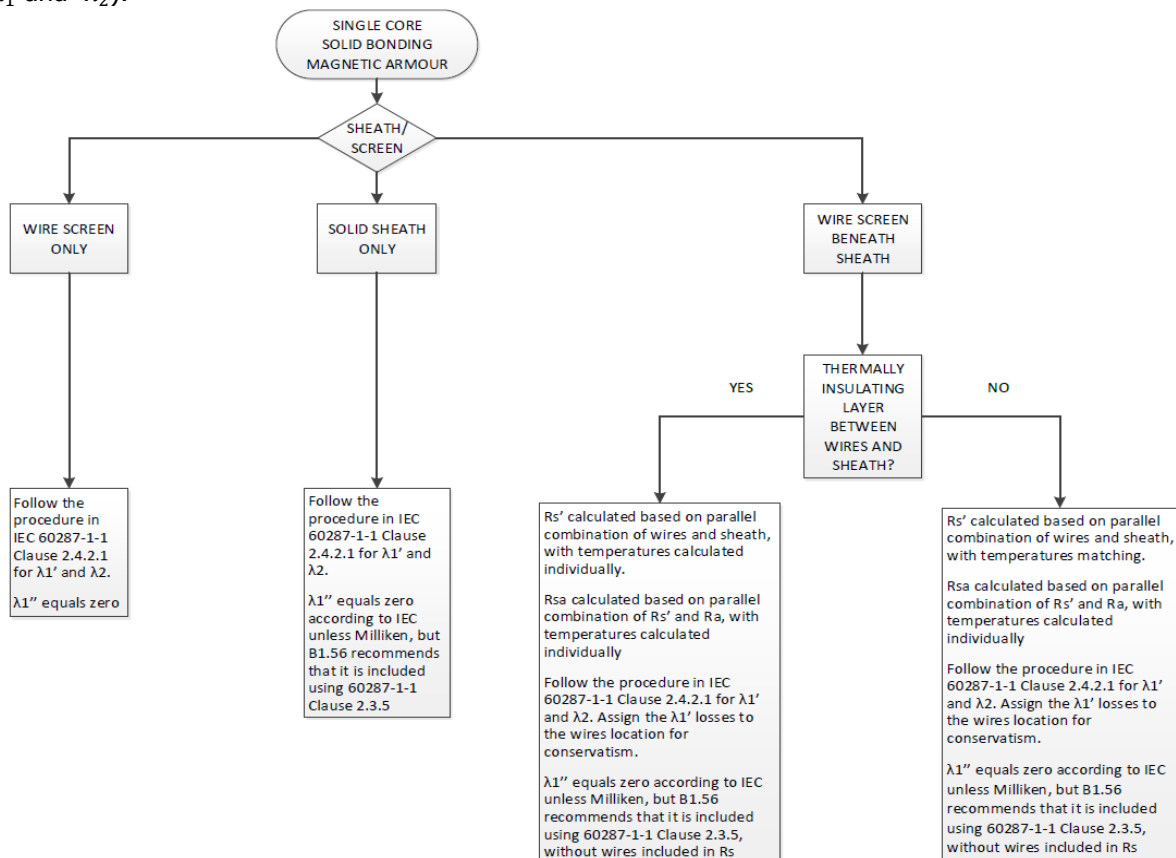


Figure 13: The loss calculations for single core cables with magnetic armour, solid bonded

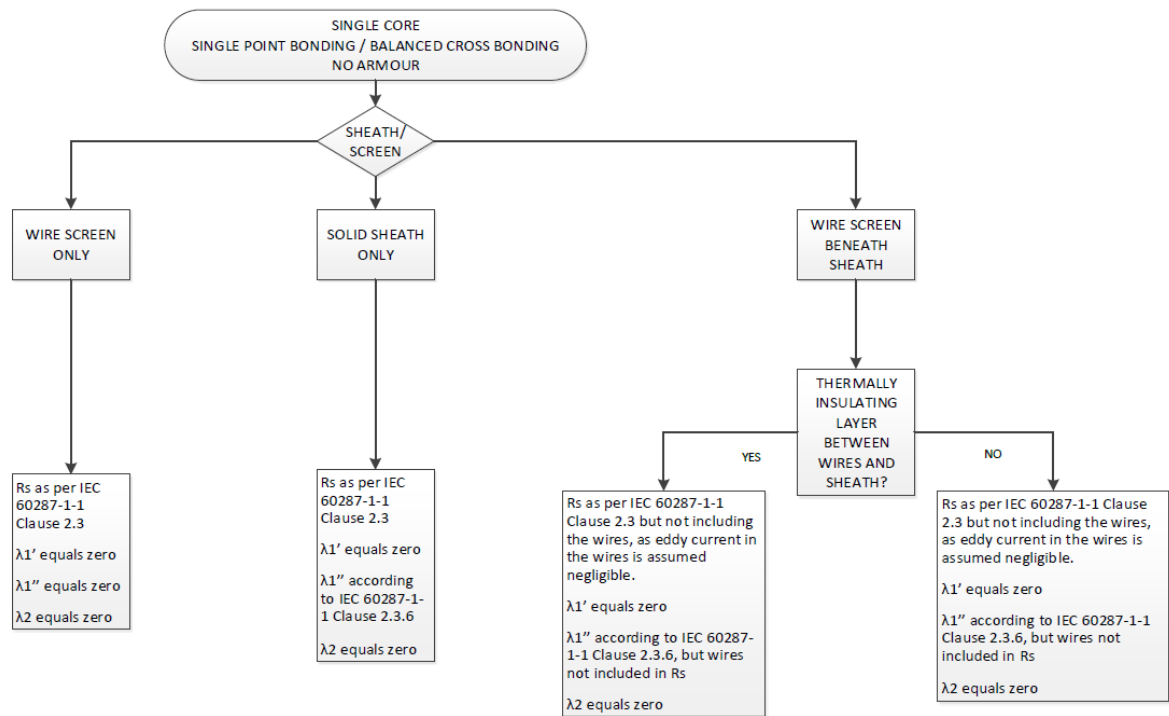


Figure 14: The loss calculations for single core cables without armour, single point bonded or balanced cross bonded

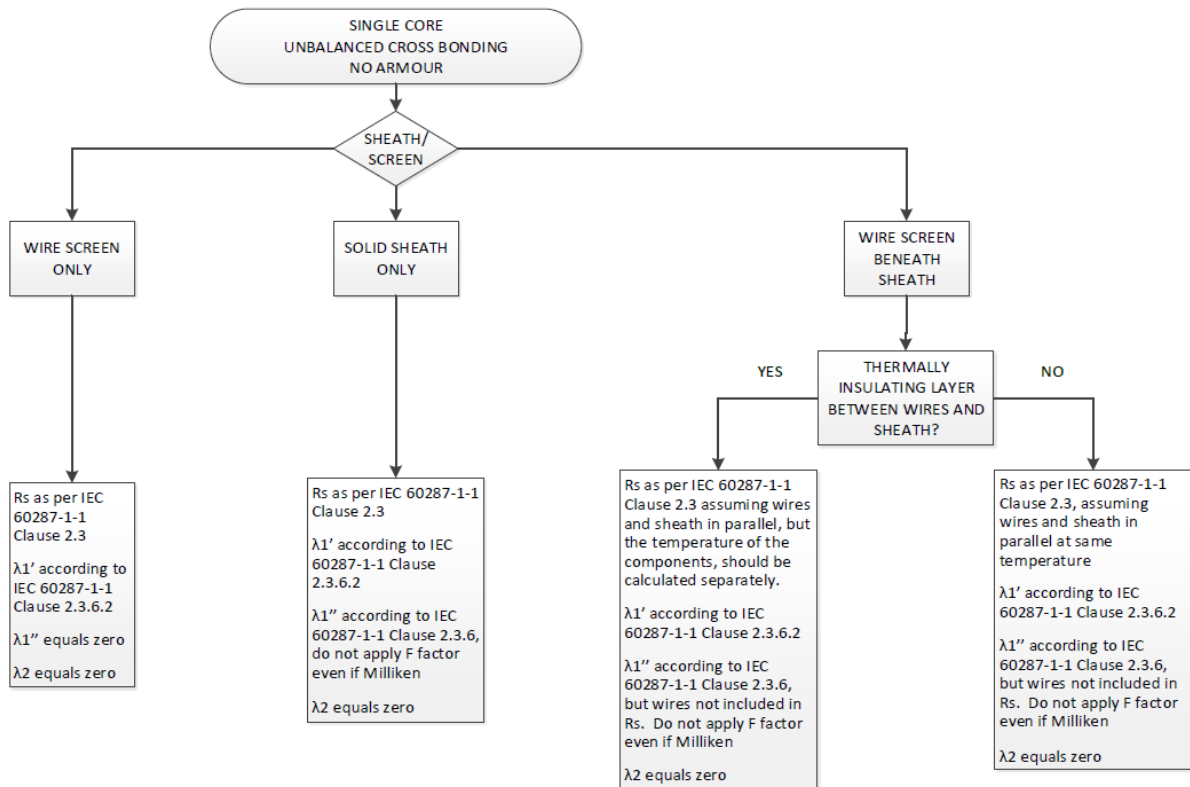


Figure 15: The loss calculations for single core cables without armour, unbalanced cross bonded

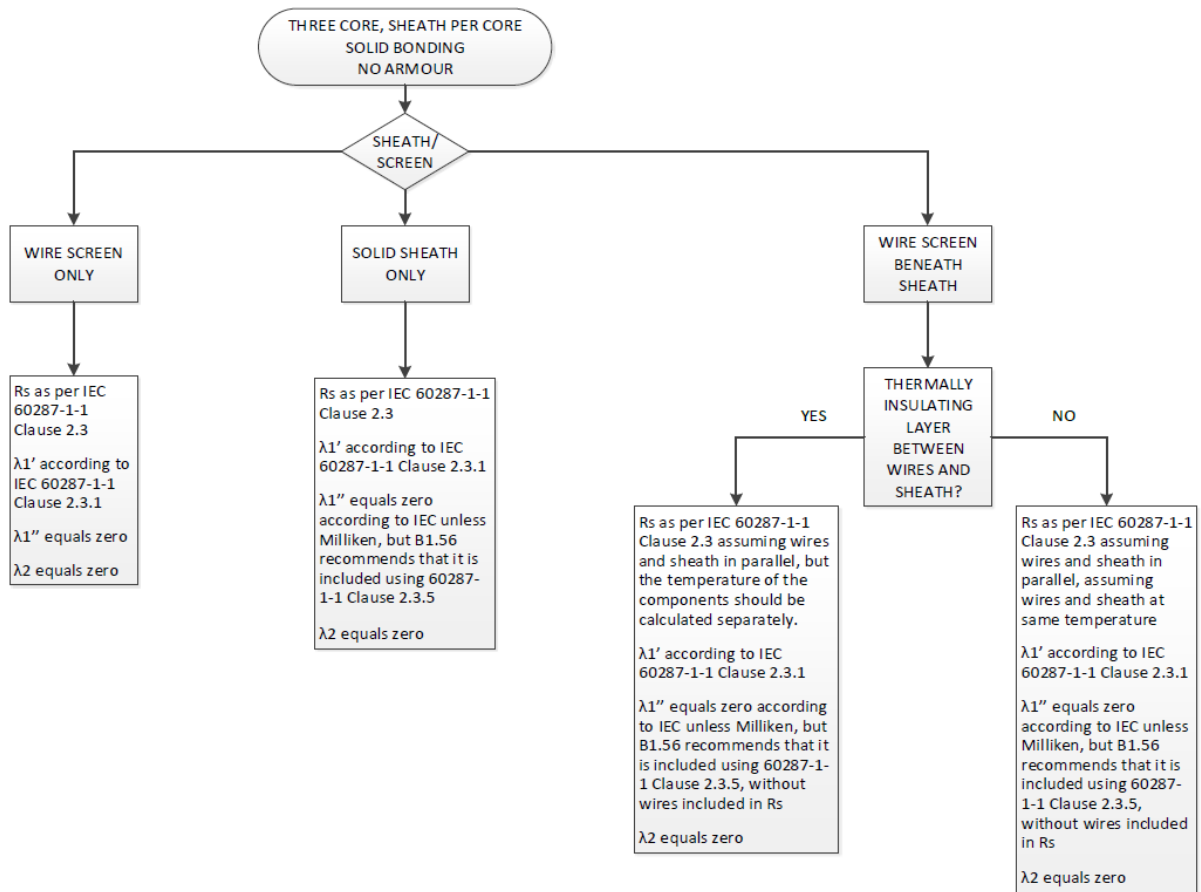


Figure 16: The loss calculations for three core cables (SL type) without armour, solid bonded

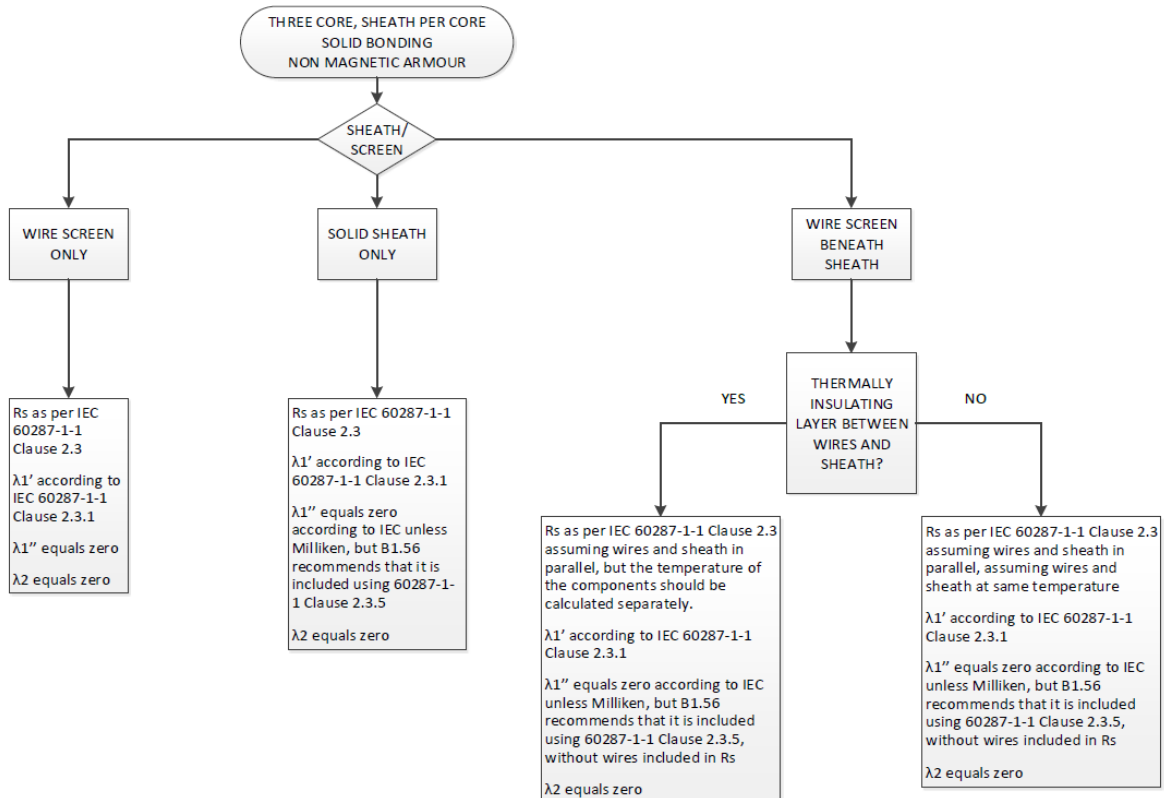


Figure 17: The loss calculations for three core cables with non-magnetic armour, solid bonded

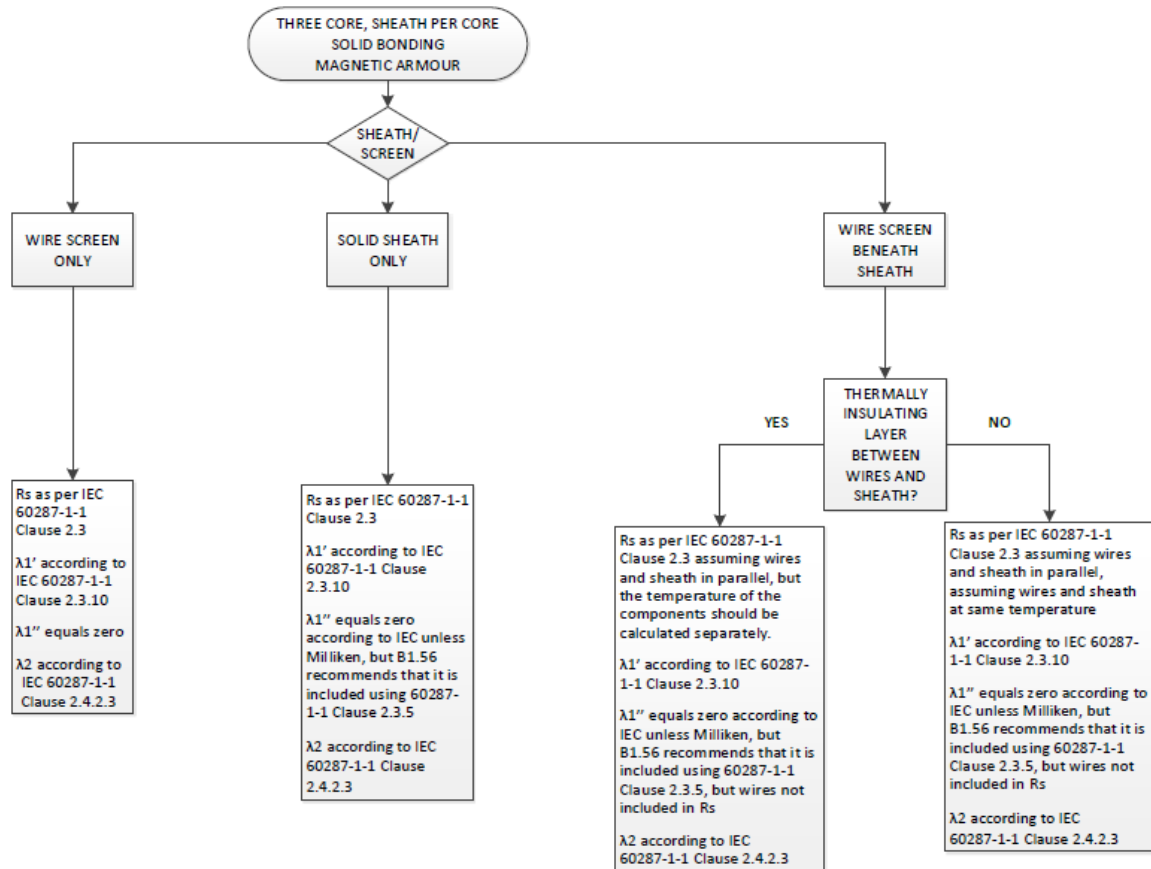


Figure 18: The loss calculations for three core cables with magnetic armour, solid bonded

Note: the λ_2 calculations referred to in Figure 18 (only) are under consideration of CIGRE WG B1.64.

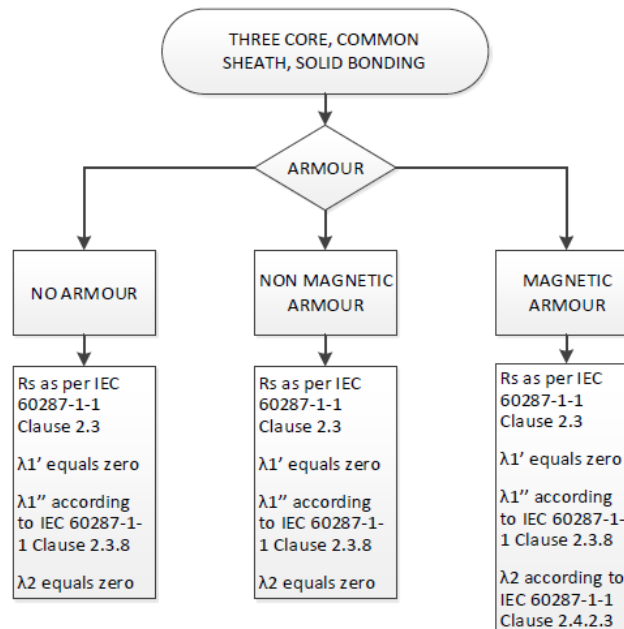


Figure 19: The loss calculations for three core cables with a common sheath, solid bonded

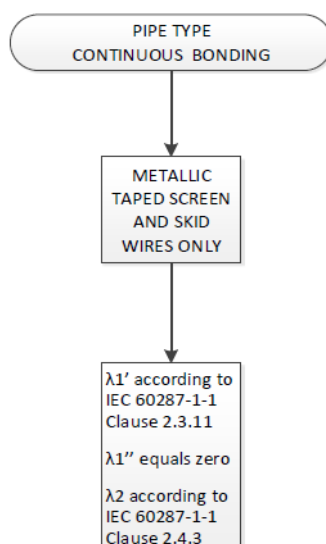


Figure 20: The loss calculations for pipe type cables, solid bonded

The following recommendations are made when performing a current rating calculation.

Guidance Point 26: Loss calculations

Issue	Clarity over what loss calculations should be used in a specific situation.
Guidance	The required loss calculations depend on the cable design. The cable design under consideration should be found in Figure 10 and then the relevant calculation method in Figure 11 to Figure 20 can be identified.
Notes	In some cases, no guidance is given. These cases are rare, very complex and not much information exists to perform any current rating calculation. In such cases, the calculations should be performed by a highly qualified specialist.
Relation to IEC	Due to the electrical and thermal separation of screens, sheaths and armours, the loss calculations can sometimes become complex. It is noted that the losses of each individual screen, sheath and armour should be calculated on the basis of the temperature of the specific screen, sheath or armour.
	This is an approach differing from the approach in IEC

Since the IEC standards give formulae for the DC resistance calculations, it is assumed that the skin effect in screens and armours is very small for most constructions. However, the IEC standard recommends an increase in the DC resistance of some armours by specifying a suitable multiplication factor.

Given the method of loss calculations, the following sections detail various aspect of metallic screen / sheath and armour issues.

2.7 Calculations for sheaths and screens

2.7.1 Wire screens and wire armour

The length of lay of the wire screens and wire armours also plays a role, and must be taken into account when calculating the electrical resistance of the screen or armour.

Guidance Point 27: Calculation of electrical resistance of a wire screen

Issue	Consideration of the lay in wire screens and wire armours
Guidance	If provided, the resistance per unit length of the wire screen as quoted by the manufacturer should be used. Otherwise, the electrical resistance for an individual phase cable can be calculated based on its geometry. Subsequently, this value must be corrected with a 'lay factor' to find the electrical resistance. Furthermore, the lay factor should be taken into account when calculating the degree of metallic covering of the cable.
Notes	The correction factor is provided in Section 2.5.
Relation to IEC	This is an addition to IEC.

Another effect is related to the condition stated in Section 4.2.4.3.3. of [3] where it must be assessed if a cable is partly metal screened or not. To calculate the percentage and with that to deduce which calculation is to be performed, the length of lay must be taken into account. If the metal covering is over 50%, consider the cable to be 'metal sheathed'. Note that this aspect influences the calculation of the thermal resistances as per [4].

Guidance Point 28: Degree of coverage of a screen

Issue	Clarity over what to do if the degree of coverage of a screen according to Section 4.2.4.3.3. of [4] is outside the 20-50% range.
Guidance	In the case coverage is more than 50%, the calculations for a metallic screened cable, section 4.2.4.3.2 of [4], should be followed. In the case that coverage is less than 20%, the calculations for a non-metallic screened cable, section 4.2.4.3.4 of [4], should be followed. In all other cases, the calculations for a part metallic screened cable, section 4.2.4.3.3. of [4] should be followed.
Notes	The lay factor should be taken into account when calculating the degree of coverage.
Relation to IEC	This is an addition to IEC.

Regularly, wire screens have a single tape applied over them in the opposite lay direction. This is a counter helix tape, for which Guidance Point 29 is applicable.

Guidance Point 29: Counter helical tapes

Issue	Consideration of counter helix tapes (a thin metal tape in an opposite lay direction over a wire screen) in current rating calculations?
Guidance	These tapes should not be taken into account.
Notes	
Relation to IEC	This is a clarification to IEC.

2.7.2 Corrugated sheaths

Similar to a length of lay, the additional amount of material resulting from the corrugation must be taken into account in a corrugation factor. The calculation procedure involves a multiplication of the electrical resistance of a smooth sheath by a corrugation factor.

There are various types of corrugation. Annular and spiralling corrugations exist, and the corrugation form or shape can vary. Figure 21 and Figure 22 illustrate different corrugation types.

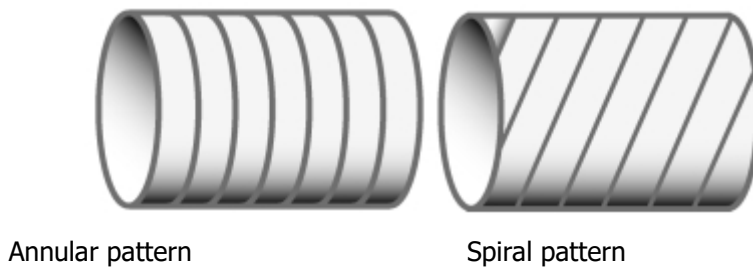


Figure 21: Annular and spiralling sheath corrugation

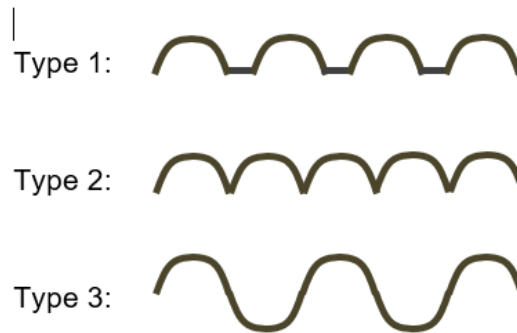


Figure 22: Various forms / shapes of corrugation

It is important to understand the general effect of corrugations on the current rating. The average heat transport over a length of cable can be taken. With this, the differences between an annular and spiralling pattern will disappear. Hence, the calculation for an annular corrugated sheath is valid for a spiralling corrugated sheath.

Figure 23 provides the definitions of width and depth of a corrugated sheath. If width and depth are not changing along the cable route, the shape of corrugation in Figure 22 indicates that type 3 and type 2 will have the same corrugation length, while type 1 will have a somewhat reduced length.

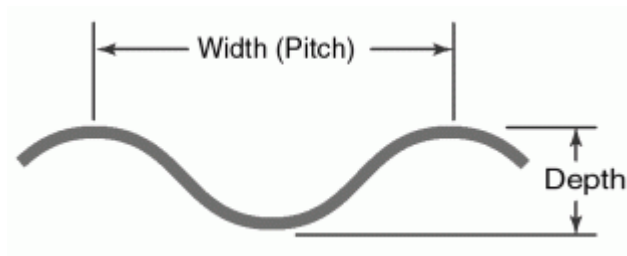


Figure 23: Definitions in case of corrugated sheaths

Denoting the width by a symbol ' L ' and the depth by ' H ', and remembering that during cable manufacture corrugations are made with circular (not sinusoidal) shapes, the expressions below show

the derivation of the corrugation factor for types 2 and 3 constructions with an ideal representation shown in Figure 24.

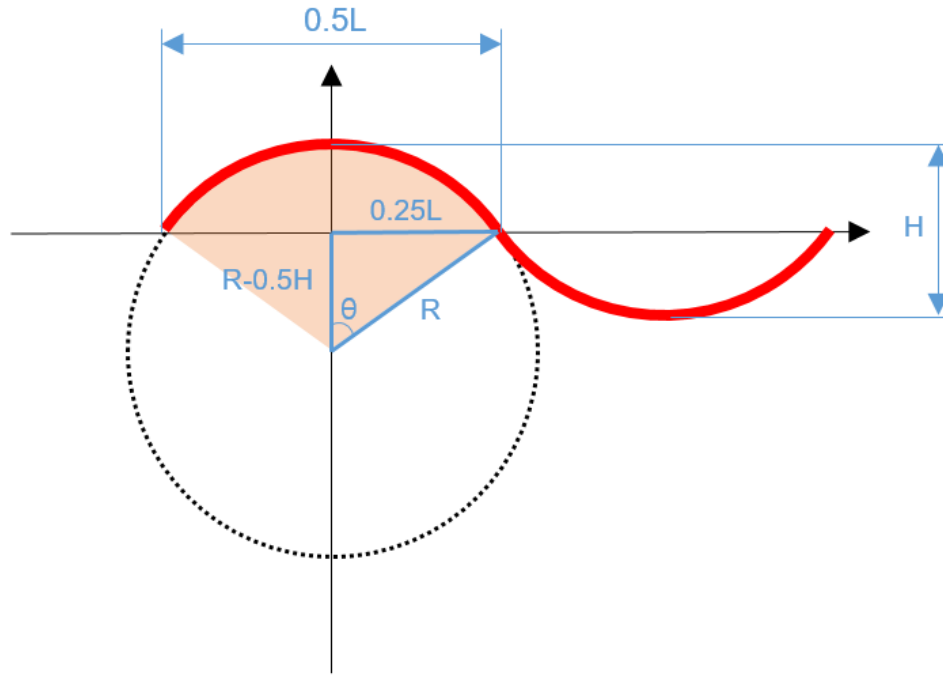


Figure 24: Close up of corrugation type 3

Based on the geometry in Figure 24 and assuming $H \leq 0.5 \cdot L$, we have:

$$R^2 = (0.25L)^2 + (R - 0.5H)^2 \rightarrow R = \frac{0.25L^2 + H^2}{4H} \quad (mm)$$

The angle of the circular sector is calculated with the following formula:

$$\theta = \sin^{-1}\left(\frac{0.25L}{R}\right) = \sin^{-1}\left(\frac{L \cdot H}{0.25L^2 + H^2}\right) \quad (rad)$$

The length of metal in one full corrugation can then be described with:

$$L_{cor} = 4 \cdot R \cdot \theta = \frac{0.25L^2 + H^2}{H} \cdot \sin^{-1}\left(\frac{L \cdot H}{0.25L^2 + H^2}\right) \quad (mm)$$

And this leads to a corrugation factor, defined as the ratio between corrugation length and pitch:

$$F_{cor} = \frac{L_{cor}}{L} = \frac{0.25L^2 + H^2}{L \cdot H} \cdot \sin^{-1}\left(\frac{L \cdot H}{0.25L^2 + H^2}\right)$$

This corrugation factor can be used to find the electrical resistance of a corrugated sheath following the next guidance point.

Guidance Point 30: Effect of corrugation

Issue	How to calculate the effect of a corrugated sheath?
Guidance	The corrugation increases the electrical resistance of the sheath with a factor F_{cor} because the length of sheath material per length of cable is increased by the corrugations. This factor must be taken into account: 1. If the cable manufacturer provides the corrugation factor, use this factor. 2. If the cable manufacturer does not provide the corrugation factor, do the following: • If the sheath resistance is provided by the manufacturer, no correction for corrugation is required. Otherwise: • Find the pitch and depth of the corrugation. If there is no depth of corrugation, calculate it by the following relation: $Depth = Outer\ diameter\ of\ sheath - Inner\ diameter\ of\ the\ sheath$ • Find the wave height by the following relation: $Height = Depth - thickness\ of\ sheath$ • Use the pitch and height to find F_{cor} with the following formula: $F_{cor} = \frac{L_{cor}}{L} = \frac{0.25L^2 + H^2}{L \cdot H} \cdot \sin^{-1} \left(\frac{L \cdot H}{0.25L^2 + H^2} \right)$ The factor F_{cor}, multiplied by the electrical resistance of a smooth sheath at the mean diameter gives the electrical resistance of the corrugated sheath. This mean diameter is the mean of the inner inside diameter and the outer outside diameter of the corrugated sheath. A typical value for F_{cor} is 1.1.
Notes	This method is correct for type 3 and type 2 corrugations. For type 1 corrugations, this method leads to a slightly higher electrical resistance. The formula assumes that $H \leq 0.5 \cdot L$
Relation to IEC	This is an addition to IEC.

2.7.3 Sheath losses in cables with Milliken conductors

In cable designs which comprise Milliken conductors, sheath losses are computed with a use of a factor F . The following guidance is offered in this case.

Guidance Point 31: Calculation of F-factor

Issue	Clarity over how to calculate the F-factor of Section 2.3.5 in [3].
Guidance	In all solid bonded (non-cross bonded) arrangements, the F-factor should be used. In cross bonded and single end bonded arrangements, the F-factor should not be used. The factor does not bind uniquely to Milliken conductors and should also be used for other conductor designs. In the calculation of eddy current losses, in cases of a combined sheath with a wire screen and a solid sheath or a foil laminate design, only the resistance of the solid sheath or foil should be used to determine R_s.
Notes	Per Guidance Point 6, we always include eddy current losses and do not neglect them, which makes this guidance point relevant for many more situations than which were addressed in IEC 60287 [3]. IEC clearly states this F factor to be applied to Milliken conductor designs only, but the reasons for the factor do not uniquely bind to Milliken conductors and are applicable to all conductor designs. The physical reason is that the circulating currents in the screen are reducing the eddy current losses. The original theory behind this was developed by Miller and can be found in [11].
Relation to IEC	This is an addition to IEC

2.8 Calculations for armour**2.8.1 Wire armour**

The losses in armour wires requires some further attention. First, the length of lay is addressed.

Guidance Point 32: Taking into account the length of lay in wire armour

Issue	Consideration of the length of lay in wire armour.
Guidance	If the electrical resistance of the wire armour is provided by the manufacturer, this electrical resistance should be used. In this case, no lay length factor needs to be applied. If the electrical resistance of the wire armour is not provided, the electrical resistance can be calculated on the basis of the number of wires, their cross section, resistivity and the lay factor of the armour wires. In both cases, the lay factor should be taken into account when calculating the degree of metallic covering of the cable.
Notes	The correction factor is provided in Guidance Point 23.
Relation to IEC	This is an addition to IEC.

In the following guidance point, the distance between the axis of a conductor and the cable centre in a three core cable is addressed.

Guidance Point 33: Distance between conductor axis and cable centre in three core cables

Issue	How to calculate the distance between the axis of a conductor and the cable centre for a three core cable?
Guidance	When calculating the armour losses for a three core cable following §2.4.2.3.1 [3] the distance between the axis of a conductor and the cable centre (c) is required. This parameter can be calculated either based on the core diameter and the assumption that the cores are in a close trefoil by: $c = \frac{\sqrt{3}}{2} D_{core}$ It could also be calculated by the assembly diameter as start point (i.e. Diameter of the circumference around all three cores without any assembly tapes or similar) using the following formula : $c = \frac{D_{assembly} - D_{core}}{2}$ These two methods might give slightly different results. If assembly diameter is available on the data sheet the second method is the recommended one as it includes the effect by the twisting of the cores that might give a small diameter increase.
Notes	
Relation to IEC	This is a clarification of IEC.

Further, a correction is made for the AC resistance of these wires and guidance is provided to deal with flat wire armour.

Guidance Point 34: Magnetic armour wire AC resistance dependency on diameter

Issue	Clarity over what should be done if the magnetic armour wires around a cable are not between 2 and 5 mm thickness as considered in Section 2.4.2.1 of [3]?
Guidance	A linear dependency of the magnetic armour wire AC resistance is proposed in the IEC standard ranging from 1.2 to 1.4 times the DC resistance when the wire diameter changes from 2 to 5 mm. Given these two points, use a linear relation to find the AC resistance for any specific wire dimension.
Notes	Use this relation within the range of 2 to 5 mm and outside of it, with the remark that wires thicker than 5 mm are common, but wires smaller than 2 mm are not common. This guidance is applicable to single and to three core cables. This subject is under consideration by CIGRE WG B1.64. The above approach is a worst case approach.
Relation to IEC	This is an addition to IEC.

Guidance Point 35: Flat wire armouring

Issue	Consideration of flat wire armouring in the AC resistance calculation.
Guidance	The DC resistance can be calculated based on the wire dimensions. In determining the AC resistance of a flat wire armouring, the smallest dimension of the cross section of a flat wire (typically the wire height) should be taken, and used to calculate the AC resistance according to guidance point 34.
Notes	This guidance is applicable to single and to three core cables. This subject is under consideration by CIGRE WG B1.64.
Relation to IEC	This is an addition to IEC.

Some specific situations may exist leading to additional guidance on the calculation of losses in armour, as shown below.

Guidance Point 36: Losses in nonmagnetic armour wires for three core cables of SL type

Issue	How should the losses be treated for a three core cable of SL type with non-magnetic armour wires?
Guidance	For non-magnetic armour wires IEC 60282-1-1 [3] gives the following guidance in §2.4.1: <i>"The general procedure is to combine the calculations of the loss in the reinforcement with that of the sheath. The formulae are given in 2.3 and the parallel combination of sheath and reinforcement resistance is used in place of the single sheath resistance R_s. The root mean square value of the sheath and reinforcement diameter replaces the mean sheath diameter d (see 2.3.11). This procedure applies to both single, twin and multicore cables.</i> <i>The value of the reinforcement resistance is dependent on the lay of the tapes as follows:</i> <i>a) If the tapes have very long lay (longitudinal tapes), the resistance is based on a cylinder having the same mass of material per unit length of cable and also the same internal diameter as the tapes.</i> <i>b) If the tapes are wound at approximately 54° to the cable axis, the resistance is twice the value calculated according to item a) above"</i> This method works well for a single core cable, but for a three core cable with wire armour this method is not possible due to lack of formula. The IEC [3] §2.3 which cover losses in screen/sheath does not have a formula for three core cables with common screen. There is a formula for common sheath but this will give too conservative values for cable with armour wires. The recommendation is therefore to assume, $\lambda_2 = 0$.
Notes	It is expected that there are some eddy currents in the armour material. They are however so small that they could be neglected. Note that a formula might be developed in future versions of IEC 60287.
Relation to IEC	This is a clarification to IEC.

Guidance Point 37: Reduction factor for armour losses of SL type cables

Issue	Which λ'_1 to use for the reduction factor for armour losses for SL type cables?
Guidance	According to paragraph 2.4.2.5 in [3] the armour losses should be reduced if the cable is of SL type (separate lead sheath on each cable core). The factor is calculated with: $\left(1 - \frac{R}{R_s} \lambda'_1\right)$ Where λ'_1 is obtained from 2.3.1 in [3]. The reduction factor is due to the reduction of magnetic field caused by the circulating current in the lead sheath of the cable. This current is affected by the presence of the armour wires and the circulating current for armoured SL type cables is there for calculated using formula from paragraph 2.3.10 in [3]. Since the circulating current is higher in the sheath the WG considers that the correct formula to use for λ'_1 in the reduction formula is the one from paragraph 2.3.10 in [3], i.e. the same λ'_1 as for the losses in the sheath.
Notes	For special cases where the sheath resistance is low, the reduction factor could become negative. For those cases the reduction factor is recommended to be set to 0.
Relation to IEC	This is a clarification of IEC.

Guidance Point 38: Loss calculations in case of cables with each core in a separate sheath and armoured

Issue	Clarity what loss calculations should be used in case of cables with each core in a separate sheath and armoured.
Guidance	In case of non-magnetic armour, use section 2.3.1 of [3]. In case of magnetic armour, use section 2.3.10 of [3], which adds a multiplication factor of 1.5 to the circulating sheath loss factor.
Notes	The factor of 1.5 should be used in cables with sheaths in general, not only for lead sheathed cables.
Relation to IEC	This is a clarification to IEC.

Guidance Point 39: Loss calculations in case of armoured cables in steel pipes and double armour constructions

Issue	Clarity over what loss calculations should be used in case of a three core armoured cable in a steel pipe, or when there are two armour layers (magnetic armour)
Guidance	In this case, two metal enclosures are present. As the first enclosure will effectively shield the magnetic fields, losses in the steel pipe or the outer armour can safely be ignored. Where there is air in between the armour and the steel pipe (an unfilled steel pipe), it is recommended to take into account the thermal resistance of the air layer in between the armour and the pipe as if the cable was located in the center of the pipe.
Notes	In case of perfect thermal contact between the armour and the pipe, the armour and the pipe attain the same temperature and hence the air in the pipe will not add to the thermal resistivity. However, perfect thermal contact is questionable in realistic cases. To remain conservative, the above guidance is proposed in which there is no thermal contact between the armour and the pipe. Reference is made to [12] and [13] illustrating the effective shielding. It is however noted that there may be constructions that shield less effectively and it is advised to consider this possibility before following the guidance, and that B1.64 may provide further guidance in their work.
Relation to IEC	This is an addition to IEC.

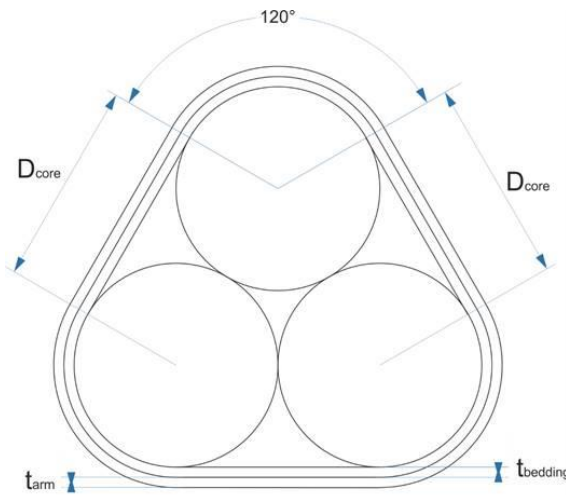
In case of wire armouring a decision has to be made over whether the wires are in contact with each other or not. Guidance Point 40 addresses this issue.

Guidance Point 40: Magnetic and electrical contact between armour wires

Issue	Consideration of whether wires are in contact or not.
Guidance	In case of physical contact between the wires, the wires are in magnetic contact with each other. Magnetic contact is easily established, and as cables with armours with a covering of ~90% of the circumference will have the armour wires regularly touching, for reasons of practicality it is proposed to consider wires in an armour layer with a covering of 90% or more in magnetic contact with each other. In case of magnetic contact, use $\mu_t=10$, in case of no magnetic contact, use $\mu_t=1$, both according to section 2.4.2.1 of [3]. In case of physical contact between the wires, electrical contact is not established, especially between armour wires. Therefore, it should always be assumed that there is no electrical contact between armour wires.
Notes	Note 3 on page 21 of [3] illustrates that there is no electrical contact between armour wires. In case it is certain that there is an electrical contact between armour wires (not envisioned), the calculations related to a continuous sheath could be used to find the losses.
Relation to IEC	This is a clarification to IEC.

In case the armour is applied over a bedding layer that is non-circular, the following guidance point is useful.

Guidance Point 41: mean circumference of armour for not round bedding layer

Issue	What is the circumference of the armour layer for a three core cable without a filler (not round)?									
Guidance	The circumference of a three core cable is the sum the edges of a triangle with rounded corners. The following figure shows the geometry where it is visible that the circumference consists of two parts. 1. Three parts of 1/3 circle with the diameter of the core plus any extra bedding material and half of the armouring wire layer thickness 2. Three times a length equal to the diameter of one core accounting for the straight edges in the figure below.  This can be written as $3D_{core} + (D_{core} + 2t_{bedding} + t_A)\pi$ Where: <table><tr><td>D_{core}</td><td>the core diameter</td><td>[mm]</td></tr><tr><td>$t_{bedding}$</td><td>the thickness of any material in-between core jacket and armour wires, e.g. tapes and beddings</td><td>[mm]</td></tr><tr><td>t_{arm}</td><td>the thickness of the armour wires</td><td>[mm]</td></tr></table>	D_{core}	the core diameter	[mm]	$t_{bedding}$	the thickness of any material in-between core jacket and armour wires, e.g. tapes and beddings	[mm]	t_{arm}	the thickness of the armour wires	[mm]
D_{core}	the core diameter	[mm]								
$t_{bedding}$	the thickness of any material in-between core jacket and armour wires, e.g. tapes and beddings	[mm]								
t_{arm}	the thickness of the armour wires	[mm]								
Notes	This is to be used for the calculation of lay length of armour wires for a three core cable without filler									
Relation to IEC	This is an addition to IEC.									

2.8.2 Reinforcing tapes

Metal tapes may be present in some power cables. They are typically used as a reinforcement. To clarify how to take these tapes correctly into account in the current rating calculations, follow Guidance Point 42 below.

Combinations of tapes and skid wires are often used in pipe type cables. For these cable types, follow both Guidance Point 42 and Guidance Point 43 below.

Guidance Point 42: Metal tapes

Issue	Consideration of metal tapes.
Guidance	Metal tapes have only an electrical effect and not a thermal effect. As such, the tapes need to be taken into account according to [3], Section 2.4.1. Both circulating and eddy current losses should be considered. In case of overlapping tapes, it is important to understand if there is electrical contact between the tapes or not. If there is no overlap, or if there is a material in between the overlapping tapes ('intercalated'), also in case of metallized tapes, then electrical currents will follow the helical tapes around the cable, and no eddy currents will be present. In this case, the electrical resistance should be calculated similar to the resistance of a wire with a lay factor: Use the dimensions, the resistivity and the lay factor to deduce the electrical resistance. Note that in the case of pipe type cables, often a lay length is used that is the width of the metal tape which is intercalated with a (metallized) polyester tape. That lay length should be used to calculate the electrical resistance. If there is electrical contact between overlapping tapes, the electric currents can flow circumferentially; hence eddy currents are present. In this case, the electrical resistance should be calculated assuming a tubular construction comprising the same amount of material as is present in the overlapping tapes. In all cases, circulating currents will be present.
Notes	Also refer to section 2.6 and Guidance Point 23.
Relation to IEC	This is an addition to IEC.

Guidance Point 43: Tapes and skid wires in pipe type cables

Issue	Consideration of tapes and skid wires in pipe type cables.
Guidance	In pipe type cables, often situations exist where there are more than one skid wire or more than one metal tape. These are taken electrically in parallel for the calculation of the circulating current losses.
Notes	Refer to Guidance Point 42 for guidance on how to calculate the electrical resistance of the tapes and the circulating and eddy current losses.
Relation to IEC	This is an addition to IEC.

2.9 Thermal resistance calculations

The following specific guidance is provided for the calculation of the thermal resistances. Additionally, it is noted that the degree of coverage of a wire screen also influences calculations of the thermal resistances. Guidance on this aspect is provided in Section 2.7.1 on wire screens.

Guidance Point 44: Compensation of T_1 in three core cables

Issue	In the case of three core cables or triplex cables, the thermal resistance between conductor and sheath, T_1 , should be calculated for a core length, not for a cable length. There is a difference between them, equal to the lay length factor calculated in Guidance Point 20.
Guidance	Divide T_1 by $F_{laylength}$ as calculated in Guidance Point 20
Notes	Partly, also T_2 is compensated for the lay length.
Relation to IEC	This is an addition to IEC.

2.10 Thermal resistance of Fillers in three core submarine cables

In the case of three core submarine cables, the thermal resistance between the sheath and armour must be calculated, but cannot be calculated according to Section 4.1.3.2 of [4] because typically each phase cable has a jacket with thickness t_j , there are fillers present, and there is a bedding with thickness t_b . In this case, it is proposed to use the following guidance.

Guidance Point 45: Calculation of T_2 in case of three core submarine cable

Issue	Clarity over what geometric factor should be used in the calculation of T ₂ for a three core cable with individual cable jackets and a bedding																					
Guidance	The calculation of T₂ is comprised of three steps: 1. Calculate the thermal resistance of the jacket around an individual phase cable with the formula given for a cylindrical layer of material (with an additional division by three included):$T_2^{\cdot} = \frac{1}{6\pi F_{lay\ length}} \rho_j \ln \left[1 + \frac{2t_j}{D_s} \right]$ With <table><tr><td>ρ_j</td><td>the thermal resistivity of the sheath</td><td>[Km/W]</td></tr><tr><td>t_j</td><td>the thickness of the jacket</td><td>[mm]</td></tr><tr><td>D_s</td><td>the outer diameter of the metallic sheath below the jacket</td><td>[mm]</td></tr><tr><td>$F_{lay\ length}$</td><td>the lay length factor (Guidance Point 20)</td><td>[-]</td></tr></table> 2. Calculate the thermal resistance of the fillers between the phase cables/cores and the bedding with the following formula:$T_2^{\cdot\cdot} = \frac{\rho_T \bar{G}}{6\pi}$ With: <table><tr><td>ρ_T</td><td>the thermal resistivity of the filler and bedding (highest of the two values in case of differences)</td><td>[Km/W]</td></tr><tr><td>G</td><td>a geometric factor, see below</td><td>[-]</td></tr></table> Note that the thermal resistance must reflect the thermal resistance of the filler and bedding in dry circumstances. Use a worst case approach to take into account combinations of materials 3. Add T₂[·] and T₂^{··} to find the total value of T₂ to be used in the calculation. A geometric factor is required. The following factor should be used: $\bar{G} = \begin{cases} 2\pi(0.00022619 + 2.11429X_b - 20.4762X_b^2) & \text{if } 0 < X_b \leq 0.03 \\ 2\pi(0.0142108 + 1.17533X_b - 4.49737X_b^2 + 10.6352X_b^3) & \text{if } 0.03 < X_b \leq 0.15 \end{cases}$ With: <table><tr><td>X_b</td><td>the thickness of material between the outer core jacket and the armour expressed as a fraction of the outer core jacket diameter</td><td>[-]</td></tr></table> The geometric factor is kept similar to what is currently inside the standard [4]. The WG considered that the jacket causes the heat flow in the filler to be different (the jacket is not an isotherm) and assumes the result is the worst case. Further investigations to the geometrical factor for three core cables are recommended, as the original experimental data on which the current factor is based, is dated and may not represent newer cable designs in the best possible way. Note: if the filler has voids which are water filled, consider the thermal properties of the filler only and do not consider the thermal properties of the water filling.	ρ_j	the thermal resistivity of the sheath	[Km/W]	t_j	the thickness of the jacket	[mm]	D_s	the outer diameter of the metallic sheath below the jacket	[mm]	$F_{lay\ length}$	the lay length factor (Guidance Point 20)	[-]	ρ_T	the thermal resistivity of the filler and bedding (highest of the two values in case of differences)	[Km/W]	G	a geometric factor, see below	[-]	X_b	the thickness of material between the outer core jacket and the armour expressed as a fraction of the outer core jacket diameter	[-]
ρ_j	the thermal resistivity of the sheath	[Km/W]																				
t_j	the thickness of the jacket	[mm]																				
D_s	the outer diameter of the metallic sheath below the jacket	[mm]																				
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ρ_T	the thermal resistivity of the filler and bedding (highest of the two values in case of differences)	[Km/W]																				
G	a geometric factor, see below	[-]																				
X_b	the thickness of material between the outer core jacket and the armour expressed as a fraction of the outer core jacket diameter	[-]																				

	If the filler has voids which are air filled also in rope filler situations, the thermal resistivity may be high. Careful consideration should be given to the value of the representative thermal resistivity. Note: Refer to [14], [15] for some latest developments.
Relation to IEC	This approach differs from the approach of IEC.

2.10.1 Thermal resistance of cables in trefoil formation

When cables are installed in trefoil, the thermal resistance T_1 and T_3 may be increased by a factor given in Sections 4.2.4.3.2 and 4.2.4.3.3. of [4]. This factor is, however, not always applicable as indicated in the guidance below.

Guidance Point 46: Thermal resistance of cables in trefoil in air

Issue	Consideration of compensation for the close proximity for cables in touching trefoil formation in air.
Guidance	In case of cables in air in trefoil formation, factors which increase T_1 or T_3 (as given in Sections 4.2.4.3.2 and 4.2.4.3.3 of [4]) must not be taken into account.
Notes	If cables are in trefoil formation, the temperature distribution in individual power cables is disturbed. However, when the cables are installed in air, the air around the power cables will reduce the disturbance. The factors are calculated for buried touching cables in trefoil formation only.
Relation to IEC	This approach differs from the approach of IEC.

Guidance Point 47: Thermal resistance of three core cables in buried pipes in trefoil

Issue	Consideration of compensation for the close proximity for cables installed in buried pipes which are in touching trefoil formation.
Guidance	In case of cables in buried pipes in trefoil formation, a factor to increase T_1 or T_3 (as given in given in Sections 4.2.4.3.2 and 4.2.4.3.3 of [4]) must not be used.
Notes	In this case, the WG actually expects a factor is required, but the factor will be different from what is currently in the IEC standard. Therefore, this issue has been highlighted to the IEC for future consideration.
Relation to IEC	This approach differs from the approach of IEC.

Guidance Point 48: Equivalent cable outer diameter in case of a non-round cable type

Issue	How to calculate the cable mean outer diameter?
Guidance	In case the cable mean outer diameter is not given, the mean outer diameter of the cable can be calculated with: $D_e = \frac{3D_{core} + (D_{core} + 2t_{bedding} + 2t_{arm})\pi}{\pi}$ Where: D_{core} the core diameter [mm] $t_{bedding}$ the thickness of any material in-between core jacket and [mm] armour wires, e.g. tapes and beddings t_{arm} the thickness of the armour wires [mm]
Notes	Also refer to Guidance Point 41
Relation to IEC	This is an addition to IEC.

2.10.2 Thermal resistance between cable(s) and duct or pipe

When cables are installed in a duct or pipe, the thermal resistance between the cable and the duct or pipe may be calculated according to section 4.2.7.2 of IEC 60287-2-1[4]. In that case, the assumption is that the cable diameter is in the range of 25 mm to 100 mm in a ducted cable, or that the equivalent diameter of the three cores in a pipe is within the range of 75 mm to 125 mm. This is a limiting assumption and it is unclear what to do outside of the named range. Therefore, the following guidance point is proposed.

Guidance Point 49: Thermal resistance between cable and duct or pipe

Issue	How to calculate the thermal resistance between cable and duct (or pipe) outside of the mentioned range of (equivalent) cable diameters.
Guidance	In case section 4.2.7.2 of [4] is to be followed, and in case the outer diameter of a cable in a duct is outside of the range of 25 mm to 100 mm or in case the equivalent diameter of the three cores in a pipe is outside the range 75 mm to 125 mm, the following guidance is provided: • There is no published information that can be referred to, but for larger cable outer diameters, the use of the U, V and Y coefficients provided by IEC [4] will be conservative. It is therefore recommended in this case to use the provided U, V and Y coefficients. • For smaller cable outer dimensions, the use of the U, V and Y coefficients will likely not be conservative and no solution to follow the IEC methodology for the calculation of the current rating exists.
Notes	The most common issue is that the cable outer diameter is exceeding the maximum range mentioned in IEC.
Relation to IEC	This is an addition to IEC.

3. Interpretation of differences

The verification of a calculation technique, tool or software may lead to finding differences between the user's (intermediate) results and the case study results reported in the remainder of this brochure. In principle, there is no major reason for these differences to exist. Note that many experts agreed to the intermediate results reported in this brochure.

Any difference should be investigated with the aim to understanding its origins and possibly correcting the calculation tool. In case of a difference, first check if all input parameters and all formulas are exactly correct. If, after correcting possible errors, the differences still exist, this may have to do with:

- the way of iterating in the calculation algorithm or the limitations set for the iteration process. Follow the first case study, reported in Section 4 to understand how the iterations are done in this brochure.
- working with limited accuracy variables or constants. Refer to section 2.2.
- differences in software compilers. It may be interesting to try a different software environment or different computer to see if the difference found earlier still exist.
- this brochure reporting a wrong result. The chance on this happening has been reduced by independent expert calculations of each case study, an additional verification round of each case study, an external expert review panel of the full work and the SC B1 review before publication. In case of still finding an issue, please notify the convener of this working group.

If the differences still exist and cannot be explained, the user could decide to allow the difference to happen and to not improve the tool. If this only leads to minute differences in the end result, this could be an explainable choice. It is however advised **not** to do this, as there are no good technical reasons to allow for the differences. Note that differences in one situation may be small, while in another, more complex installation may play a much more important role. Therefore, creating a sound basis without any differences should be the goal. Despite this advise, there may be other reasons (historic, economic) that lead to a choice of not improving the calculation technique, tool or software.

To continue to provide clarity, when verifying the calculation tool it is advised to report the verification results in detail. Any remaining differences should be explained in this verification report. When a current rating is calculated with the verified tool, this verification report should be referenced and made available, such that it is clear to all users how well the calculation tool aligns with the case studies in this report, and what the reasons for any possible differences are.

Once more it is noted that the working group agreed to present the results of case studies reported in this brochure with all significant digits. According to guidance point 2, when the current ratings are reported in normal use, it is advised to round the ampacities to the nearest 1A, 5 A or 10 A value.

4. Introductory Case study 0

4.1 Introduction

When the working group compared the calculations from 15 experts for the first time, 15 different results were obtained. Moreover, all results were different from the values that will be shown later as the best achievement. The assumptions of the first case study were not sufficiently well specified and some uncertainties were raised by a combined screen (laminated foil and wires). Guidelines were then proposed to converge towards a better consent. It was a good motivation to propose another case study: make it simpler to decrease as far as possible the risk of errors and discrepancies, and to focus on the elementary calculations only, before increasing the complexity of the analysis with a step by step approach.

The first case study has been prepared with a configuration as simple as possible, not only in terms of the installation, but also to perform the easiest calculations (small number of formulae, basic equations, parameters independent of the temperature, shared values for all cables):

- Conductor = 630 mm² in copper, suitable for the installation in a narrow trench, with a stranded design (no specific calculation vs. Milliken conductors, $k_s = k_p$).
- Metal sheath = a single laminated aluminium foil (no copper wires).
- Voltage level = 132 kV, in a range between high and extra high voltage, which could be operated in almost any transmission utility.
- Installation = firstly directly buried, to simplify the calculation of the external thermal resistance, especially in the case of touching cables in trefoil formation, for which the T_4 value is common to the three cables. No duct, neither backfill nor concrete bank: the soil is considered as homogeneous.
- Earthing mode = sheaths bonded at both ends of an electric section (with circulating currents losses, with the eddy-current losses ignored).
- Standard operating conditions, no soil drying. The maximum conductor temperature is given.

4.1.1 Variants and discussion

The initial configuration is as basic as possible and was then customised in different variants, in order to explore most of the installation and operation modes:

- Trefoil and flat geometries, with touching cables or not,
- Cables in ducts, embedded in concrete or laid in a backfill, cables in free air exposed to direct solar radiation, cables in an unfilled trough,
- Earthing mode with single point bonded cables.

A summary table is included in Section 4.11 to recapitulate the fourteen different sub-cases verified in this report (for the steady-state operation). Click on the links to jump to the appropriate sections.

Whenever possible, approximations and optimisations are discussed, as well as potential traps in the interpretation or the calculations.

Furthermore, the introductory Case Study 0 formed the opportunity for the working group to discuss approaches different from IEC, which were concluded by guidance points. The working group started from the IEC rules, then moved to a recommended framework. Variants illustrate this process. The following table is inserted for each of them and illustrates the chosen options:

Variable	Topic	IEC rule	CIGRE B1.56 recommendation
W_d	Dielectric losses	Losses neglected depending on the voltage level and insulation material ☒	Losses always taken into account GP 7 ☐
λ_1''	Eddy current losses	Losses neglected when circulating currents are considered, except for Milliken conductors ☒	Losses always taken into account GP 6 ☐
F	Reduction of eddy currents by circulating currents	Used for solid bonded cables with Milliken conductors ☒	Used for all solid bonding arrangements GP 31 ☐
T_4	External thermal resistance of buried cables or ducts	Simplified formula using $2u$ when $u > 0$ ☒	Exact formula with $u + \sqrt{u^2 - 1}$ GP 8 ☐
		Simplified formula considering equally loaded identical cables ☒	Formula considering unequally loaded cables GP 8 ☐

4.1.2 Formatted results

The values of input data are printed in blue characters, calculated results are red.

Calculated results are given with a high precision (rounded to ten significant digits for real numbers, while they are calculated on fifteen digits, corresponding to coded floating point numbers with double precision). As mentioned in Guidance Point 2, there is no physical reason to do this, but it is a way to improve the detection of potential discrepancies between verified calculations performed by the members of the working group.

The adopted decimal separator is a comma, as recommended by the ISO and the IEC standards.

Since some important parameter values are temperature dependent, many calculations are iterative. The first iteration is developed with full details. A table of the main results is then provided to follow the next iterations until a convergence is reached with a specified accuracy.

Finally, a ladder network representing the steady state thermal circuit showing the main results (electrical resistances and capacitances, losses, thermal resistances, temperature on components, hottest cable drawn in red in a pictogram showing the laying geometry) is included at the end of each sub-section.

4.2 CASE #0: Input data

4.2.1 Grid parameters

A frequency of 50 Hz is chosen to operate in most countries represented by the members of the working group.

Parameter	Symbol	Value	Unit	Selection
Voltage level	U	132	kV	User
Frequency	f	50	Hz	User

4.2.2 Cable parameters

Figure 25 shows the datasheet as used for the case study, giving the description of the cable layers (materials and geometry):

XLPE insulated cable 132 kV

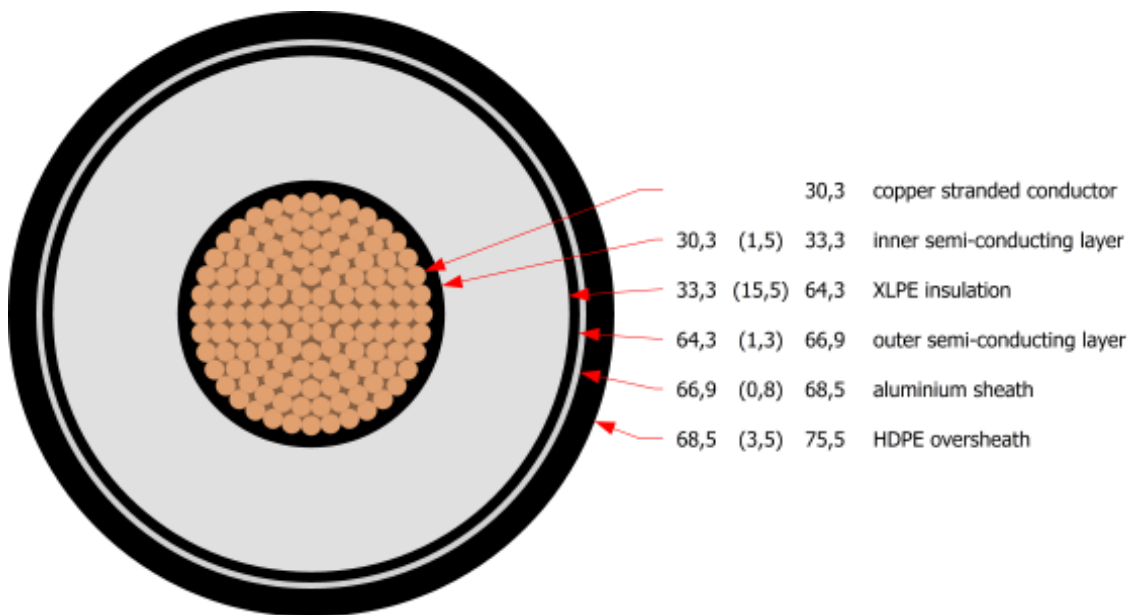
Copper cable 132 kV 630 R (aluminium)

Characteristics

Construction characteristics	
Conductor material	Copper
Insulation	XLPE (chemical)
Type of conductor	Circular compacted stranded
Metallic sheath	Welded aluminium
Dimensional characteristics	
Conductor diameter	30.3 mm
Conductor cross-section	630 mm ²
Thickness of the inner semi-conductor	1.5 mm
Average insulation thickness	15.5 mm
Thickness of the external semi-conductor	1.3 mm
Thickness of metallic sheath, approx.	.8 mm
Nominal outer diameter	75.5 mm
Screen section	170 mm ²
Average sheath thickness	3.5 mm
Approximate weight	10 kg/km

Figure 25: Cable datasheet as used for case #0

Figure 26 is the cross-sectional drawing built accordingly, assuming adjacent layers. The columns on the right side give for each layer the inner diameter, the thickness and the outer diameter respectively (mm).

**Figure 26: Cross sectional drawing of cable for case #0**

Sometimes, a felt tape is laid under the laminated aluminium sheath. This is not the case in this basic study with a minimum number of components.

4.2.2.1 Conductor

The value of the DC resistance at 20°C of a round stranded conductor for single-core cable is taken from Table 2 of the IEC 60228 standard [8].

1	8	9
Nominal cross-sectional area mm ²	Maximum resistance of conductor	
	Annealed copper conductor	
	Plain wires	Metal-coated wires
	Ω/km	Ω/km
400	0,0470	0,0475
500	0,0366	0,0369
630	0,0283	0,0286
800	0,0221	0,0224
1 000	0,0176	0,0177

$$R_o = 28,3 \cdot 10^{-6} \Omega/m$$

The values of the skin and proximity effects coefficients are selected for a round stranded conductor and extruded insulation system.

Parameter	Symbol	Value	Unit	Selection
Skin effect coefficient	k_s	1	-	[3], Table 2
Proximity effect coefficient	k_p	1	-	[3], Table 2

4.2.2.2 Insulation layer

A common XLPE insulation layer with semi-conducting layers is chosen. The values of relative permittivity and loss factor are selected for a voltage greater than 18/30 (36) kV (unfilled). The thickness is designed by the manufacturer to withstand the specified maximum electrical field when the voltage is applied between the conductor and earth.

Parameter	Symbol	Value	Unit	Selection
Relative permittivity	ϵ	2,5	-	[3], Table 3
Loss factor	$\tan \delta$	0,001	-	[3], Table 3
Thermal resistivity	ρ_i	3,5	<i>Km/W</i>	[2], Table 1

4.2.2.3 Metal sheath

An aluminium sheath is chosen as a watertight metal sheath. The thickness is designed by the manufacturer to withstand the specified short-circuit current.

Parameter	Symbol	Value	Unit	Selection
Thickness	t_s	0,8	<i>mm</i>	Manufacturer datasheet
Resistivity at 20°C	ρ_{20}	$2,84 \cdot 10^{-8}$	<i>Ω/m</i>	[3], Table 1
Temperature coefficient	α_{20}	$4,03 \cdot 10^{-3}$	<i>K⁻¹</i>	[3], Table 1

4.2.2.4 Oversheath

A common PE oversheath is chosen. The value of the thermal resistivity is selected in the 'protective coverings' section of the IEC Table. The thickness is designed by the manufacturer for a proper mechanical protection and to withstand the specified induced voltages in the metal screen.

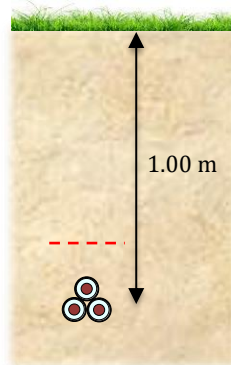
Parameter	Symbol	Value	Unit	Selection
Thermal resistivity	ρ_T	3,5	<i>Km/W</i>	[4], Table 1

4.2.3 Earthing of the metal sheaths

The case study deals with a power link whose power is moderate and hence it is compatible with bonding at both ends of an electrical section. The installation in trefoil with touching cables (tied up by a strap in helix) implies that there is no variation of the spacing between sheath bonding points.

4.2.4 Installation

In this simple case, touching cables are laid in a trefoil formation, directly buried. The depth is 1,00 m measured between the ground surface and the centre of the trefoil (which implies a short trigonometric calculation to identify the coordinates of the centre of the single-core cables). The **1,00 m** value is recommended by IEC 60287-3-1 Section 3.2.1 [16] when no information about the depth is given.



The choice of a trefoil formation is motivated by the axial symmetry, allowing calculation simplifications, compared with the flat formation leading to additional assumptions (central and adjacent cables, transposition of phases, etc.).

The depth at the bottom of the trench can be found with basic trigonometry:

$$y_{bottom} = y_{center} + \left(\frac{1}{\sqrt{3}} + 1 \right) \frac{D_e}{2}$$

Note: attention must be paid to the reference depth. The engineer often considers the centre of the circuit for the current rating calculation, while civil works involve the dimensions of the trench (bottom depth is then provided).

The (arbitrary) numbering of the cables is adopted:



For the depth $L = 1$ m at the centre of the circuit (centre of the trefoil), the coordinates of the cables are:

	1	2	3
x (m)	0,00	0,03775	-0,03775
y (m)	-0,9564100547	-1,021794973	-1,021794973

4.2.5 Operating conditions

The maximum temperature of the conductor is 90°C, relevant for the XLPE insulation. The soil is considered as homogeneous, and soil drying is not considered. The temperature on the oversheath surface may be high, but no special attention is paid on the soil interface in this theoretical case study. The ambient ground temperature and the thermal resistivity of soil are selected for a temperate climate in summer in an area with regular rainfalls.

Parameter	Symbol	Value	Unit	Selection
Maximum operating temperature of conductor	θ_c	90	$^{\circ}\text{C}$	User
Ambient ground temperature	θ_a	20	$^{\circ}\text{C}$	[16], Section 3.2.1
Thermal resistivity of soil	ρ_T	1,0	Km/W	[16], Section 3.2.2

4.3 CASE #0: Guidelines for the iterative calculation

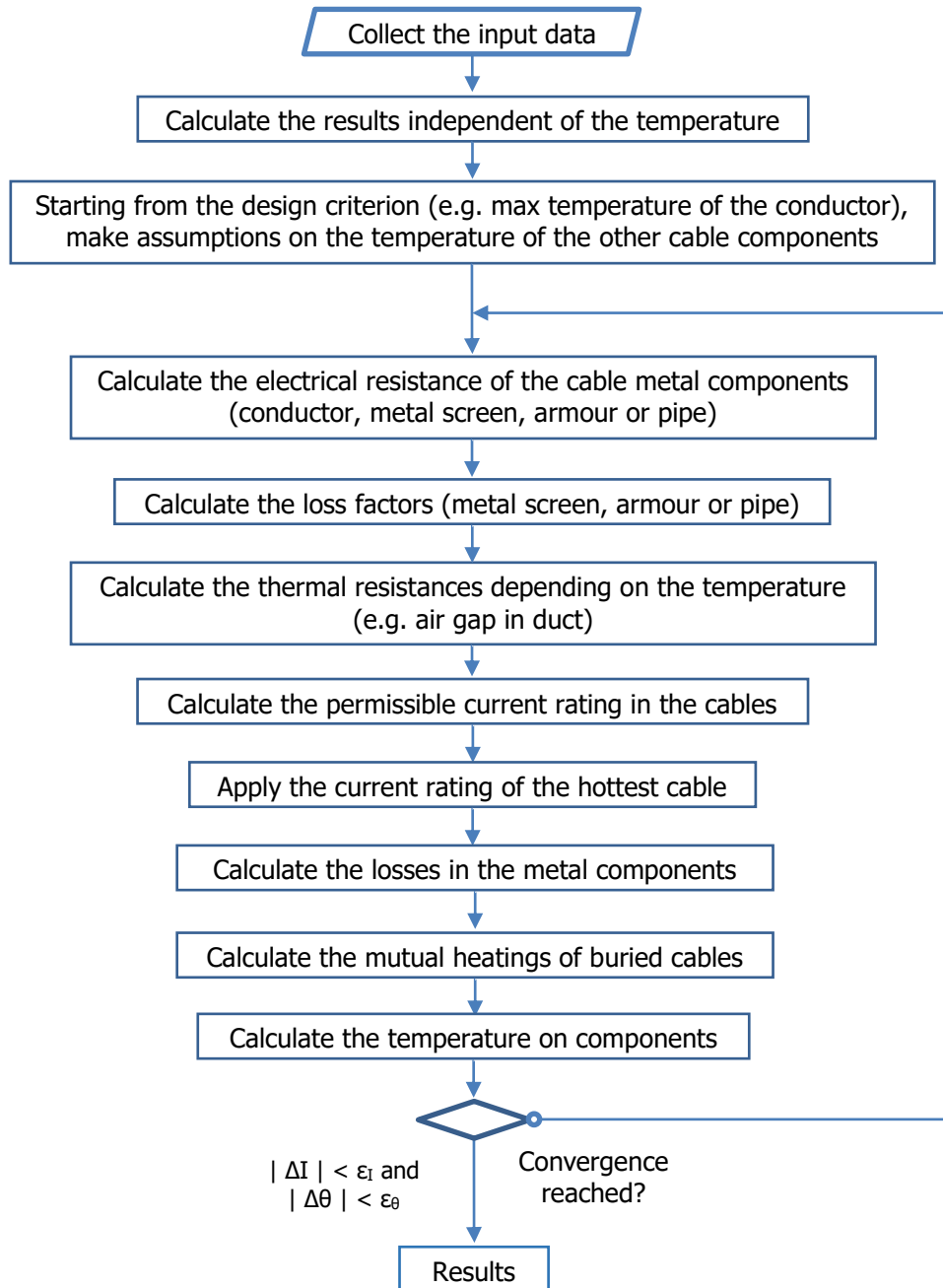
The electrical resistivity of a metal conductor varies with temperature. The temperature of each of the metal components (conductors, wires, sheathes, tapes or pipes) should be known to assess their respective resistance and dissipated losses. When the calculation starts, only the boundary temperatures are identified: the maximum conductor operating temperature and the ambient temperature of the surrounding medium (soil or air). The configuration is sometimes more complex due to additional heat sources (other power circuits, heat pipes, etc.), but the principle remains unchanged.

Since the temperature of the metal components (other than the conductor of the hottest cable) is unknown, a simple approach is to assume the maximum operating temperature for all the conductors θ , and a lower temperature for the other components (e.g. $\theta - 10^{\circ}\text{C}$ for the metal screens and sheaths, $\theta - 15^{\circ}\text{C}$ for the armours and pipes). The choice of these initial values does not affect the final result. They could be initiated with the value of the maximum conductor θ or the ambient temperature θ_a . An intuitive more logical value improves the convergence speed of the iterative process.

The electrical resistance of all the components can be calculated at the assumed temperature for the first iteration, and then be refined after the successive iterations, until the final convergence is reached, leading to the solution.

The method is summarised by the flowchart in section 4.3.1.

4.3.1 Flowchart of the iterative calculation



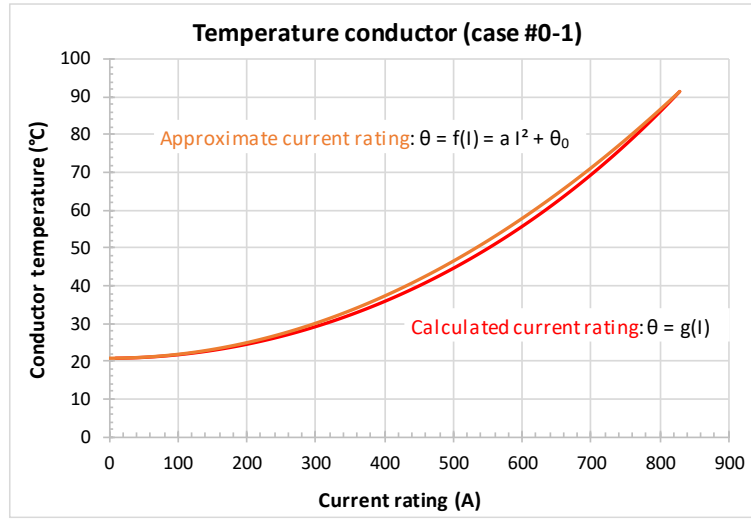
4.3.2 Criteria of convergence

The calculation process can be stopped when the difference between two consecutive values becomes lower than an accepted tolerance threshold. At the time of hand calculations (Neher and McGrath published *The Calculation of the Temperature Rise and Load Capability of Cable Systems* [1] in 1957!), the process was difficult and time-consuming. An accuracy of one amp and one degree Celsius could be acceptable. Nowadays, the use of programming and computers removes these limits.

It should be noted that the convergence test must apply on both current rating and temperature ($|\Delta I| < \epsilon_I$ and $|\Delta \theta| < \epsilon_\theta$). The test on the current rating only may indeed be insufficient. Sometimes, oscillating successive values around the final result would be stopped, while the next iteration with recalculated losses and temperatures would require to carry on the iterative process. This was observed, even on simple configurations with a single circuit.

For this case study, values of 0,001 A and 0,0001°C have been adopted, not from a physical point of view, but in order to smooth possible discrepancies (step visual effect) between close points on a curve (e.g. temperature conductor as a function of transmitted current).

A further discussion updated the relevancy of these paired values and led to Guidance Point 5. The starting point is the drawing of the current rating I (A) in terms of the conductor temperature θ_c (°C). In first order, the red curve of the calculated current rating can be approximated to the orange basic parabola arc (heating of the cable due to Joule effects $\approx R \cdot I^2$):



Two points of the parabola arc are remarkable:

- $I_0 = 0$ and $\theta_c = \theta_0 = 20,73^\circ\text{C}$: θ_0 is the ambient temperature increased by the heating due to the dielectric losses (cables are energized, but do not transmit current).
- $I_{100\%} = 821,81$ A at $\theta_{c,100\%} = 90^\circ\text{C}$ (calculated in Section 4.6.1.1): operating at the maximum conductor temperature.

It is then possible to identify the a coefficient:

$$\theta_{c,100\%} = \theta_0 + a I_{100\%}^2 \Rightarrow a = \frac{\theta_{c,100\%} - \theta_0}{I_{100\%}^2}$$

The current rating may be approximated by the minimalist following relations:

$$\theta_c = \theta_0 + a I^2 \quad I = \sqrt{\frac{\theta_c - \theta_0}{a}}$$

The maximum deviation of the approximate curve is lower than $2,1^\circ\text{C}$ at $I \approx 590$ A, which is sufficient to get an order of the link between I and θ_c expected below.

For a given temperature θ_c (e.g. $\theta_{c,100\%} = 90^\circ\text{C}$), the slope of the tangent line at the calculated point (I, θ) is the derivative of the function $f(I)$:

$$\frac{d\theta_c}{dI} = 2a I = 2 \frac{\theta_{c,100\%} - \theta_0}{I_{100\%}^2} I$$

Considering a convergence around $\theta_c = 90^\circ\text{C}$, the slope of the tangent line at the calculated point is:

$$\frac{d\theta_c}{dI} = 2 \frac{90 - 20,73}{821,81} = 0,169$$

The calculation of the derivative at the point has been performed with the other sub-cases described hereafter and led to values in the range of 0,1 to 0,2. Therefore, it was decided that the paired values of ε_I and ε_θ are linked by the relation:

$$\varepsilon_\theta = \frac{\varepsilon_I}{10}$$

For this case study, values of 0,001 A and $0,0001^\circ\text{C}$ have been adopted accordingly.

The choice remains consistent for lower maximum temperatures (in the case of paper insulated or HVDC cables, for example). The slope of the tangent line is lower, but of the same order of magnitude.

When calculations are performed and reviewed by users, it may be interesting to harmonise the level of required accuracy before any comparison.

This consideration only applies to hard-coded tools. Applications with spreadsheets have their own solver and algorithm and the user is not concerned by this item. The verification can thus start with the final result value of the temperature on each cable component and stops after a single iteration.

4.3.3 Speed of convergence

The following table shows for a few sub-cases, the required number of iterations to achieve the convergence in terms of the selected threshold of convergence on the current rating ε_I :

ε_I (A)	1	0,1	0,01	10⁻³	10 ⁻⁴	10 ⁻⁵	10 ⁻⁶	10 ⁻⁸	10 ⁻¹⁰	10 ⁻¹²
Buried cables (case #0-1, Section 4.6.4)	2	3	3	4	4	4	5	6	7	8
Touching HDPE ducts (case #0-2, Section 4.7.74.7.6)	3	3	3	4	5	5	6	7	8	9
PVC ducts in concrete (case #0-3, Section 4.8.6)	6	8	9	11	12	14	15	18	21	24
Free air + solar rad. (case #0-4, Section 4.9.7)	3	3	4	6	8	9	11	14	18	21
Unfilled trough (case #0-5, Section 4.10.74.9.7)	12	16	21	25	29	34	38	47	55	64

The shaded column stands for the selected threshold for the study ($\varepsilon_I = 0.001$ A, Guidance Point 5).

Computers with 32-bit processors using real numbers with double precision limit the convergence threshold ε_I to 10⁻¹² (four digits are already used for current ratings ≥ 1000 A).

4.4 CASE #0-1: Results independent of the temperature

4.4.1 Capacitance

The capacitance for circular conductors is given by:

$$C = \frac{\varepsilon}{18 \ln\left(\frac{D_i}{d_c}\right)} 10^{-9} \quad (\text{F/m}) \quad ([3], \text{Section 2.2})$$

E	2,5	C = 2,110766220 10⁻¹⁰ F/m
d _c	30,3 + (2 x 1,5) = 33,3 mm	
D _i	33,3 + (2 x 15,5) = 64,3 mm	

Attention must be paid to only consider the dielectric only when the values of the diameters are selected (semi-conducting layers or tapes excluded). This point is reported as a common pitfall in Section 2.4.

4.4.2 Reactance

The axial distance s between two touching cables is equal to the external diameter of the cable = $D_e = 75,5$ mm.

The mean diameter of the sheath is the external diameter of the insulation layer, including the outer semiconducting layer, increased by the thickness t_s of the aluminium foil.

The reactance is given by:

$$X = 2 \omega 10^{-7} \ln \left(\frac{2s}{d} \right) (\Omega/\text{m}) \quad ([3], \text{Section 2.3.1})$$

$$\omega = 2 \pi \times \text{frequency (1/s)}$$

f	50 Hz	X = 5,040331399 10⁻⁵ Ω/m
ω	2π × 50 = 314,159265358979 rad/s	
s	75,5 mm	
d	64,3 + (2 × 1,3) + 0,8 = 67,7 mm	

4.4.3 Electrical resistance of the metal sheath at 20°C

The resistance the cable sheath or screen at 20 °C (Ω/m) is calculated to prepare the calculation of the sheath resistance at its operating temperature R_s when the temperature of the sheath varies during the iterative process.

R_{s0} depends on the cross-sectional (mean diameter d and thickness t_s) area of the sheath and the electrical resistivity of the metal.

$$R_{s0} = \frac{\rho_{20}}{A_s} = \frac{\rho_{20}}{\pi d t_s}$$

d	67,7 mm	R_{s0} = 1,669128650 10⁻⁴ Ω/m
t _s	0,8 mm	
A _s	π × 67,7 × 0,8 = 170,14866 mm ²	
ρ ₂₀	2,84 10 ⁻⁸ Ω/m	

4.4.4 Thermal resistance between conductor and sheath

For single-core cables, the thermal resistance between one conductor and the sheath T_I is given by:

$$T_I = \frac{\rho_T}{2\pi} \ln \left[1 + \frac{2t_1}{d_c} \right] \quad ([4], \text{Section 4.1.2.1})$$

The semiconducting layers on both sides of the insulation itself must be included in the thickness t_1 because they contribute to the resistance from a thermal point of view (same polymer matrix than the insulation uncharged in carbon black).

Therefore, the inner semiconducting layer is excluded of the diameter d_c . There is no physical possible confusion, but the double use of the d_c variable with different values (see Section 4.4.1) is quite unsafe.

Note: CIGRE WG B1.56 adopted an approach so that semi-conductive XLPE layers shall be modelled as separate layers with a thermal resistivity of 2,5 Km/W as per IEC 60949 [5] (lower than the insulation layer itself).

	Inner SC layer	Insulation layer	Outer SC layer	T_I = 0,4198714890 Km/W
t ₁	1,5 mm	15,5 mm	1,3 mm	
d _c	30,3 mm	33,3 mm	64,3 mm	
ρ _T	2,5 Km/W	3,5 Km/W	2,5 Km/W	
T _{1 layer}	0,03756441990	0,3665350785	0,01577199062	

Note that according to IEC [4] Section 4.2.4.3.3, the value of T_I must be multiplied by 1,16, because in the case of part-metallic covered cables (where helically laid armour or screen wires cover from 20 % to 50 % of the cable circumference) the metal screen cannot be considered as an isotherm. But in this case study with a smooth sheath, this 1,16 factor does not apply.

4.4.5 Thermal resistance of the oversheath

The thermal resistance of the external covering T_3 is given by:

$$T_3 = \frac{1}{2\pi} \rho_T \ln \left(1 + \frac{2t_3}{D'_a} \right) \quad ([4], \text{Section 4.1.4.1})$$

In the absence of armour, we focus on the component immediately beneath the oversheath. The reference diameter D'_a is the external diameter of the metal sheath D_s (mean diameter d increased by the sheath thickness t_s).

t_3	3,5 mm	$T_3 = 0,05419960924 \text{ Km/W}$
$D'_a (D_s)$	68,5 mm	
ρ_T	3,5 Km/W	

Warning! Despite the lack of reminder in Section 4.1.4.1 of [4], the T_3 value must be increased in the case of three single-core touching cables, with metallic sheath in trefoil formation (Section 4.2.4.3.2 of [4]):

In this case, the thermal resistance of the serving over the sheath or armour, T_3 , as calculated by the method given in 4.1.4 shall be multiplied by a factor of 1,6.

Finally:

$$\mathbf{T_3 = 0,08671937479 \text{ Km/W}}$$

The factor takes into account the distortion of the heat source when the cables in trefoil are touching (it is not possible to consider the metal screen as an isotherm surface anymore). It is specified in Section 4.2.4.3.2 of [4], and therefore applies to groups of buried cables (touching) equally loaded. The working group raised the fact that the distortion of heat source should occur for touching cable in trefoil in free air as well, and to a lesser extent to touching ducts in trefoil. According to the standard, the 1,6 factor will be applied to buried cables only in the present report.

4.4.6 External thermal resistance

The case study complies with the assumption of a group of buried cables (touching) equally loaded (Section 4.2.4 of [4]), especially with three single-core cables in trefoil formation, with metal sheath. The thermal resistance T_4 of the environment of the cables is given by:

$$T_4 = \frac{1,5}{\pi} \rho_T [\ln(2u) - 0,630] \quad (\text{Section 4.2.4.3.2 of [4]})$$

L	1,00 m = 1000 mm	$T_4 = 1,594692892 \text{ Km/W}$
D_e	75,5 mm	
$u = 2L/D_e$	26,49006622	
ρ_T	1,0 Km/W	

The value is common to all cables.

Note that a specific formula applies to the case of buried touching cables, to take into account the deformation of the isotherm assumed at the surface of the cable. Therefore, the present calculation refers to the IEC (including $2u$ quantity) and not to Guidance Point 8 (see Section 2.6).

4.4.7 Dielectric losses

The dielectric loss in each phase is given by:

$$W_d = \omega C U_0^2 \tan \delta \quad (\text{W/m}) \quad ([3], \text{Section 2.2})$$

ω	314,1592653 rad/s	$W_d = 0,3851382172 \text{ W/m}$
C	$2,110766220 \cdot 10^{-10} \text{ F/m}$	
$U_0 = U/\sqrt{3}$	76210,23553 V	
$\tan \delta$	0,001	

Note: Table 3 of IEC 60287-1-1 [3] mentions that "The dielectric loss should be taken into account for values of U_0 equal or greater than 127 kV for XLPE (unfilled)". The $U = 132 \text{ kV}$ voltage level ($U_0 = 76 \text{ kV}$) means that dielectric losses could be neglected for the case study. However, because of the simplicity of the W_d expression, the calculation is systematically applied. This point is discussed in Section 4.6.1.3.

4.5 case #0-1: Results dependent on the temperature

The electrical resistance of metal components (conductor, sheath) depends on the temperature and must be reassessed at each loop in an iterative process. These results determine the value of losses and lead to a series of new temperatures for the next iteration, until convergence is reached.

The values have to be initialised by the user to launch the first iteration. The maximum operating temperature ($\theta_c = 90^\circ\text{C}$) is a good start for the temperature of the metal screens or sheaths with:

$$\theta_s = \theta_c - 10 = 80^\circ\text{C}$$

4.5.1 Results linked to the maximum operating temperature

The case study is based on a single criterion: the maximum temperature of the conductor (90°C). This parameter has not to be reconsidered and lowered to comply with possible other criteria (maximum temperature on oversheath to avoid soil drying, maximum ambient temperature in a non-ventilated confined space).

4.5.1.1 DC resistance of conductor

The DC resistance of the conductor at its maximum operating temperature θ is given by:

$$R' = R_0 [1 + \alpha_{20} (\theta - 20)] \quad (\text{Section 2.1.1 of [3]})$$

θ_c	90°C	$R' = 3,608533 \cdot 10^{-5} \Omega/\text{m}$
R_0	$28,3 \cdot 10^{-6} \Omega/\text{m}$	
α_{20}	$3,93 \cdot 10^{-3} \text{ K}^{-1}$	

4.5.1.2 Skin effect factor

The skin effect factor y_s is given by the following equations:

$$x_s^2 = \frac{8\pi f}{R'} \cdot 10^{-7} k_s \quad (\text{Section 2.1.2 of [3]})$$

$$\text{For } 0 < x_s \leq 2,8 \quad y_s = \frac{x_s^4}{192 + 0,8 x_s^4}$$

f	50 Hz	$x_s = 1,866120103$ $y_s = 0,06012412684$
k_s	1	
R'	$3,608533 \cdot 10^{-5} \Omega/m$	

4.5.1.3 Proximity effect factor

For circular conductor cables, the proximity effect factor y_p is given by the following equations:

$$x_p^2 = \frac{8\pi f}{R'} \cdot 10^{-7} k_p$$

$$y_p = \frac{x_p^4}{192 + 0,8 x_p^4} \left(\frac{d_c}{s} \right)^2 \left[0,312 \left(\frac{d_c}{s} \right)^2 + \frac{1,18}{\frac{x_p^4}{192 + 0,8 x_p^4} + 0,27} \right] \quad (\text{Section 2.1.4.1 of [3]})$$

f	50 Hz	$x_p = 1,866120103$ $y_p = 0,03510006481$
k_p	1	
R'	$3,608533 \cdot 10^{-5} \Omega/m$	
d_c	30,3 mm	
$s = D_e$	75,5 mm	

4.5.1.4 AC resistance of conductor

The AC resistance of the conductor at its maximum operating temperature is given by:

$$R = R' (1 + y_s + y_p) \quad (\text{Section 2.1 of [3]})$$

R'	$3,608533 \cdot 10^{-5} \Omega/m$	$R = 3,952152638 \cdot 10^{-5} \Omega/m$
y_s	0,060124126834	
y_p	0,03510006481	

4.5.2 First iteration

The iterative process starts with the assumptions made on the temperature of the metal screen temperature ($\theta_s = 80^\circ\text{C}$).

4.5.2.1 Resistance of sheath at operating temperature

The resistance of the sheath or screen at its operating temperature is given by:

$$R_s = R_{s0} [1 + \alpha_{20} (\theta_{sc} - 20)] \quad (\Omega/m) \quad (\text{Section 2.3 of [3]})$$

R_{s0}	$1,669128650 \cdot 10^{-4} \Omega/m$	$R_s = 2,072723957 \cdot 10^{-4} \Omega/m$
θ_s	80	
α_{20}	$4,03 \cdot 10^{-3} \text{ K}^{-1}$	

4.5.2.2 Losses caused by circulating currents

The loss factor is given by:

$$\lambda'_1 = \frac{R_s}{R} \frac{1}{1 + \left(\frac{R_s}{X} \right)^2} \quad (\text{Section 2.3.1 of [3]})$$

X	5,040331399 10^{-5} Ω/m	$\lambda_1 = 0,2928142510$
R _s	2,072723957 10^{-4} Ω/m	
R	3,952152638 10^{-5} Ω/m	

In the first instance, the IEC rule is considered. Since the conductor is not Milliken, eddy-current loss is ignored ($\lambda_1'' = 0$) and:

$$\lambda_1 = \lambda_1'$$

Then, the Guidance Point 6 and Guidance Point 31 will be applied and will illustrate how to take into account eddy current losses, even with solid-bonded cables.

4.5.2.3 Permissible current rating

The permissible current rating of an AC buried cable with no soil drying is given by:

$$I = \left[\frac{\Delta\theta - W_d [0,5 T_1 + n (T_2 + T_3 + T_4)]}{RT_1 + nR (1 + \lambda_1) T_2 + nR (1 + \lambda_1 + \lambda_2) (T_3 + T_4)} \right]^{0,5} \quad (\text{Section 1.4.1.1 of [3]})$$

The cable is unarmoured: $\lambda_2 = 0$ and $T_2 = 0$.

$\Delta\theta = \theta_c - \theta_a$	= 90-20 = 70°C	$I = 822,0666850 \text{ A}$
n	1 (single-core cable)	
R	3,952152638 10^{-5} Ω/m	
λ_1	0,2928142510	
W _d	0,3851382172 W/m	
T ₁	0,4198714890 Km/W	
T ₃	0,08671937479 Km/W	
T ₄	1,59469289 Km/W	

4.5.2.4 Losses in conductor and sheath

The losses in metal components are generated by Joule effect:

$$W_c = R \cdot I^2 \quad \text{and} \quad W_s = \lambda_1 \cdot W_c$$

R	3,952152638 10^{-5} Ω/m	$W_c = 26,70839595 \text{ W/m}$ $W_s = 7,820598957 \text{ W/m}$
λ_1	0,2928142510	
I	822,0666850 A	

4.5.2.5 Temperature on cable components

The temperature on the oversheath (jacket or external serving) can be deducted from the general expression for the temperature rise above ambient temperature (Section 1.4.1.1 of [4]):

$$\theta_j = \theta_a + n (W_c + W_s + W_d) T_4$$

Then, temperature of the metal sheath and conductor follows:

$$\theta_s = \theta_j + n (W_c + W_s + W_d) T_3$$

$$\theta_c = \theta_s + (W_c + W_d/2) T_1$$

θ_a	20°C	$\theta_j = 75,67731995^\circ\text{C}$ $\theta_s = 78,7050517^\circ\text{C}$ $\theta_c = 90^\circ\text{C}$
n	1 (single-core cable)	
W_d	0,3851382172 W/m	
W_c	26,70839595 W/m	
W_s	7,820598957 W/m	
T_1	0,4198714890 Km/W	
T_3	0,08671937479 Km/W	
T_4	1,59469289 Km/W	

The first step is completed. Further iterations have to be performed until convergence is achieved.

The assumption about the temperature of the metal sheath (round number for $\theta_s = \theta_c - 10 = 80^\circ\text{C}$) was relevant to launch the first iteration, since the result at the end of this loop is $78,7^\circ\text{C}$. The second iteration can start with the more accurate values.

Note that the symbol θ_c has been introduced here for the calculated temperature of the conductor, to remove the potential ambiguity with θ , which is the maximum operating temperature of conductor.

4.5.3 Next iterations case #0-1

The values obtained at the end of the first iteration are injected in the formulae of the next iteration in order to get a new set of results:

- Resistance of the conductor and sheath at operating temperature (R at θ_c , R_s at θ_s),
- Losses caused by circulating currents (λ_1),
- Permissible current rating (I),
- Losses in conductor and sheath (W_c , W_s),
- Temperatures on cable components (θ_c , θ_s).

For a better understanding, no improvement is applied here to accelerate the convergence.

4.5.3.1 Results

Except for the non-neglected dielectric losses, this first step is the 'as per IEC' result. Reminder of the chosen options:

Variable	Topic	IEC rule	CIGRE B1.56 recommendation
W_d	Dielectric losses	Losses neglected depending on the voltage level and insulation material ☐	Losses always taken into account GP 7 ☒
λ_1''	Eddy current losses	Losses neglected when circulating currents are considered, except for Milliken conductors ☒	Losses always taken into account GP 6 ☐
F	Reduction of eddy currents by circulating currents	Used for solid bonded cables with Milliken conductors ☒	Used for all solid bonding arrangements GP 31 ☐
T_4	External thermal resistance of buried cables or ducts	Simplified formula using $2u$ when $u > 0$ ☒	Exact formula with $u + \sqrt{u^2 - 1}$ GP 8 ☐

The results are summarised in the following table:

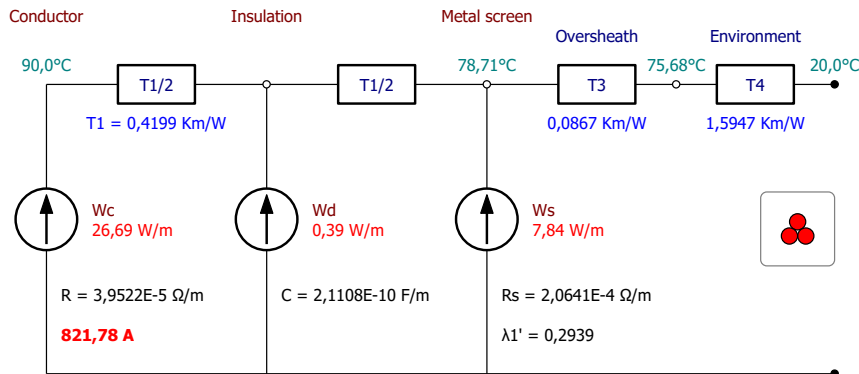
Iteration	R ($10^{-5} \Omega/\text{m}$)	R_s ($10^{-4} \Omega/\text{m}$)	λ_1	I (A)	W_c (W/m)	W_s (W/m)	θ_c (°C)	θ_s (°C)
1	3,9522	2,0727	0,29281	822,0667	26,7084	7,8206	90	78,7051
2	3,9522	2,0640	0,29391	821,7745	26,6894	7,8443	90,0000	78,7130
3	3,9522	2,0641	0,29390	821,7763	26,6895	7,8442	90,0000	78,7130
4	3,9522	2,0641	0,29390	821,7763	26,6895	7,8442	90,0000	78,7130

Ten-digit result after 7 iterations

I = 821,7763334392 A

≈ 820 A

The final values can also be reported in the following ladder network:



4.5.3.2 Comments

The case study has been designed to restrict the calculations to a minimalist set:

- The bonding of the metal sheaths at both ends, with circulating currents, lead to simple expressions (unlike the losses by eddy currents, which are neglected here to show the IEC compliant case) and the same value λ_I for the three cables in the case of the touching trefoil formation.
- The touching trefoil formation leads also to a unique value T_4 for the external thermal resistance of the soil environment.
- The straightforward configuration with no mutual heating is very stable: only four iterations are required to obtain the final permissible current rating with a good accuracy.

4.6 Case#0-1: Possible variations

4.6.1 Interpretation of IEC standards

4.6.1.1 Approximation on depth for T_4

Some approximations were useful at the time of hand calculation. For example, the calculation of the external thermal resistance T_4 of buried cables involves a parameter $u = 2L/D_e$. When the value of u exceeds 10, (the cable burial depth is one order greater than the cable radius), it is possible to replace the term $[u + \sqrt{u^2 - 1}]$ by $(2u)$.

Nowadays, there is absolutely no interest in coding this approximation on a computer. The expression with the radical in the formulae of Section 4.2.4 of [4] can be kept. Guidance Point 8 was proposed in this respect.

Note: At this stage of the technical brochure, Guidance Point 8 does not apply to the specific formulae of T_4 when metallic sheathed cables are touching. However, this item is under consideration within IEC for the next version of the 60287-2-1 Standard. The calculation is carried out here as an illustration of the low impact of the evolution:

$$T_4 = \frac{1,5}{\pi} \rho_T [\ln(2u) - 0,630] \rightarrow \frac{1,5}{\pi} \rho_T [\ln(u + \sqrt{u^2 + 1}) - 0,630]$$

Applied to the present case study of buried cables in touching trefoil, the value of T_4 obviously does not vary significantly (instead of 1,594692892 Km/W):

$$T_4 = 1,594522697 \text{ Km/W}$$

Reminder of the chosen options:

Variable	Topic	IEC rule	CIGRE B1.56 recommendation
W_d	Dielectric losses	Losses neglected depending on the voltage level and insulation material ☐	Losses always taken into account GP 7 ☒
λ_1''	Eddy current losses	Losses neglected when circulating currents are considered, except for Milliken conductors ☒	Losses always taken into account GP 6 ☐
F	Reduction of eddy currents by circulating currents	Used for solid bonded cables with Milliken conductors ☒	Used for all solid bonding arrangements GP 31 ☐
T_4	External thermal resistance of buried cables or ducts	Simplified formula using $2u$ when $u > 0$ ☐	Exact formula with $u + \sqrt{u^2 - 1}$ GP 8 ☒

The table of results becomes:

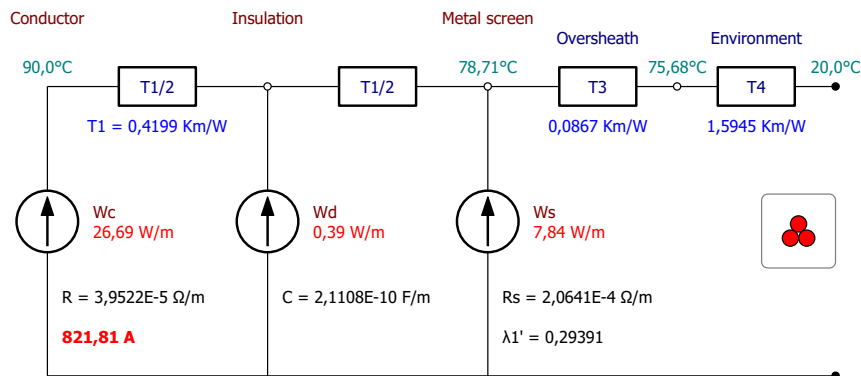
Iteration	R ($10^{-5} \Omega/\text{m}$)	R_s ($10^{-4} \Omega/\text{m}$)	λ_1	I (A)	W_c (W/m)	W_s (W/m)	θ_c (°C)	θ_s (°C)
1	3,9522	2,0727	0,29281	822,1019	26,7107	7,8213	90	78,7041
↓								
4 (exact)	3,9522	2,0641	0,29391	821,8114	26,6918	7,8449	90,0000	78,7120
4 (Section 4.5.3.1) (approx.)	3,9522	2,0641	0,29390	821,7763	26,6895	7,8442	90,0000	78,7130

Ten-digit result after 7 iterations

I = 821,8113708239 A

≈ 820 A

There is no influence on the final result (0,004 %!), but the result without approximation is still valid for shallow circuits. It is worth using it.



4.6.1.2 Approximation on eddy currents

Other quantities can be neglected according to the IEC standards:

- $g_s = 1$ and $(\beta_1 t_s)^4 / 12 \cdot 10^{-12} \approx 0$ for lead-sheathed cables in Section 2.3.6.1 of [3] for the calculation of eddy-current losses.
- And Δ_1 can Δ_2 be neglected when $m \leq 0,1$ in the same section.

Since the use of computer programmes, there is no need to neglect these quantities. Guidance Point 6 was completed in this respect.

4.6.1.3 Approximation on dielectric losses

Similarly, Table 3 of [3] gives for common insulation materials in use the value of U_0 at which the dielectric losses should be taken into account for three-core screened or single-core cables. There is no interest to implement such a table in a software application, while the coding of the full equation, whatever the voltage level, is easier and more valuable. Guidance Point 7 was proposed in this respect.

4.6.1.4 Use of physical constants

IEC gives a formula for the calculation of the capacitance C of a phase with radial electrical field ([3], Section 2.2):

$$C = \frac{\varepsilon_r}{18 \ln \left(\frac{D_i}{d_c} \right)} 10^{-9} \text{ (F/m)}$$

This formula is a good approximation (one thousandth) of the exact formula with the vacuum permittivity ε_0 ($\approx 8,854 \cdot 10^{-12}$):

$$C = \frac{2\pi\varepsilon_0\varepsilon_r}{\ln \left(\frac{D_i}{d_c} \right)} \text{ (F/m)}$$

The cable engineer could be advised to use the exact formula, especially to make it homogeneous with similar expression for two and three-core belted cables (non-radial electrical field).

4.6.1.5 Approximation on the hottest cable

When calculations were done by hand, the tedious iterative process was only performed for the hottest cable. The differentiation was effective for λ_I and T_q , but the other parameters (R , R_s , I , W_G , W_s , θ_s , θ_c) were calculated only for the hottest cable and applied to all cables. The final result could be slightly derated, but the current rating was on the safe side.

4.6.2 The present method performs the calculation for each cable. Variant with equally loaded identical cables

It may be tempting to investigate a variant of the Goldenberg formulae (1979, Section 4.2.4.3.2 of [4], used in Section 4.4.6), considering the more general expression or even that of unequally loaded cables (respectively Section 4.2.3.3.1 and Section 4.2.3.2 of [4]), involving the Kennelly assumption with heat transfer between two isothermal surfaces (soil surface, oversheath or duct surface).

But the method of images and wells for touching cables is not appropriate, the required isotherm around each cable being distorted by the proximity and contact.

4.6.3 Variant with single point bonding

The case study was specified with a bonding of metal sheaths at both ends of an electrical section (Section 4.2.3), leading to losses caused by circulating currents (Section 4.5.2.2). According to Section 2.3.1 of [3], eddy-current losses are ignored ($\lambda_1'' = 0$).

If we consider the sheaths bonded at a single point, there are no circulating currents ($\lambda_1' = 0$) and the eddy-current loss factor is given by λ_1'' , dependent on the temperature of the metal screens.

4.6.3.1 Losses caused by eddy currents

For the first iteration, a temperature of 80°C is assumed, for which the aluminium sheath resistance R_s has been calculated in Section 4.5.2.1. The same formula applied to the electrical resistivity of sheath material at operating temperature leads to:

ρ_{s0}	$2,84 \cdot 10^{-8} \Omega \cdot m$	$\rho_s = 3,526712 \cdot 10^{-8} \Omega/m$
θ_s	80	
α_{20}	$4,03 \cdot 10^{-3} K^{-1}$	

Intermediary coefficients are calculated:

$$\beta_1 = \sqrt{\frac{4 \pi \omega}{10^7 \rho_s}} \quad \text{(Section 2.3.6.1 of [3])}$$

$$m = \frac{\omega}{R_s} 10^{-7}$$

ω	$2\pi \times 50 = 314,1592654 \text{ rad/s}$	$\beta_1 = 105,8022420$ $m = 0,1515683091$
ρ_s	$3,526712 \cdot 10^{-8} \Omega/\text{m}$	
R_s	$2,072723957 \cdot 10^{-4} \Omega/\text{m}$	

$$g_s = 1 + \left(\frac{t_s}{D_s} \right)^{1,74} (\beta_1 D_s 10^{-3} - 1,6) \quad (\text{Section 2.3.6.1 of [3]})$$

β_1	105,8022420	$g_s = 1,002449783$
t_s	0,8 mm	
D_s	68,5 mm	

For cables in trefoil formation:

$$\lambda_0 = 3 \left(\frac{m^2}{1+m^2} \right) \left(\frac{d}{2s} \right)^2$$

$$\Delta_1 = (1,14 m^{2,45} + 0,33) \left(\frac{d}{2s} \right)^{(0,92 m + 1,66)} \quad (\text{Section 2.3.6.1 of [3]})$$

$$\Delta_2 = 0$$

m	0,1515683091	$\lambda_0 = 0,01354245364$ $\Delta_1 = 0,08055825544$ $\Delta_2 = 0$
s	75,5 mm	
d	$64,3 + (2 \times 1,3) + 0,8 = 67,7 \text{ mm}$	

Finally:

$$\lambda_1'' = \frac{R_s}{R} \left[g_s \lambda_0 (1 + \Delta_1 + \Delta_2) + \frac{(\beta_1 t_s)^4}{12 \times 10^{12}} \right] \quad (\text{Section 2.3.6.1 of [3]})$$

R_s	$2,072723957 \cdot 10^{-4} \Omega/\text{m}$	$\lambda_1'' = 0,07695600926$
R	$3,952152638 \cdot 10^{-5} \Omega/\text{m}$	
g_s	1,002449783	
λ_0	0,01354245364	
Δ_1	0,08055825544	
Δ_2	0	
β_1	105,8022420	
t_s	0,8 mm	

The configuration of buried touching cables in trefoil formation is interesting for the verification because IEC considers the same formula for the external thermal resistance T_4 for the three cables (Section 4.2.4.3.2 of [4] described in Section 4.4.6). With the assumption of equal sheath losses, this means that the temperature of the cable components is the same for all cables. The calculations are simplified since they rely on any one cable and duplicated for the other two.

In the case of the variant with equally loaded (Section 4.6.2) or cables in flat formation, the temperature is lower for the cable at the apex at the top of the group or for the external adjacent cables of a flat group. Therefore, the temperature dependant parameters R_s , R , g_s , λ_0 , Δ_1 and β_1 lead to two distinct values of λ_1'' .

4.6.3.2 Permissible current rating

All parameters being unchanged (except the bonding of metal screens, λ_I), most of the results of Sections 4.4 and 4.6.1.1 can be used for the first iteration, with the formula in Section 4.5.2.3.

$\Delta\theta = \theta - \theta_a$	$= 90 - 20 = 70^\circ\text{C}$	$I = 886,4253456992 \text{ A}$
n	1 (single-core cable)	
R	$3,952152638 \cdot 10^{-5} \Omega/\text{m}$	
λ_1	0,07695600926	
W_d	0,3851382172 W/m	
T_1	0,4198714890 Km/W	
T_3	0,08671937479 Km/W	
T_4	1,594692892 Km/W	

4.6.3.3 Next iterations

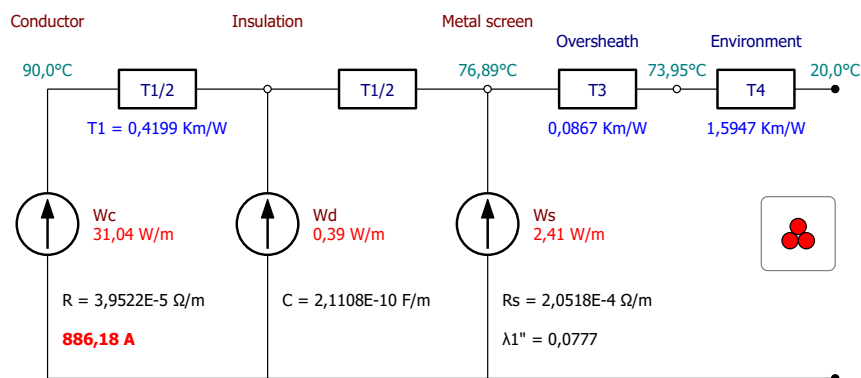
Reminder of the chosen options:

Variable	Topic	IEC rule	CIGRE B1.56 recommendation
W_d	Dielectric losses	Losses neglected depending on the voltage level and insulation material ☐	Losses always taken into account GP 7 ☒
T_4	External thermal resistance of buried cables or ducts	Simplified formula using $2u$ when $u > 0$ ☒	Exact formula with $u + \sqrt{u^2 - 1}$ GP 8 ☐

The results are shared by the three cables (common values for touching cables in trefoil) with very fast convergence:

Iteration	R ($10^{-5} \Omega/\text{m}$)	R_s ($10^{-4} \Omega/\text{m}$)	λ_1	I (A)	W_c (W/m)	W_s (W/m)	θ_c ($^\circ\text{C}$)	θ_s ($^\circ\text{C}$)
1	3,9522	2,0727	0,07696	886,4253	31,0540	2,3898	90	76,8804
2	3,9522	2,0517	0,07771	886,1747	31,0365	2,4117	90	76,8878
3	3,9522	2,0518	0,07770	886,1753	31,0365	2,4117	90	76,8878

Ten-digit result after 6 iterations	$I = 886,1752854720 \text{ A}$	$\approx 880 \text{ A}$
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4.6.4 Variant with non-neglected eddy-current losses

This option was adopted by the CIGRE WG B1.56, not to neglect the eddy-current losses when the metal sheaths are bonded at both ends of an electrical section (Guidance Point 6).

The loss factor λ_I is the sum of a loss factor λ_I' representing circulating currents and λ_I'' including eddy currents, calculated in Section 4.5.2.2 and Section 4.6.3.1, respectively. There is no particular difficulty, but at this stage, a new consideration is introduced, with the F factor calculated below.

4.6.4.1 Reduction of the eddy currents losses by the circulating currents

In case of solid bonding of the sheaths, the circulating currents in the metal screens are reducing the eddy current losses. Guidance Point 31 explains how the F factor is extended to all conductor designs, not uniquely linked to Milliken conductors.

The factor ($0 < F \leq 1$) is given by:

$$F = \frac{4 M^2 N^2 + (M + N)^2}{4 (M^2 + 1) (N^2 + 1)}$$

(Section 2.3.5 of [3])

where $M = N = \frac{R_s}{X}$ for cables in trefoil formation

For the first iteration, the calculated value is:

R_s	$2,072723957 \cdot 10^{-4} \Omega/\text{m}$	$F = 0,9441678372$
X	$5,040331399 \cdot 10^{-5} \Omega/\text{m}$	
$M = N$	4,112277137	

In the case of trefoil geometry, both values of λ_1'' and F are the same for the three cables. The eddy current loss factor is obtained by multiplying λ_1'' calculated by Section 2.3.6 [3] by F .

λ_1'' (as per [3] 2.3.6)	0,07695600926	$\lambda_1'' = 0,07265938882$
F	0,9441678372	

Finally, the loss factor λ_1 is the sum of loss factors representing circulating currents and eddy currents:

λ_1'	0,2928142510	$\lambda_1 = 0,3654736398$
λ_1''	0,07265938882	

4.6.4.2 Permissible current rating

All parameters being unchanged (except the consideration of eddy currents with solid bonding of metal screens, λ_1''), most of the results of Sections 4.4 and 4.6.1.1 can be used for the first iteration, with the formula in Section 4.5.2.3.

$\Delta\theta = \theta - \theta_a$	$= 90 - 20 = 70^\circ\text{C}$	$I = 803,3635402401 \text{ A}$
n	1 (single-core cable)	
R	$3,952152638 \cdot 10^{-5} \Omega/\text{m}$	
λ_1	0,3654736398	
W_d	0,3851382172 W/m	
T_1	0,4198714890 Km/W	
T_3	0,08671937479 Km/W	
T_4	1,594692892 Km/W	

4.6.4.3 Next iterations

Reminder of the chosen options:

Variable	Topic	IEC rule	CIGRE B1.56 recommendation
W_d	Dielectric losses	Losses neglected depending on the voltage level and insulation material ☐	Losses always taken into account GP 7 ☒
λ_1''	Eddy current losses	Losses neglected when circulating currents are considered, except for Milliken conductors ☐	Losses always taken into account GP 6 ☒
F	Reduction of eddy currents by circulating currents	Used for solid bonded cables with Milliken conductors ☐	Used for all solid bonding arrangements GP 31 ☒
T_4	External thermal resistance of buried cables or ducts	Simplified formula using $2u$ when $u > 0$ ☒	Exact formula with $u + \sqrt{u^2 - 1}$ GP 8 ☐

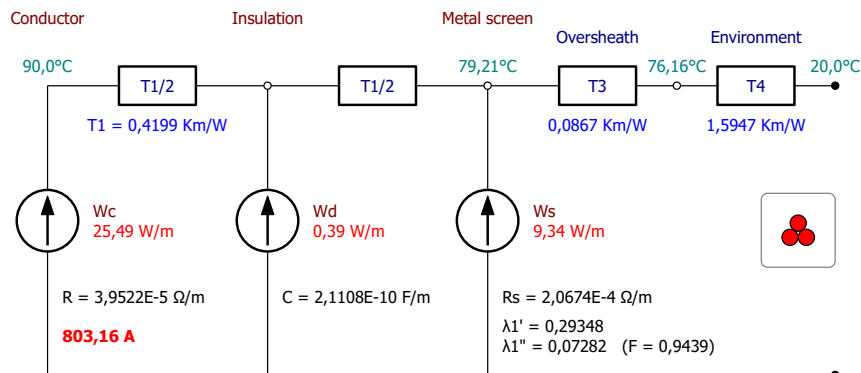
The results are shared by the three cables (common values for touching cables in trefoil):

Iteration	R ($10^{-5} \Omega/\text{m}$)	R_s ($10^{-4} \Omega/\text{m}$)	λ_1	I (A)	W_c (W/m)	W_s (W/m)	θ_c (°C)	θ_s (°C)
1	3,9522	2,0727	0,36547	803,3635	25,5069	9,3221	90	79,2095
↓								
4 ($\lambda_1 = \lambda_1' + \lambda_1''$)	3,9522	2,0674	0,36629	803,1596	25,4940	9,3383	90	79,2150
4 (Section 4.5.3.1) (λ_1'' neglected)	3,9522	2,0641	0,29390	821,7763	26,6895	7,8442	90,0000	78,7130
8 (Section 4.6.3) (single bonded)	3,9522	2,0518	0,07770	886,1753	31,0365	2,4117	90	76,8878

Ten-digit result after 7 iterations

I = 803,1595966445 A

≈ 800 A



When the convergence is reached, $\lambda_1 = \lambda_1' + \lambda_1'' = 0,29348 + 0,07282 = 0,36629$.

4.6.4.4 Comments

When possible, the bonding at a single point of the metal sheath (or crossbonding with identical electrical sections) is an improvement if we need to uprate the system (+10,3 %).

In the case of solid bonding at both ends, IEC neglects the eddy-current losses, except for Milliken conductors. But this example shows that assumption hides a derating of 2,3 %, which should be applied. Guidance Point 6 was adopted: Eddy current losses should not be neglected, and should always be taken into account.

These results have been reported in the summary table in Section 4.11.

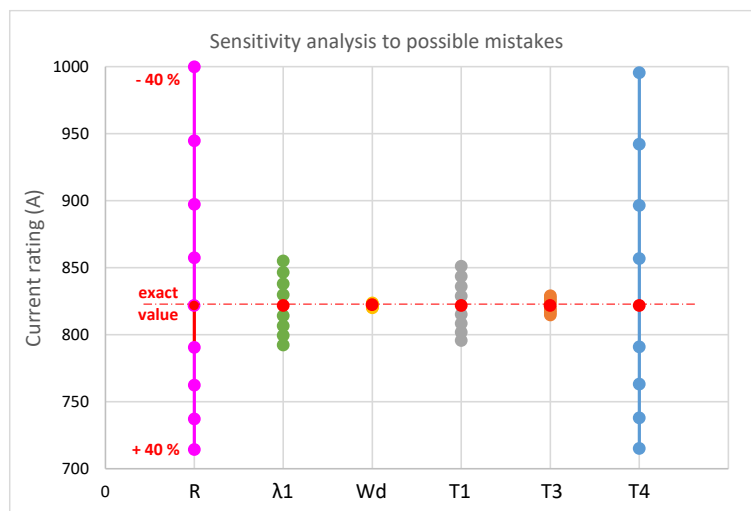
4.6.5 Sensitivity analysis to possible mistakes

Some parameters have greater influence than other on the final value of the current rating. This section presents the impact on I of a up to $\pm 40\%$ deviation on the calculated values found in the general equation below. The deviation may occur when a formula or a method is improperly implemented into a calculation tool. It is interesting to assess the impact of a programming error on the current rating.

The actual calculation was first performed (Section 4.6.1.1), and the exact values of the parameters were saved. Then, each of these values were successively modified with a 10 % step from the exact value, all things being equal, and replaced in the general equation during the iterations.

$$I = \left[\frac{\Delta\theta - W_d [0,5 T_1 + n (T_2 + T_3 + T_4)]}{RT_1 + nR (1 + \lambda_1) T_2 + nR (1 + \lambda_1 + \lambda_2) (T_3 + T_4)} \right]^{0,5} \quad (\text{Section 1.4.1.1 of [3]})$$

The present case of directly buried cables involves W_d , T_1 , T_3 , T_4 , R , and λ_1 . For each of these parameters, an under estimated value leads to an over estimated result for the current rating. The impact of the conductor AC resistance R and the external thermal resistance T_4 is very high, while a deviation/error on the dielectric losses W_d is quite negligible (for an extruded insulation).



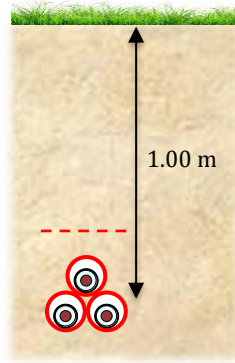
The designer therefore should take special care of the input data:

- For R : conductor DC resistance, frequency of the grid, skin and proximity effects, operating temperature, distance between conductors of a circuit.
- For T_4 : thermal resistance of the soil, laying depth and the separation of the cables.

It shall be pointed out that this chart of sensitivity analysis is case dependant. For example, an increase of the axial distance s between conductors will decrease the AC resistance R (lower proximity effect), will increase λ_1 (higher circulating currents) and will decrease the external thermal resistance T_4 (lower mutual heating). The impact of a deviation/error is unclear beforehand until the calculations are performed.

4.7 Case #0-2: Sub-case study with touching HDPE ducts

The next case study is a variation with cables pulled in HDPE (high density polyethylene) ducts instead of cables directly buried (Section 4.2.4), all other things being equal. Touching conduits are laid in a trefoil formation. This kind of installation is well adapted to rural area, with narrow trench and mechanised operations. The depth of 1,00 m measured between the ground surface and the centre of the trefoil.



The selected size of the duct is a diameter $D_o = 140$ mm and the thickness $(D_o - D_i)/2 = 10,3$ mm, leading to a standard dimension ratio $SDR = 13,6$, and a ratio between internal duct diameter and external cable diameter $D_i/D_e = 1,58 > 1,5$.

Input data (Section 4.2) and results independent of the temperature (Section 4.4) are unchanged until the thermal resistance between conductor and sheath T_i (Section 4.4.4), except for the reactance X .

It is noted that the method we apply here remains as close as possible to IEC methods although the applicability of the method of images is limited in the case of air filled ducts. This topic will be forwarded to IEC for further consideration as it is beyond the scope of this working group.

4.7.1 Results independent of the temperature

4.7.1.1 Reactance

Since the axial distance s between cables is increased (touching ducts $D_o = 140$ mm instead of touching cables $D_e = 75,5$ mm), the reactance value is modified (see equation in Section 4.4.2).

$$X = 8,920259690 \cdot 10^{-5} \Omega/m$$

4.7.1.2 Thermal resistance of the oversheath

Since the cables are not touching, the IEC formula (Section 4.1.4.1, [4]) applies, without multiplication by a factor 1.6, as already calculated in Section 4.4.5.

$$T_3 = 0,05419960924 \text{ Km/W}$$

4.7.2 External thermal resistance

The external thermal resistance of a cable in a duct consists of three parts:

- The thermal resistance of the air space between the cable surface and duct internal surface T'_4 ,
- The thermal resistance of the duct itself, T''_4 ,
- The external thermal resistance of the duct T'''_4 .

$$T_4 = T'_4 + T''_4 + T'''_4 \quad (\text{Section 4.2.7.1 of [3]})$$

4.7.2.1 Thermal resistance between cable and duct (or pipe) T'_4

The heat transfer in the air gap between the cable and the duct is assessed using three constants depending on the conduit material, given by the Table 4 of [4]:

Table 4 – Values of constants U , V and Y

Installation condition	U	V	Y
In metallic conduit	5,2	1,4	0,011
In fibre duct in air	5,2	0,83	0,006
Oil pressure pipe-type cable	0,26	0,0	0,002 6
Plastic ducts	1,87	0,312	0,003 7
Earthenware ducts	1,87	0,28	0,003 6
Water filled ducts	0,1	0,03	0,001

The mean air gap temperature value is not known yet. It must be initialised by the user to launch the first iteration. The maximum operating temperature ($\theta = 90^\circ\text{C}$) is a good start with:

$$\theta_m = \theta - 20 = 70^\circ\text{C}$$

Unlike for the directly buried cables, the external thermal resistance depends on the temperature. The value of T_4' will be updated at each iteration, as the average temperature between the external surface of the cable oversheath and the inner wall of the duct.

Finally:

$$T_4' = \frac{U}{1 + 0,1(V + Y\theta_m) D_e} \quad (\text{Section 4.2.7.2 of [4]})$$

U	1,87	$T_4' = 0,3520961015 \text{ Km/W}$
V	0,312	
Y	0,0037	
θ_m	70°C	
D_e	75,5 mm	

4.7.2.2 Thermal resistance of the duct (or pipe) itself T_4''

The thermal resistance across the wall of the duct is given by:

$$T_4'' = \frac{1}{2\pi} \rho_T \ln\left(\frac{D_o}{D_d}\right) \quad (\text{Section 4.2.7.3 of [4]})$$

D_o	140 mm	$T_4'' = 0,08866064719 \text{ Km/W}$
D_d	119,4 mm	
ρ_T	3,5 Km/W	

4.7.2.3 External thermal resistance of the duct (or pipe) T_4'''

The Section 4.2.7.4 of [4] specifies that ducts not embedded in concrete must be treated in the same way as cables. The external diameter of the duct replaces the external diameter of the cable in the formulae.

In the case of touching HDPE ducts in trefoil formation, the same formula as for touching non-metallic sheathed cables applies:

$$T_4 = \frac{1}{2\pi} \rho_T [\ln(2u) + 2 \ln(u)] \quad (\text{Section 4.2.4.3.4 of [4]})$$

*This formula is used for non-metallic sheathed cables having a screen of spaced copper wires **and for the external thermal resistance of touching ducts** (see 4.2.7.4).*

The strict application of the IEC formula is then preferred to Guidance Point 8.

L	1,00 m = 1000 mm	$T_4''' = 1,38002093961 \text{ Km/W}$
D_o	140 mm	
$u = 2L/D_o$	14,28571429	
ρ_T	1,0 Km/W	

The value is the same for all cables.

For the first iteration, the sum $T_4' + T_4'' + T_4'''$ is the same for the three cables and equal to:

$$T_4 = 1,8207776883 \text{ Km/W}$$

4.7.3 Results depending on the temperature

The process is continued in the same manner as for the case study of directly buried cables (Section 4.2.4). Since the axial distance s between cables is increased (touching ducts $D_o = 140 \text{ mm}$ instead of touching cables $D_e = 75,5 \text{ mm}$), all the values are modified.

Parameter (unit)		Values
θ	(°C)	90
$x_s = x_p$	-	1,866120103
y_s	-	0,06012412684
y_p	-	0,01010775634
R'	($10^{-5} \Omega/\text{m}$)	3,608533
R	($10^{-5} \Omega/\text{m}$)	3,86196707
R_s	($10^{-4} \Omega/\text{m}$)	2,072723957
$\lambda_1 = \lambda'_1$	-	0,8387035149

4.7.4 First iteration

4.7.4.1 Permissible current rating

From the unchanged parameters, it is possible to calculate the current rating according to Section 4.5.2.3:

$\Delta\theta = \theta - \theta_a$	= 90-20 = 70°C	$I = 680,6595349744 \text{ A}$
N	1 (single-core cable)	
R	3,86196707 $10^{-5} \Omega/\text{m}$	
λ_1	0,8387035149	
W_d	0,3851382172 W/m	
T_1	0,4198714890 Km/W	
T_3	0,05419960924 Km/W	
T_4	1,8207776883 Km/W	

4.7.4.2 Losses and temperature on cable components

During the first iteration, the expression of losses given in Section 4.5.2.4 and temperatures as per Section 4.5.2.5 are used:

Parameter (unit)		Values
I	(A)	680,65953497
W_c	(W/m)	17,892393114
W_s	(W/m)	15,006412994
θ_s	(°C)	82,406639986
θ_j	(°C)	80,602663206

A new value for the mean air gap temperature can be calculated for the next iteration, reporting the average of the jacket and duct inner wall temperatures:

$$\theta_m = \theta_j - \frac{1}{2} T'_4 (W_c + W_d + W_s)$$

$\theta_m = 74,743089686^\circ\text{C}$, which is consistent with the initial assumption of 70°C .

4.7.5 Next iterations

Reminder of the chosen options:

Variable	Topic	IEC rule	CIGRE B1.56 recommendation
W_d	Dielectric losses	Losses neglected depending on the voltage level and insulation material ☐	Losses always taken into account GP 7 ☒
λ_1''	Eddy current losses	Losses neglected when circulating currents are considered, except for Milliken conductors ☒	Losses always taken into account GP 6 ☐
F	Reduction of eddy currents by circulating currents	Used for solid bonded cables with Milliken conductors ☒	Used for all solid bonding arrangements GP 31 ☐
T_4	External thermal resistance of buried cables or ducts	Simplified formula using $2u$ when $u > 0$ ☒	Exact formula with $u + \sqrt{u^2 - 1}$ GP 8 ☐

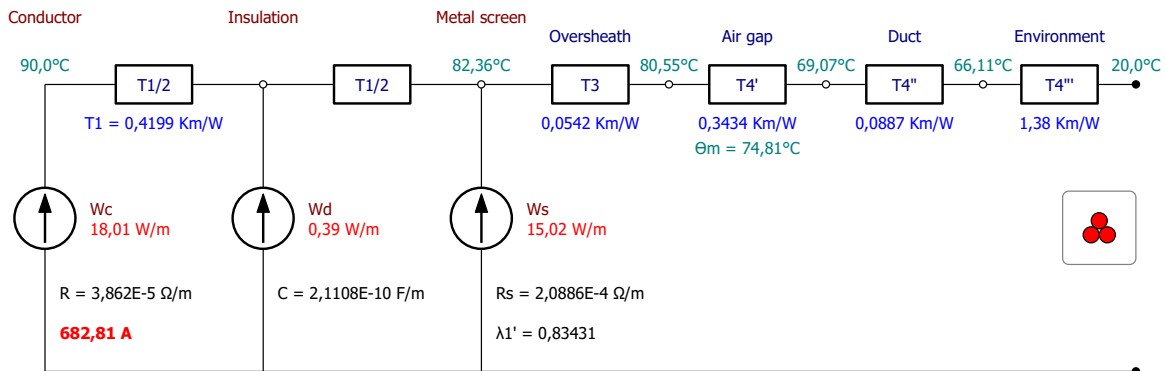
The results are summarised in the following table:

Iteration	R ($10^{-5} \Omega/\text{m}$)	R_s ($10^{-4} \Omega/\text{m}$)	λ_1	T_4 (Km/W)	I (A)	W_c (W/m)	W_s (W/m)	θ_c (°C)	θ_s (°C)
1	3,8620	2,0727	0,83870	1,8208	680,6595	17,8924	15,0064	90	82,4066
2	3,8620	2,0889	0,83422	1,8122	682,8095	18,0056	15,0206	90	82,3591
3	3,8620	2,0886	0,83430	1,8121	682,8141	18,0058	15,0224	90	82,3590
4	3,8620	2,0886	0,83431	1,8121	682,8145	18,0059	15,0224	90	82,3590

Ten-digit result after 8 iterations

I = 682,8144648700 A

≈ 680 A

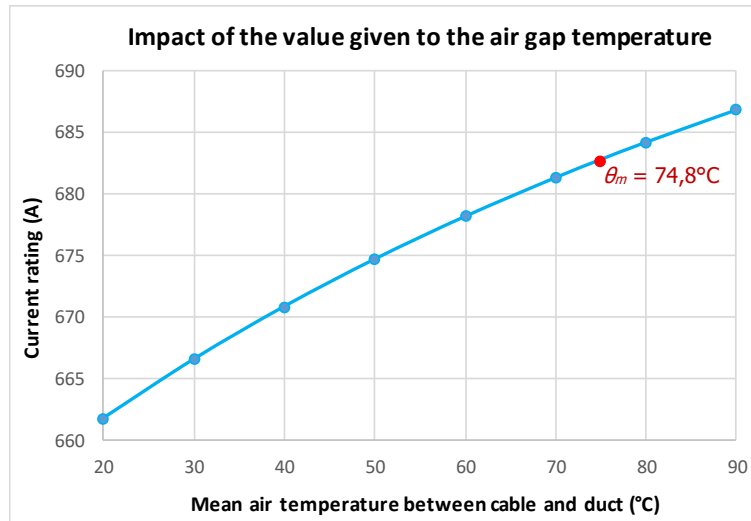


4.7.6 Discussion on the temperature of the air gap between cable and duct

The thermal resistance between the external cable surface and the inner wall of the duct T_4' involves θ_m , the mean temperature of the medium filling the space between cable and duct. IEC standard [4]

specifies that “An assumed value has to be used initially and the calculation repeated with a modified value if necessary (°C)”. The process was applied accordingly in Sections 4.7.2.1 and 4.7.4.2.

The following chart shows the impact of the initial selected value, if this value is not modified throughout the iterations. An extreme range from the ambient temperature $\theta_a = 20^\circ\text{C}$ up to the maximum conductor temperature $\theta = 90^\circ\text{C}$ is investigated:



The θ_m shall be updated during the iterations. Otherwise, a constant value given to $\theta_m \pm 15^\circ\text{C}$ leads to a difference of about 6 A (0,4%) on the current rating.

4.7.7 Variant with non-neglected eddy-current losses

Taking into account the Guidance Point 6 (not to neglect the eddy-current losses when the metal sheaths are bonded at both ends of an electrical section), the loss factor λ_I is the sum of a loss factor λ_I' representing circulating currents and λ_I'' including eddy currents, described in Section 4.5.2.2 and Section 4.6.3.1, respectively.

4.7.7.1 First iteration

Guidance Point 30 addresses the reduction of the eddy currents losses by the circulating currents, when the metal screens are solid-bonded. The calculation of the F factor has been already described in Section 4.6.4.1.

For the first iteration, the calculated value is:

R_s	$2,072723957 \cdot 10^{-4} \Omega/\text{m}$	$F = 0,8437300180$
X	$8,920259690 \cdot 10^{-5} \Omega/\text{m}$	
$M = N$	2,323613919	

In the case of trefoil geometry, both values of λ_I'' and F are the same for the three cables. The eddy current loss factor is obtained by multiplying λ_I'' calculated by Section 2.3.6 of [3] by F :

λ_I'' (as per [3] 2.3.6)	0,02177485075	$\lambda_I'' = 0,0183720952$
F	0,8437300180	

Finally, the loss factor λ_I is the sum of loss factors representing circulating currents and eddy currents:

λ_1'	0,8387035149	$\lambda_1 = 0,8570756101$
λ_1''	0,0183720952	

4.7.7.2 Permissible current rating

All parameters being unchanged (except the consideration of eddy currents with solid bonding of metal screens, λ_1''), most of the results of Sections 4.4 and 4.7.1 can be used for the first iteration, with the formula in Section 4.5.2.3.

$\Delta\theta = \theta - \theta_a$	= 90-20 = 70°C	$I = 677,6482890238 \text{ A}$
n	1 (single-core cable)	
R	3,952152638 $10^{-5} \Omega/\text{m}$	
λ_1	0,8437300180	
W_d	0,3851382172 W/m	
T_1	0,4198714890 Km/W	
T_3	0,05419960924 Km/W	
T_4	1,8207776883 Km/W	

4.7.7.3 Next iterations

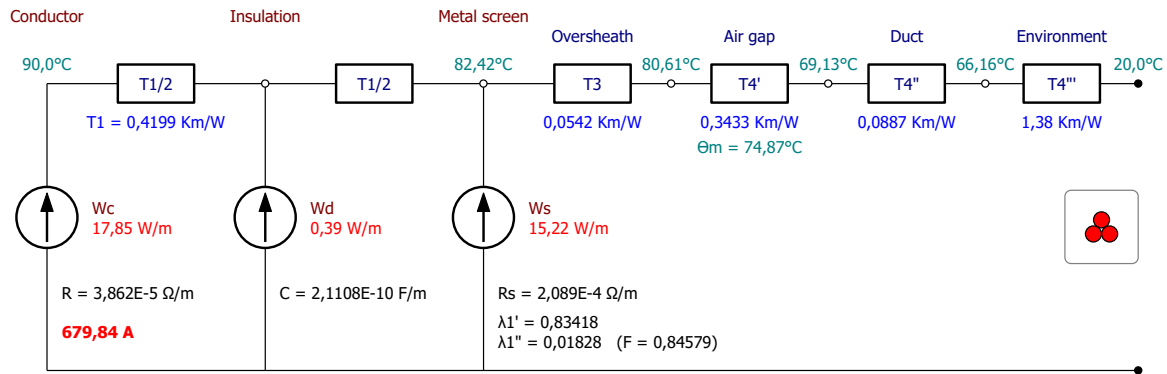
Reminder of the chosen options:

Variable	Topic	IEC rule	CIGRE B1.56 recommendation
W_d	Dielectric losses	Losses neglected depending on the voltage level and insulation material ☐	Losses always taken into account GP 7 ☒
λ_1''	Eddy current losses	Losses neglected when circulating currents are considered, except for Milliken conductors ☐	Losses always taken into account GP 6 ☒
F	Reduction of eddy currents by circulating currents	Used for solid bonded cables with Milliken conductors ☐	Used for all solid bonding arrangements GP 31 ☒
T_4	External thermal resistance of buried cables or ducts	Simplified formula using $2u$ when $u > 0$ ☒	Exact formula with $u + \sqrt{u^2 - 1}$ GP 8 ☐

The results are shared by the three cables (common values for touching ducts in trefoil):

Iteration	R ($10^{-5} \Omega/\text{m}$)	R_s ($10^{-4} \Omega/\text{m}$)	λ_1	T_4 (Km/W)	I (A)	W_c (W/m)	W_s (W/m)	θ_c (°C)	θ_s (°C)
1	3,8620	2,0727	0,85708	1,8208	677,6802	17,7344	15,1997	90	82,4730
↓									
4 ($\lambda_1 = \lambda_1' + \lambda_1''$)	3,8620	2,0890	0,85246	1,8120	679,8410	17,8494	15,2159	90	82,4247
4 (Section 4.7.5) (λ_1'' neglected)	3,8620	2,0886	0,83431	1,8121	682,8145	18,0059	15,0224	90	82,3590

Ten-digit result after 8 iterations	$I = 679,8409804502 \text{ A}$	$\approx 670 \text{ A}$
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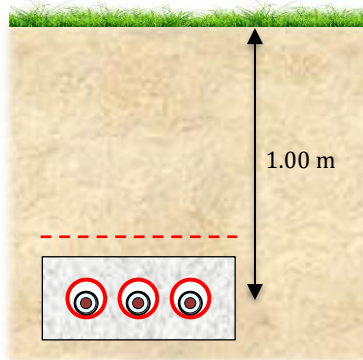
When the convergence is reached, $\lambda_I = \lambda_I' + \lambda_I'' = 0,83418 + 0,01828 = 0,85246$.

A derating of 0,5 % is found when the eddy current losses are taken into account (Guidance Point 6). It can be noted that the rounding down of the result to the nearest 10 A may be severe in this extreme example (679,8 → 670 A).

These results have been reported in the summary table in Section 4.11.

4.8 case #0-3: Sub-case study with PVC ducts in flat formation embedded in concrete

The next case study is a variation with cables pulled in PVC (polyvinyl chloride) ducts instead of cables directly buried (Section 4.2.4). Spaced conduits are laid in a flat formation, and embedded in concrete, all other things being equal. This kind of installation is well adapted to urban area, where a mechanical protection is required against potential third-party aggression. The considered depth is 1,00 m measured between the ground surface and the centre of the ducts.



The selected size of the duct is a diameter $D_o = 125$ mm and the thickness $(D_o - D_i)/2 = 3,2$ mm, leading to a standard dimension ration $SDR = 39$, and a ratio between internal duct diameter and external cable diameter $D_i/D_e = 1,57 > 1,5$.

Ducts are held by spacers (40 mm between the conduits) and the embedding is such as 100 mm of concrete is found between the ducts and the soil interface. Therefore:

- The width of the duct bank is $100 + 125 + 40 + 125 + 40 + 125 + 100 = 655$ mm.
- Its height is $100 + 125 + 100 = 325$ mm.
- The axial distance between ducts is $s = 125 + 40 = 165$ mm.
- The bottom depth of the trench is $1,00 + (125/2 + 100)/1000 = 1,1625$ m.

The thermal resistivity of the concrete is assumed to be 0,8 Km/W.

Input data (Section 4.2) and results independent of the temperature (Section 4.4) are unchanged until the thermal resistance between conductor and sheath T_I (Section 4.4.4), except for the reactance X .

4.8.1 Results independent of the temperature

4.8.1.1 Reactance

While the reactance value was the same for the three cables of a trefoil (Section 4.4.2), the case of cables in flat formation without transposition requires the following distinction:

- X is the reactance of sheath or screen per unit length of cable for two adjacent single-core cables:

$$X = 2 \times 10^{-7} \ln \left(\frac{2s}{d} \right) \quad (\Omega/\text{m}) \quad (\text{Section 2.3.3 of [3]})$$

- X_m is the mutual reactance per unit length of cable between the sheath of an outer cable and the conductors of the other two:

$$X_m = 2 \times 10^{-7} \ln (2) \quad (\Omega/\text{m}) \quad (\text{Section 2.3.3 of [3]})$$

f	50 Hz	$X = 9,95260621 \cdot 10^{-5} \Omega/\text{m}$ $X_m = 4,35517218 \cdot 10^{-5} \Omega/\text{m}$
ω	$2\pi \times 50 = 314,1592654 \text{ rad/s}$	
s	165 mm	
d	$64,3 + (2 \times 1,3) + 0,8 = 67,7 \text{ mm}$	

4.8.1.2 Thermal resistance of the duct

The thermal resistance of the duct itself T_4'' is presented in Section 4.7.2.2:

D_o	125 mm	$T_4'' = 0,05018847750 \text{ Km/W}$
D_i	118,6 mm	
ρ_T	6,0 Km/W	

4.8.1.3 External thermal resistance of the duct

The Section 4.2.7.4 of IEC 60287-2-1 [4] specifies that ducts embedded in concrete shall be considered in the same way as the cables. The external diameter of the duct replaces the external diameter of the cable in the formulae. The calculation of the thermal resistance outside the ducts is first computed assuming a uniform medium outside the ducts having a thermal resistivity equal to that of concrete, before applying a correction to account for the difference between the thermal resistivities of the concrete and soil.

In the case of non-touching ducts in flat formation, we consider the process of unequally loaded cables with the calculation of mutual heating and the formula of single isolated buried cable:

$$T_4 = \frac{1}{2\pi} \rho_T \ln \left(u + \sqrt{u^2 - 1} \right) \quad (\text{Section 4.2.2 of [4]})$$

L	1,00 m = 1000 mm	$T_4''' = 0,4411466780 \text{ Km/W}$
D_o	125 mm	
$u = 2L/D_o$	16	
ρ_c	0,8 Km/W (concrete)	

The T_4''' value is the same for all cables.

Note: The selection of the formula to calculate T_4 is consistent with Guidance Point 8. In this configuration, although cables may be equally loaded, they typically do not have equal losses. Section 4.2.3.2 of [4] (case of unequally loaded cables) is preferred to the simplified formula of Section 4.2.3.3.4 (case of three cables having unequal sheath losses, laid in a horizontal plane, equally spaced apart). This assumes that the temperature rise due to thermal mutual effects is also calculated later (Section 4.8.2.5).

4.8.1.4 Correction to the thermal resistance for the embedding in concrete

After the calculation of T_4''' , a correction is then added algebraically to take account of the difference between the thermal resistivities of concrete and soil.

Considering the quantities x and y are the shorter and longer sides (height and width), respectively, of the duct bank, the equivalent radius of the concrete bank given by:

$$\ln r_b = \frac{1}{2} \frac{x}{y} \left(\frac{4}{\pi} - \frac{x}{y} \right) \ln \left(1 + \frac{y^2}{x^2} \right) + \ln \frac{x}{2} \quad (\text{Section 4.2.7.4 of [4]})$$

Using the depth of laying to centre of duct bank L_G , and the intermediate variable $u = L_G/r_b$, the correction to the thermal resistance is given by:

$$\frac{N}{2\pi} (\rho_e - \rho_c) \ln (u + \sqrt{u^2 - 1}) \quad (\text{Section 4.2.7.4 of [4]})$$

x	Height 325 mm	$r_b = 222,1414899 \text{ mm}$
y	Width 655 mm	
L_G	1,00 m = 1000 mm	corrT₄''' = 0,2086537030 Km/W
$u = L_G/r_b$	4,501635423	
N	3 (loaded cables in the bank)	
ρ_c	0,8 Km/W (concrete)	
ρ_e	1,0 Km/W (earth around bank)	

4.8.2 Results depending on the temperature

4.8.2.1 Proximity effect factor

The axial distance between conductors is greater than for the initial case of directly buried cables (Section 4.5.1.3), and the proximity effect factor y_p must be updated:

f	50 Hz	$x_p = 1,866120103$ $y_p = 0,007268543102$
k_p	1	
R'	$3,608533 \cdot 10^{-5} \Omega/\text{m}$	
d_c	30,3 mm	
$s = D_e$	165 mm	

4.8.2.2 AC resistance of conductor

The AC resistance of the conductor at its maximum operating temperature (Section 4.5.1.4) becomes:

R'	$3,608533 \cdot 10^{-5} \Omega/\text{m}$	$R = 3,851721673 \cdot 10^{-5} \Omega/\text{m}$
y_s	0,06012412684	
y_p	0,007268543102	

4.8.2.3 Losses caused by circulating currents

For three single-core cables in flat formation, without transposition, with sheaths bonded at both ends, the resistance of the sheath or screen at its operating temperature (80°C assumed for the first iteration) has a different formula for each phase, based on the intermediate variables P and Q :

$$P = X + X_m \quad Q = X - \frac{X_m}{3} \quad (\text{Section 2.3.3 of [3]})$$

For the middle cable, the loss factor is given by:

$$\lambda'_{1m} = \frac{R_s}{R} \frac{Q^2}{R_s^2 + Q^2} \quad (\text{Section 2.3.3 of [3]})$$

For the outer cables of the flat formation, the loss factor λ'_{11} of the cable which has the greatest loss (carrying the lagging phase) and λ'_{12} of the second cable are given by:

$$\lambda'_{11} = \frac{R_s}{R} \left[\frac{0,75 P^2}{R_s^2 + P^2} + \frac{0,25 Q^2}{R_s^2 + Q^2} + \frac{2 R_s P Q X_m}{\sqrt{3 (R_s^2 + P^2) (R_s^2 + Q^2)}} \right]$$

$$\lambda'_{12} = \frac{R_s}{R} \left[\frac{0,75 P^2}{R_s^2 + P^2} + \frac{0,25 Q^2}{R_s^2 + Q^2} - \frac{2 R_s P Q X_m}{\sqrt{3 (R_s^2 + P^2) (R_s^2 + Q^2)}} \right]$$

(Section 2.3.3 of [3])

X	9,95260621 10 ⁻⁵ Ω/m	P = 14,30777839 10 ⁻⁵ Ω/m Q = 8,500882147 10 ⁻⁵ Ω/m
X _m	4,35517218 10 ⁻⁵ Ω/m	
R _s	2,072723957 10 ⁻⁴ Ω/m	λ'₁ₘ = 0,7748393697
R	3,851721673 10 ⁻⁵ Ω/m	λ'₁₁ = 1,710504596 λ'₁₂ = 1,281906611

Since the conductor is not Milliken, eddy-current loss is ignored ($\lambda_1'' = 0$) at first and:

$$\lambda_1 = \lambda_1'$$

4.8.2.4 Thermal resistance between cable and duct

Since the input data are unchanged for the assessment of the heat transfer in the air gap between the cable and the duct (same cable, same coefficient for plastic ducts), the T_4' has the same value as calculated in Section 4.7.2.1 for the first iteration (mean temperature in the duct assumed as 70°C).

$$\mathbf{T_4' = 0,3520961015 \text{ Km/W}}$$

4.8.2.5 Temperature rise due to thermal mutual effects

When the cables or ducts are not touching, the principle of superposition can be applied to unequally loaded dissimilar cables. For each cable:

- The external thermal resistance T_4 is calculated as a single isolated buried cable (Section 4.2.2 of [4]),
- The temperature rise at its surface caused by the other cables in the neighbourhood is calculated and subtracted from the value of $\Delta\theta$ used in the general equation for the rated current (Section 4.1 of [4]).

The temperature rise $\Delta\theta_p$ above ambient at the surface of the p^{th} cable, caused by the power dissipated by the other ($q - 1$) cables in the vicinity, is given by:

$$\Delta\theta_p = \Delta\theta_{1p} + \Delta\theta_{2p} + \dots \Delta\theta_{kp} + \dots \Delta\theta_{qp} \quad (\text{Section 4.2.3.2 of [4]})$$

The temperature rise $\Delta\theta_{kp}$ at the surface of the cable p produced by the power W_k dissipated in cable k is given by:

$$\Delta\theta_{kp} = \frac{1}{2\pi} \rho_T W_k \ln \left(\frac{d'_{pk}}{d_{pk}} \right) \quad (\text{Section 4.2.3.2 of [4]})$$

For the present configuration, the (arbitrary) numbering of the cables is adopted:



According to a depth $L = 1$ m at the centre of the flat circuit, the coordinates of the cables are:

	3 (outer cable lagging phase)	1 (central cable)	2 (outer cable leading phase)
x (m)	-0,165	0,00	0,165
y (m)	-1,00	-1,00	-1,00

The calculation is based on the Kennelly assumption with heat transfer between two isothermal surfaces (soil surface, oversheath or duct surface), leading to the method of images and wells ([4], reproduced in Figure 27).

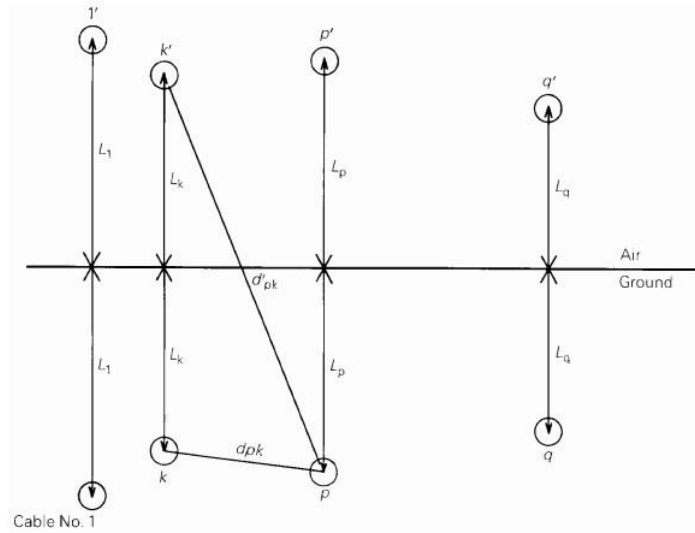


Figure 27: Illustration of the method of images

$$\frac{d'_{pk}}{d_{pk}} = \sqrt{\frac{(x'_k - x_p)^2 + (y'_k - y_p)^2}{(x_k - x_p)^2 + (y_k - y_p)^2}} = \sqrt{\frac{(x_k - x_p)^2 + (L_p + L_k)^2}{(x_k - x_p)^2 + (L_p - L_k)^2}}$$

$\frac{d'_{12}}{d_{12}} = \frac{d'_{13}}{d_{13}} = 12,1623921696$	$\frac{d'_{23}}{d_{23}} = 6,14255206097$
---	--

4.8.3 First iteration

4.8.3.1 External thermal resistance

According to Section 4.7.2, the external thermal resistance of a cable is the sum of the thermal resistance of the air space between the cable surface and duct internal surface T_4' , the thermal resistance of the duct itself T_4'' , and the external thermal resistance of the duct T_4''' . A fourth part is added algebraically to take account of the difference between the thermal resistivities of concrete and soil (Section 4.8.1.4).

T_4'	0,3520961015 Km/W	$T_4 = 1,052084960 \text{ Km/W}$
T_4''	0,05018847750 Km/W	
T_4'''	0,4411466780 Km/W	
corr T_4	0,2086537030 Km/W	

4.8.3.2 Mutual heating of adjacent cables

For the first iteration, since the current rating is not known yet, the value of the conductor losses W_c and metal screen losses W_s is set to zero. The mutual thermal effect $\Delta\theta_p$ seen by each cable is calculated as in Section 4.8.2.5 where $W_k = W_{TOT} = W_d$.

	3 (outer cable lagging phase)	1 (central cable)	2 (outer cable leading phase)
W_k (W/m)	0,3851382172		
ρ_c (Km/W)	0,8		
$\frac{d'_{pk}}{d_{pk}}$	$\frac{d'_{12}}{d_{12}} = \frac{d'_{13}}{d_{13}} = 12,1623921696$		$\frac{d'_{23}}{d_{23}} = 6,14255206097$
$\Delta\theta_p$ (K)	0,2115268420	0,2450246419	0,2115268420

Note: Attention must be paid to the selection of the thermal resistivity of the material surrounding the cables. In this instance, the ducts are embedded in concrete. The lower thermal resistivity ρ_c of the concrete applies instead of ρ_T of the soil.

4.8.3.3 Permissible current rating

From the parameters above, it is possible to calculate the current rating according to Section 4.5.2.3. The highest value of λ_1 (i.e. λ'_{11} for the outer cable with the lagging phase) determines the permissible current rating (the lowest value of I is then applied to the three phases at each iteration):

$\Delta\theta = \theta - \theta_a$	$= 90 - 20 = 70^\circ\text{C}$	$I = 725,3807555 \text{ A}$
n	1 (single-core cable)	
R	$3,851721673 \cdot 10^{-5} \Omega/\text{m}$	
λ'_{11}	1,710504596	
W_d	0,3851382172 W/m	
T_1	0,4198714890 Km/W	
T_3	0,05419960924 Km/W	
T_4	1,052084960 Km/W	
$\Delta\theta_p$	0,2115268420 K	

Note: For the first iteration, a value of current rating is still missing to initiate the calculation of the dissipated losses. The mutual thermal effect $\Delta\theta_p$ represents a very low heating due to W_d for the moment, before assessing more realistic values from the first W_c and W_s losses. Without optimised process, the mutual heating $\Delta\theta_p$ between cables is very underestimated at this stage. The first value of the current rating I is therefore overestimated, leading to high values of losses and temperatures on cable components. However, the convergence to the final value will be rapidly reached after a few iterations.

4.8.3.4 Losses and temperature of cable components

The first iteration is calculated using the expression of losses (Section 4.5.2.4) and temperatures (Section 4.5.2.5). The results are given here for the outer hottest cable (with λ'_{11}):

Parameter (unit)		3 Values (outer λ'_{11})	1 Values (middle λ'_{1m})	2 Values (outer λ'_{12})
I	(A)	725,3807555		
W_c	(W/m)	20,2668828		
W_s	(W/m)	34,66659619	15,70357870	25,98025106
$\Delta\theta_p$	(K)	22,34251986	32,4305393	24,35013899
θ_c	(°C)	112,130993	101,240519	104,5290426
θ_s	(°C)	103,5406525	92,65017824	95,93870202
θ_j	(°C)	100,5424050	90,67971894	93,41125110
$\theta_{\text{air gap}}$	(°C)	90,80367031	84,27938647	85,20173049

High resulting values of conductor temperatures θ_c above the target maximum temperature (90°C) force a significant reduction of the estimated current rating for the next iteration.

The calculation is carried on for each cable with its respective values.

4.8.4 Next iterations

Reminder of the chosen options:

Variable	Topic	IEC rule	CIGRE B1.56 recommendation
W_d	Dielectric losses	Losses neglected depending on the voltage level and insulation material ☐	Losses always taken into account GP 7 ☒
λ_1''	Eddy current losses	Losses neglected when circulating currents are considered, except for Milliken conductors ☒	Losses always taken into account GP 6 ☐
F	Reduction of eddy currents by circulating currents	Used for solid bonded cables with Milliken conductors ☒	Used for all solid bonding arrangements GP 31 ☐
T_4	External thermal resistance of buried cables or ducts	Simplified formula using $2u$ when $u > 0$ ☐	Exact formula with $u + \sqrt{u^2 - 1}$ GP 8 ☒
		Simplified formula considering equally loaded identical cables ☐	Formula considering unequally loaded cables GP 8 ☒

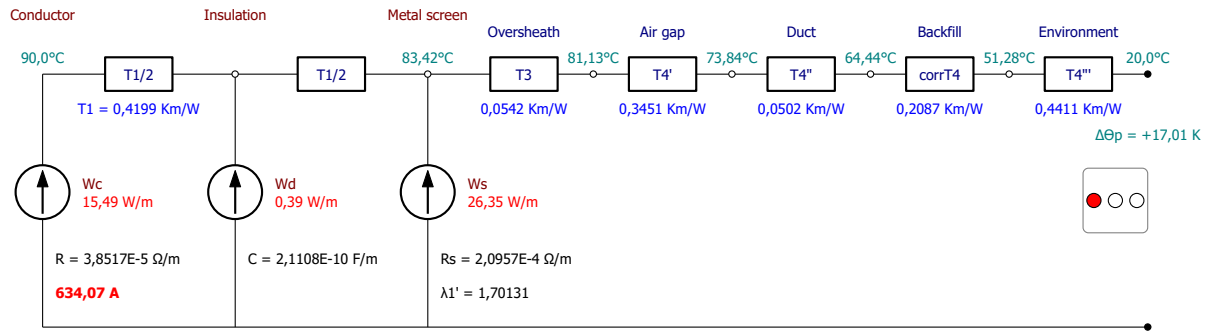
The outer cable carrying the lagging phase is the hottest cable. The results are summarised in the following table:

Iteration	R ($10^{-5} \Omega/\text{m}$)	R_s ($10^{-4} \Omega/\text{m}$)	λ_1	I (A)	W_c (W/m)	W_s (W/m)	$\Delta\theta_p$ (K)	θ_c (°C)	θ_s (°C)
1	3,8517	2,0727	1,71050	725,3808	20,2669	34,6666	22,3425	112,131	103,541
↓									
11	3,8517	2,0957	1,70131	634,0734	15,4858	26,3461	17,0076	90,0000	83,4171

Ten-digit result after 21 iterations

I = 634,0734189920 A

≈ 630 A



When the convergence is reached at the 11th iteration, the following values are obtained for the three cables:

Parameter (unit)		3 Values (outer λ'_{11})	1 Values (middle λ'_{1m})	2 Values (outer λ'_{12})
I	(A)	634,0734354		
W_c	(W/m)	15,48581225	15,14161190	15,24848955
W_s	(W/m)	26,34614380	12,12820119	19,89079441
$\Delta\theta_p$	(K)	17,00756093	24,72954692	18,55439522
θ_c	(°C)	90,00000424	81,81457122	84,35850404
θ_s	(°C)	83,41709892	75,37618581	77,87524375
θ_j	(°C)	81,12894891	73,87729825	75,94983395
$\theta_{air\ gap}$	(°C)	73,84400273	68,98250149	69,68546445

The circulating currents in the metal sheaths have a strong impact to such an extent that the central cable is not the hottest one. Their unbalanced values make the outer cable carrying the lagging phase the most prevailing heat source, leading to a maximum temperature higher than the resulting temperature of the central cable, nevertheless exposed to the proximity of the two outer cables.

4.8.5 Variant with single point bonding

The case study was specified with a bonding of metal sheaths at both ends of an electrical section (Section 4.2.3), leading to losses by circulating currents (Section 4.8.2.3). According to [3], Section 2.3.1, eddy-current losses are ignored ($\lambda_1'' = 0$).

If we consider the sheaths bonded at a single point, there are no circulating currents ($\lambda_1' = 0$) and the eddy-current loss factor is given by λ_1'' , dependent on the temperature of the metal screens.

4.8.5.1 Losses caused by eddy currents

The calculation begins with the approach and formulae used in Section 4.6.3.1, then resumes with specific items to the flat formation geometry.

For the first iteration with the assumption of a temperature of the metal screens equal to 80°C, the value of the intermediate parameters calculated in Section 4.6.3.1 is still valid:

R_s	$2,072723957 \cdot 10^{-4} \Omega/m$
R	$3,851721673 \cdot 10^{-5} \Omega/m$
m	0,1515683091
d	67,7 mm
s	165 mm

Three values are considered for λ_1'' : centre cable, outer cable leading phase and outer cable lagging phase.

Centre cable:

$$\lambda_0 = 6 \left(\frac{m^2}{1+m^2} \right) \left(\frac{d}{2s} \right)^2$$

$$\Delta_1 = 0,86 m^{3,08} \left(\frac{d}{2s} \right)^{(1,4m + 0,7)} \quad (\text{Section 2.3.6.1 (2.a) of [3]})$$

$$\Delta_2 = 0$$

λ_0	0,005670918006	$\lambda_{1m}'' = 0,0306332147$
Δ_1	0,0006070855729	
Δ_2	0	

Outer cable leading phase:

$$\lambda_0 = 1,5 \left(\frac{m^2}{1+m^2} \right) \left(\frac{d}{2s} \right)^2$$

$$\Delta_1 = 4,7 m^{0,7} \left(\frac{d}{2s} \right)^{(0,16m + 2)} \quad (\text{Section 2.3.6.1 (2.b) of [3]})$$

$$\Delta_2 = 21 m^{3,3} \left(\frac{d}{2s} \right)^{(1,47m + 5,06)}$$

λ_0	0,001417729502	$\lambda_{11}'' = 0,008059624042$
Δ_1	0,05081481774	
Δ_2	9,639437362 10^{-6}	

Outer cable lagging phase:

$$\lambda_0 = 1,5 \left(\frac{m^2}{1+m^2} \right) \left(\frac{d}{2s} \right)^2$$

$$\Delta_1 = - \frac{0,74 (m+2) m^{0,5}}{2 + (m-0,3)^2} \left(\frac{d}{2s} \right)^{(m+1)} \quad (\text{Section 2.3.6.1 (2.c) of [3]})$$

$$\Delta_2 = 0,92 m^{3,7} \left(\frac{d}{2s} \right)^{(m+2)}$$

λ_0	0,001417729502	$\lambda_{12}'' = 0,007292825617$
Δ_1	-0,04946637697	
Δ_2	2,830875144 10^{-5}	

4.8.5.2 Permissible current rating

All parameters being unchanged (except the bonding of metal screens, λ_1), most of the results of Section 4.8.1 and Section 4.8.2 can be used for the first iteration, with the formula in Section 4.8.3.3. The calculation is performed on the central cable (with the highest value of λ'_i), which will be the hottest cable of the circuit.

$\Delta\theta = \theta - \theta_a = 90 - 20 = 70^\circ\text{C}$		I = 1073,514075 A
N	1 (single-core cable)	
R	$3,851721673 \cdot 10^{-5} \Omega/\text{m}$	
λ'_{1m}	0,0306332147	
W_d	0,3851382172 W/m	
T_1	0,4198714890 Km/W	
T_3	0,05419960924 Km/W	
T_4	1,052084960 Km/W	
$\Delta\theta_p$	0,2450246419 K	

4.8.5.3 Next iterations

Reminder of the chosen options:

Variable	Topic	IEC rule	CIGRE B1.56 recommendation
W_d	Dielectric losses	Losses neglected depending on the voltage level and insulation material ☐	Losses always taken into account GP 7 ☒
T_4	External thermal resistance of buried cables or ducts	Simplified formula using $2u$ when $u > 0$ ☐	Exact formula with $u + \sqrt{u^2 - 1}$ GP 8 ☒
		Simplified formula considering equally loaded identical cables ☐	Formula considering unequally loaded cables GP 8 ☒

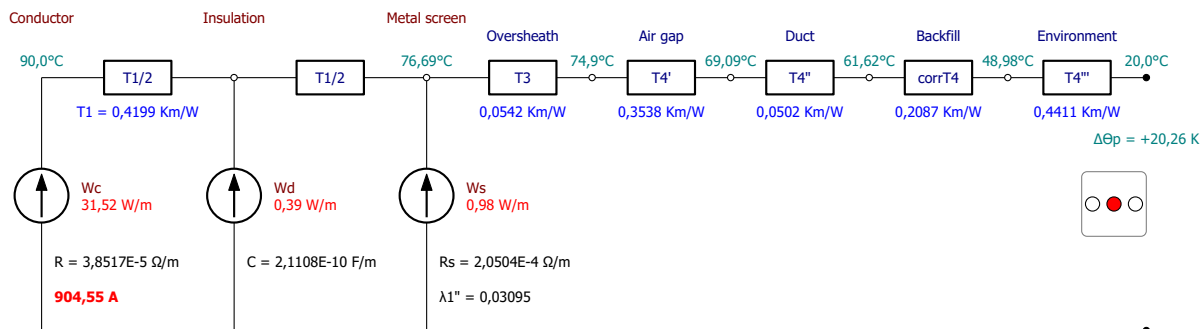
The middle cable is the hottest cable. The results are summarised in the following table:

Iteration	R ($10^{-5} \Omega/\text{m}$)	R_s ($10^{-4} \Omega/\text{m}$)	λ_1	I (A)	W_c (W/m)	W_s (W/m)	$\Delta\theta_p$ (K)	θ_c ($^\circ\text{C}$)	θ_s ($^\circ\text{C}$)
1	3,8517	2,0727	0,03063	1073,5141	44,3885	1,3598	28,7017	118,4567	99,7384
↓									
15	3,8517	2,0504	0,03095	904,5521	31,5153	0,9755	20,2609	90,0000	76,6868

Ten-digit result after 19 iterations

I = 904,5519955345 A

≈ 900 A



This time, the absence of circulating currents makes the heat dissipated by the conductors prevailing. The central cable, exposed on both sides to the vicinity of the outer cables is the hottest cable.

4.8.6 Variant with non-neglected eddy-current losses

This option was adopted by the CIGRE WG B1.56 (Guidance Point 6). The calculation is performed as in Section 4.6.4.

4.8.6.1 First iteration

Guidance Point 31 addresses the reduction of the eddy currents losses by the circulating currents when the metal screens are solid-bonded. The calculation of the F factor has been already described in Section 4.6.4.1.

The factor ($0 < F \leq 1$) is given by:

$$F = \frac{4 M^2 N^2 + (M + N)^2}{4 (M^2 + 1) (N^2 + 1)}$$

$$\text{where } \left. \begin{aligned} M &= \frac{R_s}{X + X_m} \\ N &= \frac{R_s}{X - \frac{X_m}{3}} \end{aligned} \right\} \begin{array}{l} \text{for cables in flat formation} \\ \text{with equidistant spacing} \end{array} \quad (\text{Section 2.3.5 of [3]})$$

For the first iteration, the calculated value is:

R_s	$2,072723957 \cdot 10^{-4} \Omega/\text{m}$	$F = 0,7552691592$
X	$9,95260621 \cdot 10^{-5} \Omega/\text{m}$	
X_m	$4,35517218 \cdot 10^{-5} \Omega/\text{m}$	
M	1,448669319	
N	2,438245727	

In the case of flat geometry, the value of λ_1'' is specific to each cable and F will vary with the temperature of the metal sheath R_s . The eddy current loss factor is obtained by multiplying λ_1'' calculated by Section 2.3.6 of [3] by F . The results are given for the outer cable carrying the lagging phase, which will be the hottest one:

λ_1'' (as per [3] 2.3.6)	0,007292825617	$\lambda_1'' = 0,005508046272$
F	0,7552691592	

Finally, the loss factor λ_1 is the sum of loss factors representing circulating currents and eddy currents:

λ_1'	1,710504596	$\lambda_1 = 1,716012642$
λ_1''	0,005508046272	

All parameters being unchanged (except the consideration of eddy currents with solid bonding of metal screens, λ_1''), most of the results of Sections 4.4, 4.8.1 and 4.8.2 can be used for the first iteration, with the formula in Section 4.5.2.3.

$\Delta\theta = \theta - \theta_a$	$= 90 - 20 = 70^\circ\text{C}$	$I = 724,7351165 \text{ A}$
n	1 (single-core cable)	
R	$3,851721673 \cdot 10^{-5} \Omega/\text{m}$	
λ_1	1,716012642	
W_d	0,3851382172 W/m	
T_1	0,4198714890 Km/W	
T_3	0,05419960924 Km/W	
T_4	1,052084960 Km/W	
$\Delta\theta_p$	0,2115268420 K	

4.8.6.2 Next iterations

Reminder of the chosen options:

Variable	Topic	IEC rule	CIGRE B1.56 recommendation
W_d	Dielectric losses	Losses neglected depending on the voltage level and insulation material ☐	Losses always taken into account GP 7 ☒
λ_1''	Eddy current losses	Losses neglected when circulating currents are considered, except for Milliken conductors ☐	Losses always taken into account GP 6 ☒
F	Reduction of eddy currents by circulating currents	Used for solid bonded cables with Milliken conductors ☐	Used for all solid bonding arrangements GP 31 ☒
T_4	External thermal resistance of buried cables or ducts	Simplified formula using $2u$ when $u > 0$ ☐	Exact formula with $u + \sqrt{u^2 - 1}$ GP 8 ☒
		Simplified formula considering equally loaded identical cables ☐	Formula considering unequally loaded cables GP 8 ☒

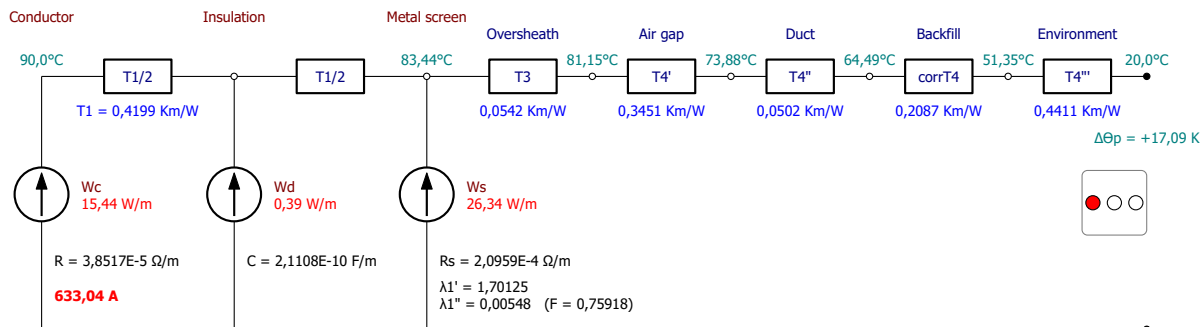
The outer cable carrying the lagging phase is the hottest cable.

Iteration	R ($10^{-5} \Omega/\text{m}$)	R_s ($10^{-4} \Omega/\text{m}$)	λ_1	I (A)	W_c (W/m)	W_s (W/m)	$\Delta\theta_p$ (K)	θ_c (°C)	θ_s (°C)
1	3,8517	2,0727	1,71601	724,7351	20,2308	34,7163	22,5251	112,314	103,743
↓									
11 ($\lambda_1 = \lambda_1' + \lambda_1''$)	3,8517	2,0959	1,70673	633,0357	15,4352	26,3437	17,0898	90,0000	83,4384
11 (Section 4.8.3.4) (λ_1'' neglected)	3,8517	2,0957	1,70131	634,0734	15,4858	26,3461	17,0076	90,0000	83,4171
15 (Section 4.8.5.3) (single bonded)	3,8517	2,0504	0,03095	904,5521	31,5153	0,9755	20,2609	90,0000	76,6868

Ten-digit result after 21 iterations

I = 633,0356512622 A

≈ 630 A



When the convergence is reached, $\lambda_1 = \lambda_1' + \lambda_1'' = 1,70125 + 0,00548 = 1,70673$.

When possible, the bonding at a single point of the metal sheath (or crossbonding with identical electrical sections) leads to improvement in the current rating of the system (+43 %).

In the case of solid bonding at both ends, IEC neglects the eddy-current losses, except for Milliken conductors. This example shows that assumption hides a slight derating of 0,22 %. (0,16 % when F factor is considered).

These results have been reported in the summary table in Section 4.11.

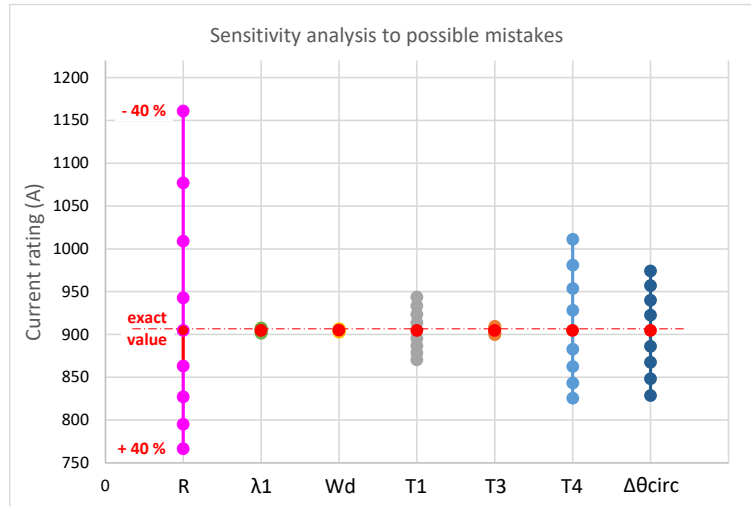
4.8.7 Sensitivity analysis to possible mistakes

The chart showing the impact of a deviation when a formula or method is improperly implemented into a calculation tool is drawn as in Section 4.6.5, with a range of $\pm 40 \%$ deviation. The present case of cables in PVC ducts embedded in concrete involves W_d , T_1 , T_3 , T_4 , R , and λ_1 , as the case of directly

buried cables did previously, but also the heating generated by the two other cables of the circuit. $\Delta\theta_{circ}$ (deducted from $\Delta\theta$) adds to the effect of T_4 , which is here the external resistance of the single duct.

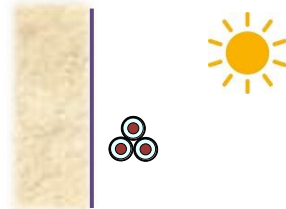
$$I = \left[\frac{\Delta\theta - W_d [0,5 T_1 + n (T_2 + T_3 + T_4)]}{RT_1 + nR (1 + \lambda_1) T_2 + nR (1 + \lambda_1 + \lambda_2) (T_3 + T_4)} \right]^{0,5} \quad (\text{Section 1.4.1.1 of [3]})$$

With the single point bonding (sub-case Section 4.8.5), there are no circulating currents. The consequence of a deviation on the eddy current losses is negligible.



4.9 case #0-4: Sub-case study with cables laid in free air directly exposed to solar radiation

The next case study is a variation with touching cables in trefoil formation, laid in free air (no thermal confinement) exposed to solar radiation.

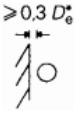
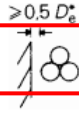


4.9.1.1 Earthing of the metal sheaths

The subcase study starts with metal sheaths bonded at both ends (solid bonding). Since the cables are installed in a touching trefoil geometry, exactly as in the first subcase of directly buried touching cables, the results of the losses caused by circulating currents can be reused for the first iteration (Section 4.5.2.2).

4.9.1.2 Installation

The cables in trefoil are installed on cleats away from a vertical wall. Table 2 of [4] gives the values for constants Z , E and g , which are required to the calculation of the heat dissipation coefficient.

No.	Installation	Z	E	d	Mode
Installation on non-continuous brackets, ladder supports or cleats, D_e^* not greater than 0,15 m					
1	Single cable ^a	0,21	3,94	0,60	
3	Three cables in trefoil	0,96	1,25	0,20	

$$Z = 0,96 \quad / \quad E = 1,25 \quad / \quad g = 0,20$$

4.9.1.3 Ambient temperature

The table of ambient temperatures at sea level (Section 3.2.1 of [16]) suggests a maximum air temperature value of 25°C for a temperate climate:

Climate	Ambient air temperature		Ambient ground temperature at a depth of 1 m	
	Min. °C	Max. °C	Min. °C	Max. °C
Tropical	25	55	25	40
Subtropical	10	40	15	30
Temperate	0	25	10	20

$$\theta_a = 25^\circ\text{C}$$

4.9.1.4 Solar radiation

When a local value cannot be obtained, it is recommended to take a value of 1000 W/m² for the intensity of solar radiation, applicable for most latitudes:

$$H = 1000 \text{ W/m}^2$$

1000 W/m² is the maximum normal surface irradiance at sea level on a clear day and can be considered on the safe side for the selected location with a moderate climate. A lower value can be assumed, depending on the season and the latitude of the studied location.

The cable surface absorption coefficient of solar radiation is given by Table 3 of [4]:

Table 3 – Absorption coefficient of solar radiation for cable surfaces

Material	σ
Bitumen/jute serving	0,8
Polychloroprene	0,8
PVC	0,6
PE	0,4
Lead	0,6

The specified value 0,4 may appear low (lower than lead especially), despite the black colour of the oversheath. But it is used in the present report as a standard without further discussion.

4.9.2 Results independent of the temperature

4.9.2.1 Heat dissipation coefficient

The heat dissipation coefficient depends on the position of the circuit in relation to the wall and the cable external diameter:

$$h = \frac{Z}{(D_e^*)^g} + E \quad (\text{Section 4.2.1.1 of [4]})$$

Z	0,96	h = 2,859466134 W/m²K^{5/4}
E	1,25	
g	0,20	
D _e *	0,0755 m	

4.9.2.2 General parameters

The values of W_d , T_1 and T_3 calculated for the first sub-case can be used.

4.9.3 First iteration

4.9.3.1 General parameters

Loss factors and thermal resistances are required for the calculation of the cable surface temperature.

Beforehand, the values of cable components have to be initialised by the user to launch the first iteration, exactly as for the first case study of directly buried cables in touching trefoil formation (Section 4.2.4). The maximum operating temperature of the conductor is $\theta = 90^\circ\text{C}$ leads to a selection of the temperature of the metal screens or sheaths of $\theta_s = \theta - 10 = 80^\circ\text{C}$. In these conditions, many values can be reused:

n	1 (single-core cable)
R	3,952152638 10^{-5} Ω/m
λ_1	0,2928142510
W_d	0,3851382172 W/m
T_1	0,4198714890 Km/W
T_3	0,05419960924 Km/W

Note: The 1,6 multiplication factor for T_3 of touching cables in trefoil formation only applies to buried cables (Section 4.2.4.3 of [4]).

4.9.3.2 Heating due to dielectric losses

$\Delta\theta_d$, which is a factor having the dimensions of temperature difference, accounts for dielectric losses:

$$\Delta\theta_d = W_d \left[\left(\frac{1}{1 + \lambda_1 + \lambda_2} - \frac{1}{2} \right) T_1 - \frac{n \lambda_2 T_2}{1 + \lambda_1 + \lambda_2} \right] \quad (\text{Section 4.2.1.1 of [4]})$$

λ_1	0,2928142510	$\Delta\theta_d = 0,04422831304$ K
W_d	0,3851382172 W/m	
T_1	0,4198714890 Km/W	

4.9.3.3 K_A factor

$$K_A = \frac{\pi D_e^* h}{(1 + \lambda_1 + \lambda_2)} \left[\frac{T_1}{n} + T_2 (1 + \lambda_1) + T_3 (1 + \lambda_1 + \lambda_2) \right] \quad (\text{Section 4.2.1.1 of [4]})$$

D_e^*	0,0755 m	$K_A = 0,2570335963 \text{ K}^{-1/4}$
h	2,859466134 W/m ² K ^{5/4}	
n	1 (single-core cable)	
λ_1	0,2928142510	
T_1	0,4198714890 Km/W	
T_3	0,05419960924 Km/W	

4.9.3.4 Heating due to direct solar radiation

$\Delta\theta_{ds}$, which is a factor having the dimensions of temperature difference, accounts for direct solar radiation:

$$\Delta\theta_{ds} = \frac{\sigma D_e^* H}{(1 + \lambda_1 + \lambda_2)} \left[\frac{T_1}{n} + T_2 (1 + \lambda_1) + T_3 (1 + \lambda_1 + \lambda_2) \right] \quad (\text{Section 4.2.1.2 of [4]})$$

D_e^*	0,0755 m	$\Delta\theta_{ds} = 11,4449804 \text{ K}$
σ	0,4	
H	1000 W/m ²	
n	1 (single-core cable)	
λ_1	0,2928142510	
T_1	0,4198714890 Km/W	
T_3	0,05419960924 Km/W	

4.9.3.5 Cable surface temperature

The excess of cable surface temperature above ambient temperature $\Delta\theta_s$ is calculated by a simple iterative method, starting with an initial value of $(\Delta\theta_s)_0^{1/4} = 2$:

$$(\Delta\theta_s)_{n+1}^{1/4} = \left[\frac{\Delta\theta + \Delta\theta_d + \Delta\theta_{ds}}{1 + K_A (\Delta\theta_s)_n^{1/4}} \right]^{0,25} \quad (\text{Section 4.2.1.2 of [4]})$$

The process reiterates until:

$$(\Delta\theta_s)_{n+1}^{1/4} - (\Delta\theta_s)_n^{1/4} \leq 0,001$$

However, the accuracy of the final result is poor if this convergence step is adopted: a deviation equal to 0,001 for $(\Delta\theta_s)^{1/4}$ means 0,18°C for $\Delta\theta_s$, which is clearly insufficient at the era of powerful laptops... Therefore, a convergence threshold of 10^{-12} is more coherent with the criterion of temperature convergence (0,0001°C) detailed in Section 4.3.2. A lower threshold would not make sense, due to the limit of float (real) numbers in double precision ($\approx 10^{-15}$). Investigations on lower thresholds than 10^{-12} show an impact on the current rating I after the tenth digit.

The first iteration leads to:

$\Delta\theta = \theta - \theta_a$	= 90-25 = 65°C	$(\Delta\theta_s)_1^{1/4} = 2,666022159 \text{ K}^{1/4}$
$\Delta\theta_d$	0,04422831304 K	
$\Delta\theta_{ds}$	11,4449804 K	
K_A	0,2570335963 K ^{-1/4}	
$(\Delta\theta_s)_0^{1/4}$	2,0 K ^{1/4}	

The following iterations show the convergence:

n	$(\Delta\theta_s)_n^{1/4}$	$ (\Delta\theta_s)_n^{1/4} - (\Delta\theta_s)_{n-1}^{1/4} $
0	2,0	-
1	2,666022159	$6,66 \cdot 10^{-1}$
2	2,59557444	$7,05 \cdot 10^{-2}$
3	2,602593757	$7,02 \cdot 10^{-3}$
4	2,601890097	$7,04 \cdot 10^{-4}$
5	2,601960594	$7,05 \cdot 10^{-5}$
6	2,601953531	$7,06 \cdot 10^{-6}$
7	2,601954238	$7,08 \cdot 10^{-7}$
8	2,60195417	$7,09 \cdot 10^{-8}$
9	2,601954174	$7,10 \cdot 10^{-9}$
10	2,601954174	$7,12 \cdot 10^{-10}$
11	2,601954174	$7,13 \cdot 10^{-11}$
12	2,601954174	$7,14 \cdot 10^{-12}$
13	2,601954174	$7,16 \cdot 10^{-13} < 10^{-12}$

4.9.3.6 External thermal resistance

Considering the previous elements, note that the external thermal resistance T_4 for cables in free air depends on temperature and must be recalculated at each iteration.

The formula of T_4 for cables directly exposed to solar radiation is the same as for cables protected, including the appropriate expression of $\Delta\theta_s$, (Section 4.9.3.5):

$$T_4 = \frac{1}{\pi D_e^* h (\Delta\theta_s)^{1/4}} \quad (\text{Section 4.2.1.1 of [4]})$$

D_e^*	0,0755 m	$T_4 = 0,5666548040 \text{ Km/W}$
h	2,859466134 W/m ² K ^{5/4}	
$(\Delta\theta_s)^{1/4}$	2,601954174 K ^{1/4}	

4.9.3.7 Permissible current rating

The general formula for the current rating calculation ([3], Section 1.4.1.1) detailed at the first case study with buried cables (Section 4.5.2.3) is modified to take into account the direct solar radiation as:

$$I = \left[\frac{\Delta\theta - W_d \left[0,5 T_1 + n (T_2 + T_3 + T_4^*) \right] - \sigma D_e^* H T_4^*}{RT_1 + nR (1 + \lambda_1) T_2 + nR (1 + \lambda_1 + \lambda_2) (T_3 + T_4^*)} \right]^{0,5} \quad (\text{Section 1.4.4.1 of [3]})$$

The cable is unarmoured: $\lambda_2 = 0$ and $T_2 = 0$.

$\Delta\theta = \theta - \theta_a$	$= 90 - 25 = 65^\circ\text{C}$	$I = 992,2203527 \text{ A}$
n	1 (single-core cable)	
R	$3,952152638 \cdot 10^{-5} \Omega/\text{m}$	
λ_1	0,2928142510	
W_d	0,3851382172 W/m	
T_1	0,4198714890 Km/W	
T_3	0,05419960924 Km/W	
T_4	0,5666548040 Km/W	
D_e^*	0,0755 m	
σ	0,4	
H	1000 W/m ²	

4.9.3.8 Losses in conductor and sheath

The losses in metal components are generated by Joule effect:

$$W_c = R \cdot I^2 \quad \text{and} \quad W_s = \lambda_1 \cdot W_c$$

R	$3,952152638 \cdot 10^{-5} \Omega/\text{m}$	$W_c = 38,90899126 \text{ W/m}$ $W_s = 11,3931071 \text{ W/m}$
λ_1	0,2928142510	
I	992,2203527 A	

4.9.3.9 Temperature on cable components

The temperature on the oversheath (jacket or external serving) can be deducted from the general expression for the temperature rise above ambient temperature ([3], Section 1.4.1.1), increased by the term which takes into account the direct solar radiation:

$$\theta_j = \theta_a + n (W_c + W_s + W_d) T_4 + \sigma D_e^* H T_4$$

The expression of the temperature of the metal sheath and conductor is unchanged from Section 4.5.2.5.

θ_a	25°C	$\theta_j = 70,83514121^\circ\text{C}$ $\theta_s = 73,58236962^\circ\text{C}$ $\theta_c = 90^\circ\text{C}$
n	1 (single-core cable)	
W_d	0,3851382172 W/m	
W_c	38,90899126 W/m	
W_s	11,3931071 W/m	
T_1	0,4198714890 Km/W	
T_3	0,05419960924 Km/W	
T_4	0,5666548040 Km/W	
D_e^*	0,0755 m	
σ	0,4	
H	1000 W/m ²	

4.9.4 Next iterations

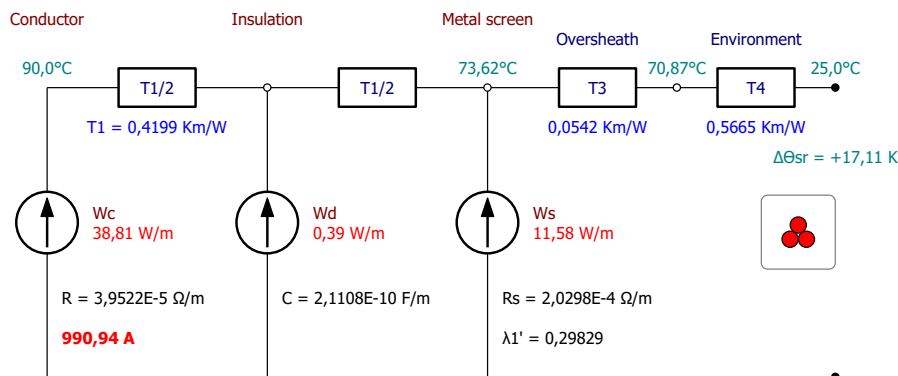
Reminder of the chosen options:

Variable	Topic	IEC rule	CIGRE B1.56 recommendation
W_d	Dielectric losses	Losses neglected depending on the voltage level and insulation material ☐	Losses always taken into account GP 7 ☒
λ_1''	Eddy current losses	Losses neglected when circulating currents are considered, except for Milliken conductors ☒	Losses always taken into account GP 6 ☐
F	Reduction of eddy currents by circulating currents	Used for solid bonded cables with Milliken conductors ☒	Used for all solid bonding arrangements GP 31 ☐

All cables are considered at the same temperature. The heating of the cable surface $\Delta\theta_{sr}$ due to the solar radiation (equal to the quantity $\sigma D_e \cdot H T_4$ to be deducted from the numerator of the permissible current rating in section 4.9.3.7) is provided in a dedicated column. The results are summarised in the following table:

Iteration	R ($10^{-5} \Omega/m$)	R_s ($10^{-4} \Omega/m$)	λ_1	T_4 (Km/W)	I (A)	W_c (W/m)	W_s (W/m)	$\Delta\theta_{sr}$ (K)	θ_c (°C)	θ_s (°C)
1	3,9522	2,0727	0,29281	0,5667	992,2204	38,9090	11,3931	17,1130	90	73,5824
2	3,9522	2,0296	0,29832	0,5665	990,8963	38,8052	11,5766	17,1094	89,9965	73,6224
3	3,9522	2,0298	0,29829	0,5665	990,9422	38,8085	11,5763	17,1096	90,0002	73,6248
4	3,9522	2,0298	0,29829	0,5665	990,9398	38,8086	11,5762	17,1095	89,9998	73,6243
5	3,9522	2,0298	0,29829	0,5665	990,9418	38,8088	11,5763	17,1095	90,0000	73,6245
6	3,9522	2,0298	0,29829	0,5665	990,9415	38,8088	11,5763	17,1095	90,0000	73,6244

Ten-digit result after 17 iterations **I = 990,9416387529 A** ≈ 990 A



With a high intensity $H = 1000 \text{ W/m}^2$, the influence of the direct solar radiation is significant, leading to a heating of the cable surface of $\Delta\theta_{sr} = 17,11 \text{ K}$.

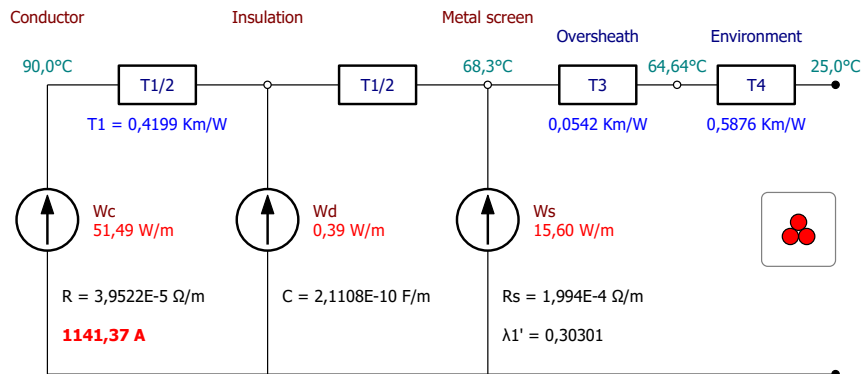
4.9.5 Variant with cables protected from direct solar radiation

The options of Section 4.9.4 are still valid. The calculation of the heating due to solar radiation is skipped (Section 4.9.3.4), which means $\Delta\theta_{ds} = 0$.

All cables are considered at the same temperature. Finally, the calculation with no solar radiation leads to:

Iteration	R ($10^{-5} \Omega/\text{m}$)	R _s ($10^{-4} \Omega/\text{m}$)	λ_1	T ₄ (K/mW)	I (A)	W _c (W/m)	W _s (W/m)	θ_c (°C)	θ_s (°C)
1	3,9522	2,0727	0,29281	0,5879	1144,1420	51,7361	15,1491	90	68,1966
↓									
4	3,9522	1,9940	0,30301	0,5876	1141,3667	51,4854	15,6005	90	68,3019

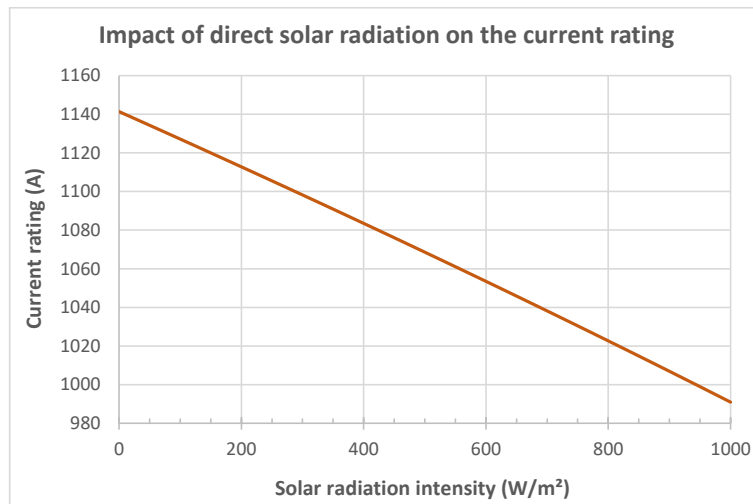
Ten-digit result after 8 iterations **I = 1141,3666669276 A** $\approx 1140 \text{ A}$



Note: Despite lower current rating (991 A instead of 1141 A) with the direct solar radiation, the temperature on the cable surface (θ_s) is warmer (70,9°C instead of 64,6°C).

The direct exposure to solar radiation leads to a derating of 13,2 %.

Another way to show this derating is to draw a chart with a range of 0-1000 W/m² of solar radiation intensity.



4.9.6 Variant with single point bonding

The case study was specified with a bonding of metal sheaths at both ends of an electrical section (Section 4.2.3), leading to losses by circulating currents (Section 4.9.3). According to Section 2.3.1 of [3], eddy-current losses are ignored ($\lambda_1'' = 0$).

By now, if we consider the sheaths bonded at a single point, there are no circulating currents ($\lambda_1' = 0$) and the eddy-current loss factor is given by λ_1'' , dependent on the temperature of the metal screens.

4.9.6.1 Eddy current losses caused

The calculation begins with the approach and formulae used in Section 4.6.3.1 and with the same geometry of installation (touching cables in trefoil formation).

For the first iteration, with the assumption of a temperature of the metal screens equal to 80°C, all the value of the parameters calculated in Section 4.6.3.1 are still valid.

$$\lambda_1'' = 0,07695600926$$

4.9.6.2 General parameters

Compared to the values in Section 4.9.3 for the solid bonding of the metal sheaths, only the value of λ_1 is modified:

n	1 (single-core cable)
R	$3,952152638 \cdot 10^{-5} \Omega/\text{m}$
λ_1	0,07695600926
W_d	0,3851382172 W/m
T_1	0,4198714890 Km/W
T_3	0,05419960924 Km/W

4.9.6.3 Heating parameters

The values of the heating due to dielectric losses $\Delta\theta_d$, the K_A factor and the heating due to direct solar radiation $\Delta\theta_{ds}$ are calculated as in Section 4.9.3.2, Section 4.9.3.3 and Section 4.9.3.4, respectively, with the new value of λ_1 :

λ_1	0,07695600926	$\Delta\theta_d = 0,06929907547 \text{ K}$ $K_A = 0,3011837996 \text{ K}^{-1/4}$ $\Delta\theta_{ds} = 13,41086433 \text{ K}$
W_d	0,3851382172 W/m	
T_1	0,4198714890 Km/W	
T_3	0,05419960924 Km/W	
D_e^*	0,0755 m	
h	$2,859466134 \text{ W/m}^2\text{K}^{5/4}$	
n	1 (single-core cable)	
σ	0,4	
H	1000 W/m ²	

4.9.6.4 Cable surface temperature

The excess of cable surface temperature above ambient temperature $\Delta\theta_s$ is calculated according to the method described in Section 4.9.3.5.

The first iteration leads to:

$\Delta\theta = \theta - \theta_a$	$= 90 - 25 = 65^\circ\text{C}$	$(\Delta\theta_s)_1^{1/4} = 2,64544926 \text{ K}^{1/4}$
$\Delta\theta_d$	0,06929907547 K	
$\Delta\theta_{ds}$	13,41086433 K	
K_A	0,3011837996 $\text{K}^{-1/4}$	
$(\Delta\theta_s)_0^{1/4}$	2,0 $\text{K}^{1/4}$	

The convergence is reached after a few iterations:

n	$(\Delta\theta_s)_n^{1/4}$	$ (\Delta\theta_s)_n^{1/4} - (\Delta\theta_s)_{n-1}^{1/4} $
0	2,0	-
1	2,645449259	$6,45 \cdot 10^{-1}$
↓		
14	2,578100084	$2,2 \cdot 10^{-13} < 10^{-12}$

4.9.6.5 External thermal resistance

According to Section 4.9.3.6:

D_e^*	0,0755 m	$T_4 = 0,5718978255 \text{ Km/W}$
h	2,859466134 $\text{W/m}^2\text{K}^{5/4}$	
$(\Delta\theta_s)^{1/4}$	2,578100084 $\text{K}^{1/4}$	

4.9.6.6 Permissible current rating

According to Section 4.9.3.7, and including the modified values of λ_1 and T_4 :

$\Delta\theta = \theta - \theta_a$	$= 90 - 25 = 65^\circ\text{C}$	$I = 1047,04269 \text{ A}$
n	1 (single-core cable)	
R	$3,952152638 \cdot 10^{-5} \Omega/\text{m}$	
λ_1	0,07695600926	
W_d	0,3851382172 W/m	
T_1	0,4198714890 Km/W	
T_3	0,05419960924 Km/W	
T_4	0,5718978255 Km/W	
D_e^*	0,0755 m	
σ	0,4	
H	1000 W/m^2	

4.9.6.7 Next iterations

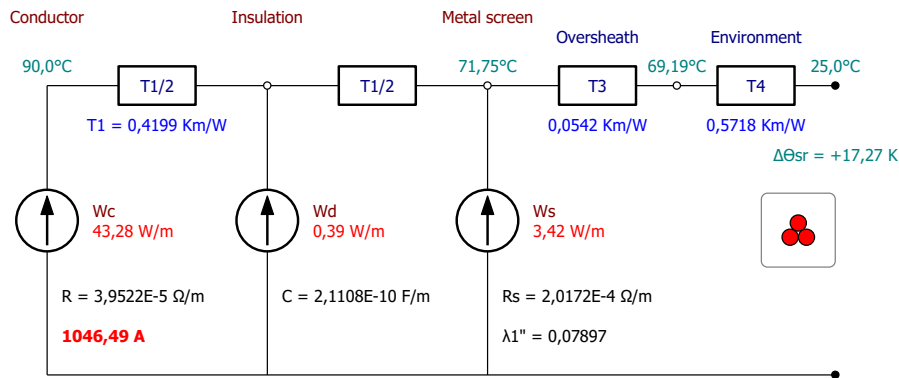
Reminder of the chosen options:

Variable	Topic	IEC rule	CIGRE B1.56 recommendation
W_d	Dielectric losses	Losses neglected depending on the voltage level and insulation material ☐	Losses always taken into account GP 7 ☒

The results are summarised in the following table:

Iteration	R ($10^{-5} \Omega/\text{m}$)	R_s ($10^{-4} \Omega/\text{m}$)	λ_1	T_4 (Km/W)	I (A)	W_c (W/m)	W_s (W/m)	θ_c (°C)	θ_s (°C)
1	3,9522	2,0727	0,07696	0,5719	1047,0427	43,3274	3,3343	90	71,7272
↓									
5	3,9522	2,0172	0,07897	0,5718	1046,4865	43,2814	3,4181	90	71,7465

Ten-digit result after 17 iterations **I = 1046,4864768318 A** $\approx 1040 \text{ A}$



4.9.7 Variant with non-neglected eddy-current losses

This option was adopted by the CIGRE WG B1.56. The calculation is performed as in Section 4.6.4, including the F factor recommended by the Guidance Point 31.

4.9.7.1 First iteration

The configuration is exactly the same as for the directly buried cables in Section 4.6.4: cables in touching trefoil. Therefore, the results of the losses in the metal sheaths can be reused as they are:

λ_1'	0,2928142510	$\lambda_1 = 0,3654736398$
F	0,9441678372	
λ_1'' (as per IEC 2.3.6)	0,07695600926	

The values of the heating due to dielectric losses $\Delta\theta_d$ and direct solar radiation $\Delta\theta_{ds}$ are calculated as in Section 4.9.3 with the new value of λ_1 :

λ_1	0,3654736398	$\Delta\theta_d = 0,03757243601 \text{ K}$ $K_A = 0,2453124400 \text{ K}^{-1/4}$ $\Delta\theta_{ds} = 10,92307042 \text{ K}$
W_d	0,3851382172 W/m	
T_1	0,4198714890 Km/W	
T_3	0,05419960924 Km/W	
D_e^*	0,0755 m	
h	2,859466134 W/m ² K ^{5/4}	
n	1 (single-core cable)	
σ	0,4	
H	1000 W/m ²	

The excess of cable surface temperature above ambient temperature $\Delta\theta_s$ is calculated by the iterative method described in Section 4.9.3.5, starting with an initial value of $(\Delta\theta_s)_0^{1/4} = 2$:

$$(\Delta\theta_s)_1^{1/4} = 2,67180689597418 \rightarrow (\Delta\theta_s)_{13}^{1/4} = 2,60878535304475$$

The external thermal resistance is given as in Section 4.9.3.6:

D_e^*	0,0755 m	$T_4 = 0,5651710021 \text{ Km/W}$
h	2,859466134 W/m ² K ^{5/4}	
$(\Delta\theta_s)^{1/4}$	2,608785353 K ^{1/4}	

The permissible current rating is calculated as in Section 4.9.3.7 with the modified values of λ_1 and T_4 :

$\Delta\theta = \theta - \theta_a$	= 90-25 = 65°C	$I = 975,6502038 \text{ A}$
n	1 (single-core cable)	
R	3,952152638 10 ⁻⁵ Ω/m	
λ_1	0,3654736398	
W_d	0,3851382172 W/m	
T_1	0,4198714890 Km/W	
T_3	0,05419960924 Km/W	
T_4	0,5651710021 Km/W	
D_e^*	0,0755 m	
σ	0,4	
H	1000 W/m ²	

Then, the losses in conductor and sheath are calculated as in Section 4.9.3.8 and the temperature on the cable components as in Section 4.9.3.9.

4.9.7.2 Next iterations

Reminder of the chosen options:

Variable	Topic	IEC rule	CIGRE B1.56 recommendation
W_d	Dielectric losses	Losses neglected depending on the voltage level and insulation material ☐	Losses always taken into account GP 7 ☒
λ_1''	Eddy current losses	Losses neglected when circulating currents are considered, except for Milliken conductors ☐	Losses always taken into account GP 6 ☒
F	Reduction of eddy currents by circulating currents	Used for solid bonded cables with Milliken conductors ☐	Used for all solid bonding arrangements GP 31 ☒

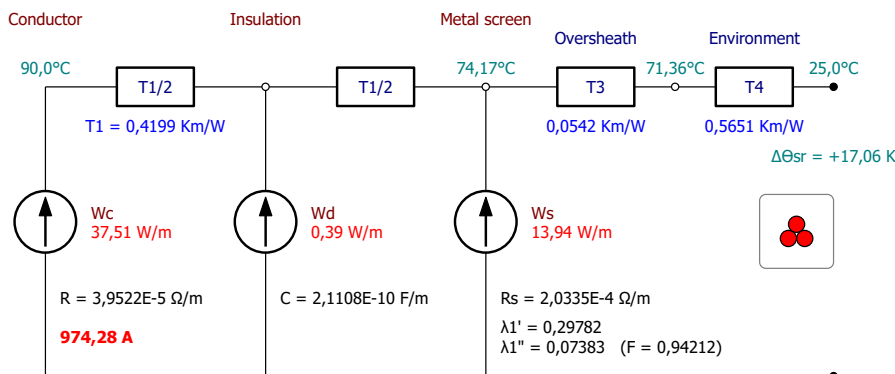
The results are summarised in the following table:

Iteration	R ($10^{-5} \Omega/\text{m}$)	R_s ($10^{-4} \Omega/\text{m}$)	λ_1	T_4 (Km/W)	I (A)	W_c (W/m)	W_s (W/m)	θ_c (°C)	θ_s (°C)
1	3,9522	2,0727	0,36547	0,5652	975,6502	37,6203	13,7492	90	74,1235
↓									
6 ($\lambda_1 = \lambda_1' + \lambda_1''$)	3,9522	2,0335	0,37165	0,5651	974,2801	37,5147	13,9423	90,0000	74,1678
6 (Section 4.9.4) (λ_1'' neglected)	3,9522	2,0298	0,29829	0,5665	990,9415	38,8088	11,5763	90,0000	73,6244
5 (Section 4.9.6.7) (single bonded)	3,9522	2,0172	0,07897	0,5718	1046,4865	43,2814	3,4181	90	71,7465
4 (Section 4.9.5) (no solar rad.)	3,9522	1,9940	0,30301	0,5876	1141,3667	51,4854	15,6005	90	68,3019

Ten-digit result after 18 iterations

I = 974,2802128208 A

≈ 970 A



When the convergence is reached, $\lambda_1 = \lambda_1' + \lambda_1'' = 0,29782 + 0,07383 = 0,37165$.

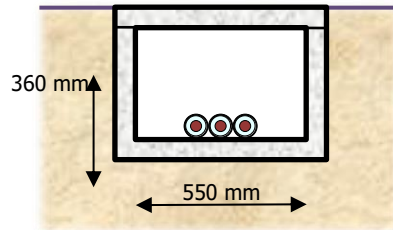
When possible, the bonding at a single point of the metal sheath (or crossbonding with identical electrical sections) is an improvement means of uprating the system (+7,4 %).

In the case of solid bonding at both ends, IEC advises to not include the eddy-current losses, except for Milliken conductors. This example shows that assumption hides a slight derating of 1,71 %.

The direct solar radiation decreases the current rating by 13,2 %. These results have been reported in the summary table in Section 4.11.

4.10 case #0-5: Sub-case study with cables in an unfilled trough

The next case is a variation with touching cables in flat formation, laid in an unfilled trough, as defined in Section 4.2.6.2 of [4]: with the top flush with the soil surface and exposed to free air. The calculation is very similar to the one of cables laid in free air. The iterative process includes an additional derating by the heating $\Delta\theta_{tr}$ inside the trough above the ambient air temperature (reflecting a thermal confinement).



The size of the trough is adopted from a catalogue of a manufacturer.

This installation in unfilled trough is mostly expected in substations with high short circuit currents, without transposition.

CIGRE Technical Brochure 640 [2] mentions some possible issues of convergence. The case study is a good opportunity to examine this issue.

4.10.1 Input data

4.10.1.1 Installation

Table 2 of [4] gives the values for constants Z , E and g , which are required for the calculation of the heat dissipation coefficient. According to the table, the coefficients refer to cables installed on cleats away from a vertical wall. In the absence of better configurations (the cables of the case study are laid on a horizontal surface), we consider the following data:

No.	Installation	Z	E	d	Mode
Installation on non-continuous brackets, ladder supports or cleats, D_e^* not greater than 0,15 m					
1	Single cable ^a	0,21	3,94	0,60	$\geq 0,3 D_e^*$
4	Three cables touching, horizontal	0,62	1,95	0,25	$\geq 0,5 D_e^*$

$$Z = 0,62 \quad / \quad E = 1,95 \quad / \quad g = 0,25$$

4.10.1.2 Ambient temperature

As well as for the previous case with cables laid in free air, the maximum air temperature of 25°C for a moderate temperate climate is selected.

The cables are protected from the direct solar radiation by the concrete cover of the trough.

4.10.2 Results independent of temperature

4.10.2.1 Reactance

Three single-core cables in flat formation, without transposition, sheaths bonded at both ends, are dealt with in Section 2.3.3 of [3]. The calculation of X (reactance of the metal screen of two adjacent single-core cables) and X_m (mutual reactance between the sheath of an outer cable and the conductors of the other two) is already described in Section 4.8.1.1.

f	50 Hz	$X = 5,040331399 \cdot 10^{-5} \Omega/\text{m}$ $X_m = 4,35517218 \cdot 10^{-5} \Omega/\text{m}$
ω	$2\pi \times 50 = 314,1592654 \text{ rad/s}$	
s	$D_e = 75,5 \text{ mm}$	
d	$64,3 + (2 \times 1,3) + 0,8 = 67,7 \text{ mm}$	

4.10.2.2 Heat dissipation coefficient

The heat dissipation coefficient depends on the position of the circuit in relation to the wall, and the cable external diameter, according to the formula expressed in Section 4.9.2.1.

Z	0,62	$h = 3,132782667 \text{ W/m}^2\text{K}^{5/4}$
E	1,95	
g	0,25	
D_e^*	0,0755 m	

4.10.3 First iteration

4.10.3.1 AC resistance of conductor

Since the assumptions of touching cables and conductors at 90°C are still valid, the results of the directly buried cables from Section 4.5.1.1 to Section 4.5.1.4 can be reused:

$$\mathbf{R = 3,952152638 \cdot 10^{-5} \Omega/m}$$

4.10.3.2 Losses caused by circulating currents

According to Section 2.3.3 of [3] and as already detailed for the cables in PVC ducts in Section 4.8.2.3, the loss factor is given for the three cables:

X	$5,040331399 \cdot 10^{-5} \Omega/\text{m}$	$P = 9,395503579 \cdot 10^{-5} \Omega/\text{m}$ $Q = 3,588607338 \cdot 10^{-5} \Omega/\text{m}$
X_m	$4,35517218 \cdot 10^{-5} \Omega/\text{m}$	
R_s	$2,072723957 \cdot 10^{-4} \Omega/\text{m}$	$\lambda'_{1m} = 0,1526334560$ $\lambda'_{11} = 0,789041528$ $\lambda'_{12} = 0,62818111$
R	$3,9521526378 \cdot 10^{-5} \Omega/\text{m}$	

Since the conductor is not Milliken, eddy-current loss is ignored ($\lambda_1'' = 0$) and:

$$\lambda_1 = \lambda_1'$$

Note: The unbalanced circulating currents in the metal sheaths will result in significant differences of cable surface temperature between the three cables of the circuit.

4.10.3.3 General parameters

Loss factors and thermal resistances are required for the calculation of the cable surface temperature.

Beforehand, the values of cable components have to be initialised by the user to launch the first iteration, exactly as for the first case study of directly buried cables in touching trefoil formation (Section 4.2.4). The maximum operating temperature of the conductor is $\theta = 90^\circ\text{C}$, coherent with a temperature of the metal screens or sheaths of $\theta_s = \theta - 10 = 80^\circ\text{C}$. In these conditions, most of the values can be reused.

While the touching trefoil formation of the previous sub-case (cables in free air Section 4.9.3.1) led to identical values for the main parameters (R , λ_1 , T_4), the non-transposed flat formation requires three parallel calculations, due to the different values of loss factor λ_1 and the temperature of the components. Only the calculation with the highest value λ'_{11} will be detailed here, but the process is the same for the two other cables involving λ'_{1m} and λ'_{12} .

n	1 (single-core cable)
R	$3,952152638 \cdot 10^{-5} \Omega/\text{m}$
λ'_{11}	0,789041528
W_d	0,3851382172 W/m
T_1	0,4198714890 Km/W
T_3	0,05419960924 Km/W

Note: The 1,6 multiplication factor for T_3 of touching cables in trefoil formation only applies to buried cables (Section 4.2.4.3 of [4]).

4.10.3.4 Heating due to dielectric losses

$\Delta\theta_d$, accounts for dielectric losses, as described in Section 4.9.3.2.

λ'_{11}	0,789041528	$\Delta\theta_d = 0,009534096771 \text{ K}$
W_d	0,3851382172 W/m	
T_1	0,4198714890 Km/W	

4.10.3.5 K_A factor

The next factor is K_A , as in Section 4.9.3.3:

D_e^*	0,0755 m	$K_A = 0,2146644525 \text{ K}^{-1/4}$
h	$3,132782667 \text{ W/m}^2\text{K}^{5/4}$	
n	1 (single-core cable)	
λ'_{11}	0,789041528	
T_1	0,4198714890 Km/W	
T_3	0,05419960924 Km/W	

4.10.3.6 Cable surface temperature

The cables are being protected by the trough cover and hence no direct solar radiation is considered. The cable surface temperature rise above ambient $\Delta\theta_s$ is calculated by the iterative method described in Section 4.9.6.4, still starting with an initial value of $(\Delta\theta_s)_0^{1/4} = 2$, not including the heating $\Delta\theta_{ds}$ due to solar radiation. IEC [4] mentions that the rating is "calculated as for a cable in free air, but the ambient temperature shall be increased by $\Delta\theta_{tr}$ ":

$$(\Delta\theta_s)_{n+1}^{1/4} = \left[\frac{\Delta\theta + \Delta\theta_d}{1 + K_A (\Delta\theta_s)_n^{1/4}} \right]^{0,25} \quad (\text{Section 4.2.1.1 of [4]})$$

At this stage of the calculation, the value of $\Delta\theta_{tr}$ is not yet known. The empirical formula giving this value requires W_{TOT} (the total power dissipated in the trough per metre length), which will be available after the calculation of the permissible current rating I (A). It is not recommended to start the first iteration with the arbitrary value $\Delta\theta_{tr} = 0 \text{ K}$, or the temperature rise due to the dielectric losses W_d . The difference with the final value (potentially several tens of degrees) could generate some instabilities and the calculation would diverge. CIGRE TB 640 [2] quotes the paper to Morello [17], which suggests an intuitive iterative approach starting with an initial guess $\Delta\theta_{tr0} = \theta_a$, the air ambient temperature above ground.

The first iteration for cable #3 (with λ'_{11}) leads to:

$\Delta\theta_{tr0}$	25 K	$(\Delta\theta_s)_1^{1/4} = 2,300158849 \text{ K}^{1/4}$
$\Delta\theta = \theta - \theta_a - \Delta\theta_{tr0}$	$= 90-25-25 = 40^\circ\text{C}$	
$\Delta\theta_d$	0,009534096771 K	
K_A	$0,2146644525 \text{ K}^{-1/4}$	
$(\Delta\theta_s)_0^{1/4}$	$2,0 \text{ K}^{1/4}$	

The convergence is reached at the 12th iteration:

n	$(\Delta\theta_s)_n^{1/4}$	$ (\Delta\theta_s)_n^{1/4} - (\Delta\theta_s)_{n-1}^{1/4} $
0	2,0	-
1	2,300158849	$3,00 \cdot 10^{-1}$
2	2,274942882	$2,52 \cdot 10^{-2}$
↓		
12	2,276851798	$3,49 \cdot 10^{-13} < 10^{-12}$

4.10.3.7 External thermal resistance

The external thermal resistance T_4 is then obtained for each cable as in Section 4.9.3.6.

D_e^*	0,0755 m	$T_4 = 0,5910689497 \text{ Km/W}$
h	$3,132782667 \text{ W/m}^2\text{K}^{5/4}$	
$(\Delta\theta_s)^{1/4}$	$2,276851798 \text{ K}^{1/4}$	

4.10.3.8 Permissible current rating

The general formula for the current rating calculation (Section 1.4.1.1 of [3]) detailed at the first case study with buried cables (Section 4.5.2.3) is modified to take into account the heating inside the unfilled trough. Indeed, everything goes as if the heating $\sigma D_e^* H T_4$ due to the direct solar radiation in the previous sub-case (Section 4.9.3.7) was replaced by the heating $\Delta\theta_{tr}$ due to thermal confinement inside the trough. IEC should add a new formula integrating $\Delta\theta_{tr}$ more explicitly.

$$I = \left[\frac{\Delta\theta - W_d [0,5 T_1 + n(T_2 + T_3 + T_4^*)] - \Delta\theta_{tr}}{RT_1 + nR(1 + \lambda_1)T_2 + nR(1 + \lambda_1 + \lambda_2)(T_3 + T_4^*)} \right]^{0,5} \quad (\text{Section 1.4.4.1 of [3]})$$

At this stage, the value of $\Delta\theta_{tr}$ is not yet known. The initial guess $\Delta\theta_{tr0} = \theta_a$ will be used again. The cable is unarmoured: $\lambda_2 = 0$ and $T_2 = 0$.

$\Delta\theta = \theta - \theta_a$	$= 90-25 = 65^\circ\text{C}$
n	1 (single-core cable)
R	$3,952152638 \cdot 10^{-5} \text{ }\Omega/\text{m}$
W_d	$0,3851382172 \text{ W/m}$
T_1	$0,4198714890 \text{ Km/W}$
T_3	$0,05419960924 \text{ Km/W}$
$\Delta\theta_{tr0}$	25 K

The calculation is performed for each cable:

	3 (outer cable lagging phase)	1 (central cable)	2 (outer cable leading phase)
λ_1	$\lambda'_{11} = 0,789041528$	$\lambda'_{1m} = 0,1526334560$	$\lambda'_{12} = 0,62818111$
T_4 (Km/W)	0,5910689497	0,6095952562	0,5945872063
I (A)	798,5023145	920,2852119	824,5853552

Predictably, the outer cable carrying the lagging phase has the greatest loss in the metal sheath and establishes the permissible current rating (798,5 A), whose value is applied to the three cables thereafter.

4.10.3.9 Losses in conductor and sheath

The losses in metal components generated by Joule effect ($W_c = R \cdot I^2$ and $W_s = \lambda_1 \cdot W_c$) are calculated with the common value of the AC conductor resistance R (because all conductors are assumed to be at 90°C before launching the first iteration):

R	3,952152638 10^{-5} Ω/m	$W_c = 25,1991602$ W/m
I	798,5023145 A	

	3 (outer cable lagging phase)	1 (central cable)	2 (outer cable leading phase)
λ_1	$\lambda'_{11} = 0,789041528$	$\lambda'_{1m} = 0,1526334560$	$\lambda'_{12} = 0,62818111$
W_s (W/m)	19,88318388	3,846234914	15,82963651

The imbalance of the circulating currents in the metal sheaths generates significantly different values of temperatures at the subsequent steps of the calculations.

4.10.3.10 Temperature on cable components

The oversheath temperature (jacket or external serving) can be calculated from the general expression for the temperature rise above ambient (Section 1.4.1.1 of [3]), increased by the term which takes into account the heating inside the trough ($W_d = 0$ since no armour):

$$\theta_j = \theta_a + n (W_c + W_s + W_d) T_4 + \Delta\theta_{tr}$$

The expressions of the temperature of the metal sheath and conductor are unchanged from Section 4.5.2.5.

θ_a	25°C
n	1 (single-core cable)
W_d	0,3851382172 W/m
W_c	25,1991602 W/m
T_1	0,4198714890 Km/W
T_3	0,05419960924 Km/W

	3 (outer cable lagging phase)	1 (central cable)	2 (outer cable leading phase)
W_s (W/m)	19,88318388	3,846234914	15,82963651
T_4 (Km/W)	0,5910689497	0,6095952562	0,5945872063
θ_j (°C)	82,40248805	73,46878455	80,15226692
θ_s (°C)	84,86680783	75,06390796	82,39688601
θ_c (°C)	95,52807103	85,72517117	93,05814922

The temperature of conductor and metal sheath will be used to update the resistance of their respective component for each cable. The first iteration leads to a high temperature of the conductor of the hottest

cable, 95,5°C much higher than the aimed 90°C. An oscillating process around the final result can be forecasted.

4.10.3.11 Heating of the air inside the trough

The temperature rise of the air in the unfilled trough above the air ambient is given by an empirical formula:

$$\Delta\theta_{tr} = \frac{W_{TOT}}{3p} \quad (\text{Section 4.2.6.2 of [4]})$$

IEC reminds that "*p is the part of the trough perimeter which is effective for heat dissipation (m). Any portion of the perimeter, which is exposed to sunlight, is, therefore, not included in the value of p.*". Since the cover of the trough is flush with the soil surface, only the vertical walls and the bottom of the trough are included in the dissipating perimeter:

$$p = 0,36 + 0,55 + 0,36 = 1,27 \text{ m}$$

The total power dissipated in the trough per metre length (W/m) is the sum of the Joule and dielectric losses for the *N* cables. *N* is the amount of cables laid into the trough, *N* = 3 in the same trough for the sub-case:

$$W_{TOT} = \sum_i^N [W_c + W_d + W_s + W_a]$$

W_d	0,3851382172 W/m
W_c	25,1991602 W/m

	3 (outer cable lagging phase)	1 (central cable)	2 (outer cable leading phase)
W_s (W/m)	19,88318388	3,846234914	15,82963651
$W_{TOT(i)}$ (W/m)	45,46748232	29,43053336	41,41393496

W_{TOT}	116,3119506 W/m	$\Delta\theta_{tr} = 30,52807103 \text{ K}$
p	1,27 m	

The initial value of $\Delta\theta_{tr} = \theta_a = 25 \text{ K}$ is a correct order of magnitude.

It can be noted that the total power dissipated in the trough per metre length W_{TOT} takes into account the respective losses of each cable calculated at their temperature.

If only one cable was considered (calculation based on the hottest cable and extended to the other ones), the result would be on the safe side but very conservative: $\Delta\theta_{tr}$ would be equal to 35,8 K and the trend would be strengthened later when keeping the highest temperatures (therefore the highest electrical resistances) for all the cables.

4.10.4 Second iteration

The following table presents the results of the second iteration, coming from the first step detailed above (especially the temperature of conductor and metal sheath of each cable):

	3 (outer cable lagging phase)	1 (central cable)	2 (outer cable leading phase)
θ_c (°C)	95,52807103	85,72517117	93,05814922
R ($10^{-5} \Omega/\text{m}$)	4,009133316	3,908160981	3,983661877
R_s ($10^{-4} \Omega/\text{m}$)	2, 105460971	2,039520898	2, 088846823
λ_1	$\lambda'_{11} = 0,7684683825$	$\lambda'_{1m} = 0,156714660$	$\lambda'_{12} = 0,6205228830$
$\Delta\theta_d$	0,01058561297	0,05894558962	0,01893361011
K_A	0,2166931927	0,3099963969	0,2327993809
$(\Delta\theta_s)^{1/4}$	2,275237340	2,07177984	2,229740436
T_4 (Km/W)	0,5914883590	0,6495750063	0,6035574273
I (A)	738,441259	841,146787	759,4491060

The calculation carries on with W_c , W_s , W_{TOT} , $\Delta\theta_{tr}$, θ_i , θ_s , and θ_c according to the same process as for the first iteration.

4.10.5 Next iterations

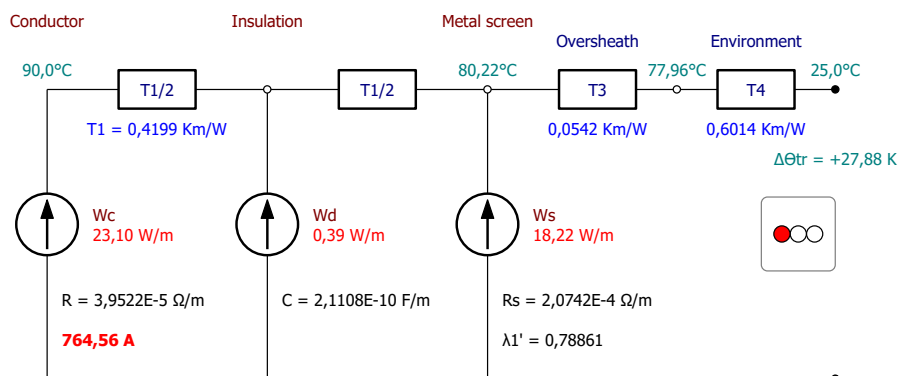
Reminder of the chosen options:

Variable	Topic	IEC rule	CIGRE B1.56 recommendation
W_d	Dielectric losses	Losses neglected depending on the voltage level and insulation material ☐	Losses always taken into account GP 7 ☒
λ_1''	Eddy current losses	Losses neglected when circulating currents are considered, except for Milliken conductors ☒	Losses always taken into account GP 6 ☐
F	Reduction of eddy currents by circulating currents	Used for solid bonded cables with Milliken conductors ☒	Used for all solid bonding arrangements GP 31 ☐

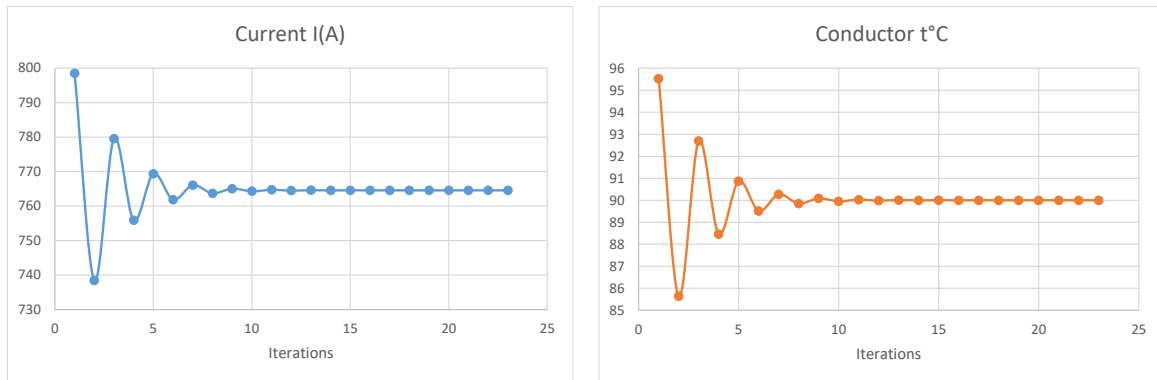
The convergence is reached after 23 iterations. The outer cable carrying the lagging phase is the hottest cable.

Iteration	R ($10^{-5} \Omega/\text{m}$)	R_s ($10^{-4} \Omega/\text{m}$)	λ_1	T_4 (Km/W)	I (A)	W_c (W/m)	W_s (W/m)	$\Delta\theta_{tr}$ (K)	θ_c (°C)	θ_s (°C)
1	3,9522	2,0727	0,78904	0,5911	798,5023	25,1991	19,8832	30,5281	95,5281	84,8668
2	4,0491	2,1055	0,76847	0,5915	738,4413	21,8616	16,8000	26,1601	85,6320	76,3721
↓										
23	3,9522	2,0742	0,78861	0,6014	764,5622	23,1025	18,2189	27,8778	90,0000	80,2191

Ten-digit result after 50 iterations	$I = 764,5620467857 \text{ A}$	$\approx 760 \text{ A}$
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The approach starting with an initial guess $\Delta\theta_{tr0} = \theta_a$ (air ambient temperature above ground) is validated since the final value $\Delta\theta_{tr}$ is not so different from it (27,9°C instead of 25°C). The initial guess $\Delta\theta_{tr0}$ is obviously an important parameter influencing the number of iterations.



Successive relaxation gives the same result with less than 10 iterations.

4.10.6 Discussion

The representativeness and reliability of the calculation are linked to several considerations:

- The covering plate of the trough does not contribute to the dissipating perimeter, and reciprocally no solar radiation is taken into account.
- Z , E and g factors are used for cables in free air (installation on non-continuous brackets, ladder supports or cleats). For lack of further information, the most probable coefficients have been selected from the most similar configuration [4].
- Cable losses for multiple circuits with mutual thermal influence shall be included, but are not described by the IEC 60287 standards. Reduction factors from IEC 60364 (*Electrical Installations for Buildings*) can be used for cables placed on the floor.

Therefore, the current rating may be lower than calculated.

4.10.7 Variant with non-neglected eddy-current losses

This option was adopted by the CIGRE WG B1.56 (Guidance Point 6). The calculation is performed as in Section 4.8.6, including the F factor recommended by the Guidance Point 31.

4.10.7.1 First iteration

The reduction of the eddy currents losses by the circulating currents, when the metal screens are solid-bonded, is taken into account by the F factor, described for cables laid in flat formation in Section 4.8.6.1. For the first iteration, the calculated value is:

R_s	$2,072723957 \cdot 10^{-4} \Omega/\text{m}$	$F = 0,884419154$
X	$5,040331399 \cdot 10^{-5} \Omega/\text{m}$	
X_m	$4,35517218 \cdot 10^{-5} \Omega/\text{m}$	
M	2,206080749	
N	5,775844950	

In the case of flat geometry, the value of λ_1'' is specific to each cable and F will vary with the temperature of the metal sheath R_s . The eddy current loss factor is obtained by multiplying λ_1'' calculated by Section 2.3.6 of [3] by F .

The results are given for the outer cable carrying the lagging phase, which will be the hottest one:

λ_1'	0,789041528	$\lambda_1 = 0,8167187646$
F	0,884419154	
λ_1'' (as per [3] 2.3.6)	0,03129425334	

The increase of the loss factor of the sheaths by the part due to the eddy current losses has an impact on the external thermal resistance, and generates a slight derating:

	3 (outer cable lagging phase)	1 (central cable)	2 (outer cable leading phase)
λ_1	0,8167187646	0,2787470317	0,6674916925
T_4 (Km/W)	0,5905175732	0,6047197253	0,5936741937
I (A)	794,2632583	891,6196640	817,9716227

With the current rating of the hottest cable applied to the three cables, and the common value of the AC conductor resistance R (all conductors being assumed at 90°C before launching the first iteration), the losses in metal components generated by Joule effect are:

R	$3,952152638 \cdot 10^{-5} \Omega/\text{m}$	$W_c = 24,93231788 \text{ W/m}$
I	794,2632583 A	

	3 (outer cable lagging phase)	1 (central cable)	2 (outer cable leading phase)
λ_1	0,8167187646	0,2787470317	0,6674916925
W_s (W/m)	20,36269186	6,94980960	16,64211506

The imbalance of the circulating currents in the metal sheaths generates significantly different values of temperatures at the subsequent steps of the calculations.

θ_a	25°C
n	1 (single-core cable)
W_d	0,3851382172 W/m
W_c	24,93231788 W/m
T_1	0,4198714890 Km/W
T_3	0,05419960924 Km/W

	3 (outer cable lagging phase)	1 (central cable)	2 (outer cable leading phase)
W_s (W/m)	20,36269186	6,94980960	16,64211506
T_4 (Km/W)	0,5905175732	0,6047197253	0,5936741937
θ_j (°C)	83,44657968	75,98430161	81,38196414
θ_s (°C)	85,922426	77,73317480	83,65615650
θ_c (°C)	96,471650	88,28239852	94,20538021

The temperature of conductor and metal sheath will be used to update the resistance of their respective component for each cable.

The total power dissipated in the trough per metre length (W/m) is the sum of the Joule and dielectric losses for the N cables of the circuit laid into the trough:

W_d	0,3851382172 W/m
W_c	24,93231788 W/m

	3 (outer cable lagging phase)	1 (central cable)	2 (outer cable leading phase)
W_s (W/m)	20,36269186	6,94980960	16,64211506
$W_{TOT(i)}$ (W/m)	45,6801480	41,95957116	41,41393496

W_{TOT}	119,90698482 W/m	$\Delta\theta_{tr} = 31,471650$ K
p	1,27 m	

The initial value of $\Delta\theta_{tr} = \theta_a = 25$ K is a correct order of magnitude.

4.10.7.2 Next iterations

The further calculations do not present any difficulty and are performed as for Sections 4.10.4 and 4.10.5.

Reminder of the chosen options:

Variable	Topic	IEC rule	CIGRE B1.56 recommendation
W_d	Dielectric losses	Losses neglected depending on the voltage level and insulation material ☐	Losses always taken into account GP 7 ☒
λ_1''	Eddy current losses	Losses neglected when circulating currents are considered, except for Milliken conductors ☐	Losses always taken into account GP 6 ☒
F	Reduction of eddy currents by circulating currents	Used for solid bonded cables with Milliken conductors ☐	Used for all solid bonding arrangements GP 31 ☒

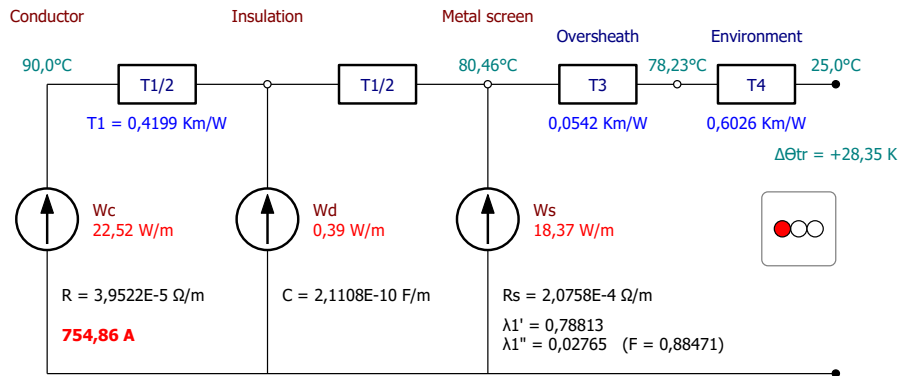
The outer cable carrying the lagging phase is the hottest cable. The results are summarised in the following table:

Iteration	R (10^{-5} Ω /m)	R_s (10^{-4} Ω /m)	λ_1	T_4 (Km/W)	I (A)	W_c (W/m)	W_s (W/m)	$\Delta\theta_{tr}$ (K)	θ_c (°C)	θ_s (°C)
1	3,9522	2,0727	0,81672	0,5905	794,2633	24,9323	20,3627	31,4716	96,4716	85,9224
2	4,0189	2,1126	0,79147	0,5910	724,0437	21,0685	16,6751	26,2515	84,7799	75,8530
↓										
25	3,9522	2,0758	0,81578	0,6026	754,8608	22,5199	18,3713	28,3529	90,0000	80,4637
23 (Section 4.10.54.9.4) (λ_1'' neglected)	3,9522	2,0742	0,78861	0,6014	764,5622	23,1025	18,2189	27,8778	90,0000	80,2191

Ten-digit result after 55 iterations

I = 754,86062387371 A

≈ 750 A



When the convergence is reached, $\lambda_I = \lambda_I' + \lambda_I'' = 0,78813 + 0,02765 = 0,81578$.

In the case of solid bonding at both ends, IEC advises to not include the eddy-current losses, except for Milliken conductors. This example shows that assumption hides a slight derating of 1,29 %.

These results have been reported in the summary table in Section 4.11.

4.11 case #0: Summary table

The following table shows the fourteen cases (for the steady-state operation) presented in this report. Click on the links to jump to the appropriate sections. The current rating value is given in red characters between brackets.

Table 4-1 Summary of the 14 different cases

Installation	Geometry	Assumption	With circulating currents	With Eddy currents
Directly buried cables	Touching cables in trefoil	Touching cables for T_4 (Section 4.2.4.3.2 of [4])	Section 4.5.3.1 (821,78 A)	
				Section 4.6.3 (886,18 A)
			Section 4.6.4 (803,16 A)	
Cables in HDPE ducts	Touching ducts in trefoil	Touching ducts for T_4''' (Section 4.2.4.3.4 of [4])	Section 4.7.5 (682,81 A)	
			Section 4.7.7 (679,84 A)	
Cables in PVC ducts embedded in concrete	Non touching ducts in flat formation	Non equally loaded cables for T_4''' (Section 4.2.3.3.1 of [4])	Section 4.8.4 (634,07 A)	
				Section 4.8.5 (904,55 A)
			Section 4.8.6 (633,04 A)	
Cables in free air	Touching cables in trefoil	Cables directly exposed to solar radiation	Section 4.9.4 (990,94 A)	
				Section 4.9.6 (1046,49 A)
			Section 4.9.7 (974,28 A)	
		Cables protected from direct solar radiation	Section 4.9.5 (1141,37 A)	
Cables in an unfilled trough	Touching cables in flat formation	Heating of the air inside the trough dissipated along a p perimeter	Section 4.10.5 (764,56 A)	
			Section 4.10.7 (754,86 A)	

When only the circulating currents are considered, the calculation deals with a solid bonded cable system (bonding at both ends), neglecting the eddy current losses. This is the standardized approach, described in IEC 60287-1-1 [3].

When only the eddy currents are considered, the calculation deals with a single point bonded cable system (bonding at one end), or cross-bonded system with balanced electrical sections. The standardized approach is also described in [3].

Lastly, the shaded cell stands for the case of solid bonding (with circulating currents) and eddy current losses non neglected (Guidance Point 6), including the F factor recommended by the Guidance Point 31.

These five cases are then more detailed in the table of results below. Results are given when the convergence is reached with $\varepsilon_I = 0,001$ A and $\varepsilon_\theta = 0,0001$. The intermediate results can be compared according to the different cable layings (geometry and environment).

Table 4-2 Overview of the 5 cases

Data common to all subcase studies:

Description of IEC symbols				Units	Case #0
Section in the report					4
Grid	voltage between conductor and screen or sheath	U_o	V		132000
	system frequency	F	Hz		50
	angular frequency of system ($2\pi f$)	Ω			314,159265
Cable	number of conductors in a cable	N			1
	external diameter of cable	D_e^*	m		0,0755
	external diameter of conductor	d_c	mm		30,3
	d.c. resistance of conductor at 20°C	R_o	Ω/m		0,0000283
	temperature coefficient of electrical resistivity at 20°C, per kelvin	α_{20}	1/K		0,00393
	factor used in calculating x_s (skin effect)	k_s			1
	factor used in calculating x_p (proximity effect)	k_p			1
Insulation	loss factor of insulation	$\tan \delta$			0,001
	relative permittivity of insulation	E			2,5
	insulation thickness between conductors and sheath	t_1	mm		15,5
	thermal resistivity of the insulation	ρ_i	Km/W		3,5
	thickness of the sheath	t_s	mm		0,8
	external diameter of metal sheath	D_s	mm		68,5
	maximum operating temperature of cable screen or sheath	θ_{sc}	°C		90
	sheath resistivity at 20°C	ρ_s	$\Omega.m$		2,84E-8
	temperature coefficient of electrical resistivity at 20°C, per kelvin	α_{20}	1/K		0,00403
	a.c. resistance of cable sheath or screen at 20°C	R_{so}	Ω/m		1,6691E-4
	mean diameter of sheath or screen	D	mm		67,7
Jacket	thickness of the serving	t_3	mm		3,5
	thermal resistivity of material	ρ_T	Km/W		3,5

Data specific to subcase studies:

Description of IEC symbols				Units	Case #0-1	Case #0-2	Case #0-3	Case #0-4	Case #0-5
	Section in the report				4.6.4	4.7.7	4.8.6	4.9.7	4.10.7
Jacket	absorption coefficient of solar radiation for the cable surface	σ						0,4	
Installation	axial separation of conductors	s	mm		75,5	140	165	75,5	75,5
Duct	internal diameter of duct	D_d	mm			119,4	118,6		
	external diameter of duct	D_o	mm			140	125		
	constant used in 4.2.7.2	U				1,87	1,87		
	constant used in 4.2.7.2	V				0,312	0,312		
	coefficient used in 4.2.7.2	Y				0,0037	0,0037		
Bank or backfill	side of duct bank ($y > x$) (see 4.2.7.4)	x	mm				325		
	side of duct bank ($y > x$) (see 4.2.7.4)	y	mm				655		
	distance from the soil surface to the centre of a duct bank	L_G	mm				1000		
	number of loaded cables in a duct bank (see 4.2.7.4)	N					3		
	thermal resistivity of earth surrounding a duct bank	ρ_e	Km/W				1		
Buried cables	thermal resistivity of concrete used for a duct bank	ρ_c	Km/W				0,8		
	depth of laying, to cable axis or centre of trefoil	L	mm		1000	1021,95	1021,55		
	thermal resistivity of moist soil	ρ_w	Km/W		1	1	1		
Unfilled trough	the part of the perimeter of the cable trough which is effective for heat dissipation (see 4.2.6.2)	p	m						1,27

Cables in free air	coefficient used in 4.2.1.1	Z					0,96	0,62
	constant used in 4.2.1.1	E					1,25	1,95
	coefficient used in 4.2.1.1 2	g					0,2	0,25
	intensity of solar radiation	H	W/m^2				1000	
Thermal environment	ambient temperature	θ_a	$^{\circ}C$	20	20	20	25	25
Insulation	capacitance per core	C	F/m	2,11077E-10	2,11077E-10	2,11077E-10	2,11077E-10	2,11077E-10
Conductor	argument of a Bessel function used to calculate skin effect	x_s		3,48240	3,48240	3,48240	3,48240	3,48240
	argument of a Bessel function used to calculate proximity effect	x_p		3,48240	3,48240	3,48240	3,48240	3,48240
	skin effect factor (see 2.1)	y_s		0,060124	0,060124	0,060124	0,060124	0,060124
	proximity effect factor (see 2.1)	y_p		0,035100	0,010108	0,007269	0,035100	0,035100
	d.c. resistance of conductor at maximum operating temperature	R'	Ω/m	3,60853E-05	3,60853E-05	3,60853E-05	3,60853E-05	3,60853E-05
	alternating current resistance of conductor at its maximum operating temperature	R	Ω/m	3,95215E-05	3,86197E-05	3,85172E-05	3,95215E-05	3,95215E-05
Metal screen or sheath	a.c. resistance of cable sheath or screen at their maximum operating temperature	R_s	Ω/m	2,06744E-04	2,08903E-04	2,09585E-04	2,03349E-04	2,07584E-04
	reactance of sheath (two-core cables and three-core cables in trefoil)	X	Ω/m	5,04033E-05	8,92026E-05	9,95261E-05	5,04033E-05	5,04033E-05
	mutual reactance between the sheath of one cable and the conductors of the other two when cables are in flat information	X_m	Ω/m			4,35517E-05		4,35517E-05
	loss factor for the middle cable (Three cables in flat formation without transposition, with sheaths bonded at both ends)	λ'_{1m}				0,023809		0,130456
	loss factor for the outer cable with the greater losses	λ'_{11}				0,006203		0,039660
	loss factor for the outer cable with the least losses	λ'_{12}				0,005479		0,027649
	coefficient defined in 2.3.5	F		0,94390	0,84579	0,75020	0,94212	0,88028
	ratio of the losses in one sheath caused by circulating currents in the sheath to the losses in one conductor	λ_1'		0,29348	0,83418	1,70125	0,29782	0,78813
	ratio of the losses in one sheath caused by eddy currents to the losses in one conductor	λ_1''		0,07282	0,01828	0,00548	0,07383	0,02765

	ratio of the total losses in metallic sheaths to the total conductor losses (or losses in one sheath to the losses in one conductor)	λ_1		0,36629	0,85246	1,70673	0,37165	0,81578
	thermal resistance per core between conductor and sheath	T_1	Km/W	0,41987	0,41987	0,41987	0,41987	0,41987
	thermal resistance of external serving	T_3	Km/W	0,08672	0,05420	0,05420	0,05420	0,05420
	mean temperature of medium between a cable and duct or pipe	θ_m	$^{\circ}C$		74,8699	73,8786		
	thermal resistance between cable and duct (or pipe)	T_4'	Km/W		0,34330	0,34506		
	thermal resistance of the duct (or pipe)	T_4''	Km/W		0,08866	0,05019		
	thermal resistance of the medium surrounding the duct (or pipe)	T_4'''	Km/W		1,38002	0,44115		
	thermal resistance of surrounding medium (ratio of cable surface temperature rise above ambient to the losses per unit length)	T_4	Km/W	1,59469	1,81198	1,04505		
	heat dissipation coefficient	h	$W/m^2K^{5/4}$				2,85947	3,13278
	coefficient used in 4.2.1	K_A					0,24437	0,21210
	factor to account for dielectric loss for calculating T_4 for cables in free air	$\Delta\theta_d$	K				0,03704	0,00820
	factor to account for both dielectric loss and direct solar radiation for calculating T_4^* for cables in free air using Figure 10	$\Delta\theta_{ds}$	K				10,8813	
	difference between the surface temperature of a cable in air and ambient temperature	$\Delta\theta_s$	K				2,6093	2,2332
	external thermal resistance in free air, adjusted for solar radiation	T_4^*	Km/W				0,5651	0,6472
Losses	dielectric losses per unit length per phase	W_d	W/m	0,3851	0,3851	0,3851	0,3851	0,3851
	losses in conductor per unit length	W_c	W/m	25,4940	17,8494	15,4352	37,5147	22,5199
	losses dissipated in sheath per unit length	W_s	W/m	9,3383	15,2159	26,3437	13,9423	18,3713
	total power dissipated in the trough per unit length	W_{TOT}	W/m					41,2764
	current in one conductor (r.m.s. value)	I	A	803,1596	679,8410	633,0357	974,2801	754,8608

Temperatures	conductor temperature	$[\theta_c]$	$^{\circ}C$	90,0000	90,0000	90,0000	90,0000	90,0000
	metal screen or sheath temperature	$[\theta_s]$	$^{\circ}C$	79,2150	82,4247	83,4384	74,1678	80,4637
	oversheath temperature	$[\theta_j]$	$^{\circ}C$	76,1609	80,6117	81,1531	71,3580	78,2265
	air gap temperature in ducts or pipes	θ_m	$^{\circ}C$		74,8699	73,8786		
	outer surface temperature of ducts or pipes	$[\theta_{od}]$	$^{\circ}C$		66,1623	64,4880		
	heating by the cables of the same circuit	$[\Delta\theta_{circ}]$	K			17,0898		
	heating by direct solar radiation	$[\Delta\theta_{solarrad}]$	K				17,0645	
	air heating in the unfilled trough	$[\Delta\theta_{tr}]$	K					28,3529

5. Case study 1: Direct buried 132kV cables

This case study consists of two parts, the 132 kV cables in direct buried trefoil formation and in direct buried flat formation. Both can be found below.

5.1 132 kV cables in direct buried trefoil formation

The selected cable type is a single core extruded cable with copper round compacted conductors, XLPE insulation, copper wires screen, aluminium laminated foil and polyethylene sheath with extruded semi-conductive layer. The datasheet is depicted in Figure 28. The current rating will be determined for the following laying configuration: cable installed in trefoil, direct buried.

5.1.1 Cable Detail

The cable details are provided in Figure 28.

Cable type	: XLPE 76/132kV 1x1200 mm ² copper			Cable data sheet
Standard	: IEC 60840			25-apr-2019

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
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Cable construction	
Conductor	: (1) Milliken compacted copper bare wires, bi-directional, water blocked
Conductorscreen	: (2) semi-conducting tape + extruded semi-conducting compound
Insulation	: (3) XLPE
Insulationscreen	: (4) semi-conducting compound
Bedding	: (5) semi-conducting swellable tape
Earthing screen	: (6) copper round wires with counter helix of copper tape
Bedding	: (7) semi-conducting swellable tape
Radial water	: (8) aluminium laminate foil
Outer sheath	: (9) black polyethylene jacket + extruded semi-conductive coating

Main dimensions and weight		Nominal thickness	Approx. Diameter
Conductor	:		43,1 mm
Conductorscreen	:		46,7 mm
Insulation	:	16,1 mm	80,5 mm
Insulationscreen	:		82,9 mm
Swelling tape	:		84,3 mm
Earthing screen	:	57x1,77 mm lay-length of 500 mm	87,8 mm
Swelling tape	:		88,4 mm
Aluminum foil (excluding glue)	:		88,8 mm
Outer sheath	:	3,6 mm	97 mm
Outer semi-conductive coating	:	0,5 mm	98 mm
Cable weight	:	appr. 18,4 kg/m	

Electrical data	
Rated voltages U ₀ /U (U _m)	: 76/132 (145) kV
Conductor	
Maximum DC resistance at 20 °C	: 0,0151 Ω/km
Maximum temperature under normal conditions	: 90 °C
Maximum temperature under short circuit conditions	: 250 °C
Permissible short circuit current during 1 sec (adiabatic 90-250 °C)	: >50 kA
Insulation	
Capacitance	: 0,25 μF/km
Charging current per core at U ₀	: 6,0 A/km
Dielectric stress on conductor screen at U ₀	: 6,2 kV/mm
Dielectric stress under insulation screen at U ₀	: 3,7 kV/mm
Dielectric loss factor at U ₀	: <10 * 10 ⁻⁴
Earthing screen	
Maximum temperature in case of short circuit	: 180 °C
Permissible short circuit current (non-adiabatic 80-180 °C)	1s : 30,2 kA

Figure 28: Cable datasheet 1x1200 mm² CU/XLPE/CWS/PE 76/132 (145) kV

5.1.2 Installation conditions

- Depth of laying of the cable 1 m from earth surface to centre of the trefoil configuration
- Close trefoil configuration
- Soil temperature 20 °C
- Soil thermal resistivity 1,0 K.m/W
- Sheaths solidly bonded
- One circuit thermally independent

5.1.3 Calculation of the conductor AC resistance at operating temperature

Table 5-1: Parameters needed to calculate the conductor AC resistance at operating temperature

Parameter	Symbol	Value	Unit	Reference
Conductor DC resistance @ 20 °C	R_o	0,0151 x 10 ⁻³	Ω/m	Cable data sheet
Conductor diameter	d_c	43,1	mm	Cable datasheet
Conductor material		Copper		Cable datasheet
Type of conductor		Milliken, bare wires, bi-directional		Cable datasheet
Skin effect coefficient	k_s	0,8	-	[3], table 2
Proximity effect coefficient	k_p	0,37	-	[3], table 2
Conductor temperature coefficient at 20 °C per Kelvin	α	3,93 x 10 ⁻³	1/K	[3], table 1
Conductor operating temperature	θ	90	°C	System design
Distance between conductor axes	s	98	mm	System design
System frequency	f	50	Hz	System design

Table 5-2: Sequential steps to calculate the conductor AC resistance at operating temperature

Step	Parameter	Symbol	Unit	Reference
1	Conductor DC resistance at operating temperature	R'	Ω/m	[3], section 2.1.1
2	Argument of a Bessel function – skin effect	x_s	-	[3], section 2.1.2
3	Skin effect factor	y_s	-	[3], section 2.1.2
4	Argument of a Bessel function – proximity effect	x_p	-	[3], section 2.1.3
5	Proximity effect factor	y_p	-	[3], section 2.1.3
6	AC resistance of conductor	R	Ω/m	[3], section 2.1

1. Conductor DC resistance at operating temperature R' :

$$R' = R_o [1 + \alpha_{20}(\theta - 20)] \quad (\Omega/m)$$

R_o	0,0151 x 10 ⁻³ Ω/m	$R' = 1,9254010000 \times 10^{-5} \Omega/m$
α_{20}	3,93 x 10 ⁻³ 1/K	
θ	90 °C	

2. Argument of a Bessel function used to calculate skin effect x_s :

$$x_s^2 = \frac{8\pi f}{R'} 10^{-7} k_s$$

f	50 Hz	$x_s^2 = 5,2213001299$
R'	1,925401 x 10 ⁻⁵ Ω/m	
k_s	0,8	

3. Skin effect factor y_s :

$$\text{for } 0 < x_s \leq 2,8 \quad \rightarrow \quad y_s = \frac{x_s^4}{192 + 0,8x_s^4}$$

Three equations are given for calculating y_s , use of these equations depends on the calculated value of x_s .

x_s^2	5,2213001299	$y_s = 0,1275058631$
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4. Argument of a Bessel function used to calculate proximity effect x_p :

$$x_p^2 = \frac{8\pi f}{R'} 10^{-7} k_p$$

f	50 Hz	$x_p^2 = 2,4148513101$
R'	$1,925401 \times 10^{-5} \Omega/\text{m}$	
k_p	0,37	

5. Proximity effect factor y_p :

$$y_p = \frac{x_p^4}{192 + 0,8x_p^4} \left(\frac{d_c}{s} \right)^2 \left[0,312 \left(\frac{d_c}{s} \right)^2 + \frac{1,18}{\frac{x_p^4}{192 + 0,8x_p^4} + 0,27} \right]$$

x_p^2	2,4148513101	$y_p = 0,0229311323$
d_c	43,1 mm	
s	98 mm	

6. AC resistance of conductor R :

$$R = R'(1 + y_s + y_p) \quad (\Omega/\text{m})$$

R'	$1,925401 \times 10^{-5} \Omega/\text{m}$	$R = 2,2150525415 \times 10^{-5} \Omega/\text{m}$
y_s	0,1275058631	
y_p	0,0229311323	

5.1.4 Dielectric losses

Table 5-3: Parameters needed to calculate the dielectric loss per unit length per phase

Parameter	Symbol	Value	Unit	Reference
External diameter of the insulation	D_i	80,5	mm	Cable data sheet
External diameter of the conductor, incl. screen	d_{sc}	46,7	mm	Cable datasheet
Relative permittivity of the insulation	ϵ	2,5	-	[3], table 3
Loss factor of the insulation	$\tan\delta$	10×10^{-4}	-	[3], table 3, note 1
Voltage to earth	U_o	$132\text{kV} / \sqrt{3}$	V	System design
System frequency	f	50	Hz	System design
Note 1: Always assume unfilled XLPE unless specified otherwise				

Table 5-4: Sequential steps to calculate the dielectric loss per unit length per phase

Step	Parameter	Symbol	Unit	Reference
1	Capacitance	C	F/m	[3], section 2.2
2	Angular frequency of system	ω	rad/s	[3], section 2.2
3	Dielectric loss per unit length in each phase	W_d	W/m	[3], section 2.2, note 1

Note 1: According to [3], table 3, the dielectric losses may be omitted in the calculation.

1. Capacitance C (for circular conductors):

$$C = \frac{\varepsilon}{18 \ln\left(\frac{D_i}{d_{sc}}\right)} 10^{-9} \quad (F/m)$$

ε	2,5	$C = 2,5506991358 \times 10^{-10} F/m$
D_i	80,5 mm	
d_{sc}	46,7 mm	

2. Angular frequency of system:

$$\omega = 2\pi f \quad (rad/s)$$

f	50 HZ	$\omega = 314,159265359 \text{ rad/s}$
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3. Dielectric loss per unit length in each phase:

$$W_d = \omega C U_o^2 \tan\delta \quad (W/m)$$

C	$2,5506991358 \times 10^{-10} F/m$	$W_d = 0,4654100053 W/m$
ω	$314,159265359 \text{ rad/s}$	
U_o	$132kV/\sqrt{3} = 7,6210235533 \times 10^4 V$	
$\tan\delta$	10×10^{-4}	

5.1.5 Loss factor for screen and laminated foil

See flow charts in Section 2.6 for details.

Table 5-5: Parameters needed to calculate the loss factor for screen and laminated foil

Parameter	Symbol	Value	Unit	Reference
AC resistance of screen at 20°C	R_{so_sc}	-	Ω/m	Cable datasheet, note 1
AC resistance of laminated foil at 20°C	R_{so_fl}	-	Ω/m	Cable datasheet, note 2
External Diameter of screen	D_{sc}	87,84	mm	Cable data sheet, note 3
External Diameter over laminated foil	D_{fl}	88,8	mm	Cable data sheet
Diameter of one copper wire of the copper wire screen	d_{cw}	1,77	mm	Cable data sheet
Number of copper wires in the copper wire screen	n_{cws}	57	-	Cable data sheet
Length of lay of copper wire screen	lay	500	mm	Cable data sheet
Thickness of the laminated foil	t_{fl}	0,2	mm	Cable data sheet
Screen resistivity at 20°C	ρ_{sc}	$1,7241 \times 10^{-8}$	Ωm	[3], table 1
Laminated foil resistivity at 20°C	ρ_{fl}	$2,84 \times 10^{-8}$	Ωm	[3], table 1
Copper wire screen temperature coefficient at 20°C per Kelvin	α_{sc}	$3,93 \times 10^{-3}$	1/K	[3], table 1
Laminated foil temperature coefficient at 20°C per Kelvin	α_{fl}	$4,03 \times 10^{-3}$	1/K	[3], table 1
Distance between conductor axes of the electrical section being considered	s	98	mm	System design
System frequency	f	50	Hz	System design
AC resistance of conductor at operating temp.	R	$2,2150525415 \times 10^{-5}$	Ω/m	Calculated
Dielectric loss per unit length in each phase	W_d	0,4654100053	W/m	Calculated
Thermal resistance between conductor and copper wire screen	T_1	0,3789088974	K.m/W	Calculated
Thermal resistance between copper wire screen and laminated foil	T_2	0,0121371405	K.m/W	Calculated
Permissible current rating	I	990,5401450554	A	Calculated
Note 1: If not given, to be calculated based on the cross section of the copper wires screen				
Note 2: If not given, to be calculated based on the cross section of the laminated foil				
Note 3: The indicated external diameter on the datasheet is 87,8 mm but this is a rounded number. Based on the copper wire diameter in the copper wires screen, the actual diameter is $84,3\text{mm} + 2 \times 1,77\text{mm} = 87,84\text{ mm}$ and this value is used in the calculations				

Table 5-6: Sequential steps to calculate the loss factor for screen and laminated foil

Step	Parameter	Symbol	Unit	Reference
1	AC Resistance of screen at 20°C	R_{so_sc}	Ω/m	-
2	AC Resistance of laminated foil at 20°C	R_{so_fl}	Ω/m	-
3	Maximum operating temperature of the cable copper wires screen, to be determined iterative	θ_{sc}	$^{\circ}C$	[3], section 2.3
4	Maximum operating temperature of the cable laminated foil, to be determined iterative	θ_{fl}	$^{\circ}C$	[3], section 2.4
5	Resistance of copper wire screen at operating temperature	R_{s_sc}	Ω/m	[3], section 2.3
6	Resistance of laminated foil at operating temperature	R_{s_fl}	Ω/m	[3], section 2.3
7	Resistance of wire screen and laminated foil at operating temperature	R_s	Ω/m	
8	Reactance per unit length of wire screen and laminated foil	X	Ω/m	[3], section 2.3.1
9	Loss factor circulating losses for wire screen and laminated foil	λ'_{1-com}	-	[3], section 2.3.1
10	Split loss factor in wire screen loss factor and laminated foil loss factor	λ'_{1-cws} λ'_{1-fl}		See note 3
11	Loss factor for eddy current losses	λ''_{1-cws} λ''_{1-fl}	-	[3], section 2.3.6.1 and note 3
12	Effect factor for Milliken conductors	F	-	[3], section
13	Add circulating losses and eddy current losses to find λ_{1-cws} and λ_{1-fl}	λ_{1-cws} λ_{1-fl}		[3], section

Note 3: To make a clear distinguishment between the wire screen losses and the foil losses, the subscript λ_{1-cws} refers to the copper wires screen losses and λ_{1-fl} refers to the aluminium laminated foil losses ratio while λ_{1-com} refers to the combined losses

1. Resistance of screen at 20°C:

Cross section of the copper wires screen, based on the wire diameter and the number of wires:

$$A_{CWS} = n_{cws} \cdot \frac{1}{4} \cdot \pi \cdot d_{cw}^2 \cdot 10^{-6} \quad (m^2)$$

n_{cws}	57	$A_{CWS} = 140,25271265 \times 10^{-6} m^2$
d_{cw}	1,77 mm	

AC Resistance of the screen at 20°C per unit length is:

$$R_{so_sc}^* = \frac{\rho_{sc}}{A_{CWS}} \quad (\Omega/m)$$

ρ_{sc}	$1,7241 \times 10^{-8} \Omega \cdot m$	$R_{so_sc}^* = 1,2292810367 \times 10^{-4} \Omega/m$
A_{CWS}	$140,25271265 \times 10^{-6} m^2$	

Correction of the AC resistance of the screen, taken into account lay-length:

$$R_{so_sc} = R_{so_sc}^* \sqrt{1 + \left(\frac{\pi(D_{sc} - d_{cw})}{lay} \right)^2} \quad (\Omega/m)$$

$R_{so_sc}^*$	$1,2292810367 \times 10^{-4} \Omega/m$	$R_{so_sc} = 1,3975243485 \times 10^{-4} \Omega/m$
D_{sc}	87,84 mm	
d_{cw}	1,77 mm	
lay	500 mm	

2. Resistance of laminated foil at 20°C:

AC Resistance of the laminated foil at 20°C per unit length is:

$$R_{so_fl} = \frac{\rho_{fl}}{t_{fl} \cdot \pi(D_{fl} - t_{fl})} \quad (\Omega/m)$$

ρ_{fl}	$2,84 \times 10^{-8} \Omega \cdot m$	$R_{so_fl} = 5,1015805686 \times 10^{-4} \Omega/m$
t_{fl}	$0,2 \times 10^{-3} m$	
D_{fl}	$88,8 \times 10^{-3} m$	

3. The operating temperature θ_{sc} (°C) of the screen is given by:

$$\theta_{sc} = \theta - (I^2 R + 0,5 W_d) T_1 \quad (^\circ C)$$

θ	90 °C	$\theta_{sc} = 81,6768374747 \text{ } ^\circ C$
I	990,5401450554 A	
R	$2,2150525415 \times 10^{-5} \Omega/m$	
W_d	0,4654100053 W/m	
T_1	0,3789088974	

Note: θ_{sc} is the operating temperature of the cable screen (°C). Since the temperature of the screen is a function of the current I, an iterative method must be used here.

4. The operating temperature θ_{fl} (°C) of the laminated foil is given by:

$$\theta_{fl} = \theta - \{(I^2 R + 0,5 W_d) T_1 + [I^2 R (1 + \lambda_{1-cws}) + W_d] n T_2\} \quad (^\circ C)$$

θ	90 °C	$\theta_{fl} = 81,2520535703 \text{ } ^\circ C$
I	990,5401450554 A	
R	$2,2150525415 \times 10^{-5} \Omega/m$	
W_d	0,4654100053 W/m	
T_1	0,3789088974 K.m/W	
λ_{1-cws}	0,5889473923	
T_2	0,0121371405 K.m/W	
n	1	

Note: θ_{fl} is the operating temperature of the laminated foil (°C). Since the temperature of the laminated foil is a function of the current I, an iterative method must be used here.

5. Copper wires screen AC resistance at operating temperature θ_{sc} :

$$R_{s_sc} = R_{so_sc} [1 + \alpha_{sc} (\theta_{sc} - 20)] \quad (\Omega/m)$$

R_{so_sc}	$1,3975243485 \times 10^{-4} \Omega/m$	$R_{s_sc} = 1,7362702352 \times 10^{-4} \Omega/m$
α_{sc}	$3,93 \times 10^{-3} 1/K$	
θ_{sc}	$81,6768374747 ^\circ C$	

6. Laminated foil AC resistance at operating temperature θ_{fl} :

$$R_{s_fl} = R_{so_fl} [1 + \alpha_{fl} (\theta_{fl} - 20)] \quad (\Omega/m)$$

R_{so_fl}	$5,1015805686 \times 10^{-4} \Omega/m$	$R_{s_fl} = 6,3608841824 \times 10^{-4} \Omega/m$
α_{fl}	$4,03 \times 10^{-3} 1/K$	
θ_{fl}	$81,2520535703 ^\circ C$	

7. Resistance of screen and laminated foil at operating temperature R_s :

$$R_s = \frac{R_{s_sc} \cdot R_{s_fl}}{R_{s_sc} + R_{s_fl}} \quad \left(\frac{\Omega}{m} \right)$$

R_{s_sc}	$1,73642702352 \times 10^{-4} \Omega/m$	$R_s = 1,3639623633 \times 10^{-4} \Omega/m$
R_{s_fl}	$6,3608841824 \times 10^{-4} \Omega/m$	

8. Calculation of the reactance X per unit length of the screen:
Mean diameter of screen and laminated foil:

$$d = \sqrt{\frac{(D_{sc} - d_{cw})^2 + (D_{fl} - t_{fl})^2}{2}} \quad (mm)$$

D_{sc}	87,84 mm	$d = 87,3441609382 \text{ mm}$
d_{cw}	1,77 mm	
D_{fl}	88,8 mm	
t_{fl}	0,2 mm	

$$X = 2\omega 10^{-7} \ln \left(\frac{2s}{d} \right) \quad (\Omega/m)$$

ω	$2\pi f = 314,159265359 \text{ rad/s}$	$X = 5,0784377532 \times 10^{-5} \Omega/m$
s	98 mm	
d	87,3441609382 mm	

9. Loss factor λ'_{1-com} for circulating losses in screen and laminated foil:

$$\lambda'_{1-com} = \left(\frac{R_s}{R} \frac{1}{1 + \left(\frac{R_s}{X} \right)^2} \right)$$

R_s	$1,3639623633 \times 10^{-4} \Omega/m$	$\lambda'_{1-com} = 0,7497067770$
R	$2,2150525415 \times 10^{-5} \Omega/m$	
X	$5,0784377532 \times 10^{-5} \Omega/m$	

10. Split loss factor in screen loss factor and laminated foil loss factor:

$$\lambda'_{1-cws} = \lambda'_{1-com} \left(\frac{R_{s_fl}}{R_{s_sc} + R_{s_fl}} \right)$$

$$\lambda'_{1-fl} = \lambda'_{1-com} \left(\frac{R_{s-sc}}{R_{s-sc} + R_{s-fl}} \right)$$

λ'_{1-com}	0,7497067770	$\lambda'_{1-cws} = 0,5889473923$ $\lambda'_{1-fl} = 0,1607593847$
R_{s-sc}	$1,7362702352 \times 10^{-4} \Omega/m$	
R_{s-fl}	$6,3608841824 \times 10^{-4} \Omega/m$	

11. Loss factor λ'_1 for eddy current losses

As per [3], section 2.3.6.1, Note 3, the eddy-current losses are considered negligible for a wire screen, thus $\lambda'_{1-cws} = 0$, see also Guidance Point 6. The eddy current losses for the laminated foil are determined hereunder:

$$m = \frac{\omega}{R_{s-fl}} 10^{-7}$$

ω	314,159265359 rad/s	$m = 0,0493892447$
R_{s-fl}	$6,3608841824 \times 10^{-4} \Omega/m$	

$$d_{fl} = D_{fl} - t_{fl} \quad (mm)$$

D_{fl}	88,8 mm	$d_{fl} = 88,6 \text{ mm}$
t_{fl}	0,2 mm	

$$\lambda_0 = 3 \left(\frac{m^2}{1 + m^2} \right) \left(\frac{d_{fl}}{2s} \right)^2$$

m	0,0493892447	$\lambda_0 = 1,4917060169 \times 10^{-3}$
d_{fl}	88,6 mm	
s	98 mm	

$$\Delta_1 = (1,14m^{2,45} + 0,33) \left(\frac{d_{fl}}{2s} \right)^{(0,92m+1,66)}$$

$$\Delta_2 = 0$$

m	0,0493892447	$\Delta_1 = 0,0853856815$ $\Delta_2 = 0$
d_{fl}	88,6 mm	
s	98 mm	

$$\beta_1 = \sqrt{\frac{4\pi\omega}{10^7 \rho_{fl} \cdot [1 + \alpha_{fl}(\theta_{fl} - 20)]}}$$

(Formula includes temperature correction of ρ_{fl} as per [3], section 2.3.6.1)

ω	314,159265359 rad/s	$\beta_1 = 105,5879429713$
ρ_{fl}	$2,84 \times 10^{-8} \Omega \cdot m$	
α_{fl}	$4,03 \times 10^{-3}$	
θ_{fl}	81,2520535703 °C	

$$g_s = 1 + \left(\frac{t_{fl}}{D_{fl}} \right)^{1,74} (\beta_1 D_{fl} 10^{-3} - 1,6)$$

D_{fl}	88,8 mm	$g_s = 1,0001924516$
t_{fl}	0,2 mm	
β_1	105,5879429713	

$$\lambda''_{1-fl} = \frac{R_{s-fl}}{R} \left[g_s \lambda_0 (1 + \Delta_1 + \Delta_2) + \frac{(\beta_1 t_{fl})^4}{12 \times 10^{12}} \right]$$

R_{s-fl}	$6,3608841824 \times 10^{-4} \Omega/m$	$\lambda''_{1-fl} = 0,0465038386$
R	$2,2150525415 \times 10^{-5} \Omega/m$	
g_s	1,0001924516	
λ_0	$1,4917060169 \times 10^{-3}$	
Δ_1	0,0853856815	
Δ_2	0	
β_1	105,5879429713	
t_{fl}	0,2 mm	

12. Effect factor for Milliken conductors:

As written in [3], section 2.3.5, eddy current losses can not be ignored for this Case and they need to be included in the sheath losses, including a correction factor F . This factor F is intended to capture the reduction in magnetic field at the metallic sheath as a result of the sheath current. For the case analysed in this chapter, this metallic sheath is composed out of two layers: copper wires screen (no eddy current losses) and aluminium laminated foil (with eddy current losses). The input parameters for calculating Factor F are the resistance and reactance of the metallic sheath. Although it could be argued that factor F should be calculated separately for the copper wires screen and for the aluminium laminated foil, it is recommended to calculate only one correction factor F for both metallic layers.

Calculation of factor 'F':

$$M = N = \frac{R_s}{X}$$

R_s	$1,3639623633 \times 10^{-4} \Omega/m$	$M = 2,6857912405$ $N = 2,6857912405$
X	$5,0784377532 \times 10^{-5} \Omega/m$	

$$F = \frac{4M^2N^2 + (M + N)^2}{4(M^2 + 1)(N^2 + 1)}$$

M	2,6857912405	$F = 0,8782488471$
N	2,6857912405	

13. Add circulating losses and eddy current losses to find λ_1 and λ_2 :

Copper wires screen losses ratio:

$$\lambda_{1-cws} = \lambda'_{1-cws} + F \cdot \lambda''_{1-cws}$$

λ'_{1-cws}	0,5889473923	$\lambda_{1-cws} = 0,5889473923$
λ''_{1-cws}	0	
F	0,8782488471	

Aluminium laminated foil losses ratio:

$$\lambda_{1-fl} = \lambda'_{1-fl} + F \cdot \lambda''_{1-fl}$$

λ'_{1-fl}	0,1607593847	$\lambda_{1-fl} = 0,2016013273$
λ''_{1-fl}	0,0465038386	
F	0,8782488471	

5.1.6 Thermal resistance T_1 between conductor and screen

Note that in the following section, sc denotes semi-conducting material.

Table 5-7: Parameters needed to calculate the thermal resistance between conductor and screen

Parameter	Symbol	Value	Unit	Reference
diameter of conductor	d_c	43,1	mm	Cable datasheet
External diameter of sc conductor screen	d_{sc}	46,7	mm	Cable datasheet
External diameter of insulation	D_i	80,5	mm	Cable data sheet
External diameter of sc insulation screen	D_{si}	82,9	mm	Cable data sheet
External diameter of sc- water blocking tape(s) under the copper wires screen	D_{uwb}	84,3	mm	Cable data sheet
Thermal resistivity of sc screen	ρ_{sc}	2,5	K.m/W	Note 1
Thermal resistivity of insulation	ρ_i	3,5	K.m/W	[4], table 1
Thermal resistivity of water blocking tape	ρ_{wb}	12	K.m/W	Note 2
Note 1: In this case: 2,5 K.m/W				
Note 2: In this case: 12 K.m/W unless more accurate values are available				

Table 5-8: Sequential steps to calculate the thermal resistance between conductor and screen

Step	Parameter	Symbol	Unit	Reference
1	Thermal resistance of sc conductor screen	T_{1-csc}	K.m/W	[4], section 4.1.2.1
2	Thermal resistance of insulation	T_{1-i}	K.m/W	[4], section 4.1.2.1
3	Thermal resistance of sc insulation screen	T_{1-isc}	K.m/W	[4], section 4.1.2.1
4	Thermal resistance of sc water blocking tape under the copper wires screen	T_{1-uwb}	K.m/W	[4], section 4.1.2.1
5	Thermal resistance between conductor and screen	T_1	K.m/W	-
6	Include correction factor	-	-	[4], section 4.2.4.3.2

1. Thermal resistance of sc conductor screen:

$$T_{1-csc} = \frac{\rho_{sc}}{2\pi} \ln\left(\frac{d_{sc}}{d_c}\right) \quad (K.m/W)$$

ρ_{sc}	2,5 K.m/W	$T_{1-csc} = 0,0319189884 \text{ K.m/W}$
d_{sc}	46,7 mm	
d_c	43,1 mm	

2. Thermal resistance of insulation:

$$T_{1-i} = \frac{\rho_i}{2\pi} \ln\left(\frac{D_i}{d_{sc}}\right) \quad (K.m/W)$$

ρ_i	3,5 K.m/W	$T_{1-i} = 0,3033167853 \text{ K.m/W}$
D_i	80,5 mm	
d_{sc}	46,7 mm	

3. Thermal resistance of sc insulation screen:

$$T_{1isc} = \frac{\rho_{sc}}{2\pi} \ln\left(\frac{D_{si}}{D_i}\right) \quad \left(K \cdot \frac{m}{W}\right)$$

ρ_{sc}	2,5 K.m/W	$T_{1isc} = 0,0116890861 \text{ K.m/W}$
D_{si}	82,9 mm	
D_i	80,5 mm	

4. Thermal resistance of sc water blocking tape under the copper wires screen:

$$T_{1uwb} = \frac{\rho_{wb}}{2\pi} \ln\left(\frac{D_{uwb}}{D_i}\right) \quad \left(K \cdot \frac{m}{W}\right)$$

ρ_{wb}	12 K.m/W	$T_{1uwb} = 0,0319840375 \text{ K.m/W}$
D_{uwb}	84,3 mm	
D_{si}	82,9 mm	

5. Thermal resistance between conductor and screen:

$$T_1 = T_{1csc} + T_{1i} + T_{1isc} + T_{1uwb} \quad (K \cdot m/W)$$

T_{1csc}	0,0319189884 K.m/W	$T_1 = 0,3789088974 \text{ K.m/W}$
T_{1i}	0,3033167853 K.m/W	
T_{1isc}	0,0116890861 K.m/W	
T_{1uwb}	0,0319840375 K.m/W	

6. Correction factor:

Although the copper wires screen covering is less than 50%, see [3], section 4.2.4.3.3, the aluminium foil is considered here as a metallic sheath and therefore no correction factor is applied for T_1 .

5.1.7 Thermal resistance T_2 between screen and laminated foil

Table 5-9: Parameters needed to calculate the thermal resistance between screen and laminated foil

Parameter	Symbol	Value	Unit	Reference
External diameter of copper wires screen	D_{sc}	87,84	mm	Cable data sheet, note 1
External diameter of sc- water blocking tape(s) over the copper wires screen	D_{owt}	88,4	mm	Cable data sheet
Thermal resistivity of water blocking tape	ρ_{wb}	12	K.m/W	Note 2
Note 1: The indicated external diameter on the datasheet is 87,8 mm but this is a rounded number. Based on the copper wire diameter in the copper wires screen, the actual diameter is 84,3mm + 2x1,77mm = 87,84 mm and this value is used in the calculations.				
Note 2: In this case: 12 K.m/W, unless more accurate values are available				

Table 5-10: Sequential steps to calculate the thermal resistance between screen and laminated foil

Step	Parameter	Symbol	Unit	Reference
1	Thermal resistance of sc water blocking tape over the copper wires screen	T_{2owbt}	K.m/W	[4], section 4.1.2.1

1. Thermal resistance of sc water blocking tape over the copper wires screen:

$$T_2 = \frac{\rho_{wb}}{2\pi} \ln \left(\frac{D_{owt}}{D_{sc}} \right) \quad (K.m/W)$$

ρ_{wb}	12 K.m/W	$T_2 = 0,0121371405 \text{ K.m/W}$
D_{owt}	88,4 mm	
D_{sc}	87,84 mm	

5.1.8 Thermal resistance T_3 of outer covering

Table 5-11: Parameters needed to calculate the thermal resistance T_3 of the outer covering

Parameter	Symbol	Value	Unit	Reference
External diameter of the aluminium foil	D_{fl}	88,8	mm	Cable data sheet
External diameter of outer covering without semi conductive layer	$D_{e.1}$	97,0	mm	Cable data sheet
External diameter semi conductive layer over outer covering	$D_{e.2}$	98,0	mm	Cable data sheet
Thermal resistivity of outer covering	ρ_{Ti}	3,5 K.m/W	K.m/W	[4], table 1
Thermal resistivity of semi conductive layer over outer covering	ρ_{Tcs}	2,5 K.m/W	K.m/W	Note 1
Note 1: In this case: 2.5 K.m/W				

Table 5-12: Sequential steps to calculate the thermal resistance T_3 of the outer covering

Step	Parameter	Symbol	Unit	Reference
1	Thermal resistance of outer covering	$T_{3.oc}$	K.m/W	[4], section 4.1.4.1
2	Thermal resistance of semi conductive layer over outer covering	$T_{3.sc}$	K.m/W	[4], section 4.1.4.1
3	Thermal resistance outer covering	T_3	K.m/W	
4	Include correction factor	-	-	[4], section 4.2.4.3.2

1. Thermal resistance of outer covering:

$$T_{3.oc} = \frac{\rho_{Ti}}{2\pi} \ln \left(\frac{D_{e.1}}{D_{fl}} \right) \quad (K.m/W)$$

ρ_{Ti}	3,5 K.m/W	$T_{3.oc} = 0,0492003872 \text{ K.m/W}$
$D_{e.1}$	97,0 mm	
D_{fl}	88,8 mm	

2. Thermal resistance of semi conductive layer over outer covering:

$$T_{3.sc} = \frac{\rho_{Tcs}}{2\pi} \ln \left(\frac{D_{e.2}}{D_{e.1}} \right) \quad (K.m/w)$$

ρ_{Tcs}	2,5 K.m/W	$T_{3.sc} = 4,0809317511 \times 10^{-3} \text{ K.m/W}$
$D_{e.1}$	97,0 mm	
$D_{e.2}$	98,0 mm	

3. Thermal resistance outer covering:

$$T_3 = T_{3.oc} + T_{3.sc} \quad (K.m/W)$$

$T_{3.oc}$	0,0492003872 K.m/W	$T_3 = 0,0532813189 \text{ K.m/W}$
$T_{3.sc}$	$4,0809317511 \times 10^{-3} \text{ K.m/W}$	

4. Correction factor:
Multiply T_3 with 1.6:

<i>factor</i>	1,6	$1.6 * T_3 = 0,0852501103 \text{ K.m/W}$
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5.1.9 External thermal resistance T_4

Table 5-13: Parameters needed to calculate the external thermal resistance T_4

Parameter	Symbol	Value	Unit	Reference
External diameter of one cable	D_{e_2}	98	mm	Cable data sheet
Distance from the surface of the ground to the cable axis	L	1000	mm	System design, note 1
Thermal resistivity of the soil	ρ_T	1,0	K.m/W	System design
Note 1: L is measured to the centre of the trefoil group				

Table 5-14: Sequential steps to calculate the external thermal resistance T_4

Step	Parameter	Symbol	Unit	Reference
1	External thermal resistance	T_4	K.m/W	[4], section 4.2.4.3.2

1. External thermal resistance:

$$u = \frac{2L}{D_{e_2}}$$

L	1000 mm	$u = 20,4081632653$
D_{e_2}	98 mm	

$$T_4 = \frac{1,5}{\pi} \rho_T [\ln(2u) - 0,630] \quad (\text{K.m/W})$$

ρ_T	1,0 K.m/W	$T_4 = 1,4701534385 \text{ K.m/W}$
u	20,4081632653	

5.1.10 Permissible current rating

Table 5-15: Parameters needed to calculate the continuous current carrying capacity

Parameter	Symbol	Value	Unit	Reference
Conductor operating temperature	θ	90	$^{\circ}\text{C}$	System design
Ambient temperature	θ_a	20	$^{\circ}\text{C}$	System design
Number of conductors in cable	n	1	-	Cable data sheet
AC resistance of conductor at operating temperature	R	$2,2150525415 \times 10^{-5}$	Ω/m	Calculated
Ratio of the total losses in metallic wire screen	λ_{1-cws}	0,5889473923	-	Calculated
Ratio of the total losses in metallic foil	λ_{1-fl}	0,2016013273	-	Calculated
Dielectric losses	W_d	0,4654100053	W/m	Calculated
Thermal resistance between conductor and metallic screen	T_1	0,3789088974	K.m/W	Calculated
Thermal resistance between metallic screen and metallic foil	T_2	0,0121371405	K.m/W	Calculated
Thermal resistance of external serving	$1,6 \cdot T_3$	0,0852501103	K.m/W	Calculated
Thermal resistance of surrounding medium	T_4	1,4701534385	K.m/W	Calculated

Table 5-16: Sequential steps to calculate the continuous current rating capacity

Step	Parameter	Symbol	Unit	Reference
1	Conductor temperature rise above ambient	$\Delta\theta$	K	[3], section 1.4.1.1
2	Permissible current rating	I	A	[3], section 1.4.1.1

1. Conductor temperature rise above ambient:

$$\Delta\theta = \theta - \theta_a$$

θ	90 °C	$\Delta\theta = 70 \text{ K}$
θ_a	20 °C	

2. Permissible current rating:

$$I = \left[\frac{\Delta\theta - W_d [0,5 \cdot T_1 + n(T_2 + 1,6 \cdot T_3 + T_4)]}{R \cdot T_1 + nR(1 + \lambda_{1-cws})T_2 + nR(1 + \lambda_{1-cws} + \lambda_{1-fl})(1,6 \cdot T_3 + T_4)} \right]^{0,5}$$

$\Delta\theta$	70 K	$I = 990,5401450554 \text{ A}$
W_d	0,4654100053 K.m/W	
T_1	0,3789088974 K.m/W	
T_2	0,0121371405 K.m/W	
$1,6 \cdot T_3$	0,0852501103 K.m/W	
T_4	1,4701534385 K.m/W	
R	$2,2150525415 \times 10^{-5} \Omega/\text{m}$	
n	1	
λ_{1-cws}	0,5889473923	
λ_{1-fl}	0,2016013273	

5.1.11 Calculation of losses

Conductor losses:

$$P_c = RI^2 \text{ (W/m)}$$

R	$2,2150525415 \times 10^{-5} \Omega/\text{m}$	$P_c = 21,7334261256 \text{ W/m}$
I	990,5401450554 A	

Copper wires screen losses:

$$P_{screen} = \lambda_{1-cws} RI^2 \text{ (W/m)}$$

R	$2,2150525415 \times 10^{-5} \Omega/\text{m}$	$P_{screen} = 12,7998446427 \text{ W/m}$
I	990,5401450554 A	
λ_{1-cws}	0,5889473923	

Aluminium laminated foil losses:

$$P_{foil} = \lambda_{1-fl} RI^2 \text{ (W/m)}$$

R	$2,2150525415 \times 10^{-5} \Omega/\text{m}$	$P_{foil} = 4,3814875547 \text{ W/m}$
I	990,5401450554 A	
λ_{1-fl}	0,2016013273	

5.2 132 kV cables in directly buried flat formation

The cable type is the same as for the trefoil configuration, see the datasheet in Figure 28. The current rating will be determined for the following laying configuration: cable installed in flat formation.

The cable temperatures for conductor and metallic sheath will be different for all three phases because of the laying configuration and the metallic sheath bonding: unbalanced cross bonding without transposition of the cables. Therefore, in the calculation example hereunder, different conductor temperatures and metallic screen temperatures are applied for all three phases.

5.2.1 Installation conditions

- Depth of laying of the cable 1 m from earth surface to centre of the cable.
- Flat formation configuration, phase spacing 500 mm.
- Soil temperature 20 °C
- Soil thermal resistivity 1,0 K.m/W
- Sheaths are cross bonded, un-balanced, cables without transposition
- One circuit thermally independent

All other relevant parameters are given in the input tables at each parameter calculations.

5.2.2 Calculation of the conductor AC resistance at operation temperature

Table 5-17: Parameters needed to calculate the conductor AC resistance at operating temperature

Parameter	Symbol	Value	Unit	Reference
Conductor DC resistance @ 20°C	R_o	$0,0151 \times 10^{-3}$	Ω/m	Cable data sheet
Conductor diameter	d_c	43,1	mm	Cable datasheet
Conductor material		Copper		Cable datasheet
Type of conductor		Milliken, bare wires, bi-directional		Cable datasheet
Skin effect coefficient	k_s	0,8		[3], table 2
Proximity effect coefficient	k_p	0,37		[3], table 2
Conductor temperature coefficient at 20°C per Kelvin	α	$3,93 \times 10^{-3}$	$1/K$	[3], table 1
Ambient temperature	θ_a	20	°C	System design
Conductor operating temperature - Middle phase - Lagging phase - Leading phase	θ_m θ_1 θ_2	90	°C	System design calculated calculated
Distance between conductor axes	s	500	mm	System design
System frequency	f	50	Hz	System design
Dielectric loss per unit length in each phase	W_d	0,4654100053	W/m	See 5.1.4
Copper wires screen loss factor for the lagging phase	λ_{1-cws_1}	$1,3832887889 \times 10^{-2}$		Calculated
Laminated foil Loss factor for the lagging phase	λ_{1-fl_1}	$4,6605201558 \times 10^{-3}$		Calculated
Copper wires screen loss factor for the leading phase	λ_{1-cws_2}	$1,0531989911 \times 10^{-2}$		Calculated
Laminated foil Loss factor for the leading phase	λ_{1-fl_2}	$3,7744294697 \times 10^{-3}$		Calculated
Thermal resistance between conductor and metallic screen	T_1	0,3789088974	K.m/W	See 5.1.6
Thermal resistance between metallic screen and metallic laminated foil	T_2	0,0121371405	K.m/W	See 5.1.7
Thermal resistance of external serving	T_3	0,0532813189	K.m/W	See 5.1.8
Thermal resistance of surrounding medium for lagging phase	T_{4_1}	0,9453832120	K.m/W	Calculated
Thermal resistance of surrounding medium for leading phase	T_{4_2}	0,9474917745	K.m/W	Calculated
Continuous current rating	I	1460,4545417538	A	Calculated
Note: Some parameters hereafter have an additional denote 'n' which stands for: m : middle phase 1 : lagging phase 2 : leading phase				

Table 5-18: Sequential steps to calculate the conductor AC resistance at operating temperature

Step	Parameter	Symbol	Unit	Reference
1	Operating temperature of each conductor	θ_n	°C	[3], section 1.4.1.1
2	Conductor DC resistance at operating temperature.	R'_n	Ω/m	[3], section 2.1.1
3	Argument of a Bessel function – skin effect	$x_{s,n}$	-	[3], section 2.1.2
4	Skin effect factor	$y_{s,n}$	-	[3], section 2.1.2
5	Argument of a Bessel function – proximity effect	$x_{p,n}$	-	[3], section 2.1.3
6	Proximity effect factor	$y_{p,n}$	-	[3], section 2.1.3
7	AC resistance of conductors	R_n	Ω/m	[3], section 2.1

1. Operating temperature of each conductor, only to be calculated for the lagging phase and leading phase (n=1 for the lagging phase and n=2 for the leading phase):

$$\theta_n = \left(I^2 R_n + \frac{1}{2} W_d \right) \cdot T_1 + \left(I^2 R_n (1 + \lambda_{1-cws_n}) + W_d \right) \cdot T_2 + \left(I^2 R_n (1 + \lambda_{1-cws_n} + \lambda_{1-fl_n}) + W_d \right) \cdot (T_3 + T_{4,n}) + \theta_a \quad (^\circ\text{C})$$

I	1460,4545417538 A	$\theta_1 = 85,0252936286 \text{ }^\circ\text{C}$ $\theta_2 = 84,9130417812 \text{ }^\circ\text{C}$
R_1	$2,1460977940 \times 10^{-5} \text{ } \Omega/m$	
R_2	$2,1455011286 \times 10^{-5} \text{ } \Omega/m$	
W_d	0,4654100053 W/m	
λ_{1-cws_1}	$1,3832887889 \times 10^{-2}$	
λ_{1-fl_1}	$4,6605201558 \times 10^{-3}$	
λ_{1-cws_2}	$1,0531989911 \times 10^{-2}$	
λ_{1-fl_2}	$3,7744294697 \times 10^{-3}$	
T_1	0,3789088974 K.m/W	
T_2	0,0121371405 K.m/W	
T_3	0,0532813189 K.m/W	
$T_{4,1}$	0,9453832120 K.m/W	
$T_{4,2}$	0,9474917745 K.m/W	

Since the conductor temperature θ_1 of the lagging and θ_2 of the leading phase is a function of the current I , an iterative method must be used here.

2. Conductor DC resistance at operating temperature R' for each phase:

$$R'_n = R_o [1 + \alpha_{20}(\theta_n - 20)] \text{ } (\Omega/m)$$

R_o	$0,0151 \times 10^{-3} \text{ } \Omega/m$	$R'_m = 1,9254010000 \times 10^{-5} \text{ } \Omega/m$ $R'_1 = 1,8958796000 \times 10^{-5} \text{ } \Omega/m$ $R'_2 = 1,8952134638 \times 10^{-5} \text{ } \Omega/m$
α_{20}	$3,93 \times 10^{-3} \text{ } 1/K$	
θ_m	90 °C	
θ_1	85,0252936286 °C	
θ_2	84,9130417812 °C	

3. Argument of a Bessel function used to calculate skin effect x_s :

$$x_{s,n}^2 = \frac{8\pi f}{R'_n} 10^{-7} k_s$$

f	50 Hz	$x_{s,m}^2 = 5,2213001299$ $x_{s,1}^2 = 5,3026028085$ $x_{s,2}^2 = 5,3044665856$
R'_m	$1,9254010000 \times 10^{-5} \Omega/m$	
R'_1	$1,8958796000 \times 10^{-5} \Omega/m$	
R'_2	$1,8952134638 \times 10^{-5} \Omega/m$	
k_s	0,8	

4. Skin effect factor y_s :

$$\text{for } 0 < x_{sn} \leq 2,8 \quad \rightarrow \quad y_{sn} = \frac{x_{sn}^4}{192 + 0,8x_{sn}^4}$$

Three equations are given for calculating y_s , use of these equations depends on the calculated value of x_s .

$x_{s,m}^2$	5,2213001299	$y_{s,m} = 0,1275058631$ $y_{s,1} = 0,1310879858$ $y_{s,2} = 0,1311704808$
$x_{s,1}^2$	5,3026028085	
$x_{s,2}^2$	5,3044665856	

5. Argument of a Bessel function used to calculate proximity effect x_p :

$$x_{p,n}^2 = \frac{8\pi f}{R'_n} 10^{-7} k_p$$

f	50 Hz	$x_{p,m}^2 = 2,4148513101$ $x_{p,1}^2 = 2,4524537989$ $x_{p,2}^2 = 2,4533157958$
R'_m	$1,9254010000 \times 10^{-5} \Omega/m$	
R'_1	$1,8958796000 \times 10^{-5} \Omega/m$	
R'_2	$1,8952134638 \times 10^{-5} \Omega/m$	
k_p	0,37	

6. Proximity effect factor $y_{p,n}$:

$$y_{p,n} = \frac{x_{p,n}^4}{192 + 0,8x_{p,n}^4} \left(\frac{d_c}{s} \right)^2 \left[0,312 \left(\frac{d_c}{s} \right)^2 + \frac{1,18}{\frac{x_{p,n}^4}{192 + 0,8x_{p,n}^4} + 0,27} \right]$$

$x_{p,m}^2$	2,4148513101	$y_{p,m} = 8,6813704842 \times 10^{-4}$ $y_{p,1} = 8,9201653011 \times 10^{-4}$ $y_{p,2} = 8,9256614522 \times 10^{-4}$
$x_{p,1}^2$	2,4524537989	
$x_{p,2}^2$	2,4533157958	
d_c	43,1 mm	
s	500 mm	

7. AC resistance of conductor R :

$$R_n = R'_n (1 + y_{s,n} + y_{p,n}) \quad (\Omega/m)$$

R'_m	$1,9254010000 \times 10^{-5} \Omega/\text{m}$	$R_m = 2,1725724283 \times 10^{-5} \Omega/\text{m}$ $R_1 = 2,1460977940 \times 10^{-5} \Omega/\text{m}$ $R_2 = 2,1455011286 \times 10^{-5} \Omega/\text{m}$
R'_1	$1,8958796000 \times 10^{-5} \Omega/\text{m}$	
R'_2	$1,8952134638 \times 10^{-5} \Omega/\text{m}$	
y_{s_m}	0,1275058631	
y_{s_1}	0,1310879858	
y_{s_2}	0,1311704808	
y_{p_m}	$8,6813704842 \times 10^{-4}$	
y_{p_1}	$8,9201653011 \times 10^{-4}$	
y_{p_2}	$8,9256614522 \times 10^{-4}$	

5.2.3 Dielectric losses

The dielectric losses are calculated in the same way as for the trefoil configuration, see section 5.1.4.
Dielectric loss per unit length in each phase:

$$W_d = \omega C U_o^2 \tan \delta \quad (W/m)$$

C	$2,5506991358 \times 10^{-10} \text{ F/m}$	$W_d = 0,4654100053 \text{ W/m}$
ω	314,159265359 rad/s	
U_o	$132\text{kV}/\sqrt{3} =$ $7,6210235533 \times 10^4 \text{ V}$	
$\tan \delta$	1×10^{-3}	

5.2.4 Loss factor for screen and laminated foil

See flow charts in Section 2.6 for details.

Table 5-19: Parameters needed to calculate the loss factor for screen and laminated foil

Parameter	Symbol	Value	Unit	Reference
AC resistance of screen at 20°C	R_{so_sc}	$1,3975243485 \times 10^{-4}$	Ω/m	See 0
AC resistance of laminated foil at 20°C	R_{so_fl}	$5,1015805686 \times 10^{-4}$	Ω/m	See 0
Correction factor for cross bonding unbalance	$\frac{p}{q}$	$\frac{1}{1,2}$		[3], section 2.3.6.2
Screen resistivity at 20°C	ρ_{sc}	$1,7241 \times 10^{-8}$	Ωm	[3], table 1
Laminated foil resistivity at 20°C	ρ_{fl}	$2,84 \times 10^{-8}$	Ωm	[3], table 1
screen temperature coefficient at 20°C per Kelvin	α_{sc}	$3,93 \times 10^{-3}$	$1/K$	[3], table 1
Laminated foil temperature coefficient at 20°C per Kelvin	α_{fl}	$4,03 \times 10^{-3}$	$1/K$	[3], table 1
Mean diameter of screen and laminated foil	d	87,3441609382	mm	See 0
Distance between conductor axes of the electrical section being considered	s	500	mm	System design
System frequency	f	50	Hz	System design
AC resistance of conductor at operating temp.	R_m R_1 R_2	$2,1725724283 \times 10^{-5}$ $2,1460977940 \times 10^{-5}$ $2,1455011286 \times 10^{-5}$	Ω/m	Calculated
Conductor operating temperature - Middle phase - Lagging phase - Leading phase	θ_m θ_1 θ_2	90 85,0252936286 84,9130417812	$^{\circ}C$	System design Calculated Calculated
Dielectric loss per unit length in each phase	W_d	0,4654100053	W/m	See 5.1.4
Copper wires screen loss factor for the lagging phase	λ_{1-cws_m}	$9,7794239160 \times 10^{-3}$		Calculated
Copper wires screen loss factor for the lagging phase	λ_{1-cws_1}	$1,3832887889 \times 10^{-2}$		Calculated
Copper wires screen loss factor for the leading phase	λ_{1-cws_2}	$1,0531989911 \times 10^{-2}$		Calculated
Thermal resistance between conductor and metallic screen	T_1	0,3789088974	$K.m/W$	Calculated
Thermal resistance between metallic screen and metallic laminated foil	T_2	0,0121371405	$K.m/W$	Calculated
Continuous current rating	I	1460,4545417538	A	Calculated
Number of current carrying conductors in one cable	n	1		See cable datasheet
Note 1: If not given, to be calculated based on the cross section of the screen Note 2: If not given, to be calculated based on the cross section of the laminated foil Note 3: Some parameters hereafter have an additional denote 'n' which refers to: m : middle phase 1 : lagging phase 2 : leading phase				

Table 5-20: Sequential steps to calculate the loss factor for screen and laminated foil

Step	Parameter	Symbol	Unit	Reference
1	Maximum operating temperature of the cable copper wires screen, to be determined iterative for all three phases	θ_{sc_n}	$^{\circ}\text{C}$	[3], section 2.3
2	Maximum operating temperature of the cable laminated foil, to be determined iterative for all three phases	θ_{fl_n}	$^{\circ}\text{C}$	[3], section 2.4
3	Resistance of screen at operating temperature	$R_{s_sc_n}$	Ω/m	[3], section 2.3
4	Resistance of laminated foil at operating temperature	$R_{s_fl_n}$	Ω/m	[3], section 2.3
5	Parallel Resistance of screen and laminated foil at operating temperature	R_{s_n}	Ω/m	-
6	Reactance per unit length of screen and laminated foil	X_{m}	Ω/m	[3], section 2.3.3
7	Correction factor for XB unbalance	-	-	[3], section 2.3.6.2
8	Loss factor circulating losses for screen and laminated foil for the middle phase, lagging phase and leading phase	λ'_{1-com_n}	-	[3], section 2.3.3, see note 1
9	Split loss factor in screen loss factor and laminated foil loss factor for middle phase, lagging phase and leading phase	λ'_{1-cws_n} λ'_{1-fl_n}		See note 1
10	Loss factor for eddy current losses for copper wires screen and laminated foil	λ''_{1-cws_n} λ''_{1-fl_n}	-	[3], section 2.3.6.1 and note 1
11	Add circulating losses and eddy current losses to find λ_1 and λ_2 for all three phases	λ_{1-cws_n} λ_{1-fl_n}		[3], section 2.3.5, see note 1
Note 1: λ_{1-cws} refers to the copper wires screen losses ratio and λ_{1-fl} refers to the aluminium foil losses ratio				

1. The operating temperatures θ_{sc_n} ($^{\circ}\text{C}$) of the screens are given by:

$$\theta_{sc_n} = \theta_n - (I^2 R_n + 0,5 W_d) T_1 \quad (^{\circ}\text{C})$$

θ_m	90 $^{\circ}\text{C}$	$\theta_{sc_m} = 72,3534172818 \text{ }^{\circ}\text{C}$ $\theta_{sc_1} = 67,5926749858 \text{ }^{\circ}\text{C}$ $\theta_{sc_2} = 67,4852452998 \text{ }^{\circ}\text{C}$
θ_1	85,0252936286 $^{\circ}\text{C}$	
θ_2	84,9130417812 $^{\circ}\text{C}$	
I	1460,4545417538 A	
R_m	$2,1725724283 \times 10^{-5} \Omega/\text{m}$	
R_1	$2,1460977940 \times 10^{-5} \Omega/\text{m}$	
R_2	$2,1455011286 \times 10^{-5} \Omega/\text{m}$	
W_d	0,4654100053 W/m	
T_1	0,3789088974	

Note: θ_{sc} is the operating temperature of the cable screen ($^{\circ}\text{C}$). Since the temperature of the screen is a function of the current I , an iterative method must be used here.

2. The operating temperatures θ_{fl_n} ($^{\circ}\text{C}$) of the laminated foil for the three phases are given by:

$$\theta_{fl_n} = \theta_n - \{ (I^2 R_n + 0,5 W_d) T_1 + [I^2 R_n (1 + \lambda_{1-cws_n}) + W_d] n T_2 \} \quad (^{\circ}\text{C})$$

θ_m	90 °C	$\theta_{fl_m} = 71,7798405812 \text{ °C}$ $\theta_{fl_1} = 67,0237669686 \text{ °C}$ $\theta_{fl_2} = 66,9183272653 \text{ °C}$
θ_1	85,0252936286 °C	
θ_2	84,9130417812 °C	
I	1460,4545417538 A	
R_m	$2,1725724283 \times 10^{-5} \text{ } \Omega/\text{m}$	
R_1	$2,1460977940 \times 10^{-5} \text{ } \Omega/\text{m}$	
R_2	$2,1455011286 \times 10^{-5} \text{ } \Omega/\text{m}$	
W_d	0,4654100053 W/m	
T_1	0,3789088974 K.m/W	
λ_{1-cws_m}	$9,7794239160 \times 10^{-3}$	
λ_{1-cws_1}	$1,3832887889 \times 10^{-2}$	
λ_{1-cws_2}	$1,0531989911 \times 10^{-2}$	
n	1	
T_2	0,0121371405 K.m/W	

Note: θ_{fl_n} is the operating temperature of the laminated foil (°C). Since the temperature of the laminated foil is a function of the current I, an iterative method must be used.

3. Copper wires screen AC resistance $R_{s_sc_n}$ at operating temperature θ_{sc_n} :

$$R_{s_sc_n} = R'_{so_sc} [1 + \alpha_{sc}(\theta_{sc_n} - 20)] \quad (\Omega/\text{m})$$

R'_{so_sc}	$1,3975243485 \times 10^{-4} \text{ } \Omega/\text{m}$	$R_{s_sc_m} = 1,6850634878 \times 10^{-4} \text{ } \Omega/\text{m}$ $R_{s_sc_1} = 1,6589162024 \times 10^{-4} \text{ } \Omega/\text{m}$ $R_{s_sc_2} = 1,6583261695 \times 10^{-4} \text{ } \Omega/\text{m}$
α_{sc}	$3,93 \times 10^{-3} \text{ 1/K}$	
θ_{sc_m}	72,3534172818 °C	
θ_{sc_1}	67,5926749858 °C	
θ_{sc_2}	67,4852452998 °C	

4. Laminated foil AC resistance $R_{s_fl_n}$ at operating temperature θ_{fl_n} :

$$R_{s_fl_n} = R_{so} [1 + \alpha_{fl}(\theta_{fl_n} - 20)] \quad \left(\frac{\Omega}{\text{m}}\right)$$

R_{so}	$5,1015805686 \times 10^{-4} \text{ } \Omega/\text{m}$	$R_{s_fl_m} = 6,1661414537 \times 10^{-4} \text{ } \Omega/\text{m}$ $R_{s_fl_1} = 6,0683595780 \times 10^{-4} \text{ } \Omega/\text{m}$ $R_{s_fl_2} = 6,0661918042 \times 10^{-4} \text{ } \Omega/\text{m}$
α_{fl}	$4,03 \times 10^{-3} \text{ 1/K}$	
θ_{fl_m}	71,7798405812 °C	
θ_{fl_1}	67,0237669686 °C	
θ_{fl_2}	66,9183272653 °C	

5. Parallel resistance R_{s_n} of screen and laminated foil at operating temperature:

$$R_{s_n} = \frac{R_{s_sc_n} \cdot R_{s_fl_n}}{R_{s_sc_n} + R_{s_fl_n}} \quad (\Omega/\text{m})$$

$R_{s\ sc\ m}$	$1,6850634878 \times 10^{-4} \Omega/m$	$R_{s\ m} = 1,3234070314 \times 10^{-4} \Omega/m$
$R_{s\ fl\ m}$	$6,1661414537 \times 10^{-4} \Omega/m$	
$R_{s\ sc\ 1}$	$1,6589162024 \times 10^{-4} \Omega/m$	$R_{s\ 1} = 1,3027747827 \times 10^{-4} \Omega/m$
$R_{s\ fl\ 1}$	$6,0683595780 \times 10^{-4} \Omega/m$	
$R_{s\ sc\ 2}$	$1,6583261695 \times 10^{-4} \Omega/m$	$R_{s\ 2} = 1,3023109859 \times 10^{-4} \Omega/m$
$R_{s\ fl\ 2}$	$6,0661918042 \times 10^{-4} \Omega/m$	

6. Calculation of the reactance X and X_m per unit length of the screen:

$$X = 2\omega 10^{-7} \ln\left(\frac{2s}{d}\right) (\Omega/m)$$

$$X_m = 2\omega 10^{-7} \ln(2) (\Omega/m)$$

ω	$2\pi f = 314,159265359 \text{ rad/s}$	$X = 1,5317771751 \times 10^{-4} \Omega/m$ $X_m = 4,3551721806 \times 10^{-5} \Omega/m$
s	500 mm	
d	87,3441609382 mm	

7. Correction factor for cross bonding unbalance:

$$ms_u = \frac{p^2 + q^2 + 1 - p - p \cdot q - q}{(p + q + 1)^2}$$

p	1	$ms_u = 3,9062500000 \times 10^{-3}$
q	1,2	

8. Loss factor circulating losses for screen and laminated foil for the middle phase, lagging phase and leading phase:

$$P = X + X_m$$

$$Q = X - \frac{X_m}{3}$$

X	$1,5317771751 \times 10^{-4} \Omega/m$	$P = 1,9672943932 \times 10^{-4}$ $Q = 1,3866047691 \times 10^{-4}$
X_{ms}	$4,3551721806 \times 10^{-5} \Omega/m$	

$$\lambda'_{1-com_m} = \frac{R_{s_m}}{R_m} \cdot \frac{Q^2}{R_{s_m}^2 + Q^2} \cdot ms_u$$

$$\lambda'_{1-com_1} = \frac{R_{s_1}}{R_1} \cdot \left[\frac{0,75P^2}{R_{s_1}^2 + P^2} + \frac{0,25Q^2}{R_{s_1}^2 + Q^2} + \frac{2R_{s_1}PQX_m}{\sqrt{3}(R_{s_1}^2 + P^2)(R_{s_1}^2 + Q^2)} \right] \cdot ms_u$$

$$\lambda'_{1-com_2} = \frac{R_{s_2}}{R_2} \cdot \left[\frac{0,75P^2}{R_{s_2}^2 + P^2} + \frac{0,25Q^2}{R_{s_2}^2 + Q^2} - \frac{2R_{s_2}PQX_m}{\sqrt{3}(R_{s_2}^2 + P^2)(R_{s_2}^2 + Q^2)} \right] \cdot ms_u$$

P	$1,9672943932 \times 10^{-4}$	$\lambda'_{1-com_m} = 1,2451913721 \times 10^{-2}$ $\lambda'_{1-com_1} = 1,7614404384 \times 10^{-2}$ $\lambda'_{1-com_2} = 1,3411139640 \times 10^{-2}$
Q	$1,3866047691 \times 10^{-4}$	
X_m	$4,3551721806 \times 10^{-5} \Omega/m$	
R_{s_m}	$1,3234070314 \times 10^{-4} \Omega/m$	
R_{s_1}	$1,3027747827 \times 10^{-4} \Omega/m$	
R_{s_2}	$1,3023109859 \times 10^{-4} \Omega/m$	
R_m	$2,1725724283 \times 10^{-5} \Omega/m$	
R_1	$2,1460977940 \times 10^{-5} \Omega/m$	
R_2	$2,1455011286 \times 10^{-5} \Omega/m$	
ms_u	$3,9062500000 \times 10^{-3}$	

9. Split loss factor in screen loss factor and laminated foil loss factor for middle phase, lagging phase and leading phase:

Screen losses of the three phases:

$$\lambda'_{1-cws_n} = \lambda'_{1-com_n} \left(\frac{R_{s_fl_n}}{R_{s_sc_n} + R_{s_fl_n}} \right)$$

Laminated foil losses of the three phase:

$$\lambda'_{1-fl_n} = \lambda'_{1-com_n} \left(\frac{R_{s_sc_n}}{R_{s_sc_n} + R_{s_fl_n}} \right)$$

λ'_{com_m}	$1,2451913721 \times 10^{-2}$	$\lambda'_{1-cws_m} = 9,7794239160 \times 10^{-3}$ $\lambda'_{1-cws_1} = 1,3832887889 \times 10^{-2}$ $\lambda'_{1-cws_2} = 1,0531989911 \times 10^{-2}$ $\lambda'_{1-fl_m} = 2,6724898052 \times 10^{-3}$ $\lambda'_{1-fl_1} = 3,7815164955 \times 10^{-3}$ $\lambda'_{1-fl_2} = 2,8791497286 \times 10^{-3}$
$R_{s_sc_m}$	$1,6850634878 \times 10^{-4} \Omega/m$	
$R_{s_fl_m}$	$6,1661414537 \times 10^{-4} \Omega/m$	
λ'_{1-com_1}	$1,7614404384 \times 10^{-2}$	
$R_{s_sc_1}$	$1,6589162024 \times 10^{-4} \Omega/m$	
$R_{s_fl_1}$	$6,0683595780 \times 10^{-4} \Omega/m$	
λ'_{1-com_2}	$1,3411139640 \times 10^{-2}$	
$R_{s_sc_2}$	$1,6583261695 \times 10^{-4} \Omega/m$	
$R_{s_fl_2}$	$6,0661918042 \times 10^{-4} \Omega/m$	

10. Loss factor for eddy current losses for copper wires screen and laminated foil only:
Eddy current losses in copper wires screen is negligible ([3], section 2.3.6.1, note 3), thus:

$$\lambda''_{1-cws_n} = 0$$

Eddy current losses in laminated foil, middle phase:

$$m_n = \frac{\omega}{R_{s_fl_n}} 10^{-7}$$

ω	314,159265359	$m_m = 5,0949085051 \times 10^{-2}$ $m_1 = 5,1770047790 \times 10^{-2}$ $m_2 = 5,1788547989 \times 10^{-2}$
$R_{s_fl_m}$	$6,1661414537 \times 10^{-4} \Omega/m$	
$R_{s_fl_1}$	$6,0683595780 \times 10^{-4} \Omega/m$	
$R_{s_fl_2}$	$6,0661918042 \times 10^{-4} \Omega/m$	

Mean diameter of laminated foil:

$$d_{fl} = D_{fl} - t_{fl} \quad (\text{mm})$$

D_{fl}	88,8 mm	$d_{fl} = 88,6 \text{ mm}$
t_{fl}	0,2 mm	

$$\lambda_{0,m} = 6 \left(\frac{m_m^2}{1 + m_m^2} \right) \left(\frac{d_{fl}}{2s} \right)^2$$

$$\lambda_{0,1} = 1,5 \left(\frac{m_1^2}{1 + m_1^2} \right) \left(\frac{d_{fl}}{2s} \right)^2$$

$$\lambda_{0,2} = 1,5 \left(\frac{m_2^2}{1 + m_2^2} \right) \left(\frac{d_{fl}}{2s} \right)^2$$

m_m	$5,0949085051 \times 10^{-2}$	$\lambda_{0,m} = 1,2194544638 \times 10^{-4}$ $\lambda_{0,1} = 3,1474107407 \times 10^{-5}$ $\lambda_{0,2} = 3,1496545999 \times 10^{-5}$
m_1	$5,1770047790 \times 10^{-2}$	
m_2	$5,1788547989 \times 10^{-2}$	
d_{fl}	88,6 mm	
s	500 mm	

$$\Delta_{1,m} = (0,86 \cdot m_m^{3,08}) \left(\frac{d_{fl}}{2s} \right)^{(1,4m_m+0,7)}$$

$$\Delta_{1,1} = - \left[\frac{0,74 \cdot (m_1 + 2) \cdot m_1^{0,5}}{2 + (m_1 - 0,3)^2} \cdot \left(\frac{d_{fl}}{2 \cdot s} \right)^{(m_1+1)} \right]$$

$$\Delta_{1,2} = 4,7 \cdot m_2^{0,7} \cdot \left(\frac{d_{fl}}{2 \cdot s} \right)^{(0,16 \cdot m_2 + 2)}$$

m_m	$5,0949085051 \times 10^{-2}$	$\Delta_{1,m} = 1,3823051666 \times 10^{-5}$ $\Delta_{1,1} = -1,3095852751 \times 10^{-2}$ $\Delta_{1,2} = 4,5520515892 \times 10^{-3}$
m_1	$5,1770047790 \times 10^{-2}$	
m_2	$5,1788547989 \times 10^{-2}$	
d_{fl}	88,6 mm	
s	500 mm	

$$\Delta_{2,m} = 0$$

$$\Delta_{2,1} = 0,92 \cdot m_1^{3,7} \cdot \left(\frac{d_{fl}}{2 \cdot s} \right)^{(m_1+2)}$$

$$\Delta_{2,2} = 21 \cdot m_2^{3,3} \cdot \left(\frac{d_{fl}}{2 \cdot s} \right)^{(1,47 \cdot m_2 + 5,06)}$$

m_m	$5,0949085051 \times 10^{-2}$	$\Delta_{2,m} = 0$ $\Delta_{2,1} = 1,1123835513 \times 10^{-7}$ $\Delta_{2,2} = 4,7105786535 \times 10^{-9}$
m_1	$5,1770047790 \times 10^{-2}$	
m_2	$5,1788547989 \times 10^{-2}$	
d_{fl}	88,6 mm	
s	500 mm	

$$\beta_{1,n} = \sqrt{\frac{4\pi\omega}{10^7 \rho_{fl} \cdot [1 + \alpha_{fl}(\theta_{fl,n} - 20)]}}$$

ω	314,159265359	$\beta_{1,m} = 107,2423523302$ $\beta_{1,1} = 108,1029187211$ $\beta_{1,2} = 108,1222324648$
ρ_{fl}	$2,84 \times 10^{-8} \Omega \cdot m$	
α_{fl}	$4,03 \times 10^{-3}$	
$\theta_{fl,m}$	71,7798405812 °C	
$\theta_{fl,1}$	67,0237669686 °C	
$\theta_{fl,2}$	66,9183272653 °C	

$$g_{s,n} = 1 + \left(\frac{t_{fl}}{D_{fl}} \right)^{1,74} (\beta_{1,n} D_{fl} 10^{-3} - 1,6)$$

D_{fl}	88,8 mm	$g_{s,m} = 1,0001960875$ $g_{s,1} = 1,0001979788$ $g_{s,2} = 1,0001980212$
t_{fl}	0,2 mm	
$\beta_{1,m}$	107,2423523302	
$\beta_{1,1}$	108,1029187211	
$\beta_{1,2}$	108,1222324648	

Loss factor $\lambda''_{1-al,n}$ for eddy current losses, laminated foil only:

$$\lambda''_{1-fl,n} = \frac{R_{s-fl,n}}{R_n} \left[g_{s,n} \lambda_{0,n} (1 + \Delta_{1,n} + \Delta_{2,n}) + \frac{(\beta_{1,n} t_{fl})^4}{12 \times 10^{12}} \right]$$

$R_{s_fl_m}$	$6,1661414537 \times 10^{-4} \Omega/m$	$\lambda''_{1-fl_m} = 3,4622526273 \times 10^{-3}$
R_m	$2,1725724283 \times 10^{-5} \Omega/m$	
g_{s_m}	1,0001960875	
λ_{0_m}	$1,2194544638 \times 10^{-4}$	
Δ_{1_m}	$1,3823051666 \times 10^{-5}$	
Δ_{2_m}	0	
β_{1_m}	107,2423523302	
$R_{s_fl_1}$	$6,0683595780 \times 10^{-4} \Omega/m$	$\lambda''_{1-fl_1} = 8,7900366024 \times 10^{-4}$
R_1	$2,1460977940 \times 10^{-5} \Omega/m$	
g_{s_1}	1,0001979788	
λ_{0_1}	$3,1474107407 \times 10^{-5}$	
Δ_{1_1}	$-1,3095852751 \times 10^{-2}$	
Δ_{2_1}	$1,1123835513 \times 10^{-7}$	
β_{1_1}	108,1029187211	
$R_{s_fl_2}$	$6,0661918042 \times 10^{-4} \Omega/m$	$\lambda''_{1-fl_2} = 8,9527974112 \times 10^{-4}$
R_2	$2,1455011286 \times 10^{-5} \Omega/m$	
g_{s_2}	1,0001980212	
λ_{0_2}	$3,1496545999 \times 10^{-5}$	
Δ_{1_2}	$4,5520515892 \times 10^{-3}$	
Δ_{2_2}	$4,7105786535 \times 10^{-9}$	
β_{1_2}	108,1222324648	
t_{fl}	0,2 mm	

11. Add circulating losses and eddy current losses to find λ_1 (screen) and λ_2 (laminated foil) for all three phases:

$$\lambda_{1-cws_n} = \lambda'_{1-cws_n} + \lambda''_{1-cws_n}$$

λ'_{1-cws_m}	$9,7794239160 \times 10^{-3}$	$\lambda_{1-cws_m} = 9,7794239160 \times 10^{-3}$
λ''_{1-cws_m}	0	
λ'_{1-cws_1}	$1,3832887889 \times 10^{-2}$	$\lambda_{1-cws_1} = 1,3832887889 \times 10^{-2}$
λ''_{1-cws_1}	0	
λ'_{1-cws_2}	$1,0531989911 \times 10^{-2}$	$\lambda_{1-cws_2} = 1,0531989911 \times 10^{-2}$
λ''_{1-cws_2}	0	

$$\lambda_{1-fl_n} = \lambda'_{1-fl_n} + \lambda''_{1-fl_n}$$

$\lambda'_{1-fl,m}$	$2,6724898052 \times 10^{-3}$	$\lambda_{1-fl,m} = 6,1347424325 \times 10^{-3}$
$\lambda''_{1-fl,m}$	$3,4622526273 \times 10^{-3}$	
$\lambda'_{1-fl,1}$	$3,7815164955 \times 10^{-3}$	$\lambda_{1-fl,1} = 4,6605201558 \times 10^{-3}$
$\lambda''_{1-fl,1}$	$8,7900366024 \times 10^{-4}$	
$\lambda'_{1-fl,2}$	$2,8791497286 \times 10^{-3}$	$\lambda_{1-fl,2} = 3,7744294697 \times 10^{-3}$
$\lambda''_{1-fl,2}$	$8,9527974112 \times 10^{-4}$	

5.2.5 Thermal resistance T_1 between conductor and screen

The thermal resistance T_1 is calculated the same way as for the trefoil configuration, see section 5.1.6. Thermal resistance between conductor and screen:

$$T_1 = T_{1_csc} + T_{1_i} + T_{1_isc} + T_{1_uwbt} \quad (K.m/W)$$

T_{1_csc}	0,0319189884 K.m/W	$T_1 = 0,3789088974 \text{ K.m/W}$
T_{1_i}	0,3033167853 K.m/W	
T_{1_isc}	0,0116890861 K.m/W	
T_{1_uwbt}	0,0319840375 K.m/W	

5.2.6 Thermal resistance T_2 between screen and Al foil

The thermal resistance T_2 is calculated the same way as for the trefoil configuration, see section 5.1.7. Thermal resistance of sc water blocking tape over the copper wires screen:

$$T_2 = \frac{\rho_{wb}}{2\pi} \ln\left(\frac{D_{owt}}{D_{sc}}\right) \quad (K.m/W)$$

ρ_{wb}	12 K.m/W	$T_2 = 0,0121371405 \text{ K.m/W}$
D_{owt}	88,4 mm	
D_{sc}	87,84 mm	

5.2.7 Thermal resistance T_3 of outer covering

The thermal resistance T_3 is calculated the same way as for the trefoil configuration, see section 5.1.8. Thermal resistance outer covering, but without the 1,6 correction factor:

$$T_3 = T_{3_oc} + T_{3_sc} \quad (K.m/W)$$

T_{3_oc}	0,0492003872 K.m/W	$T_3 = 0,0532813189 \text{ K.m/W}$
T_{3_sc}	$4,0809317511 \times 10^{-3} \text{ K.m/W}$	

5.2.8 External thermal resistance T_4 Table 5-21: Parameters needed to calculate the external thermal resistance T_4

Parameter	Symbol	Value	Unit	Reference
External diameter of one cable	D_e	98	mm	Cable data sheet
Axial separation between two adjacent cables	s	500	mm	System design
Distance from the surface of the ground to the cable axis	L	1000	mm	System design, note 1
Thermal resistivity of the soil	ρ_T	1,0 K.m/W	K.m/W	System design
Sheath loss factor for the middle cable of the group, i.e. copper wire screen	$\lambda_{1,m}$	$9,7794239160 \times 10^{-3}$	K.m/W	Calculated
Sheath loss factor for an outer cable of the group, i.e. copper wire screen	λ_{1-cws_1}	$1,3832887889 \times 10^{-2}$	K.m/W	Calculated
Sheath loss factor for the other outer cable of the group, i.e. copper wire screen	λ_{1-cws_2}	$1,0531989911 \times 10^{-2}$	K.m/W	Calculated
Sheath loss factor for the middle cable of the group, i.e. laminated foil	λ_{1-fl_m}	$6,1347424325 \times 10^{-3}$	K.m/W	Calculated
Sheath loss factor for an outer cable of the group, i.e. laminated foil	λ_{1-fl_1}	$4,6605201558 \times 10^{-3}$	K.m/W	Calculated
Sheath loss factor for the other outer cable of the group, i.e. laminated foil	λ_{1-fl_2}	$3,7744294697 \times 10^{-3}$	K.m/W	Calculated
Dielectric loss per unit length in each phase	W_d	0,4654100053	W/m	See 5.1.4
Continuous current rating	I	1460,4545417538	A	Calculated
AC resistance of conductor at operating temp.	R_m R_1 R_2	$2,1725724283 \times 10^{-5}$ $2,1460977940 \times 10^{-5}$ $2,1455011286 \times 10^{-5}$	Ω/m	Calculated

Table 5-22: Sequential steps to calculate the external thermal resistance T_4

Step	Parameter	Symbol	Unit	Reference
1	Total cable losses for each phase	W_n	W/m	
2	External thermal resistance for each phase with correction in sheath losses	T_{4_n}	K.m/W	[4], section 4.2.3.3.4

1. Total Losses per phase W_x

$$W_n = I^2 \cdot R_n \cdot (1 + \lambda_{1-cws_n} + \lambda_{1-fl_n}) + W_d \quad (W/m)$$

I	1460,4545417538 A	$W_m = 47,5422569285 \text{ W/m}$ $W_1 = 47,0866497321 \text{ W/m}$ $W_2 = 46,8820830331 \text{ W/m}$
W_d	0,4654100053 W/m	
R_m	$2,1725724283 \times 10^{-5} \Omega/m$	
R_1	$2,1460977940 \times 10^{-5} \Omega/m$	
R_2	$2,1455011286 \times 10^{-5} \Omega/m$	
λ_{1-cws_m}	$9,7794239160 \times 10^{-3}$	
λ_{1-fl_m}	$6,1347424325 \times 10^{-3}$	
λ_{1-cws_1}	$1,3832887889 \times 10^{-2}$	
λ_{1-fl_1}	$4,6605201558 \times 10^{-3}$	
λ_{1-cws_2}	$1,0531989911 \times 10^{-2}$	
λ_{1-fl_2}	$3,7744294697 \times 10^{-3}$	

2. External thermal resistance middle, leading and lagging phase $T_{4,x}$:

$$u = \frac{2L}{D_e}$$

L	1000 mm	$u = 20,4081632653$
D_e	98 mm	

$$T_{4,m} = \frac{1}{2\pi} \rho_T \left[\ln(u + \sqrt{u^2 - 1}) + \frac{W_1}{W_m} \cdot \ln \left[\frac{\sqrt{s^2 + (2L)^2}}{s} \right] + \frac{W_2}{W_m} \cdot \ln \left[\frac{\sqrt{s^2 + (2L)^2}}{s} \right] \right] \quad (K.m/W)$$

$$T_{4,1} = \frac{1}{2\pi} \rho_T \left[\ln(u + \sqrt{u^2 - 1}) + \frac{W_m}{W_1} \cdot \ln \left[\frac{\sqrt{s^2 + (2L)^2}}{s} \right] + \frac{W_2}{W_1} \cdot \ln \left[\frac{\sqrt{(2s)^2 + (2L)^2}}{2s} \right] \right] \quad (K.m/W)$$

$$T_{4,2} = \frac{1}{2\pi} \rho_T \left[\ln(u + \sqrt{u^2 - 1}) + \frac{W_m}{W_2} \cdot \ln \left[\frac{\sqrt{s^2 + (2L)^2}}{s} \right] + \frac{W_1}{W_2} \cdot \ln \left[\frac{\sqrt{(2s)^2 + (2L)^2}}{2s} \right] \right] \quad (K.m/W)$$

ρ_T	1,0 K.m/W	$T_{4,m} = 1,0358516741 \text{ K.m/W}$ $T_{4,1} = 0,9453832120 \text{ K.m/W}$ $T_{4,2} = 0,9474917745 \text{ K.m/W}$
u	20,4081632653	
L	1000 mm	
s	500 mm	
W_m	47,5422569285 W/m	
W_1	47,0866497321 W/m	
W_2	46,8820830331 W/m	

5.2.9 Permissible current rating

Table 5-23: Parameters needed to calculate the continuous current carrying capacity

Parameter	Symbol	Values	Unit	Reference
Conductor operating temperature	θ	90	$^{\circ}C$	System design
Ambient temperature	θ_a	20	$^{\circ}C$	System design
Number of conductors in cable	n	1	-	Cable data sheet
AC resistance of conductor at operating temperature	R_m	$2,1725724283 \times 10^{-5}$	Ω/m	Calculated
Ratio of the total losses in CWS	$\lambda_{1-cws,m}$	$9,7794239160 \times 10^{-3}$	-	Calculated
Ratio of the total losses in laminated foil	$\lambda_{1-fl,m}$	$6,1347424325 \times 10^{-3}$	-	Calculated
Dielectric losses	W_d	0,4654100053	W/m	See 5.1.4
Thermal resistance between conductor and metallic screen	T_1	0,3789088974	$K.m/W$	Calculated
Thermal resistance between metallic screen and laminated foil	T_2	0,0121371405	$K.m/W$	Calculated
Thermal resistance of external serving	T_3	0,0532813189	$K.m/W$	Calculated
Thermal resistance of surrounding medium, middle phase	$T_{4,m}$	1,0358516741	$K.m/W$	Calculated

Table 5-24 Sequential steps to calculate the continuous current rating capacity

Step	Parameter	Symbol	Unit	Reference
1	Conductor temperature rise above ambient	$\Delta\theta$	K	[3], section 1.4.1.1
2	Permissible current rating	I	A	[3], section 1.4.1.1

1. Conductor temperature rise above ambient:

$$\Delta\theta = \theta - \theta_a$$

θ	90 °C	$\Delta\theta = 70 \text{ K}$
θ_a	20 °C	

2. Permissible current rating:

$$I = \left[\frac{\Delta\theta - W_d [0,5 \cdot T_1 + n(T_2 + T_3 + T_{4,m})]}{R \cdot T_1 + nR(1 + \lambda_{1-cws_m})T_2 + nR(1 + \lambda_{1-cws_m} + \lambda_{1-fl_m})(T_3 + T_{4,m})} \right]^{0,5} \quad (A)$$

$\Delta\theta$	70 K	$I = 1460,4545417538 \text{ A}$
W_d	0,4654100053 K.m/W	
T_1	0,3789088974 K.m/W	
T_2	0,0121371405 K.m/W	
T_3	0,0532813189 K.m/W	
$T_{4,m}$	1,0358516741 K.m/W	
R	$2,1725724283 \times 10^{-5} \text{ } \Omega/\text{m}$	
n	1	
λ_{1-cws_m}	$9,7794239160 \times 10^{-3}$	
λ_{1-fl_m}	$6,1347424325 \times 10^{-3}$	

5.2.10 Calculation of losses

Conductor losses:

$$P_{c_n} = R_n I^2 \quad (W/m)$$

Copper wires screen losses:

$$P_{scr_n} = \lambda_{1_n} R_n I^2 \quad (W/m)$$

Laminated foil losses:

$$P_{fl_n} = \lambda_{2_n} R_n I^2 \quad (W/m)$$

I	1460,4545417538 A	
R_m	$2,1725724283 \times 10^{-5} \Omega/\text{m}$	$P_{c_m} = 46,3339394097 \text{ W/m}$ $P_{scr_m} = 0,4531725789 \text{ W/m}$ $P_{fl_m} = 0,2842802473 \text{ W/m}$
λ_{1-cws_m}	$9,7794239160 \times 10^{-3}$	
λ_{1-fl_m}	$6,1347424325 \times 10^{-3}$	
R_1	$2,1460977940 \times 10^{-5} \Omega/\text{m}$	$P_{c_1} = 45,7747093487 \text{ W/m}$ $P_{scr_1} = 0,6331964226 \text{ W/m}$ $P_{fl_1} = 0,2133339555 \text{ W/m}$
λ_{1-cws_1}	$1,3832887889 \times 10^{-2}$	
λ_{1-fl_1}	$4,6605201558 \times 10^{-3}$	
R_2	$2,1455011286 \times 10^{-5} \Omega/\text{m}$	$P_{c_2} = 45,7619829087 \text{ W/m}$ $P_{scr_2} = 0,4819647423 \text{ W/m}$ $P_{fl_2} = 0,1727253769 \text{ W/m}$
λ_{1-cws_2}	$1,0531989911 \times 10^{-2}$	
λ_{1-fl_2}	$3,7744294697 \times 10^{-3}$	

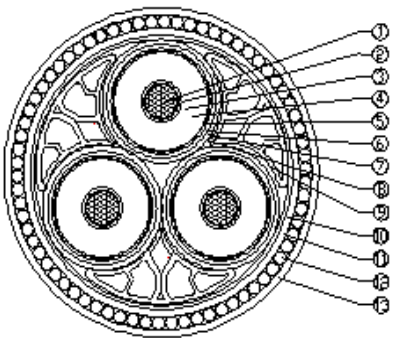
6. Case study 2: a 30kV submarine array cable

6.1 Introduction

The selected cable type is a three core submarine cable with copper round compacted conductors, XLPE insulation, lead alloy sheath, semiconducting polyethylene sheath, polypropylene yarns, galvanized steel wire armour and polypropylene yarns 3x630 mm² CU/XLPE/PB/SCPE/PPY/SWA/PPY 18/30 (36) kV according to IEC 60502 [18] where applicable. The datasheet is depicted in Figure 28. The current rating will be determined for the following laying configuration:

- Cable buried in seafloor
- Depth of laying of the cable 1 m
- Seabed soil temperature 15 °C
- Soil thermal resistivity 0,7 K.m/W
- Sheaths solidly bonded
- One circuit thermally independent

All other relevant parameters are given in the input tables at each parameter calculation.

Dimensional characteristics:		
		
		Diameter (mm)
1.	Copper round stranded compacted class 2 according to IEC 60228 of nominal cross-section equal to 630 sq.mm, longitudinally water sealed.	30 (approximate)
2.	Semiconducting tape(s) applied with overlap.	30.5 (approximate)
3.	Conductor non-metallic extruded screen: Extruded semiconducting compound bonded to inner surface of insulation.	32.5 (approximate)
4.	Insulation: XLPE super-clean according to IEC 60502 of 8.0mm nominal thickness.	48.5 (approximate)
5.	Core non-metallic extruded screen: Extruded semiconducting compound bonded to outer surface of insulation.	50.3 (approximate)
6.	Semiconducting waterblocking tape(s) applied with overlap.	52.5 (approximate)
7.	Metallic sheath: Lead alloy of 2.3 mm nominal thickness.	57.1 (approximate)
8.	Semiconducting PE sheath of 2.5 mm approximate thickness. Sheath colour: Black	62.1 (approximate)
9.	Extruded polyethylene profile fillers at the outer interstices between cores in order to give the cable a circular cross-section.	134.9 (approximate, after cabling)
10.	Binding synthetic tape(s) helically applied with overlap.	135.3 (approximate)
11.	One layer of polypropylene yarns of approximate thickness of 1 mm.	137.3 (approximate)
12.	Armour consisting of helically applied bitumen compound coated galvanized steel round wires of grade 34, class A, 6.0 mm diameter, according to EN 10257-2. Number of wires: 71, Length of lay: 1.785 mm	149.3 (approximate)
13.	Two layers of polypropylene yarns with total approximate thickness of 3.0mm. Over the inner (first) layer bitumen compound is applied. Also, the outer (second) layer shall consist of black and yellow polypropylene yarns as to form a helical yellow stripe.	Cable OD = 155.3 (approximate)

Electrical Data:		
Frequency:	50	Hz
Maximum conductor's temperature at continuous operation:	90	°C
Maximum conductor's temperature at short-circuit (5 s maximum duration):	250	°C
Maximum conductor DC resistance at 20°C:	0.0283	Ω/km

Figure 29: Cable datasheet 3x630 mm² CU/XLPE/PB/SC PE/PPY/SWA/PPY 18/30 (36) kV

It is noted that Guidance Point 45 is applied to calculate the thermal resistance of the fillers.

6.2 Calculation of lay-up factor of the cores

Table 6-1: Parameters needed to calculate the conductor AC resistance at operating temperature

Parameter	Symbol	Value	Unit	Reference
Lay length of core stranding	L_{core}	2152	mm	Cable datasheet
Core diameter	D_e	62,1	mm	Cable datasheet

Table 6-2: Sequential steps to calculate the factor for the core layup

Step	Parameter	Symbol	Unit	Reference
1	Factor for the core layup	f_{layup}	mm	

There is no formula in the IEC standards on how to take the layup of the cores into account for the resistance, capacitance and current rating.

The factor for the extra length of cores can be described with the following formula according to Appendix A and Guidance Point 23.

$$f_{layup} = \sqrt{1 + \left[\frac{\pi \cdot 1,29 \cdot D_e}{L_{core}} \right]^2}$$

L_{core}	2152 x 10 ⁻³ m	$f_{layup} = 1,0068150725$
D_e	62,1 x 10 ⁻³ m	

6.3 Calculation of the conductor AC resistance at operation temperature

Table 6-3: Parameters needed to calculate the conductor AC resistance at operating temperature

Parameter	Symbol	Value	Unit	Reference
Conductor DC resistance at 20°C	R_o	0,0283 x 10 ⁻³	Ω/m	Cable data sheet
External diameter of conductor	d_c	30	mm	Cable datasheet
Conductor material		Copper		Cable datasheet
Type of conductor		Round stranded compacted, [8]		Cable datasheet
Skin effect factor used in calculating x_s	k_s	1	-	[3], table 2
Proximity effect factor used in calculating x_p	k_p	1	-	[3], table 2
Temperature coefficient per Kelvin at 20°C	α_{20}	3,93 x 10 ⁻³	1/K	[3], table 1
Maximum operating temperature of conductor	θ	90	°C	Cable datasheet
Axial separation of conductors	s	62,1	mm	Cable datasheet
System frequency	f	50	Hz	System design

Table 6-4: Sequential steps to calculate the conductor AC resistance at operating temperature

Step	Parameter	Symbol	Unit	Reference
1	Conductor DC resistance at maximum operating temperature	R'	Ω/m	[3], section 2.1.1
2	Argument of a Bessel function used to calculate skin effect	x_s	-	[3], section 2.1.2
3	Skin effect factor	y_s	-	[3], section 2.1.2
4	Argument of a Bessel function used to calculate proximity effect	x_p	-	[3], section 2.1.4
5	Proximity effect factor	y_p	-	[3], section 2.1.4
6	AC resistance of conductor at its maximum operating temperature	R	Ω/m	[3], section 2.1

1. Conductor DC resistance at operating temperature R' :

$$R' = R_o [1 + \alpha_{20}(\theta - 20)] \quad (\Omega/m)$$

R_o	$0,0283 \times 10^{-3} \Omega/m$	$R' = 3,608533 \times 10^{-5} \Omega/m$
α_{20}	$3,93 \times 10^{-3}$	
θ	90°C	

Note: The DC resistance is given in cable datasheet and is referring to the final three core cable. According to guidance 2.5.4, the DC resistance of the conductor is not multiplied by lay-up factor of the core.

2. Argument of a Bessel function used to calculate skin effect x_s :

$$x_s^2 = \frac{8\pi f}{R'} 10^{-7} k_s$$

f	50 Hz	$x_s^2 = 3,4824042386$
R'	$3,608533 \times 10^{-5} \Omega/m$	
k_s	1	

3. Skin effect factor y_s :

$$\text{for } 0 < x_s \leq 2,8 \quad \rightarrow \quad y_s = \frac{x_s^4}{192 + 0,8x_s^4}$$

Note: three equations are given for calculating y_s , the use of these equations depends on the calculated value of x_s .

x_s^2	$3,4824042386$	$y_s = 0,0601241268$
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4. Argument of a Bessel function used to calculate proximity effect x_p :

$$x_p^2 = \frac{8\pi f}{R'} 10^{-7} k_p$$

f	50 Hz	$x_p^2 = 3,4824042386$
R'	$3,608533 \times 10^{-5} \Omega/m$	
k_p	1	

5. Proximity effect factor y_p :

$$y_p = \frac{x_p^4}{192 + 0,8x_p^4} \left(\frac{d_c}{s} \right)^2 \left[0,312 \left(\frac{d_c}{s} \right)^2 + \frac{1,18}{\frac{x_p^4}{192 + 0,8x_p^4} + 0,27} \right]$$

x_p^2	3,4824042386	$y_p = 0,0511765348$
d_c	$30 \times 10^{-3} \text{ m}$	
s	$62,1 \times 10^{-3} \text{ m}$	

6. AC resistance of conductor R :

$$R = R'(1 + 1,5 \cdot (y_s + y_p)) (\Omega/\text{m})$$

R'	$3,608533 \times 10^{-5} \Omega/\text{m}$	$R = 4,2109811659 \times 10^{-5} \Omega/\text{m}$
y_s	0,0601241268	
y_p	0,0511765348	

Note: The impact of armour wires in the conductor losses is taken into consideration with the factor 1.5 in the equation above. See Section 2.5.4 in the guidance chapter.

6.4 Dielectric losses

Table 6-5: Parameters needed to calculate the dielectric loss per unit length per phase

Parameter	Symbol	Value	Unit	Reference
Diameter over insulation	D_i	48,5	mm	Cable data sheet
External diameter of the conductor including screen	d_c	32,5	mm	Cable datasheet
Relative permittivity of the insulation	ϵ	2,5	-	[3], table 3
Loss factor of the insulation	$\tan\delta$	40×10^{-4}	-	[3], table 3, note 1
Voltage to earth	U_o	$\frac{30000}{\sqrt{3}}$	V	System design
System frequency	f	50	Hz	System design
Note 1: Assume always unfilled XLPE unless specified otherwise				

Table 6-6: Sequential steps to calculate the dielectric loss per unit length per phase

Step	Parameter	Symbol	Unit	Reference
1	Capacitance C of each core	C	F/m	[3], section 2.2
2	Angular frequency of system	ω	-	[3], section 2.2
3	Dielectric loss per unit length per phase	W_d	W/m	[3], section 2.2, note 1
Note 1: According to [3], table 3, the dielectric losses may be omitted in the calculation.				

1. Capacitance C of each core:

$$C_{core} = \frac{\epsilon}{18 \ln\left(\frac{D_i}{d_c}\right)} 10^{-9} \text{ (F/m)}$$

ε	2,5	$C_{core} = 3,4694145239 \times 10^{-10} \text{ F/m}$
D_i	$48,5 \times 10^{-3} \text{ m}$	
d_c	$32,5 \times 10^{-3} \text{ m}$	

Cable capacitance C:

$$C = C_{core} \cdot f_{lay \text{ up core}}$$

C	$3,4694145239 \times 10^{-10} \text{ F/m}$	$C = 3,4930588353 \times 10^{-10} \text{ F/m}$
$f_{lay \text{ up core}}$	1,0068150725	

2. Angular frequency of system:

$$\omega = 2\pi f \left(\frac{\text{rad}}{\text{s}} \right)$$

f	50 HZ	$\omega = 314,1592653590 \text{ rad/s}$
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3. Dielectric loss per unit length in each phase:

$$W_d = \omega C U_o^2 \tan \delta \text{ (W/m)}$$

C	$3,4930588353 \times 10^{-10} \text{ F/m}$	$W_d = 0,1316852157 \text{ W/m}$
ω	314,1592653590 rad/s	
U_o	$\frac{30000}{\sqrt{3}} \text{ V}$	
$\tan \delta$	40×10^{-4}	

6.5 Loss factor for sheath

Table 6-7: Parameters needed to calculate the loss factor for lead sheath

Parameter	Symbol	Value	Unit	Reference
AC resistance of sheath at 20°C	R_{so}	-	Ω/m	Calculated
Mean diameter of the sheath	d	54,8	mm	Cable data sheet
Cross sectional area of the metallic sheath	A_{PB}	-	mm^2	Calculated
Sheath resistivity at 20°C	ρ_{sh}	$21,4 \times 10^{-8}$	Ωm	[3], table 1
Sheath temperature coefficient at 20°C per Kelvin	α_{20}	$4,0 \times 10^{-3}$	-	[3], table 1
Distance between conductor axes	S	62,1	mm	System design
System frequency	f	50	Hz	System design
AC resistance of conductor at operating temp.	R	-	Ω/m	Calculated

Table 6-8: Sequential steps to calculate the sheath losses per unit length per phase

Step	Parameter	Symbol	Unit	Reference
1	Calculation of the cross-sectional area of the sheath			
2	AC Resistance of sheath at 20°C	R_{so}	Ω/m	-
3	Maximum operating temperature of the cable sheath, to be determined iterative	θ_{sc}	$^{\circ}C$	[3], section 2.3
4	Resistance of sheath at operating temperature	R_s	Ω/m	[3], section 2.3
5	Reactance per unit length of sheath	X	Ω/m	[3], section 2.3.1
6	Loss factor circulating losses for sheath	λ'_1	-	[3], section 2.3.10
7	Loss factor for eddy current losses	λ''_1		[3], section 2.3.6.1.
8	Loss factor of the sheath	λ_1		[3], section 2.3.

1. Calculation of cross-sectional area of the sheath:

$$A_{PB} = \pi \cdot t_{sh} \cdot (d + t_{sh})$$

d	$52,5 \times 10^{-3} \text{ m}$	$A_{PB} = 395,9663380585 \times 10^{-6} \text{ m}^2$
t_{sh}	$2,3 \times 10^{-3} \text{ m}$	

2. AC Resistance of screen at 20°C:

AC Resistance of the screen at 20°C per unit length of core is:

$$R_{so_core} = \frac{\rho_{sh}}{A_{PB}} \quad (\Omega/m)$$

ρ_{sh}	$21,4 \times 10^{-8} \Omega m$	$R_{so_core} = 5,4044998130 \times 10^{-4} \Omega/m$
A_{PB}	$395,9663380585 \times 10^{-6} \text{ m}^2$	

AC Resistance of the screen at 20°C per unit length of 3-core cable is calculated by taken into consideration the lay-up factor:

$$R_{so} = R_{so_core} \cdot f_{lay \text{ up core}}$$

R_{so_core}	$5,4044998130 \times 10^{-4} \Omega/m$	$R_{so} = 5,44133187089 \times 10^{-4} \Omega/m$
$f_{lay \text{ up core}}$	1,0068150725	

3. The operating temperature θ_{sc} ($^{\circ}C$) of the screen is given by:

$$\theta_{sc} = \theta - (I^2 R + 0,5 W_d) T_1 \quad (^{\circ}C)$$

θ	90 $^{\circ}C$	$\theta_{sc} = 79,3851216214 \text{ }^{\circ}C$
I	838,3399739336 A	
R	$4,2109811659 \times 10^{-5} \Omega/m$	
W_d	0,1316852157 W/m	
T_1	0,3578707848 K.m/W	

Note: θ_{sc} is the operating temperature of the cable sheath ($^{\circ}C$). Because the temperature of the sheath is a function of the current I, an iterative method must be used for the calculation.

4. Lead sheath AC resistance at operating temperature θ_{sh} :

$$R_s = R_{so} [1 + \alpha_{20} (\theta_{sc} - 20)] \quad (\Omega/m)$$

R_{so}	$5,44133187089 \times 10^{-4} \Omega/m$	$R_s = 6,7338684906 \times 10^{-4} \Omega/m$
α_{20}	$4,0 \times 10^{-3}$	
θ_{sc}	$79,3851216214 \text{ }^\circ\text{C}$	

5. Calculation of the reactance X_{core} per unit length of the sheath:
Mean diameter of sheath:

$$d = (D_{sh} + t_{sh}) (m)$$

D_{sh}	$52,5 \times 10^{-3} m$	$d = 54,8 \times 10^{-3} m$
t_{sh}	$2,3 \times 10^{-3} m$	

$$X_{1,core} = 2\omega 10^{-7} \ln\left(\frac{2s}{d}\right) (\Omega/m)$$

ω	$314,1592653590 \text{ rad/s}$	$X_{1,core} = 5,1409209142 \times 10^{-5} \Omega/m$
s	$62,1 \times 10^{-3} m$	
d	$54,8 \times 10^{-3} m$	

Calculation of reactance per unit length of 3-core cable is calculated by taken into consideration the lay-up factor:

$$X_1 = X_{1,core} \cdot f_{lay up core}$$

X_{core}	$5,1409209142 \times 10^{-5} \Omega/m$	$X = 5,1759566628 \times 10^{-5} \Omega/m$
$f_{lay up core}$	$1,0068150725$	

6. Loss factor λ'_1 for circulating losses of the sheath:

$$\lambda'_1 = \frac{R_s}{R} \frac{1.5}{1 + \left(\frac{R_s}{X}\right)^2}$$

R_s	$6,7338684906 \times 10^{-4} \Omega/m$	$\lambda'_1 = 0,1408854886$
R	$4,2109811659 \times 10^{-5} \Omega/m$	
X	$5,1759566628 \times 10^{-5} \Omega/m$	

Note: The impact of armour wires in the conductor losses is taken into consideration with the factor 1.5 in the equation above.

7. Loss factor λ''_1 for eddy current losses:

$$m = \frac{\omega}{R_s} 10^{-7}$$

ω	100π	$m = 0,0466536087$
R_s	$6,7338684906 \times 10^{-4} \Omega/m$	

$$\lambda_0 = 3 \left(\frac{m^2}{1 + m^2} \right) \left(\frac{d}{2s} \right)^2$$

m	$0,0466536087$	$\lambda_0 = 0,0012684268$
d	$54,8 \times 10^{-3} m$	
s	$62,1 \times 10^{-3} m$	

$$\Delta_1 = (1,14m^{2,45} + 0,33) \left(\frac{d}{2s} \right)^{(0,92m+1,66)}$$

$$\Delta_2 = 0$$

m	0,0466536087	$\Delta_1 = 0,0820762435$
d	$54,8 \times 10^{-3} \text{ m}$	
s	$62,1 \times 10^{-3} \text{ m}$	

$$\beta_1 = \sqrt{\frac{4\pi\omega}{10^7 \rho_s \cdot (1 + \alpha_s \cdot (\theta_s - 20))}}$$

ω	100π	$\beta_1 = 38,6094196564$
ρ_s	$21,4 \times 10^{-8} \Omega\text{m}$	
α_s	$4,0 \times 10^{-3} \text{ 1/K}$	
θ_{sc}	$79,3851216214 \text{ }^\circ\text{C}$	

$$g_s = 1 + \left(\frac{t_s}{D_s} \right)^{1,74} (\beta_1 D_s 10^{-3} - 1,6)$$

D_s	57,1 mm	$g_s = 1,00226113355$
t_s	2,3 mm	
β_1	38,6094196564	

$$\lambda_1'' = \frac{R_s}{R} \left[g_s \lambda_0 (1 + \Delta_1 + \Delta_2) + \frac{(\beta_1 t_s)^4}{12 \times 10^{12}} \right]$$

R_s	$6,7338684906 \times 10^{-4} \Omega/\text{m}$	$\lambda_1'' = 0,02208098453$
R	$4,2109811659 \times 10^{-5} \Omega/\text{m}$	
g_s	1,0022611336	
λ_0	0,0012684268	
Δ_1	0,08207624347	
Δ_2	0	
β_1	38,6094196564	
t_s	2,3 mm	

Calculation of factor 'F' for lead alloy sheath:

$$M = N = \frac{R_{s_PB}}{X_{core}}$$

R_{s_PB}	$6,7338684906 \times 10^{-4} \Omega/\text{m}$	$M = N = 13,0099012208$
X_{core}	$5,1759566628 \times 10^{-5} \Omega/\text{m}$	

$$F = \frac{4M^2N^2 + (M + N)^2}{4(M^2 + 1)(N^2 + 1)}$$

M	13,0099012208	$F = 0,9941265446$
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8. Add circulating losses and eddy current losses for foil to find λ_1 :

$$\lambda_1 = \lambda_1' + F \cdot \lambda_1''$$

λ'_1	0,1408854886	$\lambda_1 = 0,1628367815$
λ''_1	0,0220809845	
F	0,9941265446	

6.6 Loss factor for armour

Table 6-9: Parameters needed to calculate the loss factor for steel wire armour

Parameter	Symbol	Value	Unit	Reference
AC resistance of armour at 20°C	R_{Ao}	-	Ω/m	Calculated
Mean diameter of the armour	d_A	143,3	mm	Cable data sheet
Number of armour wires	n_1	71	-	Cable data sheet
Diameter of armour wires	d_f	6		Cable data sheet
Cross sectional area of the armour	A	-	mm ²	Calculated
Lay Length of the armour	$lay\ length_{arm}$	1785	mm	Cable data sheet
Increment factor for the estimation of the AC resistance of the armour based on the DC resistance of the armour	K	calculated	-	[3], section 2.4.2.1
Armour resistivity at 20°C	ρ_a	$13,8 \times 10^{-8}$	Ωm	[3], table 1
Armour temperature coefficient at 20°C per Kelvin	α_{20}	$4,5 \times 10^{-3}$	-	[3], table 1
Distance between conductor axes/Diameter over single core	S	62,1	mm	Cable data sheet
Diameter over assembled three cores	D_{cabl}	134,9	mm	Cable data sheet
System frequency	f	50	Hz	System design
AC resistance of conductor at operating temp.	R	-	Ω/m	Calculated

Table 6-10: Sequential steps to calculate loss factor for steel wire armour

Step	Parameter	Symbol	Unit	Reference
1	Calculation of cross-section of the armour			
2	Ratio of the length of the wires to the length of the cable	$\frac{S}{L}$	-	
3	AC Resistance of armour at 20°C	R_{Ao}	Ω/m	-
4	Maximum operating temperature of the cable armour, to be determined iterative	θ_{ar}	°C	[3], section 2.4
5	Resistance of armour at operating temperature	R_A	Ω/m	[3], section 2.4
6	Calculation of value c, which is the distance between the axis of conductors and the axis of the cable for three-core cables	c	mm	[3], section 2.4.2.3
7	Loss factor for armour	λ_2		[3], section 2.4.2.3 and 2.4.2.5

1. Calculation of cross-sectional area of the armour:

$$A = n_1 \cdot \pi \cdot \left(\frac{d_f}{2}\right)^2$$

n_1	71	$A = 2007,4777056439 \times 10^{-6} \text{ m}^2$
d	$6 \times 10^{-3} \text{ m}$	

2. Ratio of the length of the wires to the length of the cable:

$$f_{layup\ arm} = \frac{S}{L} = \sqrt{1 + \left(\frac{\pi \cdot d_A}{lay\ length_{arm}}\right)^2}$$

d_A	$143,3 \times 10^{-3}$ m	$f_{layup\ arm} = 1,0313140054$
$lay\ length_{arm}$	1785×10^{-3} m	

3. Resistance of armour at 20°C:

DC Resistance of the armour at 20°C per unit length is:

$$R_{A0}^* = \frac{\rho_a}{A} \quad (\Omega/m)$$

ρ_a	$13,8 \times 10^{-8} \Omega m$	$R_{A0}^* = 6,8742980115 \times 10^{-5} \Omega/m$
A	$2007,4777056439 \times 10^{-6} m^2$	

According to [3] section 2.4.2.1 the AC resistance of armour wire varies from about 1,2 times the DC resistance of 2 mm up to 1,4 times the DC resistance for 5 mm wires.

As this diameter exceeds the diameter of 5 mm the suitable factor is found with the use of linear interpolation as it is referred in Guidance Point 34.

$$y(x) = \frac{y_2 - y_1}{x_2 - x_1} \cdot (x - x_1) + y_1 \Rightarrow y(6) = \frac{1,4 - 1,2}{5 - 2} \cdot (6 - 2) + 1,2 \Rightarrow y(6) = 1,4666666667$$

$$R_{Ao} = k \cdot R_{A0}^* \cdot f_{layup\ arm}$$

k	1,4666666667	$R_{Ao} = 1,0398021064 \times 10^{-4} \Omega/m$
R_{A0}^*	$6,8742980115 \times 10^{-5} \Omega/m$	
$f_{layup\ arm}$	1,0313140054	

4. The operating temperature θ_{ar} (°C) of the armour is given by:

$$\theta_{ar} = \theta - (I^2 R + 0,5 W_d) T_1 - (I^2 R (1 + \lambda_1) + W_d) \cdot n \cdot T_2 \quad (^\circ C)$$

θ	90 °C	$\theta_{ar} = 71,308546665890 \text{ } ^\circ C$
I	838,3399739336 A	
R	$4,2109811659 \times 10^{-5} \Omega/m$	
W_d	0,1316852157 W/m	
T_1	0,3578707848 K.m/W	
T_2	0,0779300464 K.m/W	
n	3	

Note: θ_{ar} is the operating temperature of the cable armour (°C). Because the temperature of the screen is a function of the current I, an iterative method must be used for the calculation.

5. Steel wire armour AC resistance at operating temperature θ_{ar} :

$$R_A = R_{Ao} [1 + \alpha_{ar} (\theta_{ar} - 20)] \quad (\Omega/m)$$

R_{Ao}	$1,0398021064 \times 10^{-4} \Omega/m$	$R_A = 1,2798804135 \times 10^{-4} \Omega/m$
α_{ar}	$4,5 \times 10^{-3}$	
θ_{ar}	71,308546665890 °C	

6. Calculation of the value c which is the distance between the axis of conductors and the axis of the cable for three core cables (refer to Guidance Point 33):

$$c = \frac{D_{cabl}}{2} - \frac{S}{2} \text{ (m)}$$

S	62,1 x10 ⁻³ m	c = 36,4 x 10 ⁻³ m
D _{cabl}	134,9 x10 ⁻³ m	

7. Loss factor λ_2 of the armour, refer to Guidance Point 37:

$$\lambda_2 = 1,23 \cdot \frac{R_A}{R} \cdot \left(\frac{2c}{d_A}\right)^2 \cdot \frac{1}{\left(\frac{2,77 \cdot R_A \cdot 10^6}{\omega}\right)^2 + 1} \cdot \left(1 - \frac{R}{R_s} \cdot \lambda'_1\right)$$

R _A	1,2798804135 x10 ⁻⁴ Ω/m	λ ₂ = 0,4206526837
R	4,2109811659 x10 ⁻⁵ Ω/m	
d _A	143,3 x10 ⁻³ m	
c	36,4 x 10 ⁻³ m	
ω	314,1592653590	
R _s	6,7338684906 x 10 ⁻⁴ Ω/m	
λ' ₁	0,1408854886	

6.7 Thermal resistance T₁ between conductor and screen

Table 6-11: Parameters needed to calculate the thermal resistance between conductor and screen

Parameter	Symbol	Value	Unit	Reference
Diameter of conductor	d _c	30	mm	Cable datasheet
External diameter over the sc tape of the conductor	d _{wbcon}	30,5	mm	Cable datasheet
External diameter of sc conductor screen	d _{sc}	32,5	mm	Cable datasheet
External diameter of insulation	D _i	48,5	mm	Cable data sheet
External diameter of sc insulation screen	D _{si}	50,3	mm	Cable data sheet
External diameter of sc- water blocking tape(s)	D _{uwb}	52,5	mm	Cable data sheet
Thermal resistivity of semiconducting tapes	ρ _{sc,t}	6	K.m/W	Note 1
Thermal resistivity of sc screen	ρ _{sc}	2,5	K.m/W	Note 2
Thermal resistivity of insulation	ρ _i	3,5	K.m/W	[4], table 1
Thermal resistivity of water blocking tapes	ρ _{wb}	12	K.m/W	Note 3
Note 1: In this case: 6 K.m/W Note 2: In this case: 2,5 K.m/W Note 3: In this case: 12 K.m/W				

Table 6-12: Sequential steps to calculate the thermal resistance between conductor and screen

Step	Parameter	Symbol	Unit	Reference
1	Thermal resistance of sc tape over the conductor	$T_{1_{sc_con}}$	$K.m/W$	[4], section 4.1.2.1
2	Thermal resistance of sc conductor screen	$T_{1_{csc}}$	$K.m/W$	[4], section 4.1.2.1
3	Thermal resistance of insulation	T_{1_i}	$K.m/W$	[4], section 4.1.2.1
4	Thermal resistance of sc insulation screen	$T_{1_{isc}}$	$K.m/W$	[4], section 4.1.2.1
5	Thermal resistance of sc water blocking tape	$T_{1_{uwb}}t$	$K.m/W$	[4], section 4.1.2.1
6	Thermal resistance between conductor and screen per core length	$T_{1_{core}}$	$K.m/W$	[4], section 4.1.2.1
7	Thermal resistance between conductor and screen	T_1	$K.m/W$	[4], section 4.1.2.1

1. Thermal resistance of sc tape over conductor:

$$T_{1_{wb_con}} = \frac{\rho_{sc_t}}{2\pi} \ln\left(\frac{d_{sc}}{d_c}\right) \quad (K.m/W)$$

ρ_{sc_t}	6 K.m/W	$T_{1_{wb_con}} = 0,0157843207 \text{ K.m/W}$
d_{sc}	$30,5 \times 10^{-3} \text{ m}$	
d_c	$30 \times 10^{-3} \text{ m}$	

2. Thermal resistance of sc conductor screen:

$$T_{1_{csc}} = \frac{\rho_{sc}}{2\pi} \ln\left(\frac{d_{sc}}{d_c}\right) \quad (K.m/W)$$

ρ_{sc}	2,5 K.m/W	$T_{1_{csc}} = 0,0252711812 \text{ K.m/W}$
d_{sc}	$32,5 \times 10^{-3} \text{ m}$	
d_c	$30,5 \times 10^{-3} \text{ m}$	

3. Thermal resistance of insulation:

$$T_{1_i} = \frac{\rho_i}{2\pi} \ln\left(\frac{D_i}{D_{sc}}\right) \quad (K.m/W)$$

ρ_i	3,5 K.m/W	$T_{1_i} = 0,2229972397 \text{ K.m/W}$
D_i	$48,5 \times 10^{-3} \text{ m}$	
D_{sc}	$32,5 \times 10^{-3} \text{ m}$	

4. Thermal resistance of sc insulation screen:

$$T_{1_{isc}} = \frac{\rho_{sc}}{2\pi} \ln\left(\frac{D_{si}}{D_i}\right) \quad \left(K.\frac{m}{W}\right)$$

ρ_{sc}	2,5 K.m/W	$T_{1_{isc}} = 0,0144995243 \text{ K.m/W}$
D_{si}	$50,3 \times 10^{-3} \text{ m}$	
D_i	$48,5 \times 10^{-3} \text{ m}$	

5. Thermal resistance of sc water blocking tape:

$$T_{1_{wbt}} = \frac{\rho_{wb}}{2\pi} \ln\left(\frac{D_{si}}{D_i}\right) \quad (K.m/W)$$

ρ_{wb}	12 K.m/W	$T_{1_wbt} = 0,0817574343 \text{ K.m/W}$
D_{bst}	$52,5 \times 10^{-3} \text{ m}$	
D_{si}	$50,3 \times 10^{-3} \text{ m}$	

6. Thermal resistance between conductor and screen per core length:

$$T_{1_core} = T_{1_wb_con} + T_{1_csc} + T_{1_i} + T_{1_isc} + T_{1_wbt} \quad (K.m/W)$$

$T_{1_wb_con}$	0,0157843207 K.m/W	$T_1 = 0,3603097002 \text{ K.m/W}$
T_{1_csc}	0,0252711812 K.m/W	
T_{1_i}	0,2229972397 K.m/W	
T_{1_isc}	0,0144995243 K.m/W	
T_{1_wbt}	0,0817574343 K.m/W	

7. Thermal resistance between conductor and screen per cable length

$$T_1 = \frac{T_{1_core}}{f_{layup}}$$

T_{1_core}	0,3603097002 K.m/W	$T_1 = 0,3578707848 \text{ K.m/W}$
f_{layup}	1,0068150725	

6.8 Thermal resistance T_2 of the sheath around each core, fillers and bedding

Table 6-13: Parameters needed to calculate the thermal resistance between screen and armour

Parameter	Symbol	Value	Unit	Reference
External diameter of metallic sheath	D_S	57,1	mm	Cable data sheet
External diameter of sc PE sheath	D_e	62,1	mm	Cable data sheet
Thermal resistivity of sc PE sheath	ρ_{scPE}	2,5	K.m/W	Note 1
Geometric Factor	G			[4], section 5.6
Thermal resistivity of fillers and binding tapes	ρ_{fil_bt}	6	K.m/W	Note 2
Note 1: In this case: 2.5 K.m/W				
Note 2: In this case: 6 K.m/W				

Table 6-14: Sequential steps to calculate the thermal resistance between screen and armour

step	Parameter	Symbol	Unit	Reference
1	Thermal resistance of the sheath around each core per core length	T'_{2_core}	K.m/W	[4], section 4.1.3.1
2	Thermal resistance of the sheath around each core per cable length	T'_2	K.m/W	[4], section 4.1.3.1
3	Calculation of geometric factor G	-	-	[4], section 4.1.3.1
4	Thermal resistance of the fillers and bedding for SL-type cables	T''_2	K.m/W	[4], section 4.1.3.2
5	Thermal resistance of the sheath	T_2	K.m/W	

1. Thermal resistance of the sheath around each core per core length:

$$T'_{2_core} = \frac{\rho_{wb}}{2\pi} \ln \left(\frac{D_e}{D_S} \right) \quad (K.m/W)$$

ρ_{scPE}	2,5 K.m/W	$T'_{2_core} = 0,0333994098 \text{ K.m/W}$
D_e	$62,1 \times 10^{-3} \text{ m}$	
D_S	$57,1 \times 10^{-3} \text{ m}$	

2. Thermal resistance of the sheath around each core per cable length:

$$T'_2 = \frac{T'_{2_core}}{f_{layup}} \text{ (K.m/W)}$$

T'_{2_core}	0,0333994098	$T'_2 = 0,033173331 \text{ K.m/W}$
f_{layup}	1,0068150725	

3. Calculation of geometric factor G:

Firstly, the thickness of material between sheaths and armour is expressed as a fraction of the outer diameter of the sheath.

$$X = \frac{\frac{D_{arm} - D_{cabl}}{2}}{D_e}$$

D_{arm}	$137,3 \times 10^{-3} \text{ m}$	$X = 0,019323671498$
D_{cabl}	$134,9 \times 10^{-3} \text{ m}$	
D_e	$62,1 \times 10^{-3} \text{ m}$	

As $0 < X < 0,03$:

$$G = 2\pi(0,00022619 + 2,11429X - 20,4762 X^2)$$

X	0,019323671498	$G = 0,2100854302$
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4. Thermal resistance of fillers and bending under the armour:

$$T''_2 = \frac{\rho_T}{6\pi} G \text{ (K.m/W)}$$

ρ_T	6 K.m/W	$T''_2 = 0,0668722694 \text{ K.m/W}$
G	0,2100854302	

5. The thermal resistance T_2 is equal to:

$$T_2 = \frac{T'_2}{3} + T''_2 \text{ (K.m/W)}$$

T'_2	0,033173331	$T_2 = 0,0779300464 \text{ K.m/W}$
T''_2	0,066872269	

6.9 Thermal resistance T_3 of outer covering

Table 6-15: Parameters needed to calculate the thermal resistance T_3 of the outer covering

Parameter	Symbol	Value	Unit	Reference
External diameter of the armour	D'_a	149,3	mm	Cable data sheet
Thickness of the serving	t_3	3	mm	Cable data sheet
Thermal resistivity of outer covering	ρ_T	6	K.m/W	[4], table 1

Table 6-16: Sequential steps to calculate the thermal resistance T_3 of the outer covering

Step	Parameter	Symbol	Unit	Reference
1	Thermal resistance of outer covering	T_3	$K.m/W$	[4], section 4.1.4.1

1. Thermal resistance of outer covering

$$T_3 = \frac{\rho_T}{2\pi} \ln \left(1 + \frac{2t_3}{D'_a} \right) \quad (K.m/W)$$

ρ_T	6 K.m/W	$T_3 = 0,0376252079 \text{ K.m/W}$
D'_a	$149,3 \times 10^{-3} \text{ m}$	
t_3	$3 \times 10^{-3} \text{ m}$	

6.10 External thermal resistance T_4

Table 6-17: Parameters needed to calculate the external thermal resistance T_4

Parameter	Symbol	Value	Unit	Reference
External diameter of one cable	D_e	155,3	mm	Cable data sheet
Distance from the surface of the ground to the cable axis	L	1000	mm	System design, note 1
Thermal resistivity of the soil	ρ_T	0,7	$K.m/W$	System design

Table 6-18: Sequential steps to calculate the external thermal resistance T_4

Step	Parameter	Symbol	Unit	Reference
1	External thermal resistance	T_4	$K.m/W$	[4], section 4.2.2.

1. External thermal resistance:

$$u = \frac{2L}{D_e}$$

L	1,0 m	$u = 12,8783000644$
D_e	$155,3 \times 10^{-3} \text{ m}$	

$$T_4 = \frac{1}{2\pi} \rho_T \left[\ln(u + \sqrt{u^2 - 1}) \right] \quad (K.m/W)$$

ρ_T	0,7 K.m/W	$T_4 = 0,3617633359 \text{ K.m/W}$
u	12,8783000644	

6.11 Permissible current rating

$$I = \left[\frac{\Delta\theta - W_d [0,5 \cdot T_1 + n(T_2 + T_3 + T_4)]}{R \cdot T_1 + nR(1 + \lambda_1)T_2 + nR(1 + \lambda_1 + \lambda_2)(T_3 + T_4)} \right]^{0,5}$$

$\Delta\theta$	75 °C	$I = 838,3399739336 \text{ A}$
W_d	0,1316852157 K.m/W	
T_1	0,3578707848 K.m/W	
T_2	0,0779300464 K.m/W	
T_3	0,0376252079 K.m/W	
T_4	0,3617633359 K.m/W	
R	4,2109811659 x10 ⁻⁵ Ω/m	
n	3	
λ_1	0,1628367815	
λ_2	0,4206526837	

6.12 Calculation of losses

Conductor losses:

$$P_c = nRI^2$$

R	4,2109811659 x10 ⁻⁵ Ω/m	$P_c = 88,78608438409 \text{ W/m}$
I	838,3399739336 A	
n	3	

Screen losses:

$$P_{screen} = n\lambda_1 RI^2$$

R	4,2109811659 x10 ⁻⁵ Ω/m	$P_{screen} = 14,4576402222 \text{ W/m}$
I	838,3399739336 A	
λ_1	0,1628367815	
n	3	

Armour losses:

$$P_{arm} = n\lambda_2 RI^2$$

R	4,2109811659 x10 ⁻⁵ Ω/m	$P_{arm} = 37,3481046684 \text{ W/m}$
I	838,3399739336 A	
λ_2	0,4206526837	
n	3	

Dielectric losses:

$$W_{d,t} = 3 \cdot W_d$$

W_d	0,1316852157 W/m	$W_{d,t} = 0,3950556471 \text{ W/m}$
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7. Case study 3: A 230kV HPFF cable

7.1 Introduction

This is a single pipe-type cable circuit containing three cable cores within a steel pipe. The pipe is buried at a center-line depth of 1194mm (47in) in an envelope of special backfill. The pipe-type cable consists of a 230kV class 1266,8mm² (2500kcmil) segmented (Milliken) copper conductor with paper tapes between segments, stainless steel binder tape intercalated with a paper tape, carbon black conductor screen tapes, laminar paper-polypropylene taped insulation infused with dielectric oil, carbon black insulation screen tapes, metallized polyester tapes, stainless steel tapes intercalated with polyester tapes, and two D-shaped stainless steel skid wires. For the purposes of current rating calculations, the paper tapes, conductor binder tapes and conductor screen tapes are considered together as part of the conductor screen. The insulation screen, polyester and steel tapes are also considered together as part of the insulation screen. The data sheet for the cable is shown in Figure 30.

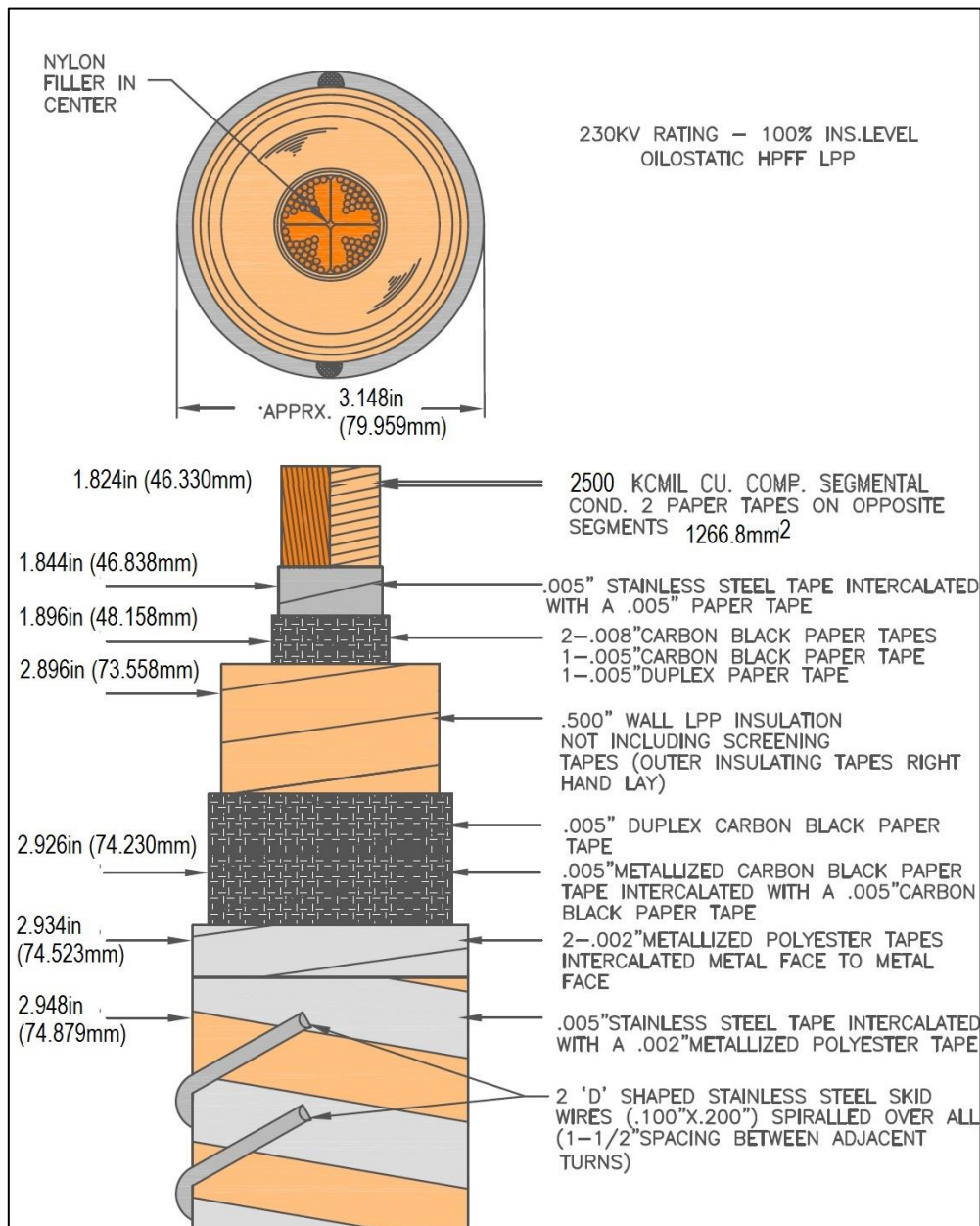


Figure 30: Cable data sheet: 1266,8mm² pipe-type cable with laminated paper polypropylene insulation and stainless steel skid wires

The cable cores are installed in cradled configuration with a 219,1mm (8,625in) carbon steel line pipe with a wall thickness of 6,25mm (0,25in) and having a 2,03mm (0,080in) polyethylene tape coating. The trench is 1000mm wide and 1350mm deep with low-thermal resistivity backfill filling the bottom 500mm of the trench around the cable pipe. The trench configuration is shown in Figure 31.

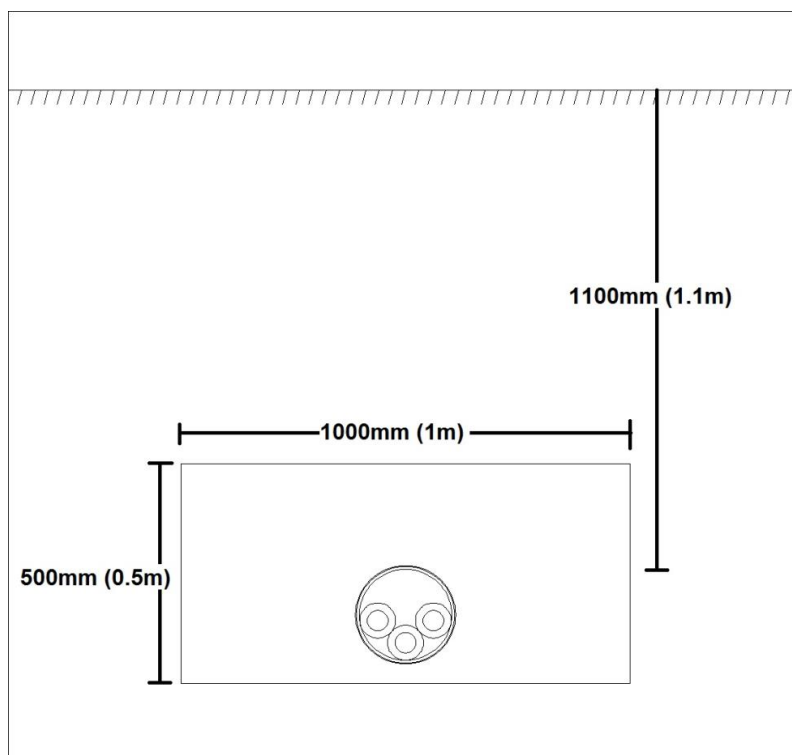


Figure 31: Pipe-type cable trench configuration and dimensions

The current rating will be determined for the following conditions:

- Native soil thermal resistivity of 1,00 K-m/Watt
- Special thermal backfill of 0,55 K-m/Watt
- Ambient soil temperature of 25°C
- Load factor of 0,80

7.2 Physical dimensions of cable system

This cable is a common type utilised in North America where dimensions are often defined in U.S. units (inches, etc.) as are illustrated on the cable data sheet shown in Figure 30. The following table summarises the physical dimensions of the cable and the metric equivalent later used for calculations.

Table 7-1: Cable system physical parameters

Dimension	Thickness (inches)	Cumulative Diameter (inches)	Thickness (mm)	Cumulative Diameter (mm)
Conductor	1,824	1,824	46,330	46,330
Conductor Screen	0,036	1,896	0,914	48,158
Insulation	0,500	2,896	12,700	73,558
Insulation Screen	0,026	2,948	0,660	74,879
Skid Wire	0,1 x 0,2	3,148	2,540 x 5,080	79,959
Skid Wire Lay	3,000	--	76,200	--
Pipe Outer	8,625	8,625	219,075	219,075
Pipe Wall	0,250	--	6,350	--
Pipe Inner		8,125	206,375	206,375
Pipe Coating	0,080	8,785	2,032	223,139*

*Earth diameter

Note about interpreting the intercalated metallized polyester tapes and metal tapes: Physically, the metal tape is both below and above the polyester tape which are helically wrapped around the outer layers of the cable. Calculation of T_1 could involve considering the polyester tape thickness as a thermal resistance above the metal tape. However, practice is to include the thickness of the polyester tape as part of the insulation thermal resistance layer, T_1 . Physically, the polyester tape would have a different thermal resistivity, but this is neglected as having a minimal impact on the T_1 thermal resistance.

In accordance with Guidance Point 42, the metal shield tape electrical resistance is calculated on the basis of the known cross-sectional area based on the tape width, tape thickness, and lay of successive turns. Because this is a helix without contact between adjacent turns, there are no eddy current losses associated with the metal tape.

In accordance with Guidance Point 43, the metallic tapes are taken electrically in parallel with the skid wires to determine the circulating current losses.

7.3 Calculation of the conductor AC resistance at operation temperature

The conductor resistance would generally be established from IEC 60228 [8], but as this is a non-standard conductor size from IEC 60228, the DC resistance will be interpolated from the two nearest conductor sizes. Absent conductor resistance tables such as IEC 60228, the conductor DC resistance may also be calculated using the conductor cross-sectional area, conductor metal electrical resistivity and an appropriate value for the conductor stranding factor (lay of conductor strands).

Table 7-2: Parameters needed to calculate the conductor AC resistance at operating temperature

Parameter	Symbol	Value	Unit	Reference
Conductor DC resistance @ 20°C (1200mm ²)	R_o	$0,0151 \times 10^{-3}$	$\frac{\Omega}{m}$	[8]
Conductor DC resistance @ 20°C (1400mm ²)	R_o	$0,0129 \times 10^{-3}$	$\frac{\Omega}{m}$	[8]
Interpolated DC resistance @ 20°C (1266,8mm ²) based on linear interpolation	R_o	$0,01436542 \times 10^{-3}$	$\frac{\Omega}{m}$	Cable data sheet
Conductor diameter	d_c	46,3296	mm	Cable data sheet
Conductor material		Copper		Cable data sheet
Type of conductor		Segmented (Milliken)		Cable data sheet
Skin effect factor	k_s	0,435	-	[3], table 2
Proximity effect factor	k_p	0,37	-	[3], table 2
Conductor temperature coefficient at 20°C per Kelvin	α	$3,93 \times 10^{-3}$	-	[3], table 1
Conductor operating temperature	θ	85	°C	Cable data sheet
Distance between conductor axes	s	$79,959 \times 10^{-3} \text{ m}$	mm	Cable data sheet (Skid wire diameter)
System frequency	f	60	Hz	System design

Table 7-3: Sequential steps to calculate the conductor AC resistance at operating temperature

Step	Parameter	Symbol	Unit	Reference
1	Conductor DC resistance at operating temperature	R'	$\frac{\Omega}{m}$	[3], section 2.1.1
2	Argument of a Bessel function – skin effect	x_s	-	[3], section 2.1.2
3	Skin effect factor	y_s	-	[3], section 2.1.2
4	Argument of a Bessel function – proximity effect	x_p	-	[3], section 2.1.3
5	Proximity effect factor	y_p	-	[3], section 2.1.3
6	AC resistance of conductor	R	$\frac{\Omega}{m}$	[3], section 2.1

1. Conductor DC resistance at operating temperature
- R'
- :

$$R' = R_o [1 + \alpha_{20}(\theta - 20)] \left(\frac{\Omega}{m} \right)$$

R_o	$0,0143653661 \times 10^{-3} \Omega/m$	$R' = 1,803499887 \times 10^{-5} \Omega/m$
α_{20}	$3,93 \times 10^{-3}$	
θ	85°C	

2. Argument of a Bessel function used to calculate skin effect
- x_s
- :

$$x_s^2 = \frac{8\pi f}{R'} 10^{-7} k_s$$

f	60 Hz	$x_s^2 = 3,6371754209$
R'	$1,803499887 \times 10^{-5} \Omega/m$	
k_s	0,435	

3. Skin effect factor
- y_s
- :

$$\text{for } 0 < x_s \leq 2,8 \quad \rightarrow \quad y_s = \frac{x_s^4}{192 + 0,8x_s^4}$$

Note: three equations are given for calculating y_s , use of these equations depends on the calculated value of x_s .

x_s^2	3,6371754209	$y_s = 0,065301776$
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4. Argument of a Bessel function used to calculate proximity effect
- x_p
- :

$$x_p^2 = \frac{8\pi f}{R'} 10^{-7} k_p$$

f	60 Hz	$x_p^4 = 9,5709143417$
R'	$1,803499887 \times 10^{-5} \Omega/m$	
k_p	0,37	

5. Proximity effect factor
- y_p
- :

$$y_p = \frac{x_p^4}{192 + 0,8x_p^4} \left(\frac{d_c}{s} \right)^2 \left[0,312 \left(\frac{d_c}{s} \right)^2 + \frac{1,18}{\frac{x_p^4}{192 + 0,8x_p^4} + 0,27} \right]$$

x_p^4	9,5709143417	$y_p = 0,0614154855$
d_c	$46,3296 \times 10^{-3} \text{ m}$	
s	$79,959 \times 10^{-3} \text{ m}$	

6. AC resistance of conductor
- R
- :

$$R = R' (1 + 1,5 \times (y_s + y_p)) \left(\frac{\Omega}{m} \right)$$

R'	$1,803438778 \times 10^{-5} \Omega/m$	$R = 2,1463017365 \times 10^{-5} \Omega/m$
y_s	0,065301776	
y_p	0,0614154855	

7.4 Dielectric losses

Table 7-4: Parameters needed to calculate the dielectric loss per unit length per phase

Parameter	Symbol	Value	Unit	Reference
External diameter of the insulation	D_i	73,558	mm	Cable data sheet
External diameter of the conductor, incl. screen	d_{csc}	48,158	mm	Cable data sheet
Relative permittivity of the insulation	ϵ	2,8	-	[3], table 3
Loss factor of the insulation	$\tan\delta$	0,0014	-	[3], table 3, note 1
System Voltage	U	230000	V	System design
Voltage to earth	U_o	132790,6	V	$\frac{U}{\sqrt{3}}$
System frequency	f	60	Hz	System design

Table 7-5: Sequential steps to calculate the dielectric loss per unit length per phase

Step	Parameter	Symbol	Unit	Reference
1	Capacitance	C	$\frac{F}{m}$	[3], section 2.2
2	Angular frequency of system	ω	-	[3], section 2.2
3	Dielectric loss per unit length in each phase	W_d	W/m	[3], section 2.2, note 1

1. Capacitance C (for circular conductors):

$$C = \frac{\epsilon}{18 \ln\left(\frac{D_i}{d_{csc}}\right)} 10^{-9} \left(\frac{F}{m}\right)$$

ϵ	2,8	$C = 3,6723655661 \times 10^{-10} \text{ F/m}$
D_i	73,558 x10 ⁻³ m	
d_{csc}	48,158 x10 ⁻³ m	

2. Angular frequency of system:

$$\omega = 2\pi f \left(\frac{\text{rad}}{\text{s}}\right)$$

f	60 HZ	$\omega = 376,9911184308 \text{ rad/s}$
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3. Dielectric loss per unit length in each phase:

$$W_d = \omega C U_o^2 \tan\delta \text{ (W/m)}$$

C	3,6723655661 x 10 ⁻¹⁰ F/m	$W_d = 3,4177435968 \text{ W/m}$
ω	376,9911184308 rad/s	
U_o	132790,6 V	
$\tan\delta$	0,0014	

7.5 Loss factor for taped screen

The resistance of the metallic screen tapes is the electrical resistivity divided by the cross-sectional area of each time times a factor to consider the helical length of a turn of one tape. When multiple tapes are included, these are taken electrically in parallel and with the skid wire(s). The cable data sheet indicates that the metallic screen tapes have a thickness of 0,127mm (0,005in). Pipe-type cables are often manufactured with tapes that are 22,225mm (0,825in) in width and are lapped 3,175mm (0,125in).

Table 7-6: Parameters needed to calculate the loss factors for taped screen

Parameter	Symbol	Value	Unit	Reference
AC resistance of screen at 20°C	R_{sc}	-	$\frac{\Omega}{m}$	Calculated
Diameter of screen tapes	D_{sc}	74,879	mm	Cable data sheet
Cross sectional area of metallic screen	A_{sc}	-	mm ²	Calculated
Thickness of the metallic screen tape	t_{sc}	0,127	mm	Cable data sheet
Metallic screen tape width	W_{sc}	22,225	mm	Industry practice
Metallic screen tapes resistivity at 20°C	ρ_{sc}	70×10^{-8}	Ωm	[3], table 1
Metallic screen tapes temperature coefficient at 20°C per Kelvin	α_{sh}	--*	-	[3], table 1
Distance between conductor axes	S	79,959	mm	Cable data sheet (Skid wire diameter)
System frequency	f	60	Hz	System design
AC resistance of screen at operating temperature	R	$2,1463017365 \times 10^{-5}$	$\frac{\Omega}{m}$	Calculated

*Stainless steel has relatively little variation in resistance with temperature over the typical operating temperature range of power cables, so the resistance at 20°C is effectively the same as the value at operating temperature. Therefore, no adjustment to the metallic screen tape resistance is necessary.

Table 7-7: Sequential steps to calculate the metallic screen tape losses per unit length per phase

Step	Parameter	Symbol	Unit	Reference
1	Calculation of the cross-sectional area of the metallic screen tape(s)			
2	Calculate helical length factor			
3	AC Resistance of screen at 20°C	R_{sc}	$\frac{\Omega}{m}$	-
4	Mutual reactance per unit length of screen and skid wire	X_m	$\frac{\Omega}{m}$	[3], section 2.3.1
5	Loss factor circulating losses for screen and skid wires	λ'_1	-	[3], section 2.3.1

1. Calculation of cross-sectional area of a metallic screen tape:

$$A_{sc} = t_{sc} \times W_{sc}$$

t_{sc}	$0,127 \times 10^{-3} m$	$A_{sc} = 2,822575 \times 10^{-6} m^2$
W_{sc}	$22,225 \times 10^{-3} m$	

2. Calculate helical length factor:

$$F_{sc} = \sqrt{1 + \left(\frac{\pi \times D_{sc}}{lay_{sc}} \right)^2} \quad (dimensionless)$$

D_{sc}	74,879 mm	$F_{sc} = 12,3889779291$
lay_{sc}	Tape width – lap = 22,225 – 3,175 = 19,050	

3. Resistance of an individual screen tape at 20°C:

AC Resistance of the screen at 20°C per unit length is:

$$R_{sc} = \frac{\rho_{sc}}{A_{sc}} F_{sc} \left(\frac{\Omega}{m} \right)$$

ρ_{sc}	$70 \times 10^{-8} \Omega m$	$R_{sc} = 3,0724726714 \frac{\Omega}{m}$
F_{sc}	12,389	
A_{sc}	$2,8222575 \times 10^{-6} m^2$	

The cable has two metallic screen tapes that will be taken electrically in parallel with the skid wires in the calculations below.

- Calculation of the reactance X per unit length of the screen tapes
This will be evaluated after determining the skid wire resistance.
- Loss factor λ'_{1sh} for circulating losses of the screen
This will be evaluated after determining the skid wire resistance.

7.6 Loss factor for skid wire

The resistance of each skid wire is the electrical resistivity of the skid wire divided by the cross-sectional area of each skid wire times a factor to consider the helical length of a single turn of a skid wire. When there is more than one skid wire, they are collectively taken electrically in parallel along with the metallic screen tapes. The cable data sheet indicates that the 'D' shaped skid wires have a width of 5,08mm and a height of 2,54mm. The lay length of one turn of each skid wire is 76.2mm.

Table 7-8: Parameters needed to calculate the skid wire(s)

Parameter	Symbol	Value	Unit	Reference
AC resistance of skid wire at 20°C	R_{sw}	-	$\frac{\Omega}{m}$	Calculated
Diameter of screen tapes	D_{sw}	79,959	mm	Cable data sheet
cross section area of skid wire	A_{sw}	-	mm ²	Calculated
Height of skid wire	H_{sw}	2,54	mm	Cable data sheet
Width of skid wire	W_{sw}	5,08	mm	Cable data sheet
Skid wire metal resistivity at 20°C	ρ_{sw}	70×10^{-8}	Ωm	[3], table 1
Skid wire temperature coefficient at 20°C per Kelvin	α_{sw}	--	-	[3], table 1
Distance between conductor axes	S	79,959	mm	Cable data sheet (Skid wire diameter)
System frequency	f	60	Hz	System design
AC resistance of screen at operating temperature	R	$2,1463017365 \times 10^{-5}$	$\frac{\Omega}{m}$	Calculated

Table 7-9: Sequential steps to calculate the skid wire losses per unit length per phase

Step	Parameter	Symbol	Unit	Reference
1	Calculation of the cross-sectional area of a single skid wire			
2	Calculate helical length factor			
3	AC Resistance of skid wire at 20°C	R_{sw}	$\frac{\Omega}{m}$	-
4	Mutual reactance per unit length of screen and skid wire	X_m	$\frac{\Omega}{m}$	[3], section 2.3.1
5	Loss factor circulating losses for screen and skid wires	λ'_1	-	[3], section 2.3.1

- Calculation of cross-sectional area of a single skid wire:

$$A_{sw} = \frac{\pi}{2} \text{radius}_{minor} \times \text{radius}_{major} \text{ (area of half an ellipse)}$$

radius_{minor}	Skid wire height = 2,54mm	
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$radius_{major}$	Skid wire width / 2 = 5,08 / 2 = 2,54mm	$A_{sw} = 10,1341495819 \times 10^{-6} \text{ m}^2$
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2. Skid wire lay factor:

$$F_{sw} = \sqrt{1 + \left(\frac{\pi \times D_{sw}}{lay_{sw}} \right)^2} \quad (\text{dimensionless})$$

D_{sw}	79,959 mm	$F_{sw} = 3,4449130312$
lay_{sw}	76,200 mm	

3. Resistance of an individual screen tape at 20°C:

AC Resistance of the screen at 20°C per unit length is:

$$R_{sw} = \frac{\rho_{sw}}{A_{sw}} F_{sw} \quad \left(\frac{\Omega}{m} \right)$$

ρ_{sw}	$70 \times 10^{-8} \Omega m$	$R_{sw} = 0,2379517988 \frac{\Omega}{m}$
F_{sw}	3,4449130312	
A_{sw}	$10,1341495819 \times 10^{-6} \text{ m}^2$	

The cable has two skid wires that will be taken electrically in parallel with the two metallic screen tapes in the calculations below. The total electrical resistance R_s of the metallic screen tapes and skid wires will now be determined. Note that there are two metallic screen tapes and two skid wires:

$$R_s = \frac{1}{\frac{n_{sc}}{R_{sc}} + \frac{n_{sw}}{R_{sw}}} \quad \left(\frac{\Omega}{m} \right)$$

R_{sc}	3,072	$R_s = 0,1104239661 \frac{\Omega}{m}$
R_{sw}	0,23	
n_{sc}	2	
n_{sw}	2	

Calculation of the mutual reactance X_m per unit length of the metallic shield and skid wire:
Mean (RMS) diameter of metallic screen and skid wire is calculated as follows:

$$d = \sqrt{\frac{\left[\frac{D_i + D_{is}}{2} \right]^2 + \left[\frac{D_{is} + D_{sw}}{2} \right]^2}{2}}$$

D_i	73,558 x10 ⁻³ m (insulation outer diameter)	$d = 75,8358846526 \times 10^{-3} \text{ m}$
D_{is}	74,879 x10 ⁻³ m (insulation screen outer diameter)	
D_{sw}	79,959 x10 ⁻³ m (skid wire outer diameter)	

4. Calculate the mutual reactance assuming cables in 'cradled' orientation within pipe:

$$X = 2\omega 10^{-7} \ln \left(\frac{2.3s}{d} \right) \quad \left(\frac{\Omega}{m} \right)$$

ω	$2\pi f = 376,9911184308 \text{ rad/s}$	$X = 6,6791820653 \times 10^{-5} \Omega/m$
s	79,959 x10 ⁻³ m (skid wire diameter)	
d	75,8358846526 x 10 ⁻³ m	

5. Loss factor λ'_1 for circulating losses of the metallic screen tapes and skid wires:

$$\lambda'_{1sh} = 1,5 \left(\frac{R_s}{R} \frac{1}{1 + \left(\frac{R_s}{X_m} \right)^2} \right)$$

R_s	0,1104239661 Ω/m	$\lambda'_1 = 0,0028234724$
R	2,1463017365 $\times 10^{-5}$ Ω/m	
X	6,6791820653 $\times 10^{-5}$ Ω/m	

7.7 Loss factor for pipe loss increment

Table 7-10: Parameters needed to calculate the incremental losses for the cable pipe

Parameter	Symbol	Value	Unit	Reference
Diameter over skid wire	D_{sw}	79,959	mm	Cable Data Sheet
Inner pipe diameter	$Pipe_{ID}$	206,375	mm	Installation Data
Conductor AC resistance	R	2,1463017365 $\times 10^{-5}$	Ω/m	Calculated

Table 7-11: Sequential steps to calculate the pipe loss increment

Step	Parameter	Symbol	Unit	Reference
1	Loss factor circulating losses for pipe	λ'_2	-	[3], section 2.4.3

1. Loss factor λ'_2 for circulating losses of the pipe, assuming cradle configuration of the cores:

$$\lambda'_2 = \frac{0,00438 \times D_{sw} + 0,00226 \times Pipe_{ID}}{R} \cdot 10^{-5}$$

D_{sw}	79,959	$\lambda'_2 = 0,3804818223$
R	2,1463017365 $\times 10^{-5}$ Ω/m	
$Pipe_{ID}$	206,375	

7.8 Thermal resistance T_1 between conductor and screen

Table 7-12: Parameters needed to calculate the thermal resistance between conductor and screen

Parameter	Symbol	Value	Unit	Reference
Diameter of conductor	d_c	46,3296	mm	Cable data sheet
External diameter of insulation screen	D_{is}	74,879	mm	Cable data sheet
Thermal resistivity of insulation	ρ_i	5,5	$K.m/W$	[4], table 1

Table 7-13: Sequential steps to calculate the thermal resistance between conductor and screen

Step	Parameter	Symbol	Unit	Reference
1	Thermal resistance between conductor and screen	T_1	$K \cdot \frac{m}{W}$	-

1. Thermal resistance of insulation:

$$T_1 = \frac{\rho_i}{2\pi} \ln \left(\frac{D_{is}}{d_c} \right) \left(K \cdot \frac{m}{W} \right)$$

ρ_i	5,5 K.m/W	
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D_{is}	74,879 x 10 ⁻³ m	$T_1 = 0,4202522806 \text{ K.m/W}$
d_c	46,3296 x 10 ⁻³ m	

7.9 Thermal resistance T_2 between the cable surface and inside of pipe

Table 7-14: Parameters needed to calculate the thermal resistance between the cable surface and inside of pipe

Parameter	Symbol	Value	Unit	Reference
Constant, oil pressure pipe-type cable	U	0,26	--	[4], table 4
Constant, oil pressure pipe-type cable	V	0	--	[4], table 4
Constant, oil pressure pipe-type cable	Y	0,0026	--	[4], table 4
Mean oil temperature	θ_m	68,1476575	°C	Iterated calculation
Number of cables within pipe	n	3	--	Cable data sheet
Diameter over insulation screen	D_{is}	74,879	mm	Cable data sheet
Effective diameter of cable	D_e	160,990	mm	$2,15 \times D_{is}$

Table 7-15: Sequential steps to calculate the thermal resistance between the cable surface and inside of pipe

Step	Parameter	Symbol	Unit	Reference
1	Thermal resistance between the cable surface and inside pipe (oil thermal resistance)	T_2	$K \cdot \frac{m}{W}$	[4], section 2.2.7.1

1. Thermal resistance between the cable surface and inside pipe (oil thermal resistance):

$$T_2 = \frac{U}{1 + 0,1(V + Y \times \theta_m) \times D_e} \left(K \cdot \frac{m}{W} \right)$$

U	0,26	$T_2 = 0,0674888479 \text{ K.m/W}$
V	0	
Y	0,0026	
θ_m	68,1476485886	
n	3	
D_e	160,990	

7.10 Thermal resistance T_3 of outer covering (pipe coating)

Table 7-16: Parameters needed to calculate the thermal resistance T_3 of the pipe coating

Parameter	Symbol	Value	Unit	Reference
External pipe diameter	D_{pipe}	219,075	mm	Cable data sheet
External diameter of outer covering	D_e	223,139	mm	Cable data sheet
Thermal resistivity of outer covering	$\rho_{coating}$	3,5	K.m/W	[4], table 1

Table 7-17: Sequential steps to calculate the thermal resistance T_3 of the outer covering

Step	Parameter	Symbol	Unit	Reference
1	Thermal resistance of outer covering	T_3	$K \cdot \frac{m}{W}$	[4], section 4.1.4.1

Thermal resistance of outer pipe coating, which is the outside of the pipe, D_{pipe} (inner surface of pipe coating) to the earth diameter, D_e (outer diameter of pipe coating):

$$T_3 = \frac{\rho_{coating}}{2\pi} \ln\left(\frac{D_e}{D_{pipe}}\right) \quad (K.m/W)$$

$\rho_{coating}$	3,5 K.m/W	$T_3 = 0,0102388601 \text{ K.m/W}$
D_{pipe}	$219,075 \times 10^{-3} \text{ m}$	
D_e	$223,139 \times 10^{-3} \text{ m}$	

7.11 External thermal resistance T_4

The calculation of the external thermal resistance includes the thermal resistance for a single cable pipe buried below ground and consideration for the correction to account for the difference between the native soil thermal resistivity outside of the cable trench and the special backfill immediately around the cable pipe.

The methodology used in the following calculation also considers the daily loss factor for AC losses using the methodology outlined in the J.H. Neher and M.H. McGrath paper on the subject of continuous cable ratings. The 'Neher/McGrath' method for considering the daily loss factor applies to the components of the environment (soil) thermal resistance falling outside of a diameter, D_x , beyond which the average heating of the soil is experienced; the daily loss factor is applied to those components.

Table 7-18: Parameters needed to calculate the external thermal resistance T_4

Parameter	Symbol	Value	Unit	Reference
External diameter of cable pipe	D_e	223,139	mm	Cable data sheet
Distance from the surface of the ground to the cable pipe axis	L	1194	mm	System design
Thermal resistivity of the soil	ρ_T	1,0	K.m/W	System design
Thermal resistivity of special backfill	ρ_b	0,6	K.m/W	System design
Daily load factor	LF	0,8	--	System design
Width of special backfill	w_b	1000	mm	Installation design
Height of special backfill	h_b	500	mm	Installation design
Depth to center of special backfill	L_b	1100	mm	Installation design
Number of pipe-type circuits in trench	N	1	--	System design
Number of cables within each pipe	n	3	--	System design

Table 7-19: Sequential steps to calculate the external thermal resistance T_4

Step	Parameter	Symbol	Unit	Reference
1	Determine effective diameter of backfill	D_b	mm	Calculation
2	Determine geometric correction factor for special backfill	G_b	--	Calculation
3	Calculate thermal diffusivity of soil	α_{soil}	mm ² / hr	Calculation
4	Diameter where average daily heating is experienced	D_x	mm	Calculation
5	Determine loss factor from load factor	μ	--	Calculation
6	External thermal resistance, single cable for ac losses	T_4	K. $\frac{m}{W}$	[4], section 4.2.2.
7	External thermal resistance, single cable for dielectric losses (100% load factor)	T_{4d}	K. $\frac{m}{W}$	[4], section 4.2.2.
8	Thermal resistance correction for native soil outside of trench for ac losses	T_{4c}	K. $\frac{m}{W}$	Calculation
9	Thermal resistance correction for native soil outside of trench for dielectric losses	T_{4c-d}	K. $\frac{m}{W}$	Calculation

1. Determine effective diameter for backfill:

$$D_b = \exp\left(\left(\frac{1}{2}\right)\left(\frac{x}{y}\right)\left(\frac{4}{\pi} - \frac{x}{y}\right)\ln\left(1 + \frac{y^2}{x^2}\right) + \ln(x)\right)$$

Note that this formula is equal to [4], section 4.2.7.4 which calculates the radius instead of the diameter.

x	500 mm (short dimension of backfill)	$D_b = 682,47668008 \text{ mm}$
y	1000 mm (long dimension of backfill)	

2. Determine geometric correction factor for special backfill:

$$G_b = \ln\left(\frac{2 \times L_b + \sqrt{4 \times L_b^2 - D_b^2}}{D_b}\right)$$

D_b	682,477668008	$G_b = 1,8386550726$
L_b	1100	

3. Thermal diffusivity of soil:

$$\alpha_{soil} = \frac{6,71 \times 10^4}{(\rho_T \times 100)^{0,8}} \quad \frac{mm^2}{hour}$$

ρ_T	1,0	$\alpha_{soil} = 1685,54757955 \frac{mm^2}{hour}$
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Note that IEC 60853-2 includes a table of thermal diffusivity values that may be applied to calculations. The method illustrated here is a method to determine the thermal diffusivity based on the native soil thermal resistivity using an empirical equation from [1].

4. Diameter beyond which average daily loading is experienced:

$$D_x = 1,02 \sqrt{\alpha_{soil} \times 24} \quad mm$$

This value is based on a calculation in the 1957 paper by Neher / McGrath [1].

α_{soil}	1685,5	$D_x = 205,1478891541$
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Note: A consideration when evaluating diameter, D_x , is that this diameter may be smaller than the outer diameter of the cable pipe. Intuitively, this seems physically inconsistent. However, the current rating method considers that all cables are line heat sources (heat producing without an actual diameter) in uniform soil. As a result, D_x being smaller than the cable pipe size is allowed. The impact of D_x being smaller than the cable pipe is that the second term in the calculation of T_4 may be negative. Intuitively, this seems inappropriate, but the calculation for the earth thermal resistance is based on considering that each of the cables is a line heat source with no diameter. The reader should also note that the earth thermal resistance, T_4 , actually consists of both terms together with the 'correction' term (discussed below) for native soil having a different thermal resistance from the special backfill in the trench, the result cannot be negative.

5. Calculate daily loss factor from daily load factor (note: the expression below was developed empirically and is applicable to transmission class cables):

$$\mu = 0,3(LF) + 0,7(LF)^2$$

LF	0,8	$\mu = 0,688$
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6. External (environment) thermal resistance for ac losses:

$$T_4 = \frac{\rho_b}{2\pi} \left[\ln\left(\frac{D_x}{D_e}\right) + \mu \times \ln\left(\frac{2 \times L + \sqrt{4 \times L^2 - D_x^2}}{D_x}\right) \right] \quad K \cdot \frac{m}{W}$$

D_e	223,139 mm	$T_4 = 0,1820833544$
D_x	205,148 mm	
μ	0,688	
L	1194 mm	
ρ_b	0,55 K.m/W	

7. External (environment) thermal resistance for dielectric losses (100% loss factor):

$$T_{4d} = \frac{\rho_b}{2\pi} \left[\ln \left(\frac{D_x}{D_e} \right) + (1,0) \times \ln \left(\frac{2 \times L + \sqrt{4 \times L^2 - D_x^2}}{D_x} \right) \right] \quad K \cdot \frac{m}{W}$$

D_e	223,139 mm	$T_{4d} = 0,2679930508$
D_x	205,148 mm	
L	1194 mm	
ρ_b	0,55 K.m/W	

8. Thermal resistivity correction for native soil outside of trench for ac losses:

$$T_{4c} = \mu \frac{\rho_T - \rho_b}{2\pi} N \times G_b \quad K \cdot \frac{m}{W}$$

ρ_T	1,0 K.m/W	$T_{4c} = 0,0905985711$
ρ_b	0,55 K.m/W	
μ	0,688	
G_b	1,8386550726	
N	1	
n	3	

Note: in situations where the native soil has a lower thermal resistivity than the backfill material, the value of T_{4c} will be negative. However, T_{4c} is considered together with T_4 to give the total earth thermal resistance, and the sum of the terms will always be positive.

9. Thermal resistivity correction for native soil outside of trench for dielectric losses:

$$T_{4c-d} = (1,0) \frac{\rho_T - \rho_b}{2\pi} N \times G_b \quad K \cdot \frac{m}{W}$$

ρ_T	1,0 K.m/W	$T_{4c-d} = 0,1316839696$
ρ_b	0,55 K.m/W	
G_b	1,8386550726	
N	1	
n	3	

7.12 Permissible current rating

$$I = \left[\frac{\Delta\theta - W_d [0,5 \cdot T_1 + n \cdot (T_2 + T_3 + T_{4d} + T_{4c-d})]}{R \cdot [T_1 + n \cdot (1 + \lambda_1) \cdot T_2 + n \cdot (1 + \lambda_1 + \lambda_2') (T_3 + T_4 + T_{4c})]} \right]^{0,5}$$

$\Delta\theta$	85 – 25 = 60 °C	$I = 1187,3557487000 \text{ A}$
W_d	3,4177435968 K.m/W	
T_1	0,4202522806 K.m/W	
T_2	0,0674888479 K.m/W	
T_3	0,0102388601 K.m/W	
T_4	0,1820833544 K.m/W	
T_{4d}	0,2679930508 K.m/W	
T_{4c}	0,0905985711 K.m/W	
T_{4c-d}	0,1316839696 K.m/W	
R	2,1463017365 x10 ⁻⁵ Ω/m	
λ_1	0,0028234724, e.g., = λ_1' (note the different notation)	
λ_2'	0,3804818223	

7.13 Calculation of losses

Conductor losses from individual cable phases:

$$P_c = RI^2$$

R	2,1463017365 x10 ⁻⁵ Ω/m	$P_c = 30,2588553658 \text{ W/m}$
I	1187,3557487 A	

Metallic screen and skid wire losses per cable:

$$P_{screen} = \lambda_1 RI^2$$

R	2,1463017365 x10 ⁻⁵ Ω/m	$P_{screen} = 0,0854350427 \text{ W/m}$
I	1187,3557487 A	
λ_1	0,0028234724	

Pipe losses caused by each cable:

$$P_{pipe} = \lambda_2 RI^2$$

R	2,1463017365 x10 ⁻⁵ Ω/m	$P_{pipe} = 11,5129444316 \text{ W/m}$
I	1187,3557487 A	
λ_2	0,3804818223	

7.14 Calculation of cable temperatures

Screen and skid wire temperature at rated temperature:

$$\theta_{screen} = \theta_{conductor} - T_1 \left(P_c + \frac{W_d}{2} \right)$$

$\theta_{conductor}$	85°C	$\theta_{screen} = 71,565489754^\circ\text{C}$
T_1	0,4202522806 K.m/W	
P_c	30,2588553658 W/m	
W_d	3,4177435968 W/m	

Mean oil temperature between cable surface and inside of pipe (assumed to be half way through the thermal resistance of the dielectric oil):

$$\theta_m = \theta_{screen} - \frac{n \cdot T_2}{2} (P_c + P_{screen} + W_d)$$

θ_{screen}	71,565489754°C	$\theta_m = 68,1476485886^\circ\text{C}$
T_2	0,0674888479 K.m/W	
P_c	30,2588553658 W/m	
P_{screen}	0,0854350427 W/m	
W_d	3,4177435968 W/m	

Earth interface temperature:

$$\theta_{ei} = \theta_{screen} - \frac{n \cdot T_2}{2} (P_c + P_{screen} + W_d) - n \cdot T_3 (P_c + P_{screen} + P_{pipe} + W_d)$$

θ_{screen}	71,565489754°C	$\theta_{ei} = 63,3391149086^\circ\text{C}$
T_2	0,0674888479 K.m/W	
T_3	0,0102388601 K.m/W	
P_c	30,2588553658 W/m	
P_{screen}	0,0854350427 W/m	
P_{pipe}	11,5129444316 W/m	
W_d	3,4177435968 W/m	

8. Case study 4: A 33kV land cable

8.1 Introduction

This example is to show an underground 33 kV land cable design with less than 50 % cover of screen wires mainly to show the use of IEC 60287.2.1 [4] section 4.2.4.3.3 for with multiplication factors for calculation of T_1 and T_3 in this special combination.

Results and intermediate results are given with 10 significant digits but full accuracy of calculation software is used for calculation.

The cable design is as shown in Figure 32 and is made as a generic design from several manufactures designs. Table 8-1 provides the cable design details and Table 8-2 provides all material dimensions and thermal resistivities. Together, these form the cable data sheet.

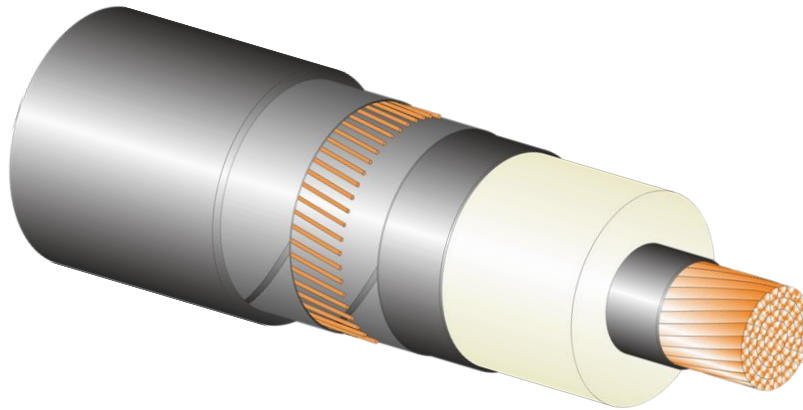


Figure 32: 33kV land cable, extruded, single core, copper wires screen

Table 8-1: 33kV land cable dimensions and design

Design aspect	Symbol	Value	Unit
Voltage Phase to phase	-	33	kV
Conductor, Cross section	-	240	mm ²
Conductor material	-	Cu	-
Conductor type	-	Round compacted, stranded	-
Conductor resistance DC at 20 °C	-	$75,4 \times 10^{-6}$	Ohm/m
Tape over conductor type/material	-	None	-
Insulation, type	-	XLPE unfilled	-
Maximum conductor temperature	θ_c	90	°C
Insulation permittivity	ϵ	2,5	-
Insulation loss angle	$\tan\delta$	0,004	-
Tape under metallic screen/sheath type	-	Semiconducting	-
Wire screen, material	-	Cu	-
Wire screen, nominal cross section	-	35	mm ²
Wire screen, Wire diameter	d_s	0,9	mm
Wire screen length of lay	L_s	240	mm
Wire screen Actual number of wires	n_s	56	#
Wire screen actual cross section (used)	A_s	35,63	mm ²
Counter helix	-	None	y/n
Outer sheath material	-	HDPE with semicon layer	-

Table 8-2: Cable material dimensions and thermal resistivities

Thickness and diameters	Thermal resistivity ρ (Km/W)	Thickness of layer t (mm)	Diameter over D (mm)
Conductor (c)	-	-	18,4
Conductor Screen (cs)	2,5	0,5	19,4
Insulation (i)	3,5	7,7	34,8
Insulation Screen (is)	2,5	0,5	35,8
Tape 1 (Semiconducting bedding) tape1	6	0,5	36,8
Wire Screen (ws)	-	0,9	38,6
Tape 2 (tape2)	6	0,3	39,2
PE sheath (incl. glue from laminate if any)	3,5	2,2	43,6
Semiconducting outer layer sheath	2,5	0,2	44,0

Note 1: Tape 1 and Tape 2 are considered as bedding tapes, not water blocking tapes, therefore the thermal resistivity for both tapes is 6,0 K.m/W

8.2 Installation conditions

The installation conditions are as shown in Table 8-3.

Table 8-3: 33kV land cable installation conditions

	Unit	Value	Symbol
Installation method	-	Buried	
Layout	-	Trefoil	
Burial depth centre of trefoil	mm	1000	L
CC distance	close or mm	Close	
Thermal resistivity of soil	K·m/W	1,0	ρ
Ambient temperature	°C	20	θ_a
Bonding		Both end	

8.3 Calculation of the conductor AC resistance at operation temperature

Table 8-4: Parameters needed to calculate the conductor AC resistance at operating temperature

Parameter	Symbol	Value	Unit	Reference
Conductor DC resistance @ 20°C	R_o	$0,0754 \times 10^{-3}$	Ω/m	Cable data sheet
Conductor diameter	d_c	18,4	mm	Cable datasheet
Conductor material		Copper		Cable datasheet
Type of conductor		Round stranded compacted, [8]		Cable datasheet
Skin effect factor	k_s	1	-	[3], table 2
Proximity effect factor	k_p	1	-	[3], table 2
Conductor temperature coefficient at 20°C per Kelvin	α	$3,93 \times 10^{-3}$	-	[3], table 1
Conductor operating temperature	θ	90	°C	Cable datasheet
Distance between conductor axes	s	44,0	mm	Cable datasheet = D_e since cable in close trefoil
System frequency	f	50	Hz	System design

Table 8-5: Sequential steps to calculate the conductor AC resistance at operating temperature

Step	Parameter	Symbol	Unit	Reference
1	Conductor DC resistance at operating temperature	R'	Ω/m	[3], section 2.1.1
2	Argument of a Bessel function – skin effect	x_s	-	[3], section 2.1.2
3	Skin effect factor	y_s	-	[3], section 2.1.2
4	Argument of a Bessel function – proximity effect	x_p	-	[3], section 2.1.3
5	Proximity effect factor	y_p	-	[3], section 2.1.3
6	AC resistance of conductor	R	Ω/m	[3], section 2.1

1. Conductor DC resistance at operating temperature R' :

$$R' = R_o [1 + \alpha_{20}(\theta - 20)] \quad (\Omega/m)$$

R_o	$0,0754 \times 10^{-3} \Omega/m$	$R' = 9,614254000 \times 10^{-5} \Omega/m$
α_{20}	$3,93 \times 10^{-3}$	
θ	90°C	

2. Argument of a Bessel function used to calculate skin effect x_s :

$$x_s^2 = \frac{8\pi f}{R'} 10^{-7} k_s$$

f	50 Hz	$x_s^2 = 1,3070562328$
R'	$9,614254000 \times 10^{-5} \Omega/m$	
k_s	1	

3. Skin effect factor y_s :

$$\text{for } 0 < x_s \leq 2,8 \quad \rightarrow \quad y_s = \frac{x_s^4}{192 + 0,8x_s^4}$$

Note: three equations are given for calculating y_s , use of these equations depends on the calculated value of x_s .

x_s^2	$1,3070562328$	$y_s = 8,835005445 \times 10^{-3}$
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4. Argument of a Bessel function used to calculate proximity effect x_p :

$$x_p^2 = \frac{8\pi f}{R'} 10^{-7} k_p$$

f	50 Hz	$x_p^2 = 1,3070562328$
R'	$9,614254000 \times 10^{-5} \Omega/m$	
k_p	1	

5. Proximity effect factor y_p :

$$y_p = \frac{x_p^4}{192 + 0,8x_p^4} \left(\frac{d_c}{s} \right)^2 \left[0,312 \left(\frac{d_c}{s} \right)^2 + \frac{1,18}{\frac{x_p^4}{192 + 0,8x_p^4} + 0,27} \right]$$

x_p^2	$1,3070562328$	$y_p = 6,622704052 \times 10^{-3}$
d_c	$18,4 \text{ mm}$	
s	$44,0 \text{ mm}$	

6. AC resistance of conductor R :

$$R = R'(1 + \gamma_s + \gamma_p)(\Omega/m)$$

R'	$9,6142500000 \times 10^{-5} \Omega/m$	$R = 9,762868345 \times 10^{-5} \Omega/m$
γ_s	$8,835005445 \times 10^{-3}$	
γ_p	$6,622704052 \times 10^{-3}$	

8.4 Dielectric losses

Dielectric losses can be neglected for this voltage level but have been included as discussed in Guidance Point 7.

Table 8-6: Parameters needed to calculate the dielectric loss per unit length per phase

Parameter	Symbol	Value	Unit	Reference
External diameter of the insulation	D_i	34,8	mm	Cable data sheet
External diameter of the conductor, incl. screen	$d_{c.sc}$	19,4	mm	Cable datasheet
Relative permittivity of the insulation	ϵ	2,5	-	[3], table 3
Loss factor of the insulation	$\tan\delta$	0,004	-	[3], table 3, note 1 (up to 35 kV)
Voltage to earth	U_o	$33kV/\sqrt{3}=19052,5588$	V	System design ($33kV/\sqrt{3}$)
System frequency	f	50	Hz	System design
Note 1: Always assume unfilled XLPE unless specified otherwise, refer to Guidance Point 18				

Table 8-7: Sequential steps to calculate the dielectric loss per unit length per phase

Step	Parameter	Symbol	Unit	Reference
1	Capacitance	C	F/m	[3], section 2.2
2	Angular frequency of system	ω	-	[3], section 2.2
3	Dielectric loss per unit length in each phase	W_d	W/m	[3], section 2.2, note 1
Note 1: According to [3], table 3, the dielectric losses may be omitted in the calculation. In this case, the dielectric losses are included, refer to Guidance Point 7				

1. Capacitance C (for circular conductors):

$$C = \frac{\epsilon}{18 \ln\left(\frac{D_i}{d_{c.sc}}\right)} 10^{-9} \quad (F/m)$$

D_i	34,8 mm	$C = 0,237683304 \times 10^{-9} F/m$
$d_{c.sc}$	19,4 mm	
ϵ	2,5	

2. Angular frequency of system:

$$\omega = 2\pi f$$

3. Dielectric loss per unit length in each phase:

$$W_d = \omega C U_o^2 \tan\delta \quad (W/m)$$

ω	$=2\pi f = 314,16 \text{ rad/s}$	$W_d = 0,10842143853 \text{ W/m}$
C	$0,237683304 \times 10^{-9} \text{ F/m}$	
U_o	$33\text{kV}/\sqrt{3} = 19052,5588 \text{ V}$	
$\tan\delta$	$0,004$	

8.5 Loss factor for Screen

Table 8-8: Parameters needed to calculate the loss factor for PB sheath

Parameter	Symbol	Value	Unit	Reference
AC resistance of sheath at 20°C	$R_{so,s}$	-	Ω/m	Calculated
Diameter under the screen	D_s	36,8	mm	Cable data sheet
Number of screen wire	n_s	56	#	Cable data sheet
Length of lay of the screen wires	L_s	240	mm	Cable data sheet
Actual cross sectional area of screen	A_s	-	mm ²	Calculated
Diameter of the screen wires	d_s	0,9	mm	Cable data sheet
Sheath resistivity at 20°C	ρ_s	$1,7241 \times 10^{-8}$	Ωm	[3], table 1
Sheath temperature coefficient at 20°C per Kelvin	α_s	$3,93 \times 10^{-3}$	1/K	[3], table 1
Distance between conductor axes	S	44,0	mm	System design
System frequency	f	50	Hz	System design
AC resistance of conductor at operating temp.	R	-	Ω/m	Calculated

Table 8-9: Sequential steps to calculate the sheath losses per unit length per phase

Step	Parameter	Symbol	Unit	Reference
1	Calculation of the cross-sectional area of the screen wires	A_s	mm ²	(geometry)
2	Calculation of the lay factor of screen wires	LF_s		(geometry)
3	AC Resistance of screen at 20°C	$R_{so,s}$	Ω/m	-
4	Maximum operating temperature of the cable screen, to be determined iterative	θ_s	°C	[3], section 2.3
5	Resistance of sheath at operating temperature	$R_{s,s}$	Ω/m	[3], section 2.3
6	Reactance per unit length of screen	X	Ω/m	[3], section 2.3.1
7	Loss factor circulating losses for sheath	$\lambda'_{1,s}$	-	[3], section 2.3.1

1. Calculation of cross-sectional area of the sheath:

$$A_s = n_s \pi \frac{d_s^2}{4}$$

n_s	56	$A_s = 35,62566069 \times 10^{-6} \text{ m}^2$
d_s	$0,9 \times 10^{-3} \text{ m}$	

2. Lay factor of the screen wires:

$$LF_s = \sqrt{1 + \frac{[\pi(D_s + d_s)]^2}{L_s^2}}$$

D_s	36,8 mm	$LF_s = 1,115138647$
d_s	0,9 mm	
L_s	240 mm	

3. Resistance of screen at 20°C:

AC Resistance of the screen at 20°C per unit length is:

$$R'_{so-s} = LF_s \frac{\rho_{sh}}{A_{PB}} \quad (\Omega/m)$$

ρ_s	$1,7241 \times 10^{-8} \Omega m$	$R'_{so-s} = 5,396701434 \times 10^{-4} \Omega/m$
A_s	$35,62566069 \times 10^{-6} m^2$	
LF_s	1,115138647	

4. The operating temperature θ_s (°C) of the screen is given by:

$$vv\theta_s = \theta - (I^2 R + 0,5W_d)T_1 \quad (^\circ C)$$

θ	90 °C	$\theta_s = 78,3859241165 \text{ } ^\circ C$
I	537,4631099368 A	
R	$9,762868345 \times 10^{-5} \Omega/m$	
W_d	0,10842143853 W/m	
T_1	0,4110322351 K.m/W	

Note: θ_{sc} is the operating temperature of the cable screen (°C). Because the temperature of the screen is a function of the current I, an iterative method must be used for the calculation.

5. Cu wires screen AC resistance at operating temperature θ_s :

$$R_{s-sh} = R'_{so-s} [1 + \alpha_{sc}(\theta_{sc} - 20)] \quad (\Omega/m)$$

R'_{so-s}	$5,396701434 \times 10^{-4} \Omega/m$	$R_s = 6,635010637 \times 10^{-4} \Omega/m$
α_s	$3,93 \times 10^{-3} 1/K$	
θ_s	78,3859241165 °C	

Note: Iteration of screen temperature was stopped at 8 significant digits.

6. Calculation of the reactance X per unit length of the screen:

Mean diameter of sheath:

$$d = (D_s + d_s) (m)$$

D_s	36,8 mm	$d = 37,7 \times 10^{-3} m$
d_s	0,9 mm	

$$X = 2\omega 10^{-7} \ln\left(\frac{2s}{d}\right) \quad (\Omega/m)$$

ω	$2\pi f = 314,159265359 \text{ rad/s}$	$X = 5,326109912 \times 10^{-5} \Omega/m$
s	$44,0 \times 10^{-3}$	
d	$37,7 \times 10^{-3} m$	

7. Loss factor λ'_{1-s} for circulating losses of the screen:

$$\lambda'_{1-s} = \frac{R_s}{R} \frac{1}{1 + \left(\frac{R_s}{X}\right)^2}$$

R_s	$6,635010635 \times 10^{-4} \Omega/\text{m}$	$\lambda'_{1-s} = 0,0435122656$
R	$9,762868345 \times 10^{-5} \Omega/\text{m}$	
X	$5,326109912 \times 10^{-5} \Omega/\text{m}$	

Since there is no sheath over the screen wires or even equalising tape there are no eddy current losses according to [3], section 2.3.3 where $\lambda''_{1-s} = 0$

Therefore:

$$\lambda_1 = \lambda'_{1-s} = 0,0435122656$$

8.6 Thermal resistance T_1 between conductor and screen

Table 8-10: Parameters needed to calculate the thermal resistance between conductor and screen

Parameter	Symbol	Unit	Reference
diameter of conductor	d_c	<i>mm</i>	Cable datasheet
External diameter of sc conductor screen	D_{sc}	<i>mm</i>	Cable datasheet
External diameter of insulation	D_i	<i>mm</i>	Cable data sheet
External diameter of sc insulation screen	D_{si}	<i>mm</i>	Cable data sheet
External diameter of sc tape(s) under the copper wires screen	D_{tuws}	<i>mm</i>	Cable data sheet
Thermal resistivity of sc screen	ρ_{sc}	<i>K.m/W</i>	Note 1
Thermal resistivity of insulation	ρ_i	<i>K.m/W</i>	[4], table 1
Thermal resistivity of sc tape	ρ_{wb}	<i>K.m/W</i>	Note 2
Lay factor of wire screen	LF_s		above
Diameter of the screen wires	d_s		Cable data sheet
Number of wires in wires screen	n_s		Cable data sheet
Note 1: In this case: 2.5 K.m/W			
Note 2: In this case: 6 K.m/W			

Table 8-11: Sequential steps to calculate the thermal resistance between conductor and screen

Step	parameter	Symbol	Unit	Reference
1	Thermal resistance of sc conductor screen	T_{1-sc}	<i>K.m/W</i>	[4], section 4.1.2.1
2	Thermal resistance of insulation	T_{1-i}	<i>K.m/W</i>	[4], section 4.1.2.1
3	Thermal resistance of sc insulation screen	T_{1-si}	<i>K.m/W</i>	[4], section 4.1.2.1
4	Thermal resistance of sc tape under the copper wires screen	T_{1-uwbt}	<i>K.m/W</i>	[4], section 4.1.2.1
5	Thermal resistance between conductor and screen	T_1	<i>K.m/W</i>	-
6	Include correction factor			[4], section 4.2.4.3.3

1. Thermal resistance of sc conductor screen:

$$T_{1-sc} = \frac{\rho_{sc}}{2\pi} \ln \left[\frac{D_{sc}}{d_c} \right]$$

ρ_{sc}	2,5 K.m/W	$T_{1_{sc}} = 0,02105715448 \text{ K.m/W}$
D_{sc}	19,4 mm	
d_c	18,4 mm	

2. Thermal resistance of insulation:

$$T_{1_i} = \frac{\rho_i}{2\pi} \ln \left[\frac{D_i}{D_{sc}} \right] \quad (K.m/W)$$

ρ_i	3,5 K.m/W	$T_{1_i} = 0,3255045049 \text{ K.m/W}$
D_i	34,8 mm	
D_{sc}	19,4 mm	

3. Thermal resistance of sc insulation screen:

$$T_{1_{si}} = \frac{\rho_{sc}}{2\pi} \ln \left[\frac{D_{si}}{D_i} \right] \quad (K.m/W)$$

ρ_{sc}	2,5 K.m/W	$T_{1_{si}} = 0,011272350425 \text{ K.m/W}$
D_{si}	35,8 mm	
D_i	34,8 mm	

4. Thermal resistance of sc tape under the copper wires screen:

$$T_{1_{uwb}} = \frac{\rho_{tusw}}{2\pi} \ln \left[\frac{D_{tuws}}{D_{si}} \right] \quad (K.m/W)$$

ρ_{tusw}	6,0 K.m/W	$T_{1_{uwb}} = 0,02630826604 \text{ K.m/W}$
D_{tuws}	36,8 mm	
D_{si}	35,8 mm	

5. Thermal resistance between conductor and screen:

$$T'_1 = T_{1_{sc}} + T_{1_i} + T_{1_{si}} + T_{1_{uwb}} \quad (K.m/W)$$

$T_{1_{sc}}$	0,02105715448 K.m/W	$T'_1 = 0,3841422758 \text{ K.m/W}$
T_{1_i}	0,3255045049 K.m/W	
$T_{1_{si}}$	0,011272350425 K.m/W	
$T_{1_{uwb}}$	0,02630826604 K.m/W	

6. Correction factor:

Calculation of degree of cover:

$$\frac{d_s \cdot n_s}{\pi(D_{tuws} + d_s)} LF_s$$

d_s	0,9 mm	47,5 %
n_s	56	
D_{tuws}	36,8 mm	
LF_s	1,115138647	

Since the degree of cover of screen wires is less than 50% it cannot be calculated as an isotherm according to IEC 60287.2.1 [4] section 4.2.4.3.3 and the calculated T'_1 must be multiplied with a factor of 1.07 (up to 35 kV) and T_3 with a factor of 1.6, the clause assumed a long length of

lay of (15 D_{tuws}), longer then in the one in the example, but there is no instructions in what is to be used in case of shorter length of lay (only $L_s/D_{tuws} = 6,2$) so the same ratio is used. Hence,

$$T_1 = 1,07 T'_1 = 1,07 \times 0,3841422758 \text{ K.m/W} = 0,4110322351 \text{ K.m/W}$$

8.7 Thermal resistance T_2 between screen and armour

$T_2 = 0$ since there is no armour or outer metallic layers.

8.8 Thermal resistance T_3 of outer covering

Table 8-12: Parameters needed to calculate the thermal resistance T_3 of the outer covering

Parameter	Symbol	Unit	Reference
External diameter of the screen wires	D_s	mm	Cable data sheet
External diameter of tape over the screen wires	D_{tosw}	mm	Cable data sheet
External diameter of outer covering without semi conductive layer	D_{oc}	mm	Cable data sheet
External diameter semi conductive layer over outer covering	D_e	mm	Cable data sheet
Thermal resistivity of sc tape over wire screen	ρ_{tows}	K.m/W	Note1
Thermal resistivity of outer covering	ρ_{oc}	K.m/W	[4], table 1
Thermal resistivity of semi conductive layer over outer covering	ρ_{oc_sc}	K.m/W	Note 2
Note 1: In this case 6 K.m/W			
Note 2: In this case: 2,5 K.m/W			

Table 8-13: Sequential steps to calculate the thermal resistance T_3 of the outer covering

Step	parameter	Symbol	Unit	Reference
1	Thermal resistance of tape over wire screen	T_{3_tows}		
2	Thermal resistance of outer covering	T_{3_oc}	K.m/W	[4], section 4.1.4.1
3	Thermal resistance of semi conductive layer over outer covering	T_{3_sc}	K.m/W	[4], section 4.1.4.1
4	Thermal resistance outer covering	T_3	K.m/W	
5	Include correction factor			[4], section 4.2.4.3.3

1. Thermal resistance of tape under outer covering:

$$T_{3_tows} = \frac{\rho_{tows}}{2\pi} \ln \left[\frac{D_{tows}}{D_s} \right]$$

ρ_{tows}	6,0 K.m/W	$T_{3_tows} = 0,014729284181 \text{ K.m/W}$
D_{tows}	39,2 mm	
D_s	38,6 mm	

2. Thermal resistance of outer covering:

$$T_{3_oc} = \frac{\rho_{oc}}{2\pi} \ln \left[\frac{D_{oc}}{D_{tows}} \right] \quad (\text{K.m/W})$$

ρ_{oc}	3,5 K.m/W	$T_{3_oc} = 0,05925838476 \text{ K.m/W}$
D_{oc}	43,6	
D_{tow}	39,2 mm	

3. Thermal resistance of semi conductive layer over outer covering:

$$T_{3_sc} = \frac{\rho_{oc_sc}}{2\pi} \ln \left[\frac{D_e}{D_{oc}} \right] \quad (\text{K.m/W})$$

ρ_{oc_sc}	2,5 K.m/W	$T_{3_oc_sc} = 0,003633699755 \text{ K.m/W}$
D_e	44,0	
D_{oc}	43,6	

4. Thermal resistance outer covering:

$$T'_3 = T_{3_tosw} + T_{3_oc} + T_{3_oc_sc} \quad (K.m/W)$$

T_{3_tosw}	0,014729284181 K.m/W	$T'_3 = 0,0776213687 \text{ K.m/W}$
T_{3_oc}	0,05925838476 K.m/W	
$T_{3_oc_sc}$	0,003633699755 K.m/W	

5. Correction factor

Multiply T_3 with 1.6 as discussed under calculation of T_1 so:

$$T_3 = 1,6 \times T'_3 = 0,12419418991 \text{ K.m/W}$$

8.9 External thermal resistance T_4

Table 8-14: Parameters needed to calculate the external thermal resistance T_4

Parameter	Symbol	Unit	Reference
External diameter of one cable	D_e	mm	Cable data sheet
Distance from the surface of the ground to the cable axis	L	mm	System design, note 1
Thermal resistivity of the soil	ρ_T	K.m/W	System design
Note 1: L is measured to the centre of the trefoil group			

Table 8-15: Sequential steps to calculate the external thermal resistance T_4

Step	parameter	Symbol	Unit	Reference
1	External thermal resistance	T_4	K.m/W	[4], section 4.2.4.3.3

1. External thermal resistance:

$$u = \frac{2L}{D_e}$$

$$T_4 = \frac{1,5}{\pi} \rho_T [\ln(2u) - 0,630] \quad (K.m/W)$$

D_e	44,0 mm	$T_4 = 1,8524966955 \text{ K.m/W}$
L	1000 mm	
ρ_T	1,0 K.m/W	

8.10 Permissible current rating

From [3] section 1.4.1.1

$$I = \left[\frac{\Delta\theta - W_d [0,5 \cdot T_1 + n(T_2 + T_3 + T_4)]}{R \cdot T_1 + nR(1 + \lambda_1)T_2 + nR(1 + \lambda_1 + \lambda_2)(T_3 + T_4)} \right]^{0,5}$$

$\Delta\theta$	90 °C-20 °C = 70 K	$I = 537,4631099368 \text{ A}$
W_d	0,10842143853 W/m	
T_1	0,4110322351 K.m/W	
T_2	0	
T_3	0,12419418991 K.m/W	
T_4	1,8524966955 K.m/W	
λ_1	0,0435122656	
λ_2	0	
R	9,762868345 x10 ⁻⁵ Ω/m	
n	1	

Found by iteration of screen temperature until calculated temperature equals temperature guess. Only the final iteration is reported.

The losses are as follows in this situation:

- Conductor losses: $w_c = R \times I^2 = 28,20166532 \text{ W/m}$ per phase
- Screen losses: $w_s = \lambda_1 \times R \times I^2 = 1,2271183509 \text{ W/m}$ per phase.

9. Case study 5: A 400kV LPOF Cable

9.1 Introduction

The selected cable type is a 400kV single core LPOF (Low Pressure Oil filled) Cable with a 2000 mm² copper conductor with corrugated aluminium sheath. The cable is installed in trefoil and in flat formation in the two sections of this Section.

9.2 400 kV LPOF cable in trefoil

9.2.1 Cable Details

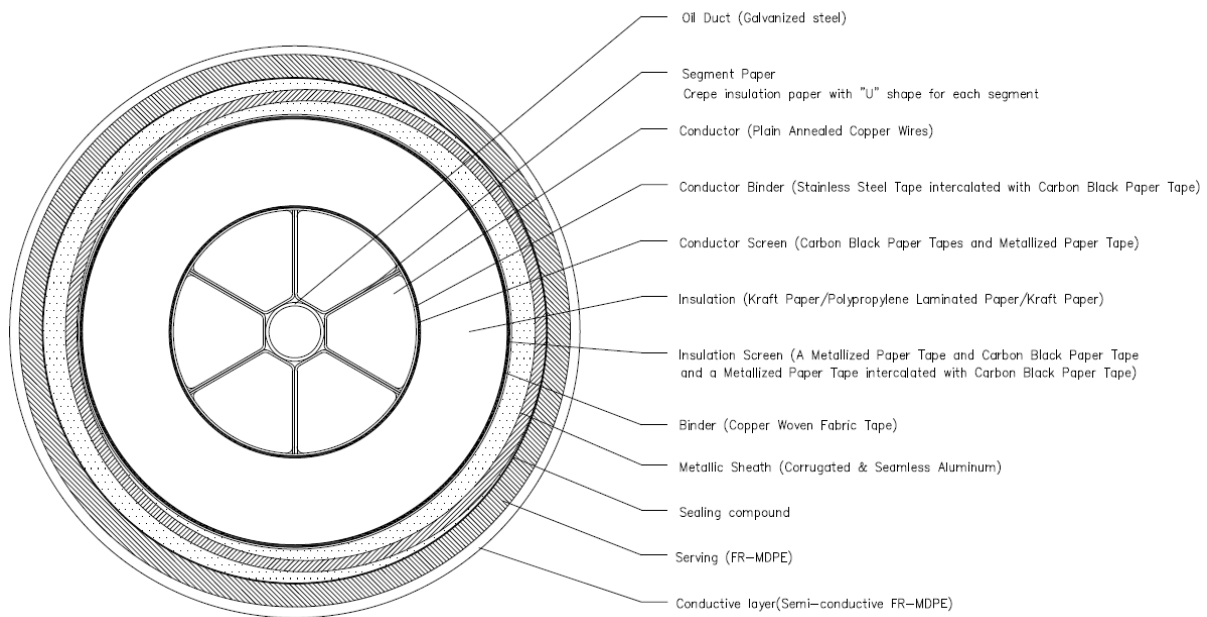


Figure 33: 400kV LPOF cable drawing (trefoil)

Table 9-1: 400kV LPOF cable dimensions

No	Description	Thickness (mm)	Details	Thermal Resistivity (K.m/W)	Nominal Diameter (mm)
1	Oil duct	Nom. 0.8	Galvanized steel	N/A	18,0
2	Conductor	-	Plain annealed copper wires	N/A	58,8
3	Conductor Binder	Nom. 0.25	Stainless Steel tape intercalated with carbon black paper tape	6,0	59,3
4	Conductor screen	Nom. 0.3	Carbon black paper tapes and metallized paper tape	5,0	60,2
5	Insulation	Nom. 0.5/20.0/4.5	Kraft paper/Polypropylene laminated paper/Kraft paper	5,0 (Kraft) 5,5 (PPLP) 5,0 (Kraft)	111,2
6	Insulation screen	Nom.0.4	A metallized paper tape and carbon black paper tape	5,0	112,4
7	Binder	Nom.0.4	Copper woven fabric tape	6,0	113,3
8	Metallic sheath	Nom.2.9	Corrugated & seamless aluminium (Wave height : Approx. 5.9mm, Pitch : Approx. 28.0mm)	N/A	131,4
9	Sealing compound	Nom. 0.2	Bitumen compound	6,0	131,8
10	Serving	Min. 5.0	FR-MDPE compound	3,5	142,8

11	Conductive layer	Min, 2.0	Semi-conductive FR-MDPE	2,5	148
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9.2.2 Installation Detail

System related parameters:

- System voltage (U) : 400kV
- Frequency : 50Hz
- Cable laying depth : 1000 mm (Center of group)
- Installation formation : Touching trefoil
- Sheath bonding method : Both ended (Solid bonding)
- Ground temperature: 20°C
- Soil thermal resistivity : 1,0 K.m/W
- Thermally independent from other circuits

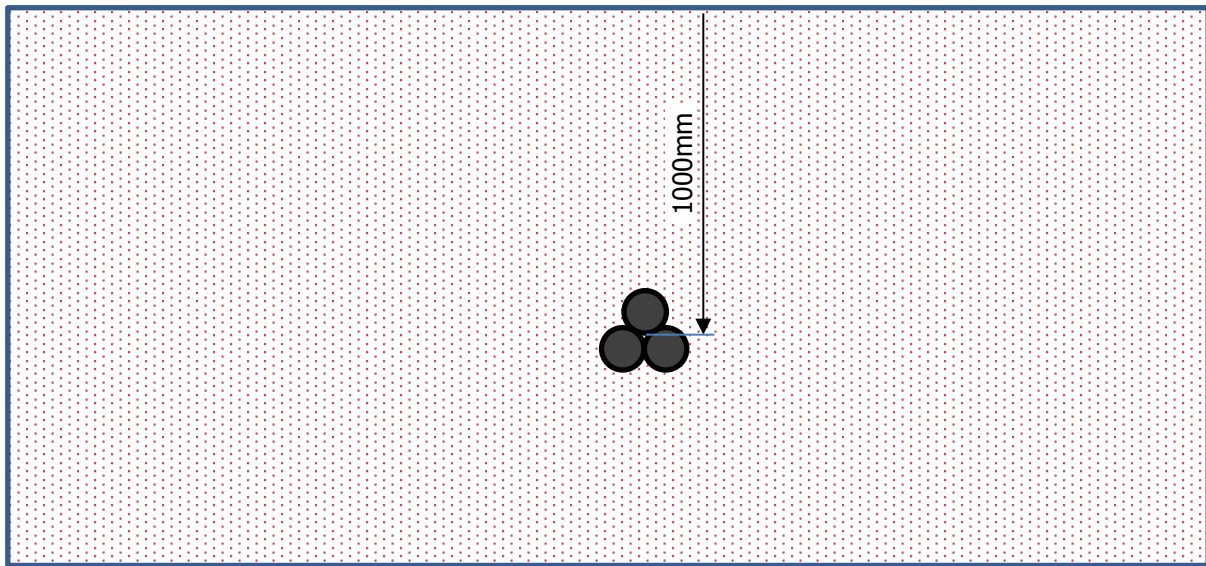


Figure 34: 400 kV LPOF cable installation detail

9.2.3 Calculation of the conductor AC resistance at operating temperature

Table 9-2: Parameters needed to calculate the conductor AC resistance at operating temperature

Parameter	Symbol	Value	Unit	Reference
Conductor DC resistance @ 20°C	R_o	$0,9 \times 10^{-5}$	Ω/m	[8] or cable datasheet
Conductor diameter	d_c	58,8	mm	Cable datasheet
Conductor material		Copper		Cable datasheet
Type of conductor		Milliken according to IEC 60228		Cable datasheet
Skin effect factor	k_s	0,435	-	[3], table 2
Proximity effect factor	k_p	0,37	-	[3], table 2
Conductor temperature coefficient at 20°C per Kelvin	α	$3,93 \times 10^{-3}$	-	[3], table 1
Conductor operating temperature	θ	85	°C	Cable datasheet
Distance between conductor axes	s	148	mm	Installation drawing / cable datasheet
System frequency	f	50	Hz	System design

Table 9-3: Sequential steps to calculate the conductor AC resistance at operating temperature

Step	Parameter	Symbol	Unit	Reference
1	Conductor DC resistance at operating temperature	R'	Ω/m	[3], section 2.1.1
2	Skin effect factor	y_s	-	[3], section 2.1.2
3	Proximity effect factor	y_p	-	[3], section 2.1.3
4	AC resistance of conductor	R	Ω/m	[3], section 2.1

1. Conductor DC resistance at operating temperature
- R'
- :

$$R' = R_o [1 + \alpha_{20}(\theta - 20)] \quad (\Omega/m)$$

R_o	$0,009 \times 10^{-3} \Omega/m$	$R' = 1,129905 \times 10^{-5} \Omega/m$
α_{20}	$3,93 \times 10^{-3}$	
θ	85°C	

2. Skin effect factor
- y_s
- :

$$x_s^2 = \frac{8\pi f}{R'} 10^{-7} k_s$$

f	50 Hz	$x_s^2 = 4,8379033788$
R'	$1,129905 \times 10^{-5} \Omega/m$	
k_s	$0,435 \text{ (Milliken, PPLP)}$	

$$\text{for } 0 < x_s \leq 2,8 \rightarrow y_s = \frac{x_s^4}{192 + 0,8x_s^4}$$

Note: three equations are given for calculating y_s , use of these equations depends on the calculated value of x_s .

x_s^2	$4,8379033788$	$y_s = 0,1110707923$
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3. Proximity effect factor
- y_p
- :

$$x_p^2 = \frac{8\pi f}{R'} 10^{-7} k_p$$

f	50 Hz	$x_p^2 = 4,1149982762$
R'	$1,129905 \times 10^{-5} \Omega/m$	
k_p	$0,37 \text{ (Milliken, PPLP)}$	

$$y_p = \frac{x_p^4}{192 + 0,8x_p^4} \left(\frac{d_c}{s} \right)^2 \left[0,312 \left(\frac{d_c}{s} \right)^2 + \frac{1,18}{\frac{x_p^4}{192 + 0,8x_p^4} + 0,27} \right]$$

x_p^2	$4,1149982762$	$y_p = 0,0441844496$
d_c	$58,8 \text{ mm}$	
s	148 mm	

4. AC resistance of conductor
- R
- :

$$R = R'(1 + y_s + y_p) \quad (\Omega/m)$$

R'	$1,129905 \times 10^{-5} \Omega/\text{m}$	$R = 1,3053287266 \times 10^{-5} \Omega/\text{m}$
γ_s	0,1110707923	
γ_p	0,0441844496	

9.2.4 Dielectric losses

Table 9-4: Parameters needed to calculate the dielectric loss per unit length per phase

Parameter	Symbol	Value	Unit	Reference
External diameter of the insulation	D_i	111,2	mm	Cable data sheet
External diameter of the conductor, incl. screen	D_{cs}	60,2	mm	Cable datasheet
Relative permittivity of the insulation	ϵ	2,8	-	[3], table 3
Loss factor of the insulation	$\tan\delta$	0,0014	-	[3], table 3, note 1
Voltage to earth	U_o	230 940,108	V	System design($U/\sqrt{3}$)
System frequency	f	50	Hz	System design
Note 1: PPL(PPLP) insulation material has been considered				

Table 9-5: Sequential steps to calculate the dielectric loss per unit length per phase

Step	Parameter	Symbol	Unit	Reference
1	Capacitance	C	F/m	[3], section 2.2
2	Angular frequency of system	ω	-	[3], section 2.2
3	Dielectric loss per unit length in each phase	W_d	W/m	[3], section 2.2

1. Capacitance C (for circular conductors):

$$C = \frac{\epsilon}{18 \ln\left(\frac{D_i}{D_{cs}}\right)} 10^{-9} \text{ (F/m)}$$

ϵ	2,8	$C = 2,5348899237 \times 10^{-10} \text{ F/m}$
D_i	111,2 mm	
D_{cs}	60,2 mm	

2. Angular frequency of system:

$$\omega = 2\pi f \text{ (rad/s)}$$

f	50 Hz	$\omega = 314,159265359 \text{ rad/s}$
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3. Dielectric loss per unit length in each phase:

$$W_d = \omega C U_o^2 \tan\delta \text{ (W/m)}$$

C	$2,5348899237 \times 10^{-10} \text{ F/m}$	$W_d = 5,9461483662 \text{ W/m}$
ω	314,159265359 rad/s	
U_o	230 940,180 V	
$\tan\delta$	0,0014	

9.2.5 Loss factor for sheath

Table 9-6: Parameters needed to calculate the loss factor for metallic sheath

Parameter	Symbol	Value	Unit	Reference
AC resistance of sheath at 20°C	R_{s0}	-	Ω/m	Calculated
Inner diameter of the sheath	d_s	113,3	mm	Cable data sheet
Outer diameter of the sheath	D_s	131,4	mm	Cable data sheet
Mean outer diameter of the sheath	D_{sm}	125,25	mm	[3], section 2.3.6.1 Note
Thickness of the sheath	t_s	2,9	mm	Cable data sheet
Cross sectional area of the metallic sheath	A_s	-	mm ²	Calculated
Sheath resistivity at 20°C	ρ_s	$2,84 \times 10^{-8}$	Ωm	[3], table 1
Sheath temperature coefficient at 20°C per Kelvin	α_s	$4,03 \times 10^{-3}$	-	[3], table 1
Distance between conductor axes	s	148	mm	System design
System frequency	f	50	Hz	System design
AC resistance of conductor at operating temp.	R	-	Ω/m	Calculated

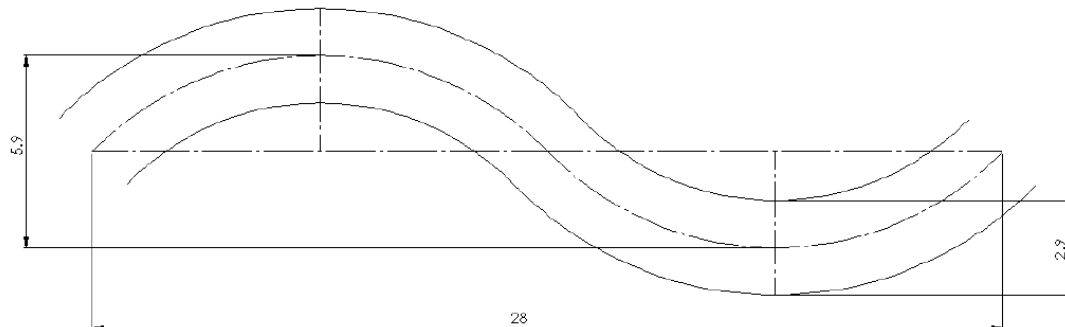
Table 9-7: Sequential steps to calculate the sheath losses per unit length per phase

Step	Parameter	Symbol	Unit	Reference
1	Corrugation factor	F_{cor}		
2	Cross sectional area of the sheath	A_s	-	
3	AC Resistance of sheath at 20°C	R_{s0}	Ω/m	-
4	Maximum operating temperature of the cable sheath, to be determined iterative	θ_s	°C	[3], section 2.3
5	Resistance of sheath at operating temperature	R_s	Ω/m	[3], section 2.3
6	Reactance per unit length of sheath	X	Ω/m	[3], section 2.3.1
7	Loss factor of circulating losses for sheath	λ'_1	-	[3], section 2.3.1
8	Effect of large segmental type conductor	F	-	[3], section 2.3.5
9	Electrical resistivity of the sheath at operating temperature	ρ_s	$\Omega \cdot m$	-
10	Loss factor of eddy current losses for sheath	λ''_1		[3], section 2.3.6.1
11	Loss factor for sheath	λ_1		[3], section 2.3

1. Corrugation factor F_{cor} (for corrugated sheath):

A detailed explanation of corrugation factor (F_{cor}) is described in Section 2.7.2 and Guidance Point 30.

According to the cable details in this document above, the wave height (H) is 5,9mm and the length of Pitch (L) is 28mm:



$$F_{cor} = \frac{0.25L^2 + H^2}{L \cdot H} \cdot \arcsin\left(\frac{L \cdot H}{0.25L^2 + H^2} \text{ rad}\right)$$

H	5,9 mm	$F_{cor} = 1,1144874069$
L	28 mm	

2. Cross sectional area of the sheath A_s :

$$A_s = \pi \left(\frac{d_s + D_s}{2} \right) \cdot t_s \cdot 10^{-6} \text{ (m}^2\text{)}$$

d_s	113,3 mm	$A_s = 1,1146841974 \times 10^{-3} \text{ m}^2$
D_s	131,4 mm	
t_s	2,9 mm	

3. AC Resistance of the sheath at 20 °C per unit length R_{s0} :

$$R_{s0} = \frac{\rho_{s0}}{A_s} \cdot F_{cor} \text{ (}\Omega/\text{m)}$$

ρ_{s0}	$2,84 \times 10^{-8} \Omega\text{m}$	$R_{s0} = 2,8394986158 \times 10^{-5} \Omega/\text{m}$
A_s	$1,1146841974 \times 10^{-3} \text{ m}^2$	
F_{cor}	1,1144874069	

4. The operating temperature θ_s (°C) of the screen θ_s :

$$\theta_s = \theta - (I^2 R + 0,5 W_d) T_1 \text{ (}^\circ\text{C)}$$

θ	85 °C	$\theta_s = 76,6050994971 \text{ }^\circ\text{C}$
I	903,6213914485 A	
R	$1,3053287266 \times 10^{-5} \Omega/\text{m}$	
W_d	5,9461483662 W/m	
T_1	0,6158458706 K.m/W	

Note: θ_s is the operating temperature of the cable screen (°C). Because the temperature of the screen is a function of the current I , an iterative method must be used for the calculation. First calculation can be done by 75°C ($\theta_c - 10^\circ\text{C}$) as an initial sheath temperature.

5. AC Resistance of the sheath at the operating temperature per unit length R_s :

$$R_s = R_{s0} \cdot [1 + \alpha_s(\theta_s - 20)] \text{ (}\Omega/\text{m)}$$

R_{s0}	$2,8394986158 \times 10^{-5} \Omega/\text{m}$	$R_s = 3,4872409255 \times 10^{-5} \Omega/\text{m}$
α_s	$4,03 \times 10^{-3}$	
θ_s	76,6050994971 °C	

6. Reactance X per unit length of the sheath:

Mean diameter of sheath:

$$d = \left(\frac{d_s + D_s}{2} \right) \text{ (mm)}$$

d_s	113,3 mm	$d = 122,35 \text{ mm}$
D_s	131,4 mm	

$$X = 2\omega 10^{-7} \ln\left(\frac{2s}{d}\right) \quad (\Omega/m)$$

ω	314,159265359 rad/s	$X = 5,5510287477 \times 10^{-5} \Omega/m$
s	148 mm	
d	122,35 mm	

7. Loss factor λ'_1 for circulating losses of the sheath:

$$\lambda'_1 = \frac{R_s}{R} \frac{1}{1 + \left(\frac{R_s}{X}\right)^2}$$

R_s	$3,4872409255 \times 10^{-5} \Omega/m$	$\lambda'_1 = 1,9155586551$
R	$1,3053287266 \times 10^{-5} \Omega/m$	
X	$5,5510287477 \times 10^{-5} \Omega/m$	

8. Effect of large segmental type conductor, F:

In case of both ended bonding system, a sheath eddy current loss factor can be ignored. However with large conductors of insulated segmental construction, the sheath loss factor cannot be ignored but shall be obtained by multiplying the value of λ'_1 by the factor F as below in accordance with the IEC 60287-1-1 [3] section 2.3.5.

$$M = N = \frac{R_s}{X}$$

R_s	$3,4872409255 \times 10^{-5} \Omega/m$	$M = N = 0,6282152523$
X	$5,5510287477 \times 10^{-5} \Omega/m$	
Note: In case of trefoil formation, M and N are identical.		

$$F = \frac{4M^2N^2 + (M + N)^2}{4(M^2 + 1)(N^2 + 1)}$$

M	0,6282152523	$F = 0,2829764868$
N	0,6282152523	

9. Electrical resistivity of the sheath at operating temperature ρ_s :

$$\rho_s = \rho_{s0} \cdot (1 + \alpha_s(\theta_s - 20)) \quad (\Omega.m)$$

ρ_{s0}	$2,84 \times 10^{-8} \Omega.m$	$\rho_s = 3,4878566848 \times 10^{-8} \Omega.m$
α_s	$4,03 \times 10^{-3}$	
θ_s	76,6050994971 °C	

10. Loss factor λ''_1 for eddy current losses of the sheath:

$$\beta_1 = \sqrt{\frac{4\pi\omega}{10^7\rho_s}}$$

ω	314,159265359 rad/s	$\beta_1 = 106,38993721$
ρ_s	$3,4878566848 \times 10^{-8} \Omega.m$	

$$g_s = 1 + \left(\frac{t_s}{D_{sm}}\right)^{1,74} (\beta_1 \cdot D_{sm} \cdot 10^{-3} - 1,6)$$

t_s	2,9 mm	$g_s = 1,0167326486$
D_{sm}	125,25 mm	
β_1	106,38993721	
Note: D_{sm} has been used instead of D_s as the cable sheath is a corrugated sheath		

$$m = \frac{\omega}{R_s} \cdot 10^{-7}$$

ω	314,159265359 rad/s	$m = 0,9008820213$
R_s	$3,4872409255 \times 10^{-5} \Omega/\text{m}$	

$$\lambda_0 = 3 \left(\frac{m^2}{1 + m^2} \right) \cdot \left(\frac{d}{2s} \right)^2$$

m	0,9008820213	$\lambda_0 = 0,2296265415$
d	122,35 mm	
s	148 mm	

$$\Delta_1 = (1,14m^{2,45} + 0,33) \cdot \left(\frac{d}{2s} \right)^{(0,92m+1,66)}$$

m	0,9008820213	$\Delta_1 = 0,1345387580$
d	122,35 mm	
s	148 mm	

$$\lambda_1'' = \frac{R_s}{R} \left[g_s \cdot \lambda_0 (1 + \Delta_1 + \Delta_2) + \frac{(\beta_1 t_s)^4}{12 \times 10^{12}} \right] \cdot F$$

R_s	$3,4872409255 \times 10^{-5} \Omega/\text{m}$	$\lambda_1'' = 0,2008153554$
R	$1,3053287266 \times 10^{-5} \Omega/\text{m}$	
g_s	1,0167326486	
λ_0	0,2296265415	
Δ_1	0,1345387580	
Δ_2	0	
β_1	106,38993721	
t_s	2,9 mm	
F	0,2829764868	

11. Loss factor λ_1 for sheath:

$$\lambda_1 = \lambda_1' + \lambda_1''$$

λ_1'	1,9155586551	$\lambda_1 = 2,1163740105$
λ_1''	0,2008153554	

9.2.6 Thermal resistance T_1 between conductor and screen

Table 9-8: Parameters needed to calculate the thermal resistance between conductor and screen

Parameter	Symbol	Value	Unit	Reference
Diameter of conductor	d_c	58,8	mm	Cable datasheet
External diameter of conductor binder	D_{cb}	59,3	mm	Cable datasheet
External diameter of conductor screen	D_{cs}	60,2	mm	Cable datasheet
Thickness of insulation of 1st Kraft	t_{i1}	0,5	mm	Cable datasheet
Thickness of insulation of PPLP	t_{i2}	-	mm	Calculated
Thickness of insulation of 2st Kraft	t_{i3}	4,5	mm	Cable datasheet
External diameter of insulation	D_i	111,2	mm	Cable data sheet
External diameter of insulation screen	D_{is}	112,4	mm	Cable data sheet
External diameter of binder (Copper woven fabric tape)	D_b	113,3	mm	Cable data sheet
Diameter of the imaginary cylinder which just touches the inside surface of the troughs of a corrugated sheath	D_{it}	113,3	mm	Cable data sheet
Diameter of the imaginary coaxial cylinder which just touches the crests of a corrugated sheath	D_{oc}	131,4	mm	Cable data sheet
Thermal resistivity of conductor binder (SS tape intercalated with carbon black paper tape)	ρ_{cb}	6,0	K.m/W	Note 1
Thermal resistivity of conductor and insulation screen	ρ_{sc}	5,0	K.m/W	[4], table 1
Thermal resistivity of insulation (Kraft)	ρ_{i1}	5,0	K.m/W	[4], table 1
Thermal resistivity of insulation (PPLP)	ρ_{i2}	5,5	K.m/W	[4], table 1
Thermal resistivity of binder (Copper woven fabric tape)	ρ_b	6,0	K.m/W	Note 2
Note 1: In this case: 6,0 K.m/W				
Note 2: In this case: 6,0 K.m/W				

Table 9-9: Sequential steps to calculate the thermal resistance between conductor and screen

Step	Parameter	Symbol	Unit	Reference
1	Thermal resistance of conductor binder	$T_{1_{cb}}$	K.m/W	[4], section 4.1.2.1
2	Thermal resistance of conductor screen	$T_{1_{cs}}$	K.m/W	[4], section 4.1.2.1
3	Thermal resistance of insulation (1 st Kraft)	$T_{1_{i1}}$	K.m/W	[4], section 4.1.2.1
4	Thermal resistance of insulation (PPLP)	$T_{1_{i2}}$	K.m/W	[4], section 4.1.2.1
5	Thermal resistance of insulation (2 nd Kraft)	$T_{1_{i3}}$	K.m/W	[4], section 4.1.2.1
6	Thermal resistance of insulation screen	$T_{1_{is}}$	K.m/W	[4], section 4.1.2.1
7	Thermal resistance of binder	T_{1_b}	K.m/W	[4], section 4.1.2.1
8	Thermal resistance of sealing compound	$T_{3_{se}}$	K.m/W	[4], section 4.1.2.1
9	Thermal resistance between conductor and screen	T_1	K.m/W	-

1. Thermal resistance of conductor binder $T_{1_{cb}}$:

$$T_{1_{cb}} = \frac{\rho_{sb}}{2\pi} \ln\left(\frac{D_{cb}}{d_c}\right) \quad (K.m/W)$$

ρ_{cb}	6,0 K.m/W	$T_{1_{cb}} = 8,0858201869 \times 10^{-3} \text{ K.m/W}$
D_{cb}	59,3 mm	
d_c	58,8 mm	

2. Thermal resistance of conductor screen
- T_{1_cs}
- :

$$T_{1_cs} = \frac{\rho_{sc}}{2\pi} \ln\left(\frac{D_{cs}}{D_{cb}}\right) \quad (K.m/W)$$

ρ_{sc}	5,0 K.m/W	$T_{1_cs} = 1,1986791392 \times 10^{-2} \text{ K.m/W}$
D_{cs}	60,2 mm	
D_{cb}	59,3 mm	

3. Thermal resistance of insulation (1
- st
- Kraft)
- T_{1_i1}
- :

$$T_{1_i1} = \frac{\rho_{i1}}{2\pi} \ln\left(1 + \frac{2 \cdot t_{i1}}{D_{cs}}\right) \quad (K.m/W)$$

ρ_{i1}	5,0 K.m/W	$T_{1_i1} = 1,3110258888 \times 10^{-2} \text{ K.m/W}$
t_{i1}	0,5 mm	
D_{cs}	60,2 mm	

4. Thermal resistance of insulation (PPLP)
- T_{1_i2}
- :

As there is a difference between diameter and thickness, we have calculated all thermal resistance in accordance with diameters. However, in case of the insulation layer, it has 3 different materials but PPLP is a major insulation. Therefore we should make up the difference nominal thickness and diameter by adding additional thickness as the PPLP.

$$D_{i1} = D_{cs} + 2t_{i1} \quad (mm)$$

D_{cs}	60,2 mm	$D_{i1} = 61,2 \text{ mm}$
t_{i1}	0,5 mm	

$$D_{i2} = D_i - 2t_{i3} \quad (mm)$$

D_i	111,2 mm	$D_{i2} = 102,2 \text{ mm}$
t_{i3}	4,5 mm	

$$T_{1_i2} = \frac{\rho_{i2}}{2\pi} \ln\left(\frac{D_{i2}}{D_{i1}}\right) \quad (K.m/W)$$

ρ_{i2}	5,5 K.m/W	$T_{1_i2} = 4,44886702325 \times 10^{-1} \text{ K.m/W}$
D_{i1}	61,2 mm	
D_{i2}	102,2 mm	

5. Thermal resistance of insulation (2
- st
- Kraft)
- T_{1_i3}
- :

$$T_{1_i3} = \frac{\rho_{i1}}{2\pi} \ln\left(1 + \frac{2 \cdot t_{i3}}{D_{i2}}\right) \quad (K.m/W)$$

ρ_{i1}	5,0 K.m/W	$T_{1_i3} = 6,7162354698 \times 10^{-2} \text{ K.m/W}$
t_{i3}	4,5 mm	
D_{i2}	102,2 mm	

6. Thermal resistance of insulation screen
- T_{1_is}
- :

$$T_{1_is} = \frac{\rho_{sc}}{2\pi} \ln\left(\frac{D_{is}}{D_i}\right) \quad (K.m/W)$$

ρ_{sc}	5,0 K.m/W	$T_{1_{is}} = 8,5414921878 \times 10^{-3} \text{ K.m/W}$
D_{is}	112,4 mm	
D_i	111,2 mm	

7. Thermal resistance of binder T_{1_b} :

In case of corrugated sheath final layer of the thermal resistance shall be calculated by the mean internal diameter of the sheath instead of the surface of the troughs of a corrugated sheath:

$$d_{sm} = \left(\frac{D_{it} + D_{oc}}{2} \right) - t_s \quad (mm)$$

D_{it}	113,3 mm	$d_{sm} = 119,45 \text{ mm}$
D_{oc}	131,4 mm	
t_s	2,9 mm	

$$T_{1_b} = \frac{\rho_b}{2\pi} \ln \left(\frac{d_{sm}}{D_{is}} \right) \quad (K.m/W)$$

ρ_b	6,0 K.m/W	$T_{1_b} = 5,8092130031 \times 10^{-2} \text{ K.m/W}$
d_{sm}	119,45 mm	
D_{is}	112,4 mm	

8. Thermal resistance between conductor and screen T_1 :

$$T_1 = T_{1_b} + T_{1_{cs}} + T_{1_{i1}} + T_{1_{i2}} + T_{1_{i3}} + T_{1_{is}} + T_{1_b} \quad (K.m/W)$$

$T_{1_{cb}}$	$8,0858201869 \times 10^{-3} \text{ K.m/W}$	$T_1 = 0,6158458706 \text{ K.m/W}$
$T_{1_{cs}}$	$1,1986791392 \times 10^{-2} \text{ K.m/W}$	
$T_{1_{i1}}$	$1,3110258888 \times 10^{-2} \text{ K.m/W}$	
$T_{1_{i2}}$	$4,4886702325 \times 10^{-1} \text{ K.m/W}$	
$T_{1_{i3}}$	$6,7162354698 \times 10^{-2} \text{ K.m/W}$	
$T_{1_{is}}$	$8,5414921878 \times 10^{-3} \text{ K.m/W}$	
T_{1_b}	$5,8092130031 \times 10^{-2} \text{ K.m/W}$	

9.2.7 Thermal resistance of outer covering (serving) T_3

Table 9-10: Parameters needed to calculate the thermal resistance T_3 of the outer covering

Parameter	Symbol	Value	Unit	Reference
Diameter of the imaginary cylinder which just touches the inside surface of the troughs of a corrugated sheath	D_{it}	113,3	mm	Cable data sheet
Diameter of the imaginary coaxial cylinder which just touches the crests of a corrugated sheath	D_{oc}	131,4	mm	Cable data sheet
Thickness of Sealing compound (Bitumen)	t_{se}	0,2	mm	Cable data sheet
External diameter of Sealing compound (Bitumen)	D_{se}	131,8	mm	Cable data sheet
External diameter of Serving (sheath)	D_j	142,8	mm	Cable data sheet
External diameter of cable	D_e	148,0	mm	Cable data sheet
Thermal resistivity of Sealing compound (Bitumen)	ρ_{se}	6,0	K.m/W	[4], table 1
Thermal resistivity of FR-MDPE	ρ_j	3,5	K.m/W	[4], table 1
Thermal resistivity of Semi-conductive FR-MDPE	ρ_{sj}	2,5	K.m/W	[4], table 1

Table 9-11: Sequential steps to calculate the thermal resistance T_3 of the outer covering

Step	Parameter	Symbol	Unit	Reference
1	Thermal resistance of Sealing compound (Bitumen)	T_{3_se}	K.m/W	[4], section 4.1.4.1
2	Thermal resistance of Serving (FR-MDPE)	T_{3_j}	K.m/W	[4], section 4.1.4.1
3	Thermal resistance of Semi-conductive FR-MDPE	T_{3_sj}	K.m/W	[4], section 4.1.4.1
4	Thermal resistance of outer covering	T_3	K.m/W	[4], section 4.1.4.1

1. Thermal resistance of Sealing compound (Bitumen) T_{3_se} :

In case of corrugated sheath, a first layer of thermal resistance shall be calculated by the mean outer diameter of the sheath instead of the crests of a corrugated sheath.

$$D_{sm} = \left(\frac{D_{it} + D_{oc}}{2} \right) + t_s \quad (\text{mm})$$

D_{it}	113,3 mm	$D_{sm} = 125,25 \text{ mm}$
D_{oc}	131,4 mm	
t_s	2,9 mm	

$$T_{3_se} = \frac{\rho_{se}}{2\pi} \ln \left(\frac{D_{se}}{D_{sm}} \right) \quad (\text{K.m/W})$$

ρ_{se}	6,0 K.m/W	$T_{3_se} = 4,8676471832 \times 10^{-2} \text{ K.m/W}$
D_{sm}	125,25 mm	
D_{se}	131,8 mm	

2. Thermal resistance of Serving (FR-MDPE) T_{3_j} :

$$T_{3_j} = \frac{\rho_j}{2\pi} \ln \left(\frac{D_j}{D_{se}} \right) \quad (\text{K.m/W})$$

ρ_j	3,5 K.m/W	$T_{3,j} = 4,4652192115 \times 10^{-2} \text{ K.m/W}$
D_j	142,8 mm	
D_{se}	131,8 mm	

3. Thermal resistance of Semi-conductive FR-MDPE $T_{3,sj}$:

$$T_{3,sj} = \frac{\rho_{sj}}{2\pi} \ln\left(\frac{D_e}{D_j}\right) \quad (K.m/W)$$

ρ_{sj}	2,5 K.m/W	$T_{3,sj} = 1,4231326194 \times 10^{-2} \text{ K.m/W}$
D_e	148 mm	
D_j	142,8 mm	

4. Thermal resistance of outer covering T_3 :

$$T_3 = 1.6 \cdot (T_{3,se} + T_{3,j} + T_{3,sj}) \quad (K.m/W)$$

$T_{3,se}$	$4,8676471832 \times 10^{-2} \text{ K.m/W}$	$T_3 = 0,1720959842 \text{ K.m/W}$
$T_{3,j}$	$4,4652192115 \times 10^{-2} \text{ K.m/W}$	
$T_{3,sj}$	$1,4231326194 \times 10^{-2} \text{ K.m/W}$	
Note: A multiplication factor of 1.6 shall be used in accordance with [4] section 4.2.4.3.2. for three single-core cables in trefoil formation.		

9.2.8 External thermal resistance T_4

Table 9-12: Parameters needed to calculate the external thermal resistance T_4

Parameter	Symbol	Value	Unit	Reference
External diameter of one cable	D_e	148	mm	Cable data sheet
Distance from the surface of the ground to the center of cable group	L	1000	mm	System design
Thermal resistivity of the soil	ρ_T	1,0	K.m/W	System design

Table 9-13: Sequential steps to calculate the external thermal resistance T_4

Step	Parameter	Symbol	Unit	Reference
1	External thermal resistance	T_4	K.m/W	[4], section 4.2.4.3.2

1. External thermal resistance T_4 :

$$u = \frac{2L}{D_e}$$

L	1000 mm	$u = 13,5135135135$
D_e	148 mm	

$$T_4 = \frac{1.5}{\pi} \cdot \rho_T \cdot [\ln(2u) - 0,630] \quad (K.m/W)$$

u	13,5135135135	$T_4 = 1,2733210478 \text{ K.m/W}$
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9.2.9 Permissible current rating

$$I = \left[\frac{\Delta\theta - W_d[0,5 \cdot T_1 + T_3 + T_4]}{R \cdot T_1 + R(1 + \lambda_1)(T_3 + T_4)} \right]^{0,5}$$

$\Delta\theta$	65 K (85 °C – 20 °C)	$I = 903,6213914485 \text{ A}$
W_d	5,9461483662 W/m	
T_1	0,6158458706 K.m/W	
T_3	0,1720959842 K.m/W	
T_4	1,2733210478 K.m/W	
R	1,3053287266 x 10 ⁻⁵ Ω/m	
λ_1	2,1163740105	
Notes :		
1. T ₂ is ignored as there is no armor in this cable		
2. n=1 , as this is a single core cable		

9.2.10 Calculation of loss

Losses in conductor per unit length:

$$W_c = I^2 R \text{ (W/m)}$$

I	903,6213914485 A	$W_c = 10,6584217855 \text{ W/m}$
R	$1,3053287266 \times 10^{-5} \Omega/\text{m}$	

Losses in metallic sheath per unit length:

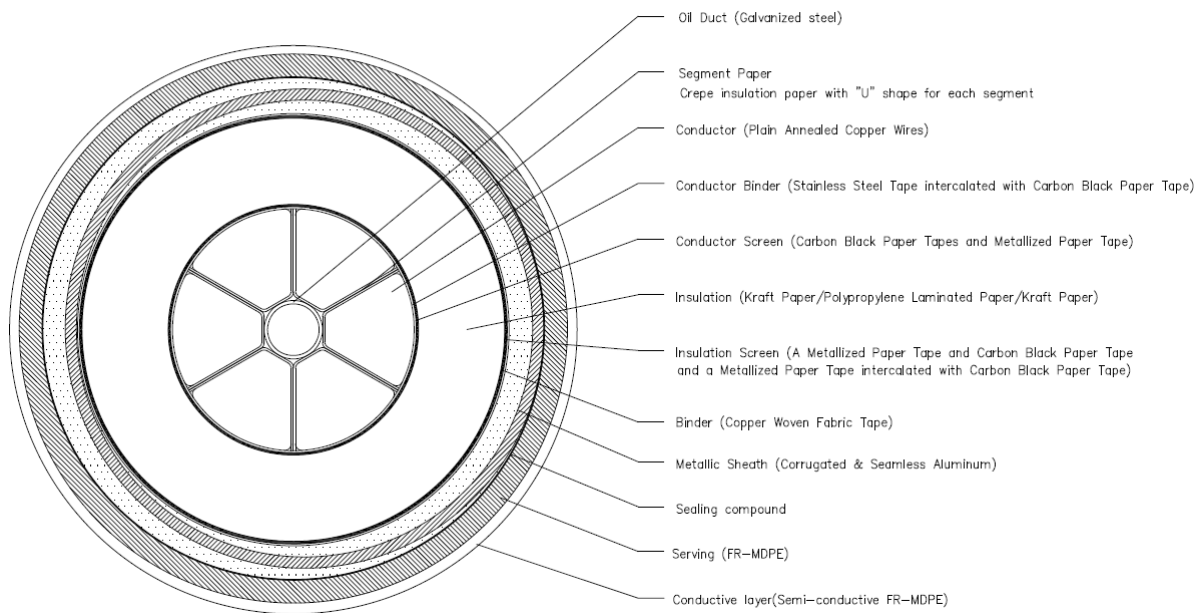
$$W_s = \lambda_1 \cdot W_c \text{ (W/m)}$$

λ_1	2,1163740105	$W_s = 22,5572068594 \text{ W/m}$
W_c	10,6584217855 W/m	

9.3 400kV LPOF cable in Flat formation

The selected cable type is a 400kV single core LPOF (Low Pressure Oil filled) Cable with a 2000 mm² copper conductor with corrugated aluminium sheath.

9.3.1 Cable Details



No	Description	Thickness (mm)	Details	Thermal Resistivity (K.m/W)	Nominal Diameter (mm)
1	Oil duct	Nom. 0.8	Galvanized steel	N/A	18,0
2	Conductor	-	Plain annealed copper wires	N/A	58,8
3	Conductor Binder	Nom. 0.25	Stainless Steel tape intercalated with carbon black paper tape	6,0	59,3
4	Conductor screen	Nom. 0.3	Carbon black paper paper tapes and metallized paper tape	5,0	60,2
5	Insulation	Nom. 0.5/20.0/4.5	Kraft paper/Polypropylene laminated paper/Kraft paper	5,0 (Kraft) 5,5 (PPLP) 5,0 (Kraft)	111,2
6	Insulation screen	Nom.0.4	A metallized paper tape and carbon black paper tape	5,0	112,4
7	Binder	Nom.0.4	Copper woven fabric tape	6,0	113,3
8	Metallic sheath	Nom.2.9	Corrugated & seamless aluminium (Wave height : Approx. 5.9mm, Pitch : Approx. 28.0mm)	N/A	131,4
9	Sealing compound	Nom. 0.2	Bitumen compound	6,0	131,8
10	Serving	Min. 5.0	FR-MDPE compound	3,5	142,8
11	Conductive layer	Min, 2.0	Semi-conductive FR-MDPE	2,5	148

9.3.2 Installation Detail

System related parameters:

- System voltage (U) : 400kV
- Frequency : 50Hz
- Cable laying depth : 1000 mm (Center of group)
- Installation formation : Spaced Flat (500mm phase spacing)
- Sheath bonding method : Cross bonding (Fully balanced between minor sections)
- Ground temperature: 20°C
- Soil thermal resistivity : 1,0 K.m/W
- Thermally independent from other circuits

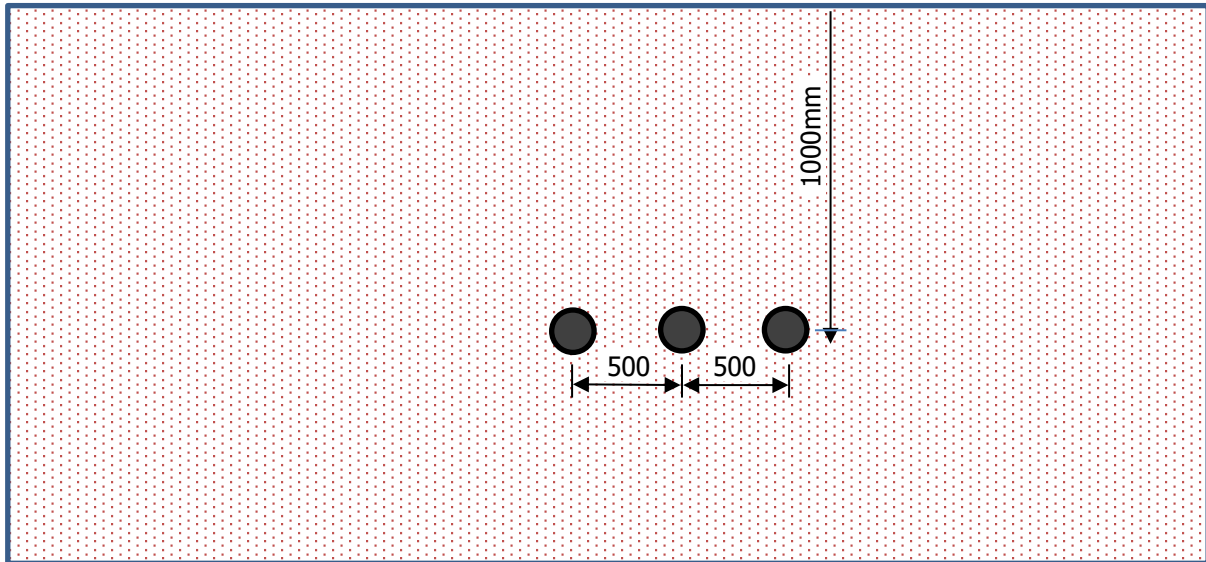


Figure 35: 400 kV LPOF cable installation detail (flat formation)

9.3.3 Calculation of the conductor AC resistance at operating temperature

Table 9-14: Parameters needed to calculate the conductor AC resistance at operating temperature

Parameter	Symbol	Value	Unit	Reference
Conductor DC resistance @ 20°C	R_o	$0,9 \times 10^{-5}$	Ω/m	[8] or cable datasheet
Conductor diameter	d_c	58,8	mm	Cable datasheet
Conductor material		Copper		Cable datasheet
Type of conductor		Milliken according to [8]		Cable datasheet
Skin effect factor	k_s	0,435	-	[3], table 2
Proximity effect factor	k_p	0,37	-	[3], table 2
Conductor temperature coefficient at 20°C per Kelvin	α	$3,93 \times 10^{-3}$	-	[3], table 1
Conductor operating temperature	θ	85	°C	Cable datasheet
Distance between conductor axes	s	500	mm	Installation drawing / cable datasheet
System frequency	f	50	Hz	System design

Table 9-15: Sequential steps to calculate the conductor AC resistance at operating temperature

Step	Parameter	Symbol	Unit	Reference
1	Conductor DC resistance at operating temperature	R'	Ω/m	[3], section 2.1.1
2	Skin effect factor	y_s	-	[3], section 2.1.2
3	Proximity effect factor	y_p	-	[3], section 2.1.3
4	AC resistance of conductor	R	Ω/m	[3], section 2.1

1. Conductor DC resistance at operating temperature
- R'
- :

$$R' = R_o [1 + \alpha_{20}(\theta - 20)] \quad (\Omega/m)$$

R_o	$0,009 \times 10^{-3} \Omega/m$	$R' = 1,129905 \times 10^{-5} \Omega/m$
α_{20}	$3,93 \times 10^{-3}$	
θ	85°C	

2. Skin effect factor
- y_s
- :

$$x_s^2 = \frac{8\pi f}{R'} 10^{-7} k_s$$

f	50 Hz	$x_s^2 = 4,8379033788$
R'	$1,129905 \times 10^{-5} \Omega/m$	
k_s	$0,435 \text{ (Milliken, PPLP)}$	

$$\text{for } 0 < x_s \leq 2,8 \rightarrow y_s = \frac{x_s^4}{192 + 0,8x_s^4}$$

Note: three equations are given for calculating y_s , use of these equations depends on the calculated value of x_s .

x_s^2	$4,8379033788$	$y_s = 0,1110707923$
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3. Proximity effect factor
- y_p
- :

$$x_p^2 = \frac{8\pi f}{R'} 10^{-7} k_p$$

f	50 Hz	$x_p^2 = 4,1149982762$
R'	$1,129905 \times 10^{-5} \Omega/m$	
k_p	$0,37 \text{ (Milliken, PPLP)}$	

$$y_p = \frac{x_p^4}{192 + 0,8x_p^4} \left(\frac{d_c}{s} \right)^2 \left[0,312 \left(\frac{d_c}{s} \right)^2 + \frac{1,18}{\frac{x_p^4}{192 + 0,8x_p^4} + 0,27} \right]$$

x_p^2	$4,1149982762$	$y_p = 0,0038200762$
d_c	$58,8 \text{ mm}$	
s	500 mm	

4. AC resistance of conductor
- R
- :

$$R = R'(1 + y_s + y_p) \quad (\Omega/m)$$

R'	$1,129905 \times 10^{-5} \Omega/\text{m}$	$R = 1,2597207668 \times 10^{-5} \Omega/\text{m}$
γ_s	0,1110707923	
γ_p	0,0038200762	

9.3.4 Dielectric losses

Table 9-16: Parameters needed to calculate the dielectric loss per unit length per phase

Parameter	Symbol	Value	Unit	Reference
External diameter of the insulation	D_i	111,2	mm	Cable data sheet
External diameter of the conductor, incl. screen	D_{cs}	60,2	mm	Cable datasheet
Relative permittivity of the insulation	ϵ	2,8	-	[3], table 3
Loss factor of the insulation	$\tan\delta$	0,0014	-	[3], table 3, note 1
Voltage to earth	U_o	230 940,108	V	System design($U/\sqrt{3}$)
System frequency	f	50	Hz	System design
Note 1: PPL(PPLP) insulation material has been considered				

Table 9-17: Sequential steps to calculate the dielectric loss per unit length per phase

Step	Parameter	Symbol	Unit	Reference
1	Capacitance	C	F/m	[3], section 2.2
2	Angular frequency of system	ω	-	[3], section 2.2
3	Dielectric loss per unit length in each phase	W_d	W/m	[3], section 2.2, note 1

1. Capacitance C (for circular conductors):

$$C = \frac{\epsilon}{18 \ln\left(\frac{D_i}{D_{cs}}\right)} 10^{-9} \text{ (F/m)}$$

ϵ	2,8	$C = 2,5348899237 \times 10^{-10} \text{ F/m}$
D_i	111,2 mm	
D_{cs}	60,2 mm	

2. Angular frequency of system:

$$\omega = 2\pi f \left(\frac{\text{rad}}{\text{s}} \right)$$

f	50 Hz	$\omega = 314,159265359 \text{ rad/s}$
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3. Dielectric loss per unit length in each phase:

$$W_d = \omega C U_o^2 \tan\delta \text{ (W/m)}$$

C	$2,5348899237 \times 10^{-10} \text{ F/m}$	$W_d = 5,9461483662 \text{ W/m}$
ω	314,159265359 rad/s	
U_o	230 940,180 V	
$\tan\delta$	0,0014	

9.3.5 Loss factor for sheath

Table 9-18: Parameters needed to calculate the loss factor for metallic sheath

Parameter	Symbol	Value	Unit	Reference
AC resistance of sheath at 20°C	R_{s0}	-	Ω/m	Calculated
Inner diameter of the sheath	d_s	113,3	mm	Cable data sheet
Outer diameter of the sheath	D_s	131,4	mm	Cable data sheet
Mean outer diameter of the sheath	D_{sm}	125,25	mm	[3], section 2.3.6.1
Thickness of the sheath	t_s	2,9	mm	Cable data sheet
Cross sectional area of the metallic sheath	A_s	-	mm ²	Calculated
Sheath resistivity at 20°C	ρ_s	$2,84 \times 10^{-8}$	Ωm	[3], table 1
Sheath temperature coefficient at 20°C per Kelvin	α_s	$4,03 \times 10^{-3}$	-	[3], table 1
Distance between conductor axes	s	148	mm	System design
System frequency	f	50	Hz	System design
AC resistance of conductor at operating temp.	R	-	Ω/m	Calculated

Table 9-19: Sequential steps to calculate the sheath losses per unit length per phase

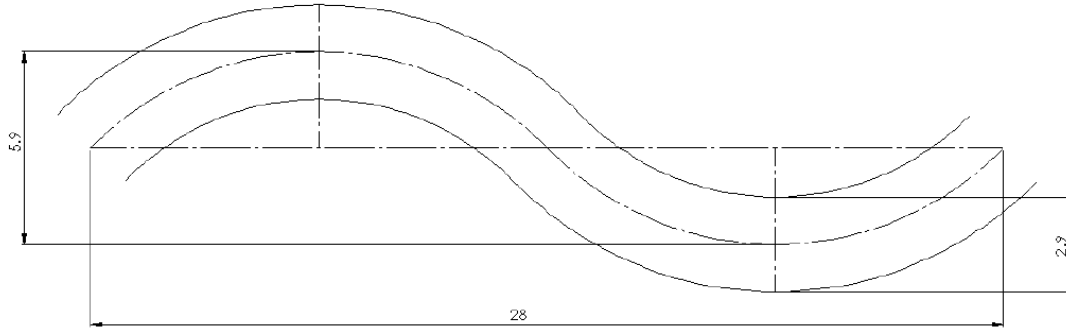
Step	Parameter	Symbol	Unit	Reference
1	Corrugation factor	F_{cor}		
2	Cross sectional area of the sheath	A_s	-	
3	AC Resistance of sheath at 20°C	R_{s0}	Ω/m	-
4	Maximum operating temperature of the cable sheath for centre cable, to be determined iterative	θ_s	°C	[3], section 2.3
5	Resistance of sheath for centre cable at operating temperature	R_s	Ω/m	[3], section 2.3
6	Electrical resistivity of the sheath for centre cable at operating temperature	ρ_s	$\Omega.m$	-
7	Loss factor of eddy current losses for sheath for centre cable	λ''_{1m}		[3], section 2.3.6.1
8	Maximum operating temperature of the cable sheath for outer cable leading phase, to be determined iterative	θ_{s1}	°C	[3], section 2.3
9	Resistance of sheath for outer cable leading phase at operating temperature	R_{s1}	Ω/m	[3], section 2.3
10	Electrical resistivity of the sheath for outer cable leading phase at operating temperature	ρ_{s1}	$\Omega.m$	-
11	Loss factor of eddy current losses for sheath for outer cable leading phase	λ''_{11}		[3], section 2.3.6.1
12	Maximum operating temperature of the cable sheath for outer cable lagging phase, to be determined iterative	θ_{s2}	°C	[3], section 2.3
13	Resistance of sheath for outer cable lagging phase at operating temperature	R_{s2}	Ω/m	[3], section 2.3
14	Electrical resistivity of the sheath for outer cable lagging phase at operating temperature	ρ_{s2}	$\Omega.m$	-
15	Loss factor of eddy current losses for sheath for outer cable lagging phase	λ''_{12}		[3], section 2.3.6.1
16	Loss factor for sheath for centre cable	λ_{1m}		[3], section 2.3
17	Loss factor for sheath for outer cable leading phase	λ_{11}		[3], section 2.3
18	Loss factor for sheath for outer cable lagging phase	λ_{12}		[3], section 2.3

Note: Loss factor circulating losses for sheath is ignored as we have considered fully balanced (minor sections) cross bonding with a transposition.

1. Corrugation factor F_{cor} (for corrugated sheath):

In case of corrugated sheath, corrugation factor shall be multiplied when calculate the AC resistance of sheath as it longer than smooth sheath. From corrugation factors it can be calculated as below if there is a wave length and pitch of corrugation.

According to the cable details in this document above, the wave height (H) is 5,9mm and the length of Pitch (L) is 28mm:



$$F_{cor} = \frac{0.25L^2 + H^2}{L \cdot H} \cdot \arcsin\left(\frac{L \cdot H}{0.25L^2 + H^2} \text{ rad}\right)$$

H	5,9 mm	$F_{cor} = 1,1144874069$
L	28 mm	

2. Cross sectional area of the sheath A_s :

$$A_s = \pi \left(\frac{d_s + D_s}{2} \right) \cdot t_s \cdot 10^{-6} \text{ (m}^2\text{)}$$

d_s	113,3 mm	$A_s = 1,1146841974 \times 10^{-3} \text{ m}^2$
D_s	131,4 mm	
t_s	2,9 mm	

3. AC Resistance of the sheath at 20°C per unit length R_{s0} :

$$R_{s0} = \frac{\rho_s}{A_s} \cdot F_{cor} \text{ (}\Omega/\text{m}\text{)}$$

ρ_s	$2,84 \times 10^{-8} \Omega\text{m}$	$R_{s0} = 2,8394986158 \times 10^{-5} \Omega/\text{m}$
A_s	$1,1146841974 \times 10^{-3} \text{ m}^2$	
F_{cor}	1,1144874069	

4. The operating temperature θ_s (°C) of the screen for centre cable θ_s :

$$\theta_s = \theta - (I^2 R + 0,5 W_d) T_1 \text{ (}^\circ\text{C}\text{)}$$

θ	85 °C	$\theta_s = 63,5506458179 \text{ }^\circ\text{C}$
I	1590,2251479060 A	
R	$1,2597207668 \times 10^{-5} \Omega/\text{m}$	
W_d	5,9461483662 W/m	
T_1	0,6158458706 K.m/W	

Note: θ_s is the operating temperature of the cable screen (°C). Because the temperature of the screen is a function of the current I , an iterative method must be used for the calculation. First calculation can be done by 75°C ($\theta_c - 10^\circ\text{C}$) as an initial sheath temperature.

5. AC Resistance of the sheath for centre cable at the operating temperature per unit length R_s :

$$R_s = R_{s0} \cdot [1 + \alpha_s(\theta_s - 20)] \quad (\Omega/\text{m})$$

R_{s0}	$2,8394986158 \times 10^{-5} \Omega/\text{m}$	$R_s = 3,3378564698 \times 10^{-5} \Omega/\text{m}$
α_s	$4,03 \times 10^{-3}$	
θ_s	$63,5506458179^\circ\text{C}$	

6. Electrical resistivity of the sheath for centre cable at operating temperature ρ_s :

$$\rho_s = \rho_{s0} \cdot (1 + \alpha_s(\theta_s - 20)) \quad (\Omega \cdot \text{m})$$

ρ_{s0}	$2,84 \times 10^{-8} \Omega \cdot \text{m}$	$\rho_s = 3,3384458515 \times 10^{-8} \Omega \cdot \text{m}$
α_s	$4,03 \times 10^{-3}$	
θ_s	$63,5506458179^\circ\text{C}$	

7. Loss factor λ''_{1m} for eddy current losses of the sheath for centre cable:

$$\beta_1 = \sqrt{\frac{4\pi\omega}{10^7 \rho_s}}$$

ω	$314,159 \text{ rad/s}$	$\beta_1 = 108,7445999943$
ρ_s	$3,3384458515 \times 10^{-8} \Omega \cdot \text{m}$	

$$g_s = 1 + \left(\frac{t_s}{D_{sm}}\right)^{1,74} (\beta_1 \cdot D_{sm} \cdot 10^{-3} - 1,6)$$

t_s	2,9 mm	$g_s = 1,0171535164$
D_{sm}	125,25 mm	
β_1	108,7445999943	
Note: D _{sm} has been used instead of D _s as the cable sheath is a corrugated sheath		

$$m = \frac{\omega}{R_s} \cdot 10^{-7}$$

ω	$314,159 \text{ rad/s}$	$m = 0,9412006424$
R_s	$3,3378564698 \times 10^{-5} \Omega/\text{m}$	

$$\lambda_{0m} = 6 \left(\frac{m^2}{1 + m^2} \right) \cdot \left(\frac{d}{2s} \right)^2$$

m	0,9412006424	$\lambda_{0m} = 0,0421904823$
d	122,35 mm	
s	500 mm	
Note: λ_0 has been calculated for centre cable only.		

$$\Delta_{1m} = 0,86m^{3,08} \cdot \left(\frac{d}{2s} \right)^{(1,4m+0,7)}$$

m	0,9412006424	$\Delta_{1m} = 0,0102923759$
d	122,35 mm	
s	500 mm	
Note: Δ_1 has been calculated for centre cable only.		

$$\lambda'_{1m} = \frac{R_s}{R} \left[g_s \cdot \lambda_{0m} (1 + \Delta_{1m} + \Delta_{2m}) + \frac{(\beta_1 t_s)^4}{12 \times 10^{12}} \right]$$

R_s	$3,3378564698 \times 10^{-5} \Omega/m$	$\lambda'_{1m} = 0,1170631205$
R	$1,2597207668 \times 10^{-5} \Omega/m$	
g_s	1,0171535164	
λ_{0m}	0,0421904823	
Δ_{1m}	0,0102923759	
Δ_{2m}	0 (Centre cable)	
β_1	108,7445999943	
t_s	2,9 mm	

8. The operating temperature θ_{s1} (°C) of the screen for outer cable leading phase θ_{s1} :

$$\theta_{s1} = \theta_1 - (I^2 R_1 + 0,5 W_d) T_1 \quad (^\circ\text{C})$$

θ_1	79,7291206273 °C	$\theta_{s1} = 58,5427968825 \text{ } ^\circ\text{C}$
I	1590,2251479060 A	
R_1	$1,2428312689 \times 10^{-5} \Omega/m$	
W_d	5,9461483662 W/m	
T_1	0,6158458706 K.m/W	

9. AC Resistance of the sheath for outer cable leading phase at the operating temperature per unit length R_{s1} :

$$R_{s1} = R_{s0} \cdot [1 + \alpha_s (\theta_{s1} - 20)] \quad (\Omega/m)$$

R_{s0}	$2,8394986158 \times 10^{-5} \Omega/m$	$R_{s1} = 3,2805507559 \times 10^{-5} \Omega/m$
α_s	$4,03 \times 10^{-3}$	
θ_{s1}	58,5427968825 °C	

10. Electrical resistivity of the sheath for outer cable leading phase at operating temperature ρ_{s1} :

$$\rho_{s1} = \rho_{s0} \cdot (1 + \alpha_s (\theta_{s1} - 20)) \quad (\Omega.m)$$

ρ_{s0}	$2,84 \times 10^{-8} \Omega.m$	$\rho_{s1} = 3,2811300189 \times 10^{-8} \Omega.m$
α_s	$4,03 \times 10^{-3}$	
θ_{s1}	58,5427968825 °C	

11. Loss factor λ'_{11} for eddy current losses of the sheath for outer cable leading phase:

$$\beta_{11} = \sqrt{\frac{4\pi\omega}{10^7 \rho_{s1}}}$$

ω	314,159 rad/s	$\beta_{11} = 109,6902808035$
ρ_{s1}	$3,2811300189 \times 10^{-8} \Omega.m$	

$$g_{s1} = 1 + \left(\frac{t_s}{D_{sm}} \right)^{1,74} (\beta_{11} \cdot D_{sm} \cdot 10^{-3} - 1,6)$$

t_s	2,9 mm	$g_{s1} = 1,0173225455$
D_{sm}	125,25 mm	
β_{11}	109,6902808035	
Note: D_{sm} has been used instead of D_s as the cable sheath is a corrugated sheath		

$$m_1 = \frac{\omega}{R_{s1}} \cdot 10^{-7}$$

ω	314,159 rad/s	$m_1 = 0,9576418374$
R_{s1}	$3,2805507559 \times 10^{-5} \Omega/\text{m}$	

$$\lambda_{01} = 1,5 \left(\frac{m_1^2}{1 + m_1^2} \right) \cdot \left(\frac{d}{2s} \right)^2$$

m_1	0,9576418374	$\lambda_{01} = 0,0107415183$
d	122,35 mm	
s	500 mm	

$$\Delta_{11} = 4,7 m_1^{0,7} \cdot \left(\frac{d}{2s} \right)^{(0,16m_1+2)}$$

m_1	0,9576418374	$\Delta_{11} = 0,0494707171$
d	122,35 mm	
s	500 mm	

$$\Delta_{21} = 21 m_1^{3,3} \cdot \left(\frac{d}{2s} \right)^{(1,47m_1+5,06)}$$

m_1	0,9576418374	$\Delta_{21} = 2,2858894134 \times 10^{-5}$
d	122,35 mm	
s	500 mm	

$$\lambda''_{11} = \frac{R_{s1}}{R_1} \left[g_{s1} \cdot \lambda_{01} (1 + \Delta_{11} + \Delta_{21}) + \frac{(\beta_{11} t_s)^4}{12 \times 10^{12}} \right]$$

R_{s1}	$3,2805507559 \times 10^{-5} \Omega/\text{m}$	$\lambda''_{11} = 0,0325240896$
R_1	$1,2428312689 \times 10^{-5} \Omega/\text{m}$	
g_{s1}	1,0173225455	
λ_{01}	0,0107415183	
Δ_{11}	0,0494707171	
Δ_{21}	$2,2858894134 \times 10^{-5}$	
β_{11}	109,6902808035	
t_s	2,9 mm	

12. The operating temperature θ_{s2} (°C) of the screen for outer cable lagging phase θ_{s2} :

$$\theta_{s2} = \theta_2 - (I^2 R_2 + 0,5 W_d) T_1 \quad (^\circ\text{C})$$

θ_2	79,6978129381 °C	$\theta_{s2} = 58,5130496389$ °C
I	1590,2251479060 A	
R_2	$1,2427310708 \times 10^{-5}$ Ω/m	
W_d	5,9461483662 W/m	
T_1	0,6158458706 K.m/W	

13. AC Resistance of the sheath for outer cable lagging phase at the operating temperature per unit length R_{s2} :

$$R_{s2} = R_{s0} \cdot [1 + \alpha_s(\theta_{s2} - 20)] \text{ (}\Omega/\text{m)}$$

R_{s0}	$2,8394986158 \times 10^{-5}$ Ω/m	$R_{s2} = 3,2802103529 \times 10^{-5}$ Ω/m
α_s	$4,03 \times 10^{-3}$	
θ_{s2}	58,5130496389 °C	

14. Electrical resistivity of the sheath for outer cable lagging phase at operating temperature ρ_{s2} :

$$\rho_{s2} = \rho_{s0} \cdot (1 + \alpha_s(\theta_{s2} - 20)) \text{ (}\Omega \cdot \text{m)}$$

ρ_{s0}	$2,84 \times 10^{-8}$ Ω.m	$\rho_{s2} = 3,2807895557 \times 10^{-8}$ Ω.m
α_s	$4,03 \times 10^{-3}$	
θ_{s2}	58,5130496389 °C	

15. Loss factor λ'_{12} for eddy current losses of the sheath for outer cable lagging phase:

$$\beta_{12} = \sqrt{\frac{4\pi\omega}{10^7 \rho_{s2}}}$$

ω	314,159 rad/s	$\beta_{12} = 109,6959721972$
ρ_{s2}	$3,2807895557 \times 10^{-8}$ Ω.m	

$$g_{s2} = 1 + \left(\frac{t_s}{D_{sm}}\right)^{1,74} (\beta_{12} \cdot D_{sm} \cdot 10^{-3} - 1,6)$$

t_s	2,9 mm	$g_{s2} = 1,0173235628$
D_{sm}	125,25 mm	
β_{12}	109,6959721972	
Note: D_{sm} has been used instead of D_s as the cable sheath is a corrugated sheath		

$$m_2 = \frac{\omega}{R_{s2}} \cdot 10^{-7}$$

ω	314,159 rad/s	$m_2 = 0,9577412165$
R_{s2}	$3,2802103529 \times 10^{-5}$ Ω/m	

$$\lambda_{02} = 1,5 \left(\frac{m_2^2}{1 + m_2^2} \right) \cdot \left(\frac{d}{2s} \right)^2$$

m_2	0,9577412165	$\lambda_{02} = 0,0107426811$
d	122,35 mm	
s	500 mm	

$$\Delta_{12} = -\frac{0,74(m_2 + 2)m^{0,5}}{2 + (m_2 - 0,3)^2} \cdot \left(\frac{d}{2s}\right)^{(m_2+1)}$$

m_2	0,9577412165	$\Delta_{12} = -0,0144047517$
d	122,35 mm	
s	500 mm	

$$\Delta_{22} = 0,92m_2^{3,7} \cdot \left(\frac{d}{2s}\right)^{(m_2+2)}$$

m_2	0,9577412165	$\Delta_{22} = 1.5695485431 \times 10^{-3}$
d	122,35 mm	
s	500 mm	

$$\lambda''_{12} = \frac{R_{s2}}{R_2} \left[g_{s2} \cdot \lambda_{02} (1 + \Delta_{12} + \Delta_{22}) + \frac{(\beta_{12} t_s)^4}{12 \times 10^{12}} \right]$$

R_{s2}	$3,2802103529 \times 10^{-5} \Omega/\text{m}$	$\lambda''_{12} = 0,0307291313$
R_2	$1,2427310708 \times 10^{-5} \Omega/\text{m}$	
g_{s2}	1,0173235628	
λ_{02}	0,0107426811	
Δ_{12}	-0,0144047517	
Δ_{22}	$1.5695485431 \times 10^{-3}$	
β_{12}	109,6959721972	
t_s	2,9 mm	

16. Loss factor λ_{1m} for sheath of centre cable:

$$\lambda_{1m} = \lambda'_{1m} + \lambda''_{1m}$$

λ'_{1m}	0	$\lambda_{1m} = 0,1170631205$
λ''_{1m}	0,1170631205	
Note: Loss factor circulating losses for sheath is ignored as we have considered fully balanced (minor sections) cross bonding.		

17. Loss factor λ_{11} for sheath of centre cable:

$$\lambda_{11} = \lambda'_{11} + \lambda''_{11}$$

λ'_{11}	0	$\lambda_{11} = 0,0325240896$
λ''_{11}	0,0325240896	
Note: Loss factor circulating losses for sheath is ignored as we have considered fully balanced (minor sections) cross bonding.		

18. Loss factor λ_{12} for sheath of centre cable:

$$\lambda_{12} = \lambda'_{12} + \lambda''_{12}$$

λ'_{12}	0	$\lambda_{12} = 0,0307291313$
λ''_{12}	0,0307291313	
Note: Loss factor circulating losses for sheath is ignored as we have considered fully balanced (minor sections) cross bonding.		

9.3.6 Thermal resistance T_1 between conductor and screen

Table 9-20: Parameters needed to calculate the thermal resistance between conductor and screen

Parameter	Symbol	Value	Unit	Reference
Diameter of conductor	d_c	58,8	mm	Cable datasheet
External diameter of conductor binder	D_{cb}	59,3	mm	Cable datasheet
External diameter of conductor screen	D_{cs}	60,2	mm	Cable datasheet
Thickness of insulation of 1st Kraft	t_{i1}	0,5	mm	Cable datasheet
Thickness of insulation of PPLP	t_{i2}	-	mm	Calculated
Thickness of insulation of 2st Kraft	t_{i3}	4,5	mm	Cable datasheet
External diameter of insulation	D_i	111,2	mm	Cable data sheet
External diameter of insulation screen	D_{is}	112,4	mm	Cable data sheet
External diameter of binder (Copper woven fabric tape)	D_b	113,3	mm	Cable data sheet
Diameter of the imaginary cylinder which just touches the inside surface of the troughs of a corrugated sheath	D_{it}	113,3	mm	Cable data sheet
Diameter of the imaginary coaxial cylinder which just touches the crests of a corrugated sheath	D_{oc}	131,4	mm	Cable data sheet
Thermal resistivity of conductor binder (SS tape intercalated with carbon black paper tape)	ρ_{cb}	6,0	K.m/W	Note 1
Thermal resistivity of conductor and insulation screen	ρ_{sc}	5,0	K.m/W	[4], table 1
Thermal resistivity of insulation (Kraft)	ρ_{i1}	5,0	K.m/W	[4], table 1
Thermal resistivity of insulation (PPLP)	ρ_{i2}	5,5	K.m/W	[4], table 1
Thermal resistivity of binder (Copper woven fabric tape)	ρ_b	6,0	K.m/W	Note 2
Note 1: In this case: 6,0 K.m/W				
Note 2: In this case: 6,0 K.m/W				

Table 9-21: Sequential steps to calculate the thermal resistance between conductor and screen

Step	Parameter	Symbol	Unit	Reference
1	Thermal resistance of conductor binder	T_{1_cb}	K.m/W	[4], section 4.1.2.1
2	Thermal resistance of conductor screen	T_{1_cs}	K.m/W	[4], section 4.1.2.1
3	Thermal resistance of insulation (1 st Kraft)	T_{1_i1}	K.m/W	[4], section 4.1.2.1
4	Thermal resistance of insulation (PPLP)	T_{1_i2}	K.m/W	[4], section 4.1.2.1
5	Thermal resistance of insulation (2 nd Kraft)	T_{1_i3}	K.m/W	[4], section 4.1.2.1
6	Thermal resistance of insulation screen	T_{1_is}	K.m/W	[4], section 4.1.2.1
7	Thermal resistance of binder	T_{1_b}	K.m/W	[4], section 4.1.2.1
8	Thermal resistance of sealing compound	T_{3_se}	K.m/W	[4], section 4.1.2.1
9	Thermal resistance between conductor and screen	T_1	K.m/W	-

1. Thermal resistance of conductor binder T_{1_cb} :

$$T_{1_cb} = \frac{\rho_{sb}}{2\pi} \ln \left(\frac{D_{cb}}{d_c} \right) \quad (\text{K.m/W})$$

ρ_{cb}	6,0 K.m/W	$T_{1_cb} = 8,0858201869 \times 10^{-3} \text{ K.m/W}$
D_{cb}	59,3 mm	
d_c	58,8 mm	

2. Thermal resistance of conductor screen
- T_{1_cs}
- :

$$T_{1_cs} = \frac{\rho_{sc}}{2\pi} \ln\left(\frac{D_{cs}}{D_{cb}}\right) \quad (K.m/W)$$

ρ_{sc}	5,0 K.m/W	$T_{1_cs} = 1,1986791392 \times 10^{-2} \text{ K.m/W}$
D_{cs}	60,2 mm	
D_{cb}	59,3 mm	

3. Thermal resistance of insulation (1
- st
- Kraft)
- T_{1_i1}
- :

$$T_{1_i1} = \frac{\rho_{i1}}{2\pi} \ln\left(1 + \frac{2 \cdot t_{i1}}{D_{cs}}\right) \quad (K.m/W)$$

ρ_{i1}	5,0 K.m/W	$T_{1_i1} = 1,3110258888 \times 10^{-2} \text{ K.m/W}$
t_{i1}	0,5 mm	
D_{cs}	60,2 mm	

4. Thermal resistance of insulation (PPLP)
- T_{1_i2}
- :

As there is a difference between diameter and thickness, we have calculated all thermal resistance in accordance with diameters. However, in case of the insulation layer, it has 3 different materials but PPLP is a major insulation. Therefore we should make up the difference nominal thickness and diameter by adding additional thickness as the PPLP.

$$D_{i1} = D_{cs} + 2t_{i1} \quad (mm)$$

D_{cs}	60,2 mm	$D_{i1} = 61,2 \text{ mm}$
t_{i1}	0,5 mm	

$$D_{i2} = D_i - 2t_{i3} \quad (mm)$$

D_i	111,2 mm	$D_{i2} = 102,2 \text{ mm}$
t_{i3}	4,5 mm	

$$T_{1_i2} = \frac{\rho_{i2}}{2\pi} \ln\left(\frac{D_{i2}}{D_{i1}}\right) \quad (K.m/W)$$

ρ_{i2}	5,5 K.m/W	$T_{1_i2} = 4,44886702325 \times 10^{-1} \text{ K.m/W}$
D_{i1}	61,2 mm	
D_{i2}	102,2 mm	

5. Thermal resistance of insulation (2
- st
- Kraft)
- T_{1_i3}
- :

$$T_{1_i3} = \frac{\rho_{i1}}{2\pi} \ln\left(1 + \frac{2 \cdot t_{i3}}{D_{i2}}\right) \quad (K.m/W)$$

ρ_{i1}	5,0 K.m/W	$T_{1_i3} = 6,7162354698 \times 10^{-2} \text{ K.m/W}$
t_{i3}	4,5 mm	
D_{i2}	102,2 mm	

6. Thermal resistance of insulation screen $T_{1_{is}}$:

$$T_{1_{is}} = \frac{\rho_{sc}}{2\pi} \ln\left(\frac{D_{is}}{D_i}\right) \quad (K.m/W)$$

ρ_{sc}	5,0 K.m/W	$T_{1_{is}} = 8,5414921878 \times 10^{-3} \text{ K.m/W}$
D_{is}	112,4 mm	
D_i	111,2 mm	

7. Thermal resistance of binder T_{1_b} :

In case of corrugated sheath final layer of thermal resistance shall be calculated by the mean internal diameter of the sheath instead of the surface of the troughs of a corrugated sheath

$$d_{sm} = \left(\frac{D_{it} + D_{oc}}{2}\right) - t_s \quad (mm)$$

D_{it}	113,3 mm	$d_{sm} = 119,45 \text{ mm}$
D_{oc}	131,4 mm	
t_s	2,9 mm	

$$T_{1_b} = \frac{\rho_b}{2\pi} \ln\left(\frac{d_{sm}}{D_{is}}\right) \quad (K.m/W)$$

ρ_b	6,0 K.m/W	$T_{1_b} = 5,8092130031 \times 10^{-2} \text{ K.m/W}$
d_{sm}	119,45 mm	
D_{is}	112,4 mm	

8. Thermal resistance between conductor and screen T_1 :

$$T_1 = T_{1_b} + T_{1_{cs}} + T_{1_{i1}} + T_{1_{i2}} + T_{1_{i3}} + T_{1_{is}} + T_{1_b} \quad (K.m/W)$$

$T_{1_{cb}}$	$8,0858201869 \times 10^{-3} \text{ K.m/W}$	$T_1 = 0,6158458706 \text{ K.m/W}$
$T_{1_{cs}}$	$1,1986791392 \times 10^{-2} \text{ K.m/W}$	
$T_{1_{i1}}$	$1,3110258888 \times 10^{-2} \text{ K.m/W}$	
$T_{1_{i2}}$	$4,4886702325 \times 10^{-1} \text{ K.m/W}$	
$T_{1_{i3}}$	$6,7162354698 \times 10^{-2} \text{ K.m/W}$	
$T_{1_{is}}$	$8,5414921878 \times 10^{-3} \text{ K.m/W}$	
T_{1_b}	$5,8092130031 \times 10^{-2} \text{ K.m/W}$	

9.3.7 Thermal resistance of outer covering (serving) T_3

Table 9-22: Parameters needed to calculate the thermal resistance T_3 of the outer covering

Parameter	Symbol	Value	Unit	Reference
Diameter of the imaginary cylinder which just touches the inside surface of the troughs of a corrugated sheath	D_{it}	113,3	mm	Cable data sheet
Diameter of the imaginary coaxial cylinder which just touches the crests of a corrugated sheath	D_{oc}	131,4	mm	Cable data sheet
Thickness of Sealing compound (Bitumen)	t_{se}	0,2	mm	Cable data sheet
External diameter of Sealing compound (Bitumen)	D_{se}	131,8	mm	Cable data sheet
External diameter of Serving (sheath)	D_j	142,8	mm	Cable data sheet
External diameter of cable	D_e	148,0	mm	Cable data sheet
Thermal resistivity of Sealing compound (Bitumen)	ρ_{se}	6,0	K.m/W	[4], table 1
Thermal resistivity of FR-MDPE	ρ_j	3,5	K.m/W	[4], table 1
Thermal resistivity of Semi-conductive FR-MDPE	ρ_{sj}	2,5	K.m/W	[4], table 1

Table 9-23: Sequential steps to calculate the thermal resistance T_3 of the outer covering

Step	Parameter	Symbol	Unit	Reference
1	Thermal resistance of Sealing compound (Bitumen)	T_{3_se}	K.m/W	[4], section 4.1.4.1
2	Thermal resistance of Serving (FR-MDPE)	T_{3_j}	K.m/W	[4], section 4.1.4.1
3	Thermal resistance of Semi-conductive FR-MDPE	T_{3_sj}	K.m/W	[4], section 4.1.4.1
4	Thermal resistance of outer covering	T_3	K.m/W	[4], section 4.1.4.1

1. Thermal resistance of Sealing compound (Bitumen) T_{3_se} :

In case of corrugated sheath, a first layer of thermal resistance shall be calculated by the mean outer diameter of the sheath instead of the crests of a corrugated sheath.

$$D_{sm} = \left(\frac{D_{it} + D_{oc}}{2} \right) + t_s \quad (mm)$$

D_{it}	113,3 mm	$D_{sm} = 125,25 \text{ mm}$
D_{oc}	131,4 mm	
t_s	2,9 mm	

$$T_{3_se} = \frac{\rho_{se}}{2\pi} \ln \left(\frac{D_{se}}{D_{sm}} \right) \quad (K.m/W)$$

ρ_{se}	6,0 K.m/W	$T_{3_se} = 4,8676471832 \times 10^{-2} \text{ K.m/W}$
D_{sm}	125,25 mm	
D_{se}	131,8 mm	

2. Thermal resistance of Serving (FR-MDPE) T_{3_j} :

$$T_{3_j} = \frac{\rho_j}{2\pi} \ln \left(\frac{D_j}{D_{se}} \right) \quad (K.m/W)$$

- $$T_{3_sj} = \frac{\rho_{sj}}{2\pi} \ln\left(\frac{D_e}{D_j}\right) \quad (K.m/W)$$

ρ_{sj}	2,5 K.m/W	$T_{3_j} = 1,4231326194 \times 10^{-2} \text{ K.m/W}$
D_e	148 mm	
D_j	142,8 mm	

- $$T_3 = T_{3_se} + T_{3_j} + T_{3_sj} \quad (K.m/W)$$

$T_{3,se}$	$4,8676471832 \times 10^{-2} \text{ K.m/W}$	$T_3 = 0,1075599901 \text{ K.m/W}$
$T_{3,j}$	$4,4652192115 \times 10^{-2} \text{ K.m/W}$	
$T_{3,sj}$	$1,4231326194 \times 10^{-2} \text{ K.m/W}$	

Diagram illustrating the geometry of a three-cable system for a bridge deck. The system consists of three cables (Cable 1, Cable 2, Cable 3) supporting a deck. The deck is represented by a horizontal line. The vertical distance from the deck to the top of Cable 1 is 1000. The vertical distance from the deck to the top of Cables 2 and 3 is 1000. The horizontal distance from the center of Cable 1 to the center of Cable 2 is 500, and the horizontal distance from the center of Cable 1 to the center of Cable 3 is 500. The total horizontal distance between the centers of Cables 2 and 3 is d_{m1} . The total horizontal distance between the centers of Cables 2 and 3 is d_{m2} . The length of Cable 1 is 2061.55. The length of Cable 2 is 2061.55. The length of Cable 3 is 2061.55. The diagram is labeled "Ground Level".

$$\Delta\theta_m = \frac{\rho_T}{2\pi} W_1 \ln\left(\frac{d'_{m1}}{d_{m1}}\right) + \frac{\rho_T}{2\pi} W_2 \ln\left(\frac{d'_{m2}}{d_{m2}}\right) \quad (\text{K})$$

W_m	41,4112641409 W/m (Shall be iterated)	$\Delta\theta_m = 17,4428254028\text{ }^{\circ}\text{C}$ (Shall be iterated)
W_1	38,7109515715 W/m (Shall be iterated)	
W_2	38,6545631129 W/m (Shall be iterated)	
d_{m1}	500 mm	
d'_{m1}	2061,5528128088 mm	
d_{m2}	500 mm	
d'_{m2}	2061,5528128088 mm	
Note: The values in this table are calculated from the initial result of permissible current rating in the previous clause. ($I = 1587,5417571179\text{ A}$). Therefore, the values shall be iterated until the current rating is converged.		

$$T_{4m} = \frac{\rho_T}{2\pi} \ln(u + \sqrt{u^2 - 1}) \quad (\text{K} \cdot \text{m}/\text{W})$$

ρ_T	1,0 K.m/W	$T_{4m} = 0,5244896315\text{ K.m/W}$
u	13,5135135135	

$$I = \left[\frac{\Delta\theta - \Delta\theta_m - W_d[0,5 \cdot T_1 + T_3 + T_{4m}]}{R \cdot T_1 + R(1 + \lambda_{1m})(T_3 + T_{4m})} \right]^{0,5}$$

Whenever calculate the temperature rise at the surface produced by different cables shall be different as the temperature of each cable should be different and it leads to different heat losses. Therefore, the calculation shall be iterated until it converged. The initial value of temperature and current rating can be used by the equally loaded result.

No of iteration	θ_{cm} ($^{\circ}\text{C}$)	θ_{c1} ($^{\circ}\text{C}$)	θ_{c2} ($^{\circ}\text{C}$)	$\Delta\theta_m$ ($^{\circ}\text{C}$)	λ_{1m}	I (A)
0	85	85	85	17,4428254028	0,1170607140	1587,5417571179
1	85	80,1377079372	80,0593073786	17,3270887711	0,1170607140	1590,8930700733
2	85	79,8178114559	79,7789059064	17,2985421429	0,1170614784	1589,9354696421
3	85	79,7199423486	79,6869479684	17,3023903042	0,1170635008	1590,3324327174
...	...	...	...	...	...	...
23	85	79,7291206272	79,6978129381	17,3007818051	0,1170631205	1590,2251479065
24	85	79,7291206273	79,6978129382	17,3007818051	0,1170631205	1590,2251479058
25	85	79,7291206272	79,6978129381	17,3007818051	0,1170631205	1590,2251479061
26	85	79,7291206273	79,6978129381	17,3007818051	0,1170631205	1590,2251479060

When the convergence is reached, the following values are obtained for the three cables.

Parameter (Unit)	Values (Outer λ_{11})	Values (Middle λ_{1m})	Values (Outer λ_{12})
I (A)	1590,2251479060		
W_c (W/m)	31,4289162416	31,8560205706	31,4263824169
W_s (W/m)	1,0221968888	3,7291651759	0,9657054325
$\Delta\theta_m$ (K)	14,2738222809	17,3007818051	14,2813819438
θ_c ($^{\circ}\text{C}$)	79,7291206273	85	79,6978129381
θ_s ($^{\circ}\text{C}$)	58,5427968825	63,5506458179	58,5130496389
θ_j ($^{\circ}\text{C}$)	54,4127878145	59,0835359302	54,3893893295

9.3.9 Calculation of losses

Losses in conductor per unit length of centre cable:

$$W_{cm} = I^2 R \text{ (W/m)}$$

I	1590,2251479060 A	$W_{cm} = 31,8560205706 \text{ W/m}$
R	$1,2597207668 \times 10^{-5} \Omega/\text{m}$	

Losses in conductor per unit length of outer cable leading phase:

$$W_{c1} = I^2 R_1 \text{ (W/m)}$$

I	1590,4245341695 A	$W_{c1} = 31,4289162416 \text{ W/m}$
R_1	$1,2428312689 \times 10^{-5} \Omega/\text{m}$	

Losses in conductor per unit length of outer cable lagging phase:

$$W_{c2} = I^2 R_2 \text{ (W/m)}$$

I	1590,4245341695 A	$W_{c2} = 31,4263824169 \text{ W/m}$
R_2	$1,2427310708 \times 10^{-5} \Omega/\text{m}$	

Losses in metallic sheath per unit length of centre cable:

$$W_{sm} = \lambda_{1m} \cdot W_{cm} \text{ (W/m)}$$

λ_{1m}	0,1170631205	$W_{sm} = 3,7291651759 \text{ W/m}$
W_{cm}	31,8560205706 W/m	

Losses in metallic sheath per unit length of outer cable leading phase:

$$W_{s1} = \lambda_{11} \cdot W_{c1} \text{ (W/m)}$$

λ_{11}	0,0325240896	$W_{s1} = 1,0221968888 \text{ W/m}$
W_{c1}	31,4289162416 W/m	

Losses in metallic sheath per unit length of outer cable lagging phase:

$$W_{s2} = \lambda_{12} \cdot W_{c2} \text{ (W/m)}$$

λ_{12}	0,0307291313	$W_{s2} = 0,9657054325 \text{ W/m}$
W_{c2}	31,4263824169 W/m	

10. Case study 6: A 400kV single core AC submarine cable circuit

10.1 Introduction

The selected cable type is a single core extruded submarine cable with copper round compacted conductor, XLPE insulation, copper wire screen, lead alloy sheath, semiconducting polyethylene sheath, steel flat wire armour and two layers of polypropylene yarns. The datasheet is depicted in Table 10-1. The current rating will be determined for the following laying configuration:

- Depth of laying is 1 m from the top of the cable
- Flat formation
- Soil temperature: 15 °C
- Soil thermal resistivity: 0,7 K.m/W
- Sheaths solidly bonded

All other relevant parameters are given in the input tables at each calculation step.

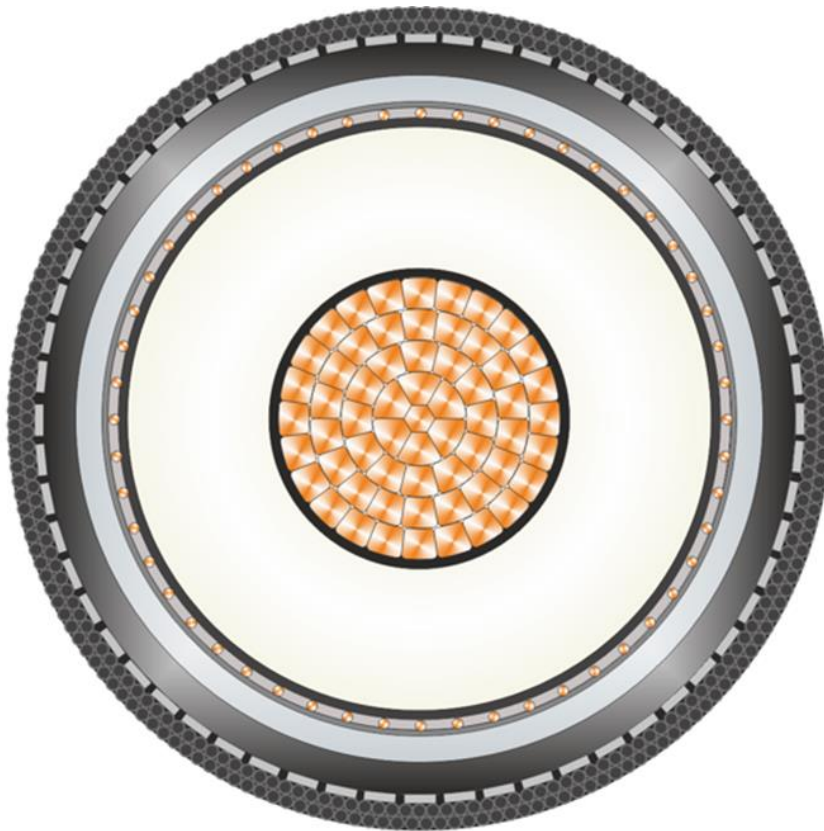


Figure 36: Indicative cable drawing – 1x800 mm² CU/XLPE/CWS/sc PE/FSWA/PPY 230/400 (420) kV

Table 10-1: Parameters needed to calculate the conductor AC resistance at operating temperature

Cable Layer	Description and diameter	Diameter (mm)
Conductor	Copper Round stranded compacted of nominal cross-section 800 sq.mm.	34,7
Conductor Tape	Semiconductive Tape	35,8
Conductor Screen	Extruded semiconductive PE	39,8
Insulation	XLPE	100,6
Insulation Screen	Extruded semiconductive PE	103,7
Tapes	Semiconductive Waterblocking Tape	104,7
Screen	Copper Wire Screen, 48 wires x 1,6 mm Ratio of the length of the wires to the length of the cable: 1,23	
Tapes	Semiconductive Waterblocking Tape	109,7
Metallic Sheath	Lead Alloy Sheath	116,9
Sheath	Semiconductive PE sheath of 3 mm	122,9
Armour	32 Galvanized Steel Flat Wires, 2,7 x 12 mm	
Outer Serving	Two layers of polypropylene yarns with total thickness 3 mm	134,3

10.2 Calculation of the conductor AC resistance at operation temperature

Table 10-2: Parameters needed to calculate the conductor AC resistance at operating temperature

Parameter	Symbol	Value	Unit	Reference
Conductor dc resistance @ 20°C	R_o	$0,0221 \times 10^{-3}$	Ω/m	[8]
Conductor diameter	d_c	34,7	mm	Cable datasheet
Conductor material		Copper		Cable datasheet
Type of conductor		Round Stranded Compacted		Cable datasheet
Skin effect factor	k_s	1	-	[3], table 2
Proximity effect factor	k_p	1	-	[3], table 2
Conductor temperature coefficient at 20°C per Kelvin	α	$3,93 \times 10^{-3}$	1/K	[3], table 1
Conductor operating temperature - Middle phase - Leading phase - Lagging phase	θ	90 To be calculated To be calculated	°C	System design
Distance between conductor axes	s	10	m	System design
System frequency	f	50	Hz	System design

Table 10-3: Sequential steps to calculate the conductor AC resistance at operating temperature

Step	Parameter	Symbol	Unit	Reference
1	Operating temperature of each conductor	θ_n	°C	[3], section 1.4.1.1.
1	Conductor DC resistance at operating temperature	R'	Ω/m	[3], section 2.1.1
2	Argument of a Bessel function – skin effect	x_s	-	[3], section 2.1.2
3	Skin effect factor	y_s	-	[3], section 2.1.2
4	Argument of a Bessel function – proximity effect	x_p	-	[3], section 2.1.4
5	Proximity effect factor	y_p	-	[3], section 2.1.4
6	AC resistance of conductor	R	Ω/m	[3], section 2.1

1. Operating temperature of each conductor, only to be calculated for the lagging phase and the leading phase (n=1 for the lagging phase and n=2 for the leading phase):

$$\theta_n = \left(I^2 R_n + \frac{1}{2} W_d \right) \cdot T_1 + (I^2 R_n (1 + \lambda_{1,n}) + W_d) \cdot T_2 + (I^2 R_n (1 + \lambda_{1,n} + \lambda_{2,n}) + W_d) \cdot (T_3 + T_{4,n}) + \theta_\alpha$$

I	1039,4347828913 A	$\theta_m = 90$ $\theta_1 = 89,7413552698$ $\theta_2 = 89,7413552421$
R _m	$3,2222847660 \times 10^{-5}$ Ohm/m	
R ₁	$3,2203115005 \times 10^{-5}$ Ohm/m	
R ₂	$3,2203115003 \times 10^{-5}$ Ohm/m	
W _d	2,50958961115 W/m	
$\lambda_{1,m}$	1,0800121517	
$\lambda_{2,m}$	1,0799516782	
$\lambda_{1,1}$	1,0798074281	
$\lambda_{2,1}$	1,0797525302	
$\lambda_{1,2}$	1,0798074266	
$\lambda_{2,2}$	1,0797525302	
T ₁	0,61888739234	
T ₂	0,05151277491	
T ₃	0,04364489953	
T _{4,m}	0,39020666356	
T _{4,1}	0,38836146837	
T _{4,2}	0,38836146837	

Since the conductor temperature of the leading and the lagging phase is a function of the current I, an iterative method must be used.

2. Conductor DC resistance at operating temperature R' :

$$R'_n = R_o [1 + \alpha_{20}(\theta_n - 20)] \text{ (}\Omega/\text{m)}$$

R _o	$0,0221 \times 10^{-3}$ Ω/m	$R'_m = 2,817971000 \times 10^{-5}$ Ω/m $R'_1 = 2,815724593 \times 10^{-5}$ Ω/m $R'_2 = 2,815724593 \times 10^{-5}$ Ω/m
α_{20}	$3,93 \times 10^{-3}$ 1/K	
θ_m	90 °C	
θ_1	89,7413552698	
θ_2	89,7413552421	

3. Argument of a Bessel function used to calculate skin effect x_s :

$$x_{s_n}^2 = \frac{8\pi f}{R'_n} 10^{-7} k_s$$

f	50 Hz	$x_{s_m}^2 = 4,4593683236$ $x_{s_1}^2 = 4,4629260425$ $x_{s_2}^2 = 4,4629260429$
R' _m	$2,817971000 \times 10^{-5}$ Ω/m	
R' ₁	$2,815724593 \times 10^{-5}$ Ω/m	
R' ₂	$2,815724593 \times 10^{-5}$ Ω/m	
k _s	1	

4. Skin effect factor y_s :

$$\text{for } 0 < x_s \leq 2,8 \quad \rightarrow \quad y_{s_n} = \frac{x_s^4}{192 + 0,8x_s^4}$$

Three equations are given for calculating y_s , use of these equations depends on the calculated value of x_s .

$x_{s,m}^2$	4,4593683236	$y_{s,m} = 0,0956475554$
$x_{s,1}^2$	4,4629260425	$y_{s,1} = 0,0957885332$
$x_{s,2}^2$	4,4629260429	$y_{s,2} = 0,0957885332$

5. Argument of a Bessel function used to calculate proximity effect x_p :

$$x_{p,n}^2 = \frac{8\pi f}{R'_n} 10^{-7} k_p$$

f	50 Hz	$x_{p,m}^2 = 4,4593683236$ $x_{p,1}^2 = 4,4629260425$ $x_{p,2}^2 = 4,4629260429$
R'_m	$2,817971000 \times 10^{-5} \Omega/\text{m}$	
R'_1	$2,815724593 \times 10^{-5} \Omega/\text{m}$	
R'_2	$2,815724593 \times 10^{-5} \Omega/\text{m}$	
k_p	1	

6. Proximity effect factor y_p :

$$y_{p,n} = \frac{x_p^4}{192 + 0,8x_p^4} \left(\frac{d_c}{s} \right)^2 \left[0,312 \left(\frac{d_c}{s} \right)^2 + \frac{1,18}{\frac{x_p^4}{192 + 0,8x_p^4} + 0,27} \right]$$

$x_{p,m}^2$	4,4593683236	$y_{p,m} = 3,71665853738 \times 10^{-6}$ $y_{p,1} = 3,72070209260 \times 10^{-6}$ $y_{p,2} = 3,72070209303 \times 10^{-6}$
$x_{p,1}^2$	4,4629260425	
$x_{p,2}^2$	4,4629260429	
d_c	$34,7 \times 10^{-3} \text{ m}$	
s	10 m	

7. AC resistance of conductor R :

$$R_n = R'_n (1 + 1,5(y_s + y_p)) \quad (\Omega/\text{m})$$

R'_m	$2,817971000 \times 10^{-5} \Omega/\text{m}$	$R_m = 3,2222847660 \times 10^{-5} \Omega/\text{m}$ $R_1 = 3,22031150048 \times 10^{-5} \Omega/\text{m}$ $R_2 = 3,22031150027 \times 10^{-5} \Omega/\text{m}$
R'_1	$2,815724593 \times 10^{-5} \Omega/\text{m}$	
R'_2	$2,815724593 \times 10^{-5} \Omega/\text{m}$	
$y_{s,m}$	0,0956475554	
$y_{s,1}$	0,0957885332	
$y_{s,2}$	0,0957885332	
$y_{p,m}$	$3,71665853738 \times 10^{-6}$	
$y_{p,1}$	$3,72070209260 \times 10^{-6}$	
$y_{p,2}$	$3,72070209303 \times 10^{-6}$	

Note: The impact of armour wires in the conductor losses is taken into consideration with the factor 1.5 in the equation above. See Guidance Point 25.

10.3 Dielectric losses

Table 10-4: Parameters needed to calculate the dielectric loss per unit length per phase

Parameter	Symbol	Value	Unit	Reference
Diameter over insulation	D_i	100,6	mm	Cable data sheet
External diameter of the conductor, including screen	d_c	39,8	mm	Cable datasheet
Relative permittivity of the insulation	ϵ	2,5	-	[3], table 3, note 1
Loss factor of the insulation	$\tan\delta$	0,001	-	[3], table 3, note 1
Voltage to earth	U_o	400 kV / $\sqrt{3}$	V	System design
System frequency	f	50	Hz	System design
Note 1: Always assume unfilled XLPE unless specified otherwise				

Table 10-5: Sequential steps to calculate the dielectric loss per unit length per phase

Step	Parameter	Symbol	Unit	Reference
1	Capacitance	C	F/m	[3], section 2.2
2	Angular frequency of system	ω	-	[3], section 2.2
3	Dielectric loss per unit length in each phase	W_d	W/m	[3], section 2.2

1. Capacitance C (for circular conductors):

$$C = \frac{\epsilon}{18 \ln\left(\frac{D_i}{d_c}\right)} 10^{-9} \text{ (F/m)}$$

ϵ	2,5	$C = 1,4978009691 \times 10^{-10} \text{ F/m}$
D_i	$100,6 \times 10^{-3} \text{ m}$	
d_c	$39,8 \times 10^{-3} \text{ m}$	

2. Angular frequency of system:

$$\omega = 2\pi f \text{ (rad/s)}$$

f	50 Hz	$\omega = 314,159265359 \text{ rad/s}$
-----	-------	--

3. Dielectric loss per unit length in each phase:

$$W_d = \omega C U_o^2 \tan\delta \text{ (W/m)}$$

C	$1,4978009691 \times 10^{-10} \text{ F/m}$	$W_d = 2,5095896112 \text{ W/m}$
ω	$314,159265359 \text{ rad/s}$	
U_o	$400\text{kV}/\sqrt{3} = 230940,1076758500 \text{ V}$	
$\tan\delta$	0,001	

10.4 Loss factor for screen, lead alloy sheath and armour

See flow chart in Section 2.6 for details.

Table 10-6: Parameters needed to calculate the loss factor for screen, lead alloy sheath and armour

Parameter	Symbol	Value	Unit	Reference
AC resistance of screen at 20°C	R_{so_sc}	-	Ω/m	
AC resistance of lead alloy sheath at 20°C	R_{so_PB}	-	Ω/m	
AC resistance of armour at 20 °C	R_{A0}	-	Ω/m	
External Diameter of lead alloy sheath	D_{PB}	116,9	mm	Cable data sheet
Number of the copper wires at the screen	N_{cws}	48		Cable data sheet
Diameter of the copper wires	d_{cws}	1,6	mm	Cable data sheet
Number of the armour wires	N_A	32		Cable data sheet
Height of the armour wires	h	2,7	mm	Cable data sheet
Width of the armour wires	w	12	mm	Cable data sheet
Thickness of lead alloy	t_{PB}	3,6	mm	Cable data sheet
Lay length factor of the screen	f_{lay_sc}	1,23		Cable data sheet
Lay length factor of the armour	f_{lay_A}	1,035		Cable data sheet
Screen resistivity at 20°C	ρ_{sc}	$1,7241 \times 10^{-8}$	Ωm	[3], table 1
Lead alloy resistivity at 20°C	ρ_{PB}	$21,4 \times 10^{-8}$	Ωm	[3], table 1
Armour resistivity at 20°C	ρ_A	$13,8 \times 10^{-8}$	Ωm	[3], table 1
Screen temperature coefficient at 20°C per Kelvin	α_{sc}	$3,93 \times 10^{-3}$	-	[3], table 1
Lead alloy temperature coefficient at 20°C per Kelvin	α_{PB}	$4,0 \times 10^{-3}$	-	[3], table 1
Steel wire temperature coefficient at 20°C per Kelvin	α_A	$4,5 \times 10^{-3}$	-	[3], table 1
Distance between conductor axes	s	10	m	System design
Geometric mean of the three spacings	s_2		m	To be calculated
Length of lay of a steel wire along the cable	p		m	To be calculated
Angle between axis of armour wire and axis of cable	β		rad	To be calculated
System frequency	f	50	Hz	System design
AC resistance of conductor at operating temp.	R'	-	Ω/m	To be calculated
Equivalent diameter of a round wire which gives the same cross-section with the flat.	d_f		m	
Mean diameter of copper wire screen	d_{cws_mean}	106,3	mm	Cable data sheet
Mean diameter of copper wire screen	d_{pb_mean}	113,3	mm	Cable data sheet

Table 10-7: Sequential steps to calculate the loss factor for screen, lead alloy sheath and armour

Step	Parameter	Symbol	Unit	Reference
1	AC Resistance of screen at 20°C	R_{so_sc}	Ω/m	-
2	AC Resistance of lead alloy sheath at 20°C	R_{so_PB}	Ω/m	-
3	AC Resistance of the armour at 20°C	R_{A0}	Ω/m	-
4	Maximum operating temperature of copper wires screen, to be determined iterative for each phase	$\theta_{sc,n}$	$^{\circ}C$	[3], section 2.3
5	Maximum operating temperature of the cable lead alloy sheath, to be determined iterative for each phase	$\theta_{PB,n}$	$^{\circ}C$	[3], section 2.3
6	Maximum operating temperature of the cable armour, to be determined iterative for each phase	$\theta_{ar,n}$	$^{\circ}C$	[3], section 2.4
7	Resistance of copper wires screen at operating temperature for each phase	$R_{sc,n}$	Ω/m	[3], section 2.3
8	Resistance of lead alloy sheath at operating temperature for each phase	$R_{sPB,n}$	Ω/m	[3], section e 2.3
9	Resistance of screen and lead alloy at operating temperature for each phase	$R_{s,n}$	Ω/m	[3], section 2.4.2.1.
10	Flat armour wire resistance at operating temperature for each phase	$R_{A,n}$	Ω/m	[3], section 2.4
11	Equivalent Resistance of screen and armour for each phase	$R_{e,n}$	Ω/m	[3], section 2.4.2.1.
12	Inductance due to the sheath	H_s	H/m	[3], section 2.4.2.1.
13	Inductance due to the steel wires	H_1 H_2 H_3	H/m	[3], section 2.4.2.1.
14	Total loss in sheath and armour for each phase	$W_{(s+A),n}$	W/m	[3], section 2.4.2.1.
15	Conductor loss for each phase	$W_{c,n}$	W/m	[3], section 2.4.2.1.
16	Loss factor in sheath and armour caused by circulating currents for each phase	$\lambda'_{1,n}$ $\lambda'_{2,n}$		[3], section 2.4.2.1.
17	Eddy current losses of the sheath for each phase	$\lambda''_{1,n}$		[3], section e 2.3.6.1.
18	Loss factor in sheath and armour for each phase	$\lambda_{1,n}$ $\lambda_{2,n}$		[3], section 2.4.3.

1. Resistance of screen at 20°C:

AC Resistance of the screen at 20°C per unit length is (see Guidance point 27):

$$R_{so_sc}^* = \frac{\rho_{sc}}{N_{cws} \cdot \pi \cdot \left(\frac{d_{cws}}{2}\right)^2} \quad (\Omega/m)$$

ρ_{sc}	$1,7241 \times 10^{-8} \Omega \cdot m$	$R_{so_sc}^* = 1,7864520663 \times 10^{-4} \Omega/m$
N_{cws}	48	
d_{cws}	$1,6 \times 10^{-3} m$	

Correction of the AC Resistance of the screen, taken into account lay-length factor:

$$R_{so_sc} = R_{so_sc}^* \cdot f_{lay_sc} (\Omega/m)$$

$R_{so_sc}^*$	$1,7864520663 \times 10^{-4} \Omega/m$	$R_{so_sc} = 2,1973360416 \times 10^{-4} \Omega/m$
f_{lay_sc}	1,23	

2. Resistance of lead alloy sheath at 20°C:

AC Resistance of lead alloy sheath at 20°C per unit length is (see Guidance Point 11):

$$R_{so_PB} = \frac{\rho_{PB}}{t_{PB} \cdot \pi (D_{PB} - t_{PB})} \quad (\Omega/m)$$

ρ_{PB}	$21,4 \times 10^{-8} \Omega\text{m}$	$R_{so_PB} = 1,6700577533 \times 10^{-4} \Omega/\text{m}$
t_{PB}	$3,6 \times 10^{-3} \text{ m}$	
D_{PB}	$116,9 \times 10^{-3} \text{ m}$	

3. Resistance of the armour at 20 °C:

$$R_A^* = \frac{\rho_A}{N_A \cdot (h \cdot w)} \quad (\Omega/\text{m})$$

ρ_A	$13,8 \times 10^{-8} \Omega\text{m}$	$R_A^* = 1,3310185185 \times 10^{-4} \Omega/\text{m}$
N_A	32	
h	$2,7 \times 10^{-3} \text{ m}$	
w	$12 \times 10^{-3} \text{ m}$	

Correction of the AC Resistance of the armour, taken into account the lay-length factor according to Guidance Point 32, Guidance point 33 and Guidance point 34.

$$y(x) = \frac{y_2 - y_1}{x_2 - x_1} \cdot (x - x_1) + y_1 \Rightarrow y(2,7) = \frac{1,4 - 1,2}{5 - 2} \cdot (2,7 - 2) + 1,2 \Rightarrow y(2,7) = 1,2466666667$$

$$R_A = R_A^* \cdot f_{lay_A} \cdot 1,2466666667 (\Omega/\text{m})$$

R_A^*	$1,3310185185 \times 10^{-4} \Omega/\text{m}$	$R_A = 1,7174131944 \times 10^{-4} \Omega/\text{m}$
f_{lay_A}	1,035	

4. The operating temperature of the copper wire screen θ_{sc} (°C) is given by:

$$\theta_{sc,n} = \theta_n - (I^2 R_n + 0,5 W_d) T_1$$

θ_m	90 °C	$\theta_{sc,m} = 67,6772551592 \text{ °C}$ $\theta_{sc,1} = 67,4318048902 \text{ °C}$ $\theta_{sc,2} = 67,4318048639 \text{ °C}$
θ_1	89,7413552698 °C	
θ_2	89,7413552421 °C	
I	1039,4347828913 A	
R_m	$3,2222847660 \times 10^{-5} \Omega/\text{m}$	
R_1	$3,2203115005 \times 10^{-5} \Omega/\text{m}$	
R_2	$3,2203115003 \times 10^{-5} \Omega/\text{m}$	
W_d	2,50958961112 W/m	
T_1	0,61888739234 K.m/W	

Note: θ_{sc} is the operating temperature of the cable screen. Because the temperature of the screen is a function of the current I, an iterative method must be used for the calculation.

5. The operating temperature θ_{PB} (°C) of the lead alloy sheath is given by:

$$\theta_{PB,n} = \theta_n - \{(I^2 R_n + 0,5 W_d) \cdot T_1 + [I^2 R_n (1 + \lambda'_1) + W_d] T_{2wb}\} \text{ (°C)}$$

θ_m	90 °C	$\theta_{PB,m} = 65,3099025381 \text{ °C}$ $\theta_{PB,1} = 65,0660723695 \text{ °C}$ $\theta_{PB,2} = 65,0660723433 \text{ °C}$
θ_1	89,7413552698 °C	
θ_2	89,7413552421 °C	
I	1039,4347828913 A	
R_m	$3,2222847660 \times 10^{-5} \text{ } \Omega/\text{m}$	
R_1	$3,2203115005 \times 10^{-5} \text{ } \Omega/\text{m}$	
R_2	$3,2203115003 \times 10^{-5} \text{ } \Omega/\text{m}$	
W_d	2,50958961112 W/m	
$\lambda'_{1,m}$	1,0799516782	
$\lambda'_{1,1}$	1,0797525302	
$\lambda'_{1,2}$	1,0797525302	
T_1	0,61888739234	
T_{2wb}	0,03159765796	

Note: θ_{PB} is the operating temperature of the lead alloy sheath (°C). Because the temperature of the lead alloy sheath is a function of the current I , an iterative method must be used for the calculation.

Note: For the calculation of the temperature of lead alloy sheath only $\lambda'_{1,1}$ is taken into account given that the eddy currents of copper wire screen are considered as negligible.

6. The operating temperature θ_{ar} (°C) of the armour wires is given by:

$$\theta_{ar,n} = \theta_n - \{(I^2 R_n + 0,5 W_d) T_1 + [I^2 R_n (1 + \lambda_{1,n}) + W_d] n T_2\} \text{ (} ^\circ\text{C)}$$

θ_m	90 °C	$\theta_{ar,m} = 63,81771817358 \text{ °C}$ $\theta_{ar,1} = 63,57491916826 \text{ °C}$ $\theta_{ar,2} = 63,57491914496 \text{ °C}$
θ_1	89,7413552698 °C	
θ_2	89,7413552421 °C	
I	1039,4347828913 A	
R_m	$3,2222847660 \times 10^{-5} \text{ } \Omega/\text{m}$	
R_1	$3,2203115005 \times 10^{-5} \text{ } \Omega/\text{m}$	
R_2	$3,2203115003 \times 10^{-5} \text{ } \Omega/\text{m}$	
W_d	2,50958961115 W/m	
T_1	0,61888739234	
$\lambda_{1,m}$	1,0800121517	
$\lambda_{1,1}$	1,0798074281	
$\lambda_{1,2}$	1,0798074266	
T_2	0,0515127749	
n	1	

Note: θ_{ar} is the operating temperature of the armour wires (°C). Because the temperature of the armour wires is a function of the current I , an iterative method must be used for the calculation.

7. Copper wires screen AC resistance at operating temperature θ_{sc} :

$$R_{s,sc,n} = R_{so,sc} [1 + \alpha_{sc} (\theta_{sc,n} - 20)] \text{ (} \Omega/\text{m)}$$

R_{so_sc}	$2,1973360416 \times 10^{-4} \Omega/m$	$R_{s_sc,m} = 2,6090544395 \times 10^{-4} \Omega/m$ $R_{s_sc,1} = 2,6069348462 \times 10^{-4} \Omega/m$ $R_{s_sc,3} = 2,6069348459 \times 10^{-4} \Omega/m$
α_{sc}	$3,93 \times 10^{-3}$	
$\theta_{sc,m}$	$67,6772551592 \text{ } ^\circ\text{C}$	
$\theta_{sc,1}$	$67,4318048902 \text{ } ^\circ\text{C}$	
$\theta_{sc,2}$	$67,4318048639 \text{ } ^\circ\text{C}$	

8. Lead alloy AC resistance at operating temperature θ_{PB} :

$$R_{s_PB,n} = R_{so_PB} [1 + \alpha_{PB} (\theta_{PB,n} - 20)] \quad (\Omega/m)$$

R_{so_PB}	$1,6700577533 \times 10^{-4} \Omega/m$	$R_{s_PB,m} = 1,9727383695 \times 10^{-4} \Omega/m$ $R_{s_PB,1} = 1,97110952764 \times 10^{-4} \Omega/m$ $R_{s_PB,2} = 1,97110952746 \times 10^{-4} \Omega/m$
α_{PB}	$4,00 \times 10^{-3}$	
$\theta_{PB,m}$	$65,3099025381 \text{ } ^\circ\text{C}$	
$\theta_{PB,1}$	$65,0660723695 \text{ } ^\circ\text{C}$	
$\theta_{PB,2}$	$65,0660723433 \text{ } ^\circ\text{C}$	

9. Resistance of screen and lead alloy at operating temperature:

$$R_{s,m} = \frac{R_{s_sc,m} \cdot R_{s_PB,m}}{R_{s_sc,m} + R_{s_PB,m}} \quad \left(\frac{\Omega}{m} \right)$$

$R_{s_sc,m}$	$2,6090544395 \times 10^{-4} \Omega/m$	$R_{s,m} = 1,1233554234 \times 10^{-4} \Omega/m$
$R_{s_PB,m}$	$1,9727383695 \times 10^{-4} \Omega/m$	

$$R_{s,1} = \frac{R_{s_sc,1} \cdot R_{s_PB,1}}{R_{s_sc,1} + R_{s_PB,1}} \quad \left(\frac{\Omega}{m} \right)$$

$R_{s_sc,1}$	$2,6069348462 \times 10^{-4} \Omega/m$	$R_{s,1} = 1,1224343177 \times 10^{-4} \Omega/m$
$R_{s_PB,1}$	$1,97110952764 \times 10^{-4} \Omega/m$	

$$R_{s,2} = \frac{R_{s_sc,2} \cdot R_{s_PB,2}}{R_{s_sc,2} + R_{s_PB,2}} \quad \left(\frac{\Omega}{m} \right)$$

$R_{s_sc,2}$	$2,6069348459 \times 10^{-4} \Omega/m$	$R_{s,2} = 1,1224343176 \times 10^{-4} \Omega/m$
$R_{s_PB,2}$	$1,97110952746 \times 10^{-4} \Omega/m$	

10. Flat armour wire resistance at operating temperature θ_A :

$$R_{s_A,n} = R_A [1 + \alpha_A (\theta_{ar,n} - 20)] \quad \left(\frac{\Omega}{m} \right)$$

R_A	$1,7174131945 \times 10^{-4} \Omega/m$	$R_{s_A,m} = 2,0560522675 \times 10^{-4} \Omega/m$ $R_{s_A,1} = 2,0541758295 \times 10^{-4} \Omega/m$ $R_{s_A,2} = 2,0541758293 \times 10^{-4} \Omega/m$
α_A	$4,5 \times 10^{-3}$	
$\theta_{ar,m}$	$63,81771817358 \text{ } ^\circ\text{C}$	
$\theta_{ar,1}$	$63,57491916826 \text{ } ^\circ\text{C}$	
$\theta_{ar,2}$	$63,57491914496 \text{ } ^\circ\text{C}$	

11. Equivalent resistance of sheath and armour in parallel:

$$R_{e,n} = \frac{R_{s_A,n} \cdot R_{s,n}}{R_{s_A,n} + R_{s,n}} \quad \left(\frac{\Omega}{m} \right)$$

$R_{s,A,m}$	$2,0560522675 \times 10^{-4} \Omega/m$	$R_{e,m} = 0,7264489773 \times 10^{-4} \Omega/m$
$R_{s,m}$	$1,1233554234 \times 10^{-4} \Omega/m$	
$R_{s,A,1}$	$2,0541758295 \times 10^{-4} \Omega/m$	$R_{e,1} = 0,7258295286 \times 10^{-4} \Omega/m$
$R_{s,1}$	$1,1224343177 \times 10^{-4} \Omega/m$	
$R_{s,A,2}$	$2,0541758293 \times 10^{-4} \Omega/m$	$R_{e,2} = 0,7258295285 \times 10^{-4} \Omega/m$
$R_{s,2}$	$1,1224343176 \times 10^{-4} \Omega/m$	

12. The inductance due to the sheath is calculated as follows:

$$H_s = 2 \cdot 10^{-7} \cdot \ln\left(\frac{2s_2}{d}\right)$$

$$s_2 = s \sqrt[3]{2} = 10 \cdot \sqrt[3]{2} = 12,5992104989 \text{ m}$$

$$d = \sqrt{\frac{d_{cwsmean}^2 + d_{pbmean}^2}{2}} = 109,8557690793 \times 10^{-3}$$

s_2	12,5992104989 m	$H_s = 1,0870736595 \times 10^{-6} \text{ H/m}$
d	109,8557690793 $\times 10^{-3}$ m	

13. The inductance due to the steel wires is calculated as follows:

$$H_1 = \pi \cdot \mu_e \cdot \left(\frac{n_1 \cdot d_f^2}{p \cdot d_A} \right) \cdot 10^{-7} \cdot \sin\beta \cdot \cos\gamma$$

An equivalent diameter of a round wire which gives the same cross-section with the flat is calculated:

$$d_f = \sqrt{\frac{4 \times (2,7 \cdot 10^{-3} \times 12 \cdot 10^{-3})}{\pi}} = 6,4228468182 \cdot 10^{-3} \text{ m}$$

$$p = \frac{\pi \cdot d_A}{\sqrt{1,035^2 - 1}} = 1478,5066531024$$

$$\beta = \tan^{-1}\left(\frac{\pi \cdot d_A}{p}\right) = 0,2608016660$$

μ_e	400	$H_1 = 1,6287818829 \times 10^{-7} \text{ H/m}$
n_1	32	
d_f	$6,4228468182 \times 10^{-3} \text{ m}$	
p	$1478,5066531024 \times 10^{-3} \text{ m}$	
d_A	$125,6 \times 10^{-3} \text{ m}$	
β	0,2608016660 rad	
γ	0,7853981634 rad	

$$H_2 = \pi \cdot \mu_e \cdot \left(\frac{n_1 \cdot d_f^2}{p \cdot d_A} \right) \cdot 10^{-7} \cdot \sin\beta \cdot \sin\gamma$$

μ_e	400	$H_2 = 1,6287818829 \times 10^{-7} \text{ H/m}$
n_1	32	
d_f	$6,4228468182 \times 10^{-3} \text{ m}$	
ρ	$1478,5066531024 \times 10^{-3} \text{ m}$	
d_A	$125,6 \times 10^{-3} \text{ m}$	
β	$0,2608016660 \text{ rad}$	
γ	$0,7853981634 \text{ rad}$	

$$H_3 = 0,4 \cdot (\mu_t \cdot \cos^2 \beta - 1) \cdot \left(\frac{d_f}{d_A} \right) \cdot 10^{-6}$$

μ_t	10	$H_3 = 1,7049399840 \times 10^{-7} \text{ H/m}$
β	$0,2608016660$	
d_f	$6,4228468182 \times 10^{-3} \text{ m}$	
d_A	$125,6 \times 10^{-3} \text{ m}$	

$$B_1 = \omega(H_s + H_1 + H_3)$$

H_s	$1,0870659570 \times 10^{-6} \text{ H/m}$	$B_1 = 0,4462462235 \times 10^{-3} \Omega/\text{m}$
H_1	$1,6287818829 \times 10^{-7} \text{ H/m}$	
H_3	$1,7049399840 \times 10^{-7} \text{ H/m}$	
ω	$314,159265359 \text{ rad/s}$	

$$B_2 = \omega H_2$$

H_2	$1,6287818827 \times 10^{-7} \text{ H/m}$	$B_2 = 0,5116969198 \times 10^{-4} \Omega/\text{m}$
ω	$314,159265359 \text{ rad/s}$	

14. Total loss in sheath and armour $W_{(s+A)}$ is given by:

$$W_{(s+A),n} = I^2 \cdot R_{e,n} \cdot \left(\frac{B_2^2 + B_1^2 + R_{e,n} \cdot B_2}{(R_{e,n} + B_2)^2 + B_1^2} \right)$$

$R_{e,m}$	$0,7275968151 \times 10^{-4} \Omega/\text{m}$	$W_{(s+A),m} = 75,1956518935 \text{ W/m}$
B_2	$0,5116969198 \times 10^{-4} \Omega/\text{m}$	
B_1	$0,4462462235 \times 10^{-3} \Omega/\text{m}$	
I_m	$1039,4347828913 \text{ A}$	
$R_{e,1}$	$0,7269762452 \times 10^{-4} \Omega/\text{m}$	$W_{(s+A),1} = 75,1357455914 \text{ W/m}$
B_2	$0,5116969198 \times 10^{-4} \Omega/\text{m}$	
B_1	$0,4462462235 \times 10^{-3} \Omega/\text{m}$	
I_1	$1039,4347828913 \text{ A}$	
$R_{e,2}$	$0,7269762452 \times 10^{-4} \Omega/\text{m}$	$W_{(s+A),2} = 75,1357455853 \text{ W/m}$
B_2	$0,5116969198 \times 10^{-4} \Omega/\text{m}$	
B_1	$0,4462462235 \times 10^{-3} \Omega/\text{m}$	
I_2	$1039,4347828913 \text{ A}$	

15. Conductor Losses W_c :

$$W_c = I^2 \cdot R$$

R_m	$3,2222847660 \times 10^{-5} \Omega/\text{m}$	$W_{c,m} = 34,8143594816 \text{ W/m}$
I	1039,4347828913 A	

R_1	$3,2203115005 \times 10^{-5} \Omega/\text{m}$	$W_{c,1} = 34,7930398339 \text{ W/m}$
I	1039,4347828913 A	

R_2	$3,2203115003 \times 10^{-5} \Omega/\text{m}$	$W_{c,2} = 34,7930398316 \text{ W/m}$
I	1039,4347828913 A	

16. Loss factor in sheath and armour caused by circulating currents may be assumed to be approximately equal, λ'_1, λ'_2 :

$$\lambda'_1 = \lambda'_2 = \frac{W_{(s+A)}}{2 W_c}$$

$W_{(s+A),m}$	72,6448977302 W/m	$\lambda'_{1,m} = \lambda'_{2,m} = 1,0799516782$
$W_{c,m}$	34,8143594816 W/m	

$W_{(s+A),1}$	72,5829528588 W/m	$\lambda'_{1,1} = \lambda'_{2,1} = 1,0797525303$
$W_{c,1}$	34,7930398339 W/m	

$W_{(s+A),2}$	72,5829528524 W/m	$\lambda'_{1,2} = \lambda'_{2,2} = 1,0797525302$
$W_{c,2}$	34,7930398316 W/m	

17. Eddy current losses (see Guidance point 6 and 31)

Loss factor $\lambda'_{1,m}$ for eddy current losses, lead alloy only, middle phase

$$m = \frac{\omega}{R_{s_{PB}}} 10^{-7}$$

ω	314,159265359	$m = 0,1592503447$
$R_{s_{PB}}$	$1,9727383695 \times 10^{-4} \Omega/\text{m}$	

$$\lambda_{0,m} = 6 \left(\frac{m^2}{1 + m^2} \right) \left(\frac{d_{pb_mean}}{2s} \right)^2$$

m	0,1592503447	$\lambda_{0,m} = 4,7625021513 \times 10^{-6}$
d_{pb_mean}	113,3 mm	
s	10000 mm	

$$\Delta_{1,m} = (0,86m^{3,08}) \left(\frac{d_{pb_mean}}{2s} \right)^{(1,4m+0,7)}$$

$$\Delta_2 = 0$$

m	0,1592503447	$\Delta_{1,m} = 2,5305842543 \times 10^{-5}$
d_{pb_mean}	113,3 mm	
s	10000 mm	

$$\beta_1 = \sqrt{\frac{4\pi\omega}{10^7 \rho_{PB} \cdot [1 + \alpha_{PB}(\theta_{PB} - 20)]}}$$

ω	314,159265359 rad/s	$\beta_1 = 39,5188211333$
ρ_{PB}	$21,4 \times 10^{-8} \Omega\text{m}$	
α_{PB}	$4,0 \times 10^{-3}$	
θ_{PB}	65,3099025381 °C	

$$g_s = 1 + \left(\frac{t_{PB}}{D_{PB}}\right)^{1,74} (\beta_1 D_{PB} 10^{-3} - 1,6)$$

D_{PB}	116,9 mm	$g_s = 1,0070784818$
t_{PB}	3,6 mm	
β_1	39,5188211333	

$$\lambda''_{1,m} = \frac{R_{s,PB}}{R} \left[g_s \lambda_{0,m} (1 + \Delta_{1,m} + \Delta_{2,m}) + \frac{(\beta_1 t_{PB})^4}{12 \times 10^{12}} \right]$$

$R_{s,PB}$	$1,9727383695 \times 10^{-4} \Omega/\text{m}$	$\lambda''_{1,m} = 2,3836589938 \times 10^{-4}$
R	$3,2222847660 \times 10^{-5} \Omega/\text{m}$	
g_s	1,0070784818	
$\lambda_{0,m}$	$4,7625021513 \times 10^{-6}$	
$\Delta_{1,m}$	$2,5305842543 \times 10^{-5}$	
$\Delta_{2,m}$	0	
β_1	39,5188211333	
t_{PB}	3,6 mm	

18. Loss factor $\lambda''_{1,1}$ for eddy current losses, lead alloy only, leading phase

$$m = \frac{\omega}{R_{s,PB}} 10^{-7}$$

ω	314,159265359	$m = 0,1593819425$
$R_{s,PB}$	$1,9711095276 \times 10^{-4} \Omega/\text{m}$	

$$\lambda_{0,1} = 1,5 \left(\frac{m^2}{1 + m^2} \right) \left(\frac{d_{PB_mean}}{2s} \right)^2$$

m	0,1593819425	$\lambda_{0,1} = 1,1925453482 \times 10^{-6}$
d_{PB_mean}	113,3 mm	
s	10000 mm	

$$\Delta_{1,1} = (4,7m^{0,7}) \left(\frac{d_{PB}}{2s} \right)^{(0,16m+2)}$$

m	0,1593819425	$\Delta_{1,1} = 3,6551833711 \times 10^{-5}$
d_{PB_mean}	113,3 mm	
s	10000 mm	

$$\Delta_{2,1} = (21m^{3,3}) \left(\frac{d_{PB}}{2s} \right)^{(1,47m+5,06)}$$

m	0,1593819425	$\Delta_{2,1} = 6,2381222232 \times 10^{-14}$
d_{PB_mean}	113,3 mm	
s	10000 mm	

$$g_s = 1 + \left(\frac{t_{PB}}{D_{PB}} \right)^{1,74} (\beta_1 D_{PB} 10^{-3} - 1,6)$$

D_{PB}	116,9 mm	$g_s = 1,007082955$
t_{PB}	3,6 mm	
β_1	39,5351461057	

$$\lambda''_{1,1} = \frac{R_{s_PB}}{R} \left[g_s \lambda_{0,1} (1 + \Delta_{1,1} + \Delta_{2,1}) + \frac{(\beta_1 t_{PB})^4}{12 \times 10^{12}} \right]$$

R_{s_PB}	$1,9711095276 \times 10^{-4} \Omega/m$	$\lambda''_{1,1} = 2,1665417336 \times 10^{-4}$
R	$3,220311500479 \times 10^{-5} \Omega/m$	
g_s	1,007082955	
$\lambda_{0,1}$	$1,1925453482 \times 10^{-6}$	
$\Delta_{1,1}$	$3,6551833711 \times 10^{-5}$	
$\Delta_{2,1}$	$6,2381222232 \times 10^{-9}$	
β_1	39,5351461057	
t_{PB}	3,6 mm	

Loss factor $\lambda''_{1,1}$ for eddy current losses, lead alloy only, lagging phase

$$m = \frac{\omega}{R_{s_PB}} 10^{-7}$$

ω	314,159265359	$m = 0,1593819425$
R_{s_PB}	$1,9711095275 \times 10^{-4} \Omega/m$	

$$\lambda_{0,2} = 1,5 \left(\frac{m^2}{1 + m^2} \right) \left(\frac{d_{PB_mean}}{2s} \right)^2$$

m	0,1593819425	$\lambda_{0,2} = 1,1925453484 \times 10^{-6}$
d_{PB_mean}	113,3 mm	
s	10000 mm	

$$\Delta_{1,2} = - \frac{0,74(m+2)m^{0,5}}{2 + (m-0,3)^2} \left(\frac{d_{PB_mean}}{2s} \right)^{(m+1)}$$

m	0,1593819425	$\Delta_{1,2} = -7,8447415210 \times 10^{-4}$
d_{PB_mean}	113,3 mm	
s	10000 mm	

$$\Delta_{2,2} = (0,92m^{3,7}) \left(\frac{d_{PB}}{2s} \right)^{(m+2)}$$

m	0,1593819425	$\Delta_{2,2} = 1,4491489325 \times 10^{-8}$
d_{PB_mean}	113,3 mm	
s	10000 mm	

$$g_s = 1 + \left(\frac{t_{PB}}{D_{PB}} \right)^{1,74} (\beta_1 D_{PB} 10^{-3} - 1,6)$$

D_{PB}	116,9 mm	$g_s = 1,007082955$
t_{PB}	3,6 mm	
β_1	39,5351461075	

$$\lambda''_{1,2} = \frac{R_{s_PB}}{R} \left[g_s \lambda_{0,2} (1 + \Delta_{1,2} + \Delta_{2,2}) + \frac{(\beta_1 t_{PB})^4}{12 \times 10^{12}} \right]$$

R_{s_PB}	$1,9711095275 \times 10^{-4} \Omega/m$	$\lambda''_{1,2} = 2,1664813805 \times 10^{-4}$
R	$3,220311500267 \times 10^{-5} \Omega/m$	
g_s	1,007082955	
$\lambda_{0,2}$	$1,1925453484 \times 10^{-6}$	
$\Delta_{1,2}$	$-7,8447415210 \times 10^{-4}$	
$\Delta_{2,2}$	$1,4491489325 \times 10^{-8}$	
β_1	39,5351461075	
t_{PB}	3,6 mm	

Note: Please refer to guidance point 6 and guidance point 31 for the calculation of eddy current losses.

19. Calculation of factor 'F' for lead alloy sheath, middle phase:

$$X_{PB} = 2 \cdot \omega \cdot 10^{-7} \cdot \ln \left(\frac{2s}{d_{PB_mean}} \right)$$

$$X_m = 2 \cdot \omega \cdot 10^{-7} \cdot \ln(2)$$

$$M = \frac{R_{s_PB}}{X_{PB} + X_m}$$

$$N = \frac{R_{s_PB}}{X_{PB} - \frac{X_m}{3}}$$

s	10000 mm	$X_{PB} = 3,2505734877 \times 10^{-4} \Omega/m$
d_{PB_mean}	113,3 mm	

R_{s_PB}	$1,9727383695 \times 10^{-4} \Omega/m$	$M = 0,535184434$ $N = 0,6352604117$
X_{PB}	$3,2505734877 \times 10^{-4} \Omega/m$	
X_m	$4,3551721806 \times 10^{-5} \Omega/m$	

$$F_m = \frac{4M^2N^2 + (M + N)^2}{4(M^2 + 1)(N^2 + 1)}$$

M	0,535184434	$F_m = 0,253700391$
N	0,6352604117	

20. Calculation of factor 'F' for lead alloy sheath, leading phase:

$$X_{PB} = 2 \cdot \omega \cdot 10^{-7} \cdot \ln \left(\frac{2s}{d_{PB_mean}} \right)$$

$$X_m = 2 \cdot \omega \cdot 10^{-7} \cdot \ln(2)$$

$$M = \frac{R_{s_PB}}{X_{PB} + X_m}$$

$$N = \frac{R_{s_PB}}{X_{PB} - \frac{X_m}{3}}$$

s	10000 mm	$X_{PB} = 3,2505734877 \times 10^{-4} \Omega/m$
d_{PB_mean}	113,3 mm	

R_{s_PB}	$1,9711095276 \times 10^{-4} \Omega/m$	$M = 0,5347425457$ $N = 0,6347358927$
X_{PB}	$3,2505734877 \times 10^{-4} \Omega/m$	
X_m	$4,3551721806 \times 10^{-5} \Omega/m$	

$$F_n = \frac{4M^2N^2 + (M + N)^2}{4(M^2 + 1)(N^2 + 1)}$$

M	0,5347425457	$F_1 = 0,2533894613$
N	0,6347358927	

21. Calculation of factor 'F' for lead alloy sheath, lagging phase:

$$X_{PB} = 2 \cdot \omega \cdot 10^{-7} \cdot \ln \left(\frac{2s}{d_{PB_mean}} \right)$$

$$X_m = 2 \cdot \omega \cdot 10^{-7} \cdot \ln(2)$$

$$M = \frac{R_{s_PB}}{X_{PB} + X_m}$$

$$N = \frac{R_{s_PB}}{X_{PB} - \frac{X_m}{3}}$$

s	10000 mm	$X_{PB} = 3,2505734877 \times 10^{-4} \Omega/m$
d_{PB_mean}	113,3 mm	

R_{s_PB}	$1,9711095275 \times 10^{-4} \Omega/m$	$M = 0,5347425456$ $N = 0,6347358926$
X_{PB}	$3,2505734877 \times 10^{-4} \Omega/m$	
X_m	$4,3551721806 \times 10^{-5} \Omega/m$	

$$F_n = \frac{4M^2N^2 + (M + N)^2}{4(M^2 + 1)(N^2 + 1)}$$

M	0,5347425456	$F_2 = 0,2533894613$
N	0,6347358926	

22. Loss factor in sheath and armour may be assumed to be approximately equal, λ_1, λ_2 :

$$\lambda_{1,n} = \lambda'_{1,n} + F_n \cdot \lambda''_{1,n}$$

$\lambda'_{1,m}$	1,0799516782	$\lambda_{1,m} = 1,0800121517$
$\lambda''_{1,m}$	$2,3836589938 \times 10^{-4}$	
F_m	0,253700391	

$\lambda'_{1,1}$	1,0797525303	$\lambda_{1,1} = 1,0798074281$
$\lambda''_{1,1}$	$2,1665417336 \times 10^{-4}$	
F_1	0,2533894613	

$\lambda'_{1,2}$	1,0797525302	$\lambda_{1,2} = 1,07980742659$
$\lambda''_{1,2}$	$2,1664813805 \times 10^{-4}$	
F_2	0,2533894613	

$$\lambda_{2,n} = \lambda'_{2,n} + \lambda''_{2,n}$$

Eddy current losses can be neglected:

$\lambda'_{2,m}$	1,0799516782	$\lambda_{2,m} = 1,0799516782$
$\lambda''_{2,m}$	0	
$\lambda'_{2,1}$	1,0797525303	$\lambda_{2,1} = 1,0797525303$
$\lambda''_{2,1}$	0	
$\lambda'_{2,2}$	1,0797525302	$\lambda_{2,2} = 1,0797525302$
$\lambda''_{2,2}$	0	

10.5 Thermal resistance T_1 between conductor and screen

Table 10-8: Parameters needed to calculate the thermal resistance between conductor and screen

Parameter	Symbol	Value	Unit	Reference
Diameter of conductor	d_c	34,7	mm	Cable datasheet
External diameter of sc- water blocking tape(s) over the conductor	d_{sct}	35,8	mm	Cable datasheet
External diameter of sc conductor screen	d_{sc}	39,8	mm	Cable datasheet
External diameter of insulation	d_i	100,6	mm	Cable data sheet
External diameter of sc insulation screen	d_{sci}	103,7	mm	Cable data sheet
External diameter of sc- water blocking tape(s) under the copper wires screen	d_{uwb}	104,7	mm	Cable data sheet
Thermal resistivity of sc screen	ρ_{sc}	2,5	K.m/W	Note 1
Thermal resistivity of insulation	ρ_i	3,5	K.m/W	[4], table 1
Thermal resistivity of water blocking tape over the conductor	ρ_{wbc}	6	K.m/W	Note 2
Thermal resistivity of water blocking tape	ρ_{wb}	12	K.m/W	Note 3
Note 1: In this case: 2,5 K.m/W				
Note 2: In this case: 6 K.m/W				
Note 3: In this case: 12 K.m/W				

Table 10-9: Sequential steps to calculate the thermal resistance between conductor and screen T1

Step	Parameter	Symbol	Unit	Reference
1	Thermal resistance of sc water blocking tape over the conductor	$T_{1_{sct}}$	$K.m/W$	[4], section 4.1.2.1
2	Thermal resistance of sc conductor screen	$T_{1_{csc}}$	$K.m/W$	[4], section 4.1.2.1
3	Thermal resistance of insulation	T_{1_i}	$K.m/W$	[4], section 4.1.2.1
4	Thermal resistance of sc insulation screen	$T_{1_{isc}}$	$K.m/W$	[4], section 4.1.2.1
5	Thermal resistance of sc water blocking tape under the copper wires screen	$T_{1_{uwb}}$	$K.m/W$	[4], section 4.1.2.1
7	Thermal resistance between conductor and copper wire screen	T_1	$K.m/W$	-

1. Thermal resistance of water blocking tape over the conductor:

$$T_{1_{sct}} = \frac{\rho_{wbc}}{2\pi} \ln\left(\frac{d_{sct}}{d_c}\right) \quad (K.m/W)$$

ρ_{wbc}	6 K.m/W	$T_{1_{cst}} = 0,0298016419 \text{ K.m/W}$
d_{sct}	$35,8 \times 10^{-3} \text{ m}$	
d_c	$34,7 \times 10^{-3} \text{ m}$	

2. Thermal resistance of sc conductor screen:

$$T_{1_{csc}} = \frac{\rho_{sc}}{2\pi} \ln\left(\frac{d_{sc}}{d_{sct}}\right) \quad \left(K.\frac{m}{W}\right)$$

ρ_{sc}	2,5 K.m/W	$T_{1_{csc}} = 0,0421438385 \text{ K.m/W}$
d_{sc}	$39,8 \times 10^{-3} \text{ m}$	
d_{sct}	$35,8 \times 10^{-3} \text{ m}$	

3. Thermal resistance of insulation:

$$T_{1_i} = \frac{\rho_i}{2\pi} \ln\left(\frac{d_i}{d_{sc}}\right) \quad (K.m/W)$$

ρ_i	3,5 K.m/W	$T_{1_i} = 0,5165371623 \text{ K.m/W}$
d_i	$100,6 \times 10^{-3} \text{ m}$	
d_{sc}	$39,8 \times 10^{-3} \text{ m}$	

4. Thermal resistance of sc insulation screen:

$$T_{1_{isc}} = \frac{\rho_{sc}}{2\pi} \ln\left(\frac{d_{sci}}{d_i}\right) \quad (K.m/W)$$

ρ_{wb}	2,5 K.m/W	$T_{1_{isc}} = 0,0120758246 \text{ K.m/W}$
d_{sci}	$103,7 \times 10^{-3} \text{ m}$	
d_i	$100,6 \times 10^{-3} \text{ m}$	

5. Thermal resistance of water blocking tape under the copper wire screen:

$$T_{1_{uwb}} = \frac{\rho_{wb}}{2\pi} \ln\left(\frac{d_{uwb}}{d_{sci}}\right) \quad \left(K.\frac{m}{W}\right)$$

ρ_{wb}	12 K.m/W	$T_{1_uwb} = 0,0183289249 \text{ K.m/W}$
d_{uwb}	$104,7 \times 10^{-3} \text{ m}$	
d_{sci}	$103,7 \times 10^{-3} \text{ m}$	

6. Thermal resistance between conductor and under copper wire screen:

$$T_{1_} = T_{1_sct} + T_{1_csc} + T_{1_i} + T_{1_isc} + T_{1_uwb} \quad (K.m/W)$$

T_{1_sct}	0,0298016419 K.m/W	$T_{1_} = 0,6188873923 \text{ K.m/W}$
T_{1_csc}	0,0421438385 K.m/W	
T_{1_i}	0,5165371623 K.m/W	
T_{1_isc}	0,0120758246 K.m/W	
T_{1_uwb}	0,0183289249 K.m/W	

10.6 Thermal resistance T_2 between lead alloy sheath and armour

Table 10-10: Parameters needed to calculate the thermal resistance between lead alloy sheath and armour

Parameter	Symbol	Value	Unit	Reference
External diameter of copper wires screen	d_{sc}	107,9	mm	Cable data sheet
External diameter of sc- water blocking tape(s) over the copper wires screen	d_{wbt}	109,7	mm	Cable data sheet
External diameter over metallic sheath	d_{PB}	116,9	mm	Cable data sheet
External diameter over sc PE sheath	d_{scPE}	122,9	mm	Cable data sheet
Thermal resistivity of sc PE	ρ_{sc}	2,5	K.m/W	Note 1
Note 1: In this case: 2,5 K.m/W				

Table 10-11: Sequential steps to calculate the thermal resistance between lead alloy sheath and armour

Step	Parameter	Symbol	Unit	Reference
1	Thermal resistance of sc water blocking tape over the copper wires screen	T_{2_owbt}	K.m/W	[4], section 4.1.3.1
2	Thermal resistance of sc PE over the lead alloy sheath:	T_{2_PE}	K.m/W	[4], section 4.1.3.1
3.	Thermal resistance of sc PE sheath over the lead alloy sheath	T_2	K.m/W	[4], section 4.1.3.1

1. Thermal resistance of sc water blocking tape over the copper wires screen

$$T_{2_wb} = \frac{\rho_{wb}}{2\pi} \ln\left(\frac{d_{wbt}}{d_{sc}}\right) \quad (K.m/W)$$

ρ_{wb}	12 K.m/W	$T_{2_wb} = 0,0315976580 \text{ K.m/W}$
d_{wbt}	$109,7 \times 10^{-3} \text{ m}$	
d_{sc}	$107,9 \times 10^{-3} \text{ m}$	

2. Thermal resistance of sc PE over the lead alloy sheath:

$$T_{2_PE} = \frac{\rho_{sc}}{2\pi} \ln\left(\frac{d_{scPE}}{d_{PB}}\right) \quad (K.m/W)$$

ρ_{sc}	2,5 K.m/W	$T_{2_PE} = 0,0199151170 \text{ K.m/W}$
d_{scPE}	$122,9 \times 10^{-3} \text{ m}$	
d_{PB}	$116,9 \times 10^{-3} \text{ m}$	

3. Thermal resistance between copper wire screen and armour

$$T_2 = T_{2_wb} + T_{2_PE}$$

T_{2_wb}	0,0315976580 K.m/W	$T_2 = 0,0515127749 \text{ K.m/W}$
T_{2_PE}	0,0199151170 K.m/W	

10.7 Thermal resistance T_3 of outer covering

Table 10-12: Parameters needed to calculate the thermal resistance T_3 of the outer covering

Parameter	Symbol	Value	Unit	Reference
External diameter of the armour	d_A	128,3	mm	Cable data sheet
External diameter of outer covering	d_{oc}	134,3	mm	Cable data sheet
Thermal resistivity of outer covering	ρ_{oc}	6	K.m/W	[4], table 1

Table 10-13: Sequential steps to calculate the thermal resistance T_3 of the outer covering

Step	Parameter	Symbol	Unit	Reference
1	Thermal resistance of outer covering	T_3	K.m/W	[4], section 4.1.4.1

1. Thermal resistance of outer covering:

$$T_3 = \frac{\rho_{oc}}{2\pi} \ln \left(\frac{d_{oc}}{d_A} \right) \quad (\text{K.m/W})$$

ρ_{oc}	6 K.m/W	$T_3 = 0,0436448995 \text{ K.m/W}$
d_{oc}	$134,3 \times 10^{-3} \text{ m}$	
d_A	$128,3 \times 10^{-3} \text{ m}$	

10.8 External thermal resistance T_4

Table 10-14: Parameters needed to calculate the external thermal resistance T_4

Parameter	Symbol	Value	Unit	Reference
External diameter of one cable	D_e	134,3	mm	Cable data sheet
Distance from the surface of the ground to the cable axis	L	1,06715	m	System design, note 1
Thermal resistivity of the soil	ρ_T	0.7	K.m/W	System design

Table 10-15: Sequential steps to calculate the external thermal resistance T_4

Step	Parameter	Symbol	Unit	Reference
1	External thermal resistance	T_4	K.m/W	[4]

1. External thermal resistance:

$$u = \frac{2L}{D_e}$$

L	1,06715 m	$u = 15,8920327625$
D_e	$134,3 \times 10^{-3}$ m	

$$T_{4,m} = \frac{1}{2\pi} \rho_T \left[\ln(u + \sqrt{u^2 - 1}) + \frac{W_1}{W_m} \cdot \ln \left[\frac{\sqrt{s^2 + (2L)^2}}{s} \right] + \frac{W_2}{W_m} \cdot \ln \left[\frac{\sqrt{s^2 + (2L)^2}}{s} \right] \right] \quad (K.m/W)$$

$$T_{4,1} = \frac{1}{2\pi} \rho_T \left[\ln(u + \sqrt{u^2 - 1}) + \frac{W_m}{W_1} \cdot \ln \left[\frac{\sqrt{s^2 + (2L)^2}}{s} \right] + \frac{W_2}{W_1} \cdot \ln \left[\frac{\sqrt{(2s)^2 + (2L)^2}}{2s} \right] \right] \quad (K.m/W)$$

$$T_{4,2} = \frac{1}{2\pi} \rho_T \left[\ln(u + \sqrt{u^2 - 1}) + \frac{W_m}{W_2} \cdot \ln \left[\frac{\sqrt{s^2 + (2L)^2}}{s} \right] + \frac{W_1}{W_2} \cdot \ln \left[\frac{\sqrt{(2s)^2 + (2L)^2}}{2s} \right] \right] \quad (K.m/W)$$

ρ_T	0,7 K.m/W	$T_{4,m} = 0,3902066636$ K.m/W $T_{4,1} = 0,3883614684$ K.m/W $T_{4,2} = 0,3883614684$ K.m/W
u	15,8920327625	
L	1067,15 mm	
s	10000 mm	
W_m	112,52170633313 W/m	
W_1	112,4402851007 W/m	
W_2	112,44028503907 W/m	

10.9 Permissible current rating

Table 10-16: Input parameters needed to calculate the continuous current carrying capacity

Parameter	Symbol	Value	Unit	Reference
Conductor operating temperature	θ_m	90	$^{\circ}\text{C}$	System design
Ambient temperature	θ_{amb}	15	$^{\circ}\text{C}$	System design
Number of conductors in cable	n	1	-	Cable data sheet
AC resistance of conductor at operating temperature	R_m	$3,2222847662 \times 10^{-5}$	Ω/m	To be calculated
Ratio of the total losses in metallic screen	$\lambda_{1,m}$	1,0820606109	-	To be calculated
Ratio of the total losses in metallic foil	$\lambda_{2,m}$	1,0818229333	-	To be calculated
Dielectric losses	W_d	2,5095896112	W/m	To be calculated
Thermal resistance between conductor and metallic screen	T_1	0,6504850503	$K.m/W$	To be calculated
Thermal resistance between copper wire screen and armour	T_2	0,0199151170	$K.m/W$	To be calculated
Thermal resistance of external serving	T_3	0,0436448995	$K.m/W$	To be calculated
Thermal resistance of surrounding medium	$T_{4,m}$	0,3902065743	$K.m/W$	To be calculated

Table 10-17: Sequential steps to calculate the continuous current rating capacity

Step	Parameter	Symbol	Unit	Reference
1	Conductor temperature rise above ambient	$\Delta\theta$	K	[3], section 1.4.1.1
2	Permissible current rating	I	A	[3], section 1.4.1.1

1. Conductor temperature rise above ambient:

$$\Delta\theta = \theta - \theta_{amb}$$

θ	90 °C	$\Delta\theta = 75$ °C
θ_{amb_e}	15 °C	

2. Permissible current rating:

$$I = \left[\frac{\Delta\theta_m - W_d [0,5 \cdot T_1 + n(T_2 + T_3 + T_{4,m})]}{R_m \cdot T_1 + nRR_m(1 + \lambda_{1,m})T_2 + nR_m(1 + \lambda_{1,m} + \lambda_{2,m})(T_3 + T_{4,m})} \right]^{0,5}$$

$\Delta\theta_m$	75 °C	$I = 1039,4347828913$ A
W_d	2,50958961115 W/m	
T_1	0,6188873923 K.m/W	
T_2	0,0515127749 K.m/W	
T_3	0,0436448995 K.m/W	
$T_{4,m}$	0,3902066636 K.m/W	
R_m	$3,2222847660 \times 10^{-5}$ Ω/m	
n	1	
$\lambda_{1,m}$	1,0800121517	
$\lambda_{2,m}$	1,0799516782	

10.10 Calculation of losses

Conductor losses at middle phase:

$$P_c = R_m I^2$$

R_m	$3,2222847660 \times 10^{-5}$ Ω/m	$P_c = 34,8143594816$ W/m
I	1039,4347828913 A	

Copper wires screen and lead alloy sheath losses at middle phase:

$$P_s = \lambda_{1,m} R_m I^2$$

R_m	$3,2222847660 \times 10^{-5}$ Ω/m	$P_s = 37,5999312937$ W/m
I	1039,4347828913 A	
$\lambda_{1,m}$	1,0800121517	

Armour losses at middle phase:

$$P_{arm} = \lambda_{2,m} R_m I^2$$

R_m	$3,2222847660 \times 10^{-5}$ Ω/m	$P_{arm} = 37,5978259467$ W/m
I	1039,4347828913 A	
$\lambda_{2,m}$	1,0799516782	

Conductor losses at leading phase:

$$P_c = R I^2$$

R_1	$3,2203115005 \times 10^{-5}$ Ω/m	$P_c = 34,7930398339$ W/m
I	1039,4347828913 A	

Copper wires screen and lead alloy sheath losses at leading phase:

$$P_s = \lambda_{1,1} R I^2$$

R_1	$3,2203115005 \times 10^{-5} \text{ } \Omega/\text{m}$	$P_s = 37,5697828600 \text{ W/m}$
I	1039,4347828913 A	
$\lambda_{1,1}$	1,0798074281	

Armour losses at leading phase:

$$P_{arm} = \lambda_{2,1} R I^2$$

R_1	$3,2203115005 \times 10^{-5} \text{ } \Omega/\text{m}$	$P_{arm} = 37,5678727957 \text{ W/m}$
I	1039,4347828913 A	
$\lambda_{2,1}$	1,0797525302	

Conductor losses at lagging phase:

$$P_c = R I^2$$

R_2	$3,2203115003 \times 10^{-5} \text{ } \Omega/\text{m}$	$P_c = 34,7930398316 \text{ W/m}$
I	1039,4347828913 A	

Copper wires screen and lead alloy sheath losses at lagging phase:

$$P_s = \lambda_{1,2} R I^2$$

R_2	$3,2203115003 \times 10^{-5} \text{ } \Omega/\text{m}$	$P_s = 37,56978280369 \text{ W/m}$
I	1039,4347828913 A	
$\lambda_{1,2}$	1,07980742659	

Armour losses at lagging phase:

$$P_{arm} = \lambda_{2,2} R I^2$$

R_2	$3,2203115003 \times 10^{-5} \text{ } \Omega/\text{m}$	$P_{arm} = 37,5678727926 \text{ W/m}$
I	1039,4347828913 A	
$\lambda_{2,2}$	1,0797525302	

11. Case study 7: A 320kV HVDC submarine bipole

11.1 Introduction

The selected cable type is a single core, XLPE submarine 320 kV HVDC cable with a 2000 mm² copper conductor with lead sheath and armouring. An indicative cable drawing is depicted in Figure 37, the datasheet is shown in Table 11-1. The maximum permissible temperature drop across the insulation is 10.7 K and is only valid for this cable at the voltage level given. It must be noted that the maximum permissible temperature drop across the insulation and in the conductor, are determined during cable type testing. The maximum conductor temperature is 70 °C.

System related parameters - general:

▪ System voltage	:	320kV
▪ System operation	:	HVDC
▪ Number of poles	:	Bipolar system
▪ Soil thermal resistivity	:	0.8 K.m/W
▪ Soil temperature	:	15 °C

The current rating will be determined for the following laying configurations (see Figure 38):

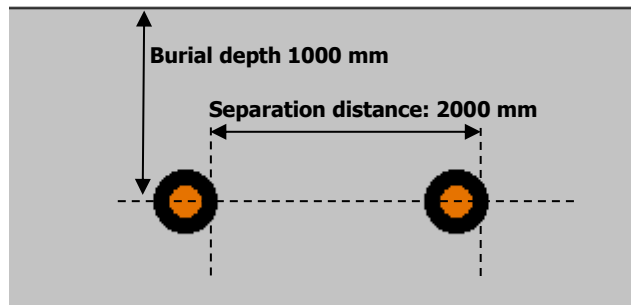
- Direct buried at seafloor
- Laying depth– centre circuit is 1000 mm
- Axial separation distance: 2000 mm



Figure 37: 320kV HVDC indicative cable drawing

Table 11-1: Cable data sheet 320kV HVDC 2000 mm² CU/XLPE/PB/SC PE/SWA/PP, Outer diameter is 123,4 mm

Element	Construction	Thickness (mm)	Outer diameter (mm)
Conductor	profile/copper, longitudinal water sealed	-	51,6
Conductor tape	Semi-conductive		52,9
Conductor screen	Semi-conductive PE	2,0	
Insulation	XLPE	16,0	
Insulation screen	Semi-conductive PE	1,5	
Semi-conductive tape	Semi-conductive		93,5
Lead sheath	Lead alloy	2,3	
Protection sheath	Semi-conductive PE	3,1	
Armour bedding	Semi-conductive bedding		104,6
Steel wires	Galvanized steel + bitumen	2,5	
Plastic nylon tape			110,4
Steel wires	Galvanized steel + bitumen	2,5	
Outer serving	PP yarn + bitumen	4,0	

**Figure 38: Installation of two poles under seafloor**

11.2 Introduction to rating calculation of HVDC cables

For the rating calculation of HVDC cables two limiting factors are considered. The first is the maximum permissible temperature in the conductor related to the maximum operating temperature of the insulation. The second factor is the maximum permissible electric field at the insulation screen when the cable carries the load current. Figure 39 shows a typical field distribution at different loading conditions. As can be seen, the electric field at the insulation screen (outer radius of the insulation) is increasing as the load increases. This can be explained by the fact that the field distribution is mainly determined by the conductivity of the insulation for a given voltage. Since the conductivity is temperature dependent, the electric field distribution also becomes temperature dependent. It can be shown that the electric field at the insulation screen can be taken into account as a function of temperature drop across the insulation ($\Delta\theta_{ins}$). It should be noted that the temperature drop limit might be reached in a specific condition where the other limit of the conductor temperature is not yet reached. It should be noted that the given temperature drop is verified during type testing and is only valid for the given voltage.

Since the voltage and current are not varying with time, there are no dielectric losses ($W_d = 0$) as well as no induced losses in the metallic sheath and armour wires ($\lambda_1 = \lambda_2 = 0$). However, there are always some high frequency harmonics due to the switching operation of the power electronic converters. The impact of these harmonics is assumed negligible and is not considered in this document, see also [Guidance Point 10](#).

Generally, the rated current of HVDC cables is calculated providing the constraints below:

$$\Delta\theta_{ins} = R' I^2 \cdot T_{1,i} \leq \Delta\theta_{ins,max}$$

$$\theta = R' I^2 [T_1 + nT_2 + n(T_3 + T_4)] + \theta_a \leq \theta_{c,max}$$

Note that $T_{1,i}$ is the thermal resistance of the insulation material only (excluding semiconducting screens, refer to Table 11-5).

This is normally calculated with a simple approach as explained below:

1. Assume that the conductor temperature θ is reached to its maximum permissible value ($\theta_{c,max}$)
2. Calculate the current with the equation below as given in IEC 60287-1-1 [3]:

$$I = \left[\frac{\theta_{c,max} - \theta_a}{R' \cdot T_1 + nR'T_2 + nR'(T_3 + T_4)} \right]^{0,5}$$

3. Calculate the temperature drop across the insulation ($\Delta\theta_{ins}$) through equation below:

$$\Delta\theta_{ins} = R'I^2T_{1,i}$$

4. If $\Delta\theta_{ins} < \Delta\theta_{ins,max}$, the current obtained is thermally limited by conductor temperature and go to step 6.

If $\Delta\theta_{ins} > \Delta\theta_{ins,max}$, then the rated current is limited by the $\Delta\theta_{ins}$. This means that at the rated current the temperature drop must be limited to $\Delta\theta_{ins,max}$ ($\Delta\theta_{ins} = \Delta\theta_{ins,max}$). Follow step 5.

5. Following the equation given in step 3, the conductor losses can be stated with:

$$R'I^2 = \frac{\Delta\theta_{ins,max}}{T_{1,i}}$$

then the conductor temperature can be calculated using equation in step 1:

$$\theta - \theta_a = R'I^2[T_1 + nT_2 + n(T_3 + T_4)] \rightarrow \theta = \frac{\Delta\theta_{ins,max}}{T_{1,i}}[T_1 + nT_2 + n(T_3 + T_4)] + \theta_a$$

6. Finish.

Note: In case T_4 is temperature dependent such as installation in free air, etc., the equations become nonlinear and iterative methods should be used.

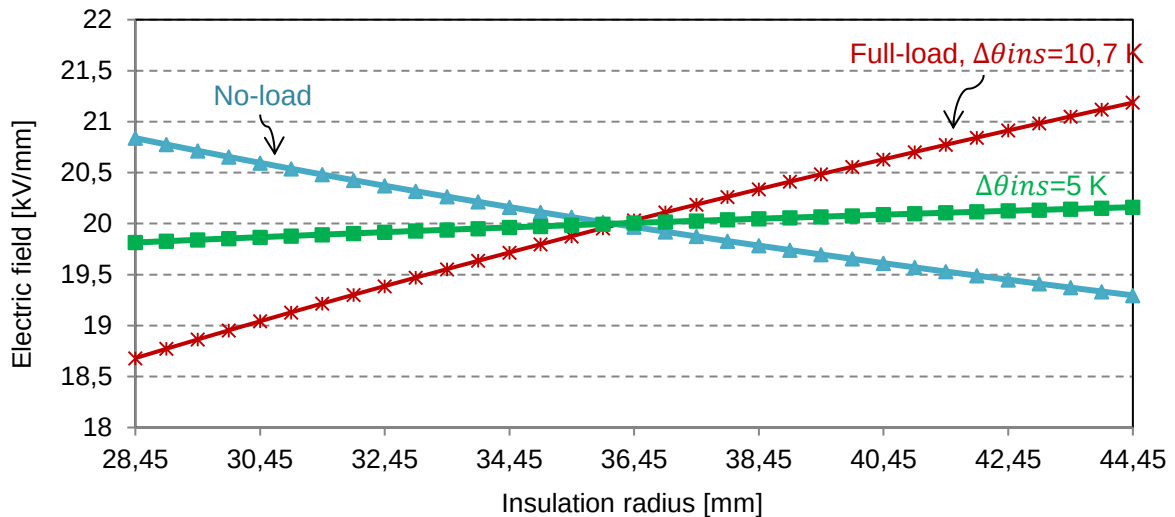


Figure 39: Distribution of electric field across the insulation at no-load and full-load conditions, specific for this cable

11.3 Calculation of the conductor DC resistance at operation temperature

Table 11-2: Parameters for calculating the conductor resistance at operation temperature

Parameter	Symbol	Value	Unit	Reference
Conductor DC resistance @ 20°C	R_o	$9,0 \times 10^{-6}$	Ω/m	[8]
Conductor diameter	D_c	51,6	mm	Cable datasheet
Conductor material		Copper		Cable datasheet
Type of conductor		Shaped stranded compacted per [8]		Cable datasheet
Conductor temperature coefficient at 20°C per Kelvin	α_{20}	$3,93 \times 10^{-3}$	1/K	[3], table 1
Conductor operating temperature	θ	70	°C	Cable datasheet

Table 11-3: Sequential steps to calculate DC resistance at operating temperature

Step	Parameter	Symbol	Unit	Reference
1	Conductor DC resistance at maximum operating temperature	R'	Ω/m	[3], section 2.1.1

1. Conductor DC resistance at maximum operating temperature R' :

$$R' = R_o [1 + \alpha_{20}(\theta - 20)] \text{ } (\Omega/m)$$

R_o	$9,0 \times 10^{-6} \Omega/m$	$R' = 1,0768500000 \times 10^{-5} \Omega/m$
α_{20}	$3,93 \times 10^{-3} 1/K$	
θ	70 °C	

11.4 Thermal resistance T_1 between conductor and the screen

Table 11-4: Parameters needed to calculate the T_1

Parameter	Symbol	Value	Unit	Reference
Diameter of conductor	D_c	51,6	mm	Cable datasheet
External diameter over sc water blocking tape of the conductor	D_{wb_con}	52,9	mm	Cable datasheet
External diameter of sc conductor screen	D_{sc}	56,9	mm	Cable datasheet
External diameter of insulation	D_i	88,9	mm	Cable datasheet
External diameter of sc insulation screen	D_{si}	91,9	mm	Cable datasheet
External diameter of sc water blocking tape of the insulation	D_{wb_ins}	93,5	mm	Cable datasheet
Thermal resistivity of sc screen	ρ_{sc}	2,5	K.m/W	Table 2-3
Thermal resistivity of insulation	ρ_i	3,5	K.m/W	[4], table 1
Thermal resistivity of water blocking tapes	ρ_{wb}	12	K.m/W	Table 2-3

Table 11-5: Sequential steps to calculate the T_1

Step	Parameter	Symbol	Unit	Reference
1	Thermal resistance of sc water blocking tape over the conductor	$T_{1_wb_con}$	$K.m/W$	[4], section 4.1.2.1
2	Thermal resistance of sc conductor screen	T_{1_csc}	$K.m/W$	[4], section 4.1.2.1
3	Thermal resistance of insulation	T_{1_i}	$K.m/W$	[4], section 4.1.2.1
4	Thermal resistance of sc insulation screen	T_{1_isc}	$K.m/W$	[4], section 4.1.2.1
5	Thermal resistance of sc water blocking tape under the copper wires screen	$T_{1_wb_ins}$	$K.m/W$	[4], section 4.1.2.1
6	Thermal resistance between conductor and screen	T_1	$K.m/W$	-

1. Thermal resistance of sc water blocking tape over conductor:

$$T_{1_wb_con} = \frac{\rho_{wb}}{2\pi} \ln\left(\frac{D_{wb_con}}{D_c}\right) \quad (K.m/W)$$

ρ_{wb}	12 K.m/W	$T_{1_wb_con} = 0,0475204824 \text{ K.m/W}$
D_{wb_con}	52,9 mm	
D_c	51,6 mm	

2. Thermal resistance of sc conductor screen:

$$T_{1_csc} = \frac{\rho_{sc}}{2\pi} \ln\left(\frac{D_{sc}}{D_{wb_con}}\right) \quad (K.m/W)$$

ρ_{sc}	2,5 K.m/W	$T_{1_csc} = 0,0290028062 \text{ K.m/W}$
D_{sc}	56,9 mm	
D_{wb_con}	52,9 mm	

3. Thermal resistance of insulation:

$$T_{1_i} = \frac{\rho_i}{2\pi} \ln\left(\frac{D_i}{D_{sc}}\right) \quad (K.m/W)$$

ρ_i	3,5 K.m/W	$T_{1_i} = 0,2485616337 \text{ K.m/W}$
D_i	88,9 mm	
D_{sc}	56,9 mm	

4. Thermal resistance of sc insulation screen:

$$T_{1_isc} = \frac{\rho_{sc}}{2\pi} \ln\left(\frac{D_{si}}{D_i}\right) \quad \left(K.\frac{m}{W}\right)$$

ρ_{sc}	2,5 K.m/W	$T_{1_isc} = 0,0132054385 \text{ K.m/W}$
D_{si}	91,9 mm	
D_i	88,9 mm	

5. Thermal resistance of sc water blocking tape under the copper wires screen:

$$T_{1_wb_ins} = \frac{\rho_{wb}}{2\pi} \ln\left(\frac{D_{wb_ins}}{D_{si}}\right) \quad \left(K.\frac{m}{W}\right)$$

ρ_{wb}	12 K.m/W	$T_{1_wb_ins} = 0,0329649490 \text{ K.m/W}$
D_{wb_ins}	93,5 mm	
D_{si}	91,9 mm	

6. Thermal resistance between conductor and screen:

$$T_1 = T_{1_wb_con} + T_{1_csc} + T_{1_i} + T_{1_isc} + T_{1_wb_ins} \quad (K.m/W)$$

$T_{1_wb_con}$	0,0475204824 K.m/W	$T_1 = 0,3712553097 \text{ K.m/W}$
T_{1_csc}	0,0290028062 K.m/W	
T_{1_i}	0,2485616337 K.m/W	
T_{1_isc}	0,0132054385 K.m/W	
$T_{1_wb_ins}$	0,0329649490 K.m/W	

11.5 Thermal resistance T_2 between lead sheath and the armour

It should be noted that the armouring layers are considered as isothermal layers with no thermal resistances.

Table 11-6: Parameters needed to calculate the T_2

Parameter	Symbol	Value	Unit	Reference
External diameter of lead sheath	D_{Lead}	98,1	mm	Cable datasheet
External diameter of sc PE sheath	$D_{sc\ PE}$	104,3	mm	Cable datasheet
External diameter of sc bedding	$D_{sc\ BED}$	104,6	mm	Cable datasheet
External diameter of first armor layer	$D_{Arm.1}$	109,6	mm	Cable datasheet
External diameter of plastic nylon tape	D_{n_tape}	110,4	mm	Cable datasheet
Thermal resistivity of sc PE sheath	$\rho_{sc\ PE}$	2,5	K.m/W	Table 2.3
Thermal resistivity of sc bedding	$\rho_{sc\ Bed}$	6,0	K.m/W	Note 1
Thermal resistivity of plastic nylon tape	$\rho_{sc\ n_tape}$	6,0	K.m/W	Note 1

Note 1: The armour wires are well soaked in bitumen which penetrates in the tape. Combination of nylon tape and bitumen results in high thermal resistivity that is assumed to be 6,0 K.m/W. See Section 2.10 and Guidance Point 17 for further discussion

Table 11-7: Sequential steps to calculate the T_2

Step	Parameter	Symbol	Unit	Reference
1	Thermal resistance of sc PE sheath	$T_{2_sc\ PE}$	K.m/W	[4], section 4.1.3.1
2	Thermal resistance of sc bedding	$T_{2_sc\ BED}$	K.m/W	[4], section 4.1.3.1
3	Thermal resistance of plastic nylon tape	$T_{2_n\ tape}$	K.m/W	[4], section 4.1.3.1
4	Thermal resistance between lead and armor	T_2	K.m/W	-

1. Thermal resistance of sc PE sheath:

$$T_{2_sc\ PE} = \frac{\rho_{sc\ PE}}{2\pi} \ln \left(\frac{D_{sc\ PE}}{D_{Lead}} \right) \quad (K.m/W)$$

$\rho_{sc\ PE}$	2,5 K.m/W	$T_{2_sc\ PE} = 0,0243841270 \text{ K.m/W}$
$D_{sc\ PE}$	104,3 mm	
D_{Lead}	98,1 mm	

2. Thermal resistance of sc bedding:

$$T_{2_sc\ BED} = \frac{\rho_{sc\ BED}}{2\pi} \ln \left(\frac{D_{sc\ BED}}{D_{sc\ PE}} \right) \quad (K.m/W)$$

$\rho_{sc\ BED}$	6,0 K.m/W	$T_{2_sc\ BED} = 0,0027427391\text{ K.m/W}$
$D_{sc\ BED}$	104,6 mm	
$D_{sc\ PE}$	104,3 mm	

3. Thermal resistance of nylon tape:

$$T_{2_n\ tape} = \frac{\rho_{sc\ n_tape}}{2\pi} \ln\left(\frac{D_{n_tape}}{D_{Arm_1}}\right) \quad (K.m/W)$$

$\rho_{sc\ n_tape}$	6,0 K.m/W	$T_{2_n\ tape} = 0,0069449736\text{ K.m/W}$
D_{Arm_1}	109,6 mm	
D_{n_tape}	110,4 mm	

4. Thermal resistance between lead sheath and the armour:

$$T_2 = T_{2_sc\ PE} + T_{2_sc\ BED} + T_{2_n\ tape} \quad (K.m/W)$$

$T_{2_sc\ PE}$	0,0243841270 K.m/W	$T_2 = 0,0340718397\text{ K.m/W}$
$T_{2_sc\ BED}$	0,0027427391 K.m/W	
$T_{2_n\ tape}$	0,0069449736 K.m/W	

11.6 Thermal resistance T_3 outer serving

Table 11-8: Parameters needed to calculate the T_3

Parameter	Symbol	Value	Unit	Reference
External diameter of second armor layer	D_{Arm_2}	115,4	mm	Cable datasheet
Outer diameter of the cable	D_e	123,4	mm	Cable datasheet
Thermal resistivity of PP-yarn	ρ_{pp}	6,0	mm	[4], table 1 (compounded jute and fibrous material)

Table 11-9: Sequential steps to calculate the T_3

Step	Parameter	Symbol	Unit	Reference
1	Thermal resistance of PP-yarn	T_3	K.m/W	[4], section 4.1.4.1

1. Thermal resistance of PP-yarn:

$$T_3 = \frac{\rho_{PP}}{2\pi} \ln\left(\frac{D_e}{D_{Arm_2}}\right) \quad (K.m/W)$$

ρ_{PP}	6,0 K.m/W	$T_3 = 0,0640058386\text{ K.m/W}$
D_{Arm_2}	115,4 mm	
D_e	123,4 mm	

11.7 External thermal resistance T_4

Table 11-10: Parameters needed to calculate the external thermal resistance T_4

Parameter	Symbol	Value	Unit	Reference
External diameter of one cable	D_e	123,4	mm	Cable datasheet
Distance from the surface of the ground to the cable axis	L	1000	mm	System design
Thermal resistivity of the soil	ρ_T	0,8	K.m/W	System design
Axial separation between cables	s_1	2000,0	mm	System design

Table 11-11: Sequential steps to calculate the external thermal resistance T_4

Step	Parameter	Symbol	Unit	Reference
1	External thermal resistance	T_4	K.m/W	[4] section 4.2.3.3.2, note 1
Note 1: This equation results from the general equation of [4], section 4.2.3.3.1				

1. External thermal resistance:

$$u = \frac{2L}{D_e}$$

L	1000 mm	$u = 16,2074554295$
D_e	123,4 mm	

$$T_4 = \frac{1}{2\pi} \rho_T \left[\ln(u + \sqrt{u^2 - 1}) + \frac{1}{2} \ln \left[1 + \left(\frac{2L}{s_1} \right)^2 \right] \right] \quad (\text{K.m/W})$$

ρ_T	0,8 K.m/W	$T_4 = 0,4869172368 \text{ K.m/W}$
u	16,2074554295	
L	1000 mm	
s_1	2000 mm	

11.8 Permissible current rating-thermally limited

$$I = \left[\frac{\theta - \theta_a}{R' \cdot T_1 + nR'T_2 + nR'(T_3 + T_4)} \right]^{0,5}$$

$\theta - \theta_a$	55 K	$I = 2311,0958805122 \text{ A}$
R'	0,0000107685 Ω/m	
T_1	0,3712553097 K.m/W	
T_2	0,0340718397 K.m/W	
T_3	0,0640058386 K.m/W	
T_4	0,4869172368 K.m/W	
n	1	

11.9 Calculation of temperature drop across the insulation

$$\Delta\theta_{ins} = R' I^2 T_{1_i}$$

R'	0,0000107685 Ω/m	$\Delta\theta_{ins} = 14,2963520433$ °C Note 1
I	2311,0958805122 A	
T_{1_i}	0,2485616337 K.m/W	

Note 1: The temperature drop is greater than maximum permissible 10,7 K, thus the current is field limited.

11.10 Calculation of field limited conductor temperature:

$$\theta = \frac{\Delta\theta_{ins,max}}{T_{1_i}} [T_1 + nT_2 + n(T_3 + T_4)] + \theta_a$$

$\Delta\theta_{ins,max}$	10,7 K	$\theta = 56,1643472556$ °C
T_1	0,3712553097 K.m/W	
T_2	0,0340718397 K.m/W	
T_3	0,0640058386 K.m/W	
T_4	0,4869172368 K.m/W	
T_{1_i}	0,2485616337 K.m/W	
n	1	
θ_a	15 °C	

11.11 Permissible current rating-field limited

Table 11-12: Sequential steps of current rating calculation-field limited

Step	Parameter	Symbol	Unit	Reference
1	DC resistance of conductor at conductor temperature	R'	Ω/m	[3], section 2.1.1
2	Current rating	I	A	See Section 11.2

- Conductor DC resistance at maximum operating temperature R' :

$$R' = R_o [1 + \alpha_{20}(\theta - 20)] \text{ (}\Omega/m\text{)}$$

R_o	$9,0 \times 10^{-6} \Omega/m$	$R' = 1,0279132962 \times 10^{-5} \Omega/m$
α_{20}	$3,93 \times 10^{-3} 1/K$	
θ	56,1643472556 °C	

- Current rating:

$$I^2 = \frac{\Delta\theta_{ins,max}}{R' T_{1_i}}$$

$\Delta\theta_{ins,max}$	10,7 K	$I = 2046,4286136895$ A
R'	$1,0279132962 \times 10^{-5} \Omega/m$	
T_{1_i}	0,2485616337 K.m/W	

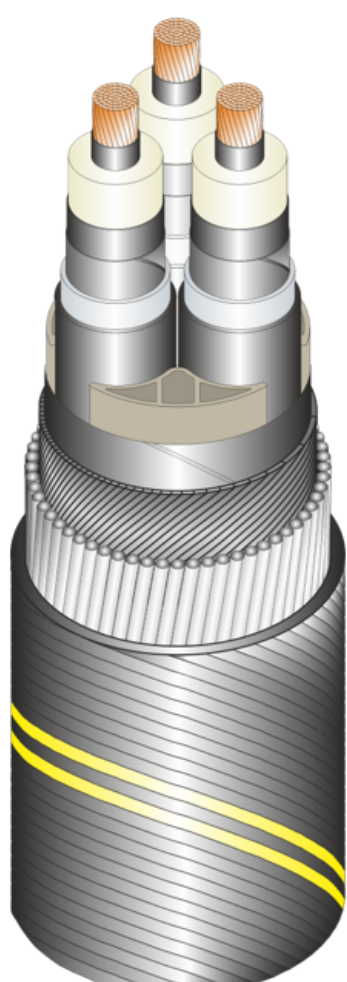
12. Case study 8: A 220kV 3-core submarine export cable

12.1 Introduction

The selected cable type is a three-core submarine cable with copper round compacted conductors, XLPE insulation, lead alloy sheath, semiconducting polyethylene sheath, tape bedding, stainless steel wire armour and polypropylene yarns 3x1000 mm² 220 kV [18] where applicable. The datasheet is depicted in Figure 40. The current rating will be determined for the following laying configuration:

- Cable buried in the seafloor
- Depth of laying of the cable 1 m
- Seabed soil temperature 15 °C
- Soil thermal resistivity 0.7 K·m/W
- Sheaths solidly bonded
- One circuit thermally independent

All other relevant parameters are given in the input tables at each calculation step.



Designation:	FXBLTV 3x1000 mm ²	
Rated voltage	220	kV
Impulse level	1050	kV
Conductor		
Type/material	round & compacted / Cooper	
Longitudinal water sealed	yes	
Cross-section	1000 mm ²	
Diameter	38.6	mm
Conductor tape		
Material	semi-conductive tape	
Thickness	0.2	mm
Conductor screen		
Material	semi-conductive PE	
Thickness	1.5	mm
Insulation		
Material	XLPE	
Thickness	23.0	
Insulation screen		
Material	semi-conductive PE	
Thickness	1.4	mm
Longitudinal water barrier		
Material	swelling tape	
Thickness	0.7	mm
Metallic sheath		
Material	lead alloy	
Thickness	3.6	mm
Protective sheath		
Material	semi-conductive PE	
Thickness	2.6	mm
Assembling		
Material 1	polymeric fillers	
Diameter	226	mm
Length of lay	2750 (left)	mm
Cable core binder		
Material	polymeric tapes	
Thickness	0.3	mm
Armour bedding		
Material	polymeric tapes	
Thickness	0.2	mm
Armour		
Material 1	stainless steel wires	
Material 2	bitumen	
Wire diameter	5.0	mm
Number of wires	133	
Length of lay	2900 (right)	mm
Outer cover		
Material 1	polypropylene yarns	
Material 2	bitumen	
Thickness	2 x 2	mm
Complete cable		
Diameter approx.	245	mm
Weight approx.	113	kg/m

Figure 40: Cable datasheet 220kV 3x1000mm² Submarine

12.2 Core layup effect

Table 12-1: Parameters needed to calculate the core layup effect

Parameter	Symbol	Value	Unit	Reference
Conductor diameter	d_c	38,6	mm	Cable datasheet
Outer diameter of one core	D_{core}	104,6	mm	Cable datasheet
Lay length of core stranding	L_{core}	2750	mm	Cable datasheet
Core diameter	d_e	104,6	mm	Cable datasheet

Table 12-2: Sequential steps to calculate the core layup effect

Step	Parameter	Symbol	Unit	Reference
1	Factor for the core layup	f_{layup}	mm	Guidance Point 23

1. Calculation of layup factor:

$$f_{layup} = \sqrt{1 + \left[\frac{\pi(1,29D_{core})}{L_{core}} \right]^2}$$

D_{core}	104,6 mm	$f_{layup} = 1,0118110892$
L_{core}	2750 mm	

Note: In this case study, the electrical resistance is given per meter of core contrary to IEC 60228 [8]. The case thus illustrates how to calculate the current rating using Guidance Point 23. The length of lay factor is used to adjust resistances, capacitance and thermal resistances to account for the layup of the cores.

12.3 Calculation of the conductor AC resistance at operation temperature

Table 12-3: Parameters needed to calculate the conductor AC resistance at operating temperature

Parameter	Symbol	Value	Unit	Reference
Conductor DC resistance @ 20°C	R_o	$0,0176 \times 10^{-3}$	Ω/m	
Conductor diameter	d_c	38,6	mm	Cable datasheet
Conductor material		Copper		Cable datasheet
Type of conductor		Round stranded compacted		Cable datasheet
Skin effect factor	k_s	1	-	[3], table 2
Proximity effect factor	k_p	1	-	[3], table 2
Conductor temperature coefficient at 20°C per Kelvin	α_{20}	$3,93 \times 10^{-3}$	1/K	[3], table 1
Conductor operating temperature	θ	90	°C	Cable datasheet
Distance between conductor axes	s	104,6	mm	Cable datasheet
System frequency	f	50	Hz	System design

Table 12-4: Sequential steps to calculate the conductor AC resistance at operating temperature

Step	Parameter	Symbol	Unit	Reference
1	Conductor DC resistance at operating temperature	R'	Ω/m	[3], section 2.1.1
2	Argument of a Bessel function – skin effect	x_s	-	[3], section 2.1.2
3	Skin effect factor	y_s	-	[3], section 2.1.2
4	Argument of a Bessel function – proximity effect	x_p	-	[3], section 2.1.3
5	Proximity effect factor	y_p	-	[3], section 2.1.3
6	AC resistance of conductor	R	Ω/m	[3], section 2.1
7	Adjust AC resistance with the core layup factor	R	Ω/m	

1. Conductor DC resistance at operating temperature R'

$$R' = R_o [1 + \alpha_{20}(\theta - 20)] \quad (\Omega/m)$$

R_o	$0,0176 \times 10^{-3} \Omega/m$	$R' = 2,244176 \times 10^{-5} \Omega/m$
α_{20}	$3,93 \times 10^{-3} 1/K$	
θ	$90^\circ C$	

Note: This resistance is assumed to be per meter of core. The resistance is in step 7 compensated for the extra length due to layup.

2. Argument of a Bessel function used to calculate skin effect x_s

$$x_s^2 = \frac{8\pi f}{R'} 10^{-7} k_s$$

f	50 Hz	$x_s^2 = 5,5995477246$
R'	$2,244176 \times 10^{-5} \Omega/m$	
k_s	1	

3. Skin effect factor y_s

$$x_s = 2,3663363507 \rightarrow \text{for } 0 < x_s \leq 2,8 \rightarrow y_s = \frac{x_s^4}{192 + 0,8x_s^4}$$

Note: three equations are given for calculating y_s , use of these equations depends on the calculated value of x_s .

x_s^2	5,5995477246	$y_s = 0,1444369104$
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4. Argument of a Bessel function used to calculate proximity effect x_p :

$$x_p^2 = \frac{8\pi f}{R'} 10^{-7} k_p$$

f	50 Hz	$x_p^2 = 5,5995477246$
R'	$2,244176 \times 10^{-5} \Omega/m$	
k_p	1	

5. Proximity effect factor y_p :

$$y_p = \frac{x_p^4}{192 + 0,8x_p^4} \left(\frac{d_c}{s} \right)^2 \left[0,312 \left(\frac{d_c}{s} \right)^2 + \frac{1,18}{\frac{x_p^4}{192 + 0,8x_p^4} + 0,27} \right]$$

x_p^2	5,5995477246	$y_p = 0,05683893091$
d_c	38,6 mm	
s	104,6 mm	

6. AC resistance of conductor R per meter of conductor:

$$R_{core} = R'(1 + y_s + y_p) \quad (\Omega/m)$$

R'	$2,244176 \times 10^{-5} \Omega/m$	$R_{core} = 2,6958744125 \times 10^{-5} \Omega/m$
y_s	0,1444369104	
y_p	0,05683893091	

7. Adjusting AC resistance due to lay of cores

$$R = f_{layup} \cdot R_{core} \quad (\Omega/m)$$

R_{core}	$2,6958744125 \times 10^{-5} \Omega/m$	$R = 2,7277156257 \times 10^{-5} \Omega/m$
f_{layup}	1,0118110892	

Note: The resistance, R_{core} , is per m of core, due to the layup of the three cores the resistance per m cable will be larger, this is compensated here by multiplying with the layup factor.

12.4 Dielectric losses

Table 12-5: Parameters needed to calculate the dielectric loss per unit length per phase

Parameter	Symbol	Value	Unit	Reference
External diameter of the insulation	D_i	88,0	mm	Cable datasheet
External diameter of the conductor, incl. screen	$d_{c.sc}$	42,0	mm	Cable datasheet
Relative permittivity of the insulation	ϵ	2,5	-	[3], table 3
Loss factor of the insulation	$\tan\delta$	1×10^{-3}	-	[3], table 3, note 1
Voltage to earth	U_o	$220\,000/\sqrt{3}$	V	System design
System frequency	f	50	Hz	System design

Note 1: Always assume unfilled XLPE unless specified otherwise

Table 12-6: Sequential steps to calculate the dielectric loss per unit length per phase

Step	Parameter	Symbol	Unit	Reference
1	Capacitance per m of core	C_{core}	F/m	[3], section 2.2
2	Adjust capacitance with the core layup factor	C	F/m	Guidance Point 23
3	Angular frequency of system	ω	Rad/s	[3], section 2.2
4	Dielectric loss per unit length in each phase	W_d	W/m	[3], section 2.2

1. Capacitance C (for circular conductors):

$$C_{core} = \frac{\epsilon}{18 \ln\left(\frac{D_i}{d_{c.sc}}\right)} 10^{-9} \quad (F/m)$$

ϵ	2,5	$C_{core} = 1,8777213536 \times 10^{-10} F/m$
D_i	88,0 mm	
$d_{c.sc}$	42,0 mm	

Note: The capacitance is per m of core, due to the layup of the three cores will the capacitance per m cable be larger and is compensated in next step.

2. Adjusting capacitance due to lay of cores

$$C = f_{layup} \cdot C_{core} \quad (\Omega/m)$$

C_{core}	$1,8777213536 \times 10^{-10}$ F/m	$C = 1,8998992881 \times 10^{-10}$ F/m
f_{layup}	1,0118110892	

3. Angular frequency of system:

$$\omega = 2\pi f \left(\frac{rad}{s} \right)$$

f	50 Hz	$\omega = 100\pi$ rad/s
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4. Dielectric loss per unit length in each phase:

$$W_d = \omega C U_o^2 \tan \delta \quad (W/m)$$

C	$1,8998992881 \times 10^{-10}$ F/m	$W_d = 0,9629518229$ W/m
ω	100π rad/s	
U_o	$220\,000/\sqrt{3}$ V	
$\tan \delta$	1×10^{-3}	

12.5 Loss factor for sheath

Table 12-7: Parameters needed to calculate the loss factor for PB sheath

Parameter	Symbol	Value	Unit	Reference
External diameter of metal sheath	D_s	99,4	mm	Cable datasheet
Thickness of the sheath	t_s	3,6	mm	Cable datasheet
Sheath resistivity at 20°C	ρ_s	$21,4 \times 10^{-8}$	Ωm	[3], table 1
Sheath temperature coefficient at 20°C per Kelvin	α_s	$4,0 \times 10^{-3}$	1/K	[3], table 1
Axial separation of conductors	s	104,6	mm	System design
System frequency	f	50	Hz	System design
AC resistance of conductor at operating temp.	R	$2,7277156257 \times 10^{-5}$	Ω/m	Calculated

Table 12-8: Sequential steps to calculate the sheath losses per unit length per phase

Step	Parameter	Symbol	Unit	Reference
1	Calculation of the cross-sectional area of the sheath	A_s	mm ²	
2	AC Resistance of sheath at 20°C	R_{so}	Ω/m	[3], section 2.3
3	Maximum operating temperature of the cable sheath, to be determined iterative	θ_{sc}	°C	[3], section 2.3
4	Resistance of sheath at operating temperature	R_s	Ω/m	[3], section 2.3
5	Reactance per unit length of sheath	X	Ω/m	[3], section 2.3.1
6	Loss factor circulating losses for sheath	λ_1'	-	[3], section 2.3.1
7	Loss factor for eddy current losses	λ_1''	-	[3], section 2.3.6.1

1. Calculation of cross-sectional area of the sheath:

$$A_s = \pi \cdot t_s \cdot (D_s - t_s)$$

D_s	99,4 mm	$A_s = 1083,4724743701 \text{ mm}^2$
t_s	3,6 mm	

2. Resistance of screen at 20°C:

AC Resistance of the screen at 20°C per unit length is:

$$R_{so} = \frac{\rho_s}{A_s} f_{layup} \quad (\Omega/m)$$

ρ_s	$21,4 \times 10^{-8} \Omega m$	$R_{so} = 1,9984593814 \times 10^{-4} \Omega/m$
A_s	$1083,4724743701 \times 10^{-6} \text{ m}^2$	
f_{layup}	1,0118110892	

3. PB sheath AC resistance at estimated temperature θ_{sh} :

The resistance of the lead sheath depends on the temperature of the sheath that is dependent on the unknown current. For this reason, an estimated temperature must be used as initial value and then an iterative process to find final value. Initial value is set to 70 °C.

$$R_s = R_{so} [1 + \alpha_s (\theta_{sc} - 20)] \quad \left(\frac{\Omega}{m} \right)$$

R_{so}	$1,9984593814 \times 10^{-4} \Omega/m$	$R_s = 2,3981512577 \times 10^{-4} \Omega/m$
α_s	$4,0 \times 10^{-3}$	
θ_{sc}	70 °C	

4. Calculation of the reactance X per unit length of the sheath:

Mean diameter of sheath:

$$d = D_s - t_s \quad (m)$$

D_s	99,4 mm	$d = 95,8 \text{ mm}$
t_s	3,6 mm	

$$X = 2\omega 10^{-7} \ln \left(\frac{2s}{d} \right) f_{layup} \quad (\Omega/m)$$

ω	$100\pi \text{ rad/s}$	$X_{core} = 4,9653050279 \times 10^{-5} \Omega/m$
s	104,6 mm	
d	95,8 mm	
f_{layup}	1,0118110892	

5. Loss factor λ'_1 for circulating losses of the sheath:

$$\lambda'_1 = \frac{R_s}{R} \frac{1}{1 + \left(\frac{R_s}{X} \right)^2}$$

R_s	$2,3981512577 \times 10^{-4} \Omega/m$	$\lambda'_1 = 0,3613987411$
R	$2,7277156257 \times 10^{-5} \Omega/m$	
X	$4,9653050279 \times 10^{-5} \Omega/m$	

6. Loss factor λ''_1 for eddy current losses:

$$m = \frac{\omega}{R_s} 10^{-7}$$

ω	100π	$m = 0,1310006049$
R_s	$2,3981512577 \times 10^{-4} \Omega/\text{m}$	

$$\lambda_0 = 3 \left(\frac{m^2}{1 + m^2} \right) \left(\frac{d}{2s} \right)^2$$

m	$0,1310006049$	$\lambda_0 = 0,01061416285$
d	$95,8 \text{ mm}$	
s	$104,6 \text{ mm}$	

$$\Delta_1 = (1,14m^{2,45} + 0,33) \left(\frac{d}{2s} \right)^{(0,92m+1,66)}$$

m	$0,1310006049$	$\Delta_1 = 0,08409364691$
d	$95,8 \text{ mm}$	
s	$104,6 \text{ mm}$	

$$\Delta_2 = 0$$

$$\rho_s = \rho(1 + \alpha_s(\theta_{sc} - 20))$$

ρ	$21,4 \times 10^{-8} \Omega\text{m}$	$\rho_s = 2,5680 \times 10^{-7} \Omega\cdot\text{m}$
α_s	$4,0 \times 10^{-3}$	
θ_{sc}	$70 \text{ }^\circ\text{C}$	

$$\beta_1 = \sqrt{\frac{4\pi\omega}{10^7 \rho_s}}$$

ω	100π	$\beta_1 = 39,2086923756$
ρ_s	$2,5680 \times 10^{-7} \Omega\text{m}$	

$$g_s = 1 + \left(\frac{t_s}{D_s} \right)^{1,74} (\beta_1 D_s 10^{-3} - 1,6)$$

D_s	$99,4 \text{ mm}$	$g_s = 1,0071406883$
t_s	$3,6 \text{ mm}$	
β_1	$39,2086923756$	

$$\lambda_1'' = \frac{R_s}{R} \left[g_s \lambda_0 (1 + \Delta_1 + \Delta_2) + \frac{(\beta_1 t_s)^4}{12 \times 10^{12}} \right]$$

R_s	$2,3981512577 \times 10^{-4} \Omega/\text{m}$	$\lambda_1'' = 0,1021781544$
R	$2,7277156257 \times 10^{-5} \Omega/\text{m}$	
g_s	$1,0071406883$	
λ_0	$0,01061416285$	
Δ_1	$0,08409364691$	

Δ_2	0	
β_1	39,2086923756	
t_s	3,6 mm	

7. Effect factor for Eddy currents:

$$M = N = \frac{R_s}{X}$$

R_s	$2,3981512577 \times 10^{-4} \Omega/\text{m}$	$M = 4,8298165857$ $N = 4,8298165857$
X	$4,9653050279 \times 10^{-5} \Omega/\text{m}$	

$$F = \frac{4M^2N^2 + (M + N)^2}{4(M^2 + 1)(N^2 + 1)}$$

M	4,8298165857	$F = 0,9588936273$
N	4,8298165857	

8. Add circulating losses and eddy current losses for foil to find λ_1 :

$$\lambda_1 = \lambda_1' + F \cdot \lambda_1''$$

λ_1'	0,3613987411	$\lambda_1 = 0,4593767223$
λ_1''	0,1021781544	
F	0,9588936273	

12.6 Loss factor for armour

Armour losses in cables with non magnetic wire armouring can be neglected according to Guidance Point 26.

$$\lambda_2 = 0$$

12.7 Thermal resistance T_1 between conductor and screen

Table 12-9: Parameters needed to calculate the thermal resistance between conductor and screen

Parameter	Symbol	Value	Unit	Reference
Diameter of conductor	d_c	38,6	mm	Cable datasheet
External diameter over the sc tape of the conductor	d_{cond_t}	39,0	mm	Cable datasheet
External diameter of sc conductor screen	d_{sc}	42,0	mm	Cable datasheet
External diameter of insulation	D_i	88,0	mm	Cable datasheet
External diameter of sc insulation screen	D_{si}	90,8	mm	Cable datasheet
External diameter of sc- water blocking tape(s) under the copper wires screen	D_{uwb}	92,2	mm	Cable datasheet
Thermal resistivity of sc screen	ρ_{sc}	2,5	$K \cdot m/W$	Note 1
Thermal resistivity of insulation	ρ_i	3,5	$K \cdot m/W$	[4], table 1
Thermal resistivity of water blocking tapes	ρ_{wb}	12	$K \cdot m/W$	Note 1
Thermal resistivity of conductor tape	ρ_t	6	$K \cdot m/W$	Note 1
Note 1: See guidance for details about thermal resistivity				

Table 12-10: Sequential steps to calculate the thermal resistance between conductor and screen

Step	Parameter	Symbol	Unit	Reference
1	Thermal resistance of conductor tape	$T_{1_cond_t}$	$K \cdot m/W$	[4], section 4.1.2.1
2	Thermal resistance of sc conductor screen	T_{1_csc}	$K \cdot m/W$	[4], section 4.1.2.1
3	Thermal resistance of insulation	T_{1_i}	$K \cdot m/W$	[4], section 4.1.2.1
4	Thermal resistance of sc insulation screen	T_{1_isc}	$K \cdot m/W$	[4], section 4.1.2.1
5	Thermal resistance of sc water blocking tape under the copper wires screen	T_{1_uwbt}	$K \cdot m/W$	[4], section 4.1.2.1
6	Thermal resistance between conductor and screen	T_{1_core}	$K \cdot m/W$	-
7	Adjust Thermal resistance with the core layup factor	T_1	$K \cdot m/W$	Guidance Point 23

1. Thermal resistance of the conductor tape:

$$T_{1_cond_t} = \frac{\rho_t}{2\pi} \ln\left(\frac{d_{cond_t}}{d_c}\right) \quad (K \cdot m/W)$$

ρ_t	6,0 K·m/W	$T_{1_cond_t} = 0,009844722848 \text{ K} \cdot \text{m/W}$
d_{cond_t}	39,0 mm	
d_c	38,6 mm	

2. Thermal resistance of sc conductor screen:

$$T_{1_csc} = \frac{\rho_{sc}}{2\pi} \ln\left(\frac{d_{sc}}{d_{cond_t}}\right) \quad (K \cdot m/W)$$

ρ_{sc}	2,5 K·m/W	$T_{1_csc} = 0,02948662523 \text{ K} \cdot \text{m/W}$
d_{sc}	42,0 mm	
d_{cond_t}	39,0 mm	

3. Thermal resistance of insulation:

$$T_{1_i} = \frac{\rho_i}{2\pi} \ln\left(\frac{D_i}{d_{sc}}\right) \quad \left(K \cdot \frac{m}{W}\right)$$

ρ_i	3,5 K·m/W	$T_{1_i} = 0,4120259168 \text{ K} \cdot \text{m/W}$
D_i	88,0 mm	
d_{sc}	42,0 mm	

4. Thermal resistance of sc insulation screen:

$$T_{1_isc} = \frac{\rho_{sc}}{2\pi} \ln\left(\frac{D_{si}}{D_i}\right) \quad (K \cdot m/W)$$

ρ_{sc}	2,5 K·m/W	$T_{1_isc} = 0,01246281528 \text{ K} \cdot \text{m/W}$
D_{si}	90,8 mm	
D_i	88,0 mm	

5. Thermal resistance of sc water blocking tape under the lead sheath:

$$T_{1_uwbt} = \frac{\rho_{wb}}{2\pi} \ln\left(\frac{D_{uwbt}}{D_{si}}\right) \quad (K \cdot m/W)$$

ρ_{wb}	12 K·m/W	$T_{1_uwb} = 0,02922246130 \text{ K·m/W}$
D_{uwb}	92,2 mm	
D_{si}	90,8 mm	

6. Thermal resistance between conductor and screen:

$$T_{1_core} = T_{1_cond_t} + T_{1_csc} + T_{1_i} + T_{1_isc} + T_{1_uwb} \quad (K \cdot m/W)$$

$T_{1_cond_t}$	0,009844722848 K·m/W	$T_{1_core} = 0,4930425415 \text{ K·m/W}$
T_{1_csc}	0,02948662523 K·m/W	
T_{1_i}	0,4120259168 K·m/W	
T_{1_isc}	0,01246281528 K·m/W	
T_{1_uwb}	0,02922246130 K·m/W	

7. Thermal resistance between conductor and screen compensated for the lay up of core:

$$T_1 = \frac{T_{1_core}}{f_{layup}} \quad (K \cdot m/W)$$

T_{1_core}	0,4930425415 K·m/W	$T_1 = 0,48728714947 \text{ K·m/W}$
f_{layup}	1,0118110892	

12.8 Thermal resistance T_2 between screen and armour

Table 12-11: Parameters needed to calculate the thermal resistance of the sheath around each core

Parameter	Symbol	Value	Unit	Reference
External diameter of metallic sheath	D_{PB}	99,4	mm	Cable datasheet
External diameter of sc PE sheath	$D_{sc\ PE}$	104,6	mm	Cable datasheet
Inner diameter of armour	D_{arm}	227	mm	Cable datasheet
Diameter of stranded cores	D_{cabl}	226	mm	Cable datasheet
Diameter of cable core	D_s	104,6	mm	Cable datasheet
Thermal resistivity of sc PE sheath	ρ_{scPE}	2,5	K·m/W	Note 1
Thermal resistivity of fillers and binding tapes	ρ_{fil_bt}	6	K·m/W	Note 2
Note 1: See guidance for details about thermal resistivity				
Note 2: The fillers are made of PE with thermal resistivity 3.5 K·m/W. The fillers has some air or water filled voids which might have a higher thermal resistivity than the filler itself. The binding tape has a thermal resistivity of 6 K·m/W. The highest thermal resistivity of the solid materials are used for this combined material. The voids in the fillers are considered to be covered by the use of 6 K·m/W instead of 3.5 K·m/W for the whole filler part.				

Table 12-12: Sequential steps to calculate the thermal resistance of the sheath around each core

step	Parameter	Symbol	Unit	Reference
1	Thermal resistance of the sheath around each core	T_{2_scPE}	K · m/W	[4], section 4.1.2.1
2	Calculation of geometric factor	G	-	[4], section 5.6
3	Thermal resistance of the fillers and bedding for SL-type cables	T_{2_fill}	K · m/W	[4], section 4.1.3.2
4	Thermal resistance between screen and armour	T_2	K · m/W	-

1. Thermal resistance of the sheath around each core:

$$T_{2_scPE} = \frac{\rho_{scPE}}{2\pi} \ln\left(\frac{D_{scPE}}{D_{PB}}\right) \quad (K \cdot m/W)$$

ρ_{scPE}	2,5 K·m/W	$T_{2_scPE} = 0,02028884852 \text{ K·m/W}$
D_{scPE}	104,6 mm	
D_{PB}	99,4 mm	

2. Calculation of geometric factor G:

Firstly, the thickness of material between sheaths and armour is expressed as a fraction of the outer diameter of the sheath:

$$X = \frac{\frac{D_{arm} - D_{cabl}}{2}}{D_s}$$

D_{arm}	227 mm	$X = 0,004780114723$
D_{cabl}	226 mm	
D_s	104,6 mm	

The upper curve is used from IEC 60287-2-1 [4] Figure 6, see Guidance Point 45 for more details about when to choose which curve. With $0 < X < 0,03$ the G factor can be estimated with the following formula ([4], section 5.6):

$$G = 2\pi(0,00022619 + 2,11429X - 20,4762X^2)$$

X	0,004780114723	$G = 0,06198279299$
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3. Thermal resistance of the between outer sheath of the core to the armour:

$$T_{2_fill} = \frac{\rho_{fil_bt}}{6\pi} G \quad (K \cdot m/W)$$

ρ_{fil_bt}	6 K·m/W	$T_{2_fill} = 0,01972973578 \text{ K·m/W}$
G	0,06198279299	

4. Thermal resistance between screen and armour:

The total thermal resistance for T_2 is the sum of T_{2_scPE} and T_{2_fill} . However, T_{2_scPE} will only have the losses from one core while T_{2_fill} should have the losses for all phases. T_{2_scPE} should also be divided by f_{layup} to compensate for the effect of the extra loss due to core layup.

$$T_2 = \frac{1}{nf_{layup}} T_{2_scPE} + T_{2_fill}$$

T_{2_scPE}	0,02028884852 K·m/W	$T_2 = 0,02641373992 \text{ K·m/W}$
T_{2_fill}	0,01972973578 K·m/W	
n	3	
f_{layup}	1,0118110892	

12.9 Thermal resistance T_3 of outer covering

Table 12-13: Parameters needed to calculate the thermal resistance T_3 of the outer covering

Parameter	Symbol	Value	Unit	Reference
External diameter of the armour	D'_a	237	mm	Cable datasheet
External diameter of outer covering	D_e	245	mm	Cable datasheet
Thermal resistivity of outer covering	ρ_{Ti}	6	K·m/W	[4], table 1

Table 12-14: Sequential steps to calculate the thermal resistance T_3 of the outer covering

Step	Parameter	Symbol	Unit	Reference
1	Thermal resistance of outer covering	T_3	K·m/W	[4], section 4.1.4.1

1. Thermal resistance of outer covering:

$$T_3 = \frac{\rho_{Ti}}{2\pi} \ln \left(\frac{D_e}{D'_a} \right) \quad (K \cdot m/W)$$

ρ_{Ti}	6 K·m/W	$T_3 = 0,03170182109 \text{ K·m/W}$
D_e	237 mm	
D'_a	245 mm	

12.10 External thermal resistance T_4

Table 12-15: Parameters needed to calculate the external thermal resistance T_4

Parameter	Symbol	Value	Unit	Reference
External diameter of one cable	D_e	245	mm	Cable datasheet
Distance from the surface of the ground to the cable axis	L	1000	mm	System design, note 1
Thermal resistivity of the soil	ρ_T	0,7	K·m/W	System design

Table 12-16: Sequential steps to calculate the external thermal resistance T_4

Step	Parameter	Symbol	Unit	Reference
1	External thermal resistance	T_4	K·m/W	[4], section 4.2.2.

1. External thermal resistance:

$$u = \frac{2L}{D_e}$$

L	1000 mm	$u = 8,1632653061$
D_e	245 mm	

$$T_4 = \frac{1}{2\pi} \rho_T \ln \left(u + \sqrt{u^2 - 1} \right) \quad (K \cdot m/W)$$

ρ_T	0,7 K·m/W	$T_4 = 0,3107202651 \text{ K·m/W}$
u	8,1632653061	

12.11 Current rating

Table 12-17: Parameters needed to calculate the continuous current carrying capacity

Parameter	Symbol	Value	Unit	Reference
Conductor operating temperature	θ	90	$^{\circ}\text{C}$	System design
Ambient temperature	θ_{amb}	15	$^{\circ}\text{C}$	System design
Number of conductors in cable	n	3	-	Cable datasheet
AC resistance of conductor at operating temperature	R	$2,7277156257 \times 10^{-5}$	Ω/m	See the previous calculation of R , Section
Ratio of the total losses in metallic screen	λ_1	0,4593767223	-	See the previous calculation of λ_1 Section
Ratio of the total losses in metallic foil	λ_2	0	-	See Section
Dielectric losses	W_d	0,9629518229	W/m	See the previous calculation of W_d Section
Thermal resistance between conductor and metallic screen	T_1	0,48728714947	$\text{K}\cdot\text{m}/\text{W}$	See the previous calculation of T_1 Section
Thermal resistance between metallic screen and metallic foil	T_2	0,02641373992	$\text{K}\cdot\text{m}/\text{W}$	See the previous calculation of T_2 Section
Thermal resistance of external serving	T_3	0,03170182109	$\text{K}\cdot\text{m}/\text{W}$	See the previous calculation of T_3 Section
Thermal resistance of surrounding medium	T_4	0,3107202651	$\text{K}\cdot\text{m}/\text{W}$	See the previous calculation of T_4 Section

Table 12-18: Sequential steps to calculate the continuous current rating capacity

Step	Parameter	Symbol	Unit	Reference
1	Conductor temperature rise above ambient	$\Delta\theta$	K	[3], section 1.4.1.1
2	Current rating	I	A	[3], section 1.4.1.1

1. Conductor temperature rise above ambient:

$$\Delta\theta = \theta - \theta_{amb}$$

θ	90 $^{\circ}\text{C}$	$\Delta\theta = 75 \text{ K}$
θ_{amb}	15 $^{\circ}\text{C}$	

2. Current rating:

$$I = \left[\frac{\Delta\theta - W_d[0,5 \cdot T_1 + n(T_2 + T_3 + T_4)]}{R \cdot T_1 + nR(1 + \lambda_1)T_2 + nR(1 + \lambda_1 + \lambda_2)(T_3 + T_4)} \right]^{0.5}$$

$\Delta\theta$	75 $^{\circ}\text{C}$	$I = 1133,7238746986 \text{ A}$
W_d	0,96295182288 W/m	
T_1	0,48728714947 $\text{K}\cdot\text{m}/\text{W}$	
T_2	0,02641373992 $\text{K}\cdot\text{m}/\text{W}$	
T_3	0,03170182109 $\text{K}\cdot\text{m}/\text{W}$	

T_4	0,3107202651 K·m/W	
R	$2,7277156257 \times 10^{-5} \Omega/\text{m}$	
n	3	
λ_1	0,4593767223	
λ_2	0	

Note: This is the result of the first calculation with assumed sheath temperature. In the following Sections this will be used to calculate the sheath temperature and with an iterative approach reach the final current rating.

12.12 Iteration

An iterative method is required to find the final rated current. The sheath temperature is updated based on the calculated current, followed by the screen losses and then a new rated current is calculated, this is repeated until the result is convergent.

1. The operating temperature θ_{sc} (°C) of the screen is given by:

$$\theta_{sc} = \theta - (I^2 R + 0,5 W_d) T_1 \quad (^\circ\text{C})$$

θ	90 °C	$\theta_{sc} = 72,6810261001^\circ\text{C}$
I	1133,7238746986 A	
R	$2,7277156257 \times 10^{-5} \Omega/\text{m}$	
W_d	0,96295182288 W/m	
T_1	0,48728714947 K·m/W	

2. PB sheath AC resistance at operating temperature θ_{sc}

$$R_s = R'_{so} [1 + \alpha_s (\theta_{sc} - 20)] \quad (\Omega/\text{m})$$

R'_{so}	$1,9984593814 \times 10^{-4} \Omega/\text{m}$	$R_s = 2,4195829447 \times 10^{-4} \Omega/\text{m}$
α_s	$4,0 \times 10^{-3} 1/\text{K}$	
θ_{sc}	72,6810261001 °C	

3. Loss factor λ'_1 for circulating losses of the sheath:

$$\lambda'_1 = \frac{R_s}{R} \frac{1}{1 + \left(\frac{R_s}{X}\right)^2}$$

R_s	$2,4195829447 \times 10^{-4} \Omega/\text{m}$	$\lambda'_1 = 0,3584574925$
R	$2,7277156257 \times 10^{-5} \Omega/\text{m}$	
X	$4,9653050279 \times 10^{-5} \Omega/\text{m}$	

4. Loss factor λ''_1 for eddy current losses, lead sheath only,

$$m = \frac{\omega}{R_s} 10^{-7}$$

ω	100π	$m = 0,1298402545$
R_s	$2,4195829447 \times 10^{-4} \Omega/\text{m}$	

$$\lambda_0 = 3 \left(\frac{m^2}{1 + m^2} \right) \left(\frac{d}{2s} \right)^2$$

m	0,1298402545	$\lambda_0 = 0,01043006726$
d	95,8 mm	
s	104,6 mm	

$$\Delta_1 = (1,14m^{2,45} + 0,33) \left(\frac{d}{2s} \right)^{(0,92m+1,66)}$$

m	0,1298402545	$\Delta_1 = 0,08412168597$
d	95,8 mm	
s	104,6 mm	

$$\Delta_2 = 0$$

$$\rho_s = \rho(1 + \alpha_s(\theta_{sc} - 20))$$

ρ	$21,4 \times 10^{-8} \Omega\text{m}$	$\rho_s = 2,5909495834 \times 10^{-7} \Omega\cdot\text{m}$
α_s	$4,0 \times 10^{-3}$	
θ_{sc}	72,6810261001 °C	

$$\beta_1 = \sqrt{\frac{4\pi\omega}{10^7 \rho_s}}$$

ω	100π	$\beta_1 = 39,0346587696$
ρ_s	$2,5909495834 \times 10^{-7} \Omega\text{m}$	

$$g_s = 1 + \left(\frac{t_s}{D_s} \right)^{1,74} (\beta_1 D_s 10^{-3} - 1,6)$$

D_s	99,4 mm	$g_s = 1,0070869191$
t_s	3,6 mm	
β_1	39,0346587696	

$$\lambda_1'' = \frac{R_s}{R} \left[g_s \lambda_0 (1 + \Delta_1 + \Delta_2) + \frac{(\beta_1 t_s)^4}{12 \times 10^{12}} \right]$$

R_s	$2,4195829447 \times 10^{-4} \Omega/\text{m}$	$\lambda_1'' = 0,1013003792$
R	$2,7277156257 \times 10^{-5} \Omega/\text{m}$	
g_s	1,0070869191	
λ_0	0,01043006726	
Δ_1	0,08412168597	
Δ_2	0	
β_1	39,0346587696	
t_s	3,6 mm	

$$M = N = \frac{R_s}{X}$$

R_s	$2,4195829447 \times 10^{-4} \Omega/\text{m}$	$M = 4,8729794668$ $N = 4,8729794668$
X	$4,9653050279 \times 10^{-5} \Omega/\text{m}$	

$$F = \frac{4M^2N^2 + (M + N)^2}{4(M^2 + 1)(N^2 + 1)}$$

M	4,8729794668	$F = 0,9595893125$
N	4,8729794668	

5. Add circulating losses and eddy current losses for foil to find λ_1 :

$$\lambda_1 = \lambda_1' + F \cdot \lambda_1''$$

λ_1'	0,3584574925	$\lambda_1 = 0,4556642538$
λ_1''	0,1013003792	
F	0,9595893125	

6. New permissible current rating:

$$I = \left[\frac{\Delta\theta - W_d [0,5 \cdot T_1 + n(T_2 + T_3 + T_4)]}{R \cdot T_1 + nR(1 + \lambda_1)T_2 + nR(1 + \lambda_1 + \lambda_2)(T_3 + T_4)} \right]^{0.5}$$

$\Delta\theta$	75 °C	$I = 1134,8332499345 \text{ A}$
W_d	0,96295182288 W/m	
T_1	0,48728714947 K·m/W	
T_2	0,02641373992 K·m/W	
T_3	0,03170182109 K·m/W	
T_4	0,3107202651 K·m/W	
R	$2,7277156257 \times 10^{-5} \Omega/\text{m}$	
n	3	
λ_1	0,4556642538	
λ_2	0	

7. New iteration

The iteration is continued with restart at step 1 until convergence is reached at $\theta_{sc} = 72,6479847905125$ °C. That temperature results in the following temperature dependent intermediate results:

θ_{sc}	72,6479847905125°C
R_s	$2,4193188179 \times 10^{-4} \Omega/\text{m}$
λ_1'	0,3584934634
m	0,1298544297
λ_0	0,01043230700
Δ_1	0,08412134024
ρ_s	$2,5906667498 \times 10^{-7} \Omega \cdot \text{m}$
β_1	39,0367894976
g_s	1,0070875774
λ_1''	0,1013111067
F	0,9595808450
λ_1	0,4557096608

12.13 Final result

Final permissible current rating:

$$I = \left[\frac{\Delta\theta - W_d[0,5 \cdot T_1 + n(T_2 + T_3 + T_4)]}{R \cdot T_1 + nR(1 + \lambda_1)T_2 + nR(1 + \lambda_1 + \lambda_2)(T_3 + T_4)} \right]^{0.5}$$

$\Delta\theta$	75 °C	$I = 1134,8196615454\text{A}$
W_d	0,96295182288 W/m	
T_1	0,48728714947 K·m/W	
T_2	0,02641373992 K·m/W	
T_3	0,03170182109 K·m/W	
T_4	0,3107202651 K·m/W	
R	$2,7277156257 \times 10^{-5} \Omega/\text{m}$	
n	3	
λ_1	0,4557096608	
λ_2	0	

12.14 Calculation of losses

The following losses are per phase, to get the total losses in the cable they should be multiplied with $n=3$.

Conductor losses:

$$W_c = RI^2$$

R	$2,7277156257 \times 10^{-5} \Omega/\text{m}$	$W_c = 35,1279491032 \text{ W/m}$
I	1134,8196615454 A	

Lead sheath losses:

$$W_s = \lambda_1 RI^2$$

R	$2,7277156257 \times 10^{-5} \Omega/\text{m}$	$W_s = 16,0081457714 \text{ W/m}$
I	1134,8196615454 A	
λ_1	0,4557096608	

Dielectric losses:

-	See the previous calculation of W_d Section	$W_d = 0,96295182288 \text{ W/m}$
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Armour losses:

$$W_a = \lambda_2 RI^2$$

R	$2,7277156257 \times 10^{-5} \Omega/\text{m}$	$W_a = 0 \text{ W/m}$
I	1134,8196615454 A	
λ_2	0	

13. Case study 9: A 110kV retrofitted cable

13.1 Introduction

The selected cable type is a XLPE-insulated three-core cable with round stranded copper conductor, aluminium-polyethylene-laminated sheath and steel wire armour in a PE-covered steel pipe. The cable drawing is shown in Figure 41 and associated data sheet in Figure 42.

The current rating will be determined for the following laying configuration:

- Cable and pipe buried in earth
- Depth of laying of the pipe 1 m
- Soil temperature 20°C
- Thermal resistivity of soil 1 Km/W
- Pipe dimensions 152 mm x 7 mm with 4 mm PE-covering
- One circuit thermally independent

All other relevant parameters are given in the input tables at each parameter calculations.

Assumptions:

- Losses in pipe due to presence of armour neglected, refer to Guidance Point 39.
- Loss factor for sheath calculated according with [3] section 2.3.10, refer to Guidance Point 26.
- Loss factor for armour calculated according with [3] section 2.4.2.3.1 and section 2.4.2.5, refer to Guidance Point 26.



Figure 41: Generic drawing of a 110kV cable with steel wire armour used for retrofitting

2. Design data

conductor		
structure	round, stranded	
material	copper	
cross section	400 mm ²	
diameter	approx.	23,8 mm
conductor screen		
material	conductive XLPE	
thickness of extruded layer	min.	0,3 mm
diameter	approx.	25,4 mm
insulation		
material	XLPE	
nominal thickness		10,0 mm
min. thickness at any point		9,0 mm
diameter	approx.	45,4 mm
insulation screen		
material	conductive XLPE	
thickness of extruded layer	min.	0,3 mm
diameter	approx.	47,0 mm
bedding and longitudinal water barrier		
material	semi-conducting and swelling tapes	
diameter	approx.	47,8 mm
screen / metal sheath (radial water barrier)		
structure	aluminium-polyethylene-laminated sheath	
material	longitudinally applied aluminium tape	
thickness	approx.	0,15 mm
diameter	approx.	48,1 mm
outer sheath		
material	PE	
nominal thickness		2,0 mm
min. thickness at any point		1,6 mm
diameter	approx.	52,1 mm
cabling		
three cores touching		
diameter	approx.	113 mm
bedding		
material	jute tapes	
diameter	approx.	114,7 mm
armour		
structure	flat wires	
material	galvanized steel	
number of wires	42	
width of wire	7,0 mm	
thickness of wire	1,2 mm	
diameter	approx.	117 mm

Figure 42: Cable data for a 110kV cable with steel wire armour used for retrofitting

13.2 Calculation of the conductor AC resistance at operation temperature

Table 13-1: Parameters needed to calculate the conductor AC resistance at operating temperature

Parameter	Symbol	Value	Unit	Reference
Conductor DC resistance @ 20°C	R_o	$4,70 \times 10^{-5}$	Ω/m	[8]
Conductor diameter	d_c	23,8	mm	Cable datasheet
Conductor material		Copper		Cable datasheet
Type of conductor		Round stranded compacted according to [8]		Cable datasheet
Skin effect factor	k_s	1	-	[3], table 2
Proximity effect factor	k_p	1	-	[3], table 2
Conductor temperature coefficient at 20°C per Kelvin	α_{20}	$3,93 \times 10^{-3}$	1/K	[3], table 1
Conductor operating temperature	θ	90	°C	System design
Distance between conductor axes	s	52,1	mm	Cable datasheet
System frequency	f	50	Hz	System design

Table 13-2: Sequential steps to calculate the conductor AC resistance at operating temperature

Step	Parameter	Symbol	Unit	Reference
1	Conductor DC resistance at operating temperature	R'	Ω/m	[3], section 2.1.1
2	Argument of a Bessel function – skin effect	x_s	-	[3], section 2.1.2
3	Skin effect factor	y_s	-	[3], section 2.1.2
4	Argument of a Bessel function – proximity effect	x_p	-	[3], section 2.1.3
5	Proximity effect factor	y_p	-	[3], section 2.1.3
6	AC resistance of conductor	R	Ω/m	[3], section 2.1

1. Conductor DC resistance at operating temperature R' :

$$R' = R_o [1 + \alpha_{20}(\theta - 20)] \quad (\Omega/m)$$

R_o	$4,70 \times 10^{-5} \Omega/m$	$R' = 5,99297 \times 10^{-5} \Omega/m$
α_{20}	$3,93 \times 10^{-3} 1/K$	
θ	90 °C	

2. Argument of a Bessel function used to calculate skin effect x_s :

$$x_s = \sqrt{\frac{8\pi f}{R'} 10^{-7} k_s}$$

f	50 Hz	$x_s = 1,4480510743$
R'	$0,0599297 \times 10^{-3} \Omega/m$	
k_s	1	

3. Skin effect factor y_s :

$$\text{for } 0 < x_s \leq 2,8 \rightarrow y_s = \frac{x_s^4}{192 + 0,8x_s^4}$$

Note: three equations are given for calculating y_s , use of these equations depends on the calculated value of x_s .

x_s	1,4480510743	$y_s = 0,0224879590$
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4. Argument of a Bessel function used to calculate proximity effect x_p :

$$x_p = \frac{8\pi f}{R'} 10^{-7} k_p$$

f	50 Hz	$x_p = 1,4480510743$
R'	$0,0599297 \times 10^{-3} \Omega/\text{m}$	
k_p	1	

5. Proximity effect factor y_p :

$$y_p = \frac{x_p^4}{192 + 0,8x_p^4} \left(\frac{d_c}{s} \right)^2 \left[0,312 \left(\frac{d_c}{s} \right)^2 + \frac{1,18}{\frac{x_p^4}{192 + 0,8x_p^4} + 0,27} \right]$$

x_p	1,4480510743	$y_p = 0,0192377982$
d_c	23,8 mm	
s	52,1 mm	

6. AC resistance of conductor R per meter of conductor:

$$R = R' (1 + 1,5(y_s + y_p)) \quad (\Omega/\text{m})$$

R'	$0,0599297 \times 10^{-3} \Omega/\text{m}$	$R = 6,3680618159 \times 10^{-5} \Omega/\text{m}$
y_s	0,0224879590	
y_p	0,0192377982	

Note: The resistance is per meter of complete cable as per IEC 60228 [8].

13.3 Calculation of Core layup factor

Table 13-3: Parameters needed to calculate the core layup effect

Parameter	Symbol	Value	Unit	Reference
Outer diameter of one core	D_{core}	52,1	mm	Cable datasheet
Lay length of core stranding	L_{core}	2237	mm	Additional information

Table 13-4: Sequential steps to calculate the core layup effect

Step	Parameter	Symbol	Unit	Reference
1	Factor for the core layup	f_{layup}	mm	Guidance Point 23

1. Calculation of layup factor:

$$f_{layup} = \sqrt{1 + \left[\frac{\pi(1,29D_{core})}{L_{core}} \right]^2}$$

D_{core}	52,1 mm	$f_{layup} = 1,0044445582$
L_{core}	2237 mm	

13.4 Dielectric losses

Table 13-5: Parameters needed to calculate the dielectric loss per unit length per phase

Parameter	Symbol	Value	Unit	Reference
External diameter of the insulation	D_i	45,4	mm	Cable datasheet
External diameter of the conductor, incl. screen	d_{sc}	25,4	mm	Cable datasheet
Relative permittivity of the insulation	ϵ	2,5	-	[3], table 3
Loss factor of the insulation	$\tan\delta$	1×10^{-3}	-	[3], table 3, note 1
Voltage to earth	U_o	$110\,000/\sqrt{3}$	V	System design
System frequency	f	50	Hz	System design
Note 1: Always assume unfilled XLPE unless specified otherwise				

Table 13-6: Sequential steps to calculate the dielectric loss per unit length per phase

Step	Parameter	Symbol	Unit	Reference
1	Capacitance	C	F/m	[3], section 2.2
2	Angular frequency of system	ω	Rad/s	[3], section 2.2
3	Dielectric loss per unit length in each phase	W_d	W/m	[3], section 2.2

1. Capacitance C_{core} (for circular conductors) without considering lay length:

$$C_{core} = \frac{\epsilon}{18 \ln\left(\frac{D_i}{d_{sc}}\right)} 10^{-9} \text{ (F/m)}$$

ϵ	2,5	$C = 0,2391490253 \times 10^{-9} \text{ F/m}$
D_i	45,4 mm	
d_{sc}	25,4 mm	

- Capacitance C (for circular conductors) with considering lay length:

$$C = f_{layup} \frac{\epsilon}{18 \ln\left(\frac{D_i}{d_{sc}}\right)} 10^{-9} \text{ (F/m)}$$

f_{layup}	1,0044445582	$C = 0,2402119371 \times 10^{-9} \text{ F/m}$
ϵ	2,5	
D_i	45,4 mm	
d_{sc}	25,4 mm	

2. Angular frequency of system:

$$\omega = 2\pi f \text{ (rad/s)}$$

f	50 Hz	$\omega = 100\pi \text{ rad/s}$
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3. Dielectric loss per unit length in each phase with considering lay length:

$$W_d = \omega C U_o^2 \tan\delta \text{ (W/m)}$$

C	$0,2402119371 \times 10^{-9} \text{ F/m}$	$W_d = 0,3043747162 \text{ W/m}$
ω	$100\pi \text{ rad/s}$	
U_o	$110000/\sqrt{3} \text{ V}$	
$\tan\delta$	1×10^{-3}	

13.5 Loss factor for sheath

Table 13-7: Parameters needed to calculate the loss factor for aluminium sheath

Parameter	Symbol	Value	Unit	Reference
External diameter of metal sheath	D_s	48,1	mm	Cable datasheet
Thickness of the sheath	t_s	0,15	mm	Cable datasheet
Sheath resistivity at 20°C	ρ_s	$2,84 \times 10^{-8}$	Ωm	[3], table 1
Sheath temperature coefficient at 20°C per Kelvin	α_s	$4,03 \times 10^{-3}$	1/K	[3], table 1
Axial separation of conductors	s	52,1	mm	System design
System frequency	f	50	Hz	System design
AC resistance of conductor at operating temp.	R	$6,3680618159 \times 10^{-5}$	Ω/m	Calculated

Table 13-8: Sequential steps to calculate the sheath losses per unit length per phase

Step	Parameter	Symbol	Unit	Reference
1	Calculation of the cross-sectional area of the sheath	A_s	mm^2	
2	AC Resistance of sheath at 20°C	R_{so}	Ω/m	[3], section 2.3
3	Resistance of sheath at operating temperature	R_s	Ω/m	[3], section 2.3
4	Reactance per unit length of sheath	X	Ω/m	[3], section 2.3.1
5	Loss factor circulating losses for sheath	λ'_1	-	[3], section 2.3.10
6	Loss factor for eddy current losses	λ''_1	-	[3], section 2.3.4
7	Add circulating losses and eddy current losses for foil to find λ_1	λ_1	-	[3], section 2.3

1. Calculation of cross-sectional area of the sheath:

$$A_s = \pi \cdot t_s \cdot (D_s - t_s)$$

D_s	48,1 mm	$A_s = 22,5959051609 \text{ mm}^2$
t_s	0,15 mm	

2. Resistance of sheath at 20°C:
AC Resistance of the sheath at 20°C per unit length is:

$$R_{so} = \frac{\rho_s}{A_s} f_{layup} \quad (\Omega/\text{m})$$

ρ_s	$2,84 \times 10^{-8} \Omega\text{m}$	$R_{so} = 1,2624511056 \times 10^{-3} \Omega/\text{m}$
A_s	$22,5959051609 \times 10^{-6} \text{ m}^2$	
f_{layup}	1,0044445582	

3. Resistance of sheath at operating temperature:

The resistance of the aluminium sheath depends on the temperature of the sheath that is dependent on the unknown current. For this reason, an estimated temperature must be used as initial value and then an iterative process to find final value. Initial value is set to 80 °C.

$$R_s = R_{so}[1 + \alpha_s(\theta_s - 20)] \quad (\Omega/m)$$

R_{so}	$1,2624511056 \times 10^{-3} \Omega/m$	$R_s = 1,5677117829 \times 10^{-3} \Omega/m$
α_s	$4,03 \times 10^{-3}$	
θ_s	80	

4. Calculation of the reactance X per unit length of the sheath:

Mean diameter of sheath:

$$d = D_s - t_s \quad (m)$$

D_s	48,1 mm	$d = 47,95 \text{ mm}$
t_s	0,15 mm	

$$X = 2\omega 10^{-7} \ln\left(\frac{2s}{d}\right) f_{layup} \quad (\Omega/m)$$

ω	$100\pi \text{ rad/s}$	$X = 4,8983900309 \times 10^{-5} \Omega/m$
s	52,1 mm	
d	47,95 mm	
f_{layup}	1,0044445582	

5. Loss factor λ'_1 for circulating losses of the sheath:

$$\lambda'_1 = \frac{R_s}{R} \frac{1,5}{1 + \left(\frac{R_s}{X}\right)^2}$$

R_s	$1,5677117829 \times 10^{-3} \Omega/m$	$\lambda'_1 = 0,03601643351$
R	$6,3680618159 \times 10^{-5} \Omega/m$	
X	$4,8983900309 \times 10^{-5} \Omega/m$	

9. Loss factor λ''_1 for eddy current losses:

$$m = \frac{\omega}{R_s} 10^{-7}$$

ω	100π	$m = 0,02003935090$
R_s	$1,5677117829 \times 10^{-3} \Omega/m$	

$$\lambda_0 = 3 \left(\frac{m^2}{1 + m^2} \right) \left(\frac{d}{2s} \right)^2$$

m	0,02003935090	$\lambda_0 = 0,0002550092730$
d	47,95 mm	
s	52,1 mm	

$$\Delta_1 = (1,14m^{2,45} + 0,33) \left(\frac{d}{2s} \right)^{(0,92m+1,66)}$$

m	0,02003935090	
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d	47,95 mm	$\Delta_1 = 0,08971227149$
s	52,1 mm	

$$\Delta_2 = 0$$

$$\rho_s = \rho(1 + \alpha_s(\theta_s - 20))$$

ρ	$2,84 \times 10^{-8} \Omega\text{m}$	$\rho_s = 3,526712 \times 10^{-8} \Omega\text{m}$
α_s	$4,03 \times 10^{-3}$	
θ_s	80 °C	

$$\beta_1 = \sqrt{\frac{4\pi\omega}{10^7 \rho_s}}$$

ω	100π	$\beta_1 = 105,8022419697$
ρ_s	$3,526712 \times 10^{-8} \Omega\text{m}$	

$$g_s = 1 + \left(\frac{t_s}{D_s}\right)^{1,74} (\beta_1 D_s 10^{-3} - 1,6)$$

D_s	48,1 mm	$g_s = 1,0001521172$
t_s	0,15 mm	
β_1	105,8022419697	

$$\lambda_1'' = \frac{R_s}{R} \left[g_s \lambda_0 (1 + \Delta_1 + \Delta_2) + \frac{(\beta_1 t_s)^4}{12 \times 10^{12}} \right]$$

R_s	$1,5677117829 \times 10^{-3} \Omega/\text{m}$	$\lambda_1'' = 0,00684228384$
R	$6,3680618159 \times 10^{-5} \Omega/\text{m}$	
g_s	1,0001521172	
λ_0	0,0002550092730	
Δ_1	0,08971227149	
Δ_2	0	
β_1	105,8022419697	
t_s	0,15 mm	

Note that the F-factor is applied in the next section.

$$M = N = \frac{R_s}{X}$$

R_s	$1,5677117829 \times 10^{-3} \Omega/\text{m}$	$M = 32,0046336251$ $N = 32,0046336251$
X	$4,8983900309 \times 10^{-5} \Omega/\text{m}$	

$$F = \frac{4M^2N^2 + (M + N)^2}{4(M^2 + 1)(N^2 + 1)}$$

M	32,0046336251	$F = 0,9990246724$
N	32,0046336251	

6. Add circulating losses and eddy current losses for foil to find λ_1 :

$$\lambda_1 = \lambda_1' + F\lambda_1''$$

λ_1'	0,03601643351	$\lambda_1 = 0,04285204388$
λ_1''	0,00683561037	
F	0,9990159919	

13.6 Loss factor for armour

Table 13-9: Parameters needed to calculate the loss factor for armour

Parameter	Symbol	Value	Unit	Reference
Amount of amour wires	n_A	42	-	Cable datasheet
Thickness of amour wires	t_A	1,2	mm	Cable datasheet
Width of amour wires	w_A	7,0	mm	Cable datasheet
Diameter under amour wires	D_A	114,7	mm	Cable datasheet
Length of lay of amour wires	L_A	1053	mm	Additional Value
Amour resistivity at 20°C	ρ_A	$13,8 \times 10^{-8}$	Ωm	[3], table 1
Amour temperature coefficient at 20°C per Kelvin	α_A	$4,5 \times 10^{-3}$	1/K	[3], table 1
Diameter over complete cable core	D_{core}	52,1	mm	Cable datasheet
Diameter after cabling	$D_{cabling}$	113	mm	Cable datasheet
System frequency	f	50	Hz	System design
AC resistance of conductor at operating temp.	R	$6,3680618159 \times 10^{-5}$	Ω/m	Calculated

Table 13-10: Sequential steps to calculate the amour losses per unit length per phase

Step	Parameter	Symbol	Unit	Reference
1	Calculation of the cross-sectional area of amour	A_A	mm ²	
2	Mean circumference of amour	M_A	mm	
3	Elongation factor of amour	$f_{layup\ arm}$	-	
4	DC Resistance of amour at 20°C	R_{Adc}	Ω/m	[3], section 2.4
5	AC Resistance of amour at 20°C	R_{A0}	Ω/m	[3], section 2.4
6	Resistance of amour at operating temperature	R_A	Ω/m	[3], section 2.4
7	Mean diameter of amour	d_A	mm	
8	Distance between the axis of a conductor and the cable center	c	mm	
9	Amour losses λ_2	λ_2	-	[3], section 2.4.2.3.1 and section 2.4.2.5

1. Calculation of cross-sectional area of armour:

$$A_A = n_A t_A w_A$$

n_A	42	$A_A = 352,8 \text{ mm}^2$
t_A	1,2 mm	
w_A	7,0 mm	

2. Mean circumference of armour:

As the do not have a filler it will not be round but triangular with rounded edges. The circumference of the armour is therefore described with the following formula. See Guidance Point 41.

$$M_A = 3D_{core} + (D_{core} + 2t_{bedding} + t_A)\pi \text{ (mm)}$$

D_{core}	52,1 mm	$M_A = 329,0875959474 \text{ mm}$
$t_{bedding}$	0,85 mm (deduced from Figure 42)	
t_A	1,2 mm	

3. Elongation Factor of armour:

$$f_{layup \text{ arm}} = \sqrt{1 + \left(\frac{M_A}{L_A}\right)^2}$$

M_A	329,0875959474 mm	$f_{layup \text{ arm}} = 1,0476980224$
L_A	1053 mm	

4. Resistance of armour at 20°C:

DC Resistance of armour at 20°C per unit length:

$$R_{Adc} = \frac{\rho_A}{A_A} f_{layup \text{ arm}} \quad (\Omega/m)$$

ρ_A	$13,8 \times 10^{-8} \Omega m$	$R_{Adc} = 4,0981385229 \times 10^{-4} \Omega/m$
A_A	$352,8 \times 10^{-6} \text{ m}^2$	
$f_{layup \text{ arm}}$	1,0476980224	

5. AC resistance of armour at 20°C:

According to [3], section 2.4.2.1 the AC resistance of armour wire varies from about 1,2 times the DC resistance of 2 mm wire diameter up to 1,4 times the DC resistance for 5 mm wire diameter.

For any other wire diameter interpolation and extrapolation should be used according to Guidance Point 34. Formula for that can be derived from:

$$k(t_{wire}) = \frac{y_2 - y_1}{x_2 - x_1} \cdot (t_{wire} - x_1) + y_1 = \frac{1,4 - 1,2}{5 - 2} \cdot (t_{wire} - 2) + 1,2 \Leftrightarrow$$

$$k(t_{wire}) = \frac{t_{wire} + 16}{15}$$

In this case the armour is made of flat wires 1,2x7mm. According to Guidance Point 35 the smallest diameter should be used for this calculations, i.e 1.2mm:

$$k = \frac{t_{wire} + 16}{15}$$

t_{wire}	1,2	$k = 1,1466666667$
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$$R_{Ao} = k \cdot R_{Adc}$$

k	1,1466666667	$R_{Ao} = 4,6991988396 \times 10^{-4} \Omega/m$
R_{Adc}	$4,0981385229 \times 10^{-4} \Omega/m$	

6. Resistance of armour at operating temperature:

The resistance of the armour depends on the temperature of the armour that is dependent on the unknown current. For this reason, an estimated temperature must be used as initial value and then an iterative process to find final value. Initial value is set to 75 °C.

$$R_A = R_{Ao} [1 + \alpha_A (\theta_A - 20)] \quad (\Omega/m)$$

R_{Ao}	$4,6991988396 \times 10^{-4} \Omega/m$	$R_A = 5,8622505523 \times 10^{-4} \Omega/m$
α_A	$4,5 \times 10^{-3}$	
θ_A	75°C	

7. Mean diameter of armour:

$$d_A = \frac{M_A}{\pi} \text{ (mm)}$$

M_A	329,0875959474 mm	$d_A = 104,7518352105 \text{ mm}$
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8. Distance between the axis of a conductor and the cable center:

$$c = \frac{d_{cabling} - D_{core}}{2}$$

$D_{cabling}$	113 mm	$c = 30,45 \text{ mm}$
D_{core}	52,1 mm	

9. Loss factor λ_2 for losses of the armour:

$$\lambda_2 = 1,23 \frac{R_A}{R} \left(\frac{2c}{d_A} \right)^2 \frac{1}{1 + \left(\frac{2,77 R_A}{\omega} \right)^2} \left(1 - \frac{R}{R_s} \lambda'_1 \right)$$

R_A	$5,8622505523 \times 10^{-4} \Omega/m$	$\lambda_2 = 0,1378766242$
R	$6,3680618159 \times 10^{-5} \Omega/m$	
c	30,45 mm	
d_A	104,7518352105 mm	
ω	$100\pi \text{ rad/s}$	
λ'_1	0,03601643351	
R_s	$1,5677117829 \times 10^{-3} \Omega/m$	

13.7 Thermal resistance T_1 between conductor and screen

Table 13-11: Parameters needed to calculate the thermal resistance between conductor and screen

Parameter	Symbol	Value	Unit	Reference
Diameter of conductor	d_c	23,8	mm	Cable datasheet
External diameter of sc conductor screen	d_{sc}	25,4	mm	Cable datasheet
External diameter of insulation	D_i	45,4	mm	Cable datasheet
External diameter of sc insulation screen	D_{si}	47,0	mm	Cable datasheet
External diameter of sc- water blocking tape(s) under aluminium foil	D_{uwb}	47,8	mm	Cable datasheet
Thermal resistivity of sc screen	ρ_{sc}	2,5	$K \cdot m/W$	Note 1
Thermal resistivity of insulation	ρ_i	3,5	$K \cdot m/W$	[4], table 1
Thermal resistivity of water blocking tapes	ρ_{wb}	12	$K \cdot m/W$	Note 1
Note 1: See guidance for details about thermal resistivity				

Table 13-12: Sequential steps to calculate the thermal resistance between conductor and screen per unit length per phase

Step	Parameter	Symbol	Unit	Reference
1	Thermal resistance of conductor tape	$T_{1_cond_t}$	$K \cdot m/W$	[4], section 4.1.2.1
2	Thermal resistance of sc conductor screen	T_{1_csc}	$K \cdot m/W$	[4], section 4.1.2.1
3	Thermal resistance of insulation	T_{1_i}	$K \cdot m/W$	[4], section 4.1.2.1
4	Thermal resistance of sc insulation screen	T_{1_isc}	$K \cdot m/W$	[4], section 4.1.2.1
5	Thermal resistance of sc water blocking tape under the copper wires screen	T_{1_uwb}	$K \cdot m/W$	[4], section 4.1.2.1
6	Thermal resistance between conductor and screen	T_1	$K \cdot m/W$	-

1. Thermal resistance of the conductor tape (no conductor tape over conductor):

$$T_{1_cond_t} = 0 \quad (K \cdot m/W)$$

2. Thermal resistance of sc conductor screen:

$$T_{1_csc} = \frac{\rho_{sc}}{2\pi} \ln \left(\frac{d_{sc}}{d_c} \right) \quad (K \cdot m/W)$$

ρ_{sc}	2,5 K·m/W	$T_{1_csc} = 0,02588798124 \text{ K} \cdot \text{m/W}$
d_{sc}	25,4 mm	
d_c	23,8 mm	

3. Thermal resistance of insulation:

$$T_{1_i} = \frac{\rho_i}{2\pi} \ln \left(\frac{D_i}{d_{sc}} \right) \quad (K \cdot m/W)$$

ρ_i	3,5 K·m/W	$T_{1_i} = 0,3235095193 \text{ K} \cdot \text{m/W}$
D_i	45,4 mm	
d_{sc}	25,4 mm	

4. Thermal resistance of sc insulation screen:

$$T_{1_isc} = \frac{\rho_{sc}}{2\pi} \ln \left(\frac{D_{si}}{D_i} \right) \quad (K \cdot m/W)$$

ρ_{sc}	2,5 K·m/W	$T_{1_isc} = 0,01378102625 \text{ K} \cdot \text{m/W}$
D_{si}	47,0 mm	
D_i	45,4 mm	

5. Thermal resistance of sc water blocking tape under aluminium foil:

$$T_{1_uwb} = \frac{\rho_{wb}}{2\pi} \ln \left(\frac{D_{uwb}}{D_{si}} \right) \quad \left(K \cdot \frac{m}{W} \right)$$

ρ_{wb}	12 K·m/W	$T_{1_uwb} = 0,03223467772 \text{ K} \cdot \text{m/W}$
D_{uwb}	47,8 mm	
D_{si}	47,0 mm	

6. Thermal resistance between conductor and screen:

$$T_{1_core} = T_{1_cond_t} + T_{1_csc} + T_{1_i} + T_{1_isc} + T_{1_uwbt} \quad (K \cdot m/W)$$

$T_{1_cond_t}$	0 K·m/W	$T_{1_core} = 0,39541320454 \text{ K} \cdot \text{m/W}$
T_{1_csc}	0,02588798124 K·m/W	
T_{1_i}	0,3235095193 K·m/W	
T_{1_isc}	0,01378102625 K·m/W	
T_{1_uwbt}	0,03223467772 K·m/W	

5. Thermal resistance between conductor and screen per cable length

$$T_1 = \frac{T_{1_core}}{f_{layup}}$$

T_{1_core}	0,39541320454 K·m/W	$T_1 = 0,3936635440 \text{ K} \cdot \text{m/W}$
f_{layup}	1,0044445582	

13.8 Thermal resistance T_2 between sheath and armour

Table 13-13: Parameters needed to calculate the thermal resistance between sheath and armour

Parameter	Symbol	Value	Unit	Reference
Diameter over Al sheath	D_s	48,1	mm	Cable datasheet
Diameter of PE sheath	D_{core}	52,1	mm	Cable datasheet
Inner diameter of armour	D_{arm}	114,7	mm	Cable datasheet
Diameter of stranded cores	D_{cabl}	113	mm	Cable datasheet
Thermal resistivity of PE	ρ_{PE}	3,5	K·m/W	Note 1
Thermal resistivity of bedding	ρ_{bed}	6	K·m/W	Note 2
Note 1: See guidance for details about thermal resistivity				
Note 2: For void and binding tape and bedding, the resistivity of bedding and tape, 6 K·m/W, is used				

Table 13-14: Sequential steps to calculate the thermal resistance of the sheath around each core

Step	Parameter	Symbol	Unit	Reference
1	Thermal resistance of the sheath around each core	T_{2_PE}	K · m/W	[4], section 4.1.2.1
2	Calculation of geometric factor G	G	-	[4], section 5.6
3	Thermal resistance of the fillers and bedding for SL-type cables	T_{2_fill}	K · m/W	[4], section 4.1.3.2
4	Thermal resistance between screen and armour	T_2	K · m/W	-

1. Thermal resistance of the sheath around each core:

$$T_{2_PE} = \frac{\rho_{PE}}{2\pi} \ln \left(\frac{D_{PE}}{D_{PB}} \right) \quad (K \cdot m/W)$$

ρ_{PE}	3,5 K·m/W	$T_{2_PE} = 0,04449808373 \text{ K} \cdot \text{m/W}$
D_{PE}	52,1 mm	
D_{Al}	48,1 mm	

2. Calculation of geometric factor G:

Firstly, the thickness of material between sheaths and armour is expressed as a fraction of the outer diameter of the core sheath.

$$X = \frac{\frac{D_{arm} - D_{cabl}}{2}}{D_{jacket}}$$

D_{arm}	114,7 mm	$X = 0,01631477927$
D_{cabl}	113 mm	
D_{jacket}	52,1 mm	

The upper curve is used from [3], Figure 6, see guidance for more details about when to choose which curve.

With $0 < X < 0,03$ the G factor can be estimated with the following formula (section 5.6 of [4]):

$$G = 2\pi(0,00022619 + 2,11429X - 20,4762X^2)$$

X	0,01631477927	$G = 0,1839099215$
-----	---------------	--------------------

3. Thermal resistance of the between outer sheath of the core to the armour:

$$T_{2_fill} = \frac{\rho_{bed}}{6\pi} G \quad (K \cdot m/W)$$

ρ_T	6 K·m/W	$T_{2_fill} = 0,05854034619 \text{ K·m/W}$
G	0,1839099215	

4. Thermal resistance between screen and armour:

The total thermal resistance for T_2 is the sum of T_{2_PE} and T_{2_fill} . However, T_{2_PE} will only have the losses from one core while T_{2_fill} should have the losses for all phases. T_{2_PE} should also be divided by f_{layup} to compensate for the effect of the extra loss due to core layup.

$$T_2 = \frac{1}{nf_{layup}} T_{2_PE} + T_{2_fill}$$

T_{2_PE}	0,04449808373 K·m/W	$T_2 = 0,07330740743 \text{ K·m/W}$
T_{2_fill}	0,05854034619 K·m/W	
n	3	
f_{layup}	1,0044445582	

13.9 Thermal resistance T_3 of outer covering

The cable has no outer covering, therefore T_3 is:

$$T_3 = 0 \quad (K \cdot m/W)$$

Note that the covering of the pipe is considered as part of T_4 in section 13.10.

13.10 External thermal resistance T_4

Table 13-15: Parameters needed to calculate the thermal resistance T_4

Parameter	Symbol	Value	Unit	Reference
Factor U	U	5,2	-	[4], table 4
Factor V	V	1,4	-	[4], table 4
Factor Y	Y	0,011	-	[4], table 4
Temperature of air	θ_A	75 °C	°C	See iterative calculation
Temperature of pipe	θ_P	70 °C	°C	See iterative calculation
External diameter of cable	D_e	117	mm	Cable datasheet
Outer diameter of steel pipe	D_{inner}	146	mm	Cable datasheet
Outer diameter of steel pipe over PE covering	D_{outer}	152	mm	Cable datasheet
Distance from the surface of the ground to the cable axis	L	1000	mm	System design
Thermal resistivity of PE covering of steel pipe	ρ_{T_PE}	3,5	K·m/W	[4], table 1
Thermal resistivity of the soil	ρ_{T_soil}	1	K·m/W	System design

Note that the nearest U,V and Y factors are selected from IEC 60287 to represent the heat transfer between the cable and the pipe although in this specific case the U, V and Y factors may actually be different because of the special shape of the cable and the material usage.

Table 13-16: Sequential steps to calculate the thermal resistance T_4

Step	Parameter	Symbol	Unit	Reference
1	Calculate equivalent cable diameter	D_e	m	
1	Thermal resistance between cable and steel pipe	$T'_4(\theta_m)$	K · m/W	[4], section 4.2.7.2
2	Thermal resistance of PE-covering of steel pipe	T''_4	K · m/W	[4], section 4.1.2.1
3	Thermal resistance of soil	T'''_4	K · m/W	[4], section 4.2.2
4	Thermal resistance T_4	T_4	K · m/W	[4], section 4.2.2.

1. Equivalent cable diameter

This cable do not have a filler and is therefore not round. An equivalent diameter is calculated using the following formula based on the circumference according to Guidance Point 48.

$$D_e = \frac{3D_{core} + (D_{core} + 2t_{bedding} + 2t_A)\pi}{\pi} \text{ (mm)}$$

D_{core}	52,1 mm	$D_e = 105,9518352105 \text{ mm}$
$t_{bedding}$	0,85 mm	
t_A	1,2 mm	

Note that this diameter is outside of the range of cable diameters of section 4.2.7.2 of [4]. In this case, Guidance Point 49 brings a solution by recommending to use the U, V and Y values as provided by IEC.

2. Thermal resistance between cable and steel pipe:

The thermal resistance between cable and steel pipe depends on the temperature of the air inside that is dependent on the unknown current. For this reason, an estimated temperature must be used as initial value and then an iterative process to find final value. Initial value is set to 70 °C.

Thermal resistance between cable and steel pipe:

$$T'_4 = \frac{U}{1 + 0,1(V + Y\theta_m)D_e}$$

U	5,2	$T'_4 = 0,2167429942 \text{ K}\cdot\text{m}/\text{W}$
V	1,4	
Y	0,011	
θ_m	70 °C	
D_e	105,9518352105 mm	

3. Thermal resistance of PE-covering of steel pipe:

$$T''_4 = \frac{\rho_{T_{PE}}}{2\pi} \ln\left(\frac{D_{outer}}{D_{inner}}\right) \left(K \cdot \frac{m}{W}\right)$$

$\rho_{T_{PE}}$	3,5 K·m/W	$T''_4 = 0,0224342654 \text{ K}\cdot\text{m}/\text{W}$
D_{outer}	152 mm	
D_{inner}	146 mm	

4. Thermal resistance of soil:

$$u = \frac{2L}{D_{outer}}$$

L	1000 mm	$u = 13,1578947368$
D_{outer}	152 mm	

$$T'''_4 = \frac{1}{2\pi} \rho_{soil} \ln\left(u + \sqrt{u^2 - 1}\right) \left(K \cdot m/W\right)$$

ρ_{soil}	1 K·m/W	$T'''_4 = 0,5202332609 \text{ K}\cdot\text{m}/\text{W}$
u	13,1578947368	

5. Thermal resistance T_4 :

$$T_4 = T'_4 + T''_4 + T'''_4 \left(K \cdot m/W\right)$$

T'_4	0,2167429942 K·m/W	$T_4 = 0,7594105205 \text{ K}\cdot\text{m}/\text{W}$
T''_4	0,0224342654 K·m/W	
T'''_4	0,5204635801 K·m/W	

13.11 Current rating

Table 13-17: Parameters needed to calculate the continuous current carrying capacity

Parameter	Symbol	Value	Unit	Reference
Conductor operating temperature	θ	90	$^{\circ}\text{C}$	System design
Ambient temperature	θ_{amb}	20	$^{\circ}\text{C}$	System design
Number of conductors in cable	n	3	-	Cable datasheet
AC resistance of conductor at operating temperature	R	$6,3680618159 \times 10^{-5}$	Ω/m	See the previous calculation of R ,
Ratio of the total losses in metallic screen	λ_1	0,04285204388	-	See the previous calculation of λ_1
Ratio of the total losses in metallic foil	λ_2	0,1378766242	-	See the previous calculation of λ_2
Dielectric losses	W_d	0,3043747162	W/m	See the previous calculation of W_d
Thermal resistance between conductor and metallic screen	T_1	0,3936635440	$\text{K}\cdot\text{m}/\text{W}$	See the previous calculation of T_1
Thermal resistance between metallic screen and metallic foil	T_2	0,07330740743	$\text{K}\cdot\text{m}/\text{W}$	See the previous calculation of T_2
Thermal resistance of external serving	T_3	0	$\text{K}\cdot\text{m}/\text{W}$	See the previous calculation of T_3
Thermal resistance of surrounding medium	T_4	0,7594105205	$\text{K}\cdot\text{m}/\text{W}$	See the previous calculation of T_4

Table 13-18: Sequential steps to calculate the continuous current rating capacity

Step	Parameter	Symbol	Unit	Reference
1	Conductor temperature rise above ambient	$\Delta\theta$	K	[3], section 1.4.1.1
2	Current rating	I	A	[3], section 1.4.1.1

1. Conductor temperature rise above ambient:

$$\Delta\theta = \theta - \theta_{amb}$$

θ	90 $^{\circ}\text{C}$	$\Delta\theta = 70 \text{ K}$
θ_{amb}	20 $^{\circ}\text{C}$	

2. Current rating:

$$I = \left[\frac{\Delta\theta - W_d[0,5 \cdot T_1 + n(T_2 + T_3 + T_4)]}{R \cdot T_1 + nR(1 + \lambda_1)T_2 + nR(1 + \lambda_1 + \lambda_2)(T_3 + T_4)} \right]^{0.5}$$

$\Delta\theta$	70 $^{\circ}\text{C}$	$I = 572,6326816418 \text{ A}$
W_d	0,3043747162 W/m	
T_1	0,3936635440 $\text{K}\cdot\text{m}/\text{W}$	
T_2	0,07330740743 $\text{K}\cdot\text{m}/\text{W}$	
T_3	0 $\text{K}\cdot\text{m}/\text{W}$	
T_4	0,7594105205 $\text{K}\cdot\text{m}/\text{W}$	
R	$6,3680618159 \times 10^{-5} \Omega/\text{m}$	
n	3	
λ_1	0,04285204388	
λ_2	0,1378766242	

Note: This is the result of the first calculation with assumed temperatures of sheath, armour and air in pipe. In the following sections this will be used to calculate the temperatures and with an iterative approach reach the final current rating.

13.12 Iterative calculation of Current rating

An iterative method is required to find the final rated current. The sheath, armour and air inside pipe temperatures are updated based on the calculated current, followed by the sheath and armour losses and thermal resistance of air in pipe. After that is a new rated current is calculated, this is repeated until the result is convergent.

1. The operating temperature θ_s (°C) of the screen is given by:

$$\theta_s = \theta - (I^2 R + 0,5W_d)T_1 \quad (^\circ\text{C})$$

θ	90 °C	$\theta_s = 81,7198449859 \text{ } ^\circ\text{C}$
I	572,6326816418 A	
R	$6,3680618159 \times 10^{-5} \Omega/\text{m}$	
W_d	0,3043747162 W/m	
T_1	0,3936635440 K·m/W	

2. The operating temperature θ_A (°C) of the armour is given by:

$$\theta_A = \theta - (I^2 R + 0,5W_d)T_1 - (I^2 R(1 + \lambda_1) + W_d)nT_2 \quad (^\circ\text{C})$$

θ	90 °C	$\theta_A = 76,8638344692 \text{ } ^\circ\text{C}$
I	572,6326816418A	
R	$6,3680618159 \times 10^{-5} \Omega/\text{m}$	
W_d	0,3043747162 W/m	
T_1	0,3936635440 K·m/W	
T_2	0,07330740743 K·m/W	
n	3	
λ_1	0,04285204388	

3. The operating temperature θ_m (°C) of the air in the pipe is given by:

$$\theta_m = \theta - (I^2 R + 0,5W_d)T_1 - n(I^2 R(1 + \lambda_1) + W_d)T_2 - n(I^2 R(1 + \lambda_1 + \lambda_2) + W_d)\left(T_3 + \frac{T'_4}{2}\right) \quad (^\circ\text{C})$$

θ	90 °C	$\theta_m = 68,7490945436 \text{ } ^\circ\text{C}$
I	572,6326816418A	
R	$6,3680618159 \times 10^{-5} \Omega/\text{m}$	
W_d	0,3043747162 W/m	
T_1	0,3936635440 K·m/W	
T_2	0,07330740743 K·m/W	
T_3	0 K·m/W	
T'_4	0,2167429942 K·m/W	
n	3	
λ_1	0,04285204388	
λ_2	0,1378766242	

4. New screen loss factor is calculated:

a. Resistance of sheath at operating temperature:

$$R_s = R_{so} [1 + \alpha_s (\theta_s - 20)] \quad (\Omega/m)$$

R_{so}	$1,2624511056 \times 10^{-3} \Omega/m$	$R_s = 1,5764618004 \times 10^{-3} \Omega/m$
α_s	$4,03 \times 10^{-3} 1/K$	
θ_s	$81,7198449859 \text{ }^\circ\text{C}$	

b. Loss factor λ'_1 for circulating losses of the sheath:

$$\lambda'_1 = \frac{R_s}{R} \frac{1,5}{1 + \left(\frac{R_s}{X}\right)^2}$$

R_s	$1,5764618004 \times 10^{-3} \Omega/m$	$\lambda'_1 = 0,03581691407$
R	$6,3680618159 \times 10^{-5} \Omega/m$	
X	$4,8983900309 \times 10^{-5} \Omega/m$	

c. Loss factor λ''_1 for eddy current losses:

$$m = \frac{\omega}{R_s} 10^{-7}$$

ω	100π	$m = 0,01992812419$
R_s	$1,5764618004 \times 10^{-3} \Omega/m$	

$$\lambda_0 = 3 \left(\frac{m^2}{1 + m^2} \right) \left(\frac{d}{2s} \right)^2$$

m	$0,01992812419$	$\lambda_0 = 0,0002521874351$
d	$47,95 \text{ mm}$	
s	$52,1 \text{ mm}$	

$$\Delta_1 = (1,14m^{2,45} + 0,33) \left(\frac{d}{2s} \right)^{(0,92m+1,66)}$$

m	$0,01992812419$	$\Delta_1 = 0,08971910688$
d	$47,95 \text{ mm}$	
s	$52,1 \text{ mm}$	

$$\Delta_2 = 0$$

$$\rho_s = \rho (1 + \alpha_s (\theta_s - 20))$$

ρ	$2,84 \times 10^{-8} \Omega m$	$\rho_s = 3,5463959698 \times 10^{-8} \Omega \cdot m$
α_s	$4,03 \times 10^{-3} 1/K$	
θ_s	$81,7198449859 \text{ }^\circ\text{C}$	

$$\beta_1 = \sqrt{\frac{4\pi\omega}{10^7 \rho_s}}$$

ω	100π	$\beta_1 = 105,5082102203$
ρ_s	$3,5463959698 \times 10^{-8} \Omega \cdot m$	

$$g_s = 1 + \left(\frac{t_s}{D_s}\right)^{1,74} (\beta_1 D_s 10^{-3} - 1,6)$$

D_s	48,1 mm	$g_s = 1,0001515006$
t_s	0,15 mm	
β_1	105,5082102203	

$$\lambda_1'' = \frac{R_s}{R} \left[g_s \lambda_0 (1 + \Delta_1 + \Delta_2) + \frac{(\beta_1 t_s)^4}{12 \times 10^{12}} \right]$$

R_s	$1,5764618004 \times 10^{-3} \Omega/m$	$\lambda_1'' = 0,00680437504$
R	$6,3680618159 \times 10^{-5} \Omega /m$	
g_s	1,0001515006	
λ_0	0,0002521874351	
Δ_1	0,08971910688	
Δ_2	0	
β_1	105,5082102203	
t_s	0,15 mm	

$$M = N = \frac{R_s}{X}$$

R_s	$1,5764618004 \times 10^{-3} \Omega/m$	$M = 32,1832641010$ $N = 32,1832641010$
X	$4,8983900309 \times 10^{-5} \Omega/m$	

$$F = \frac{4M^2N^2 + (M + N)^2}{4(M^2 + 1)(N^2 + 1)}$$

M	32,1832641010	$F = 0,9990354589$
N	32,1832641010	

d. Add circulating losses and eddy current losses for foil to find λ_1 :

$$\lambda_1 = \lambda_1' + F\lambda_1''$$

λ_1'	0,03581691407	$\lambda_1 = 0,04261472601$
λ_1''	0,00680437504	
F	0,9990354589	

5. New armour loss factor is calculated:

a. Resistance of armour at operating temperature:

$$R_A = R_{Ao} [1 + \alpha_A (\theta_A - 20)] \quad (\Omega/m)$$

R_{Ao}	$4,699198840 \times 10^{-4} \Omega/m$	$R_A = 5,9016639318 \times 10^{-4} \Omega /m$
α_A	$4,5 \times 10^{-3} 1/K$	
θ_A	76,8638344692 °C	

b. Loss factor λ_2 for circulating losses of the sheath:

$$\lambda_2 = 1,23 \frac{R_A}{R} \left(\frac{2c}{d_A} \right)^2 \frac{1}{1 + \left(\frac{2,77R_A}{\omega} \right)^2} \left(1 - \frac{R}{R_s} \lambda_1' \right)$$

R_A	$5,9016639318 \times 10^{-4} \Omega / \text{m}$	$\lambda_2 = 0,1370238656$
R	$6,3680618159 \times 10^{-5} \Omega / \text{m}$	
c	30,45 mm	
d_A	104,7518352105 mm	
ω	$100\pi \text{ rad/s}$	
λ_1'	0,03581691407	
R_s	$1,5764618004 \times 10^{-3} \Omega / \text{m}$	

6. New external thermal resistance is calculated:

a. Thermal resistance between cable and steel pipe:

$$T_4'(\theta_m) = \frac{U}{1 + 0,1(V + Y\theta_m)D_e}$$

U	5,2	$T_4' = 0,2180681275 \text{ K}\cdot\text{m/W}$
V	1,4	
Y	0,011	
θ_m	68,7754685168 °C	
D_e	105,9518352105 mm	

b. Thermal resistance T_4 :

$$T_4 = T_4' + T_4'' + T_4''' \quad (\text{K}\cdot\text{m/W})$$

T_4'	0,2180681275 K·m/W	$T_4 = 0,7607356539 \text{ K}\cdot\text{m/W}$
T_4''	0,0224342654 K·m/W	
T_4'''	0,5204635801 K·m/W	

7. Current rating:

$$I = \left[\frac{\Delta\theta - W_d[0,5 \cdot T_1 + n(T_2 + T_3 + T_4)]}{R \cdot T_1 + nR(1 + \lambda_1)T_2 + nR(1 + \lambda_1 + \lambda_2)(T_3 + T_4)} \right]^{0.5}$$

$\Delta\theta$	70 °C	$I = 572,4416208561 \text{ A}$
W_d	0,3043747162 W/m	
T_1	0,3936635440 K·m/W	
T_2	0,07330740743 K·m/W	
T_3	0 K·m/W	
T_4	0,7607356539 K·m/W	
R	$6,3680618159 \times 10^{-5} \Omega / \text{m}$	
n	3	
λ_1	0,04261472601	
λ_2	0,1370238656	

8. Iteration

Restart at step 1 until convergence is reached.

Convergence is reached at $I=571,4247945717$ A with the following intermediate results:

θ_s	81,7255586727 °C
R_s	$1,5764908698 \times 10^{-3} \Omega/\text{m}$
λ'_1	0,03581625490
m	0,01992775673
λ_0	0,0002521781385
Δ_1	0,08971912946
ρ_s	$3,5464613641 \times 10^{-8} \Omega \cdot \text{m}$
β_1	105,5072374667
g_s	1,0001514985
λ''_1	0,00680424980
F	0,9990354945
λ_1	0,04261394196
θ_A	76,8739695915 °C
R_A	$5,9018782531 \times 10^{-4} \Omega/\text{m}$
λ_2	0,1370192514
θ_M	68,7216049836 °C
T'_4	0,2180974302 K·m/W
T_4	0,7607649566 K·m/W

13.13 Final result

Final permissible current rating:

$$I = \left[\frac{\Delta\theta - W_d[0,5 \cdot T_1 + n(T_2 + T_3 + T_4)]}{R \cdot T_1 + nR(1 + \lambda_1)T_2 + nR(1 + \lambda_1 + \lambda_2)(T_3 + T_4)} \right]^{0.5}$$

$\Delta\theta$	70 °C	$I = 572,4336357055$ A
W_d	0,3043747162 W/m	
T_1	0,3936635440 K·m/W	
T_2	0,07330740743 K·m/W	
T_3	0 K·m/W	
T_4	0,7607649566 K·m/W	
R	$6,3680618159 \times 10^{-5} \Omega/\text{m}$	
n	3	
λ_1	0,04261394196	
λ_2	0,1370192514	

13.14 Calculation of losses

The following losses are per phase, to get the total losses in the cable they should be multiplied with $n=3$.

Conductor losses:

$$W_c = RI^2$$

R	$6,3680618159 \times 10^{-5} \Omega/\text{m}$	$W_c = 20,8668819794 \text{ W/m}$
I	$572,4336357055 \text{ A}$	

Lead sheath losses:

$$W_s = \lambda_1 RI^2$$

R	$6,3680618159 \times 10^{-5} \Omega/\text{m}$	$W_s = 0,8892200977 \text{ W/m}$
I	$572,4336357055 \text{ A}$	
λ_1	$0,04261394196$	

Dielectric losses:

-	See the previous calculation of W_D	$W_d = 0,3043747162 \text{ W/m}$
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Armour losses:

$$W_a = \lambda_2 RI^2$$

R	$6,3680618159 \times 10^{-5} \Omega/\text{m}$	$W_a = 2,8591645483 \text{ W/m}$
I	$572,4336357055 \text{ A}$	
λ_2	$0,1370192514$	

14. Case study 10: A 10kV three core PILC cable

14.1 Introduction

The selected cable type is a three-core paper insulated cable with aluminium sector shaped compacted conductor, mass impregnated paper insulation (unscreened), common lead sheath with steel tape armour and PVC outer sheath. The datasheet is depicted in Figure 43. The current rating will be determined for the following laying configuration:

- Cable installed direct buried.
- Depth of laying of the cable 1 m from earth surface to centre of the cable
- Soil temperature 15 °C
- Soil thermal resistivity 1,0 K.m/W
- Sheath solidly bonded
- One circuit thermally independent

All other relevant parameters are given in the input tables at each calculation step.

Cable type	:	PILC 8/10 kV 3x95Al	Cable data sheet	
Article number	:			
Standard	:	NEN 3172	18-feb-2019	

Cable construction				
Conductor	:	(1)	stranded compacted aluminium, sector shaped	
Insulation	:	(2)	impregnated paper	
Assembly	:	(3)	stranded cores, with paper fillers in the interstices	
Belt	:	(4)	impregnated paper	
Metallic sheath	:	(5)	lead	
Bedding layer	:	(6)	bitumized tapes and fibrous tape	
Armouring	:	(7)	galvanized steel tapes, 2x(50x0.8), gap : 25mm	
Outer Sheath	:	(8)	PVC sheath	

Main dimensions and weight		Nominal thickness		Diameter
Conductor, (height)	:	(1)		9,5 mm
Insulation	:			
- between two conductors	:	(2,3)	4,7 mm	
- between conductor and sheath	:	(4)	3,9 mm	
Metallic sheath	:	(5)	2,0 mm	40,0 mm
Bedding	:	(6)		46,0 mm
Armouring	:	(7)	1,6 mm	49,2
Outer sheath	:	(8)	2,8 mm	55,0 mm
Cable weight	:	appr.	6,3 kg/m	

Electrical data				
Rated voltages U_0/U (U_m)	:			8/10 (12) kV
Conductor				
Maximum DC resistance at 20° C	:			0,32 Ω /km
Maximum temperature under normal conditions	:			50 °C

Figure 43: Cable datasheet 3x95 mm² AL/PILC/PVC 8/10 (12) kV

14.2 Calculation of the conductor AC resistance at operation temperature

Table 14-1: Parameters needed to calculate the conductor AC resistance at operating temperature

Parameter	Symbol	Value	Unit	Reference
Conductor DC resistance @ 20°C	R_o	$0,320 \times 10^{-3}$	$\frac{\Omega}{m}$	Cable data sheet
Conductor diameter	d_c	9,5	mm	Cable datasheet
Conductor material		Aluminium		Cable datasheet
Type of conductor		sector shaped		Cable datasheet
Skin effect coefficient	k_s	1	-	[3], table 2
Proximity effect coefficient	k_p	0,8	-	[3], table 2
Conductor temperature coefficient at 20°C per Kelvin	α_{20}	$4,03 \times 10^{-3}$	-	[3], table 1
Conductor operating temperature	θ	50	°C	System design
Thickness of insulation between conductors	t	4,7	mm	Cable datasheet
System frequency	f	50	Hz	System design

Table 14-2: Sequential steps to calculate the conductor AC resistance at operating temperature

Step	Parameter	Symbol	Unit	Reference
1	Conductor DC resistance at operating temperature	R'	$\frac{\Omega}{m}$	[3], section 2.1.1
2	Argument of a Bessel function – skin effect	x_s	-	[3], section 2.1.2
3	Skin effect coefficient	y_s	-	[3], section 2.1.2
4	Determined diameter of equivalent circular conductor of the same cross sectional area and degree of compaction	d_x	mm	[3], section 2.1.4.2
5	Determined axial separation of conductors	s	mm	[3], section 2.1.4.2
6	Argument of a Bessel function – proximity effect	x_p	-	[3], section 2.1.4.1
7	Proximity effect coefficient	y_p	-	[3], section 2.1.4.1 [3], section 2.1.4.2
8	AC resistance of conductor at operating temperature	R	$\frac{\Omega}{m}$	[3], section 2.1.5

1. Conductor DC resistance at operating temperature R' :

$$R' = R_o [1 + \alpha_{20}(\theta - 20)] \left(\frac{\Omega}{m} \right)$$

R_o	$0,32 \times 10^{-3} \Omega/m$	$R' = 3,58688 \times 10^{-4} \Omega/m$
α_{20}	$4,03 \times 10^{-3}$	
θ	50°C	

2. Argument of a Bessel function used to calculate skin effect x_s :

$$x_s^2 = \frac{8\pi f}{R'} 10^{-7} k_s$$

f	50 Hz	$x_s^2 = 0,3503426547$
R'	$3,58688 \times 10^{-4} \Omega/m$	

k_s	1	
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3. Skin effect factor y_s :

$$\text{for } 0 < x_s \leq 2,8 \rightarrow y_s = \frac{x_s^4}{192 + 0,8x_s^4}$$

Three equations are given for calculating y_s . The use of these equations depends on the calculated value of x_s .

x_s^2	0,3503426547	$y_s = 6,3894394042 \times 10^{-4}$
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4. Determined diameter of equivalent circular conductor of the same cross sectional area and degree of compaction:

As this parameter is difficult to obtain because not all dimensions are known of the sector shaped conductor, the diameter of an equivalent circular conductor is taken from [8], Table C.2 with:

- minimum diameter of 95 mm² circular conductor: 11,0mm
- maximum diameter of 95 mm² circular conductor: 12,0mm

The chosen equivalent circular conductor diameter d_x influences the following parameters:

- the AC resistance of the conductor
- the armour losses factor, λ_2 , in the steel tapes
- the dielectric losses 'W_d'
- the thermal resistance per core between conductor and sheath

In order to select the appropriate equivalent conductor diameter, for finding the worst case current rating, is not strictly linked to the smallest (11,0 mm) or largest (12,0 mm): the value may lay between these two extremes. In this case, the smallest diameter is selected as this value gives the lowest rating.

$$d_c = d_x \text{ (mm)}$$

d_x	11,0 mm	$d_c = 11,0 \text{ mm}$
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5. Distance between conductor axes:

$$s = d_x + t \text{ (mm)}$$

d_x	11,0 mm	$s = 15,7 \text{ mm}$
t	4,7 mm	

6. Argument of a Bessel function used to calculate proximity effect x_p :

$$x_p^2 = \frac{8\pi f}{R'} 10^{-7} k_p$$

f	50 Hz	$x_p^2 = 0,2802741238$
R'	$3,58688 \times 10^{-4} \Omega/\text{m}$	
k_p	0,8	

7. Proximity effect factor y_p :

$$y_p = \frac{x_p^4}{192 + 0,8x_p^4} \left(\frac{d_c}{s} \right)^2 \left[0,312 \left(\frac{d_c}{s} \right)^2 + \frac{1,18}{\frac{x_p^4}{192 + 0,8x_p^4} + 0,27} \right] \times \frac{2}{3}$$

Note: as per [3] section 2.1.4.2, the value of y_p shall be two-thirds of the value calculated according to [3] section 2.1.4.1.

x_p^2	0,2802741238	$y_p = 6,0458825885 \times 10^{-4}$
d_c	11,0 mm	
s	15,7 mm	

8. AC resistance of conductor R :

$$R = R'(1 + 1,5(y_s + y_p)) \quad (\Omega/m)$$

R'	$3,58688 \times 10^{-4} \Omega/m$	$R = 3,5935706012 \times 10^{-4} \Omega/m$
y_s	$6,3894394042 \times 10^{-4}$	
y_p	$6,0458825885 \times 10^{-4}$	

Note: as noted in Guidance Point 25 for all armoured cable the skin effect factor and the proximity effect factor shall be corrected with a factor of 1,5. Inclusion of the layfactor for the conductor resistance is not needed as the conductor DC resistance @ 20°C is already given in the datasheet in Ω/km for the completed cable, thus already compensated by the conductor laylength.

14.3 Dielectric losses

Table 14-3: Parameters needed to calculate the dielectric loss per unit length per phase

Parameter	Symbol	Value	Unit	Reference
Equivalent circular conductor diam.	d_x	11,0	mm	Calculated
Insulation thickness between conductors	t_i	4,7	mm	Cable datasheet
External diameter of the lead sheath	D_s	40,0	mm	Cable datasheet
Nominal thickness of the lead sheath	t_s	2,0	mm	Cable datasheet
Relative permittivity of the insulation	ϵ	4,0	-	[3], table 3
Loss factor of the insulation	$\tan\delta$	0,01	-	[3], table 3
Voltage to earth	U_o	10 kV / $\sqrt{3}$	V	System design
System frequency	f	50	Hz	System design

Table 14-4: Sequential steps to calculate the dielectric loss per unit length per phase

Step	Parameter	Symbol	Unit	Reference
1	Circumscribing radius of three sector-shaped conductors	r_1	Mm	[3], list of symbols
2	Capacitance	C	$\frac{F}{m}$	See reference list, [7] and Guidance Point 19
3	Angular frequency of system	ω	-	[3], section 2.2
4	Dielectric loss per unit length in each phase	W_d	W/m	[3], section 2.2, note 1

Note 1: According to [3] table 3, the dielectric losses may be omitted in the calculation.

1. Circumscribing radius of three-sector shaped conductors

$$r_1 = \frac{D_s - 2 \cdot t_s - 2 \cdot t_{i2}}{2}$$

D_s	40,0 mm	$r_1 = 14,1 \text{ mm}$
t_s	2,0 mm	
t_{i2}	3,9 mm	

2. Capacitance C (for circular conductors):
Distance between the axes of conductors and the axis of the cable:

$$c = 0,55 r_1 + 0,29 t_{i1} \text{ (mm)}$$

r_1	14,1 mm	$c = 9,118 \text{ mm}$
t_{i1}	4,7 mm	

Diameter over belted insulation:

$$d_a = D_s - 2t_s \text{ (mm)}$$

D_s	40,0 mm	$d_a = 36,0 \text{ mm}$
t_s	2,0 mm	

$$C = \frac{\epsilon}{18 \cdot \ln \left[\frac{3 \cdot c^2 \cdot \left(\left(\frac{d_a}{2} \right)^2 - c^2 \right)^3}{\left(\frac{d_x}{2} \right)^2 \cdot \left[\left(\frac{d_a}{2} \right)^6 - c^6 \right]} \right]} \cdot 10^{-9} \left(\frac{F}{m} \right)$$

ϵ	4,0	$C = 3,5926331043 \times 10^{-10} \text{ F/m}$
c	9,118 mm	
d_x	11,0 mm	
d_a	36,0 mm	

3. Angular frequency of system:

$$\omega = 2\pi f \text{ (rad/s)}$$

f	50 Hz	$\omega = 314,159265359 \text{ rad/s}$
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4. Dielectric loss per unit length in each phase:

$$W_d = \omega C U_o^2 \tan \delta \text{ (W/m)}$$

C	$3,5926331043 \times 10^{-10} \text{ F/m}$	$W_d = 0,0376219659 \text{ W/m}$
ω	$314,159265359 \text{ rad/s}$	
U_o	$10000 \text{ V} / \sqrt{3} = 5,7735026919 \times 10^3$	
$\tan \delta$	0,01	

Inclusion of the layfactor is not considered here as it is assumed that the compensation for this has already been included in the capacitance calculation formula.

14.4 Loss factor for lead sheath

See Guidance Point 26 for details.

Table 14-5: Parameters needed to calculate the loss factor for lead sheath

Parameter	Symbol	Value	Unit	Reference
External diameter of lead sheath	D_s	40,0	mm	Cable data sheet
Thickness of the lead sheath	t_s	2,0	mm	Cable data sheet
Lead sheath resistivity at 20°C	ρ_s	21,4	Ωm	[3], table 1
Lead sheath temperature coefficient at 20°C per Kelvin	α_s	4,00	-	[3], table 1
Insulation thickness between conductors	t_{i1}	4,7	mm	Cable datasheet
Insulation thickness between conductor and metallic sheath	t_{i2}	3,9	mm	Cable datasheet
Diameter over armour bedding layer	D_b	46,0	mm	Cable datasheet
Armour tape width	w_A	50	mm	Cable datasheet
Armour tape thickness	t_A	0,8	mm	Cable datasheet
Armour tape gap	w_g	25	mm	Cable datasheet
Number of armour tapes	N	2	-	Cable datasheet
Permeability steel tape of armour	μ	300	-	[3], section 2.3.9
System frequency	f	50	Hz	System design
AC resistance of conductor at operating temp.	R	$3,5935706012 \times 10^{-4}$	Ω/m	Calculated
Dielectric losses per phase	W_d	0,0376219659	W/m	Calculated
Thermal resistance between conductor and screen	T_1	0,8045689810	K.m/W	Calculated
Permissible current rating	I	165,7415608133	A	Calculated

Table 14-6: Sequential steps to calculate the loss factor for lead sheath per phase

Step	Parameter	Symbol	Unit	Reference
1	Resistance of lead sheath at 20°C	R_{so}	Ω/m	-
2	Maximum operating temperature of the lead sheath, to be determined iterative	θ_{sc}	°C	[3], section 2.3
3	Resistance of lead sheath at operating temperature	R_s	Ω/m	[3], section 2.3
4	Loss factor circulating losses for lead sheath	λ'_1	-	[3], section 2.3.8
5	Loss factor for eddy current losses for lead sheath	λ''_1	-	[3], section 2.3.8
6	Additional factor because of the steel tape armour	f_A	-	[3], section 2.3.9
7	Determine total loss factor for lead sheath	λ_1	-	[3]

1. Resistance of lead sheath at 20°C:

Cross section of the lead sheath:

$$S = \pi \cdot t_s \cdot (D_s - t_s) \text{ (mm}^2\text{)}$$

t_s	2,0 mm	$S = 2,3876104167 \times 10^2 \text{ mm}^2$
D_s	40,0 mm	

Lead sheath resistance at 20°C:

$$R_{so} = \frac{\rho_s}{S} \text{ (}\Omega/m\text{)}$$

ρ_s	$21,4 \times 10^{-8} \Omega \cdot m$	$R_{so} = 8,9629362689 \times 10^{-4} \Omega/m$
S	$2,3876104167 \times 10^2 \text{ mm}^2$	

2. The operating temperature θ_{sc} (°C) of the lead sheath is given by:

$$\theta_{sc} = \theta - (I^2 R + 0,5 W_d) T_1 \quad (^\circ\text{C})$$

θ	50 °C	$\theta_{sc} = 42,0424550288 \text{ } ^\circ\text{C}$
I	165,7415608133 A	
R	$3,5935706012 \times 10^{-4} \Omega/m$	
W_d	0,0376219659 W/m	
T_1	0,804568981 K.m/W	

Note: θ_{sc} is the operating temperature of the cable lead sheath (°C). Because the temperature of the screen is a function of the current I , an iterative method must be used.

3. Cable lead sheath AC resistance at operating temperature θ_{sc} :

$$R_s = R_{so} [1 + \alpha_s (\theta_{sc} - 20)] \quad \left(\frac{\Omega}{m} \right)$$

R_{so}	$8,9629362689 \times 10^{-4} \Omega/m$	$R_s = 9,7531967474 \times 10^{-4} \Omega/m$
α_s	$4,0 \times 10^{-3}$	
θ_{sc}	42,0424550288 °C	

4. Loss factor λ'_1 for circulating losses lead sheath:

$$\lambda'_1 = 0$$

5. Loss factor λ''_1 for eddy current losses lead sheath:

Mean diameter of the lead sheath:

$$d = D_s - t_s \text{ (mm)}$$

D_s	40,0 mm	$d = 38,0 \text{ mm}$
t_s	2,0 mm	

$$\lambda''_1 = 0,94 \cdot \frac{R_s}{R} \cdot \left(\frac{2 \cdot r_1 + t_{i1}}{d} \right)^2 \cdot \frac{1}{1 + \left(\frac{R_s}{\omega} \cdot 10^7 \right)^2}$$

$$\omega = 2 \cdot \pi \cdot f$$

R_s	$9,7531967474 \times 10^{-4} \Omega/m$	$\lambda''_1 = 1,9821141851 \times 10^{-3}$
R	$3,5935706012 \times 10^{-4} \Omega/m$	
r_1	14,1 mm	
t_{i1}	4,7 mm	
d	38,0 mm	
ω	314,159265359	
f	50 Hz	

6. Additional factor because of the steel tape armour:

Cross section of the steel tape armour:

$$A = \frac{w_A}{w_A + w_g} \pi t_A N [D_b + N t_A] \quad (\text{mm}^2)$$

w_A	50 mm	$A_1 = 159,5091309983 \text{ mm}^2$
w_g	25 mm	
t_A	0,8 mm	
N	2	
D_b	46,0 mm	

Equivalent thickness of armour:

$$\delta = \frac{A}{\pi \cdot d_A}$$

A	159,5091309983 mm ²	$\delta = 1,0666666667 \text{ mm}$
d_A	47,6 mm	

Additional factor f_A for λ'_1 because of the steel tape armouring:

$$f_A = \left[1 + \left(\frac{d}{d_A} \right)^2 \cdot \frac{1}{1 + \frac{d_A}{\mu \cdot \delta}} \right]^2$$

d_A	47,6 mm	$f_A = 2,4173685240$
d	38,0 mm	
μ	300	
δ	1,0666666667 mm	

7. Add circulating losses and eddy current losses for foil to find λ_1 :

$$\lambda_1 = \lambda'_1 + f_A \cdot \lambda''_1$$

λ'_1	0	$\lambda_1 = 4,7915004419 \times 10^{-3}$
λ''_1	$1,9821141851 \times 10^{-3}$	
f_A	2,417368524	

14.5 Loss factor for the armouring.

See Guidance Point 26 for details.

Table 14-7: Parameters needed to calculate the loss factor for armouring layer

Parameter	Symbol	Value	Unit	Reference
Equivalent conductor diameter	d_x	11,0	mm	Calculated
Distance between conductor axes	s	15,7	mm	Calculated
Insulation thickness between conductors	t_{i1}	4,7	mm	Cable datasheet
Equivalent thickness of armour	δ	1,0666666667	mm	Calculated
Permeability steel tape of armour	μ	300	-	[3], section 2.3.9
System frequency	f	50	Hz	System design
AC resistance of conductor at operating temp.	R	$3,5935706012 \times 10^{-4}$	Ω/m	Calculated

Table 14-8: Sequential steps to calculate the loss factor for armouring layer

Step	Parameter	Symbol	Unit	Reference
1	Factor	k	-	[3], section 2.4.2.4
2	Hysteresis loss of steel tape armour	λ'_2	-	[3], section 2.4.2.4
3	Eddy current loss for steel tape armour	λ''_2		[3], section 2.4.2.4
4	Total armour losses	λ_2		[3], section 2.4.2.4

1. Factor k :

$$k = \frac{1}{1 + \frac{d_A}{\mu \cdot \delta}}$$

d_A	47,6 mm	$k = 0,8705114255$
μ	300	
δ	1,0666666667 mm	

2. Hysteresis loss of steel tape armour:

$$\lambda'_2 = \frac{s^2 \cdot k^2 \cdot 10^{-7}}{R \cdot d_A \cdot \delta}$$

s	15,7 mm	$\lambda'_2 = 1,0237322812 \times 10^{-3}$
k	0,8705114255	
R	$3,5935706012 \times 10^{-4} \text{ } \Omega/\text{m}$	
d_A	47,6 mm	
δ	$1,0666666667 \times 10^{-3} \text{ m}$	

3. Eddy current loss for steel tape armour:

$$\lambda''_2 = \frac{2,25 \cdot s^2 \cdot k^2 \cdot \delta \cdot 10^{-8}}{R \cdot d_A}$$

s	15,7 mm	$\lambda''_2 = 2,6207546399 \times 10^{-4}$
k	0,8705114255	
R	$3,5935706012 \times 10^{-4} \text{ } \Omega/\text{m}$	
d_A	47,6 mm	
δ	1,0666666667 mm	

4. Total armour losses:

$$\lambda_2 = \lambda'_2 + \lambda''_2$$

λ'_2	$1,0237322812 \times 10^{-3}$	$\lambda_2 = 1,2858077452 \times 10^{-3}$
λ''_2	$2,6207546399 \times 10^{-4}$	

14.6 Thermal resistance T_1 between conductor and screen

Table 14-9: Parameters needed to calculate the thermal resistance between conductor and screen

Parameter	Symbol	Value	Unit	Reference
Diameter of conductor	d_x	11,0	mm	Calculated
External diameter of lead sheath	D_s	40,0	mm	Cable data sheet
Thickness of the lead sheath	t_s	2,0	mm	Cable data sheet
Diameter over belted insulation	d_a	36,0	mm	Calculated
Radius of the circle circumscribing the conductors	r_1	14,1	mm	Calculated
Insulation thickness between conductors	t_{i1}	4,7	mm	Cable datasheet
Insulation thickness between conductor and metallic sheath	t_{i2}	3,9	mm	Cable datasheet
thermal resistivity of insulation	ρ_i	6	K.m/W	[4], table 1

Table 14-10: Sequential steps to calculate the thermal resistance between conductor and screen

Step	Parameter	Symbol	Unit	Reference
1	Diameter over belted insulation	d_a	mm	[4], section 4.1.2.2.6
2	radius of the circle circumscribing the conductors	r_1	mm	[4], section 4.1.2.2.6
3	Factor	F_2	—	[4], section 4.1.2.2.6
4	Factor	G	—	[4], section 4.1.2.2.6
5	Thermal resistance between conductor and screen	T_1	K.m/W	[4], section 4.1.2.2.1

1. Diameter over belted insulation:

$$d_a = D_s - 2 \cdot t_s \text{ (mm)}$$

D_s	40,0 mm	$d_a = 36,0 \text{ mm}$
t_s	2,0 mm	

2. Radius of the circle circumscribing the conductors:

$$r_1 = \frac{D_s - 2 \cdot t_s - 2 \cdot t_{i2}}{2} \text{ (mm)}$$

D_s	40,0 mm	$r_1 = 14,1 \text{ mm}$
t_s	2,0 mm	
t_{i2}	3,9 mm	

3. Factor F_2 :

$$F_2 = 1 + \frac{3 \cdot t_{i1}}{2 \cdot \pi \cdot (d_x + t_{i1}) - t_{i1}}$$

d_x	11,0 mm	$F_2 = 1,1500862048$
t_{i1}	4,7 mm	

4. Factor G :

$$G = 3 \cdot F_2 \cdot \ln\left(\frac{d_a}{2 \cdot r_1}\right)$$

F_2	1,1500862048	$G = 0,8425426667$
d_a	36,0 mm	
r_1	14,1 mm	

5. Thermal resistance between conductor and metallic sheath:

$$T_1 = \frac{\rho_i}{2 \cdot \pi} \cdot G \quad (K \cdot m/W)$$

ρ_i	6,0 K.m/W	$T_1 = 0,804568981 \text{ K.m/W}$
G	0,8425426667	

T_1 is not corrected with a layfactor, as it is calculated with a geometric factor and it is therefore assumed that the laylength has already been taken into account.

14.7 Thermal resistance T_2 between sheath and armour

Table 14-11: Parameters needed to calculate the thermal resistance between metallic sheath and armour

Parameter	Symbol	Value	Unit	Reference
Diameter over bedding layer	D_b	46,0	mm	Cable data sheet
Diameter over lead sheath	D_s	40,0	mm	Cable data sheet
Thermal resistivity of bedding layer	ρ_b	6,0	K.m/W	[4], table 1

Table 14-12: Sequential steps to calculate the thermal resistance between metallic sheath and armour

Step	Parameter	Symbol	Unit	Reference
1	Thermal resistance of bedding layer	T_2	K.m/W	[4], section 4.1.3.1

1. Thermal resistance of bedding layer underneath the armour layer:

$$T_2 = \frac{\rho_b}{2\pi} \ln\left(\frac{D_b}{D_s}\right) \quad (K \cdot \frac{m}{W})$$

ρ_b	6,0 K.m/W	$T_2 = 0,1334628239 \text{ K.m/W}$
D_b	46,0 mm	
D_s	40,0 mm	

14.8 Thermal resistance T_3 of outer covering

Table 14-13: Parameters needed to calculate the thermal resistance T_3 of the outer covering

Parameter	Symbol	Value	Unit	Reference
External diameter of amour layer	D_a	49,2	mm	Cable data sheet
External diameter of outer covering	D_e	55,0	mm	Cable data sheet
Thermal resistivity of outer covering	ρ_T	5,0	K.m/W	[4], table 1

Table 14-14: Sequential steps to calculate the thermal resistance T_3 of the outer covering

Step	Parameter	Symbol	Unit	Reference
1	Thermal resistance of outer covering	T_3	K.m/W	[4], section 4.1.4.1

1. Thermal resistance of outer covering:

$$T_3 = \frac{\rho_T}{2\pi} \ln\left(\frac{D_e}{D_a}\right) \quad (K \cdot m/W)$$

ρ_T	5,0 K.m/W	$T_3 = 0,0886807855 \text{ K.m/W}$
D_e	55,0 mm	
D_a	49,2 mm	

14.9 External thermal resistance T_4

Table 14-15: Parameters needed to calculate the external thermal resistance T_4

Parameter	Symbol	Value	Unit	Reference
External diameter of the cable	D_e	55	mm	Cable data sheet
Distance from the surface of the ground to the cable axis	L	1000	mm	System design
Thermal resistivity of the soil	ρ_T	1,0	K.m/W	System design

Table 14-16: Sequential steps to calculate to calculate the external thermal resistance T_4

Step	Parameter	Symbol	Unit	Reference
1	External thermal resistance	T_4	K.m/W	[4], section 4.2.2

1. External thermal resistance:

$$u = \frac{2L}{D_e}$$

L	1000 mm	$u = 36,3636363636$
D_e	55 mm	

$$T_4 = \frac{\rho_T}{2 \cdot \pi} \cdot \ln(u + \sqrt{u^2 - 1}) \quad (\text{K.m/W})$$

ρ_T	1,0 K.m/W	$T_4 = 0,6822220147 \text{ K.m/W}$
u	36,3636363636	

14.10 Permissible current rating

Table 14-17: Parameters needed to calculate the continuous current carrying capacity

Parameter	Symbol	Value	Unit	Reference
Conductor operating temperature	θ	50	°C	System design
Ambient temperature	θ_{amb}	15	°C	System design
Number of conductors in cable	n	3	-	Cable data sheet
AC resistance of conductor at operating temperature	R	$3,5935706012 \times 10^{-4}$	Ω/m	Calculated
Ratio of the total losses in metallic screen	λ_1	$4,7915004419 \times 10^{-3}$	-	Calculated
Ratio of the total losses in metallic foil	λ_2	$1,2858077452 \times 10^{-3}$	-	Calculated
Dielectric losses	W_d	0,0376219659	W/m	Calculated
Thermal resistance between conductor and metallic screen	T_1	0,8045689810	K.m/W	Calculated
Thermal resistance between metallic screen and metallic foil	T_2	0,1334628239	K.m/W	Calculated
Thermal resistance of external serving	T_3	0,0886807855	K.m/W	Calculated
Thermal resistance of surrounding medium	T_4	0,6822220147	K.m/W	Calculated

Table 14-18: Sequential steps to calculate the continuous current rating capacity

Step	Parameter	Symbol	Unit	Reference
1	Conductor temperature rise above ambient	$\Delta\theta$	K	[3], section 1.4.1.1
2	Permissible current rating	I	A	[3], section 1.4.1.1

1. Conductor temperature rise above ambient:

$$\Delta\theta = \theta - \theta_{amb}$$

θ	50 °C	$\Delta\theta = 35 \text{ K}$
θ_{amb}	15 °C	

2. Permissible current rating:

$$I = \left[\frac{\Delta\theta - W_d [0,5 \cdot T_1 + n(T_2 + T_3 + T_4)]}{R \cdot T_1 + nR(1 + \lambda_1)T_2 + nR(1 + \lambda_1 + \lambda_2)(T_3 + T_4)} \right]^{0,5}$$

$\Delta\theta$	35 K	$I = 165,7415608133 \text{ A}$
W_d	0,0376219659 W/m	
T_1	0,8045689810 K.m/W	
T_2	0,1334628239 K.m/W	
T_3	0,0886807855 K.m/W	
T_4	0,6822220147 K.m/W	
R	$3,5935706012 \times 10^{-4} \Omega/\text{m}$	
n	3	
λ_1	$4,7915004419 \times 10^{-3}$	
λ_2	$1,2858077452 \times 10^{-3}$	

14.11 Calculation of losses

Conductor losses:

$$P_c = RI^2 \text{ (W/m)}$$

R	$3,5935706012 \times 10^{-4} \Omega/\text{m}$	$P_c = 9,8716336641 \text{ W/m}$
I	165,7415608133 A	

Dielectric losses:

-	See the previous calculation of W_d , section 14.3	$W_d = 0,0376219659 \text{ W/m}$
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Metallic sheath losses:

$$P_{sheath} = \lambda_1 RI^2 \text{ (W/m)}$$

R	$3,5935706012 \times 10^{-4} \Omega/\text{m}$	$P_{sheath} = 0,0472999371 \text{ W/m}$
I	165,7415608133 A	
λ_1	$4,7915004419 \times 10^{-3}$	

Armouring tape losses:

$$P_{tape} = \lambda_2 R I^2 \text{ (W/m)}$$

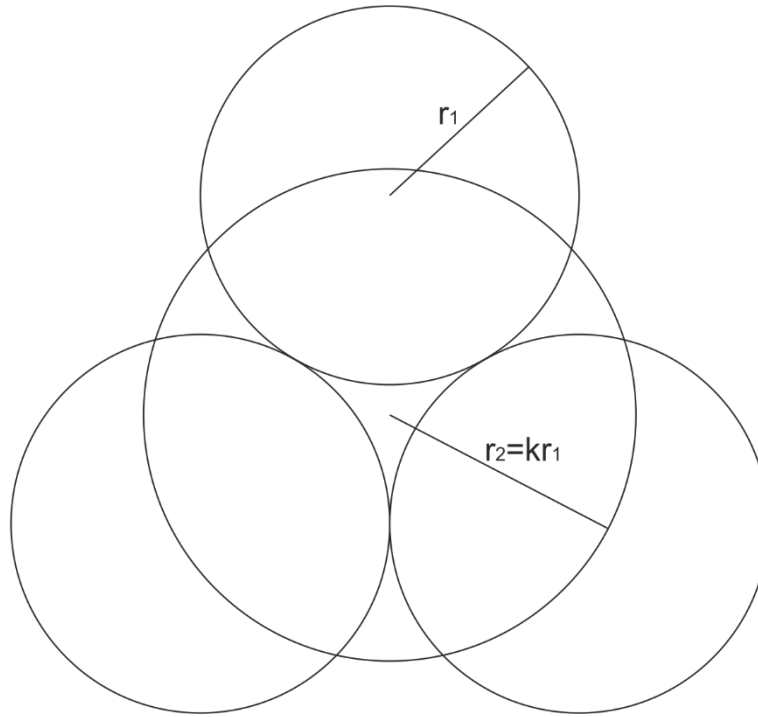
R	$3,5935706012 \times 10^{-4} \Omega/\text{m}$	$P_{tape} = 0,012693023 \text{ W/m}$
I	$165,7415608133 \text{ A}$	
λ_2	$1,2858077452 \times 10^{-3}$	

APPENDIX A. Mechanical neutral line deduction

The twisting of three cores are assumed to occur at the mechanical neutral line which is the location where a centre circle divides the cores into 2 equal large parts.

Assume that: $r_2 = kr_1$

[0,0] in the coordinate system is in the centre of the threecore cable.



The circle of one core can then be described as:

$$r_1^2 = x^2 + \left(y - \frac{2}{\sqrt{3}}r_1\right)^2$$

And the circle of the mechanical neutral line can be described as:

$$k^2 r_1^2 = x^2 + y^2$$

The Y coordinate where these two lines meet is:

$$k^2 r_1^2 - y^2 = r_1^2 - \left(y - \frac{2}{\sqrt{3}}r_1\right)^2 \rightarrow$$

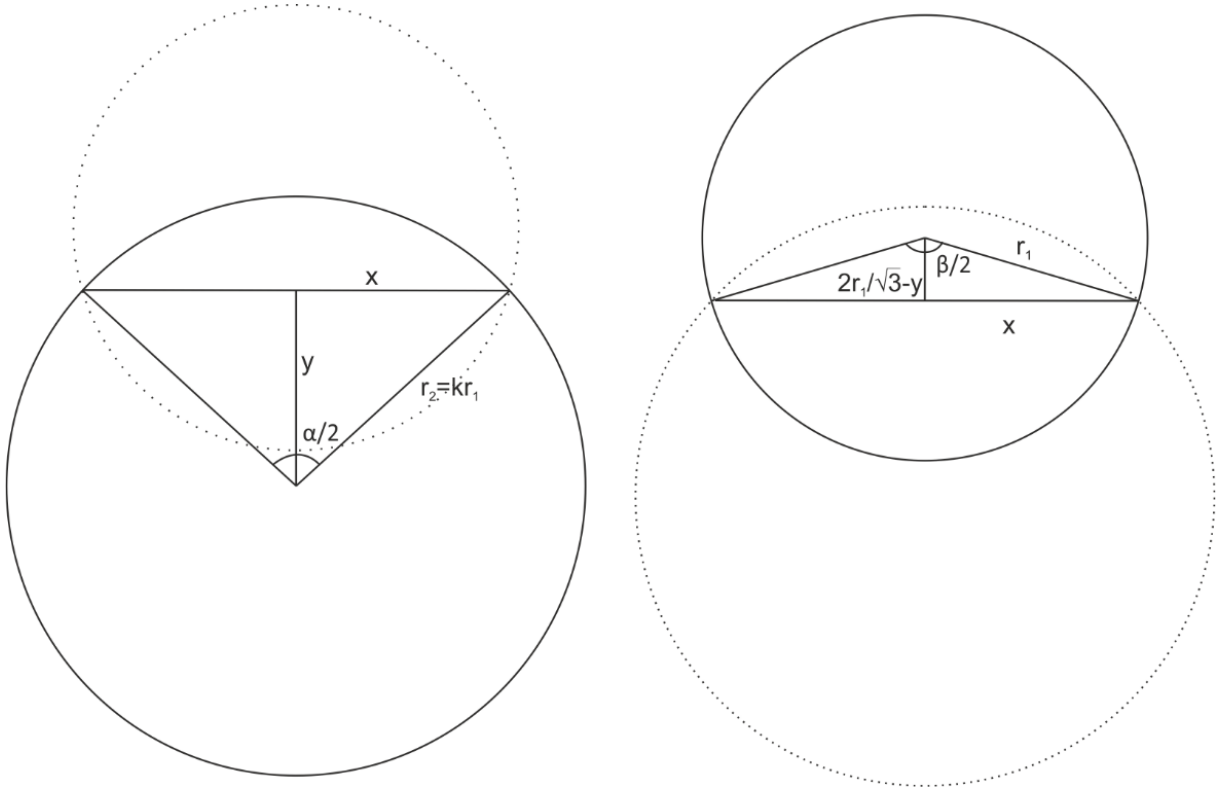
$$y = \frac{\sqrt{3}}{4} \left(k^2 + \frac{1}{3}\right) r_1$$

And the X coordinate is:

$$k^2 r_1^2 = x^2 + \left(\frac{\sqrt{3}}{4} \left(k^2 + \frac{1}{3}\right) r_1\right)^2 \rightarrow$$

$$x = \frac{\sqrt{3}}{4} r_1 \sqrt{-k^4 + \frac{14}{3}k^2 - \frac{1}{9}}$$

Within these circles we can then define some triangles with angles α and β .

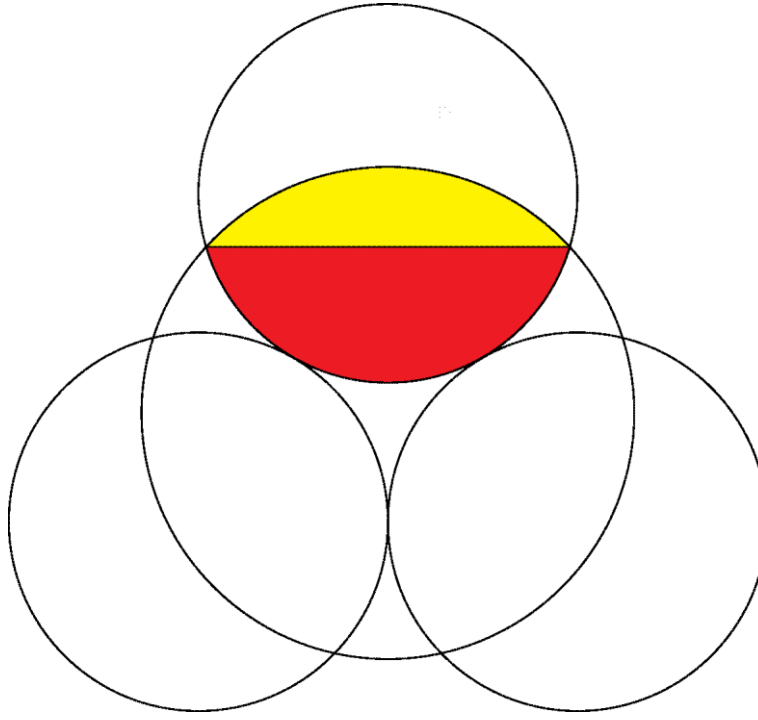


The angles can be described as:

$$\alpha = 2 \sin^{-1} \frac{x}{kr_1} = 2 \sin^{-1} \left[\frac{\sqrt{3}}{4k} \sqrt{-k^4 + \frac{14}{3}k^2 - \frac{1}{9}} \right]$$

$$\beta = 2 \sin^{-1} \frac{x}{r_1} = 2 \sin^{-1} \left[\frac{\sqrt{3}}{4} \sqrt{-k^4 + \frac{14}{3}k^2 - \frac{1}{9}} \right]$$

The area of the intersected part of one core can be calculated using circle segment for the two areas marked with red and yellow in below figure.



The area of a circle segment is calculated with area of the circle sector minus the area of the triangle from circle centre to the base of the circle segment. Using the known dimensions of the circles and triangles from previous figure an equation for these areas can be setup.

The total area of the circle segments should equal half of the area of the core circle, which gives us the following equation:

$$r_2^2 \pi \cdot \frac{\alpha}{2\pi} - 2 \frac{xy}{2} + r_1^2 \pi \cdot \frac{\beta}{2\pi} - 2 \frac{x \left(\frac{2}{\sqrt{3}} r_1 - y \right)}{2} = \frac{1}{2} \pi r_1^2$$

Replacing $r_2 = kr_1$:

$$k^2 r_1^2 \pi \cdot \frac{\alpha}{2\pi} - 2 \frac{xy}{2} + r_1^2 \pi \cdot \frac{\beta}{2\pi} - 2 \frac{x \left(\frac{2}{\sqrt{3}} r_1 - y \right)}{2} = \frac{1}{2} \pi r_1^2$$

$$\Leftrightarrow$$

$$k^2 r_1^2 \cdot \alpha + r_1^2 \cdot \beta - x \frac{4}{\sqrt{3}} r_1 = \pi r_1^2$$

Inserting x from previous definition gives:

$$k^2 r_1^2 \cdot \alpha + r_1^2 \cdot \beta - \frac{\sqrt{3}}{4} r_1 \sqrt{-k^4 + \frac{14}{3} k^2 - \frac{1}{9}} \cdot \frac{4}{\sqrt{3}} r_1 = \pi r_1^2$$

$$\Leftrightarrow$$

$$k^2 \cdot \alpha + \beta - \sqrt{-k^4 + \frac{14}{3} k^2 - \frac{1}{9}} = \pi$$

And inserting α and β gives:

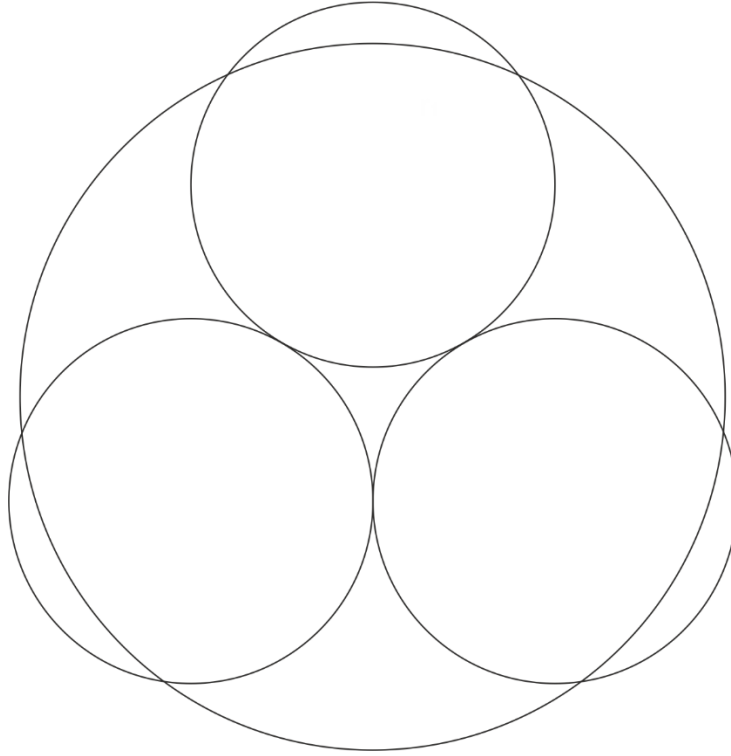
$$2k^2 \sin^{-1} \left[\frac{\sqrt{3}}{4k} \sqrt{-k^4 + \frac{14}{3} k^2 - \frac{1}{9}} \right] + 2 \sin^{-1} \left[\frac{\sqrt{3}}{4} \sqrt{-k^4 + \frac{14}{3} k^2 - \frac{1}{9}} \right] - \sqrt{-k^4 + \frac{14}{3} k^2 - \frac{1}{9}} = \pi$$

This equation can be numerical solved to:

$$k_1 = 1,293735422394694$$

$$k_2 = 1,937608650679144$$

k_2 results in a $y > \frac{2}{\sqrt{3}}r_1$ which is not a correct solution for this problem. See the figure below that shows how the circle is placed using k_2 .



The mechanical neutral line therefore is placed at $r_2 = 1,29r_1$

15. References

The following references have been used in this technical brochure:

- [1] A. Neher, M. McGrath, "The Calculation of the Temperature Rise and Load Capability of Cable Systems", *Transactions of the American Institute of Electrical Engineers. Part III – Power Apparatus and Systems*, vol. 76, no. 3, pp. 752-764, April 1957.
- [2] CIGRE "TB 640, A Guide for Rating Calculations of Insulated Cables", CIGRE, 2015-12.
- [3] "IEC 60287-1-1, Electric Cables - Calculation of the Current Rating - Part 1-1: Current Rating Equations (100 % load factor) and Calculation of Losses – General", IEC, 2014-11.
- [4] "IEC 60287-2-1, Electric Cables – Calculation of the Current Rating – Part 2-1: Thermal Resistance – Calculation of Thermal Resistance, Edition 2.0", IEC, 2015-2.
- [5] "IEC 60949 + Amendment : Calculation of Thermally Permissible Short-Circuit Currents, taking into account Non-Adiabatic Heating Effects", IEC, 2008.
- [6] D. Wald, G. Anders, "Presentation at ICC 2015 Tucson Fall meeting "Where the 3,5 came from ?"," 2015.
- [7] L. Heinhold, "Power Cables and Their Application: Part 1 Section 22", University of California, Wiley, ISBN 3800915359, 1996.
- [8] "IEC 60228, Conductors of Insulated Cables, Edition 3.0", IEC, 2004-11.
- [9] "IEC TR 62602, Conductors of Insulated Cables – Data for AWG and KCMIL sizes", IEC, 2009.
- [10] CIGRE, "TB 272, Large Cross-Sections and Composite Screens Design", CIGRE, 2005-6.
- [11] K. Miller, "Thesis "Sheath Currents, Sheath Losses, Induced Sheath Voltages and Apparent Conductor Impedances of Metal Sheathed Cables Carrying Alternating Currents"," University of Illinois, 1929.
- [12] K. Goddard and *et al*, "Induced Losses in Three-Core SL-Type High-Voltage Cables," *IEEE Transactions on Power Delivery*, vol. 30, No 3, 2015.
- [13] D. Palmgren and *et al*, "Armour Loss in Three-core Submarine XLPE Cables", *Jicable 2011 paper 177*, 2011.
- [14] D. Chatzipetros, J. Pilgrim, "Review of the Accuracy of Single-Core Equivalent Thermal Model (SCETM) for Offshore Wind Farm (OWF) Cables", in *IEEE*.
- [15] L. Ramirez, W. Kamara, G. Anders, "Thermal Analysis of 3-core SL-type Cables with Jacket around each Core using the IEC Standard", *Jicable 2019 paper A2.3*, 2019.
- [16] "IEC 60287-3-1, Electric Cables - Calculation of the Current Rating - Part 3-1: Sections on Operating Conditions – Reference Operating Conditions and Selection of Cable Type", IEC, 1999.
- [17] Morello, "The Calculation of the Current Flow in New Power Transmission Cables", *L'Elettrotecnica*, Vol. 46, No. 1, p.2, 1959.
- [18] "IEC 60502-2 Edition 3, Power cables with Extruded Insulation and their Accessories for Rated Voltages from 1 kV ($U_m=1,2\text{kV}$) up to 30 kV ($U_m = 36\text{kV}$) – Part 2 : Cables for Rated Voltages from 6kV ($U_m=7,2\text{ kV}$) up to 30 kV ($U_m = 36\text{ kV}$)", IEC, 2014.



ISBN : 978-2-85873-585-3